

Full Original Tail Return

Proof Supplement

Conductor transport, proper-source values, and the complete action

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This companion contains the complete full-return calculation, the four accepted original-tail proofs, and complete immediate provider arguments. The cumulative source retains the full transitive proof corpus. Original coordinates, masses, phases, periods, quotient values and endpoint weights are preserved. The source and formatting manifests record the unchanged accepted 811-page predecessor and each exact source interval used here. Human mathematical sources are cited at the relevant uses; the historical provenance reconstruction remains open with its assigned owner. This local successor has no newly assigned deposited DOI.

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1 The full original tail through conductor transport and all receiving values

This calculation receives the explicit original relation and simultaneous flag matrices NTG/SLC/STF/NFI into the original IPR, CPQ, NFR and CAI objects. It preserves the programme's recorded origin and the distinct creators and dates in its two deposited source notes [H19, H20]. The exact internal maps are stated with their source locators, while the human mathematical dependencies are cited at their uses below. All statements concern the original simple-quartet and fixed-period domain specified in each section. No additional multiplicity or asymptotic hypothesis is inserted.

1.1 The original data and the two exact tail matrices

Fix the actual simple quartet, its period and logarithm branch before the degree limit. Retain

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad L = q - v, \quad m = L - q', \quad s \in \{1, j\}. \quad (\text{FTR1})$$

The complete conductor is $E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{s,u,w} z}$, with $\beta_{a,u,w} = a/2 + (2u - a)\delta + i(2w - a)\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, and $\mu_d = E_A^{(d)}(0)$, $v = \text{ord}_0 E_A$, $\mu_v \neq 0$. The relations, including their scalar factors, are $g_r = \mathcal{T}_A(\chi_j S^r)$, $\ell_r = \mu_v \binom{q+r}{v}$ and $u_r = g_r/\ell_r$. Use the unchanged native line $S' = j/2 - 4 + iy$ and measure

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{FTR2})$$

The orthogonal polynomials, full norms and phases are exactly NTG3: $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$ and $h_{n,s} = (2\pi)^{s/2} n! (s/2)_n$. This is the stated Meixner–Pollaczek convention, with its factor $dy = 2dx$ and mass checked in HPS1–4 and NTG3 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. NTG4–12 and SLC3–20 evaluate every entry of the original graph $X_r = [x_0, \dots, x_r]$ and invertible functional matrix A . In particular the positive polynomial modification in that entry calculation is the complete confluent determinant of Szegő, including the original leading cofactor and both rows at every root [H24, Theorem 2.5, pp. 29–30]; the full substitution and its polynomial proof are SLC1–21. None of those factors is changed in this receiving calculation.

Let $\Omega_r = I_m + X_r X_r^*$, $\Omega_{-1} = I_m$, and keep any integer $0 \leq R < q$. For every $R \leq t \leq q$ put

$$G = \overline{\Omega_R}, \quad H_1 = A^* G A = D_{q+R}, \quad H = \text{diag}(H_1, 4), \quad Y_t = [x_{R+1}, \dots, x_t], \quad Z_t = G^{-1/2} \overline{Y_t}, \\ \mathcal{Z}_t = \text{diag}(Z_t, 4), \quad \mathcal{N}_t = \mathcal{Z}_t \mathcal{Z}_t^*, \quad \mathcal{U} = \mathbb{C}^{4m}. \quad (\text{FTR3})$$

Here $D_N = D_{j,N-v}^{(s)}$ is the original native analytic quotient form, and $\mathcal{D}_N = \text{diag}(D_N, 4)$. For each fixed original flag $X \subset \mathcal{U}$ with independent frame X_0 , define

$$\widehat{Z}_X = \text{diag}(G^{1/2} A, 4) X_0 (X_0^* H X_0)^{-1/2}, \quad \widehat{P}_X = \widehat{Z}_X \widehat{Z}_X^*, \\ \widehat{K}_{X,t} = \mathcal{N}_t^{1/2} \widehat{P}_X \mathcal{N}_t^{1/2}, \quad f_t(X) = \log \det(I_{4m} + \widehat{K}_{X,t}). \quad (\text{FTR4})$$

An empty frame has projector zero and determinant one. The square roots are the unique positive Hermitian square roots; all inverses are on the indicated positive forms. The $\widehat{P}_X$ notation in FTR4 means the explicit unitary change of flag coordinates in NFI7, and does not denote the later conductor-transported metric. The latter will have a superscript tr below.

For clarity the exact coordinate map behind FTR4 is $U_4 = \text{diag}(G^{1/2} A H_1^{-1/2}, 4)$, with inverse U_4^* . The row permutation and phases are NFI3's $\mathcal{R}_t = \text{diag}(\Theta_t^*, 4) \Pi_t$, $\Theta_t = \text{diag}_{r=R+1}^t (-i^{-(L+r)})$. NFI4 proves $\mathcal{R}_t C_t = \mathcal{Z}_t^* U_4$ directly from $\rho_{L+r}/\sqrt{1 + \zeta_{L+r}} = -i^{-(L+r)} x_r^\top A$. Thus no phase or transpose is inferred from a determinant equality. The range isometry in NFI9 identifies every nonzero eigenvalue in FTR4, with multiplicity, with its original stacked-row compression; its inverse

is the adjoint on the image, and both complementary kernels have zero contribution to the two logarithmic functions. It follows in the same original frame that

$$f_t(X) = \log \frac{\det(X_0^* \mathcal{D}_{q+t} X_0)}{\det(X_0^* H X_0)}. \quad (\text{FTR5})$$

This is precisely STF6 and NFI8–9, now with all input entries supplied by SLC16–21.

1.2 Every original quotient and both allocations

For a fixed original pair $K \subset Y$ choose its original kernel frame X_K and fixed lifts X_Q of its quotient values. Set

$$\begin{aligned} Z &= X_Q - X_K(X_K^* H X_K)^{-1} X_K^* H X_Q, & Q_0 &= Z^* H Z, \\ E &= H^{1/2} Z Q_0^{-1/2}, & \hat{E} &= U_4 E, \\ Q_t &= Q_0^{1/2} \left[I + \hat{E}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \hat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \hat{E} \right] Q_0^{1/2}, \\ \tau_{Y/K} &= \sum_{t=q-1, q} \log \frac{\det Q_t}{\det Q_0} = \sum_{t=q-1, q} \{f_t(Y) - f_t(K)\}. \end{aligned} \quad (\text{FTR6})$$

For $K = 0$ omit its kernel term; for $Y = K$ all quotient operators are the unique operators on the zero space and its determinant is one. These are the actual unchanged quotient values. Indeed every representative of x has transformed coordinate $EQ_0^{1/2}x + Z_K a$, where Z_K frames $H^{1/2}K$ orthonormally. Its new norm is $\|Q_0^{1/2}x\|^2 + \|a\|^2 + \|C_t EQ_0^{1/2}x + C_t Z_K a\|^2$. Completing the square gives $a = -(I + Z_K^* C_t^* C_t Z_K)^{-1} Z_K^* C_t^* C_t EQ_0^{1/2}x$. Substitution followed by the actual unitary row map proves the matrix in FTR6. In the fixed frame $[X_K, X_Q]$, block elimination gives $\det \text{Gram}(Y) = \det \text{Gram}(K) \det Q$, proving its determinant ratio. This is the complex Hermitian form of the full Schur minimum and factorization in Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]; all positive blocks, value fibres and cross terms were just specified. In particular $Q_t \succeq Q_0$ and $\tau_{Y/K} \geq 0$.

Use the original CPQ maps without choosing new subspaces:

$$\begin{aligned} B &= \text{im } B_{\text{cof}}, & I &= \text{im}(B_{\text{cof}} J_0), & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J, & e &= 0, 1. \end{aligned} \quad (\text{FTR7})$$

The exact pairs and resulting tail returns are

original form	(Y, K)	tail
Q_b^E	(W, W_i)	$\tau_b := \tau_{W/W_i}$
$Q_{\text{rem}, e}$	$(\mathcal{U}, T)$	$\tau_{\text{rem}} := \tau_{\mathcal{U}/T}$
$G_{s, e}$	(S, K_c)	$\tau_s := \tau_{S/K_c}$
A_{inv}	(I, K_i)	$\tau_A := \tau_{I/K_i}$
H_{inv}	(W_i, J)	$\tau_I := \tau_{W_i/J}$
$G_{t, e}$	(T, W)	$\tau_t := \tau_{T/W}$

(FTR8)

The first four are the same-value identifications CPQ7–10 and STF14; the fifth is proved below on the fixed original values U_R . The last is CPQ7's connecting target quotient. For each pair the complete metric, including its original frame factors, is FTR6. Only same-frame ratios annihilate these constant factors.

Define

$$\begin{aligned}
\tau_K &= \frac{1}{4}\tau_{\mathcal{U}/J}, & \tau_R &= \frac{1}{4} \sum_{t=q-1,q} f_t(J), & \tau_{\text{total}} &= \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_R)^2}, \\
(X_1, \dots, X_8) &= (T, K_c, W_i, K_i, W, S, J, I), \\
(\sigma_1, \dots, \sigma_8) &= (1, 1, 1, 1, -1, -1, -1, -1), \\
\tau_L &= \sum_{t=q-1,q} \sum_{a=1}^8 \sigma_a f_t(X_a).
\end{aligned} \tag{FTR9}$$

All eight positions are retained when subspaces coincide. Substituting FTR8 and cancelling identical terms proves the three simultaneous equalities

$$\begin{aligned}
\tau_L &= \tau_t - \tau_s + \tau_I - \tau_A, & 4\tau_K &= \tau_L + \tau_b + \tau_{\text{rem}} + \tau_s + \tau_A, \\
\tau_K + \tau_R &= \tau_{\text{total}}.
\end{aligned} \tag{FTR10}$$

The last equality also follows from NFI6 applied at both endpoints: $f_t(\mathcal{U}) = 4 \log(\det \Omega_t / \det \Omega_R)$. The two allocations here are the original value quotient and original kernel restriction, not selected fractions of their sum.

For a log determinant $h(N)$ in any fixed original value frame set

$$\begin{aligned}
\mathcal{R}h &= -h(q-1) - h(q) + h(2q-1) + h(2q), \\
\mathcal{R}_{\leq R}h &= 2h(q+R) - h(q) - h(q-1), \\
\mathcal{R}_{> R}h &= h(2q-1) + h(2q) - 2h(q+R).
\end{aligned} \tag{FTR11}$$

They sum exactly, and they are the original weighted increments with weights one at $r = 0, q$ and two at $0 < r < q$. For each form in FTR8 write $p_Y = \mathcal{R}_{\leq R} \log \det Q_Y$; write p_K, p_R for the original single-copy allocations and p_L for the signed original prefix. Then the complete native returns have the exact expressions

$$\begin{aligned}
V_Y &= p_Y + \tau_Y, & S_K &= p_K + \tau_K, & S_R &= p_R + \tau_R, & \mathcal{L}_e &= p_L + \tau_L, \\
4p_K &= p_L + p_b + p_{\text{rem}} + p_s + p_A, & p_K + p_R &= p_{\text{total}}.
\end{aligned} \tag{FTR12}$$

IPR3 proves telescoping and IPR4–8 proves these original fibre identities at each part of the window. Positivity of the same increment gives, with the literal NIB/IPR budget $\mathfrak{E}_{s,R} = \sum_{r=0}^R \vartheta_r E_{j,s,r}$,

$$\begin{aligned}
|p_L| &\leq 4\mathfrak{E}_{s,R}, & 0 &\leq p_Y \leq 4\mathfrak{E}_{s,R} \quad (Y = b, \text{rem}, s, A, I, t), \\
0 &\leq p_K, p_R \leq \mathfrak{E}_{s,R}, & 0 &\leq p_K + p_R \leq \mathfrak{E}_{s,R}.
\end{aligned} \tag{FTR13}$$

For the two added pairs I, t , this bound follows by applying the same complete minimum to their inclusions in the four-copy ambient space. Explicitly, extend the fixed pair frame to a full frame $C = [X_K, X_Q, X_{\text{out}}]$. Two full Schur eliminations give $|\det C|^2 \det F = \det(F|_K) \det Q_{Y/K,F} \det Q_{\mathcal{U}/Y,F}$ for each positive metric F . Under one positive update each same-value factor is nondecreasing: restriction preserves the quadratic inequality and minimization over the identical affine fibre preserves it as well. Their determinant ratios therefore are at least one and their product is the full ambient ratio. Each quotient ratio is thus at most that ambient ratio. The latter is $(1 + a_r)^4$ by the original four-copy update. Taking logs and the original weights proves FTR13. This argument retains the original pair and does not substitute a space of equal rank.

1.3 Directed conductor transport at the actual evaluation sites

Let a superscript tr mean the identical coefficient and value maps endowed with NCT's conductor-transported metric. For any one restriction or attained quotient Y of actual rank r_Y , write

$$d_Y(N) = \log \det Q_Y^{\text{tr}}(N) - \log \det Q_Y(N), \quad -l_Y(N) \leq d_Y(N) \leq u_Y(N). \tag{FTR14}$$

Here $l_Y(N) = \mathcal{L}_{j,N-v,s,r_Y}^{[4]}$ and $u_Y(N) = \mathcal{U}_{j,N-v,s,r_Y}^{[4]}$ are precisely NCT19–20 at any fixed admissible original SGT radius. Explicitly, with $h_4 = \min(r_Y, 4H_\rho)$, they are

$$\begin{aligned} u_Y(N) &= 4r_Y L_{s,N-v,v} + 2r_Y \log^+ F_{\rho,N-v}, \\ l_Y(N) &= 4r_Y L_{s,N-v,v} + 2h_4 \log^+ G_{0,N-v} + 2(r_Y - h_4) \log^+ V_{\rho,N-v}^{\text{cap}}. \end{aligned} \quad (\text{FTR15})$$

Every radius, principal part, exceptional rank, residue and diagonal factor in these already evaluated NCT constants is kept. They are zero at rank zero. The transport is the actual complex-linear functional map $\Lambda_F \mapsto \Lambda_{E_A F}$, in its ordinary- transpose matrix, together with its complete lower quotient, as in NCT11–14. In particular it applies to the same quotient values in FTR6, rather than to arbitrarily chosen Euclidean flag frames.

For $Z \in \{\text{full}, \leq R, > R\}$ let $\eta_Y^Z = \mathcal{R}_Z d_Y$, using $\mathcal{R}$ for “full”. The following directed costs preserve all literal endpoint signs:

Z	$E_Y^{Z,-}$	$E_Y^{Z,+}$
full	$u_Y(q-1) + u_Y(q) + l_Y(2q-1) + l_Y(2q)$	$l_Y(q-1) + l_Y(q) + u_Y(2q-1) + u_Y(2q)$
$\leq R$	$2l_Y(q+R) + u_Y(q) + u_Y(q-1)$	$2u_Y(q+R) + l_Y(q) + l_Y(q-1)$
$> R$	$l_Y(2q-1) + l_Y(2q) + 2u_Y(q+R)$	$u_Y(2q-1) + u_Y(2q) + 2l_Y(q+R)$

(FTR16)

Then $-E_Y^{Z,-} \leq \eta_Y^Z \leq E_Y^{Z,+}$ and $\eta_Y^{\text{full}} = \eta_Y^{\leq R} + \eta_Y^{> R}$ exactly. Proof: substitute FTR14 in each term of FTR11. A negative coefficient exchanges the roles of its two endpoint costs; coefficient two multiplies its actual cost by two. This proves every table entry, without adding per-row errors or assuming independent extremizers. At repeated sites one may combine their actual coefficients first; the displayed uncombined nonnegative costs remain valid.

For the signed eight-flag expression, write $\eta_{X,\text{res}}^Z$ for the residual of the restriction form on the subspace X , and put $\eta_L^Z = \sum_a \sigma_a \eta_{X_a,\text{res}}^Z$ and

$$E_L^{Z,-} = \sum_{\sigma_a=1} E_{X_a}^{Z,-} + \sum_{\sigma_a=-1} E_{X_a}^{Z,+}, \quad E_L^{Z,+} = \sum_{\sigma_a=1} E_{X_a}^{Z,+} + \sum_{\sigma_a=-1} E_{X_a}^{Z,-}. \quad (\text{FTR17})$$

The $E_{X_a}^{Z,\pm}$ in this display are the costs for those restriction forms. These bounds keep all eight labelled occurrences. Define $\eta_{\text{tar}}^Z = 4(S_K^{Z,\text{tr}} - S_K^Z)$ by applying FTR14–16 directly to the full four-copy target quotient of rank $4t_j$; define $\eta_J^Z = \eta_{J,\text{res}}^Z = 4(S_R^{Z,\text{tr}} - S_R^Z)$ by applying them directly to the actual four-copy kernel restriction. The transported and native determinant factorizations at the same sites give

$$\begin{aligned} \eta_{\text{tar}}^Z &= \eta_L^Z + \eta_b^Z + \eta_{\text{rem}}^Z + \eta_s^Z + \eta_A^Z, & \eta_{\text{tar}}^Z + \eta_J^Z &= \eta_{\mathcal{U},\text{res}}^Z, \\ \eta_L^Z &= \eta_t^Z - \eta_s^Z + \eta_I^Z - \eta_A^Z. \end{aligned} \quad (\text{FTR18})$$

Here η_I is the quotient residual for H_{inv} in FTR8; it is distinct from the restriction residual $\eta_{I,\text{res}}$. Similarly η_{tar} denotes the full target quotient, while $\eta_{T,\text{res}}$ is one of the eight restriction residuals. These are equalities of the actual residuals. For example their first line follows by subtracting the full CPQ identity for the native metric from that for the transported metric and applying the same $\mathcal{R}_Z$. Thus the direct target interval in FTR16 can be intersected with the interval supplied by its first identity; their extremizers have not been chosen separately.

The resulting complete returns are explicitly

$$\begin{aligned} V_Y^{\text{tr}} &= \tau_Y + p_Y + \eta_Y^{\text{full}}, & \mathcal{L}_e^{\text{tr}} &= \tau_L + p_L + \eta_L^{\text{full}}, \\ 4S_K^{\text{tr}} &= 4\tau_K + 4p_K + \eta_{\text{tar}}^{\text{full}}, & 4S_R^{\text{tr}} &= 4\tau_R + 4p_R + \eta_J^{\text{full}}. \end{aligned} \quad (\text{FTR19})$$

In particular each V_Y^{tr} lies in the finite interval $[\tau_Y - E_Y^{\text{full},-}, \tau_Y + 4\mathfrak{E}_{s,R} + E_Y^{\text{full},+}]$, intersected with $[0, \infty)$. The direct allocation intervals are obtained by using $4\mathfrak{E}_{s,R}$ and dividing their four-copy endpoints by four. The exact formulas FTR18–19 retain the correlation stronger than these separate enclosures.

1.4 The complete transported threshold profile from the common matrices

Define the original finite signed count

$$\mathcal{B}_{e,s}(R) = \sum_{t=q-1,q} \int_1^\infty \sum_{a=1}^8 \sigma_a \# \{ \lambda_b(\hat{K}_{X_{a,t}}) > u \} \frac{du}{u}. \quad (\text{FTR20})$$

Every matrix is the full original-data FTR3–4 matrix. Each integral terminates past its finite largest eigenvalue. Set $h(x) = \log(1+x) - \log \max(1, x)$ and retain the actual finite residual

$$\epsilon_{e,s}(R) = \sum_{t=q-1,q} \sum_{a=1}^8 \sigma_a \sum_b h(\lambda_b(\hat{K}_{X_{a,t}})). \quad (\text{FTR21})$$

The elementary identity $\log \max(1, \lambda) = \int_1^\infty \mathbf{1}_{u < \lambda} du/u$ proves $\tau_L = \mathcal{B}_{e,s}(R) + \epsilon_{e,s}(R)$ exactly. Let $d_t = \text{rank } \mathcal{N}_t$. The two actual nested chains $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$ imply

$$|\epsilon_{e,s}(R)| \leq \varepsilon_R := 2(d_{q-1} + d_q) \log 2 \leq 16m \log 2. \quad (\text{FTR22})$$

Here is the scalar error argument. Write $h = u - v$ with $u(x) = \log(1 + \min(x, 1))$ and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both are increasing from zero and bounded by $\log 2$. For four ordered eigenvalues, the sum of the first and last disjoint increments of each function is between zero and $\log 2$, by adding its middle nonnegative increment. Their difference lies between $-\log 2$ and $\log 2$. The two flag chains therefore cost $2 \log 2$ for each of the common d_t range coordinates. Matrix order gives the eigenvalue order by the minimum–maximum characterization of Hermitian eigenvalues, whose elementary proof is included in NCT18. This proves FTR22, including singular matrices and coincident flags.

Let $\mathcal{P}_e^{\text{tr}}$ be the original transported successive-row threshold sum NPK10, with its original endpoint weights. Define its actual difference from its own determinant return by $\chi_e^{\text{tr}} = \mathcal{P}_e^{\text{tr}} - \mathcal{L}_e^{\text{tr}}$. At each row the same argument uses four eigenvalues and two chains, so its cost is $8 \log 2$. The original weights sum to $2q$, hence

$$|\chi_e^{\text{tr}}| \leq 16q \log 2, \quad \boxed{\mathcal{P}_e^{\text{tr}} = \mathcal{B}_{e,s}(R) + p_L + \epsilon_{e,s}(R) + \eta_L^{\text{full}} + \chi_e^{\text{tr}}}. \quad (\text{FTR23})$$

This compares the complete transported row sum directly with the native simultaneous matrices. In particular its directed enclosure is

$$\begin{aligned} -4\mathfrak{E}_{s,R} - \varepsilon_R - E_L^{\text{full},-} - 16q \log 2 &\leq \mathcal{P}_e^{\text{tr}} - \mathcal{B}_{e,s}(R) \\ &\leq 4\mathfrak{E}_{s,R} + \varepsilon_R + E_L^{\text{full},+} + 16q \log 2. \end{aligned} \quad (\text{FTR24})$$

The bound for $\mathcal{L}_e^{\text{tr}} - \mathcal{B}_{e,s}(R)$ omits the last $16q \log 2$ on each side. The original native threshold-to-determinant error is not paid a second time in FTR24. Each residual in FTR23 has its specified actual expression; there is no assertion that individual endpoint and row eigenvalues are equal. Replacing “full” by either partial operator in FTR18–19 gives the corresponding complete partial identities.

At the already established $R_j = \lfloor j^{3/2}/\log j \rfloor$ with $R_j < q$, IPR9 gives $\mathfrak{E}_{s,R_j} = O_{\text{actual}}(j^{5/2} + j^3/\log j) = o(kq_k)$. The finite transport bounds may be replaced by the infimum of their valid endpoint intervals over admissible radii; equivalently use NPK11’s absolute bounds for the asymptotic statement. For each fixed radius NCT21 first takes the degree limit, and only afterwards admissible radii tend to one. Since all actual flag and quotient ranks here are $O(j)$, this yields $o(kq_k)$ for every four-site or three-site radius-infimum transport cost. No fixed-radius cost is asserted to be little-oh. Since $m = O(j)$ and $q = O(j^2)$, FTR22–24 prove for each original order and primitive index that

$$\mathcal{P}_e^{\text{tr}} - \mathcal{B}_{e,s}(R_j) = o(kq_k), \quad \mathcal{L}_e^{\text{tr}} - \mathcal{B}_{e,s}(R_j) = o(kq_k). \quad (\text{FTR25})$$

The exact quantity on the right of FTR20 remains the original correlated spectral count. This proof evaluates the full receiving map and finite costs; it assigns no new asymptotic coefficient to that count. The independently retained proper source and complete arithmetic action are treated next through their actual value maps.

All prefixes in this receiver also have an explicit finite graph formula, rather than merely the bounds FTR13. Set $\mathcal{A} = \text{diag}(A, 4)$ and $\mathcal{O}_r = \text{diag}(\overline{\Omega_r}, 4)$. For every unchanged restriction frame X_0 , put

$$p_{X,\text{res}} = 2 \log \det(X_0^* \mathcal{A}^* \mathcal{O}_R \mathcal{A} X_0) - \log \det(X_0^* \mathcal{A}^* \mathcal{O}_0 \mathcal{A} X_0) - \log \det(X_0^* \mathcal{A}^* \mathcal{A} X_0). \quad (\text{FTR26})$$

The target cutoffs $q+R, q, q-1$ correspond to native graph indices $R, 0, -1$, respectively, so NTG20 and FTR11 prove this formula including $R = 0$. Every quotient prefix is the difference $p_{Y,\text{res}} - p_{K,\text{res}}$ in its fixed pair; in particular p_I in FTR12–13 equals $p_{W_i,\text{res}} - p_{J,\text{res}}$. Every signed prefix is the original eight-term combination. Frame constants cancel because the coefficients in FTR26 sum to zero. Thus every prefix in FTR19 and FTR23 is itself evaluated in the same original finite coefficients.

For $R = R_j$ and $t = q - 1, q$ let $\mathfrak{U}_{R,t}$ be the minimum of the three explicitly evaluated bounds NTG31:

$$\mathfrak{U}_{R,t} = \min \left\{ \sum_{a=R+1}^t E_{j,s,a}, \quad d_0 \log \left(1 + \frac{\sum_{a=R+1}^t x_a^* \Omega_R^{-1} x_a}{d_0} \right), \right. \\ \left. \sum_{a=R+1}^t \log(1 + x_a^* \Omega_R^{-1} x_a) - \Delta_{R,t} \right\}, \quad d_0 = \min(m, t - R), \quad (\text{FTR27})$$

where $\Delta_{R,t}$ is the full adjacent-pair correction NTG28, including each actual $x_a^* \Omega_R^{-1} x_b$. At an empty endpoint set $\mathfrak{U}_{R,R} = 0$. Define $\mathfrak{U}_R = \mathfrak{U}_{R,q-1} + \mathfrak{U}_{R,q}$. The complete Schur determinant NTG25 proves $0 \leq \tau_{\text{total}} \leq \mathfrak{U}_R$. Each quotient ratio is at most its four-copy ambient ratio, as proved for FTR13, and each allocation is nonnegative with sum τ_{total} . The original two-chain signed estimate NIB16 therefore yields simultaneously

$$0 \leq \tau_K, \tau_R \leq \mathfrak{U}_R, \quad 0 \leq \tau_Y \leq 4\mathfrak{U}_R \quad (Y = b, \text{rem}, s, A, I, t), \\ |\tau_L| \leq 4\mathfrak{U}_R, \quad |\mathcal{B}_{e,s}(R)| \leq 4\mathfrak{U}_R + \varepsilon_R. \quad (\text{FTR28})$$

For the signed bound, the same-chain disjoint-increment argument applied to $\log(1+x)$ at an original row bounds the absolute signed increment by $4\log(1+a_r)$; summing the tail and using NTG32 proves it directly. The count bound follows from FTR21–22. These estimates may be inserted in the directed intervals FTR19 and FTR24 without deleting any of their correlated residuals. No assertion that $\mathfrak{U}_R = o(kq_k)$ is used here.

Finally retain both original source orders, with a subscript 1 or j on each actual matrix and residual. For every positive quotient in FTR8 the exact order difference is

$$V_{Y,j}^{\text{tr}} - V_{Y,1}^{\text{tr}} = (\tau_{Y,j} - \tau_{Y,1}) + (p_{Y,j} - p_{Y,1}) + (\eta_{Y,j}^{\text{full}} - \eta_{Y,1}^{\text{full}}). \quad (\text{FTR29})$$

Its excess above the displayed tail difference lies in

$$[-4\mathfrak{E}_{1,R} - E_{Y,j}^{\text{full},-} - E_{Y,1}^{\text{full},+}, \\ 4\mathfrak{E}_{j,R} + E_{Y,j}^{\text{full},+} + E_{Y,1}^{\text{full},-}]. \quad (\text{FTR30})$$

This follows by subtracting the two actual FTR19 equalities, using nonnegative prefixes FTR13 and directed errors FTR16. For the signed profile use its two actual FTR23 identities instead; the prefix difference has absolute bound $4(\mathfrak{E}_{j,R} + \mathfrak{E}_{1,R})$, its two simultaneous errors cost $\varepsilon_{R,j} + \varepsilon_{R,1}$, its directed transport costs have the same subtracted signs as FTR30, and the two original threshold residuals cost $32q \log 2$. This propagates the new matrices through the earlier two-order receivers without treating a known difference as an absolute value.

2 The original invariant value fibre and its full proper-source return

This calculation propagates the original ITL1–6, NFR1–12, CPQ2–20 and EFP19,28–31 through the simultaneous matrices STF4–18 and the exact relation-graph map NFI1–10. The classical quadratic minimum and block determinant identity used at each quotient are those in Boyd and Vandenberghe, Appendix A.5.5, p. 650, equations (A.13)–(A.14), and Appendix C.4.1, p. 673, equations (C.4)–(C.5) [H23]. The complex Hermitian calculation, every fibre and every map needed here are given below. The original programme maps and results remain attributed to their cited ITL/NFR/CPQ/EFP and NTG/STF/NFI providers; the classical citation supplies the finite linear algebra operation.

2.1 The unchanged domain and actual quotient values

Retain the original hypothetical simple quartet, five-orbit period, logarithm branch, complete nonreduced conductor E_A , its order $v = \text{ord}_0 E_A$ and the full nonzero derivative $\mu_v = E_A^{(v)}(0)$. The data and original rank-valid domain are those of ITL1–3 and CPQ1–3, including the actual period-domain condition $|u| \geq R_*$ wherever it is required by those maps. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_a &= (a + 1)^2, & q'_a &= (a - 7)^2, & m_a &= q_a - v - q'_a, \\ c'_a &= a/2 - 4, & \alpha_A &= v!/\mu_v, & p = 0 &: (s_0, s_1) = (1, 1), & p = 1 &: (s_0, s_1) = (k, j), \\ q &= q_j, & N &\in [q - 1, 2q] \cap \mathbb{Z}, & D &= N - v. \end{aligned} \quad (\text{PRD1})$$

The simple-quartet domain fixes q_a as displayed; no change of quartet multiplicity is made. Its conductor order v is retained even when the selected period makes it nonzero. Every source-order identity below holds separately for $p = 0, 1$. Each connecting identity holds separately for both primitive indices $e = 0, 1$.

Use the full original coefficient multiplier, maps and spaces

$$\begin{aligned} \mathcal{V} &= \mathbb{C}^{m_k}, \quad \mathcal{U} = \mathbb{C}^{4m_j}, \quad V = \mathbb{C}^{j_{0,k}}, \quad J_0 : V \longrightarrow \mathcal{V}, \\ T_a &= \alpha_A N_a Z_{\text{exp},a}, \quad M_i = T_j^{-1} m_{i,k} T_k, \quad m_{i,k}[f] = [x_i^4 f], \quad B_{\text{cof}} = \text{stack}_{i=1}^4 M_i, \\ F &= B_{\text{cof}} J_0 : V \longrightarrow \mathcal{U}, \quad \mathbb{B} = \text{diag}(B_j, 4) : \mathcal{U} \longrightarrow \mathbb{C}^{4t_j}, \quad J = \ker \mathbb{B}, \\ X &= \mathbb{B}F, \quad E = \ker X, \quad I = \text{im } F, \quad K_i = I \cap J, \quad W_i = I + J. \end{aligned} \quad (\text{PRD2})$$

The maps F and J_0 are injective and B_j is onto by the original ITL2–3/CPQ2 proof. Their ordinary-transpose coefficient duality is unchanged; stars below are complex adjoints only in Hermitian metric expressions. In particular the original maps $N_a, Z_{\text{exp},a}$ and the multiplier $m_{i,k}$ remain in PRD2.

Choose exactly the original fixed ITL4 frame $Z = [Z_E, Z_R]$ of V , where Z_E spans E . Write

$$e_E = \dim E, \quad r_I = \dim V - e_E, \quad F_E = FZ_E, \quad F_R = FZ_R, \quad U_R = XZ_R = \mathbb{B}F_R. \quad (\text{PRD3})$$

The frame F_R is the original coefficient lift of U_R . The frame U_R is injective: if $XZ_R x = 0$, then $Z_R x \in E$, while the complement property of Z forces $x = 0$. Also

$$FE = K_i, \quad I = FE \oplus \text{im } F_R, \quad W_i = J \oplus \text{im } F_R. \quad (\text{PRD4})$$

Indeed $Fa \in J$ is equivalent to $Xa = 0$, which is $a \in E$. This proves the first equality. Decomposing $a = Z_E b + Z_R x$ proves the second, using injectivity of F ; adding J proves the third, and its directness follows by applying $\mathbb{B}$.

For complete value maps let $L_R = (U_R^* U_R)^{-1} U_R^*$, the fixed left inverse on $\text{im } U_R$. The two maps and their inverses are

$$\begin{aligned} I/K_i &\longrightarrow \mathbb{C}^{r_I}, & [i] &\longmapsto L_R \mathbb{B}i, & x &\longmapsto [F_R x] & (\text{inverse}), \\ W_i/J &\longrightarrow \mathbb{C}^{r_I}, & [w] &\longmapsto L_R \mathbb{B}w, & x &\longmapsto [F_R x] & (\text{inverse}). \end{aligned} \quad (\text{PRD5})$$

These rules are well defined because $\mathbb{B}$ kills the indicated kernels; PRD4 proves that they are mutual inverses. Composing either with $x \mapsto U_R x$ identifies its values with the very same original invariant image in $\mathbb{C}^{4t_j}$. The inclusion $I \hookrightarrow W_i$ therefore induces the exact isomorphism

$$\iota_I : I/K_i \longrightarrow W_i/J, \quad [i] \longmapsto [i]. \quad (\text{PRD6})$$

Its inverse sends $[i+j]$ to $[i]$: if two decompositions represent the same class, their I components differ by an element of $I \cap J = K_i$. Both maps are the identity on the specified $\mathbb{C}^{r_I}$ values. This proves their relationship before any metrics or determinant ratios are introduced.

2.2 All three full metrics and their attained lifts

Put

$$\mathcal{D}_N^p = \text{diag}(D_{j,N-v}^{(s_1)}, 4), \quad G_N^p = J_0^* D_{k,N-v}^{(s_0)} J_0. \quad (\text{PRD7})$$

Each $D_{a,D}^{(s)}$ is the original positive coefficient metric after its entire lower-evaluation minimum. More explicitly, with the original lower and analytic-numerator rows through degree D ,

$$D_{a,D}^{(s)} = T_{a,D}^* [I - V_{a,D} (V_{a,D}^* V_{a,D})^{-1} V_{a,D}^*] T_{a,D}. \quad (\text{PRD8})$$

This is the MNT/DNE form used in ITL1 and STF2. Its rows retain the original centre c'_a , conductor division moments and phases, and the full squared norms $h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$. PRD8 requires the original degree- k rows when $a = k$ and degree- j rows when $a = j$. Positivity and full lower rank on PRD1 are the original MNT/DNE rank results; no new rank assumption is introduced.

Here is the precise complex Hermitian version of the classical minimum used below [H23, Appendix A.5.5, p. 650]. For a positive Hermitian P , an independent kernel frame K_0 and value lifts L_0 independent modulo that kernel, set

$$\begin{aligned} \text{Sch}(P; K_0, L_0) &= L_0^* P L_0 - L_0^* P K_0 (K_0^* P K_0)^{-1} K_0^* P L_0, \\ z_P(x) &= L_0 x - K_0 (K_0^* P K_0)^{-1} K_0^* P L_0 x. \end{aligned} \quad (\text{PRD9})$$

Every representative is $L_0 x + K_0 a$. Its squared norm is

$$(a + (K_0^* P K_0)^{-1} K_0^* P L_0 x)^* (K_0^* P K_0) (a + (K_0^* P K_0)^{-1} K_0^* P L_0 x) + x^* \text{Sch}(P; K_0, L_0) x.$$

The first term has a unique zero, proving PRD9's attained minimum and lift. A triangular change of determinant one from $[K_0, L_0]$ to $[K_0, z_P]$ diagonalizes its Gram by blocks; therefore its determinant is $\det(K_0^* P K_0) \det \text{Sch}(P; K_0, L_0)$. This proves the complex form of the cited Schur operation with the actual affine fibre and all cross terms retained.

Choose any fixed independent frame J_* for the entire J . The three forms on the same original Z_R values are exactly

$$\begin{aligned} Q_N^p &= \text{Sch}(G_N^p; Z_E, Z_R), \\ A_N^p &= \text{Sch}(\mathcal{D}_N^p; F_E, F_R), \\ H_N^p &= \text{Sch}(\mathcal{D}_N^p; J_*, F_R) = U_R^* (\mathbb{B}(\mathcal{D}_N^p)^{-1} \mathbb{B}^*)^{-1} U_R. \end{aligned} \quad (\text{PRD10})$$

Thus A_N^p is precisely A_{inv}^p for (I, K_i) , and H_N^p is precisely H_{inv}^p for (W_i, J) . To prove the last equality directly, for a target value y the vector

$$z = (\mathcal{D}_N^p)^{-1} \mathbb{B}^* (\mathbb{B}(\mathcal{D}_N^p)^{-1} \mathbb{B}^*)^{-1} y$$

satisfies $\mathbb{B}z = y$ and is $\mathcal{D}_N^p$ -orthogonal to J . Every other lift differs from it by J , so its norm is the full minimum. Taking $y = U_R x$ proves PRD10 and agrees with the third attained lift in PRD9. Because $K_i \subset J$, the two target-order minima also give $H_N^p \preceq A_N^p$. The same-valued metric map

in PRD6 is exactly this contraction; the independent source metric is Q_N^p with the first formula in PRD10.

In particular, every entry of Q_N^p is the finite original degree- k expression

$$Q_N^p = Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R - Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_E (Z_E^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_E)^{-1} Z_E^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R. \quad (\text{PRD11})$$

For $E = 0$ every term containing its frame is absent. For $r_I = 0$ all value forms are the unique operators on the zero space, their determinants are 1 and all value returns below are 0. These conventions apply throughout, including PRD5's empty maps.

2.3 Literal two-endpoint matrices on those same values

Fix an original integer $0 \leq R < q$, put $N_* = q + R$ and $\mathcal{H} = \mathcal{D}_{N_*}^p$. Use the already calculated NFI matrices

$$\mathcal{Z}_t = \text{diag}(\overline{\Omega_R}^{-1/2} \overline{Y}_t, 4), \quad Y_t = [x_{R+1}, \dots, x_t], \quad t = q-1, q, \quad (\text{PRD12})$$

including the empty tail at $R = t = q-1$. These x_r, Ω_R are the full NTG relation coefficients at target degree j and source order s_1 ; the exact phase/permutation maps NFI3–4 are retained. Let U_4 be the NFI1 unitary. For either original pair $(Y, K) = (W_i, J)$ or (I, K_i) , use the value lifts F_R and respectively the kernel frames J_* or F_E , and define

$$\begin{aligned} Z_{Y/K} &= F_R - K_*(K_*^* \mathcal{H} K_*)^{-1} K_*^* \mathcal{H} F_R, & Q_{Y/K,*} &= Z_{Y/K}^* \mathcal{H} Z_{Y/K}, \\ E_{Y/K} &= \mathcal{H}^{1/2} Z_{Y/K} Q_{Y/K,*}^{-1/2}, & \hat{E}_{Y/K} &= U_4 E_{Y/K}, & \hat{P}_K &= U_4 P_K U_4^*, \\ \mathcal{J}_{Y/K,t} &= I_{r_I} + \hat{E}_{Y/K}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \hat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \hat{E}_{Y/K}. \end{aligned} \quad (\text{PRD13})$$

The original P_K is STF5's projector onto $\mathcal{H}^{1/2} K$. Every displayed kernel term is omitted when the kernel is zero. The same complete quadratic minimum PRD9, now after stacking the actual tail rows, gives STF10; conjugating its row space by the explicit unitary NFI3 gives

$$H_{q+t}^p = (H_{N_*}^p)^{1/2} \mathcal{J}_{W_i/J,t} (H_{N_*}^p)^{1/2}, \quad A_{q+t}^p = (A_{N_*}^p)^{1/2} \mathcal{J}_{I/K_i,t} (A_{N_*}^p)^{1/2}. \quad (\text{PRD14})$$

For clarity, the minimization giving the middle factor has the literal form $\min_a \{\|a\|^2 + \|y + Ba\|^2\}$, with $B = C_t Z_K$ and $y = C_t E_{Y/K} Q_{Y/K,*}^{1/2} x$. Here C_t is the STF4 stack of the actual original row updates, right-multiplied by $\mathcal{H}^{-1/2}$, and Z_K is any orthonormal column frame of $\mathcal{H}^{1/2} K$. Its minimizer is $-(I + B^* B)^{-1} B^* y$; multiplication proves $I - B(I + B^* B)^{-1} B^* = (I + B B^*)^{-1}$, yielding PRD13. The NFI3 row conjugation changes C_t into $\mathcal{Z}_t^* U_4$. This proves PRD14 with every cross term, including singular tail matrices. Its $Q_{Y/K,*}^{1/2}$ factors remain on both sides in the original quotient-value space; they are not scalar masses.

Let $F_t(Y)$ retain exactly STF5/NFI8's restriction log ratio. Applying the block determinant identity proved after PRD9 to the fixed frame $[K_*, F_R]$ at the two cutoffs gives

$$\begin{aligned} \log \det \mathcal{J}_{W_i/J,t} &= F_t(W_i) - F_t(J), \\ \log \det \mathcal{J}_{I/K_i,t} &= F_t(I) - F_t(K_i). \end{aligned} \quad (\text{PRD15})$$

Every fixed frame determinant cancels in its own two-cutoff ratio. Define the two exact tail values

$$\tau_I^p = \sum_{t=q-1,q} \log \det \mathcal{J}_{W_i/J,t}, \quad \tau_A^p = \sum_{t=q-1,q} \log \det \mathcal{J}_{I/K_i,t}. \quad (\text{PRD16})$$

They are nonnegative because PRD13 is the identity plus a positive semidefinite matrix. This is the same value for the target invariant form whether obtained by its complete J minimum or by its restriction to the original image U_R .

2.4 The full relative return, with the degree- k source intact

For any value form Z_N let

$$\Pi_Z^p = 2 \log \det Z_{N_*}^p - \log \det Z_q^p - \log \det Z_{q-1}^p. \quad (\text{PRD17})$$

Use this definition for $Z = H, A$, writing their prefixes as Π_I^p, Π_A^p . Direct addition of PRD14's two logarithmic determinants and PRD17 gives

$$V_I^p = \Pi_I^p + \tau_I^p, \quad V_A^p = \Pi_A^p + \tau_A^p. \quad (\text{PRD18})$$

These are the original four-endpoint returns with signs $(-1, -1, +1, +1)$ at $N = (q-1, q, 2q-1, 2q)$, not new windows. Define the proper-source term directly from PRD11 at those four original arguments:

$$V_Q^p = \log \frac{\det Q_{2q-1}^p \det Q_{2q}^p}{\det Q_{q-1}^p \det Q_q^p}, \quad \mathcal{R}_{X,N}^p = (Q_N^p)^{-1/2} H_N^p (Q_N^p)^{-1/2}. \quad (\text{PRD19})$$

Thus at each upper endpoint its complete relative positive operator is

$$\mathcal{R}_{X,q+t}^p = (Q_{q+t}^p)^{-1/2} (H_{N_*}^p)^{1/2} \mathcal{J}_{W_i/J,t} (H_{N_*}^p)^{1/2} (Q_{q+t}^p)^{-1/2}. \quad (\text{PRD20})$$

The order of these generally noncommuting matrices is part of the formula. Taking determinants, with the exact inverse square root on both sides, proves the full finite receiving identities

$$\begin{aligned} \mathcal{M}^p &= \mathcal{R} \log \det \mathcal{R}_X^p = \Pi_I^p + \tau_I^p - V_Q^p, \\ \mathcal{R} \log \frac{\det A_{\text{inv}}^p}{\det Q_{\text{inv}}^p} &= \Pi_A^p + \tau_A^p - V_Q^p, \\ \mathcal{R} \log \omega_{\text{inv}}^p &= (\Pi_I^p - \Pi_A^p) + (\tau_I^p - \tau_A^p). \end{aligned} \quad (\text{PRD21})$$

The first line is the NFR matched return, the second its actual same-value source correction, and the third the original last target contraction. Their equality uses the proved maps PRD5–6, not an identification of the degree- k and degree- j metrics.

The proper source in every term of PRD19 is evaluated at

$$(D_0, D_1, D_2, D_3) = (q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v). \quad (\text{PRD22})$$

Its degree is k , centre $c'_k = k/2 - 4$, and order $s_0 = 1$ or k . Writing $b = q_j - q_k = 8k + 24$ proves that its two low cutoffs are the standard degree- k low cutoffs plus b , and the two high cutoffs are the standard degree- k high cutoffs plus $2b$. The target matrices in PRD12–16 instead retain degree j , centre $c'_j = j/2 - 4$ and order $s_1 = 1$ or j . Consequently PRD21 keeps the exact same Z_R values while explicitly retaining both original polynomial row families and all four source displacements. No estimate for V_Q^p has been inserted in this identity.

The exact target-prefix cost can also be retained as a finite interval. For any original $K \subset Y \subset \mathcal{U}$, restriction and minimum over the fixed fibre preserve the increasing metric $\mathcal{D}_{N_*}^p$. Complete a fixed frame for K , then the specified lifts for Y/K , then lifts for $\mathcal{U}/Y$. Successive Schur factorization expresses the ambient determinant increment as the sum of three nonnegative log determinant increments. Thus the quotient increment lies between zero and the ambient increment. For either pair in PRD13 the latter is $4 \log(1 + a_r)$ by the four-copy original IPR1 update. With the original weights $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ inside the window, this proves

$$0 \leq \Pi_I^p, \Pi_A^p \leq 4 \sum_{r=0}^R \vartheta_r \log(1 + a_r) \leq 4 \mathfrak{E}_{s_1, R}. \quad (\text{PRD23})$$

The last finite constant is exactly the already proved IPR2/NIB14 budget. In particular, writing $Z = I$ or A and interpreting its left side respectively as $\mathcal{M}^p$ or $\mathcal{R} \log(\det A_{\text{inv}}^p / \det Q_{\text{inv}}^p)$, the full finite interval is

$$\tau_Z^p - V_Q^p \leq \Pi_Z^p + \tau_Z^p - V_Q^p \leq \tau_Z^p - V_Q^p + 4\mathfrak{E}_{s_1, R}. \quad (\text{PRD24})$$

It has the original independent proper-source term on both sides. At the already established $R_j = \lfloor j^{3/2} / \log j \rfloor$ on its original $R_j < q$ domain, this width is $o(kq_k)$ by IPR9. This bounds the removed target prefix; it does not assign an asymptotic value to either retained term.

2.5 The connecting receiver and all compatibility terms

Use precisely the STF13/CPQ flags S, K_c, W, T and the original connecting columns at primitive index e . Their quotient pairs for $G_t, G_s, A_{\text{inv}}, H_{\text{inv}}$ are respectively $(T, W), (S, K_c), (I, K_i), (W_i, J)$. Applying PRD15 to these pairs and substituting the actual eight flags of STF16 gives, by literal cancellation of all eight labelled occurrences,

$$\mathcal{L}_e^{p, > R} = \tau_I^p + \tau_t^p - \tau_s^p - \tau_A^p, \quad \mathcal{L}_e^{p, \leq R} = \Pi_I^p + \Pi_t^p - \Pi_s^p - \Pi_A^p. \quad (\text{PRD25})$$

Here $\tau_t = \sum_{h=q-1, q} [F_h(T) - F_h(W)]$ and $\tau_s = \sum_{h=q-1, q} [F_h(S) - F_h(K_c)]$. Prefixes are the original PRD17 values of G_t, G_s . Coincident flags retain their positions and signs. The prefix identity follows from the same fixed determinant identities because its evaluation coefficients sum to zero. Adding the two equalities proves exactly the NFR7 receiver

$$\mathcal{M}^p = \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p = \Pi_I^p + \tau_I^p - V_Q^p. \quad (\text{PRD26})$$

In particular the independent V_Q^p does not enter the four-target-copy CPQ allocation identity. It enters through the proved same-value relative operator PRD19–20.

Finally let $\mathcal{C}^p$ retain NFR10/EFP31's complete original compatibility return, including its source-complement projections, and retain the actual endpoint errors of NFR2:

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p. \end{aligned} \quad (\text{PRD27})$$

NFR5's exact equality is $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Substituting all three lines of PRD21, before estimating any term, proves the sharpened receiving expression

$$\begin{aligned} \mathcal{C}^p &= \Pi_A^p + \tau_A^p - V_Q^p - \mathcal{E}^p, \\ \Pi_A^p + \tau_A^p - V_Q^p - E_p^+ &\leq \mathcal{C}^p \leq \Pi_A^p + \tau_A^p - V_Q^p - E_p^-. \end{aligned} \quad (\text{PRD28})$$

This preserves the four actual compatibility constants with the two low signs reversed. It also proves the exact relation between the complete compatibility return and the proper pullback correction without dropping their finite error.

All existing source-order intervals therefore receive these expressions by literal substitution. For example, on EFP29's already proved finite guards, with Δ meaning order one minus order zero,

$$r_I(\Phi_j^- - \Phi_k^+) \leq \Delta(\Pi_I + \tau_I - V_Q) \leq r_I(\Phi_j^+ - \Phi_k^-). \quad (\text{PRD29})$$

The degree- k constants here are evaluated at PRD22, exactly as EFP28–29 prescribe, and their target constants remain degree j . For EFP31 its same unchanged full compatibility interval applies to $\Delta(\Pi_A + \tau_A - V_Q - \mathcal{E})$ by PRD28. No guard is enlarged: EFP28's sufficient $k \geq 37$ and actual QOT root tests, and the separate EOR tests wherever used, remain those of the source theorem. Outside those comparison guards, the exact finite matrix identities PRD1–28 retain their original rank-valid domain.

3 The analytic tail inside the complete original action

This continuation preserves the programme's recorded human origin and the separate recorded creators and dates of its deposited source notes [H19, H20]. Its new use of the accepted NTG/STF/NFI identities is the complete original-action return proved below. The exact packet morphisms extend the saved typed-action calculation; the full source lineage is recorded in the accompanying private use ledger. Every new finite Hermitian factorization is proved here, with the classical Schur references cited where used.

All statements here retain the hypothetical original simple quartet, the fixed actual period and branch, every conductor coefficient, and the original source masses. Write the actual packet degree as a until its value is specified. In the current NTG/STF/NFI calculation it is $a = j = k + 4$, not k . Set

$$\begin{aligned} q &= q_a = (a + 1)^2, & q' &= q'_a = (a - 7)^2, \\ m &= q - q' - v, & N_i &= (q - 1, q, 2q - 1, 2q)_i, & D_i &= N_i - v, \\ s &\in \{1, a\}, & a &= 4l + 1 \geq 9. \end{aligned} \tag{TAC1}$$

The original source centres are $c_0 = a/2$ for the whole numerator packet and $c_1 = a/2 - 4$ for the divided analytic functionals. The common Gamma measure and orthonormal polynomial phases are precisely

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, & p_{1,s} &= y, & p_{n+1,s} &= yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, & \phi_{n,s,c}(S) &= p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}. \end{aligned} \tag{TAC1a}$$

The classical Meixner–Pollaczek dictionary is $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$; the substitution, including $dy = 2d(y/2)$, gives the displayed mass and norms, as calculated in NTG3 and HPS1–5 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. The symbol m in TAC1 is the analytic coefficient dimension; CAI's quartet multiplicity is exactly one here. This calculation does not extend the NTG result to a multiple quartet.

3.1 Both stages of the analytic conductor quotient

Here are the complete maps from the original packet. Put

$$\begin{aligned} \beta_{a,u,w} &= a/2 + (2u-a)\delta + i(2w-a)\gamma, & \chi_a(S) &= \prod_{u,w=0}^a (S - \beta_{a,u,w}), \\ \chi_-(S') &= \chi_{a-8}(S'), & E &= \mathbb{C}[S]/(\chi_a), & E_- &= \mathbb{C}[S']/(\chi_-), \\ E_A(z) &= \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, & \mu_n &= E_A^{(n)}(0), & v = \text{ord}_0 E_A &\leq 80, & \mu_v &\neq 0, \\ (\mathcal{T}_A P)(S') &= \sum_{r,t=0}^8 a_{rt} P(S' + \beta_{8,r,t}), & \bar{\mathcal{T}}_A : E &\longrightarrow E_-, & [P] &\longmapsto [\mathcal{T}_A P], \\ L_v &= \{[P] : \deg P < v\} \subset E, & V_A &= \ker \bar{\mathcal{T}}_A. \end{aligned} \tag{TAC1b}$$

The complete coefficients of E_A remain as in the original fixed period; no repeated contribution is removed. Distinctness of the grid roots follows from $\delta > 0$ and $\gamma > 0$ by comparing real and imaginary parts. For every lower root $\beta_{a-8,u,w}$, $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. Consequently $\chi_- \mid \mathcal{T}_A(\chi_a P)$, proving that the packet arrow in TAC1b is well defined. Its matrix in upper and lower root-value coordinates is

$$C_{(u,w),(i,j)} = \begin{cases} a_{i-u,j-w}, & 0 \leq i-u, j-w \leq 8, \\ 0, & \text{otherwise.} \end{cases} \tag{TAC1c}$$

Indeed evaluation of the complete shift sum gives this entry, with all contributions retained. For a lower coefficient column η ,

$$\mathcal{N}_{C^\top \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}.$$

If this vanishes, the lower sum vanishes on an open disk on which $E_A \neq 0$, hence identically; its distinct-root Vandermonde then kills η . Thus $C^\top$ is injective and C is onto. In particular $\dim V_A = q - q'$. The polynomial identity

$$\mathcal{T}_A S^n = \sum_{d=0}^n \binom{n}{d} \mu_d S'^{n-d}$$

shows that the exact polynomial kernel is $\mathbb{C}[S]_{<v}$ and that the degree of every nonzero image of degree $n \geq v$ is $n - v$, with leading coefficient multiplied by $\binom{n}{v} \mu_v$. Triangular elimination proves surjectivity onto every retained target polynomial space. Since $v < q$, L_v injects into E , and $L_v \subset V_A$.

For any $N \geq q - 1$, put $D = N - v$ and $b = N - q \geq -1$; negative polynomial degree means the zero space. Define the full relation map and both quotient spaces by

$$\begin{aligned} \mathcal{U}_b h &= \mathcal{T}_A(\chi_a h) / \chi_-, & X_N &= \mathbb{C}[S']_{\leq D} / \mathcal{T}_A(\chi_a \mathbb{C}[S]_{\leq b}), \\ Y_N &= \mathbb{C}[S']_{\leq D-q'} / \mathcal{U}_b \mathbb{C}[S]_{\leq b}, \\ \kappa_N : E/L_v &\longrightarrow X_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P], \\ \theta_N : V_A/L_v &\longrightarrow Y_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P / \chi_-]. \end{aligned} \tag{TAC1d}$$

Choose a polynomial representative of degree at most N in these formulas. Changing it by $\chi_a h$ changes the two images by their respective displayed relation; changing it by a low polynomial changes neither image. Surjectivity of $\mathcal{T}_A$ proves surjectivity of κ_N . If its value is zero, subtract the corresponding $\chi_a h$; the result has degree below v , proving injectivity. The same argument after multiplying a prescribed Y_N representative by χ_- proves both surjectivity and injectivity of θ_N . Every required preimage has degree at most N , by the proved exact degree drop. Thus these are proved isomorphisms, with inverse given by any such preimage followed by its remainder class modulo L_v .

Multiplication by χ_- and reduction modulo χ_- give the commutative diagram of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & V_A/L_v & \longrightarrow & E/L_v & \xrightarrow{\overline{\mathcal{T}}_A} & E_- \longrightarrow 0 \\ & & \theta_N \downarrow & & \kappa_N \downarrow & & \parallel \\ 0 & \longrightarrow & Y_N & \xrightarrow{\chi_-} & X_N & \xrightarrow{\text{mod } \chi_-} & E_- \longrightarrow 0. \end{array} \tag{TAC1e}$$

The bottom left map is injective: $\chi_- g = \mathcal{T}_A(\chi_a h)$ is equivalent to $g = \mathcal{U}_b h$ in the polynomial integral domain. Its image is the kernel of the bottom right map, since every polynomial vanishing at all lower roots is divisible by χ_- . That right map is onto because $D \geq q - 1 - v \geq q' - 1$. This proves each exactness claim and the commutativity on actual representatives. Finally $\mathcal{U}_b$ is injective: for nonzero h of degree t , its image has degree $q + t - v - q' = m + t$ and leading coefficient $\text{lc}(h) \binom{q+t}{v} \mu_v \neq 0$. Subtracting the actual polynomial dimensions gives

$$\dim(E/L_v) = \dim X_N = q - v, \quad \dim(V_A/L_v) = \dim Y_N = q - v - q' = m. \tag{TAC1f}$$

This displays both stages with their maps and their complete relation spaces. In the target application $a = j = k + 4$, the NTG coefficient metric has the latter dimension $m = 16j - 48 - v$.

The algebraic pairing identifying its analytic dual is equally exact. Write $\mathbb{V}[n, \beta] = \beta^n$ and let p be the monomial remainder coefficient column. The isomorphism

$$\Phi : \mathbb{C}^q \longrightarrow E^\vee, \quad \Phi(\nu)([P]) = \sum_{\beta} \nu_{\beta} P(\beta) = (\mathbb{V}\nu)^\top p \tag{TAC1g}$$

is complex linear; the invertible Vandermonde gives its inverse. The first v moment conditions say exactly $\Phi(\mathcal{N}_v) = \text{ann}(L_v)$. For $\eta \in \mathbb{C}^{q'}$, $\Phi(C^\top \eta)([P]) = \sum_{\beta'} \eta_{\beta'} (\mathcal{T}_A P)(\beta')$, the actual dual of the onto conductor. Its image is therefore $\text{ann}(V_A)$: one inclusion follows by evaluation, and both dimensions are q' . Restriction of functionals induces

$$\Psi : \mathcal{N}_v / \text{im } C^\top \xrightarrow{\sim} (V_A / L_v)^\vee, \quad [\nu] \mapsto ([P] + L_v \mapsto \sum_{\beta} \nu_{\beta} P(\beta)). \quad (\text{TAC1h})$$

Adding a low polynomial or a conductor coefficient changes that value by zero. Its kernel is exactly the stated conductor image. Any functional on V_A / L_v extends linearly from V_A to E while still vanishing on L_v ; its unique Φ representative lies in $\mathcal{N}_v$. This proves surjectivity without a metric choice. For $F = \mathcal{N}_v / E_A$, analyticity follows from the first v vanishing moments. The exact finite Leibniz identity is

$$\begin{aligned} \Lambda_F(\mathcal{T}_A P) &= \Lambda_{E_A F}(P) = \Phi(\nu)([P]), \\ \ell_F(g) &= \Lambda_F(\chi_- g), \quad \ell_F(\theta_N([P] + L_v)) = \Psi([\nu])([P] + L_v). \end{aligned} \quad (\text{TAC1i})$$

For proof on $P = S^n$, expand the complete derivative product as $\sum_{d=v}^n \binom{n}{d} \mu_d F^{(n-d)}(0)$; finite linearity proves the identity for every retained polynomial. On a relation $g = \mathcal{U}_b h$ its value is $\Phi(\nu)([\chi_a h]) = 0$. Thus it is a well-defined dual pairing on Y_N . A lower exponential has zero pairing because $\chi_-(\beta') = 0$; conversely TAC1h and the isomorphism θ_N identify exactly that denominator via $C^\top$. Every scalar is one in TAC1d–i. The original $\alpha_A = v!/\mu_v$ remains in the separate primary-coordinate maps of NFR1 and is not inserted into these unscaled conductor maps. These arguments prove the CEP3–6,42–43 and NCT3–11 bridge on its original objects and coordinates.

3.2 The exact subquotient and its two complementary factors

Let $\mathbb{V}$ have entries $\mathbb{V}[n, \beta] = \beta^n$, $0 \leq n < q$, in the original upper-root order. For $\nu \in \mathbb{C}^q$ put $\mathcal{N}_\nu(z) = \sum_{\beta} \nu_{\beta} e^{\beta z}$. Define

$$\begin{aligned} \mathcal{N}_v &= \{\nu : (\mathbb{V}\nu)_n = 0 \ (0 \leq n < v)\}, \\ C &= C_a, \quad Z = Z_{\text{exp},a}, \quad Z_v = \mathbb{V}^{-1} P_{<v}, \\ F &= [C^\top, Z, Z_v], \quad U = [C^\top, Z], \end{aligned} \quad (\text{TAC2})$$

where $P_{<v}$ inserts the first v standard coordinate vectors. NCT6–7 proves that $C^\top$ is injective and that U is a frame of $\mathcal{N}_v$, with respective column counts q' and m . The first v moment rows vanish on U and form the identity on Z_v . Thus F is invertible: a relation among its columns first has zero Z_v coefficient, and then has zero U coefficient by independence. The spaces and dimensions are therefore exactly

$$0 \subset \text{im } C^\top \subset \mathcal{N}_v \subset \mathbb{C}^q, \quad q = q' + m + v. \quad (\text{TAC3})$$

Let $G_N^{(s)}$ be the full CAI2 remainder-fibre Gram on $E_a = \mathbb{C}[S]/(\chi_a)$ with Gamma source order s and centre c_0 . It has the original monomial remainder frame. Let

$$\mathbf{E}_{N,s}[n, \beta] = \phi_{n,s,c_0}(\beta), \quad K_N^{(s)} = \mathbf{E}_{N,s}^* \mathbf{E}_{N,s}.$$

This is the complete upper observation matrix NCT8, at $0 \leq n \leq N$. The complex-linear functional associated with ν sends a remainder coefficient vector x to $(\mathbb{V}\nu)^\top x$. Its squared dual norm is $(\mathbb{V}\nu)^* (G_N^{(s)})^{-1} (\mathbb{V}\nu)$: indeed the representing vector in the Hermitian metric $G_N^{(s)}$ is $(G_N^{(s)})^{-1} \overline{\mathbb{V}\nu}$, whose norm has that value after complex conjugation of its real squared norm. Computing the same norm in the original orthonormal polynomial basis gives $\nu^* K_N^{(s)} \nu$. Equality for every ν proves

$$K_N^{(s)} = \mathbb{V}^* \overline{(G_N^{(s)})^{-1} \mathbb{V}}, \quad \det K_N^{(s)} = \frac{|\det \mathbb{V}|^2}{\det G_N^{(s)}}. \quad (\text{TAC4})$$

The ordinary transpose in the functional pairing and the conjugated inverse in TAC4 are both essential. This identity is also NCT8a factored through the original remainder map. It does not identify an arithmetic source with a Gamma source.

Suppress s temporarily and define the actual complementary forms

$$\begin{aligned} K_{C,N} &= \overline{C} K_N C^\top, \\ \widehat{D}_N &= Z^* \{K_N - K_N C^\top K_{C,N}^{-1} \overline{C} K_N\} Z, \\ Q_{v,N} &= Z_v^* \{K_N - K_N U (U^* K_N U)^{-1} U^* K_N\} Z_v. \end{aligned} \tag{TAC5}$$

Here $\widehat{D}_N$ is exactly NCT8's $\widehat{D}_{a,N-v}^{(s)}$, not a new m -dimensional quotient. The metric $K_{C,N}$ is the restriction to every conductor column; $Q_{v,N}$ is the full attained quotient $\mathbb{C}^q / \mathcal{N}_v$ in the fixed values Z_v . In particular its minimization includes both the conductor columns and all analytic columns.

For clarity, the determinant factorization is proved in these frames. In $F^* K_N F$, subtract from each of the last $m+v$ column blocks its K_N -orthogonal projection onto $C^\top$, and make the conjugate row operations. These triangular operations have determinant one. They split off $K_{C,N}$. Next subtract from the final block its projection onto the remaining m columns. This splits off $\widehat{D}_N$; minimizing in succession over those two subspaces is minimizing over their sum $\mathcal{N}_v$, so the last block is precisely $Q_{v,N}$. Consequently

$$\det K_{C,N} \det \widehat{D}_N \det Q_{v,N} = |\det F|^2 \det K_N = \frac{|\det F|^2 |\det \mathbb{V}|^2}{\det G_N^{(s)}}. \tag{TAC6}$$

The fixed determinant factor can also be evaluated in the exact DNE-selected minor. Let P_v insert the final $q-v$ moment coordinates, put $J_v = P_v^\top \mathbb{V} C^\top$, and retain the first nonzero q' -row minor indexed by $\mathcal{I}$ and the complementary insertion $E_{\mathcal{J}}$. Then $Z = \mathbb{V}^{-1} P_v E_{\mathcal{J}}$ and

$$\mathbb{V} F = [P_v J_v, P_v E_{\mathcal{J}}, P_{<v}], \quad |\det F|^2 |\det \mathbb{V}|^2 = |\det [J_v, E_{\mathcal{J}}]|^2 = |\det (J_v)_{\mathcal{I}}|^2. \tag{TAC6a}$$

Permute the first v rows to meet the last v columns, then expand on those identity columns. Expanding the remaining determinant on the complementary coordinate insertion leaves exactly the stated minor with a permutation sign. Its absolute square removes the sign and retains the actual nonzero minor; it is not assigned the value one.

The two outer factors have explicit primal meanings. With $T = C \mathbb{V}^\top$, the onto conductor in monomial-to-lower-root coordinates, and $G = G_N^{(s)}$, put

$$G_{C,N} = (T G^{-1} T^*)^{-1}, \quad G_{L,N} = P_{<v}^* G P_{<v}.$$

The original value minimum for T is attained by $G^{-1} T^* (T G^{-1} T^*)^{-1}$: its composition with T is the identity and it is G -orthogonal to $\ker T$. This proves the first formula as the full quotient metric. Direct substitution in TAC4 then proves

$$K_{C,N} = \overline{G_{C,N}^{-1}}, \quad Q_{v,N} = \overline{G_{L,N}^{-1}}. \tag{TAC6b}$$

For the second equality the value map on the dual coefficient packet is $T_v = P_{<v}^\top \mathbb{V}$. Its kernel is $\mathcal{N}_v$ and $T_v Z_v = I_v$, so the attained quotient is $(T_v K_N^{-1} T_v^*)^{-1}$. Substituting TAC4 gives precisely the second formula. This proof includes every direction of the full vanishing-moment kernel.

The middle factor is the dual metric on the actual primal subquotient V_A / L_v , with the restriction of G to V_A followed by its attained quotient by L_v . To prove this, a functional on a Hilbert subspace extends with minimum norm by composition with the orthogonal projection. Every other extension differs by a functional on the orthogonal complement, and their squared norms add. If the original functional vanishes on L_v , that minimum extension still vanishes on L_v ; all other admissible extensions differ by $\text{ann}(V_A)$. This is exactly TAC5's middle minimum under TAC1g-i. Let $H_{W,N}$ be the primal subquotient Gram transported by θ_N to the fixed low monomial frame

of Y_N , and retain DNE4's complete pairing matrix A_ℓ . The complex-linear dual norm calculation used in TAC4 gives

$$\widehat{D}_N = A_\ell^* \overline{H_{W,N}^{-1}} A_\ell, \quad H_{W,N} = \overline{A_\ell \widehat{D}_N^{-1} A_\ell^*}. \quad (\text{TAC6c})$$

The pairing matrix is invertible by TAC1h–i. This proves both displayed formulas and their exact pairing coefficient. NCT22 gives this same transported Gram. Its native lower-source counterpart replaces $\widehat{D}_N$ by $D_{a,N-v}^{(s)}$; TAC9 retains their entire metric difference.

All positive inverses exist because $K_N > 0$ and the stated frames are independent. If $v = 0$, Z_v and $Q_{v,N}$ are the unique empty maps and $\det Q_{v,N} = 1$. No limiting inverse is required. This is the usual Schur factorization, with its complete proof just given; compare Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23].

The native-to-numerator map in this factorization is explicit. For $D = N - v$, set

$$\mathcal{Y}_D = \{\Lambda_{\mathcal{N}_v/E_A} |_{\mathbb{C}[S']_{\leq D}} : \nu \in \mathcal{N}_v\}, \quad \mathcal{X}_D = \text{span}\{\text{ev}_{\beta_{a-8,u,w}}\} \subset (\mathbb{C}[S']_{\leq D})^\vee.$$

Here $\mathcal{X}_D \subset \mathcal{Y}_D$ by TAC1c. The complete finite-jet product identity TAC1i implies injectivity of the map from $\mathcal{N}_v$ to $\mathcal{Y}_D$: zero derivatives through degree D make every numerator derivative through N zero, and the first q such derivatives form the invertible upper Vandermonde. Thus these spaces have dimensions $q - v$ and q' respectively. For $F_b = \mathcal{N}_{Z_{e_b}}/E_A$, NCT3–11 and TAC1i give

$$[F] \mapsto [E_A F] : \mathcal{Y}_D / \mathcal{X}_D \xrightarrow{\sim} \mathcal{N}_v / \text{im } C^\top. \quad (\text{TAC7})$$

On a monomial it is the exact Leibniz identity

$$\Lambda_F(\mathcal{T}_A S^n) = \sum_{d=v}^n \binom{n}{d} \mu_d F^{(n-d)}(0) = (E_A F)^{(n)}(0), \quad (\mathcal{T}_A P)(S') = \sum_{u,w} a_{uw} P(S' + \beta_{8,u,w}).$$

The nonzero coefficient $\binom{v+r}{v} \mu_v$ recursively recovers each derivative of F , so this is an isomorphism on the retained finite jets. It takes the entire lower-evaluation space to the entire conductor-multiple space. Its orthonormal finite-jet matrix is the ordinary transpose $A_{D,s}^\top$ of NCT3's unscaled conductor map. The two quotient metrics on the same x values are the original $D_{a,D}^{(s)}$ and $\widehat{D}_{a,D}^{(s)}$; they are not declared equal.

These are exact paired vector-space maps, which need no action invariance of either intermediate subspace. The full dual multiplication action in root coefficients is $\text{diag}(\beta)$ and in the complete frame F it is $F^{-1} \text{diag}(\beta) F$, with all its blocks retained. The conductor's exact action defect is

$$\mathcal{T}_A(SP) - S' \mathcal{T}_A P = \sum_{u,w} a_{uw} \beta_{8,u,w} P(S' + \beta_{8,u,w}), \quad (E_A F)' = E_A F' + E_A' F. \quad (\text{TAC8})$$

Both identities follow term by term from the original shift sum and the product rule. The action calculation below uses the whole packet CAI action; it does not discard TAC8 or manufacture an induced action on a noninvariant intermediate quotient.

The complete primal action can be recovered from these dual frames, including every block crossing this flag. This provides an exact action calculation even when the intermediate subspaces have an action defect. For any positive original packet Gram G in its monomial frame set $K = \mathbb{V}^* \overline{G^{-1}} \mathbb{V}$, and use all the minima TAC5 in that K . Define

$$H = C^\top, \quad J = Z - H(H^* K H)^{-1} H^* K Z, \quad (\text{TAC8a})$$

$$R_\perp = Z_v - U(U^* K U)^{-1} U^* K Z_v, \quad B = [H, J, R_\perp], \quad D = \text{diag}(K_C, \widehat{D}, Q_v).$$

All three block columns are pairwise K -orthogonal, as checked in the proof of TAC6. Thus $B^* K B = D$ and $B^{-1} = D^{-1} B^* K$. Let $D_\beta = \text{diag}_\beta(\beta)$, and index these block columns by $u = 1, 2, 3$, of widths q', m, v . The full action and every one of its nine blocks are

$$A = B^{-1} D_\beta B, \quad A_{uv} = D_u^{-1} B_u^* K D_\beta B_v \quad (1 \leq u, v \leq 3). \quad (\text{TAC8b})$$

This equality follows from the stated inverse, so no off-diagonal block is suppressed. For example the action of a middle-block vector z is exactly $(A_{12}z, A_{22}z, A_{32}z)$ in this frame, which explicitly records both components outside the middle block.

Let $A = M_S$ on E . Put $P = (\nabla B)^{-\top}$, a complete primal monomial coefficient frame. The pairing is precisely $B^\top \nabla^\top P = I_q$. Applying the root action to the functional argument shows that the primal multiplication matrix in P is $A^\top$: for arbitrary functional and primal columns x, y , the pairing after primal multiplication equals $x^\top A^\top y$, which is also the pairing after applying the dual action to x . Moreover direct inversion gives

$$P^*GP = \overline{D}^{-1}, \quad P^*(A^*G + GA - aG)P = \overline{A}\overline{D}^{-1} + \overline{D}^{-1}A^\top - a\overline{D}^{-1}. \quad (\text{TAC8c})$$

For the first equality, the conjugate inverse of its left side is $(\nabla B)^* \overline{G}^{-1} (\nabla B) = D$. For the second, use $AP = PA^\top$ and substitute the first equality. These computations prove exact pairing and action compatibility, with all phases and blocks. Empty blocks are the unique empty maps. Taking $G = G_N^{\text{ar}}$ in TAC8a–c gives the complete original CAI arithmetic control; taking $G = G_N^{(s)}$ gives the original Gamma control. The section matrices and factors are recomputed in the chosen G ; the arithmetic and Gamma forms have not been identified. The explicit arithmetic volume correction in TAC13 remains in the action below.

3.3 Whole-volume return with the native factor inserted

Use the original dual sign operator $\mathcal{R}f = -f(q-1) - f(q) + f(2q-1) + f(2q)$ and put

$$\begin{aligned} \mathcal{C}_{a,s} &= \mathcal{R} \log \det K_{C,N}^{(s)}, & \mathcal{V}_{a,s} &= \mathcal{R} \log \det Q_{v,N}^{(s)}, & \mathcal{T}_{a,s} &= \mathcal{R} \log \det D_{a,N-v}^{(s)}, \\ \epsilon_{a,N,s} &= \log \det [(D_{a,N-v}^{(s)})^{-1/2} \widehat{D}_{a,N-v}^{(s)} (D_{a,N-v}^{(s)})^{-1/2}], & \Xi_{a,s} &= \mathcal{R} \epsilon_{a,N,s}. \end{aligned} \quad (\text{TAC9})$$

TAC6b identifies the last return with the earlier literal low-polynomial return, including its orientation:

$$\begin{aligned} \mathcal{V}_{a,s} &= -\mathcal{R} \log \det G_{L,N}^{(s)} \\ &= \log \frac{\det G_{L,q-1}^{(s)} \det G_{L,q}^{(s)}}{\det G_{L,2q-1}^{(s)} \det G_{L,2q}^{(s)}} = F_{L_a}^{(s)}, \quad L_a = L_v = \mathbb{C}[S]_{<v} \subset E. \end{aligned} \quad (\text{TAC9a})$$

Indeed $G_{L,N}^{(s)} = P_{<v}^* G_N^{(s)} P_{<v} > 0$ has positive real determinant, so TAC6b gives $\log \det Q_{v,N}^{(s)} = -\log \det G_{L,N}^{(s)}$ at every cutoff. Applying the displayed four signs proves all equalities. At $v = 0$ both logarithms are zero by the empty-determinant convention. Thus the low factor in this complete packet decomposition is exactly the original FEX/NSR32 low return used again in TAC17.

Applying $\mathcal{R}$ to TAC6 cancels the two fixed frame constants and, because the original CAI volume uses the opposite signs on $\log \det G_N^{(s)}$, proves the exact whole-volume decomposition

$$\boxed{\begin{aligned} \mathcal{B}_a^{(s)} &= \mathcal{C}_{a,s} + \mathcal{T}_{a,s} + \mathcal{V}_{a,s} + \Xi_{a,s} \\ &= \mathcal{C}_{a,s} + \mathcal{T}_{a,s} + F_{L_a}^{(s)} + \Xi_{a,s}. \end{aligned}} \quad (\text{TAC10})$$

The q' -dimensional conductor restriction and v -dimensional final quotient are thus the actual complementary factors of the m -dimensional analytic quotient. They are not CPQ's four complementary returns; those four decompose a different, four-copy target allocation.

At each original cutoff $K_{N+1} - K_N \succeq 0$. Restriction preserves this inequality, and taking the minimum of the two quadratic forms on each identical affine fibre preserves it as well. Thus $K_{C,N}$, $\widehat{D}_N$ and $Q_{v,N}$ increase as positive forms. Their $\mathcal{R}$ returns are nonnegative, by the original telescoping weights. This observation does not evaluate their growing-degree values or change the signed defect $\Xi_{a,s}$.

The finite NCT20 constants retain the exact source, rank, cutoff and radius. Restore the radius argument suppressed in NCT19, and let

$$\begin{aligned}\mathcal{A}_A &= \{\varrho \in (0, 1) : g(t) \neq 0 \text{ for every } |t| = \varrho\}, \\ g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), \quad w(t) = -i \arctan t, \quad g(0) = (-i)^v \mu_v / v!, \\ L_i &= \inf_{\varrho \in \mathcal{A}_A} \mathcal{L}_{a, D_i, s, m}^{[1]}(\varrho), \quad U_i = \inf_{\varrho \in \mathcal{A}_A} \mathcal{U}_{a, D_i, s, m}^{[1]}(\varrho).\end{aligned}\tag{TAC10a}$$

Every retained principal part and its full pole order remain in the NCT16–19 constants. The symbol is analytic in the open unit disk and $g(0) \neq 0$. Its zeros are isolated; on each smaller closed disk they are finite, so their radii form a countable set. Thus $\mathcal{A}_A$ is nonempty and contains radii arbitrarily close to one. The displayed infima are finite and nonnegative, since each fixed admissible radius has finite nonnegative constants. Attainment of these infima is not required or asserted. For every admissible radius NCT20 gives $-\mathcal{L}(\varrho) \leq \epsilon_{a, N_i, s} \leq \mathcal{U}(\varrho)$. Taking the supremum of the left lower bounds and the infimum of the right upper bounds proves $-L_i \leq \epsilon_{a, N_i, s} \leq U_i$. Define the directed whole-return allowances

$$\Xi_{a, s}^- = -U_0 - U_1 - L_2 - L_3, \quad \Xi_{a, s}^+ = L_0 + L_1 + U_2 + U_3.\tag{TAC11}$$

Negating the two low inequalities and adding the two high inequalities proves $\Xi_{a, s}^- \leq \Xi_{a, s} \leq \Xi_{a, s}^+$. This bounds the same actual four corrections; it does not assign independent values to their correlated matrices.

Here is the order of limits for these allowances. For a fixed admissible ϱ , the NCT19 term $4mL_{s, D_i, v}$ is $O(a^2 \log(a+1))$, because $m \leq 16a$, $s \leq a$ and $D_i \leq 3a^2$ for $a \geq 9$. The exceptional inverse contribution has rank at most the fixed H_ϱ and is $O_\varrho(a^2)$. The remaining logarithms obey NCT21's bound $D_i \log(1/\varrho) + O_\varrho(\log(a+1))$ and have coefficient at most $2m$. Consequently, for either original source order and for each endpoint,

$$0 \leq \limsup_{a \rightarrow \infty} \frac{L_i}{a^3} \leq 96 \log(1/\varrho), \quad 0 \leq \limsup_{a \rightarrow \infty} \frac{U_i}{a^3} \leq 96 \log(1/\varrho).\tag{TAC11a}$$

For proof, the infimum defining L_i or U_i is bounded above by its value at this one fixed radius, and the preceding estimates apply with that radius held fixed throughout the degree limit. Only after taking this limsup choose admissible radii approaching one. The right sides tend to zero; therefore $L_i, U_i = o(a^3) = o(aq_a)$ and $\Xi_{a, s}^-, \Xi_{a, s}^+ = o(aq_a)$. This last statement concerns the radius-infimum allowances TAC10a–11. A particular fixed-radius allowance is retained with its exact finite value whenever it is used; no zero leading coefficient for that fixed-radius bound is asserted. No q' rank is inserted into the comparison; its complete metric remains separately in TAC10.

Now take $a = j = k + 4$ and an original $0 \leq R < q$. NTG18–25 and NFI5–6 give the following exact scalar computed by the two tail matrices:

$$\begin{aligned}\mathcal{T}_{a, s, R} &:= \sum_{t \in \{q-1, q\}} \log \frac{\det \Omega_t}{\det \Omega_R} \\ &= \frac{1}{4} \sum_{t \in \{q-1, q\}} \log \det(I_{4m} + M_t) = \mathcal{T}_{a, s}^{>R}, \\ \mathcal{T}_{a, s} &= \mathcal{T}_{a, s}^{\leq R} + \mathcal{T}_{a, s, R}, \quad 0 \leq \mathcal{T}_{a, s}^{\leq R} \leq \mathfrak{E}_{s, R}.\end{aligned}\tag{TAC12}$$

The coefficient $1/4$ is forced by NFI6's four copies; each $\log \det \Omega$ ratio already represents the single native target. The two endpoint expressions are exactly $f(2q-1) + f(2q) - 2f(q+R)$, so no extra last-row weight is inserted. At $R = q-1$ the first term is zero, consistently with the empty stack. IPR2–3 and IPR6 prove the last two statements. At $R = \lfloor j^{3/2} / \log j \rfloor$ on its stated range, IPR9 gives $\mathfrak{E}_{s, R} = O_{\text{actual}}(j^{5/2} + j^3 / \log j) = o(kq_k)$. No value for $\mathcal{T}_{a, s, R}$ is assigned by this identity.

3.4 Exact complete action and finite receiving interval

For this same packet degree a , keep every CAI13 argument and define the actual arithmetic source correction

$$\delta_{a,s}^{\text{ar}} := \mathcal{B}_a^{\text{ar}} - \mathcal{B}_a^{(s)} = \begin{cases} \delta_a^\sigma, & s = 1, \\ \mathcal{H}_a + \delta_a^\sigma, & s = a. \end{cases} \quad (\text{TAC13})$$

Both equalities follow from CAI5, with no change of measure. In the literal notation $d_n = V_n^{\text{ar}}/V_{n-1}^{\text{ar}}$, CAI12–15 has

$$\begin{aligned} W_a &= \log \frac{\omega_{2q-1,a}^{\text{ar}} \omega_{2q,a}^{\text{ar}}}{\omega_{q-1,a}^{\text{ar}} \omega_{q,a}^{\text{ar}}}, \\ P_{a,-} &= -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \\ P_{a,+} &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n), \\ J_a &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N = \mathcal{B}_a^{\text{ar}} + W_a - P_{a,-} - P_{a,+}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \text{ inside.} \end{aligned} \quad (\text{TAC14})$$

The weight sum is $2q$. The factor 2 before the weighted logarithms means the exterior benchmark is $4q \log(\delta q(a+1)/2)$. Substituting TAC10–13 into this already proved identity gives

$$\begin{aligned} J_a &= \mathcal{C}_{a,s} + F_{L_a}^{(s)} + \delta_{a,s}^{\text{ar}} + \Xi_{a,s} + \mathcal{T}_{a,s}^{\leq R} + \mathcal{J}_{a,s,R} \\ &\quad + W_a - P_{a,-} - P_{a,+}. \end{aligned}$$

(TAC15)

Every term has the original a -packet and window. For the current new matrices $a = j$; TAC15 does not substitute the target matrices at j for the independent degree- k proper-source metric of NFR1.

The original W/P results are consumed without rederivation. CAI29 and CAI28 give $-E_{\text{hi}} \leq W_a - \mathcal{W}_q^\sigma \leq E_{\text{lo}}$ and $0 \leq P_{a,-} + P_{a,+} \leq \Pi_{h,a}$, with the complete constants of those formulas, evaluated at a and its q_a . Set the actual retained centre

$$A_{a,s,R} = \mathcal{C}_{a,s} + F_{L_a}^{(s)} + \delta_{a,s}^{\text{ar}} + \mathcal{J}_{a,s,R} + \mathcal{W}_q^\sigma.$$

Adding the four directed bounds, without choosing independent values for any actual correlated determinant, gives the finite enclosure

$$\begin{aligned} A_{a,s,R} + \Xi_{a,s}^- - E_{\text{hi}} - \Pi_{h,a} &\leq J_a, \\ J_a &\leq A_{a,s,R} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}. \end{aligned}$$

(TAC16)

On ECT's stronger finite guards its already proved penalty cap may replace $\Pi_{h,a}$; no new penalty theorem is asserted here. The exact classical-source window remains CAI31's

$$\mathcal{W}_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}.$$

Equivalently, CAI32–35 supplies $C(q) = 4q \log q + (8 \log 2 - 4)q$ and its full finite errors $-\log 2$, η_q , E_{lo} , E_{hi} and $\Pi_{h,a}$ in TAC15–16. The new statement has not replaced $\delta_{a,s}^{\text{ar}}$ or either complementary volume by an unproved estimate.

3.5 Why the allocation enters twice and cancels once

Use the same $a = j$ and order pair that defines the CPQ/STF terms. Put

$$Z_e^X = \mathcal{L}_e^X + V_b^X + V_{\text{rem}}^X + V_s^X + V_A^X, \quad X \in \{\leq R, > R\}.$$

IPR7 and STF18 give $Z_e^X = 4S_{K,a}^{X,(s)}$. The exact FEX/RQB33/NSR32 correction is

$$\begin{aligned} \delta_{K,a,s} &= \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_a}^{(s)} + F_{(K_a+L_a)/K_a}^{(s)}, \quad \sigma = (1, 1, -1, -1), \quad L_a = \mathbb{C}[S]_{<v}, \\ F_{K,a}^{(s)} &= \frac{1}{4}(Z_e^{\leq R} + Z_e^{>R}) - \delta_{K,a,s}. \end{aligned} \quad (\text{TAC17})$$

Here $e_{N_i,s}$ is the actual conductor source Jacobian and both displayed low metrics remain in their original primal frames. At source $s = a$, substitution into CAI13/15 gives, term for term,

$$\begin{aligned} \Psi_a &= \frac{1}{4}(Z_e^{\leq R} + Z_e^{>R}) - \delta_{K,a,a} + \mathcal{H}_a + \delta_a^\sigma, \\ C_a^{\text{budget}} &= \mathcal{B}_a^{(a)} - \frac{1}{4}(Z_e^{\leq R} + Z_e^{>R}) + \delta_{K,a,a} + W_a - P_{a,-} - P_{a,+}, \\ \Psi_a + C_a^{\text{budget}} &= \mathcal{B}_a^{(a)} + \mathcal{H}_a + \delta_a^\sigma + W_a - P_{a,-} - P_{a,+} = J_a. \end{aligned} \quad (\text{TAC18})$$

Thus the coefficient of every one of the five CPQ tail terms in Ψ_a is $+1/4$ and in its matching complete budget is $-1/4$. The two coefficients of $\delta_{K,a,a}$ are -1 and $+1$. They cancel as exact quantities. TAC15 instead inserts the total analytic factor with coefficient one into the proved whole-volume factorization, retaining its two genuinely complementary factors. These are different exact substitutions, and TAC18 explains their compatibility without double-counting a boundary term.

For the independent NFR source keep, at its original displaced cutoffs,

$$\mathcal{M}^p = \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p, \quad V_A^p - V_Q^p = \mathcal{R} \log \det[(Q_{\text{inv}}^p)^{-1/2} A_{\text{inv}}^p (Q_{\text{inv}}^p)^{-1/2}]. \quad (\text{TAC19})$$

These are NFR7's identities on the same Z_R values. The degree- k source, original centre, source orders and NFR2 compatibility error remain. The present j -packet action decomposition does not delete them or turn a target quotient into that proper-source form.

Finally, CAI11/37 gives on this same complete original arithmetic class

$$4q \log L_{h,a} \leq J_a, \quad L_{h,a} = \delta q(a+1)/2, \quad L_{h,a} \leq \min_N \epsilon_N \leq \exp(J_a/(4q)). \quad (\text{TAC20})$$

Combining this with TAC16 yields the explicit necessary inequality

$$4q \log L_{h,a} \leq A_{a,s,R} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}.$$

The unevaluated terms in this right side are their literal actual matrices and arithmetic correction. No smaller numerical upper bound than the benchmark has been proved here. The known canonical lower bounds, and a large weighted mean, do not by themselves contradict TAC20. When only a specified complete volume is transported and $W_a, P_{a,-}, P_{a,+}$ keep these arguments, direct subtraction of TAC14 gives exactly $\hat{J}_a - J_a = \hat{\mathcal{B}}_a - \mathcal{B}_a$. An analytic-factor substitution qualifies for that identity only through TAC6–15 with all other factors retained.

3.6 The same finite joint control in the complete action

In TAC15–16 put $a = j$ and $R = R_j$ on the already stated $R_j < q_j$ domain. The original single-copy tail there is exactly $\mathcal{T}_{j,s,R} = \tau_{\text{total}}$ by FTR9–10 and TAC12. Consequently the three simultaneous finite estimates retained in FTR27–28 give, on the same full original arithmetic packet,

$$J_j \leq \mathcal{C}_{j,s} + \mathcal{V}_{j,s} + \delta_{j,s}^{\text{ar}} + \mathcal{W}_{q_j}^\sigma + \mathfrak{U}_R + \Xi_{j,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}. \quad (\text{FTR31})$$

To prove this, substitute the exact TAC15 equality, then apply $\mathcal{T}_{j,s,R} \leq \mathfrak{U}_R$ from FTR28, $\Xi_{j,s} \leq \Xi_{j,s}^+$ from TAC11, $0 \leq \mathcal{T}_{j,s}^{\leq R} \leq \mathfrak{E}_{s,R}$ from TAC12, $W_j \leq \mathcal{W}_{q_j}^\sigma + E_{\text{lo}}$ from CAI29, and $P_{j,-} + P_{j,+} \geq 0$ from the actual CAI penalties. Every term has coefficient one in TAC15. The exact two complementary packet volumes and arithmetic correction remain on the right of FTR31. The finer attained value τ_{total} may always be retained in place of its upper bound, giving TAC16's sharper exact-data centre.

The four CPQ complements enter the allocation through FTR10 and TAC17 with coefficient $1/4$; at TAC18's stated order $s = j$, they enter the matching budget with coefficient $-1/4$. They are therefore retained with both coefficients before cancellation; they do not replace either packet complement in FTR31. The independently computed relative proper-source returns remain PRD21 and PRD28 at their degree- k cutoffs. This identifies the actual receiving maps for all three uses of the same tail calculation, with their original domains, values and coefficients.

Programme origin and human mathematical sources. The Split-Zero construction and programme were developed in KokunoYumeto’s project with AI assistance, with the originating notes documented in [H19, H20]. This uses the expressly authorized public pseudonym and the documented project provenance; it makes no assertion of global priority. The historical deposited creator fields are preserved in the citations. The present continuation was derived and checked with AI assistance within that programme. Its classical polynomial-modification input is Szegő’s Theorem 2.5, with his credit to Christoffel retained; its scalar orthogonal-polynomial formulas use the authored DLMF chapter and exact parameter dictionary; its Schur minima use the directly read Boyd–Vandenberghe derivations with the complex Hermitian specialization proved here. These human sources are cited at their actual uses. Project-provenance citations do not replace those mathematical source citations.

4 The original native tail as an attained relation Gram matrix

4.1 Original data and the classical tools being used

Fix the original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, its nonzero amplitude, its original period, its logarithm branch, and its complete nonreduced conductor. The period is fixed before any degree limit. Retain

$$\begin{aligned}
 k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad c' = j/2 - 4, \\
 \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma, \quad \chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad \chi_- = \chi_{j-8}, \\
 E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_d = E_A^{(d)}(0) = \sum_{u,w=0}^8 a_{uw} \beta_{8,u,w}^d, \quad v = \text{ord}_0 E_A, \quad \mu_v \neq 0, \\
 L &= q - v, \quad m = L - q' = 16j - 48 - v > 0, \quad s \in \{1, j\}.
 \end{aligned} \tag{NTG1}$$

The original bounds on v and the original admissible period domain remain in force; in particular no exceptional finite-period fibre is replaced by $v = 0$. Every μ_d is a derivative, not a Taylor coefficient. The accepted original relation calculation NIB1–9 supplies

$$\begin{aligned}
 (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \quad g_t = \mathcal{T}_A(\chi_j S^t), \\
 \ell_t &= \mu_v \binom{q+t}{v}, \quad u_t = g_t / \ell_t, \quad \deg u_t = L + t, \quad \text{lc } u_t = 1, \quad \chi_- \mid u_t.
 \end{aligned} \tag{NTG2}$$

It also proves that $u_0, \dots, u_t$ span the complete original functional annihilator at degree $L + t$. We keep all original analytic columns $F_b = N_b / E_A$ of NSO12, $1 \leq b \leq m$, with the original complex-linear functional convention $\Lambda_F(S'^d) = F^{(d)}(0)$.

The scalar measure and polynomial recurrence are

$$\begin{aligned}
 dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1 + s/2) p_{n-1,s}, \\
 h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S') = p_{n,s}((S' - c')/i) / \sqrt{h_{n,s}}, \\
 \langle P, Q \rangle_s &= \int \overline{P(c' + iy)} Q(c' + iy) dm_s(y).
 \end{aligned} \tag{NTG3}$$

Inner products are conjugate-linear in the first argument. These Gamma-weight orthogonal polynomials belong to the classical Meixner–Pollaczek theory, not to a new family introduced here. One retained historical source record is Josef Meixner, *Orthogonale Polynomsysteme mit einer besonderen Gestalt der erzeugenden Funktion*, J. London Math. Soc. 9 (1934), 6–13, doi:10.1112/jlms/sl-9.1.6. The precise recurrence, phase and mass used in this calculation are the displayed original NTG3; its orthonormality is retained from NIB2 and the cumulative source’s full Gamma calculation. The directly checked modern formula reference is NIST DLMF, 18.19.8, 18.19.9, and 18.22.8 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. The exact parameter map is $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$. Substitution in the displayed DLMF recurrence gives NTG3. Its norm under the original density is

$$(n!)^2 \frac{M_s 2^{s/2}}{2\pi\Gamma(s/2)} \frac{2\pi\Gamma(n + s/2)}{2^{s/2} n!} = M_s n! (s/2)_n.$$

The first factor of 2 in the density conversion is the Jacobian $dy = 2 dx$, so this dictionary also checks the original mass. DLMF is used as a verified formula reference; the Meixner record preserves a historical attribution trail whose exact relation to these displayed formulas has not been verified here from the complete article. It is not used as a checked theorem provider.

The orthogonal projections and attained least-squares minima below use the classical Gram–Schmidt construction. The retained historical record includes Erhard Schmidt, *Zur Theorie der linearen und nichtlinearen Integralgleichungen. I*, Math. Ann. 63 (1907), 433–476, doi:10.1007/BF01449770, with the earlier J. P. Gram attribution retained for the historical audit to trace in the original pages. The block determinant eliminations are Schur-complement calculations; their retained historical trail includes Issai Schur, *Über Potenzreihen, die im Innern des Einheitskreises beschränkt sind*, J. Reine Angew. Math. 147 (1917), 205–232, doi:10.1515/crll.1917.147.205.

Every finite projection, determinant identity and inequality needed below is proved explicitly. The contribution here is their application to NTG1–3 with the full original conductor and functional frames, including the precise native phase. For the actual minimum and determinant identities we also cite the directly read derivations in Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]; the formulas below provide the complete complex Hermitian specialization used in this programme. The publisher records for Schmidt and Schur were checked for their bibliographic metadata. Their complete historical proofs were not read as external theorem providers in this calculation; the finite proofs below are self-contained.

4.2 Finite evaluation of every coefficient and every pairing

Write $\chi_j(S) = \sum_{a=0}^q c_a S^a$, with $c_q = 1$ and all coefficients obtained from the original root product. Direct binomial expansion gives, for $0 \leq h \leq q + t$,

$$[S'^h]_{g_t} = \sum_{\substack{0 \leq a \leq q \\ a+t \geq h}} c_a \binom{a+t}{h} \mu_{a+t-h}. \quad (\text{NTG4})$$

Thus every coefficient above degree $L + t$ vanishes because $\mu_d = 0$ for $d < v$, and the coefficient at degree $L + t$ is exactly ℓ_t . This verifies the derivative factorials in the original translation map. It does not replace that map by a derivative of order v .

For each integer $d \geq 0$ let $J_s^{(d)}$ be the matrix on indices $0, \dots, d$ with zero diagonal and off-diagonal entries $J_s^{(d)}[n, n+1] = J_s^{(d)}[n+1, n] = \sqrt{(n+1)(n+s/2)}$. Use the convention that the zeroth power is the identity. The exact moments are

$$M_{d,s} := \int y^d dm_s(y) = M_s((J_s^{(d)})^d)_{00}. \quad (\text{NTG5})$$

Indeed multiplication by y in the orthonormal basis of NTG3 has exactly those adjacent entries. A path of d multiplication steps starting at index zero never exceeds index d . Taking its inner product with the constant unit polynomial $1/\sqrt{M_s}$ proves NTG5. In particular it preserves the entire original mass.

Put $\mathfrak{H}_s[h, b] = \langle S'^h, S'^b \rangle_s$. Expansion on the original vertical line gives the completely finite formula

$$\mathfrak{H}_s[h, b] = \sum_{\alpha=0}^h \sum_{\beta=0}^b \binom{h}{\alpha} \binom{b}{\beta} (c')^{h+b-\alpha-\beta} (-i)^\alpha i^\beta M_{\alpha+\beta,s}. \quad (\text{NTG6})$$

Consequently, for arbitrary polynomials P, Q with coefficient columns p, q , their inner product is $p^* \mathfrak{H}_s q$ at their finite degree cutoff. This formula evaluates every cross pairing below without quadrature, unlisted constants, or an asymptotic replacement of the Gamma density.

Set

$$B_s[a, b] = \langle \chi - S'^a, \chi - S'^b \rangle_s. \quad (\text{NTG7})$$

NTG6 and the full lower root coefficients calculate its entries. Every finite leading principal block is strictly positive: a nonzero polynomial has positive squared integral against the strictly

positive Gamma density outside its finite zero set. For $b \geq 1$, let $B_{<b} = (B_s[a, d])_{a, d < b}$ and $z_b = (B_s[a, b])_{a < b}$. Define

$$\begin{aligned} \pi_b^-(S') &= S'^b - \sum_{a < b} S'^a (B_{<b}^{-1} z_b)_a, & \nu_{b,s}^- &= B_s[b, b] - z_b^* B_{<b}^{-1} z_b, \\ \pi_0^- &= 1, & \nu_{0,s}^- &= B_s[0, 0], & e_b(S') &= \chi_-(S') \pi_b^-(S') / \sqrt{\nu_{b,s}^-}. \end{aligned} \quad (\text{NTG8})$$

Completing the coefficient square proves that π_b^- is the unique monic minimizer of degree b in the lower weighted norm and $\nu_{b,s}^-$ is its strictly positive squared norm. Its residual is orthogonal to every preceding monomial. It follows by successive degrees that the e_b are orthonormal, and $e_0, \dots, e_d$ span the complete ideal $\chi_- \mathbb{C}[S']_{\leq d}$. This is the actual Gram–Schmidt operation just cited, with its entire finite coefficient map given in NTG7–8.

For later use the analytic entries are also a finite calculation from the same nonreduced conductor. Write $n_{d,b} = N_b^{(d)}(0) = \sum_{\lambda} Z_{\lambda b} \beta_{j,\lambda}^d$ with the original numerator frame Z , and $f_{d,b} = F_b^{(d)}(0)$. The original vanishing conditions $n_{d,b} = 0$ for $d < v$ make N_b/E_A analytic at zero. Leibniz's formula gives the recursion

$$f_{d,b} = \frac{n_{d+v,b} - \sum_{h=v+1}^{d+v} \binom{d+v}{h} \mu_h f_{d+v-h,b}}{\binom{d+v}{v} \mu_v} \quad (d \geq 0). \quad (\text{NTG9})$$

Each index in the sum is smaller than d , so the recursion starts with $f_{0,b} = n_{v,b}/\mu_v$ and determines every required derivative in finitely many operations. If $e_a = \sum_h e_{ha} S'^h$, then $\Lambda_{F_b}(e_a) = \sum_h e_{ha} f_{h,b}$, with no conjugation of this complex-linear functional value.

4.3 The actual relation graph and its full attained Gram

For $Q \geq 0$ form the coefficient matrix of $u_0, \dots, u_Q$ in the original orthonormal lower ideal basis:

$$\begin{aligned} \mathcal{R}_Q[b, t] &= \langle e_b, u_t \rangle_s, \quad 0 \leq b \leq m + Q, \quad 0 \leq t \leq Q, \\ \mathcal{R}_Q &= \begin{bmatrix} \mathcal{L}_Q \\ H_Q \end{bmatrix}, \quad \mathcal{L}_Q[b, t] = \mathcal{R}_Q[b, t] \quad (b < m), \quad H_Q[a, t] = \mathcal{R}_Q[m + a, t]. \end{aligned} \quad (\text{NTG10})$$

NTG4,6,8 evaluate every entry. If $a > t$, e_{m+a} is orthogonal to u_t by degree. For $a = t$, the difference $u_t - \chi_- \pi_{m+t}^-$ has smaller degree in the same lower ideal, so

$$H_Q[a, t] = 0 \quad (a > t), \quad H_Q[t, t] = \sqrt{\nu_{m+t,s}^-} > 0. \quad (\text{NTG11})$$

In particular H_Q is invertible and is upper triangular in the unchanged degree order. Define

$$X_Q = \mathcal{L}_Q H_Q^{-1} = [x_0, \dots, x_Q], \quad \Omega_Q = I_m + X_Q X_Q^*, \quad X_{-1} \text{ empty}, \quad \Omega_{-1} = I_m. \quad (\text{NTG12})$$

The first $Q+1$ columns of X_{Q+1} equal X_Q . To prove this, partition H_{Q+1} as $\begin{bmatrix} H_Q & h \\ 0 & d \end{bmatrix}$; its first $Q+1$ inverse columns are $\begin{bmatrix} H_Q^{-1} \\ 0 \end{bmatrix}$. Multiplication by $[\mathcal{L}_Q, \ell]$ proves the assertion. Thus the columns x_t are actual compatible original-data columns and are not selected anew at each cutoff.

In the coefficient frame $e_0, \dots, e_{m+Q}$ the complete functional relation space has the exact graph frame

$$\mathcal{R}_Q H_Q^{-1} = \begin{bmatrix} X_Q \\ I_{Q+1} \end{bmatrix}. \quad (\text{NTG13})$$

Both sides are injective maps from $\mathbb{C}^{Q+1}$ into the same original ideal. If c is the original u_t coefficient column and z is its high-degree coordinate column, the two mutually inverse coordinate

maps are $z = H_Q c$ and $c = H_Q^{-1} z$. The inverse of the graph map itself is simply high-degree coordinate projection. Thus this calculation preserves every original relation.

Let $G_Q = (\langle u_a, u_b \rangle_s)_{0 \leq a, b \leq Q}$. Orthonormality and NTG13 give

$$G_Q = H_Q^*(I_{Q+1} + X_Q^* X_Q) H_Q, \quad (NTG14)$$

$$\frac{\det G_Q}{\prod_{t=0}^Q \nu_{m+t,s}^-} = \det(I_{Q+1} + X_Q^* X_Q) = \det \Omega_Q.$$

For completeness the last determinant equality follows by eliminating either diagonal identity block in $\begin{bmatrix} I_m & X_Q \\ -X_Q^* & I_{Q+1} \end{bmatrix}$. These two eliminations use block unitriangular factors of determinant one. This proves the identity without an assumption on singular values or ranks. The scalar denominator in NTG14 is exactly $|\det H_Q|^2$ by NTG11; none of the lower-ideal minima has been replaced by a trial polynomial.

Successively completing the squares of G_Q identifies its last Schur complement with the original $\eta_{Q,s}$ of NIB5. Induction gives $\det G_Q = \prod_{t=0}^Q \eta_{t,s}$, including $Q = 0$. NIB9 and NTG14 therefore imply

$$\log \frac{\det \Omega_Q}{\det \Omega_R} = \sum_{t=R+1}^Q \log \frac{\eta_{t,s}}{\nu_{m+t,s}^-} = \sum_{t=R+1}^Q \log(1 + r_{L+t}) \quad (-1 \leq R < Q). \quad (NTG15)$$

This is a relation-Gram identity on the actual source, with dimension $m = O(j)$ on its left. No asymptotic bound for this left side is inferred merely from that dimension.

4.4 The literal analytic quotient metric and the native phase

Define the square matrix and its succeeding rows by

$$A[a, b] = \Lambda_{F_b}(e_a) \quad (0 \leq a < m, \ 1 \leq b \leq m), \quad F_{hi}[t, b] = \Lambda_{F_b}(e_{m+t}). \quad (NTG16)$$

All entries are evaluated by NTG8–9. The matrix A is invertible. Indeed an ideal polynomial of degree at most $q' + m - 1 = L - 1$ annihilated by all F_b also vanishes at all lower nodes. The full original functional map at degree $L - 1$ is invertible by NIB4's Vandermonde argument. Thus that polynomial is zero. In the m -dimensional basis $e_0, \dots, e_{m-1}$ its functional map is $A^\top$, which is consequently injective and hence invertible.

Applying each complex-linear functional to NTG13 gives, with ordinary transpose,

$$X_Q^\top A + F_{hi} = 0. \quad (NTG17)$$

Here F_{hi} contains rows $0, \dots, Q$. All signs, conjugations and the complete lower ideal are fixed by this actual polynomial equation.

Let D_n be the original native quotient Gram after observing rows $0, \dots, n$:

$$D_n = T_n^*(I - V_n(V_n^* V_n)^{-1} V_n^*) T_n, \quad n \geq L - 1. \quad (NTG18)$$

The columns of V_n are all original lower evaluations and those of T_n are all original analytic NSO functionals evaluated on $\phi_0, \dots, \phi_n$. Their full ranks follow from the same original Vandermonde argument. Let C_n be the matrix of coefficients of $e_0, \dots, e_{n-q'}$ in that orthonormal native basis. Then $C_n^* C_n = I$ and $V_n^\top C_n = 0$. Polynomial division by χ_- and the dimension count show that these are all vectors in $\ker V_n^\top$. Thus

$$I - V_n(V_n^* V_n)^{-1} V_n^* = \overline{C_n} C_n^\top. \quad (NTG19)$$

Both sides are the orthogonal projector onto $\ker V_n^* = \overline{\ker V_n^\top}$; the identity also follows directly by applying them to their image and its orthogonal complement. Since $C_n^\top T_n$ is the matrix of actual functional values on the e_b , substitution of NTG17 into NTG18 proves the full matrix equality

$$D_{L+Q} = A^*(I_m + \overline{X_Q} X_Q^\top) A = A^* \overline{\Omega_Q} A, \quad D_{L-1} = A^* A. \quad (NTG20)$$

This is equality in the unchanged analytic coefficient frame, not only a determinant equality. It proves the exact map to every original restriction and quotient: in four copies use $\text{diag}(A, 4)$ on the original coefficient frame, retain $\text{diag}(\overline{\Omega_Q}, 4)$ as its metric, and perform the original restriction or Schur minimum on the original affine coefficient fibre. The conjugation in NTG20 is forced by the complex-linear functional pairing in NTG17 and NTG19.

There is also an exact row phase. Put $n = L + t$ and write v_n, t_n for the appended lower and analytic rows. At cutoff $n - 1$, retain

$$\zeta_n = v_n(V_{n-1}^* V_{n-1})^{-1} v_n^*, \quad \rho_n = t_n - v_n(V_{n-1}^* V_{n-1})^{-1} V_{n-1}^* T_{n-1}. \quad (\text{NTG21})$$

The last native coefficient of e_{m+t} is $c_n = i^n \sqrt{h_{n,s}/\nu_{m+t,s}^-} = i^n / \sqrt{1 + \zeta_n}$ by its leading coefficient and NIB8. Its old coefficient column is the minimum-norm solution of the lower evaluation equation, because it is orthogonal to the old lower ideal. This solution is $-c_n \overline{V_{n-1}} (V_{n-1}^T \overline{V_{n-1}})^{-1} v_n^T$. Multiplying its ordinary transpose by T_{n-1} and adding $c_n t_n$ gives $\Lambda_F(e_{m+t}) = c_n \rho_n$. Equation NTG17 therefore proves

$$\boxed{\frac{\rho_n}{\sqrt{1 + \zeta_n}} = -i^{-n} x_t^T A, \quad r_n = x_t^* \Omega_{t-1}^{-1} x_t.} \quad (\text{NTG22})$$

For the last equality substitute NTG20 and the row equality into $r_n = \rho_n D_{n-1}^{-1} \rho_n^* / (1 + \zeta_n)$. It yields $x_t^T \overline{\Omega_{t-1}^{-1}} x_t$, the conjugate of the real nonnegative number $x_t^* \Omega_{t-1}^{-1} x_t$. This supplies the original phase-sensitive row to each of the eight fixed flag compressions; zeros and all eigenvalues stay present.

4.5 Joint minimization of the complete later relations

Fix the accepted prefix endpoint $R = R_j = \lfloor j^{3/2} / \log j \rfloor < q$ and any $R < Q \leq q$. Write $X_Q = [X_R, Y]$, with $Y = [x_{R+1}, \dots, x_Q]$, and partition the upper triangular matrix of NTG11 as

$$H_Q = \begin{bmatrix} H_R & H_{12} \\ 0 & H_{22} \end{bmatrix}. \quad (\text{NTG23})$$

Let $S_{R,Q}$ be the full attained Gram on the later relation coefficients after minimizing over every earlier coefficient:

$$z^* S_{R,Q} z = \min_{a \in \mathbb{C}^{R+1}} \left\| \sum_{t=0}^R a_t u_t + \sum_{t=R+1}^Q z_{t-R-1} u_t \right\|_s^2. \quad (\text{NTG24})$$

The minimum is attained uniquely since the preceding relation Gram is positive. In the graph coordinates NTG13, the preceding columns are $[X_R; I_{R+1}; 0]$ and the new columns are $[Y; 0; I_{Q-R}]$. Their two diagonal Grams are $I + X_R^* X_R$ and $I + Y^* Y$, and their cross Gram is $X_R^* Y$. Completing the entire preceding square gives

$$\begin{aligned} S_{R,Q} &= H_{22}^* (I_{Q-R} + Y^* \Omega_R^{-1} Y) H_{22}, \\ \frac{\det S_{R,Q}}{\prod_{t=R+1}^Q \nu_{m+t,s}^-} &= \det(I_{Q-R} + Y^* \Omega_R^{-1} Y) = \frac{\det \Omega_Q}{\det \Omega_R}. \end{aligned} \quad (\text{NTG25})$$

The H_{12} component is in the preceding relation span and is minimized over, so it disappears only through the displayed full minimum. To verify its inverse factor, direct multiplication proves $I - X_R(I + X_R^* X_R)^{-1} X_R^* = (I + X_R X_R^*)^{-1}$. Conjugating by Y proves the first line. The determinant identities follow by the same two identity-block eliminations used in NTG14. This calculation jointly minimizes every later linear combination and preserves every off-diagonal pairing between those combinations.

4.6 Finite analytic improvement from the retained cross pairings

All the following quantities have already been evaluated by the finite original-data operations NTG4–12:

$$\begin{aligned} a_t &= x_t^* \Omega_R^{-1} x_t, \quad c_{th} = x_t^* \Omega_R^{-1} x_h, \quad R < t, h \leq Q, \\ d &= Q - R, \quad d_0 = \min(m, d), \quad T_{R,Q} = \sum_{t=R+1}^Q a_t. \end{aligned} \tag{NTG26}$$

The exact unweighted tail value $Z_{R,Q} := \log \det \Omega_Q - \log \det \Omega_R$ satisfies

$$0 \leq Z_{R,Q} \leq d_0 \log \left(1 + \frac{T_{R,Q}}{d_0} \right). \tag{NTG27}$$

Indeed the nonzero eigenvalues of $\Omega_R^{-1/2} Y Y^* \Omega_R^{-1/2}$ are nonnegative, number at most d_0 , and sum to $T_{R,Q}$. The determinant equality NTG25 identifies $Z_{R,Q}$ with the sum of $\log(1 + \lambda)$ over those eigenvalues. Append zeros to reach d_0 terms. The elementary concavity inequality for $\log$, proved by its negative second derivative and finite weighted averaging, gives NTG27. This bound charges the actual dimension m rather than the number of remaining rows when $Q - R > m$. It does not bound $T_{R,Q}$ by an unproved small number.

There is an explicit second improvement that measures the retained interaction of distinct later relations. Partition the actual consecutive indices $R + 1, \dots, Q$ into adjacent pairs and at most one final singleton; denote the pairs by $\mathcal{P}$. Define

$$\Delta_{R,Q} := - \sum_{(t,h) \in \mathcal{P}} \log \left(1 - \frac{|c_{th}|^2}{(1 + a_t)(1 + a_h)} \right) \geq 0. \tag{NTG28}$$

Every term is finite. Minimizing $\|x_h - z x_t\|_{\Omega_R^{-1}}^2$ over complex z gives the Cauchy–Schwarz inequality in this exact positive metric (if $x_t = 0$ the mixed pairing is already zero). Thus $|c_{th}|^2 \leq a_t a_h < (1 + a_t)(1 + a_h)$. It follows that

$$Z_{R,Q} \leq \sum_{t=R+1}^Q \log(1 + a_t) - \Delta_{R,Q}. \tag{NTG29}$$

Here is the determinant proof, including the full interaction discarded by a scalar bound. The positive matrix $M = I + Y^* \Omega_R^{-1} Y$ has diagonal entries $1 + a_t$ and off-diagonal entries c_{th} . For a two-block partition, elimination gives $\det M = \det M_{11} \det(M_{22} - M_{21} M_{11}^{-1} M_{12}) \leq \det M_{11} \det M_{22}$. The inequality follows since the second positive factor is bounded above by M_{22} ; conjugating by $M_{22}^{-1/2}$ gives eigenvalues at most one and therefore a determinant at most one. Applying this successively to the prescribed pairs bounds $\det M$ by the product of the actual pair determinants and the possible singleton entry. Each pair determinant is exactly $(1 + a_t)(1 + a_h) - |c_{th}|^2$. Taking logarithms proves NTG29.

This is the familiar determinant inequality associated with Hadamard and its block refinements, applied to an explicitly computed original Gram. The retained historical source record is Jacques Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. (2) 17 (1893), 240–246; a scan of the original article is preserved at the primary-article archive. This archive’s bibliographic metadata was consulted here; its full original proof was not used as a verified theorem. The proof above supplies the precise positive-block version used here rather than importing an unstated variant.

For the first two remaining indices $a = R + 1$, $b = R + 2$, the entire joint calculation is especially explicit:

$$\begin{aligned} \log(1 + r_{L+a}) &= \log(1 + a_a), \\ r_{L+b} &= a_b - \frac{|c_{ab}|^2}{1 + a_a}, \\ Z_{R,R+2} &= \log((1 + a_a)(1 + a_b) - |c_{ab}|^2). \end{aligned} \tag{NTG30}$$

The inverse identity $(\Omega_R + x_a x_a^*)^{-1} = \Omega_R^{-1} - \Omega_R^{-1} x_a x_a^* \Omega_R^{-1} / (1 + a_a)$ is verified by multiplication; substituting it in NTG22 proves the second line and the other lines follow. Thus the calculated correction for the second later row is the exact nonnegative quantity $|c_{ab}|^2 / (1 + a_a)$. It vanishes precisely when the displayed original cross pairing vanishes. No strict positivity is asserted without evaluating that original pairing, and none is needed for NTG25–30.

The older scalar bound NIB14 remains valid for every row. Combining it with the two joint bounds gives the fully finite original-data estimate

$$0 \leq Z_{R,Q} \leq \min \left\{ \sum_{t=R+1}^Q E_{j,s,t}, d_0 \log \left(1 + \frac{T_{R,Q}}{d_0} \right), \sum_{t=R+1}^Q \log(1 + a_t) - \Delta_{R,Q} \right\}, \quad (\text{NTG31})$$

Every entry in the new two bounds is a finite expression in the actual roots, all a_{uw} , the exact μ_d , the complete Gamma moments and the attained lower minima. The sum of the old scalar estimates is retained as a separate bound, so these new finite bounds cannot degrade the previously proved one. The exact formula NTG25 is stronger still when its finite determinant is used.

4.7 The surviving original tail and its precise receiving scope

The original tail has weights two at $R+1, \dots, q-1$ and weight one at q . Its scalar value is therefore

$$\begin{aligned} \mathcal{T}_{j,s}(R) &:= 2Z_{R,q-1} + Z_{q-1,q} = Z_{R,q-1} + Z_{R,q}, \\ &= \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_R)^2} \geq 0. \end{aligned} \quad (\text{NTG32})$$

Here $Z_{R,R} = 0$ when needed. For the accepted $j \geq 13$ both $q-1$ and q are larger than R_j . Applying NTG31 separately at $Q = q-1, q$ gives an explicit joint finite upper estimate for this entire unchanged tail. By NIB16 the absolute signed native loss tail, the absolute signed threshold tail, and each of the four nonnegative CPQ complementary tails are bounded by $4\mathcal{T}_{j,s}(R)$. Their exact values retain their original eight fixed flags and their respective quotient minima, all supplied with the same matrix NTG20 and row NTG22. The signs remain positive at T_e, K_c, W_i, K_i and negative at W, S, J, I ; the two nested flags and the full CPQ identity remain unchanged.

NIB20 already controls the prefix by $o(kq_k)$. The unresolved asymptotic size here is the calculated matrix expression NTG32, equivalently the correlated original-data columns x_t beyond R_j . A small rank or the original scalar Christoffel bounds alone do not assign its leading coefficient: its exact map to those norms is NTG10–14, retaining the full block $\mathcal{L}_Q$ and the complete triangular cross block H_Q . Equations NTG24–31 calculate the simultaneous minimization and its finite improvement; they do not assume a missing asymptotic estimate. Both orders $s = 1, j$, the fixed period, every original root and conductor coefficient, and all masses and phases occur throughout.

4.8 Exact accepted source locators

The original input definitions and prior proofs are NIB1–20 in `ORIGINAL_NATIVE_INTERIOR_BOUND.tex`, SHA256 `bcc07cec41f4c8e94362947b420d06a8973967a5cfc13abb42c9413657c7d7de`, and IGI1–30 in `ORIGINAL_GRID_INTERIOR_INTERFACE.tex`, SHA256 `4311bbc13a0f16f2585cd05755e61cf8bf4e72853af903d2172f55253072b36b`. These internal pins identify the exact original objects used; they do not replace the point-of-use human provenance above. In particular NIB4 supplies completeness of the original functional annihilator, NIB8–9 supplies the exact native row ratio, NIB16 supplies the unchanged eight-flag and complementary bounds, and NIB18–20 supplies the accepted superlinear prefix. IGI7,12 records the same complex-linear dual convention that produces the conjugation in NTG20.

5 Szegő's confluent modification formula on the full original lower grid

5.1 The primary source and its exact original-data dictionary

The human source used here is Gábor Szegő, *Orthogonal Polynomials*, fourth edition, AMS Colloquium Publications 23 (1975), Theorem 2.5 and equations (2.5.1)–(2.5.4), printed pages 29–30 [H24, Theorem 2.5, pp. 29–30]. The publisher's book-level identifier is doi:10.1090/coll/023; the fourth-edition date is printed on the controlling local copy's edition-history page. The source explicitly credits E. B. Christoffel's earlier special case. Theorem 2.5 represents the orthogonal polynomial for a nonnegative polynomial modification of a positive distribution by a determinant of consecutive original orthogonal polynomials. For a repeated multiplier root it specifies the value and successive derivative rows. It does not assert that this determinant is monic or orthonormal; its proof establishes nonvanishing of the leading cofactor by positivity. The calculation below retains that cofactor and proves the complete specialization in the original coordinates.

Keep all NTG1–3 data, including the fixed original period, the complete nonreduced conductor and actual $\mu_d = E_A^{(d)}(0)$, the two source orders $s = 1, j$, and

$$\begin{aligned} a &= j - 8, \quad c' = a/2, \quad d = q' = (a + 1)^2, \quad D = 2d, \quad q = (j + 1)^2, \quad L = q - v, \quad m = L - d, \\ z_{u,w} &= (2w - a)\gamma - i(2u - a)\delta = (\beta_{a,u,w} - c')/i, \quad 0 \leq u, w \leq a, \\ Q_-(y) &= \prod_{u,w=0}^a (y - z_{u,w}) = i^{-d} \chi_-(c' + iy), \quad \rho(y) = Q_-(y)^2. \end{aligned} \tag{SLC1}$$

The index order is the original lexicographic order of (u, w) . The roots are distinct because real and imaginary parts separately determine w and u . The permutation $(u, w) \mapsto (a - u, w)$ conjugates the roots; hence Q_- has real coefficients. No root is real because a is odd, so $2u - a \neq 0$ and $\delta > 0$. In particular

$$\rho(y) > 0 \quad (y \in \mathbb{R}), \quad |\chi_-(c' + iy)|^2 = \rho(y). \tag{SLC2}$$

The source distribution is literally the original dm_s , the old monic polynomials are the $p_{n,s}$ of NTG3, and the multiplier is the degree- D polynomial ρ . It has each original $z_{u,w}$ as a root of multiplicity two. The original Gamma density is strictly positive and has all polynomial moments; the finite exponential upper bound NIB11 also proves this directly. Thus all integrals used here are finite. No finite-interval approximation is needed.

Szegő's displayed old polynomials are orthonormal. In the present determinant we use the original monic $p_{n,s}$ instead. This is the exact invertible column map multiplying its l th old column by $\sqrt{h_{b+l,s}}$, so the whole determinant is multiplied by $\prod_{l=0}^D \sqrt{h_{b+l,s}}$. Every derivative row is multiplied by the same column factor. The denominator cofactor changes by its own corresponding product, and we retain those factors through the original $h_{n,s} = M_s n! (s/2)_n$ below. The direct proof that follows also verifies this monic presentation without suppressing any column or mass factor.

5.2 The full confluent determinant and its leading cofactor

For each $b \geq 0$ form the D by $D + 1$ matrix

$$\begin{aligned} \mathcal{M}_{b,s}[(\lambda, 0), l] &= p_{b+l,s}(z_\lambda), \quad \mathcal{M}_{b,s}[(\lambda, 1), l] = p'_{b+l,s}(z_\lambda), \\ 0 &\leq l \leq D. \end{aligned} \tag{SLC3}$$

For each root in the stated order, its value row precedes its first-derivative row. These are derivatives of order one, without an inserted factorial or derivative in S' . Define every signed cofactor

and the complete polynomial

$$\begin{aligned} C_{b,l} &= (-1)^l \det \mathcal{M}_{b,s}^{\widehat{l}}, & \alpha_b &= C_{b,D}, \\ \mathcal{F}_b(y) &= \det \begin{bmatrix} (p_{b+l,s}(y))_{l=0}^D \\ \mathcal{M}_{b,s} \end{bmatrix} = \sum_{l=0}^D C_{b,l} p_{b+l,s}(y). \end{aligned} \quad (\text{SLC4})$$

The superscript $\widehat{l}$ removes exactly column l , leaving all root and derivative rows in their original order. The sign is the first-row Laplace sign $(-1)^{1+(l+1)}$. The determinant block notation in SLC4 means its first row followed by all D rows of SLC3.

The leading cofactor α_b is nonzero. Since $D = 2d$ is even, it is the determinant of columns $0, \dots, D-1$ of $\mathcal{M}_{b,s}$. If it vanished, a nonzero coefficient column would give a nonzero polynomial $H = \sum_{l=0}^{D-1} v_l p_{b+l,s}$ vanishing with its first derivative at every z_λ . Distinct-root division gives $H = \rho G$ with $\deg G \leq b-1$. For $b=0$ degree alone makes $H=0$, a contradiction. For $b \geq 1$, every $p_{b+l,s}$ is orthogonal in dm_s to every polynomial of degree below b . Taking that polynomial to be G in the conjugate-linear first argument yields

$$0 = \int_{\mathbb{R}} \overline{G(y)} H(y) dm_s(y) = \int_{\mathbb{R}} \rho(y) |G(y)|^2 dm_s(y). \quad (\text{SLC5})$$

The strict positivity in SLC2 implies $G=0$: a nonzero polynomial is nonzero on a nonempty real interval with positive measure. This contradicts independence of the distinct-degree monic polynomials in H . This is the complex-coefficient version of the positivity argument in Szegő's (2.5.4), preserving the repeated roots and all derivative rows.

At $y = z_\lambda$, the first row of $\mathcal{F}_b$ repeats its value row, so $\mathcal{F}_b(z_\lambda) = 0$. Differentiating only its first row shows that $\mathcal{F}'_b(z_\lambda) = 0$ by the corresponding derivative row. Thus $\rho \mid \mathcal{F}_b$. Its degree is precisely $b+D$, with leading coefficient $\alpha_b \neq 0$, because only the last monic column in SLC4 has this degree. Hence

$$P_b^-(y) = \frac{\mathcal{F}_b(y)}{\alpha_b \rho(y)} \quad (\text{SLC6})$$

is a monic polynomial of degree b . For every polynomial h of degree below b , $\int \bar{h} \rho P_b^- dm_s = \alpha_b^{-1} \int \bar{h} \mathcal{F}_b dm_s = 0$ by SLC4 and the old orthogonality. Consequently P_b^- is the unique monic minimizer for the positive measure ρdm_s : adding any smaller-degree polynomial adds its positive squared norm, by the just proved orthogonality. Uniqueness and the real density imply that P_b^- has real coefficients, since coefficientwise conjugation gives another monic polynomial with the same orthogonality and norm.

The exact monic coordinate map and its inverse are

$$\pi_b^-(S') = i^b P_b^-((S' - c')/i), \quad P_b^-(y) = i^{-b} \pi_b^-(c' + iy). \quad (\text{SLC7})$$

The leading coefficient in S' is one. These inverse maps carry the same complete monic affine fibre to the one in NTG8; they multiply values on the original line by the phase i^b of modulus one. Thus π_b^- is exactly the original NTG8 lower minimizer, and

$$\begin{aligned} \nu_{b,s}^- &= \int \rho(y) |P_b^-(y)|^2 dm_s(y), \\ e_b(c' + iy) &= \frac{i^{d+b} Q_-(y) P_b^-(y)}{\sqrt{\nu_{b,s}^-}}. \end{aligned} \quad (\text{SLC8})$$

Every displayed phase comes from SLC1 and SLC7.

5.3 The norm, the cofactor sign and every polynomial coefficient

Both endpoint cofactors in SLC4 are explicit. Since D is even, deleting column zero gives

$$C_{b,0} = \det[p_{b+1,s}, \dots, p_{b+D,s}]_{\text{jets}} = \alpha_{b+1}. \quad (\text{SLC9})$$

Using $P_b^- = p_{b,s} +$ a polynomial of degree below b , old orthogonality gives $\int \overline{P_b^-} p_{b,s} dm_s = h_{b,s}$. All terms $p_{b+l,s}$ with $l > 0$ are orthogonal to P_b^- of degree b . Multiplying SLC6 by $\rho \overline{P_b^-}$ and integrating therefore yields

$$\boxed{\nu_{b,s}^- = h_{b,s} \frac{C_{b,0}}{\alpha_b} = M_s b! (s/2)_b \frac{\alpha_{b+1}}{\alpha_b} > 0.} \quad (\text{SLC10})$$

In particular the actual cofactor ratio is positive real; no absolute value is substituted for a determinant.

The initial cofactor is

$$\alpha_0 = \prod_{\lambda < \mu} (z_\mu - z_\lambda)^4 > 0. \quad (\text{SLC11})$$

Here is a direct verification of its sign and value. The old columns $p_{0,s}, \dots, p_{D-1,s}$ are obtained from $1, y, \dots, y^{D-1}$ by a triangular coefficient matrix with diagonal one, so this changes no determinant. For each root use the two nearby evaluation rows at z_λ and $z_\lambda + \epsilon_\lambda$. Subtract the first from the second and divide the latter by ϵ_λ . The ordinary Vandermonde determinant, divided by $\prod_\lambda \epsilon_\lambda$, tends as the nonzero ϵ_λ tend to zero to exactly the value-and-first-derivative determinant in SLC3. Each cross-root pair contributes four copies of $z_\mu - z_\lambda$; each within-root factor divided by ϵ_λ contributes one. This proves the product. The root conjugation permutation sends the Vandermonde $\Delta = \prod_{\lambda < \mu} (z_\mu - z_\lambda)$ to $\overline{\Delta} = \pm \Delta$. Hence Δ is real or purely imaginary, and $\Delta^4 > 0$ since $\Delta \neq 0$. This proves SLC11. SLC10 now proves $\alpha_b > 0$ for every b by induction. Complex determinant rows and their original order have been retained throughout.

For a complete coefficient evaluation write $p_{n,s}(y) = \sum_h p_{n,h} y^h$ and $\rho(y) = \sum_{\ell=0}^D \rho_\ell y^\ell$, with $\rho_D = 1$. The old coefficients follow directly from the exact NTG3 recurrence, so SLC3 and SLC4 are finite expressions in the original roots and s . If $\mathcal{F}_b = \sum_h \mathfrak{f}_{b,h} y^h$ and $P_b^- = \sum_{h=0}^b P_{b,h} y^h$, then

$$\begin{aligned} \mathfrak{f}_{b,h} &= \sum_{l=0}^D C_{b,l} p_{b+l,h}, \quad P_{b,b} = 1, \\ P_{b,h} &= \frac{\mathfrak{f}_{b,D+h}}{\alpha_b} - \sum_{r=h+1}^b P_{b,r} \rho_{D+h-r} \quad (h = b-1, b-2, \dots, 0), \\ [S'^h] \pi_b^- &= \sum_{r=h}^b i^{b-r} P_{b,r} \binom{r}{h} (-c')^{r-h}. \end{aligned} \quad (\text{SLC12})$$

An index outside the degree range of a polynomial has coefficient zero. The middle equation is downward division of the already proved identity $\mathcal{F}_b / \alpha_b = \rho P_b^-$ at degree $D+h$; the last is binomial expansion of SLC7. Thus the whole modified polynomial, including its cross-pairing coefficients, is evaluated.

5.4 Evaluation of every original relation coefficient

Keep the actual $g_t = \mathcal{T}_A(\chi_j S^t)$, $\ell_t = \mu_v \binom{q+t}{v}$ and $u_t = g_t / \ell_t$, without changing a single shift or conductor coefficient. Exact root addition $\beta_{a,u,w} + \beta_{8,r,h} = \beta_{j,u+r,w+h}$ shows separately in every translation summand that g_t vanishes at each lower root. The roots are distinct, so define its exact polynomial quotient

$$G_t(S') = g_t(S') / \chi_-(S'), \quad \deg G_t = m + t, \quad \text{lc } G_t = \ell_t. \quad (\text{SLC13})$$

This is polynomial division on the original object, not division of its norm or deletion of any root.

Here are all coefficients needed for that division. Let $g_{t,h} = [S''^h]g_t$ be the finite sum NTG4: $g_{t,h} = \sum_{r+t \geq h} c_r \binom{r+t}{h} \mu_{r+t-h}$, where $\chi_j = \sum c_r S^r$. Write $\chi_- = \sum_{h=0}^d \chi_h S''^h$, $G_t = \sum_{h=0}^{B_t} \gamma_{t,h} S''^h$, $B_t = m + t$. Then

$$\begin{aligned} \gamma_{t,B_t} &= \ell_t, & \gamma_{t,h} &= g_{t,d+h} - \sum_{r=h+1}^{B_t} \gamma_{t,r} \chi_{d+h-r} \quad (h = B_t - 1, \dots, 0), \\ G_t(c' + iy) &= \sum_{h=0}^{B_t} d_{t,h} y^h, & d_{t,h} &= i^h \sum_{r=h}^{B_t} \gamma_{t,r} \binom{r}{h} (c')^{r-h}. \end{aligned} \tag{SLC14}$$

The zero coefficient convention in SLC12 applies here too. The vanishing remainder follows from the proved divisibility, and the formulas give its unique quotient and its exact original-line phase.

Expand this complete quotient in the old Gamma polynomial basis, with all finite coefficients retained:

$$\begin{aligned} G_t(c' + iy) &= \sum_{n=0}^{B_t} \kappa_{n,t} p_{n,s}(y), \\ \kappa_{n,t} &= \frac{1}{h_{n,s}} \sum_{h=0}^{B_t} \sum_{\alpha=0}^n d_{t,h} p_{n,\alpha} M_{h+\alpha,s}, & M_{r,s} &= M_s((J_s^{(r)})^r)_{00}. \end{aligned} \tag{SLC15}$$

The moments and $J_s^{(r)}$ are exactly NTG5. The old polynomials have real coefficients, so $\int p_{n,s} G_t(c' + iy) dm_s = h_{n,s} \kappa_{n,t}$. Old orthogonality proves both this identity and the finite expansion; it gives zero for $n > B_t$. In particular $\kappa_{B_t,t} = i^{B_t} \ell_t$ by the leading coefficient.

The actual NTG relation entry, in its original conjugate-linear first argument, is now

$$\mathcal{R}_Q[b, t] = \langle e_b, u_t \rangle_s = \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-} \alpha_b} \sum_{l=0}^{\min(D, B_t - b)} C_{b,l} h_{b+l,s} \kappa_{b+l,t}.$$

(SLC16)

For $b > B_t$ the sum is empty and the entry is zero. To prove the formula, use SLC8 and $g_t(c' + iy) = i^d Q_-(y) G_t(c' + iy)$. Since Q_- and P_b^- are real on the integration line, this gives

$$\begin{aligned} \langle e_b, u_t \rangle_s &= \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-}} \int \rho(y) P_b^-(y) G_t(c' + iy) dm_s(y) \\ &= \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-} \alpha_b} \int \mathcal{F}_b(y) G_t(c' + iy) dm_s(y). \end{aligned}$$

The factor i^{-b} is the exact product $i^{-(d+b)} i^d$ of the two original-line phases. Expanding $\mathcal{F}_b$ by SLC4 and using SLC15 proves SLC16. No conjugation of ℓ_t or $\kappa_{n,t}$ is permitted: they belong to the second, complex-linear argument. All the $2d$ confluent rows and the original leading cofactor remain in its evaluated numerator.

At $b = B_t = m + t$ only $l = 0$ survives; substituting $\kappa_{B_t,t} = i^{B_t} \ell_t$ and SLC9–10 gives

$$\mathcal{R}_Q[m + t, t] = \frac{C_{B_t,0} h_{B_t,s}}{\sqrt{\nu_{B_t,s}^-} \alpha_{B_t}} = \sqrt{\nu_{m+t,s}^-}. \tag{SLC17}$$

Together with the empty-sum zero for $b > m + t$, this recovers the exact positive triangular diagonal of NTG11 directly from the confluent determinant. Every entry with $b < m + t$ remains the complete sum SLC16; scalar norm ratios alone would not give it.

5.5 The full original NTG graph and analytic frame receive the formula

For every $Q \geq 0$ set, exactly as in NTG10–12,

$$\begin{aligned} \mathcal{L}_Q[b, t] &= \mathcal{R}_Q[b, t] \quad (0 \leq b < m), \\ H_Q[a, t] &= \mathcal{R}_Q[m + a, t] \quad (0 \leq a, t \leq Q), \quad X_Q = \mathcal{L}_Q H_Q^{-1} = [x_0, \dots, x_Q]. \end{aligned} \quad (\text{SLC18})$$

Each entry in both blocks is now the actual determinant and moment sum SLC16. For a direct finite column evaluation, the same invertible triangular equation is

$$x_t = \frac{\mathcal{L}_Q[:, t] - \sum_{a=0}^{t-1} x_a H_Q[a, t]}{\sqrt{\nu_{m+t, s}^-}}, \quad t = 0, \dots, Q. \quad (\text{SLC19})$$

Indeed $X_Q H_Q = \mathcal{L}_Q$ and SLC17 give this by successive columns. The denominator is strictly positive. This recurrence includes every preceding column and every earlier/later cross pairing. It is the same inverse map as NTG12, with an explicit evaluation, so it proves compatibility of the resulting X_Q as Q increases without any new choice of frame.

The complete lower-basis coefficients used by the original analytic frame can also be read off, rather than being reconstructed from its determinant. From SLC12 and NTG8,

$$\begin{aligned} e_{hb} := [S^h]e_b &= \frac{1}{\sqrt{\nu_{b, s}^-}} \sum_{r=0}^d \chi_r \sum_{0 \leq h-r \leq \ell \leq b} i^{b-\ell} P_{b, \ell} \binom{\ell}{h-r} (-c')^{\ell-(h-r)}, \\ A[b, c] &= \sum_{h=0}^{d+b} e_{hb} f_{h, c} \quad (0 \leq b < m). \end{aligned} \quad (\text{SLC20})$$

An empty inner sum is zero. Here the exact original derivatives $f_{h, c} = F_c^{(h)}(0)$ are evaluated by NTG9, with all μ_r and all numerator coefficients in their original frame. Multiplying $\chi_- \pi_b^-$ and applying the complex-linear functional proves SLC20 term by term. Thus the fixed matrix A and every column x_t have received the same complete Szegő polynomial, not only its squared norm.

It follows by the already proved NTG20,22 that these explicit entries give exactly

$$\begin{aligned} \Omega_Q &= I_m + X_Q X_Q^*, \quad D_{L+Q} = A^* \overline{\Omega_Q} A, \\ \frac{\rho_{L+t}}{\sqrt{1 + \zeta_{L+t}}} &= -i^{-(L+t)} x_t^\top A. \end{aligned} \quad (\text{SLC21})$$

The common lower-ideal denominator, source order, fixed period, $\mu_v \binom{q+t}{v}$, every root, derivative row and native phase have all survived the map SLC3–21. It consequently evaluates the input matrices for the complete original eight flags and CPQ quotient minima in their unchanged coefficient frames. It changes no endpoint or weight and assigns no asymptotic coefficient to the growing determinant.

5.6 Primary-source read receipt and accepted inputs

The controlling PDF read for Theorem 2.5 is `Orthogonal Polynomials. AMS Colloquium Publications, 1975 Szego.pdf`, SHA256 988969e6bb6ad590c3fc09a119aa69f7f8aee87063c9137615f8ff9ff7529fea. Physical PDF pages 38–39 contain printed pages 29–30; the full theorem and proof were extracted and read, and printed page 30 was visually inspected, including the determinant, derivative-row instruction and leading-cofactor argument. The continuation on printed page 31 was also read. The confluent determinant is explicitly attributed here to that human source, with its own credit to Christoffel preserved.

The unchanged input proof is `ORIGINAL_NATIVE_TAIL_GRAM.tex`, NTG1–32, SHA256 a7a109771c368f7bced035bd2f8e0f457e01607801b215bed1546d0a97dd4699. This is the citation-revised

receiving copy; its mathematics is unchanged from the authored `daf4d2df5d3a4a9955b0b6bc1da8f3c7e70166b89de4041b9afd984e37a32b2b` input. The determinant polynomial $\mathcal{F}_b$ above is distinct from the original analytic germ $F_c = N_c/E_A$ in SLC20. NTG3 supplies the exact original Gamma parameters and its checked Meixner–Pollaczek formula dictionary; NTG4–9 supplies the original conductor coefficients, moments and analytic derivative recursion; NTG10–22 supplies the full graph and phase-sensitive native receiving maps. The present calculation replaces none of those objects. It adds a primary-source-backed finite evaluation of their complete lower polynomials and cross coefficients. No growing-degree asymptotic claim is made.

6 The simultaneous original tail and its eight coefficient flags

The Schur minimum and determinant factorization used below are classical: see Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]. We prove the required complex Hermitian formulas explicitly. The application is to the unchanged NIB/IPR rows and the original CPQ flags; no generic subspace of the same dimension replaces a programme map.

Fix the original simple quartet, its period and logarithm branch, and its complete conductor. Retain

$$j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad L = q - v, \quad m = L - q', \quad s \in \{1, j\}, \quad c' = j/2 - 4. \quad (\text{STF1})$$

Here $k = 4l + 1 \geq 9$, $v = \text{ord}_0 E_A$ and the full nonzero derivative $\mu_v = E_A^{(v)}(0)$ remain those of NIB1. In particular $m > 0$ on its original domain. The complete native basis is $\phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{M_s n! (s/2)_n}$, with $M_s = (2\pi)^{s/2}$ and the original Meixner–Pollaczek dictionary HPS1–4 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. For all original lower roots $\beta_{j-8,u,w}$, put $V_d[n, (u, w)] = \phi_n(\beta_{j-8,u,w})$; for each original analytic numerator functional Λ_{F_b} put $T_d[n, b] = \Lambda_{F_b}(\phi_n)$, $0 \leq n \leq d$, $1 \leq b \leq m$. The functional pairing is complex-linear, as in NIB; inner products are conjugate-linear in their first entry. The original target form is

$$D_N = T_{N-v}^*(I - V_{N-v}(V_{N-v}^* V_{N-v})^{-1} V_{N-v}^*) T_{N-v}, \quad \mathcal{D}_N = \text{diag}(D_N, 4). \quad (\text{STF2})$$

All inverses exist on $q - 1 \leq N \leq 2q$ by the full original rank proof NIB4. In particular the original masses and phases occur in every entry of these matrices.

At $N = q + r$, $n = L + r$, let $V_0 = V_{n-1}$, $T_0 = T_{n-1}$, and write v_n, t_n for their original next rows. Retain

$$\zeta_n = v_n(V_0^* V_0)^{-1} v_n^*, \quad \rho_n = t_n - v_n(V_0^* V_0)^{-1} V_0^* T_0, \quad U_r = \text{diag}(\rho_n / \sqrt{1 + \zeta_n}, 4). \quad (\text{STF3})$$

IPR1 proves the exact update $\mathcal{D}_{q+r} = \mathcal{D}_{q+r-1} + U_r^* U_r$. The old lower Gram, conductor and analytic functionals in STF3 are retained at their actual changing cutoff.

6.1 Two matrices for the entire original tail

Choose any original integer $0 \leq R < q$ and let $N_0 = q + R$, $H = \mathcal{D}_{N_0}$. For $t \in \{q - 1, q\}$, define the actual stacked rows and relative forms

$$A_t = \begin{bmatrix} U_{R+1} \\ \vdots \\ U_t \end{bmatrix}, \quad C_t = A_t H^{-1/2}, \quad M_t = C_t^* C_t, \quad \mathcal{D}_{q+t} = H^{1/2} (I_{4m} + M_t) H^{1/2}. \quad (\text{STF4})$$

An empty stack gives C_t with zero rows and $M_t = 0$. Summing STF3 proves STF4, including $t = R = q - 1$. The map $z \mapsto H^{1/2} z$ is an isometry from the original coefficient space with metric H to Euclidean $\mathbb{C}^{4m}$, with inverse $H^{-1/2}$. Formula STF4 records both directions and the entire original H ; it does not divide a source measure by its mass. The matrices M_t have size $4m$, independent of the number of tail rows.

For any fixed original subspace X with independent column frame X_0 , set

$$Z_X = H^{1/2} X_0 (X_0^* H X_0)^{-1/2}, \quad P_X = Z_X Z_X^*, \quad K_{X,t} = C_t P_X C_t^*, \quad F_t(X) = \log \det(I + K_{X,t}). \quad (\text{STF5})$$

For $X = 0$ use empty Z_X , $P_X = 0$ and $F_t(X) = 0$. Since $Z_X^* Z_X = I$, P_X is exactly the orthogonal projector onto $H^{1/2} X$. Direct multiplication of STF4 gives

$$\frac{\det(X_0^* \mathcal{D}_{q+t} X_0)}{\det(X_0^* H X_0)} = \det(I + Z_X^* M_t Z_X) = \det(I + K_{X,t}). \quad (\text{STF6})$$

For the last equality, elimination of either diagonal identity block in $\begin{bmatrix} I & B \\ -A & I \end{bmatrix}$ proves $\det(I + AB) = \det(I + BA)$ for rectangular A, B . Apply it with $A = C_t Z_X$ and $B = Z_X^* C_t^*$. Thus every original fixed-frame determinant cancels only in its own same-frame ratio. No cutoff-dependent frame is substituted.

The nonzero spectrum of $K_{X,t}$ can be calculated in $4m$ dimensions:

$$\tilde{K}_{X,t} = M_t^{1/2} P_X M_t^{1/2}. \quad (\text{STF7})$$

Here is the exact map, including singular M_t . On $\text{im } M_t^{1/2}$ define $W_t(M_t^{1/2} z) = C_t z$. It is well defined since $\|C_t z\|^2 = z^* M_t z = \|M_t^{1/2} z\|^2$, and is an isometry onto $\text{im } C_t$. Extend it by zero on $\ker M_t$. Then $C_t = W_t M_t^{1/2}$, and $K_{X,t} = W_t \tilde{K}_{X,t} W_t^*$. Both operators vanish on the corresponding complementary kernels; on their ranges this is a unitary equivalence with inverse W_t^* . It preserves every nonzero eigenvalue with multiplicity. The zero eigenvalues contribute zero to both functions $\log(1+x)$ and $\log \max(1, x)$ used below.

6.2 Each original quotient, with its cross terms

Let $K \subset Y$ be fixed original coefficient spaces. Choose independent X_K for K and fixed lifts X_Q of the specified quotient values, so $[X_K, X_Q]$ frames Y . Let Q_0 be its exact quotient metric in H , and retain the attained base representative

$$Z = X_Q - X_K(X_K^* H X_K)^{-1} X_K^* H X_Q, \quad Q_0 = Z^* H Z, \quad E = H^{1/2} Z Q_0^{-1/2}. \quad (\text{STF8})$$

For $K = 0$ omit its entire term. For $Y = K$ the quotient value space is zero-dimensional: its determinant is 1, its log determinant is 0, and its inverse and square root are the unique operators on that space. These conventions also apply to empty frames throughout. The columns of E are orthonormal, orthogonal to $H^{1/2} K$, and span its orthogonal complement in $H^{1/2} Y$. Thus $EE^* = P_Y - P_K$. Let Z_K be an orthonormal frame for $H^{1/2} K$. Every vector representing quotient value x has transformed coordinate $E Q_0^{1/2} x + Z_K a$. Its new quadratic form is

$$\|Q_0^{1/2} x\|^2 + \|a\|^2 + \|C_t E Q_0^{1/2} x + C_t Z_K a\|^2. \quad (\text{STF9})$$

For any matrix B and vector y , differentiation or square completion gives the unique minimum of $\|a\|^2 + \|y + Ba\|^2$ at $a = -(I + B^* B)^{-1} B^* y$. Its value is $y^*(I + BB^*)^{-1} y$: multiplication verifies $I - B(I + B^* B)^{-1} B^* = (I + BB^*)^{-1}$. Applying this complete minimum to STF9 proves the literal original quotient form at the later endpoint,

$$Q_t = Q_0^{1/2} [I + E^* C_t^* (I + C_t P_K C_t^*)^{-1} C_t E] Q_0^{1/2}. \quad (\text{STF10})$$

In particular $Q_t \succeq Q_0$ and no kernel direction or cross pairing has been removed without minimizing over it. The expression inside STF10 also has the finite coefficient-space realization

$$C_t^* (I + C_t P_K C_t^*)^{-1} C_t = M_t^{1/2} (I + M_t^{1/2} P_K M_t^{1/2})^{-1} M_t^{1/2}. \quad (\text{STF11})$$

To prove it, use the isometry W_t just constructed. On its range the two inverses are intertwined; on the complementary range both surrounding C_t factors vanish. This proves STF11 even at rank loss.

Block elimination in the fixed frame $[X_K, X_Q]$ gives $\det \text{Gram}(Y) = \det \text{Gram}(K) \det Q$. Taking its ratio at $q + t$ and N_0 , all frame constants cancel, yielding

$$\log \frac{\det Q_t}{\det Q_0} = F_t(Y) - F_t(K) \geq 0. \quad (\text{STF12})$$

Consequently STF10 evaluates every complementary tail at the same two matrices M_{q-1}, M_q , preserving the dependence between the quotients.

6.3 The eight flags and the exact allocation identity

Use precisely the CPQ coefficient maps on $\mathcal{U} = \mathbb{C}^{4m}$:

$$\begin{aligned} B &= \text{im } B_{\text{cof}}, & I &= \text{im}(B_{\text{cof}}J_0), & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J, & e &\in \{0, 1\}. \end{aligned} \quad (\text{STF13})$$

Since $B \subset S$, this T is the same $S + W$ as CPQ. The original inclusions give $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$. The actual pairs for the four CPQ complementary forms are

$$\begin{array}{c|cccc} \text{form} & Q_b^E & Q_{\text{rem},e} & G_{s,e} & A_{\text{inv}} \\ \hline (Y, K) & (W, W_i) & (\mathcal{U}, T) & (S, K_c) & (I, K_i). \end{array} \quad (\text{STF14})$$

These identifications are the exact same-value CPQ maps, not merely dimension equalities: W/J identifies with the original image of Φ , W_i/J with its invariant image; $\mathcal{U}/T$ is CPQ8's remaining target quotient; S/K_c is CPQ7's connecting source quotient; and I/K_i is the source pullback quotient in CPQ9. The corresponding fixed lift changes are independent of cutoff and therefore cancel in STF12. All their original value metrics are recovered by STF8–10.

For the logarithm of a restriction determinant in a fixed frame, $f_X(N) = \log \det(X_0^* \mathcal{D}_N X_0)$, IPR3 gives exactly

$$\mathcal{R}_{>R} f_X = F_{q-1}(X) + F_q(X). \quad (\text{STF15})$$

Indeed both terms on the right subtract the same $f_X(q + R)$. It follows that the original signed tail is

$$\mathcal{L}_e^{>R} = \sum_{t=q-1, q}^8 \sum_{a=1}^8 \sigma_a F_t(X_a), \quad (\text{STF16})$$

$$(X_1, \dots, X_8) = (T, K_c, W_i, K_i, W, S, J, I), \quad (\sigma_1, \dots, \sigma_8) = (1, 1, 1, 1, -1, -1, -1, -1).$$

The tuple consists of labelled occurrences: coincident subspaces are retained in every position with its indicated sign. All flag sums below use this tuple, including when a quotient is zero-dimensional. Every complementary tail is $\sum_{t=q-1, q} [F_t(Y) - F_t(K)]$ using its exact pair in STF14. The four-copy kernel restriction and target quotient allocations are

$$4S_R^{>R} = \sum_{t=q-1, q} F_t(J), \quad 4S_K^{>R} = \sum_{t=q-1, q} [F_t(\mathcal{U}) - F_t(J)]. \quad (\text{STF17})$$

The latter is the minimum over the actual full kernel of $\text{diag}(B_j, 4)$. Substitution of STF14 into STF16 cancels the same terms algebraically, proving

$$4S_K^{>R} = \mathcal{L}_e^{>R} + V_b^{>R} + V_{\text{rem}}^{>R} + V_s^{>R} + V_A^{>R}, \quad S_K^{>R} + S_R^{>R} = \mathcal{T}^{>R}. \quad (\text{STF18})$$

Thus the joint calculation supplies every term of IPR7 through common finite matrices and its exact quotient operations.

6.4 Finite comparison with the original row-threshold profile

Define the simultaneous endpoint threshold value

$$\mathcal{B}_e^{>R} = \sum_{t=q-1, q}^8 \sum_{a=1}^8 \sigma_a \sum_b \log \max(1, \lambda_b(K_{X_{a,t}})), \quad d_t = \text{rank } C_t \leq 4m. \quad (\text{STF19})$$

All matrices $K_{X,t}$ are supported on the same d_t -dimensional space $\text{im } C_t$. Compress to it by the exact isometry in STF7. Nested X give ordered projectors and ordered $K_{X,t}$, hence ordered eigenvalues: for each test subspace every Rayleigh quotient of the larger Hermitian matrix dominates

the smaller one, and successive minima and maxima preserve that inequality. This proves the required eigenvalue ordering without assuming a common eigenbasis.

For $x \geq 0$, put $h(x) = \log(1+x) - \log \max(1, x) = u(x) - v(x)$, where $u(x) = \log(1 + \min(x, 1))$ and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both u, v are increasing from zero and at most $\log 2$. Along any four-element ordered eigenvalue chain, the sum of the first and last disjoint increments of either function lies in $[0, \log 2]$: adding its omitted middle nonnegative increment proves the upper bound. The corresponding sum of h increments lies in $[-\log 2, \log 2]$. The two chains in STF13 therefore give a difference bounded by $2 \log 2$ per eigenvalue. Summing over the common range proves

$$|\mathcal{L}_e^{>R} - \mathcal{B}_e^{>R}| \leq 2(d_{q-1} + d_q) \log 2 \leq 16m \log 2. \quad (\text{STF20})$$

This is a bound for the simultaneous endpoint representation, whose eigenvalues differ from the original successive four-row compressions. The exact map to the latter scalar profile is as follows. At each original row the same two-chain argument uses four eigenvalues, hence its signed logarithmic and threshold increments differ by at most $8 \log 2$. The tail weights sum to $2(q - R) - 1$, so the original row-threshold tail $\mathcal{P}_e^{>R}$ satisfies

$$|\mathcal{P}_e^{>R} - \mathcal{B}_e^{>R}| \leq [8\{2(q - R) - 1\} + 2(d_{q-1} + d_q)] \log 2. \quad (\text{STF21})$$

This proves the relation between the two threshold constructions with their precise finite discrepancy; it asserts no equality of individual row and endpoint eigenvalues.

For a positive semidefinite finite matrix let $n_X(t, a) = \#\{\lambda_b(K_{X,t}) > a\}$, counting multiplicity. The elementary identity $\log \max(1, \lambda) = \int_1^\infty \mathbf{1}_{a < \lambda} da/a$ and the finite eigenvalue sum give the exact remaining integral

$$\mathcal{B}_e^{>R} = \sum_{t=q-1, q} \int_1^\infty \sum_{b=1}^8 \sigma_b n_{X_b}(t, a) \frac{da}{a}. \quad (\text{STF22})$$

No limiting interchange is needed; each integrand vanishes beyond the largest eigenvalue of its finite matrices. It is the signed spectral count of the original correlated flag compressions.

Finally choose only the already-proved original prefix $R = R_j = \lfloor j^{3/2}/\log j \rfloor < q$. NIB19–20 and IPR8 bound the removed signed and threshold prefixes by $4\mathfrak{E}_{s, R_j} = o(kq_k)$. Since $m = 16j - 48 - v$ and $q = (j + 1)^2$, STF20–21 are respectively $O(j)$ and $O(j^2)$. Thus the full signed and original threshold returns each differ from the exact finite expression STF22 by $o(kq_k)$, with explicit finite errors given by the prefix budget and STF20–21. This evaluates the complete source-to-expression map and its error; the growing-degree value of STF22 itself is still to be calculated. Both source orders, both primitive indices, the independent proper source and the full action retain their existing domains and receiving identities.

7 The original relation graph supplies the simultaneous mixed flags

This calculation joins the original-data relation formulas NTG4–22 to the simultaneous restriction and quotient formulas STF4–22. The finite Hermitian operations are proved below; the Schur quotient operations being transported are those of Boyd and Vandenberghe, Appendix A.5.5, p. 650, and C.4.1, p. 673 [H23], with their complex specialization proved in STF8–12. Every scalar polynomial and norm retains the Meixner–Pollaczek parameter and mass dictionary of NTG3 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)].

Use NTG’s original A , x_r and Ω_R , and choose an integer $0 \leq R < q$. The symbol D_n^{nat} in this section denotes the matrix called D_n in NTG18. STF2’s target-cutoff convention is $D_N = D_{N-v}^{\text{nat}}$; therefore $N = q + r$ corresponds to $n = L + r$. Define, with the unique positive Hermitian square roots,

$$\begin{aligned} G &= \overline{\Omega_R}, & H_1 &= A^*GA = D_{q+R}, & H &= \text{diag}(H_1, 4), \\ U &= G^{1/2}AH_1^{-1/2}, & U_4 &= \text{diag}(U, 4). \end{aligned} \quad (\text{NFI1})$$

All matrices in the inverses are positive definite by NTG12,16,20. Direct multiplication gives $U^*U = I_m$. It is square and injective, so its inverse is U^* and $UU^* = I_m$ as well. In particular U_4 is a unitary map between the two stated Euclidean coordinate spaces, not an alteration of the original source measure.

For $R \leq t \leq q$ set $d = t - R$ and retain

$$\begin{aligned} Y_t &= [x_{R+1}, \dots, x_t], & Z_t &= G^{-1/2}\overline{Y_t}, \\ B_t &= Z_t^*U = Y_t^T AH_1^{-1/2}, & \Theta_t &= \text{diag}_{s=R+1}^t(-i^{-(L+r)}). \end{aligned} \quad (\text{NFI2})$$

Here Y_t, Z_t are m by d , B_t is d by m , and Θ_t is unitary of size d . For $d = 0$ each displayed rectangular matrix is the corresponding empty map. The equality for B_t follows from $(G^{-1/2})^* = G^{-1/2}$ and the formula for U ; it keeps the ordinary transpose required by the original complex-linear functionals.

Extend STF4’s endpoint notation to every $R \leq t \leq q$ by $C_t = [U_{R+1}^T, \dots, U_t^T]^T H^{-1/2}$, where the bracket means the vertical stack of the original row-block matrices U_r , and use the empty stack for $t = R$. It agrees with STF4 at its two stated endpoints. These four-copy original row updates are stacked by degree. Write their row coordinates as (r, b) , with $R + 1 \leq r \leq t$ and $1 \leq b \leq 4$, and let Π_t be the permutation unitary sending coordinate (r, b) to (b, r) . Thus it changes only the stated row ordering. Set

$$\mathcal{R}_t = \text{diag}(\Theta_t^*, 4)\Pi_t, \quad \mathcal{Z}_t = \text{diag}(Z_t, 4). \quad (\text{NFI3})$$

NTG22 gives the full original row $\rho_{L+r}/\sqrt{1 + \zeta_{L+r}} = -i^{-(L+r)}x_r^T A$. Multiplying on the right by $H_1^{-1/2}$, stacking and using the defined permutation proves, entry by entry,

$$\Pi_t C_t = \text{diag}(\Theta_t B_t, 4), \quad \boxed{\mathcal{R}_t C_t = \mathcal{Z}_t^* U_4}. \quad (\text{NFI4})$$

The inverse row map is $\mathcal{R}_t^*$. Thus the original native phase has been recorded in an explicit invertible map before taking any Gram matrix. No phase is inferred merely from equality of norms.

Taking the Gram of NFI4 gives the exact coefficient-space relation

$$\mathcal{N}_t = \mathcal{Z}_t \mathcal{Z}_t^* = \text{diag}(G^{-1/2}\overline{Y_t} Y_t^T G^{-1/2}, 4), \quad M_t = U_4^* \mathcal{N}_t U_4. \quad (\text{NFI5})$$

This is an equality in the original STF coordinates after the specified unitary map. For example,

$$\det(I_{4m} + M_t) = \left(\frac{\det \Omega_t}{\det \Omega_R} \right)^4. \quad (\text{NFI6})$$

To prove NFI6, $\Omega_t = \Omega_R + Y_t Y_t^*$, so conjugating by $\Omega_R^{-1/2}$ gives $\det(I + \Omega_R^{-1/2} Y_t Y_t^* \Omega_R^{-1/2}) = \det \Omega_t / \det \Omega_R$. Complex conjugation gives the determinant of the single block of $I + \mathcal{N}_t$; both determinants are positive real numbers. There are four blocks. This also proves the empty-tail case, with ratio 1. It is consistent with both allocations in STF17–18, without selecting either allocation.

For every original coefficient subspace X , including those in the labelled eight-flag tuple STF16, retain its fixed independent frame X_0 and put

$$\widehat{Z}_X = \text{diag}(G^{1/2} A, 4) X_0 (X_0^* H X_0)^{-1/2}, \quad \widehat{P}_X = \widehat{Z}_X \widehat{Z}_X^*. \quad (\text{NFI7})$$

If $X = 0$ use the empty frame. NFI1 and STF5 give $\widehat{Z}_X = U_4 Z_X$, hence $\widehat{Z}_X^* \widehat{Z}_X = I$ and $\widehat{P}_X = U_4 P_X U_4^*$. This supplies the exact transported flag with its full original conductor-dependent coefficient matrix A . Its inverse is U_4^* on the transformed Euclidean space, followed by $H^{-1/2}$ when recovering original coefficient vectors. Consequently NFI4 proves

$$\mathcal{R}_t K_{X,t} \mathcal{R}_t^* = \mathcal{Z}_t^* \widehat{P}_X \mathcal{Z}_t, \quad F_t(X) = \log \det(I + \mathcal{Z}_t^* \widehat{P}_X \mathcal{Z}_t). \quad (\text{NFI8})$$

The determinant equality uses a unitary conjugation of the entire row space, including its zero eigenvalues. Its version on the fixed $4m$ -dimensional coefficient space is

$$\widehat{K}_{X,t} = \mathcal{N}_t^{1/2} \widehat{P}_X \mathcal{N}_t^{1/2}. \quad (\text{NFI9})$$

For completeness define the map $\mathcal{W}_t(\mathcal{N}_t^{1/2} z) = \mathcal{Z}_t^* z$ on $\text{im } \mathcal{N}_t^{1/2}$, extending it by zero on the kernel. It is well defined and isometric because $\|\mathcal{Z}_t^* z\|^2 = z^* \mathcal{N}_t z = \|\mathcal{N}_t^{1/2} z\|^2$. Then the right side of NFI8 equals $\mathcal{W}_t \widehat{K}_{X,t} \mathcal{W}_t^*$. The inverse on the image is $\mathcal{W}_t^*$; its complementary kernel contributes only zero eigenvalues. This proves the precise spectral correspondence used by both logarithmic functions in STF, including rank loss and coincident flags.

There is a corresponding formula for every original quotient $K \subset Y$. Retain Q_0 and E from STF8 and set $\widehat{E} = U_4 E$; then $\widehat{E} \widehat{E}^* = \widehat{P}_Y - \widehat{P}_K$. Substituting NFI4 into the complete attained minimum STF10 gives

$$Q_t = Q_0^{1/2} \left[I + \widehat{E}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \widehat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \widehat{E} \right] Q_0^{1/2}. \quad (\text{NFI10})$$

Indeed the middle inverse is carried into STF10's inverse by the unitary $\mathcal{R}_t$, and its two adjacent factors carry E into $\widehat{E}$. This is an equality of matrices on the original quotient value space, with its original Q_0 on both sides. All cross terms survive the attained minimum. It applies to the four actual pairs in STF14 and to the target quotient $(\mathcal{U}, J)$; the empty quotient conventions remain those of STF8. Thus it transfers every complement and both allocations, rather than just their sum of determinants.

Finally use the eight labelled occurrences (X_a, σ_a) of STF16, retaining repeated subspaces separately. The remaining signed count in STF22 is now the fully original-data expression

$$\mathcal{B}_e^{>R} = \sum_{t=q-1,q} \int_1^\infty \sum_{a=1}^8 \sigma_a \# \{ \lambda_b(\widehat{K}_{X_a,t}) > u \} \frac{du}{u}. \quad (\text{NFI11})$$

All eigenvalues count multiplicities. The matrices in NFI11 are obtained from the finite coefficients NTG4–12, the original analytic recursion NTG9, and the actual CPQ frames in NFI7. The equality follows from the proved unitary and partial-isometry maps, not from a comparison of dimensions. With the accepted prefix R_j , the finite discrepancy bounds STF20–21 and its prefix budget apply without change. NFI11 gives the full coefficient formula for the growing-degree quantity; it does not assert a large-degree estimate for that quantity.

8 Propagation through the original receiving maps

The full original quotient metrics at the common tail base. The new simultaneous formula in STF8–12 evaluates these same CPQ affine value fibres after all preceding native rows have been retained. The exact pairs are (W, W_i) , $(\mathcal{U}, T)$, (S, K_c) and (I, K_i) , with the original value coordinates and lift factors preserved as in STF14. NFI10 transports each attained matrix through the full original analytic frame A and relation matrix X_t , including every cross term. The Schur minimum is the classical operation of Boyd and Vandenberghe, Appendix A.5.5, p. 650 [H23]; STF9–12 proves its exact complex Hermitian specialization on these fibres. The eight signed terms are a labelled tuple, so opposite labels are retained even when their subspaces coincide. Substitution in the original CPQ identity gives STF18 with both allocations and all four complements. A fixed lift determinant cancels only in its own same-frame endpoint ratio; no absolute frame factor or proper-source comparison is set to one.

Continuation of the original prefix and its full relation tail. The row estimate NIB14 and the proved prefix NIB18–20 now receive the complete original relation calculation in Sections 4, 5 and 7. In the native-cutoff convention $n = L + t$, the literal analytic Gram is $D_n^{\text{nat}} = A^* \Omega_t A$ and its new row is $\rho_n / \sqrt{1 + \zeta_n} = -i^{-n} x_t^T A$. NTG16–22 proves these equalities in the original functional frame; SLC3–20 calculates every entry from the full original root and conductor data using Szegő’s confluent modification formula [H24, Theorem 2.5, pp. 29–30]. For the retained prefix $R = R_j$, the earlier scalar tail budget can use the minimum of all three proved bounds in NTG31, with its retained cross-pair correction, and the full weighted tail is exactly $\log \det \Omega_{q-1} + \log \det \Omega_q - 2 \log \det \Omega_R$. The proof is NTG14–15 followed by the unchanged endpoint weights, as displayed in NTG32. The earlier prefix estimate remains valid; this new formula supplies the complete surviving correlated tail, whose growing-degree value remains the next calculation.

Simultaneous evaluation at the two original tail endpoints. Section 6 now evaluates every occurrence in IPR3–7 using the same base $N_0 = q + R$ and the two actual matrices M_{q-1}, M_q of size $4m$. For a fixed original space X , its exact restriction return is $F_{q-1}(X) + F_q(X)$ by STF6,15. For any one of the four CPQ quotient pairs, the attained value matrix is STF10 and its return is the difference of those same restriction returns by STF12. This proves the formulas for both allocations and all complementary returns in STF17–18 without changing IPR’s weights or source orders. The original signed tail differs from the simultaneous signed spectral count by at most $2(d_{q-1} + d_q) \log 2 \leq 16m \log 2$; its original successive-row threshold version has the additional explicit error $8\{2(q - R) - 1\} \log 2$, as proved in STF20–22. NFI1–11 gives the exact phase, permutation and unitary maps from the original relation coefficients to these flag matrices. At the already proved R_j , both discrepancies and the removed prefix are $o(kq_k)$. This does not assign a value to the remaining signed count, and it leaves the independent proper source and complete action at their original receiving identities.

A The original finite Gamma translation on every small-rank plane

Fix the original simple quartet, its actual five-orbit period and logarithm branch. Keep every coefficient of the original nonreduced conductor:

$$\begin{aligned}
0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad \beta_* = \tfrac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \\
E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \mu_m = E_A^{(m)}(0), \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad \mu_v \neq 0, \quad (\text{SGT1}) \\
k = 4l + 1 \geq 9, \quad s \in \{1, k\}, \quad c_0 = k/2, \quad c_1 = k/2 - 4.
\end{aligned}$$

The coefficients, symbol, order and first nonzero moment are fixed as k varies. No particular coefficient is assumed nonzero except through the stated exact order. Every result below is for this original operator and these two original source orders. An arbitrary integer $N \geq 0$ is allowed in the finite statements.

A.1 Full Gamma measure, recurrence and coefficient isometry

Write $b = s/2 > 0$ and retain

$$\begin{aligned}
M_s &= (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\
\|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad (\text{SGT2}) \\
p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+b)p_{n-1,s}.
\end{aligned}$$

These are exactly the measure and recurrence in PDS6 and DNE7. For completeness their generating and norm identities are derived here. The recurrence, multiplied by $t^n/n!$ and summed, gives $(1+t^2)\partial_t F = (y-bt)F$, $F(y,0) = 1$. The unique formal solution, also holomorphic for $|t| < 1$ at fixed y , is

$$F(y,t) = \sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1+t^2)^{-b/2} \exp(y \arctan t). \quad (\text{SGT3})$$

Both branches take their value zero or one at $t = 0$ as appropriate.

Here is a direct check of the full mass and norms. With $a = s/4$, the Euler gamma integrals and the change of variables $r = u + v_0$, $x = \log(u/v_0)$ in their product give

$$|\Gamma(a + i\xi)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{i\xi x} (2 \cosh(x/2))^{-2a} dx.$$

The integrand as a function of x is analytic for $|\Im x| < \pi$. Shifting the integration line by any fixed amount of modulus less than π is justified by its exponential decay at both real ends. It bounds this Fourier transform by a constant times $e^{-h|\xi|}$ for each $0 < h < \pi$. Fourier inversion is therefore absolutely convergent; it may also be obtained by inserting a Gaussian factor and passing to the limit by dominated convergence. Substituting $y = 2\xi$ and the exact multiplier in SGT2 gives

$$\int_{\mathbb{R}} e^{iuy} dm_s(y) = M_s (\cosh u)^{-b}, \quad \int_{\mathbb{R}} e^{uy} dm_s(y) = M_s (\cos u)^{-b} \quad (|u| < \pi/2). \quad (\text{SGT4})$$

For the second equality, the preceding exponential bounds give analytic continuation to the indicated strip; equality on the real Fourier axis then proves it there. In particular the mass is exactly M_s . For sufficiently small real t, z , SGT3-4 and $\cos(\arctan t + \arctan z) = (1 - tz)/\sqrt{(1+t^2)(1+z^2)}$ imply

$$\int_{\mathbb{R}} F(y,t) F(y,z) dm_s(y) = M_s (1 - tz)^{-b}.$$

The exponential moment bound justifies differentiating under the integral in a neighborhood of $(0, 0)$. Comparing coefficients proves

$$\int_{\mathbb{R}} p_{n,s}(y) p_{m,s}(y) dm_s(y) = \mathbf{1}_{n=m} h_{n,s}, \quad h_{n,s} = M_s n! (b)_n, \quad (SGT5)$$

$$\phi_{n,s,c}(S) = \frac{p_{n,s}((S-c)/i)}{\sqrt{h_{n,s}}}, \quad \rho_n = \sqrt{\frac{n!}{(b)_n}} > 0.$$

The $p_{n,s}$ are real monic polynomials by their recurrence. Thus these ϕ_n are orthonormal for the complex Hilbert norm in SGT2 and span each indicated polynomial degree range. Their leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$, not $1/\sqrt{h_{n,s}}$.

A.2 The original quotient map and its exact Toeplitz matrix

Let $\mathcal{P}_m = \mathbb{C}[S]_{\leq m}$, with $\mathcal{P}_{-1} = 0$. The original translation is

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad \mathcal{T}_A S^m = \sum_{j=0}^m \binom{m}{j} \mu_j S'^{m-j}. \quad (SGT6)$$

The second equality is the binomial theorem followed by the exact definition of every moment. If $m \geq v$ and P has leading coefficient $p_m \neq 0$, then $\mathcal{T}_A P$ has degree $m - v$ and leading coefficient $p_m \binom{m}{v} \mu_v \neq 0$. If $m < v$ its image is zero. Consequently the following map is an isomorphism, with an onto representative map and the displayed full kernel:

$$\begin{aligned} \ker(\mathcal{T}_A : \mathcal{P}_{N+v} \longrightarrow \mathcal{P}_N) &= \mathcal{P}_{v-1}, \\ A_{N,s} : (\mathcal{P}_{N+v}/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\xrightarrow{\simeq} (\mathbb{C}[S']_{\leq N}, \|\cdot\|_{s,c_1}), \\ \| [P] \|_{s,c_0,\text{quot}}^2 &= \min_{L \in \mathcal{P}_{v-1}} \|P + L\|_{s,c_0}^2. \end{aligned} \quad (SGT7)$$

Surjectivity follows successively by degree from the nonzero diagonal coefficients in SGT6. The orthogonal representative of this quotient is precisely the span of $\phi_{v,s,c_0}, \dots, \phi_{N+v,s,c_0}$. It is not a quotient by an omitted arithmetic relation.

On the entire open unit disk put

$$w(t) = -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v! \neq 0. \quad (SGT8)$$

The value at zero is removable by the exact order v . For $|t| < 1$, $|w(t)| \leq \arctanh |t|$ by the arctangent series. This proves holomorphy of the specified g . Applying the finite sum SGT6 to SGT3, with its polynomial argument, gives exactly

$$\begin{aligned} \sum_{m \geq 0} (\mathcal{T}_A p_{m,s}((S - c_0)/i))(S') \frac{t^m}{m!} &= (1 + t^2)^{-b/2} e^{(S' - c_0)w(t)} E_A(w(t)) \\ &= t^v g(t) (1 + t^2)^{-b/2} e^{(S' - c_1)w(t)}. \end{aligned} \quad (SGT9)$$

The center factor is e^{-4w} because $c_0 - c_1 = 4$. No inverse of $\mathcal{T}_A$, or scalar α_A , has been inserted in this identity.

For a holomorphic germ $u(t) = \sum_{j \geq 0} u_j t^j$ define $T_N(u)_{r,n} = u_{n-r}$ for $0 \leq r \leq n \leq N$, and zero for $r > n$. Coefficients of SGT9 and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give the complete matrix of SGT7 in its two orthonormal frames:

$$\begin{aligned} A_{N,s} &= D_{\text{out}}^{-1} T_N(g) D_{\text{in}}, \\ D_{\text{out}} &= \text{diag}(\rho_0, \dots, \rho_N), \quad D_{\text{in}} = \text{diag}(\rho_v, \dots, \rho_{N+v}), \\ (A_{N,s})_{r,n} &= \begin{cases} g_{n-r} \rho_{n+v} / \rho_r, & r \leq n, \\ 0, & r > n. \end{cases} \end{aligned} \quad (SGT10)$$

Each source and target frame contains $1/\sqrt{M_s}$. Those identical factors cancel only in this coefficient matrix, since s is unchanged. All actual squared norms still use the full SGT2 measure. In particular, for $N = 0$ the matrix is the literal scalar $(-i)^v \mu_v \rho_v / v!$.

A.3 A proved original-data zero-free disk

The following constants use every original coefficient and shift:

$$\begin{aligned} A_{\text{abs}} &= \sum_{a,b} |a_{ab}|, & B &= \max\{1, \max_{a,b} |\beta_{8,a,b}|\}, & C_v &= \frac{A_{\text{abs}} B^v}{|\mu_v|} \geq 1, \\ R_0 &= \frac{1}{B} \min\{1, (v+1)/(2 \exp(1) C_v)\}, & a_0 &= \min\{1/2, \tanh(R_0/2)\}, & & \text{(SGT11)} \\ d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)} \geq 8/9, & M_0 &= \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}. \end{aligned}$$

All are finite, R_0, a_0, d_0, M_0 are positive, and $a_0 < 1$. Indeed $|\mu_v| \leq A_{\text{abs}} B^v$ proves the stated bound on C_v . Define the entire removable quotient $B_\infty(z) = v! E_A(z)/(\mu_v z^v)$, with value one at zero. Its full Taylor remainder is bounded by

$$|B_\infty(z) - 1| \leq \frac{C_v B |z|}{v+1} e^{B|z|} \leq \frac{1}{2} \quad (|z| \leq R_0). \quad \text{(SGT12)}$$

For the first inequality expand each exponential, use $|\mu_m| \leq A_{\text{abs}} B^m$ and $v!/(v+1+j)! \leq 1/((v+1)j!)$, and sum the exponential series. The two inequalities $BR_0 \leq 1$ and $C_v BR_0/(v+1) \leq 1/(2 \exp(1))$ prove the second. This is the complete RDS11 disk proof for the unchanged symbol.

On $|t| \leq a_0$ we have $|w(t)| \leq R_0/2$ and, also at zero,

$$\left| \frac{\arctan t}{t} - 1 \right| \leq \sum_{j \geq 1} \frac{a_0^{2j}}{2j+1} \leq \frac{a_0^2}{3(1-a_0^2)}.$$

Thus $|w(t)/t| \geq d_0$. Factoring SGT8 as $g(t) = (\mu_v/v!)(w(t)/t)^v e^{-4w(t)} B_\infty(w(t))$ shows that it has no zero on that entire closed disk and proves

$$|g(t)| \geq \frac{|\mu_v|}{2v!} d_0^v e^{-2R_0}, \quad |1/g(t)| \leq M_0 \quad (|t| \leq a_0). \quad \text{(SGT13)}$$

This supplies the zero-free disk and the inverse bound from the actual finite symbol rather than assuming them.

A.4 Every actual pole and its finite-rank Toeplitz contribution

Call $0 < a < 1$ admissible when g has no zero on $|t| = a$. Zeros of a nonzero holomorphic function are isolated; therefore there are only finitely many in each compact subdisk. There are admissible radii arbitrarily close to one. For a fixed such a , let $\mathcal{Z}_a$ be the exact zeros ζ of g with $|\zeta| < a$, and let m_ζ be their orders. Zero is not among them. Retain every multiplicity and define the actual Laurent coefficients

$$\begin{aligned} H_a &= \sum_{\zeta \in \mathcal{Z}_a} m_\zeta, & A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \left(\frac{(t - \zeta)^{m_\zeta}}{g(t)} \right) \Big|_{t=\zeta}, & 1 \leq h \leq m_\zeta, \\ f &= 1/g, & f_a(t) &= f(t) - \sum_{\zeta \in \mathcal{Z}_a} \sum_{h=1}^{m_\zeta} \frac{A_{\zeta,h}}{(t - \zeta)^h}. \end{aligned} \quad \text{(SGT14)}$$

The factors t^{-v} and e^{-4w} in g create no nonzero zero. Moreover $w'(t) = -i/(1+t^2) \neq 0$ and $e^{-2w(t)} = (1+it)/(1-it)$ proves injectivity of w on the disk. Thus these poles are precisely the symbol zeros already identified in PEC3–5, under this exact change of variable, with their same orders. The extra center factor is fully retained in every $A_{\zeta,h}$. The function f_a is holomorphic on a neighborhood of the closed a disk, since all its interior principal parts have been subtracted and no boundary zero exists.

Here are completely specified finite constants on that circle:

$$\begin{aligned}
W_a &= \operatorname{arctanh} a, \quad m_{E,a} = \min_{|t|=a} |E_A(w(t))| > 0, \quad G_a = A_{\text{abs}} a^{-v} e^{(B+4)W_a}, \\
M_a &= \frac{a^v e^{4W_a}}{m_{E,a}} + \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(a - |\zeta|)^h}, \quad P_a = \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(1 - |\zeta|)^h}, \\
S_N(a) &= \sum_{j=0}^N a^{-j} = a^{-N} \frac{1 - a^{N+1}}{1 - a}.
\end{aligned} \tag{SGT15}$$

Empty pole sums are zero. The minimum is the same actual finite-circle quantity used in PEC4, and is positive by continuity and admissibility. It is not a new uniform zero-free assertion. The triangle inequality on that circle gives $|g| \leq G_a$ and $|f_a| \leq M_a$. Every constant in SGT15 is defined by the original fixed function and its exact finitely many interior poles; none depends on k, N or s .

For one term $A/(t - \zeta)^h$, its Taylor coefficients at zero are

$$c_n = A(-1)^h \zeta^{-(n+h)} \binom{n+h-1}{h-1} \quad (n \geq 0). \tag{SGT16}$$

Extend the binomial as the polynomial $\binom{x+h-1}{h-1} = \prod_{\ell=1}^{h-1} (x + \ell)/(h-1)!$ to every integer x , also for $h = 1$ by the empty product. The full matrix $C_{i,j} = c_{j-i}$, $0 \leq i, j \leq N$, has rank at most h : after multiplying rows by ζ^{-i} and columns by ζ^j , its entries are a polynomial of total degree at most $h-1$ in $j-i$. Expanding that polynomial in powers of i gives at most h column vectors, independently of N . Equivalently its column space is contained in $\operatorname{span}\{(\zeta^i i^d)_i : 0 \leq d < h\}$. For all terms belonging to one pole, these nested spaces give combined rank at most m_ζ , rather than the sum of the integers h .

For a lower-diagonal gap $\ell = i - j \geq 1$, direct substitution in the polynomial binomial gives

$$c_{-\ell} = \begin{cases} 0, & 1 \leq \ell < h, \\ -A\zeta^{\ell-h} \binom{\ell-1}{h-1}, & \ell \geq h. \end{cases} \tag{SGT17}$$

For $\ell \geq h$ each factor in the binomial is negative, and their $(-1)^{h-1}$ sign combines with $(-1)^h$ to give the displayed minus sign. For $1 \leq \ell < h$ one factor is zero. Consequently the upper triangular Taylor matrix of this pole is $C + R$, where R is strictly lower triangular and its gap- ℓ coefficient is $A\zeta^{\ell-h} \binom{\ell-1}{h-1}$ for $\ell \geq h$. If J is the upper shift with $J_{i,i+1} = 1$, then $(J^*)^\ell$ has norm at most one. The absolutely convergent binomial series therefore proves

$$\|R\| \leq |A| \sum_{\ell=h}^{\infty} |\zeta|^{\ell-h} \binom{\ell-1}{h-1} = \frac{|A|}{(1 - |\zeta|)^h}. \tag{SGT18}$$

The series identity follows by differentiating the geometric series $h-1$ times, or multiplying h copies of it and collecting coefficients. All powers and signs are those of the original pole; no principal part has been deleted from the actual Toeplitz inverse.

Since $T_N(u) = \sum_{j=0}^N u_j J^j$, multiplication of these finite matrices is truncated power-series multiplication. In particular $T_N(g)^{-1} = T_N(f)$, because $g(0) \neq 0$ and $gf = 1$ as germs. Cauchy's coefficient estimate, SGT13 and SGT15–18 prove

$$\begin{aligned}
\|T_N(g)\| &\leq F_{a,N} := G_a S_N(a), \\
\|T_N(f)\| &\leq G_{0,N} := M_0 S_N(a_0), \\
T_N(f) &= C_{a,N} + R_{a,N}, \quad \operatorname{rank} C_{a,N} \leq H_a, \quad \|R_{a,N}\| \leq V_{a,N} := M_a S_N(a) + P_a.
\end{aligned} \tag{SGT19}$$

Indeed sum the coefficient estimates times $\|J^j\| \leq 1$ for the analytic terms, and sum the exact pole ranks and lower-triangular remainders just proved. The total inverse matrix is unchanged.

A.5 Finite exterior bounds, including the expensive directions

Write $d = N + 1$ and arrange singular values in decreasing order. If a matrix C has rank at most H , then $\ker C$ has codimension at most H . For $x \in \ker C$, $\|(C + R)x\| \leq \|R\|\|x\|$. The variational description of singular values therefore gives $\sigma_j(C + R) \leq \|R\|$ for $j > H$. To verify this description directly, diagonalize $(C + R)^*(C + R)$: the complement of its first $j - 1$ eigenvectors attains the upper bound, and every subspace of codimension $j - 1$ intersects the first j eigenvectors for the reverse inequality. Applying this argument to SGT19 and retaining the separate global bound gives

$$h = \min\{r, H_a\}, \quad 0 \leq r \leq d, \quad V_{a,N}^{\text{cap}} = \min\{G_{0,N}, V_{a,N}\},$$

$$\left\| \bigwedge^r T_N(g) \right\| \leq F_{a,N}^r, \quad \left\| \bigwedge^r T_N(f) \right\| \leq G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}. \quad (\text{SGT20})$$

The norm of the r th exterior power is the product of the r largest singular values: this follows by applying a singular-value decomposition to the orthonormal wedge bases. Only at most H_a directions use the zero-free-radius bound. If $H_a > d$, then $h = r$ and there is no tail factor. Every constant is positive; the empty exterior power has norm one.

Retain the exact diagonal constants from the original norms:

$$\alpha_+ = \left(\max_{0 \leq n \leq N} \rho_{n+v} \right) \left(\max_{0 \leq n \leq N} \rho_n^{-1} \right), \quad \alpha_- = \left(\max_{0 \leq n \leq N} \rho_{n+v}^{-1} \right) \left(\max_{0 \leq n \leq N} \rho_n \right),$$

$$L_{s,N,v} = \frac{1}{2} \max\{1, s/2 - 1\} \mathfrak{H}_{N+v}, \quad \mathfrak{H}_m = \sum_{j=1}^m 1/j, \quad \mathfrak{H}_0 = 0. \quad (\text{SGT21})$$

For $s = k \geq 9$, the product defining ρ_n and $\log(1+x) \leq x$ imply $|\log \rho_n| \leq (b-1)\mathfrak{H}_n/2$. For $s = 1$, its logarithm is positive and $\log((j+1)/(j+1/2)) \leq 1/(2j+1) \leq 1/(j+1)$ gives $|\log \rho_n| \leq \mathfrak{H}_n/2$. Thus $|\log \rho_n| \leq L_{s,N,v}$ for all $0 \leq n \leq N+v$, including $n = 0$. In particular $|\log \alpha_{\pm}| \leq 2L_{s,N,v}$. Applying the two diagonal maps in SGT10 on exterior powers gives

$$\left\| \bigwedge^r A_{N,s} \right\| \leq \alpha_+^r F_{a,N}^r,$$

$$\left\| \bigwedge^r A_{N,s}^{-1} \right\| \leq \alpha_-^r G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}. \quad (\text{SGT22})$$

This retains the rectangular degree shift $n \mapsto n + v$ exactly.

Let $[P_1], \dots, [P_r]$ be any independent original classes in SGT7 and use that same ordered frame in both Grams:

$$H_{\text{in}} = (\langle [P_i], [P_j] \rangle_{s, c_0, \text{quot}})_{i,j}, \quad H_{\text{out}} = (\langle \mathcal{T}_A P_i, \mathcal{T}_A P_j \rangle_{s, c_1})_{i,j}, \quad \Delta_{N,s}(\mathbf{P}) = \log \frac{\det H_{\text{out}}}{\det H_{\text{in}}}. \quad (\text{SGT23})$$

The inner products are conjugate-linear in their first variable. Both Grams are positive, by independence and SGT7. Exterior volumes and their inverse give the finite, potentially asymmetric bounds

$$-2r \log \alpha_- - 2h \log G_{0,N} - 2(r-h) \log V_{a,N}^{\text{cap}} \leq \Delta_{N,s}(\mathbf{P})$$

$$\leq 2r \log \alpha_+ + 2r \log F_{a,N}. \quad (\text{SGT24})$$

The classical Gram/exterior identity used here is recorded in Lyons [H3, §4, (4.1), p. 11]; the source and image frames below specify its exact application to SGT24.

For proof, express the classes in their SGT7 orthonormal quotient frame as columns X . Their squared wedge norm is $\det(X^*X)$, and the squared norm of their image wedge is $\det(X^*A_{N,s}^*A_{N,s}X)$. Apply SGT22 to that wedge and then to its image with $A_{N,s}^{-1}$. A change of the original ordered frame by a matrix U multiplies both determinants by $|\det U|^2$; the ratio therefore retains the original frame and is frame invariant. No specially conditioned plane has been selected.

For a nonnegative enclosure write $\log^+ x = \max\{0, \log x\}$ and set

$$\begin{aligned}\mathcal{U}_{a,N,s,r} &= 4rL_{s,N,v} + 2r \log^+ F_{a,N}, \\ \mathcal{L}_{a,N,s,r} &= 4rL_{s,N,v} + 2h \log^+ G_{0,N} + 2(r-h) \log^+ V_{a,N}^{\text{cap}}.\end{aligned}\tag{SGT25}$$

Then $-\mathcal{L} \leq \Delta_{N,s}(\mathbf{P}) \leq \mathcal{U}$. These are finite explicit constants even for $r = 0$, with value zero for that empty Gram ratio. All maxima, minima, poles and residues in the constants refer to the fixed original symbol SGT1.

A.6 Uniform small-rank volume distortion and the actual PDS frame

Fix finite constants $C_N, C_r > 0$. Uniformly over $0 \leq N \leq C_N k^2$, $0 \leq r \leq \min\{C_r k, N+1\}$, $s = 1$ or k , and every original rank- r plane in SGT7,

$$\sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \longrightarrow 0 \quad (k = 4l+1 \longrightarrow \infty).\tag{SGT26}$$

Here the supremum is over precisely those finite-degree and rank domains, including all coefficient choices for the planes. We give both limiting estimates. For each fixed admissible a , $S_N(a) \leq (N+1)a^{-N}$ and $a^{-N} \geq 1$, so

$$\begin{aligned}\log^+ F_{a,N} &\leq N \log(1/a) + \log(N+1) + \log^+ G_a, \\ \log^+ V_{a,N}^{\text{cap}} &\leq \log^+ V_{a,N} \leq N \log(1/a) + \log(N+1) + \log^+(M_a + P_a), \\ \log^+ G_{0,N} &\leq N \log(1/a_0) + \log(N+1) + \log^+ M_0.\end{aligned}\tag{SGT27}$$

Also $\mathfrak{H}_m \leq 1 + \log(m+1)$, by integral comparison. The common diagonal error $4rL$ is therefore $O_{C_N, C_r, v}(k^2 \log(k+1))$. The global inverse term has $h \leq H_a$, which is fixed before the k limit, so it is $O_{\text{actual}, a, C_N}(k^2 + \log(k+1))$. It follows from SGT25 and SGT27 that

$$\limsup_{k \rightarrow \infty} \sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \leq 2C_r C_N \log(1/a).\tag{SGT28}$$

The constants in this step may depend on this fixed a . Only after this limit do we take admissible radii arbitrarily close to one, as proved before SGT14. The right side then tends to zero. This proves SGT26 without assuming a zero-free unit disk or a uniform pole count as a tends to one.

For the original PDS quotient set

$$\begin{aligned}q_k &= (k+1)^2, \quad Q = (k+5)^2, \quad e \in \{0, 1\}, \\ N_e &= Q - v - 1 + e, \quad \ell_e = 8k + 23 + e, \quad d_e = \ell_e + 1 = 8k + 24 + e, \\ P_a(S) &= \chi_k(S)S^a, \quad r_a(S') = \mathcal{T}_A P_a(S'), \quad 0 \leq a \leq \ell_e, \\ \chi_k(S) &= \prod_{u,w=0}^k (S - \beta_{k,u,w}).\end{aligned}\tag{SGT29}$$

Then $q_k + \ell_e = N_e + v = Q - 1 + e$. A nonzero linear combination of the P_a has degree at least $q_k > v - 1$ because it is χ_k times a nonzero polynomial. Thus their classes modulo $\mathcal{P}_{v-1}$ are independent, of exactly the original rank d_e , without a minor or genericity assumption.

Let X_e be their full column matrix in the original c_0 orthonormal frame of degrees $0, \dots, N_e + v$, and write $P_{<v}$ and $P_{\geq v}$ for the row selections in that frame. The exact source and image Grams are

$$\begin{aligned}Y_e &= P_{\geq v} X_e, \quad H_{\text{src},e} = Y_e^* Y_e = X_e^* X_e - (P_{<v} X_e)^* (P_{<v} X_e) > 0, \\ C_e &= A_{N_e, s} Y_e, \quad H_e = C_e^* C_e, \quad -\mathcal{L}_{a, N_e, s, d_e} \leq \log \frac{\det H_e}{\det H_{\text{src},e}} \leq \mathcal{U}_{a, N_e, s, d_e}.\end{aligned}\tag{SGT30}$$

The first equality is the attained full quotient minimum over the entire original $\mathcal{P}_{v-1}$. The next equality is the coefficient map SGT10, since those removed low columns are killed by $\mathcal{T}_A$. The resulting C_e, H_e are exactly PDS7, including the target center c_1 and both original orders. For $k \geq 9$, $Q \leq (14k/9)^2 < 3k^2$, so $N_e \leq 3k^2$ and $d_e \leq 11k$. Hence SGT26 applies directly:

$$\log \frac{\det H_e}{\det H_{\text{src},e}} = o(k^3) = o(kq_k) \quad (e = 0, 1, \quad s = 1, k), \quad (\text{SGT31})$$

uniformly over those four original source/endpoint choices. The full original mass has not been removed from either Gram.

The precise dual statement also retains all PDS conjugations: for the complex-linear functional $x^\top u$ on a Gram H , its squared dual norm is $x^* \overline{H^{-1}} x$. Thus the same coefficient coordinates carry

$$G_e = \overline{H_e^{-1}}, \quad G_{\text{src},e} = \overline{H_{\text{src},e}^{-1}}, \quad \log \frac{\det G_e}{\det G_{\text{src},e}} = -\log \frac{\det H_e}{\det H_{\text{src},e}}. \quad (\text{SGT32})$$

This is an equality of the original full quotient-source dual metrics, not a statement about a later connecting-target Gram.

For the endpoint $e = 1$ and the original four PDS8 rows, retain the same full rank-four matrix W and choose any basis Z of $\ker W^\top$. The polynomial quotient by those four columns, in the dual coordinates $a = Z^\top f$, has metric $(Z^\top H^{-1} \bar{Z})^{-1}$ by minimizing $f^* H f$. Consequently it is the conjugate inverse of the restriction $Z^* \overline{H^{-1}} Z$, exactly as in PDS9. For the paired forms $H_e, H_{\text{src},e}$, fixed complementary frames or the same attained Schur decomposition therefore give

$$\begin{aligned} \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} &= -\Delta_{N_e,s}(\mathbf{P}) + \log \frac{\det(W^* H_e W)}{\det(W^* H_{\text{src},e} W)}, \\ -\mathcal{U}_{a,N_e,s,d_e} - \mathcal{L}_{a,N_e,s,4} &\leq \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} \leq \mathcal{L}_{a,N_e,s,d_e} + \mathcal{U}_{a,N_e,s,4}. \end{aligned} \quad (\text{SGT33})$$

Indeed $\ker Z^\top = \text{im } W$. If J is any fixed right inverse of $Z^\top$, then $[W, J]$ is invertible, and block Schur factorization gives $\det((Z^\top H^{-1} \bar{Z})^{-1}) = |\det[W, J]|^2 \det H / \det(W^* H W)$. Taking the ratio cancels this exact frame determinant and proves the first equality. The W columns correspond to the four independent original polynomials $\chi_k R_i$ modulo $\mathcal{P}_{v-1}$; SGT24–25 applied to that actual rank-four plane prove the interval. At $e = 1$ this is the original primitive source, and its determinant distortion is again $o(kq_k)$. At $e = 0$ the primitive condition is empty and SGT32 already gives its full statement.

The earlier PEC1–18 theorem uses actual numerator-jet cancellation at symbol zeros to control a positive dual spectrum. Here SGT7–10 first proves the polynomial source/target morphism, and SGT16–22 retains all principal parts in an exact finite-rank Toeplitz summand, giving both directions on every small-rank plane. Neither statement replaces the original lower quotient, the four primitive rows, or the connecting-target metric. The original-data zero-free disk is the RDS11 disk with its full proof repeated in SGT11–13.

B The connecting image inside the full remaining target quotient

This calculation uses the complete original CIT, RQB and NSR objects. The fixed hypothetical simple quartet, five-orbit period, logarithm branch, conductor E_A , order $v = \text{ord}_0 E_A$, nonzero moment μ_v and scalar $\alpha_A = v!/\mu_v$ remain those of the original construction. Put $k = 4l + 1 \geq 9$, $j = k + 4$, $q_a = (a + 1)^2$, $q'_a = (a - 7)^2$, $m_a = 16a - 48 - v$, $t_a = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$. Use the original rank-valid domain, including $|u| \geq R_*$. The original source pairs are $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (s_0, s_1) = (k, j)$. Every form below has its actual pair p ; we suppress this index only in identities valid separately for both.

At each original target cutoff $q_j - 1 \leq N \leq 2q_j$, write $D = N - v$. The native coefficient form is

$$\mathcal{D}_N = \text{diag}(D_{j,D}^{(s_1)}, 4), \quad \mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j). \quad (\text{CPQ1})$$

Here $D_{j,D}$ is the full analytic-numerator form after minimizing over all original lower evaluations, as in DNE and CIT19. Its rows use $M_s = (2\pi)^{s/2}$ and $h_{n,s} = M_s n! (s/2)_n$, the original centres, division moments and phases. Nothing in this calculation changes those rows or replaces their lower Gram. Fixed low-cycle coefficient flags are used at all these later cutoffs.

B.1 All coefficient maps and their kernels

Retain

$$\begin{aligned} \mathcal{U} &= \mathbb{C}^{4m_j}, \quad \mathcal{V} = \mathbb{C}^{m_k}, \quad B = B_{\text{cof}} : \mathcal{V} \longrightarrow \mathcal{U}, \\ \mathbb{B} &= \text{diag}(B_j, 4) : \mathcal{U} \longrightarrow \mathbb{C}^{4t_j}, \quad \mathbb{J} = \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \\ \mathcal{B} &= \text{im } B, \quad \mathcal{S}_e = \text{im}[B, F_e], \quad W = \mathbb{J} + \mathcal{B}, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \end{aligned} \quad (\text{CPQ2})$$

B is the original injective coefficient multiplier and B_j is onto with its full original invariant kernel. The columns F_e , for $e = 0, 1$, are the actual CIT7 connecting columns, determined at $N = q_j - 1 + e$ by the complete Taylor equation with independent target lower corrections. They give an isomorphism $\mathbb{C}^{r_e} \rightarrow \mathcal{S}_e/\mathcal{B}$, $x \mapsto [F_e x]$, with $r_e = 8k + 24 - 3e$. Their independence and equality in the two source orders are proved in CIT9 and NSR5.

Use the original full target frame

$$C_T = [U_R, U_b, U_o], \quad U_R = \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad \text{im}[U_R, U_b] = \text{im } \Phi. \quad (\text{CPQ3})$$

It is the same invertible RQB27 frame. Its block dimensions are $r_I = j_{0,k} - \epsilon$, $r_b = t_k - \ell + \epsilon$ and $r_o = 4t_j - m_k + \ell$, where $\ell = \dim \ker \Phi$ and $\epsilon = \dim \ker(\Phi J_0)$. Define the actual quotient-coordinate map and its connecting matrix by

$$\pi_o = [0 \ 0 \ I_{r_o}] C_T^{-1} \mathbb{B}, \quad E_e = \pi_o F_e : \mathbb{C}^{r_e} \longrightarrow \mathbb{C}^{r_o}. \quad (\text{CPQ4})$$

The map π_o is onto and $\ker \pi_o = W$. Indeed its value is zero exactly when $\mathbb{B}z \in \text{im } \Phi$, equivalently $\mathbb{B}(z - Bx) = 0$ for some $x \in \mathcal{V}$. Its onto property uses the onto map $\mathbb{B}$ and the full last block of C_T . Consequently the maps

$$\mathcal{U}/W \longrightarrow \mathbb{C}^{r_o}, \quad [z] \mapsto \pi_o z, \quad \mathcal{S}_e/K_e \longrightarrow \text{im } E_e, \quad [z] \mapsto \pi_o z \quad (\text{CPQ5})$$

are isomorphisms; the latter identifies T_e/W with the same image. The exact sequence in the original connecting coordinates is

$$0 \longrightarrow \ker E_e \longrightarrow \mathbb{C}^{r_e} \xrightarrow{E_e} \text{im } E_e \longrightarrow 0, \quad \ker E_e \simeq K_e/\mathcal{B}, \quad c_e = \text{rank } E_e, \quad h_e = r_e - c_e. \quad (\text{CPQ6})$$

For its asserted kernel isomorphism, $F_e x \in W$ means that $F_e x - Bb \in \mathbb{J}$ for some b . Its class lies in $K_e/\mathcal{B}$, and every such class has a unique original connecting coordinate x . This proves both injectivity and onto without removing any conductor or invariant kernel. In particular $c_e = \text{rank}[\Phi, \mathbb{B}F_e] - \text{rank } \Phi$, and $0 \leq c_e \leq \min(r_e, r_o)$.

Choose fixed independent columns Z_h spanning $\ker E_e$ and fixed complement columns Z_c such that $C_P = [Z_h, Z_c]$ is invertible. Set $A_e = E_e Z_c$ and choose R_e^o so that $C_o = [A_e, R_e^o]$ is an invertible r_o -square matrix. These choices concern only the actual fixed coefficient maps. They are made once and used at every N and both orders. Their determinants are retained below. Empty blocks have their ordinary zero-dimensional meanings and determinant one.

B.2 The two original source and target minima

Let $H_e(N) = H_{\text{conn},e}(N)$ be CIT18's full connecting Gram on $\mathcal{S}_e/\mathcal{B}$ in F_e coordinates. It is the minimum of $(F_e x + Bb)^* \mathcal{D}_N(F_e x + Bb)$ over all b . The source kernel restriction and source quotient are exactly

$$\begin{aligned} G_{h,e} &= Z_h^* H_e Z_h, \\ G_{s,e} &= Z_c^* H_e Z_c - Z_c^* H_e Z_h (Z_h^* H_e Z_h)^{-1} Z_h^* H_e Z_c, \\ x^* G_{s,e} x &= \min_{z,b} (F_e Z_c x + F_e Z_h z + Bb)^* \mathcal{D}_N(F_e Z_c x + F_e Z_h z + Bb), \\ |\det C_P|^2 \det H_e &= \det G_{h,e} \det G_{s,e}. \end{aligned} \tag{CPQ7}$$

The kernel fibre in the displayed minimum is exactly K_e by CPQ6. Thus $G_{s,e}$ is the original metric on $\mathcal{S}_e/K_e$ in the values $[F_e Z_c x]$. Block elimination in $C_P^* H_e C_P$ proves its formula and the full determinant identity. Positive definiteness follows from independence in these quotient values.

Write $Q_o(N)$ for the original RQB remaining quotient on $\mathbb{C}^{4t_j}/\text{im } \Phi$, in the lifts U_o . The map CPQ5 is an isometry from $\mathcal{U}/W$ with its attained $\mathcal{D}_N$ metric to this form. To prove it, first minimize over $\mathbb{J}$, giving exactly $\text{diag}(H_{B,j,D}, 4)$ on $\mathbb{B}z$, where $H_{B,j,D} = (B_j D_{j,D}^{-1} B_j^*)^{-1}$. Then minimize over $\text{im } \Phi$. The combined fibre is W ; this is precisely the definition of Q_o in the frame C_T .

It follows that the original connecting target metric and its remaining target quotient are

$$\begin{aligned} G_{t,e} &= A_e^* Q_o A_e, \\ Q_{\text{rem},e} &= (R_e^o)^* Q_o R_e^o - (R_e^o)^* Q_o A_e (A_e^* Q_o A_e)^{-1} A_e^* Q_o R_e^o, \\ x^* G_{t,e} x &= \min_{w \in W} (F_e Z_c x + w)^* \mathcal{D}_N(F_e Z_c x + w), \\ |\det C_o|^2 \det Q_o &= \det G_{t,e} \det Q_{\text{rem},e}, \quad \omega_{\text{conn},e} = \det G_{t,e} / \det G_{s,e}. \end{aligned} \tag{CPQ8}$$

For the target restriction, every representative of a value in T_e/W lies in T_e because $W \subset T_e$. Thus the ambient minimum over W is its intrinsic quotient minimum. Substitution of $\pi_o F_e Z_c = A_e$ proves the first and third lines. The second line is the complete minimum over the connecting image inside Q_o . Block elimination in $C_o^* Q_o C_o$ proves the determinant identity. The last equality uses the identical value coordinates on both sides of the original CIT21 contraction. Its inverse and frame factors have therefore been established, rather than deduced from equality of dimensions. The new quotient has rank $r_o - c_e$ and represents exactly $\mathcal{U}/T_e$.

B.3 Positive returns and the formerly signed remainder

For the forms in CPQ7–8 set

$$V_h = \mathcal{R} \log \det G_{h,e}, \quad V_s = \mathcal{R} \log \det G_{s,e}, \quad V_t = \mathcal{R} \log \det G_{t,e}, \quad V_{\text{rem}} = \mathcal{R} \log \det Q_{\text{rem},e}. \tag{CPQ9}$$

Keep the original $V_I = \mathcal{R} \log \det H_{\text{inv}}$, $V_b = \mathcal{R} \log \det Q_b^E$, $V_o = \mathcal{R} \log \det Q_o$ and $V_A = \mathcal{R} \log \det A_{\text{inv}}$, with the full invariant source fibre $FZ_R x + FE$, $F = BJ_0$ and $E = \ker(\Phi J_0)$. In particular A_{inv} uses the target order s_1 and is not replaced by the independent source form Q_{inv} of NSR25.

All eight returns $V_h, V_s, V_t, V_{\text{rem}}, V_I, V_b, V_o, V_A$ are nonnegative. Indeed the original row extension increases $\mathcal{D}_N$. Restriction preserves this order. Minimizing two ordered forms over the same full affine fibre preserves it too. Every displayed form is such a restriction or quotient on a fixed

flag. Its successive log-determinant increments are nonnegative, and $\mathcal{R}$ is their sum with weights one at both ends and two at each interior row. This proves the assertion for these particular forms, including their complete gauges.

Write $V_{\text{conn}} = \mathcal{R} \log \det H_e$. CPQ7–8 and the original RQB target identity give

$$\begin{aligned} V_{\text{conn}} &= V_h + V_s, & V_o &= V_t + V_{\text{rem}}, \\ 4\widehat{F}_j^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ \mathcal{R} \log \omega_{\text{conn},e} &= V_t - V_s, & \mathcal{L}_e &:= \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}}) = V_I + V_t - V_s - V_A. \end{aligned} \quad (\text{CPQ10})$$

All fixed frame determinants in CPQ7–8 cancel only here, since $\mathcal{R}1 = 0$. No varying minimum representative is treated as a fixed frame. Substitution into CIT30's original identity $\mathcal{L}_e = \mathcal{P}_e^{\text{rest}} - V_b$ now evaluates its entire signed Gram combination as

$$\boxed{\mathcal{P}_e^{\text{rest}} = 4\widehat{F}_j^{(s_1)} - V_{\text{rem}} - V_s - V_A, \quad \mathcal{L}_e = 4\widehat{F}_j^{(s_1)} - V_b - V_{\text{rem}} - V_s - V_A.} \quad (\text{CPQ11})$$

This equality retains both new proper quotients. In particular neither Q_o nor the source quotient has been discarded. It expresses the original signed combination by the complete target return and three specified nonnegative returns.

For completeness the fixed factors before taking the return can also be made explicit. Use the original CIT constants $c_{\text{conn},e}$ and C_W , and the fixed change $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$. The original expression $P(N) = \log g_{T_e} + \log g_{K_e} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}}$ satisfies

$$\begin{aligned} P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem},e} - \log \det G_{s,e} - \log \det A_{\text{inv}} + \log \mathfrak{c}_e, \\ \mathfrak{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}. \end{aligned} \quad (\text{CPQ12})$$

To verify every factor, CIT26 gives $P = \log \omega_{\text{conn}} + \log \omega_{\text{inv}} + \log \det Q_b^{\text{CIT}} - \log c_{\text{conn},e} - 2 \log |\det C_W|$. Substitute CPQ8 and $|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}} \det Q_b^E \det Q_o$. Then substitute $\det G_{t,e} = |\det C_o|^2 \det Q_o / \det Q_{\text{rem},e}$. These substitutions give CPQ12 with the displayed positive constant. Its return is zero, which proves CPQ11 in the original absolute frames as well.

The following directed intervals follow from these exact nonnegative quantities:

$$\begin{aligned} V_I - V_s - V_A &\leq \mathcal{L}_e \leq 4\widehat{F}_j^{(s_1)} - V_b - V_s - V_A, \\ V_I + V_b - V_s - V_A &\leq \mathcal{P}_e^{\text{rest}} \leq 4\widehat{F}_j^{(s_1)} - V_s - V_A, \\ V_I - V_{\text{conn}} - V_A &\leq \mathcal{L}_e \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A. \end{aligned} \quad (\text{CPQ13})$$

The width of each of the first two intervals is exactly V_o ; the location of its actual value is fixed by the retained V_{rem} . The first follows from $0 \leq V_t \leq V_o = 4\widehat{F}_j - V_I - V_b$ in CPQ10. The second adds V_b . For the third use $V_s \leq V_{\text{conn}}$ on the lower side and the original RQB35 cap, on its proved domain, on the upper. These are finite bounds on the original scalar. They do not assign an uncomputed asymptotic value to a positive return.

B.4 Actual row formulas for each new factor

The new factors can be evaluated by the original arithmetic rows. Here is a full common formula specifying their minima. Let Y be any of the fixed coefficient subspaces and $K \subset Y$ the kernel in one of the following pairs:

$$(Y, K) = (K_e, \mathcal{B}), (\mathcal{S}_e, K_e), (T_e, W), (\mathcal{U}, T_e), (\text{im } F, FE). \quad (\text{CPQ14})$$

They represent respectively $G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}$. Choose independent coefficient columns X_K for K and lifts X_Q for the specified quotient values. The lifts are respectively $F_e Z_h, F_e Z_c, F_e Z_c$,

any fixed lifts of $U_o R_e^o$ under $\mathbb{B}$, and FZ_R . In the fourth pair the coordinate map to $\mathcal{U}/T_e$ is the last quotient in CPQ8.

Let $\mathcal{V}_N$ be the full original four-copy lower-evaluation matrix through D , and $\mathcal{T}_N$ the four-copy original analytic-numerator matrix through D . Form the raw matrices

$$Z_{K,N} = [\mathcal{V}_N, \mathcal{T}_N X_K], \quad Z_{Y,N} = [Z_{K,N}, \mathcal{T}_N X_Q], \quad G_K^{\text{raw}} = Z_{K,N}^* Z_{K,N}. \quad (\text{CPQ15})$$

Their columns are independent: the full lower frame is independent, and the DNE coefficient Gram is positive after its minimum; the remaining fixed coefficient columns are independent. Thus all the following inverses exist, including at the first cutoff, with the empty-block convention. Sequential elimination gives exactly the quotient form $Q = \text{Gram}_{\mathcal{D}_N}(Y/K)$ in these lifts and $\det Q = \det(Z_{Y,N}^* Z_{Y,N}) / \det G_K^{\text{raw}}$.

At the next original degree $n = D + 1$, append the four rows

$$a = [\text{diag}(v_n, 4), \text{diag}(t_n, 4)X_K], \quad b = \text{diag}(t_n, 4)X_Q, \quad C = (G_K^{\text{raw}})^{-1} Z_{K,N}^* \mathcal{T}_N X_Q, \quad R = b - aC. \quad (\text{CPQ16})$$

These are literally CIT19's v_n, t_n , with the complete arithmetic recursion

$$\begin{aligned} f_{a,m} &= \frac{\nu_{a,m+v} - \sum_{d=v+1}^{m+v} \binom{m+v}{d} \mu_d f_{a,m+v-d}}{\binom{m+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2)b_{n-1,m}, \\ (t_n)_a &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} f_{a,m}, \quad (v_n)_{u,w} = h_{n,s_1}^{-1/2} \sum_m b_{n,m} \beta_{j-8,u,w}^m. \end{aligned} \quad (\text{CPQ17})$$

The symbol $b_{n,m}$ in CPQ17 is the original polynomial coefficient, distinct from the new-row block b in CPQ16. Arrays outside their original polynomial range are zero. The divisor μ_v and the full old lower and coefficient Gram occur explicitly. The row formula remains valid for every allowed zero row and for both prescribed source orders.

Put $L = a(G_K^{\text{raw}})^{-1}a^*$. The exact next quotient is

$$Q^+ = Q + R^*(I_4 + L)^{-1}R, \quad \gamma_{Y/K,n} = \log \det(I_4 + L + RQ^{-1}R^*) - \log \det(I_4 + L) \geq 0. \quad (\text{CPQ18})$$

To prove the first equality, subtract the old attained minimum and change the old kernel coefficient by η . The remaining minimum is $\eta^* G_K^{\text{raw}} \eta + \|Rx - a\eta\|^2$. Solving its positive normal equations gives the residual $x^* R^*(I_4 + L)^{-1}Rx$. The determinant lemma gives the second equality. For its sign, conjugate by $(I_4 + L)^{-1/2}$; the added form is positive. Thus each of the five returns in CPQ14 is the sum of these particular γ 's, with endpoint weights one and all interior weights two. This supplies the original row calculation for all terms of CPQ11, including the previously unspecified remainder, without a new row model.

B.5 Back substitution into both source orders and the allocation

The ranks of $(G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}})$ are respectively $(h_e, c_e, c_e, r_o - c_e, r_I)$. NSO's full native same-fibre comparison therefore gives, for each of their returns V_X , the finite interval

$$-\text{rank}(X)\Omega_j^{\text{hi}} \leq V_X^{p=1} - V_X^{p=0} \leq \text{rank}(X)\Omega_j^{\text{lo}}. \quad (\text{CPQ19})$$

The cutoffs inside Ω_j are exactly those in CPQ1 and NSR2. The same coefficient frames occur in both orders by CPQ4 and NSR5. Their determinants therefore cancel in these ratios. Adding the two correlated decompositions gives precisely $\Delta V_o = \Delta V_t + \Delta V_{\text{rem}}$ and $\Delta V_{\text{conn}} = \Delta V_h + \Delta V_s$, preserving the full earlier NSR intervals. All ranks are $O(k)$ on the fixed actual programme, so the new individual order differences are $O_{\text{actual}}(k^2 \log q_j) = o(kq_k)$.

Finally the original degree- j allocation obeys $S_{K,j}^{(s_1)} = \widehat{F}_j^{(s_1)}$ and $S_{R,j}^{(s_1)} = \mathcal{T}_{j,s_1} - \widehat{F}_j^{(s_1)}$ by the exact HJR/NSR31 inverse-metric calculation. Hence CPQ10 gives the complete positive refinement of that kernel allocation:

$$\begin{aligned} 4S_{K,j}^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, & 4S_{R,j}^{(s_1)} &= 4\mathcal{T}_{j,s_1} - V_I - V_b - V_t - V_{\text{rem}}, \\ 4S_{K,j}^{(s_1)} &= \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \end{aligned} \tag{CPQ20}$$

The target degree is $j = k + 4$ in this identity. The independent degree- k source quotient Q_{inv} and the proper-source correction in NFR retain their own centre, displaced cutoffs and source order. In particular replacing V_A by $\mathcal{R} \log \det Q_{\text{inv}}$ is not part of CPQ20. The exact relation is $V_A = \mathcal{R} \log \det Q_{\text{inv}} + \mathcal{R} \log(\det A_{\text{inv}} / \det Q_{\text{inv}})$; both forms use the same lifts Z_R , so this ratio is the determinant of their actual relative positive operator. ITL/NFR retains its full proper-source comparison. CPQ20 thus carries the new quotient refinement into the original allocation while preserving that remaining source comparison and every positive term which must enter a quantitative estimate.

Finite equilibrium comparison for all five attained quotients. Retain CPQ2–8’s fixed maps and frames in both actual source pairs, and evaluate EFP11–15 at the original target degree $j = k + 4$ and $D_i = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$. For the five forms in CPQ14, whose ranks are exactly $(h_e, c_e, c_e, r_o - c_e, r_I)$, the improved finite receiver is

$$\text{rank}(X)\Phi_j^- \leq V_X^{(k,j)} - V_X^{(1,1)} \leq \text{rank}(X)\Phi_j^+, \quad X = G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}. \tag{EFR2}$$

To verify its types, the value maps in order are $K_e/\mathcal{B}$, S/K_e , T_e/W , $\mathcal{U}/T_e$ and $\text{im}(BJ_0)/(BJ_0E)$, in the unchanged lifts $F_e Z_h, F_e Z_c, F_e Z_o, U_o R_e^o$ (lifted under $\mathbb{B}$), and $BJ_0 Z_R$. Each is the full original affine minimum EFP13. Thus EFP14 and the four signed additions EFP16 prove EFR2. The identities $V_{\text{conn}} = V_h + V_s$ and $V_o = V_t + V_{\text{rem}}$ remain equalities of these same quantities. The determinants $|\det C_P|^2, |\det C_o|^2, |\det C_T|^2$, the full CPQ12 $\mathfrak{c}_e$, and the original lower-Gram inverse in CPQ18 remain in EFP21–22 before their exact return cancellation. The independent degree- k Q_{inv} is related to A_{inv} by the unchanged relative determinant in CPQ20 and NFR7.

C The complete root-form comparison at growing degree ratios

Scope and audited conclusion. This section proves equations (2)–(13) of the received submission, including all their finite constants, on the original stipulated complete-order packet. The verdict on that scope is *proved as written*, with the displayed guards retained. The accepted EIQ equilibrium and ECL evaluations are stated below in their exact original domain and normalization; the new full-form argument is proved in full here. No assertion about arithmetic monic minima, allowance asymptotics, shifted windows, an intrinsic mixed allocation, or a period evaluation is made here. In particular no original mass, period, root, or multiplicity is altered.

C.1 Original objects and exact dependency statements

Fix $0 < \delta < 1/2$, $\gamma > 2$, and the positive integer complete order m_{pkt} . Retain the original integer subsequence $k = 4l + 1 \geq 5$ and put

$$e_k = 1 + k(m_{\text{pkt}} - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \quad R = R_k = k\sqrt{\delta^2 + \gamma^2}.$$

The root polynomial and its exact real-coordinate image are

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^{e_k}, \\ Q(y) &= i^{-q} \chi(c + iy) = \prod_{a,b=0}^k (y - \zeta_{ab})^{e_k}, \quad \zeta_{ab} = (2b - k)\gamma - i(2a - k)\delta. \end{aligned}$$

The symbol m_{pkt} here denotes exactly the received packet order m ; the subscript merely distinguishes it from the degree of a parity block. All q roots, with their complete multiplicities, occur in opposite and conjugate pairs. They satisfy $|\zeta_{ab}| \leq R$.

The measure remains

$$d\sigma(y) = s(y) dy, \quad s(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}.$$

Its accepted original density bounds are, on all of $\mathbb{R}$,

$$c_{\Gamma} e^{-\pi|y|/2} (1 + |y|)^{-1/2} \leq s(y) \leq C_{\Gamma} e^{-\pi|y|/2} (1 + |y|)^{-1/2}, \quad c_{\Gamma} = \sqrt{2}e^{-7/6}, \quad C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{GRF1})$$

These are AMT5/TW4, restated in RGC2; no new constants are substituted. For completeness the mass follows directly from the Gamma integral. Writing $f_a(x) = e^{ax-e^x}$ gives $\Gamma(a + it) = \int_{\mathbb{R}} f_a(x) e^{itx} dx$ for $a > 0$. Plancherel and $z = e^x$ therefore give

$$\int_{\mathbb{R}} |\Gamma(a + it)|^2 \frac{dt}{2\pi} = \int_{\mathbb{R}} e^{2ax-2e^x} dx = \frac{\Gamma(2a)}{2^{2a}}.$$

Setting $a = 1/4$ and $y = 2t$ proves $\sigma(\mathbb{R}) = 2\Gamma(1/2)/\sqrt{2} = \sqrt{2\pi}$.

Here is the precise accepted equilibrium theorem used. For every $\alpha > 0$, $\beta > 0$, let $V(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. The EIQ probability measure has support $[u^2, v^2]$, $0 < u < v$, where

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad vE(\kappa) = \frac{(\alpha + 2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v^2, \quad (\text{GRF2})$$

and

$$K(\kappa) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - \kappa^2 \sin^2 \theta}}, \quad E(\kappa) = \int_0^{\pi/2} \sqrt{1 - \kappa^2 \sin^2 \theta} d\theta.$$

The density, extended by zero off this interval, is

$$\rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x), \quad h(x) = \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x + s)\sqrt{(u^2 + s)(v^2 + s)}} ds > 0. \quad (\text{GRF3})$$

It is bounded and continuous, has total mass one, and its logarithmic potential $P^{\text{eq}}(z) = \int_{u^2}^{v^2} \log|z - x| \rho(x) dx$ is continuous on $\mathbb{C}$. With the original Robin constant λ ,

$$V(x) - 2P^{\text{eq}}(x) = \lambda \quad (u^2 \leq x \leq v^2), \quad V(x) - 2P^{\text{eq}}(x) > \lambda \quad (0 < x < u^2 \text{ or } x > v^2). \quad (\text{GRF4})$$

These statements are EIQ1–19 on the entire positive parameter quadrant. Writing $w = (u + v)/2$, $z_0 = uv$, and $c_{\text{cap}} = (v^2 - u^2)/4$, the accepted ECL6 and ECL14 give exactly

$$\lambda = 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}, \quad P^{\text{eq}}(0) = 2(\alpha + 1) \log w - \alpha \log z_0 - 2. \quad (\text{GRF5})$$

The use of these already proved dependencies does not assume compactness of a moving family of parameters. All estimates below use the actual α, β, u, v .

C.2 The complete root polynomial

Lemma 1. *Put*

$$U_2 = \frac{k(k+2)}{3}, \quad U_4 = \frac{k(k+2)(3k^2+6k-4)}{15},$$

$$D_2 = qU_2(\gamma^2 - \delta^2), \quad M_4 = q[(\delta^4 + \gamma^4)U_4 + 2\delta^2\gamma^2U_2^2].$$

For real $|y| > R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} = -\frac{D_2}{y^2} + E_Q(y), \quad |E_Q(y)| \leq \frac{M_4}{2y^4(1 - R^2/y^2)}. \quad (\text{GRF6})$$

For real $|y| \geq 2R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} \leq -\frac{13D_2}{30y^2} < 0. \quad (\text{GRF7})$$

For every real y one has

$$|Q(y)|^2 \leq (y^2 + R^2)^q. \quad (\text{GRF8})$$

Proof. The original odd values of k give the literal positive real factorization

$$Q(y) = \prod_{a,b \in \{1,3,\dots,k\}} [((y - b\gamma)^2 + a^2\delta^2)((y + b\gamma)^2 + a^2\delta^2)]^{e_k} > 0.$$

No root is real, and this also shows that Q is even. Finite differentiation of

$$\sum_{a=0}^k e^{(2a-k)t} = \frac{\sinh((k+1)t)}{\sinh t}$$

at $t = 0$, using the removable value there, gives zero odd moments and

$$\frac{1}{k+1} \sum_{a=0}^k (2a-k)^2 = U_2, \quad \frac{1}{k+1} \sum_{a=0}^k (2a-k)^4 = U_4.$$

For example division of the Taylor series gives the coefficients $U_2/2$ and $U_4/24$ in the series of the displayed ratio divided by $k+1$. Expanding ζ_{ab}^2 and $|\zeta_{ab}|^4 = [(2b-k)^2\gamma^2 + (2a-k)^2\delta^2]^2$ therefore proves

$$\sum_{a,b} e_k \zeta_{ab}^2 = D_2, \quad \sum_{a,b} e_k |\zeta_{ab}|^4 = M_4 \leq qR^4.$$

For $|y| > R$ the series for $\log(1 - \zeta_{ab}/y)$ converges absolutely. Taking twice the real part, opposite-root pairing cancels its odd terms, and conjugate pairing makes its even sums real. Hence

$$\log \frac{|Q(y)|^2}{y^{2q}} = - \sum_{j \geq 1} \frac{\sum_{a,b} e_k \zeta_{ab}^{2j}}{j y^{2j}}.$$

This is the real logarithm of a positive number. Since $\sum e_k |\zeta_{ab}|^{2j} \leq M_4 R^{2j-4}$ for $j \geq 2$ and $1/j \leq 1/2$, summing the geometric majorant proves (GRF6).

The packet restrictions give $\delta^2/\gamma^2 < 1/16$, so

$$\frac{D_2}{qR^2} = \frac{k+2}{3k} \frac{\gamma^2 - \delta^2}{\gamma^2 + \delta^2} \geq \frac{5}{17} > 0.$$

At $|y| \geq 2R$, the error divided by D_2/y^2 is at most

$$\frac{qR^2}{D_2} \frac{R^2/y^2}{2(1 - R^2/y^2)} \leq \frac{17}{5} \frac{1/4}{2(3/4)} = \frac{17}{30}.$$

Subtracting this from the negative main term proves (GRF7). Finally pair each root with its opposite, with multiplicities. Every factor satisfies $|(y - \zeta)(y + \zeta)| = |y^2 - \zeta^2| \leq y^2 + R^2$. There are exactly $q/2$ such factors in Q , and squaring their product proves (GRF8). $\square$

C.3 The exact parity coordinate and moving evaluation bound

Let r be an integer, set $m_\varepsilon = \lfloor (r - \varepsilon)/2 \rfloor$ for $\varepsilon \in \{0, 1\}$, and impose $m_0, m_1 \geq 1$. This guard is equivalent to $r \geq 3$ and is retained throughout the finite theorem. Every complex polynomial p of degree at most r has the unique decomposition

$$p(y) = P_0(y^2/q^2) + (y/q)P_1(y^2/q^2), \quad \deg P_\varepsilon \leq m_\varepsilon.$$

Indeed, if $p(y) = \sum_{j=0}^r a_j y^j$, then $P_\varepsilon(x) = \sum_{j \geq 0} a_{2j+\varepsilon} q^{2j+\varepsilon} x^j$, with only indices $2j + \varepsilon \leq r$ included. This gives both the map and its inverse on coefficient spaces. Define, for $y > 0$,

$$\mathfrak{r}(y) = \frac{s(y)e^{\pi y/2}\sqrt{y}}{\sqrt{2}}.$$

For each block let

$$\alpha_\varepsilon = \frac{q + \varepsilon - 3/4}{m_\varepsilon} > 0, \quad \beta_\varepsilon = \frac{\pi q}{2m_\varepsilon} > 0, \quad V_\varepsilon(x) = \beta_\varepsilon \sqrt{x} - \alpha_\varepsilon \log x, \quad (\text{GRF9})$$

and

$$\|P\|_\varepsilon^2 = \int_0^\infty |P(x)|^2 e^{-m_\varepsilon V_\varepsilon(x)} \mathfrak{r}(q\sqrt{x}) dx.$$

Denote its actual equilibrium data by $u_\varepsilon, v_\varepsilon, P_\varepsilon^{\text{eq}}, \lambda_\varepsilon$. Since s and $|Q|^2$ are even, the two half-lines cancel the cross term $2 \operatorname{Re}(P_0 \bar{P}_1) y/q$ for both norms. Using $y = q\sqrt{x}$, $dy = q dx/(2\sqrt{x})$ on the positive half-line gives

$$\mathcal{N}_0[p] := \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \|P_\varepsilon\|_\varepsilon^2. \quad (\text{GRF10})$$

In particular the power of x in the integrand is $q + \varepsilon - 3/4$, exactly as in received equation (5). With $H_Q(x) = q^{-2q} |Q(q\sqrt{x})|^2$, the other norm is exactly

$$\mathcal{N}_\chi[p] := \int_{\mathbb{R}} |Q(y)p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \int_0^\infty |P_\varepsilon(x)|^2 H_Q(x) x^{\varepsilon-3/4} e^{-\pi q \sqrt{x}/2} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRF11})$$

All integrals converge by (GRF1).

For each block put

$$\begin{aligned}\Sigma_\varepsilon &= m_\varepsilon(2P_\varepsilon^{\text{eq}}(0) + \lambda_\varepsilon), & b_\varepsilon &= \min \left\{ \frac{u_\varepsilon}{2}, \frac{1}{4}e^{-\Sigma_\varepsilon/(2q)} \right\} > 0, \\ \Lambda_\varepsilon &= \frac{\alpha_\varepsilon}{u_\varepsilon^2} + \frac{\beta_\varepsilon}{2u_\varepsilon}, & r_+ &= \frac{C_\Gamma}{\sqrt{2}}, & r_{-, \varepsilon} &= \frac{c_\Gamma}{\sqrt{2}} \left(\frac{qu_\varepsilon}{1 + qu_\varepsilon} \right)^{1/2} > 0.\end{aligned}\tag{GRF12}$$

The original density bounds imply $\mathfrak{r}(y) \leq r_+$ for all $y > 0$ and $\mathfrak{r}(q\sqrt{x}) \geq r_{-, \varepsilon}$ on the actual support. Indeed the extra factor is $\sqrt{y/(1+y)}$, which increases in y . Also $|V'_\varepsilon(x)| \leq \Lambda_\varepsilon$ on that support.

Adding the two exact expressions in (GRF5) yields, for each parameter pair and its original w, z_0 ,

$$2P^{\text{eq}}(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z_0} \right) + 2 \log \frac{v+u}{v-u}.\tag{GRF13}$$

Here $4 \log w - 2 \log c_{\text{cap}} = 2 \log((v+u)/(v-u))$ and the constants give $-4 + 2(\alpha + 2) = 2\alpha$. Thus the -2 in each occurrence of $P^{\text{eq}}(0)$ is retained.

Lemma 2. Assume $m = m_\varepsilon \geq 1$ and $m(v_\varepsilon^2 - u_\varepsilon^2) \geq 2$. For every complex polynomial P of degree at most m and every $0 \leq x \leq b_\varepsilon^2$,

$$|P(x)|^2 \leq \frac{32e^{3\Lambda_\varepsilon/2}m^4}{r_{-, \varepsilon}} e^{\Sigma_\varepsilon} \|P\|_\varepsilon^2.\tag{GRF14}$$

Proof. Suppress the block subscripts only within this proof, retaining their actual values. If $P = 0$ the result holds. Otherwise let $M = \max_{[u^2, v^2]} |P(x)|e^{-mV(x)/2} > 0$, attained at x_0 . At least one side of x_0 in this interval has length at least $1/m$, because its full width is at least $2/m$. Take the closed one-sided interval J of length $1/m$ from x_0 . For $x \in J$,

$$|P(x)| \leq M e^{mV(x)/2} \leq e^{\Lambda/2} |P(x_0)|.$$

For any interval J and every complex polynomial of degree at most m ,

$$\|P'\|_{\infty, J} \leq \frac{4m^3}{|J|} \|P\|_{\infty, J}.\tag{GRF15}$$

To prove this, map J affinely to $[-1, 1]$ and expand $P = \sum_{j=0}^m c_j T_j$, where $T_j(\cos \theta) = \cos(j\theta)$. Cosine orthogonality gives $|c_j| \leq 2\|P\|_\infty$ for $j \geq 1$, also for complex c_j . The finite geometric-sum identity for $\sin(j\theta)/\sin \theta$ gives its modulus at most j . Thus $|T'_j| \leq j^2$, including the endpoints by continuity. Since $\sum_{j=1}^m j^2 \leq m^3$ for every integer $m \geq 1$, the derivative bound on $[-1, 1]$ is $2m^3\|P\|_\infty$; the affine derivative factor $2/|J|$ proves (GRF15).

On J , this gives $|P'| \leq 4m^4 e^{\Lambda/2} |P(x_0)|$. Consequently on the segment from x_0 of length $d = (8e^{\Lambda/2}m^4)^{-1} \leq 1/m$, the fundamental theorem of calculus and the triangle inequality imply $|P| \geq |P(x_0)|/2$. On the same segment $e^{-mV(x)} \geq e^{-\Lambda} e^{-mV(x_0)}$. Using the actual lower density r_- on it proves

$$\|P\|_\varepsilon^2 \geq r_- d \frac{|P(x_0)|^2}{4} e^{-\Lambda} e^{-mV(x_0)} = \frac{r_- M^2}{32e^{3\Lambda/2}m^4}.$$

For the passage off the support, continuity on the whole complex plane can be checked directly from the accepted bounded density. If its bound is K_ρ , then for $0 < a < 1$ and every complex z ,

$$\int_{|z-t|<a} -\log |z-t| \rho(t) dt \leq 2K_\rho a(1 - \log a).$$

Indeed the real length of $\{t : |z-t| < s\}$ is at most $2s$; integrating the level sets of the negative logarithm gives the stated bound. It tends uniformly to zero as $a \downarrow 0$. Truncation of this singularity gives continuous potentials converging locally uniformly, while the positive logarithmic part is

continuous on compact sets. Thus P^{eq} is continuous on $\mathbb{C}$. Outside the support its integrands are harmonic and uniformly differentiable on compact subsets, so P^{eq} is harmonic there. Now consider $f(z) = \log |P(z)| - mP^{\text{eq}}(z)$ on $\mathbb{C} \setminus [u^2, v^2]$. The second term is harmonic there and the first is subharmonic, including its value $-\infty$ at zeros. Continuity of P^{eq} and (GRF4) give boundary upper values $f \leq \log M + m\lambda/2$. This inequality also bounds the entire exterior, including infinity: if $n = \deg P \leq m$ and $w = 1/z$, then near $w = 0$

$$f(1/w) = (m - n) \log |w| + \log |w^n P(1/w)| - m \int \log |1 - tw| \rho(t) dt.$$

The last two terms are harmonic near zero because their arguments are nonzero there. For $n = m$ the function extends harmonically at infinity; for $n < m$ it extends subharmonically with value $-\infty$. Thus it is subharmonic on the complement of the interval in the Riemann sphere. Its boundary values have the stated upper bound. The maximum principle applies on this compact closure: an upper value greater than the boundary bound would attain an interior maximum, forcing constancy on the connected exterior, contrary to the boundary bound. Hence

$$|P(z)|^2 \leq M^2 e^{m(2P^{\text{eq}}(z) + \lambda)}.$$

For $0 \leq x < u^2$, every support point t satisfies $0 < t - x \leq t$, so $\log |x - t| \leq \log t$ and $P^{\text{eq}}(x) \leq P^{\text{eq}}(0)$. Since $b \leq u/2$, this applies throughout $[0, b^2]$. Substituting the proved estimate for M^2 yields (GRF14). $\square$

C.4 The entire inner region and the finite full-form theorem

Let $\varrho = R/q$. Impose, for both blocks,

$$m_\varepsilon \geq 1, \quad m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \geq 2, \quad \varrho \leq b_\varepsilon/2. \quad (\text{GRF16})$$

Set

$$\eta_\varepsilon = \frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4 b_\varepsilon^{2\varepsilon+1/2}}{r_{-, \varepsilon}(\varepsilon + 1/4)} 8^{-q}, \quad \eta = \max(\eta_0, \eta_1). \quad (\text{GRF17})$$

On $[0, b_\varepsilon^2]$, (GRF8) gives $H_Q(x) \leq (x + \varrho^2)^q$. The cutoff definition and the third guard give

$$e^{\Sigma_\varepsilon} (b_\varepsilon^2 + \varrho^2)^q \leq e^{\Sigma_\varepsilon} (5b_\varepsilon^2/4)^q \leq (5/64)^q \leq 8^{-q}.$$

The same estimate holds with x^q instead of $H_Q(x)$. Multiplying (GRF14) by the remaining factors in (GRF11), and using $e^{-\pi q \sqrt{x}/2} \leq 1$ and $\mathfrak{r}(q\sqrt{x}) \leq r_+$, bounds each entire inner block by

$$\frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4}{r_{-, \varepsilon}} 8^{-q} \|P_\varepsilon\|_\varepsilon^2 \int_0^{b_\varepsilon^2} x^{\varepsilon-3/4} dx = \eta_\varepsilon \|P_\varepsilon\|_\varepsilon^2.$$

The integral is finite for both $\varepsilon = 0, 1$, and its exact value is $b_\varepsilon^{2\varepsilon+1/2}/(\varepsilon + 1/4)$. Thus this estimate includes the endpoint zero and every neighborhood of it.

Define the finite quantity

$$C_\chi = \max_{\varepsilon=0,1} \left\{ \frac{D_2}{q^2 b_\varepsilon^2} + \frac{M_4}{2q^4 b_\varepsilon^4 (1 - R^2/(q^2 b_\varepsilon^2))} \right\}. \quad (\text{GRF18})$$

Its denominators are positive, at least $3/4$ in the parenthesis, by (GRF16).

Theorem 3. *Under (GRF16) and $\eta < 1$, for every complex polynomial p of degree at most r ,*

$$e^{-C_\chi} (1 - \eta) \mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1 + \eta) \mathcal{N}_0[p]. \quad (\text{GRF19})$$

Proof. In block ε , the outer interval $x \geq b_\varepsilon^2$ corresponds to $|y| \geq qb_\varepsilon \geq 2R$. By (GRF6) and monotonicity of its displayed majorants there, the density ratio $H_Q(x)/x^q$ is at least e^{-C_χ} . By (GRF7) the same ratio is at most 1. Write $N_\varepsilon = \|P_\varepsilon\|_\varepsilon^2$. The proved homogeneous inner bound implies its homogeneous outer contribution is at least $(1 - \eta_\varepsilon)N_\varepsilon$. Positivity of the root inner contribution then gives the lower bound $e^{-C_\chi}(1 - \eta_\varepsilon)N_\varepsilon$ for the root block. For the upper bound, its outer contribution is at most N_ε and its entire inner contribution is at most $\eta_\varepsilon N_\varepsilon$. Sum the two block inequalities with the common, unchanged factor $\sqrt{2}q^{2q+1/2}$ of (GRF10)–(GRF11). The result is (GRF19). No assumption on the location of polynomial mass and no monic minimization has been used. $\square$

C.5 Proof of the broad growing-ratio range

Theorem 4. *Keep the fixed original packet and let $k = 4l + 1 \rightarrow \infty$. Let $r = r_k$ be an integer, $t = r/q$, and suppose*

$$t \rightarrow \infty, \quad \log t = o(q/R). \quad (\text{GRF20})$$

Then both blocks eventually satisfy every guard in Theorem 3. More precisely,

$$u_\varepsilon \sim \frac{2}{\log t}, \quad v_\varepsilon \sim 2t, \quad b_\varepsilon \sim \frac{u_\varepsilon}{2e} \sim \frac{1}{e \log t}, \quad (\text{GRF21})$$

$$C_\chi = O_h\left(\frac{R^2}{q}(1 + \log t)^2\right) = o(q), \quad \eta = \exp[-q \log 8 + o(q)]. \quad (\text{GRF22})$$

The corresponding original reference endpoint u_t at $(\alpha, \beta) = (2/t, \pi/t)$ has $u_t \sim 2/\log t$, so $b_\varepsilon \sim u_t/(2e)$ and $qb_\varepsilon \sim q/(e \log t)$.

Proof. All limits below hold for each of the two possible block indices, so are uniform over that finite set. Put

$$c_\varepsilon = \varepsilon - 3/4, \quad d_\varepsilon = r - 2m_\varepsilon \in \{\varepsilon, \varepsilon + 1\}, \quad A_\varepsilon = 1 + c_\varepsilon/q, \quad B_\varepsilon = A_\varepsilon + 2m_\varepsilon/q.$$

These are exact identities; in particular $B_\varepsilon = 1 + t + (c_\varepsilon - d_\varepsilon)/q$. The endpoint equations (GRF2) at the actual parameters become

$$u_\varepsilon K(\kappa_\varepsilon) = 2A_\varepsilon, \quad v_\varepsilon E(\kappa_\varepsilon) = 2B_\varepsilon. \quad (\text{GRF23})$$

Here $A_\varepsilon \rightarrow 1$, $B_\varepsilon/t \rightarrow 1$, and $m_\varepsilon \sim qt/2$. Thus $m_\varepsilon \geq 1$ eventually.

We prove the endpoint asymptotics by the defining elliptic integrals, without importing a uniform remainder estimate. Write $s = u/v \in (0, 1)$, $L = \log(1/s)$, and suppress the block index for this argument. The substitution $z = \tan \theta$ gives exactly

$$K(\sqrt{1 - s^2}) = \int_0^\infty \frac{dz}{\sqrt{(1 + z^2)(1 + s^2 z^2)}}.$$

The integral on $[0, 1]$ is at most 1, and that on $[1/s, \infty)$ is at most $\int_{1/s}^\infty dz/(sz^2) = 1$. On $[1, 1/s]$ the integrand is $z^{-1}(1 + z^{-2})^{-1/2}(1 + s^2 z^2)^{-1/2}$. For nonnegative a, b , $0 \leq 1 - (1 + a)^{-1/2}(1 + b)^{-1/2} \leq (a + b)/2$; this follows by applying $(1 + x)^{-1/2} \geq 1 - x/2$ separately and using $1 - AB = (1 - A) + A(1 - B)$ with $0 < A \leq 1$. The difference of the middle integral from $\int_1^{1/s} dz/z = L$ has magnitude at most

$$\frac{1}{2} \int_1^{1/s} (z^{-3} + s^2 z) dz = \frac{1 - s^2}{2} \leq \frac{1}{2}.$$

Consequently

$$L - \frac{1}{2} \leq K(\sqrt{1 - s^2}) \leq L + 2. \quad (\text{GRF24})$$

The defining E integral has $1 \leq E \leq \pi/2$ and $E \rightarrow 1$ as $s \rightarrow 0$, by dominated convergence to the integral of $\cos \theta$. By (GRF23),

$$\frac{sK}{E} = \frac{A}{B} \rightarrow 0. \quad (\text{GRF25})$$

Since $K \geq \pi/2$ and $E \leq \pi/2$, this implies $s \rightarrow 0$ and $L \rightarrow \infty$. Taking logarithms in (GRF25) gives

$$L = \log(B/A) + \log K - \log E = \log t + \log L + O(1).$$

Here (GRF24) gives $\log K = \log L + o(1)$ and $\log(B/A) = \log t + o(1)$. As $L \rightarrow \infty$, division of $\log t = L - \log L + O(1)$ by L proves $L \sim \log t$ and then $K \sim \log t$. Now (GRF23) yields

$$u \sim 2/\log t, \quad v \sim 2t, \quad s = O(1/(t \log t)). \quad (\text{GRF26})$$

The same proof with $A = 1$, $B = 1 + t$ is exactly the claimed statement for u_t at $(2/t, \pi/t)$.

The exact expression (GRF13) gives

$$\frac{\Sigma_\varepsilon}{2q} = A_\varepsilon \left[1 + \log \frac{1 + s_\varepsilon}{2u_\varepsilon} \right] + \frac{m_\varepsilon}{q} \log \frac{1 + s_\varepsilon}{1 - s_\varepsilon}. \quad (\text{GRF27})$$

For $0 < s \leq 1/2$, $0 \leq \log((1+s)/(1-s)) \leq 4s$, by integration of $2/(1-s^2)$. Thus the last term is $O(1/\log t) = o(1)$ by (GRF26). Also $\log(1+s) = o(1)$. Since $A_\varepsilon - 1 = O(1/q)$ and $1 + |\log u_\varepsilon| = O(1 + \log \log t)$, the error from replacing A_ε by 1 is $O((1 + \log \log t)/q) = o(1)$. The last conclusion follows from (GRF20): for large k , $\log t \leq q/R \leq q$, so $\log \log t = O(\log q) = o(q)$. Therefore

$$\frac{\Sigma_\varepsilon}{2q} = 1 + \log \frac{1}{2u_\varepsilon} + o(1), \quad \frac{1}{4} e^{-\Sigma_\varepsilon/(2q)} = \frac{u_\varepsilon}{2e} (1 + o(1)).$$

Since $1/e < 1$, this second quantity is eventually the smaller member in (GRF12), proving (GRF21).

We now verify every remaining finite guard and error. The support width satisfies $v_\varepsilon^2 - u_\varepsilon^2 \sim 4t^2$, so

$$m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \sim 2qt^3 \rightarrow \infty.$$

Furthermore

$$\frac{R}{qb_\varepsilon} \sim \frac{eR \log t}{q} \rightarrow 0,$$

which proves the cutoff guard. The exact parameters satisfy $\alpha_\varepsilon = O(1/t)$ and $\beta_\varepsilon = O(1/t)$, whence

$$\Lambda_\varepsilon = O((\log t)^2/t + \log t/t) = O((\log t)^2/t) = o(1).$$

Since $q/\log t \rightarrow \infty$, one has $qu_\varepsilon \rightarrow \infty$ and $r_{-, \varepsilon} \rightarrow c_\Gamma/\sqrt{2} > 0$. Taking logarithms in the exact (GRF17) gives

$$\begin{aligned} \log \eta_\varepsilon = & -q \log 8 + 4 \log m_\varepsilon + (2\varepsilon + 1/2) \log b_\varepsilon + \frac{3}{2} \Lambda_\varepsilon \\ & + \log \frac{32r_+}{r_{-, \varepsilon}(\varepsilon + 1/4)}. \end{aligned}$$

Here $\log m_\varepsilon = \log q + \log t + O(1) = o(q)$: $\log q = o(q)$ and (GRF20), with $R \rightarrow \infty$, gives $\log t = o(q)$. Also $\log b_\varepsilon = -\log \log t + O(1) = o(q)$, and all other displayed terms are $o(q)$. Both block errors are therefore $\exp[-q \log 8 + o(q)]$; their maximum has the same form and is eventually less than 1.

Finally $D_2 \leq qR^2$ by its root-sum representation and $M_4 \leq qR^4$. The two terms in (GRF18) are bounded by

$$\frac{R^2}{qb_\varepsilon^2} \left(1 + \frac{R^2/(q^2 b_\varepsilon^2)}{2(1 - R^2/(q^2 b_\varepsilon^2))} \right).$$

The parenthesis tends to 1 and $b_\varepsilon^{-2} = O((1 + \log t)^2)$. This proves the first estimate in (GRF22). Dividing it by q and using $R \log t/q \rightarrow 0$ proves $C_\chi = o(q)$. $\square$

C.6 Source pins and exact audit boundary

The received text was read in full before reconstruction. Its SHA256 is

b89f991d351b9a8b7cc2cf76768eb9997d79c50e8df03929fe82ea427e4a60ef.

The source file `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex` in `output/Split_Zero_Web_Continuation_CURRENT` has SHA256

9926c5b5670777548f15f3fd626a4237d2c10a26834f51ba2b466a3b3a349656.

The identical immutable receiving copy is stored in `work/growing_ratio_math_integration_20260917/baseline_692`. The exact inputs inspected there are EIQ1–33, lines 11082–11539; OOQ7 and its complete proof, lines 13597–13641; ECL6, lines 18491–18505, and ECL14, lines 18664–18674; PCRP1–8, lines 57079–57169; RGC1–2, lines 57416–57447; and the earlier explicitly reference-only endpoint statement UCT40–42, lines 56636–56660. The related `02_FULL_GAMMA_PROOFS.tex` has SHA256

c226f7110ed7929b62c17b43bc6759d089a72e766fca387ebffcd80723a3c08;

its original mass and density declarations HMR19–20 are at lines 15889–15902. The complete EIQ provider is also present there at lines 1897–2344; its ECL provider begins at line 18675.

The direct elementary elliptic-integral proof in Theorem 4 does not rely on the “standard complete-elliptic expansions” cited by the earlier UCT40 paragraph. The evaluation proof independently reconstructs OOQ7 with the actual lower density and derivative constant, rather than using its earlier compact-parameter supremum. Thus no compact-parameter bound is extrapolated to $(\alpha_\varepsilon, \beta_\varepsilon) \rightarrow (0, 0)$. The two guards $m_\varepsilon \geq 1$ are essential to the displayed definitions; finite degrees $r < 3$ are outside this finite theorem, without affecting the eventually growing-ratio conclusion.

D The reference norm at its actual moving parameters

The coordinate throughout is the literal coordinate $y = q\sqrt{x}$ on the positive half-line. The underlying measure is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{GRE1})$$

It is not divided by its mass. Fix $q > 0$, an integer $m \geq 1$, and $\alpha, \beta > 0$. Write $\mu = \mu_{\alpha, \beta}$ for the probability measure constructed in the complete EIQ proof included in the accompanying complete source, with support $[a, b] = [u^2, v^2]$, $0 < u < v$, logarithmic potential $P(x) = \int \log|x - s| d\mu(s)$, and Robin constant λ in the convention

$$V(x) = \beta\sqrt{x} - \alpha \log x, \quad D(x) = V(x) - 2P(x) - \lambda \geq 0 \quad (x > 0). \quad (\text{GRE2})$$

Equality holds on $[a, b]$ and is strict outside it. The complete EIQ argument, included in the accompanying complete source, constructs the endpoints for every point of $(0, \infty)^2$, proves positivity and mass one, computes the Cauchy transform, and proves this inequality on both components of the exterior. In particular the inequality is available near zero as well as at infinity. Put

$$W = v^2 - u^2, \quad c = W/4, \quad w = (u + v)/2, \quad z = uv. \quad (\text{GRE3})$$

Here c is logarithmic capacity, and z is a positive scalar, not a complex integration variable.

D.1 Exact Robin and logarithmic-moment identities

The complete ECL1–14 proof is included in the accompanying complete source. Its moving-support derivative has the following precise test-function meaning:

$$\frac{\partial}{\partial \alpha} \int f(x) d\mu_{\alpha, \beta}(x) = \frac{1}{2} \int_a^b f(x) \left(1 - \frac{z}{x}\right) \frac{dx}{\pi \sqrt{(b-x)(x-a)}}. \quad (\text{GRE4})$$

It holds at fixed β for every continuously differentiable f on a neighbourhood of the support. To track the endpoint terms, write $p(x) = h(x)/(2\pi)$ and $\rho(x) = p(x)\sqrt{(b-x)(x-a)}$. Differentiation at the actual moving endpoints gives the signed density

$$\left[\dot{p}(x)\sqrt{(b-x)(x-a)} + \frac{p(x)\{\dot{b}(x-a) - \dot{a}(b-x)\}}{2\sqrt{(b-x)(x-a)}} \right] \mathbf{1}_{(a,b)}(x). \quad (\text{GRE5})$$

The proof changes variables to $x = a + (b-a)s$, $0 \leq s \leq 1$, differentiates there, and integrates the term containing f' by parts. The boundary term is proportional to $\sqrt{s(1-s)}$ and vanishes. The remaining derivative is integrable, bounded by a constant times $[s(1-s)]^{-1/2}$. Thus no endpoint atom is present and no moving-support term was dropped. The complete transform and zero-mass logarithmic-energy calculation in ECL7–10 identifies (GRE5) exactly with (GRE4).

In particular, setting $\mathcal{L} = P(0) = \int \log x d\mu(x)$ gives

$$\begin{aligned} \lambda &= 2(\alpha + 2) - 2\alpha \log w - 2 \log c, \\ \mathcal{L} &= 2(\alpha + 1) \log w - \alpha \log z - 2, \\ \lambda_\alpha &= -2 \log w, \quad \mathcal{L}_\alpha = 2 \log w - \log z. \end{aligned} \quad (\text{GRE6})$$

The constant -2 in the logarithmic moment remains. Direct addition, using $w^2/c = (u+v)/(v-u)$, yields

$$\boxed{2P(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z}\right) + 2 \log \frac{v+u}{v-u}}. \quad (\text{GRE7})$$

This is the received equation (6), with the original potential and Robin sign convention.

For clarity, the second Robin derivative also retains all support movement. For $\beta_0 > 0$ put $c_0 = \beta_0/\beta$. The dilation $x \mapsto c_0^2 x$ pushes μ_{α, β_0} to $\mu_{\alpha, \beta}$: on every probability measure its energy changes by the constant $-2(\alpha + 1) \log c_0$, so the proved unique minimizer is carried to the proved unique minimizer. At corresponding points its potential is $2 \log c_0 + P_{\alpha, \beta_0}(x)$, while its external field is $V_{\alpha, \beta_0}(x) - 2\alpha \log c_0$. Consequently

$$\lambda(\alpha, \beta) = \lambda(\alpha, \beta_0) + 2(\alpha + 2) \log(\beta/\beta_0), \quad \boxed{\lambda_\beta = \frac{2(\alpha + 2)}{\beta}}. \quad (\text{GRE8})$$

The distinct identity $\int \sqrt{x} d\mu = 2(\alpha + 1)/\beta$ concerns the moment, not the Robin derivative. It follows by differentiating the energy of the competitor $x \mapsto c_0^2 x$ at $c_0 = 1$ as in EIQ31.

D.2 Original Gamma constants, mass, and monic factorials

Retain exactly

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \quad \mathfrak{r}(y) = \frac{\sigma(y)e^{\pi y/2}\sqrt{y}}{\sqrt{2}} \quad (y > 0). \quad (\text{GRE9})$$

The identity rederived here is Binet's integral formula in the human-authored Gamma chapter of Askey and Roy [H22, eq. (5.9.10_2)], valid for $\Re z > 0$. Its integration variable t is the present s , and its kernel is exactly

$$\frac{1}{s} \left(\frac{1}{e^s - 1} - \frac{1}{s} + \frac{1}{2} \right) = \frac{\coth(s/2) - 2/s}{2s} = K(s).$$

The holomorphic logarithm of Γ is the branch real on the positive axis; $\log z$ is principal on this half-plane. At $z = \zeta = 1/4 + iy/2$ the local BI17–19 bound gives $|R(\zeta)| \leq (12\Re\zeta)^{-1} = 1/3$. With the retained density $\sigma(y) = |\Gamma(\zeta)|^2/(2\pi)$, twice the real part gives GRE10. The explicit constants in GRE9–11 are then obtained by the displayed local inequalities.

The complete derivation of Binet's identity from the defining Gamma integral is reproduced in the accompanying complete source as BI1–26. Its remainder kernel is $K(s) = (\coth(s/2) - 2/s)/(2s)$ with $0 < K(s) < 1/12$. Therefore at $\zeta = 1/4 + iy/2$ its remainder satisfies $|R(\zeta)| \leq 1/3$, and twice the real part of that exact identity gives, with $Y = |y|$,

$$\sigma(y) = |\zeta|^{-1/2} \exp\{-Y \arctan(2Y) - 1/2 + 2\Re R(\zeta)\}. \quad (\text{GRE10})$$

For $Y > 0$, $Y \arctan(1/(2Y)) \leq 1/2$, whereas $\arctan(2Y) \leq \pi/2$; the same resulting inequalities hold at $Y = 0$. Thus the exponent in (GRE10) lies between $-\pi Y/2 - 7/6$ and $-\pi Y/2 + 2/3$. Also

$$\frac{(1+Y)^2}{4} - |\zeta|^2 = \frac{3+8Y}{16}, \quad |\zeta|^2 - \frac{(1+Y)^2}{20} = \frac{(4Y-1)^2}{80}.$$

Taking the appropriate positive fourth roots proves

$$c_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2}. \quad (\text{GRE11})$$

In particular all polynomial moments are finite and

$$r_+ = C_\Gamma/\sqrt{2}, \quad r_- = (c_\Gamma/\sqrt{2})\sqrt{\frac{qu}{1+qu}}, \quad \mathfrak{r}(y) \leq r_+, \quad \mathfrak{r}(q\sqrt{x}) \geq r_- \quad (a \leq x \leq b). \quad (\text{GRE12})$$

The lower estimate follows from the increasing function $y/(1+y)$.

For completeness, the mass and factorial monic norms can be recovered without selecting another measure. The beta identity, proved in BI13, with $a > 0$ and the substitution $x = \log(t/(1-t))$, gives

$$\frac{\Gamma(a+i\eta)\Gamma(a-i\eta)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{i\eta x} (2 \cosh(x/2))^{-2a} dx. \quad (\text{GRE13})$$

The beta integrand after substitution is absolutely integrable. At $a = 1/4$, Fourier inversion therefore gives

$$\int_{\mathbb{R}} e^{i\xi y} d\sigma(y) = \sqrt{2\pi}(\cosh \xi)^{-1/2}. \quad (\text{GRE14})$$

Here Fourier inversion can be justified directly by inserting a Gaussian factor $e^{-\delta\eta^2}$ in the inversion integral, using Fubini, and evaluating the Gaussian Fourier integral. The result is convolution with the probability density $(4\pi\delta)^{-1/2}e^{-x^2/(4\delta)}$, which tends to the continuous beta density at each point: split at $|x| = \varepsilon$, use uniform continuity on the first piece and the Gaussian tail on the second. On the Fourier side dominated convergence is valid by (GRE11), with $y = 2\eta$. This proves the inversion used in (GRE14). Both sides of (GRE14) extend holomorphically to $|\Im \xi| < \pi/2$: for the left side (GRE11) gives locally uniform exponential domination; for the right side the branch is positive at zero and has no zero of $\cosh \xi$ in the strip. The identity theorem consequently gives

$$\int e^{\theta y} d\sigma(y) = \sqrt{2\pi}(\cos \theta)^{-1/2} \quad (|\theta| < \pi/2). \quad (\text{GRE15})$$

This retains mass $\sqrt{2\pi}$ at $\theta = 0$.

Define monic polynomials p_n by the convergent generating series

$$F(y, z) = (1 + z^2)^{-1/4} e^{y \arctan z} = \sum_{n \geq 0} p_n(y) \frac{z^n}{n!}.$$

The coefficient of y^n in p_n is one, because $\arctan z = z + O(z^3)$ and the other prefactor has value one at zero. For sufficiently small real z, w , (GRE15) and $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$ give the exact generating identity

$$\int F(y, z) F(y, w) d\sigma(y) = \sqrt{2\pi}(1 - zw)^{-1/2}.$$

Termwise differentiation at zero is justified by (GRE11) and exponential domination in a neighbourhood of $(z, w) = (0, 0)$. Hence

$$\int p_n p_j d\sigma = \mathbf{1}_{n=j} \sqrt{2\pi} n! (1/2)_n, \quad \gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} \frac{(2n)!}{4^n}. \quad (\text{GRE16})$$

For a complex monic degree- n polynomial, subtraction of p_n leaves degree at most $n-1$, orthogonal to p_n . Pythagoras proves that γ_n is precisely the monic minimum; the product equality in (GRE16) follows by separating the even and odd factors of $(2n)!$.

D.3 A global density bound and a full tail integral

The actual positive density in EIQ10–11 obeys

$$0 < h(x) \leq \frac{\alpha}{xuv}, \quad \rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x).$$

For $u^2 \leq x \leq v^2$, $(v^2 - x)(x - u^2) \leq v^2 x$. It follows, including the zero endpoint values, that

$$\boxed{\|\rho\|_{\infty} \leq \mathcal{M} := \frac{\alpha}{2\pi u^2}}. \quad (\text{GRE17})$$

This is the received equation (16).

Set, at these very parameters,

$$C_{\lambda} = 2 \log 2 + \max(\lambda, 0),$$

$$T_* = \max \left\{ 1, v, \left(\frac{8(\alpha + 2)}{\beta} \right)^2, \frac{4C_{\lambda}}{\beta} \right\}, \quad J_* = T_*^2 + \frac{8}{\beta^2}. \quad (\text{GRE18})$$

For $s \geq \max\{1, v\}$ and every support point y , $|s^2 - y| \leq s^2 + v^2 \leq 2s^2$. The logarithm is integrable even if $s = v$, so integration gives $P(s^2) \leq 2 \log s + \log 2$. Therefore

$$D(s^2) \geq \beta s - 2(\alpha + 2) \log s - C_\lambda.$$

For $s \geq 1$, $\log s \leq \sqrt{s}$: the maximum of $\log s / \sqrt{s}$ is $2/e < 1$, obtained by direct differentiation. If $s \geq T_*$, then $2(\alpha + 2) \log s \leq 2(\alpha + 2) \sqrt{s} \leq \beta s / 4$ and $C_\lambda \leq \beta s / 4$. Consequently

$$D(s^2) \geq \beta s / 2 \quad (s \geq T_*), \quad \int_0^\infty e^{-mD(x)} dx \leq T_*^2 + \int_0^\infty 2s e^{-\beta s / 2} ds = J_*. \quad (\text{GRE19})$$

The first interval uses $D \geq 0$, and the tail uses $m \geq 1$. The integral starts at zero. All constants are finite at each point of the full positive parameter quadrant.

D.4 Arcsine lower bound and explicit quantile upper bound

Define the actual monic minimum

$$h_{m,q}(\alpha, \beta) = \min_{p \in \mathbb{C}[x], p \text{ monic}, \deg p = m} \int_0^\infty |p(x)|^2 e^{-mV(x)} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRE20})$$

The integrals are finite: the weight near zero is at most $r_+ x^{m\alpha}$ and at infinity at most $r_+ x^{m\alpha} e^{-m\beta\sqrt{x}}$. Its density is positive on $(0, \infty)$, so its finite coefficient Gram matrix is positive definite. Completing the square in that finite matrix proves existence and uniqueness of the monic minimizer.

Let $d\nu = dx / (\pi \sqrt{(b-x)(x-a)})$ be arcsine probability on $[a, b]$. Its potential satisfies

$$\int \log |x - \zeta| d\nu(x) \geq \log c \quad (\zeta \in \mathbb{C}), \quad \int \log |x - y| d\nu(x) = \log c \quad (y \in [a, b]). \quad (\text{GRE21})$$

To prove both statements rootwise, put $C = (a + b)/2$, $d = W/2$ and $x = C + d \cos \theta$. The polynomial $Z^2 - 2(\zeta - C)Z/d + 1$ has roots z_1, z_2 with product one. On the unit circle, $|x - \zeta| = (d/2)|Z - z_1||Z - z_2|$. The circle mean of $\log |Z - z_j|$ equals $\log \max\{1, |z_j|\}$. For $|z_j| < 1$ this follows by integrating the absolutely convergent logarithmic series, for $|z_j| > 1$ by factoring $|z_j|$, and for $|z_j| = 1$ by a limit dominated near the single singular angle by a constant plus $|\log |\theta||$. The dependence on $\cos \theta$ identifies the whole-circle mean with the arcsine integral. Both roots have modulus one precisely when $\zeta \in [a, b]$, proving the equality; their two log-max terms are nonnegative, proving the inequality. All logarithmic integrals just used are absolutely integrable.

For a monic degree- m polynomial p , factorization and (GRE21) give $\int \log |p| d\nu \geq m \log c$. Integrate the Euler equality against ν . Fubini is justified by the bounded equilibrium density and the integrable arcsine logarithmic singularity, and gives

$$\int V d\nu = 2 \log c + \lambda. \quad (\text{GRE22})$$

The arcsine entropy is exactly

$$\int \log \frac{d\nu}{dx} d\nu = -\log(\pi c). \quad (\text{GRE23})$$

Indeed $\sqrt{(b-x)(x-a)} = (W/2) \sin \theta$, and $\pi^{-1} \int_0^\pi \log \sin \theta d\theta = -\log 2$. For the latter identity, put $I = \int_0^{\pi/2} \log \sin \theta d\theta$. Symmetry gives the same cosine integral, while doubling the argument in $\log \sin \theta + \log \cos \theta = \log \sin(2\theta) - \log 2$ gives $2I = I - (\pi/2) \log 2$. Endpoint logarithms are integrable.

Probability Jensen applied to $|p|^2 e^{-mV} / (d\nu/dx)$, whose logarithm is integrable, now proves

$$\int_a^b |p|^2 e^{-mV} dx \geq \exp \left\{ 2 \int \log |p| d\nu - m \int V d\nu - \int \log(d\nu/dx) d\nu \right\} \geq \pi c e^{-m\lambda}. \quad (\text{GRE24})$$

Multiplication by the lower remainder bound in (GRE12) gives the claimed lower monic bound.

For the upper bound take the unique midpoint quantiles $x_i \in (a, b)$ with $\mu((0, x_i]) = (i-1/2)/m$, $1 \leq i \leq m$, and define $p_m(x) = \prod_{i=1}^m (x - x_i)$. The distribution function of its empirical probability measure differs from that of μ by at most $1/(2m)$, and hence by at most $1/m$. For fixed $x > 0$ put

$$\delta = 1/m, \quad f_x(y) = \log \max\{|x - y|, \delta\}.$$

This is continuous and of bounded variation on $[a, b]$. If x lies inside $[a, b]$, its decreasing and increasing pieces each have variation at most $\log(1 + W/\delta)$. If x lies outside the interval it is monotone and its endpoint ratio after clipping is at most $1 + W/\delta$. Thus, in all cases, $\text{Var}_{[a, b]} f_x \leq 2 \log(1 + mW)$. Stieltjes integration by parts against the zero-mass empirical error has zero endpoint term, and gives

$$\sum_{i=1}^m f_x(x_i) \leq m \int f_x d\mu + 2 \log(1 + mW). \quad (\text{GRE25})$$

By (GRE17),

$$\int (f_x(y) - \log|x - y|) d\mu(y) \leq \mathcal{M} \int_{-\delta}^{\delta} \log(\delta/|s|) ds = 2\mathcal{M}\delta. \quad (\text{GRE26})$$

Each nodal logarithm is at most its clipped value; at a root its value is $-\infty$, where the resulting inequality is automatic. Combining (GRE25)–(GRE26), then multiplying by e^{-mV} , yields the global positive-half-line estimate

$$|p_m(x)|^2 e^{-mV(x)} \leq e^{4\mathcal{M}} (1 + mW)^4 e^{-m\lambda} e^{-mD(x)} \quad (x > 0). \quad (\text{GRE27})$$

Integrating it with (GRE12) and (GRE19) proves the complete finite norm theorem

$$\boxed{c_- e^{-m\lambda} \leq h_{m,q}(\alpha, \beta) \leq c_+ e^{-m\lambda}, \quad c_- = r_- \pi W/4, \quad c_+ = r_+ J_* e^{4\mathcal{M}} (1 + mW)^4.} \quad (\text{GRE28})$$

This is the received (17). Neither its proof nor its constants require a compact parameter rectangle, an endpoint lower density estimate, or a lower bound on quantile-cell lengths.

D.5 Exact parity map, shifts, and factorial cancellation

Let $q \geq 1$ be an integer and $r = 2m + \varepsilon$ an integer with $m \geq 1$, $\varepsilon \in \{0, 1\}$. Define the full homogeneous reference monic minimum

$$\nu_r^0 = \min_{p \text{ monic}, \deg p = r} \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y). \quad (\text{GRE29})$$

An even positive weight makes the even and odd coefficient spaces orthogonal. For any monic degree- r polynomial, the component with parity opposite to r contributes a nonnegative separate squared norm and has no leading term, so the unique monic minimizer has parity r . The exact bijection between its parity fibre and the monic degree- m fibre is

$$p(y) = q^r (y/q)^\varepsilon P(y^2/q^2). \quad (\text{GRE30})$$

The leading coefficient is $q^{r-\varepsilon-2m} = 1$, and expansion in powers proves surjectivity and injectivity. Both half-lines contribute equally. Substitution $y = q\sqrt{x}$ has Jacobian $q/(2\sqrt{x})$, and (GRE9) gives exactly

$$\boxed{\nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{m,q}(\alpha, \beta), \quad \alpha = \frac{q + \varepsilon - 3/4}{m}, \quad \beta = \frac{\pi q}{2m}.} \quad (\text{GRE31})$$

Every mass and power, including the quarter shift, is retained.

Put $t = r/q$, $\alpha_0 = 2/t$, $\beta_0 = \pi/t$, and $\lambda_0 = \lambda(\alpha_0, \beta_0)$. The differences are exactly

$$A := m(\alpha - \alpha_0) = \varepsilon - \frac{3}{4} + \frac{q\varepsilon}{r}, \quad B := m(\beta - \beta_0) = \frac{\pi q\varepsilon}{2r}. \quad (\text{GRE32})$$

Let $(\alpha_\theta, \beta_\theta) = (\alpha_0, \beta_0) + \theta(\alpha - \alpha_0, \beta - \beta_0)$, $0 \leq \theta \leq 1$, and let w_θ be its actual midpoint $(u_\theta + v_\theta)/2$. All these parameter pairs are positive. The full support derivatives in (GRE6),(GRE8) and the fundamental theorem of calculus give

$$\begin{aligned}\Delta &:= m\lambda(\alpha, \beta) - \frac{r}{2}\lambda_0 \\ &= \int_0^1 \left[-2A \log w_\theta + B \frac{2(\alpha_\theta + 2)}{\beta_\theta} \right] d\theta - \frac{\varepsilon}{2}\lambda_0.\end{aligned}\tag{GRE33}$$

Thus the finite, explicitly evaluable parameter bound is

$$|\Delta| \leq B_{\text{par}} := 2|A| \int_0^1 |\log w_\theta| d\theta + 2|B| \int_0^1 \frac{\alpha_\theta + 2}{\beta_\theta} d\theta + \frac{\varepsilon}{2}|\lambda_0|.\tag{GRE34}$$

No fixed-endpoint differentiation is used in this formula.

Define the original profile in its exact Robin form

$$\psi(t) = (1+t)(1 - \log(1+t)) - \frac{t}{4}\lambda(2/t, \pi/t).\tag{GRE35}$$

To identify it explicitly, let $\kappa \in (0, 1)$ be the EIQ modulus and $s = \sqrt{1 - \kappa^2}$. The endpoint equations at these actual parameters give

$$u_t K = 2, \quad v_t E = 2(1+t), \quad 1+t = \frac{E}{sK}, \quad w_t = \frac{(1+t)(1+s)}{E}, \quad c_t = \frac{(1+t)^2 \kappa^2}{E^2}.$$

Inserting these formulas into (GRE6),(GRE35), with no endpoint change, gives

$$\psi(t) = \log \frac{1+s}{E} + t \log \frac{\kappa}{E}.\tag{GRE36}$$

This is precisely the original UCT26 profile.

Write $n = q + r$. BI26 evaluated at $2n$ and the recurrence $\Gamma(2n+1) = 2n\Gamma(2n)$ give

$$\log(2n)! = (2n + \tfrac{1}{2}) \log(2n) - 2n + \tfrac{1}{2} \log(2\pi) + R_{2n}, \quad 0 < R_{2n} < \frac{1}{24n}.$$

Combine this with (GRE16),(GRE31). Exact cancellation gives

$$\log \frac{\nu_r^0}{\gamma_{q+r}} = 2(q+r)(1 - \log(1+t)) + \log h_{m,q} - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}.\tag{GRE37}$$

Equivalently, for $X = \log h_{m,q} + m\lambda$,

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(t) + X - \Delta - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}, \quad \log c_- \leq X \leq \log c_+}.\tag{GRE38}$$

The useful finite endpoints, before any asymptotic replacement, are also exactly

$$\log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_- \leq \log \frac{\nu_r^0}{\gamma_{q+r}} \leq \log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_+.\tag{GRE39}$$

D.6 Uniform size of the retained errors

The elementary elliptic proof included in the accompanying complete source establishes, uniformly also along the segment in (GRE33),

$$u_\theta \sim \frac{2}{\log t}, \quad v_\theta \sim 2t, \quad W \sim 4t^2, \quad w_\theta \sim t, \quad |\lambda_0| + |\lambda| = O(1 + \log t)\tag{GRE40}$$

as $q \rightarrow \infty$, $t \rightarrow \infty$. The segment has $\alpha_\theta = O(1/t)$ and $\beta_\theta \asymp 1/t$. Equations (GRE32)–(GRE34) consequently give $|\Delta| = O(1 + \log t)$, since $A = O(1)$ and $B = O(1/t)$. The same support estimates in the full finite constants give

$$\mathcal{M} = O((\log t)^2/t), \quad T_* = O(t^2), \quad J_* = O(t^4). \quad (\text{GRE41})$$

For T_* , its third entry is $O(t^2)$, its fourth is $O(t(1 + \log t)) = O(t^2)$, and its other entries are $O(t)$. These statements concern the actual endpoints, rather than compact rectangle extrema.

For the packet range of the received source, let $q = [1 + k(M - 1)](k + 1)^2$, with the fixed original multiplicity $M \geq 1$, $R_k = k\sqrt{\delta^2 + \gamma^2}$, and $\log t = o(q/R_k)$. Then $q/\log t \rightarrow \infty$, hence $qu \rightarrow \infty$ and r_- is bounded above and below by fixed positive constants. Thus $c_- \asymp t^2$, and

$$|\log c_-| + |\log c_+| = O(\log(q + r)). \quad (\text{GRE42})$$

Indeed $\mathcal{M} = O(1)$, $\log J_* = O(1 + \log t)$ and $mW = O(rt^2)$; since $q \geq 1$, $t = r/q \leq r$, so $\log(1 + rt^2) = O(\log(q + r))$. Lower bounds for J_* and $1 + mW$ are one, so this also bounds the absolute logarithm of c_+ , not only its positive part. The positive lower bound of r_+ is fixed. Similarly (GRE40) and $r_- \asymp 1$ bound both signs of $\log c_-$.

Finally (GRE38), (GRE34), (GRE42) and $0 < R_{2(q+r)} < 1/(24(q + r))$ prove

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(r/q) + O(\log(q + r))}. \quad (\text{GRE43})$$

The constants are uniform along every such growing-ratio sequence and for either parity. This is the received equation (18). All finite content is already specified by (GRE18), (GRE28), (GRE33), and (GRE38)–(GRE39); no unspecified equilibrium norm theorem occurs.

E Original arithmetic transfer and the complete shifted action

This section proves the arithmetic receiving calculation for the moving equilibrium comparison. It retains the stipulated complete zero quartet, all source masses and divided jets, the original remainder quotient, and both adjacent contraction factors. The moving-cutoff full-form theorem and the moving-parameter reference-norm theorem are the analytic results proved in the companion sections. Their exact conclusions are displayed where used; none of their constants is replaced by a compact-parameter constant. The new calculation here is the exact arithmetic transfer, its degree-uniform asymptotic consequences, and its original complete-action receiver.

E.1 Objects, coefficient maps, and all-degree arithmetic constants

Fix the original complete quartet data

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{N}_{>0}, \quad h(z) = \prod_{\zeta \in \{1/2 \pm \delta \pm i\gamma\}} (z - \zeta)^m, \quad v_h = 2\xi/h.$$

The zero order at each centre is exactly the stated m , so the entire quotient v_h is nonzero at every centre. Along the original tensor subsequence put

$$\begin{aligned} k = 4\ell + 1 \geq 5, \quad c = k/2, \quad e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \quad R_k = k\sqrt{\delta^2 + \gamma^2}, \\ \lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}, \\ Q(y) = i^{-q}\chi(c + iy), \quad E = \mathbb{C}[S]/(\chi), \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ d\mu_k(y) = w_h^{*k}(y) dy, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \end{aligned} \tag{GRA1}$$

The masses are $(\int w_h)^k$ and $\sqrt{2\pi}$, respectively. For $P \in \mathcal{P}_j(S) = \mathbb{C}[S]_{\leq j}$ the arithmetic squared norm is $\|P\|_{\text{ar}}^2 = \int |P(c + iy)|^2 d\mu_k(y)$; the corresponding source Gamma norm uses $d\sigma$ in the same coordinate. Inner products are conjugate-linear in their first argument. Every polynomial coefficient below may be complex.

The precise coefficient maps are

$$U_j P(y) = i^{-j} P(c + iy), \quad U_j^{-1} p(S) = i^j p((S - c)/i), \quad U_{q+j} M_\chi = M_Q U_j. \tag{GRA2}$$

Direct substitution proves both inverse identities and the multiplication identity. The leading coefficient of a polynomial of degree j is unchanged. The leading coefficient of a polynomial of degree $a < j$ is multiplied by i^{a-j} , which is retained. Thus U_j bijects the two degree- j monic affine spaces, and its scalar phase has modulus one in each pulled-back integral norm. No Euclidean norm of a coefficient vector is asserted to be invariant under translation.

Write $\alpha_* = \pi/2$ to distinguish this fixed tail constant from the moving equilibrium parameters. The original AMT/TW constants are

$$\begin{aligned} p_h = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M_h = 21 + 4m, \\ \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_{\alpha_*} = \int_{\mathbb{R}} e^{\alpha_* y} w_h(y) dy, \\ c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha_*(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p_h+1} M_{\alpha_*}^{k-1}, \\ D_d = [1 + 4(d + M_h)^2]^{M_h}, \quad L_{k,d} = \log(A_k D_d / a_k). \end{aligned} \tag{GRA3}$$

Here c_h and C_h^{tilt} are exactly the positive original TW7 convolution lower-envelope and BT12 compact/tail constants. Their already proved defining estimates, with the full numerator $2\xi/h$, are

$$\begin{aligned} w_h^{*k}(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha_*(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ w_h^{*k}(y) &\leq k C_h^{\text{tilt}} M_{\alpha_*}^{k-1} e^{-\alpha_*|y|} (1+|y|/k)^{-p_h}, \\ c_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2} &\leq d\sigma/dy \leq C_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2}. \end{aligned} \quad (\text{GRA4})$$

These bounds hold for every convolution order $k \geq 3$. In particular there is no upper polynomial-degree restriction in the following argument.

The first and third estimates, $k-2+|y| \leq (k-2)(1+|y|)$, and $(1+|y|)^{B_h} \leq 2^{B_h/2}(1+y^2)^{M_h}$ give $w_h^{*k} \geq a_k(1+y^2)^{-M_h} d\sigma/dy$. The second and third, together with $1+|y|/k \geq (1+|y|)/k$ and $p_h > 1/2$, give $w_h^{*k} \leq A_k d\sigma/dy$. For the original Gamma orthonormal polynomials e_n one has

$$ye_n = \sqrt{(n+1)(n+1/2)}e_{n+1} + \sqrt{n(n-1/2)}e_{n-1}.$$

The two shifts on polynomials of degree at most d have norms at most $d+1$ and d , respectively. Hence $\|yf\|_\sigma \leq 2(d+1)\|f\|_\sigma$. Applying this inequality successively gives, for $0 \leq j \leq M_h$, $\|y^j f\|_\sigma^2 \leq [4(d+M_h)^2]^j \|f\|_\sigma^2$. Expansion of $(1+y^2)^{M_h}$ therefore bounds its weighted squared norm by $D_d \|f\|_\sigma^2$. Cauchy–Schwarz gives

$$\|f\|_\sigma^4 \leq \left(\int |f|^2 (1+y^2)^{-M_h} d\sigma \right) \left(\int |f|^2 (1+y^2)^{M_h} d\sigma \right).$$

Combining these inequalities and using the bijective substitution $S = c+iy$ proves, on the complete coefficient space and at every $d \geq 0$,

$$\frac{a_k}{D_d} H_d^\sigma \preceq H_d^{\text{ar}} \preceq A_k H_d^\sigma. \quad (\text{GRA5})$$

Both forms are positive definite by their positive densities, and all moments are finite by their exponential upper bounds. Applying (GRA5) to a nonzero vector proves $L_{k,d} \geq 0$. Its dependence on degree is explicit. Subtracting the logarithms in (GRA3) gives

$$\begin{aligned} \log(A_k/a_k) &= k\mathfrak{s}_h + (p_h+1)\log k + B_h \log(k-2) + \mathfrak{c}_h, \\ \mathfrak{s}_h &= \alpha_* + \log(M_{\alpha_*}/\vartheta_h), \\ \mathfrak{c}_h &= -\log M_{\alpha_*} + 3\log \vartheta_h - 3\alpha_* + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \\ L_{k,d} &= O_h(k + \log(d+1)). \end{aligned} \quad (\text{GRA6})$$

The constant implicit here is independent of d and k .

E.2 Full forms, fixed affine fibres, and arithmetic monic minima

For $p \in \mathcal{P}_r(y)$ retain the two full forms

$$\mathcal{N}_0[p] = \int y^{2q} |p(y)|^2 d\sigma(y), \quad \mathcal{N}_\chi[p] = \int |Q(y)p(y)|^2 d\sigma(y).$$

The companion moving-cutoff theorem gives, with its actual finite constants $C_\chi(k, r)$ and $\eta(k, r)$ and all its displayed guards,

$$e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1+\eta)\mathcal{N}_0[p], \quad 0 \leq \eta < 1. \quad (\text{GRA7})$$

It is the original Q on the entire real line. In particular this theorem does not remove the inner integration interval. Applying (GRA5) at $d = q+r$ to $\chi U_r^{-1}p$ and using (GRA2) proves the exactly typed form of the arithmetic receiver:

$$\frac{a_k}{D_{q+r}} e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \|\chi U_r^{-1}p\|_{\text{ar}}^2 \leq A_k(1+\eta)\mathcal{N}_0[p]. \quad (\text{GRA8})$$

Indeed the middle integrand after substitution is exactly $|Q(y)p(y)|^2$. This specifies the coordinate implicit in the received notation $\|\chi p\|_{\text{ar}}$.

For completeness the same constants act on every unchanged affine coefficient fibre. Let $C : \mathcal{P}_r(y) \rightarrow F$ be a specified complex linear map and $z \in \text{im } C$. Each positive form has a unique minimum on $Cp = z$: after choosing one lift, minimization is a strictly positive quadratic problem on $\ker C$. Apply each inequality of (GRA8) to every vector of this same nonempty fibre and take the infima. This proves the two inequalities for the resulting minimum forms, also at $z = 0$. In original coordinates the observation is CU_r , and U_r is the exact fibre bijection. Restriction follows by applying the form inequality to every vector in the specified subspace. No mixed matrix entry is set equal to zero by either operation.

Let ω_d^{ar} be the degree- d monic arithmetic squared norm, and retain the original Gamma monic squared norm

$$\gamma_d = \sqrt{2\pi} d!(1/2)_d = \sqrt{2\pi} (2d)!/4^d.$$

The mass is its degree-zero value; the Gamma recurrence gives $\gamma_d/\gamma_{d-1} = d(d-1/2)$. The monic bijection U_d and (GRA5) give

$$\frac{a_k}{D_d} \gamma_d \leq \omega_d^{\text{ar}} \leq A_k \gamma_d. \quad (\text{GRA9})$$

Define the original relation minima and ratios by

$$\nu_r^{\text{ar}} = \min_{P(S) \text{ monic, deg } P=r} \|\chi P\|_{\text{ar}}^2, \quad \nu_r^0 = \min_{p(y) \text{ monic, deg } p=r} \mathcal{N}_0[p], \quad T_r = \nu_r^{\text{ar}} / \omega_{q+r}^{\text{ar}}.$$

Taking these identical monic affine minima in (GRA8), then dividing by the opposite endpoints of (GRA9), proves

$$\begin{aligned} \log T_r &\geq \log(\nu_r^0 / \gamma_{q+r}) - L_{k,q+r} - C_\chi + \log(1 - \eta), \\ \log T_r &\leq \log(\nu_r^0 / \gamma_{q+r}) + L_{k,q+r} + \log(1 + \eta). \end{aligned} \quad (\text{GRA10})$$

The denominator is the actual arithmetic monic minimum throughout.

Here are the complete finite scalar endpoints obtained from the companion reference theorem. Write $r = 2n_r + \varepsilon_r$ with $\varepsilon_r \in \{0, 1\}$, and retain its actual moving parameters, support, Robin constant λ_r , and positive constants $c_{-,r}$, $c_{+,r}$. That theorem proves

$$c_{-,r} e^{-n_r \lambda_r} \leq h_{n_r, q} \leq c_{+,r} e^{-n_r \lambda_r}, \quad \nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{n_r, q}.$$

Consequently, with every constant evaluated at this r ,

$$\begin{aligned} Z_{q,r} &= \log \frac{\sqrt{2} q^{2(q+r)+1/2}}{\gamma_{q+r}} - n_r \lambda_r, \\ \underline{E}_r &= Z_{q,r} + \log c_{-,r} - L_{k,q+r} - C_\chi(k, r) + \log(1 - \eta(k, r)), \\ \overline{E}_r &= Z_{q,r} + \log c_{+,r} + L_{k,q+r} + \log(1 + \eta(k, r)), \\ \underline{E}_r &\leq \log T_r \leq \overline{E}_r. \end{aligned} \quad (\text{GRA11})$$

These are usable finite endpoints wherever the analytic theorem's finite guards hold. They are obtained by inequalities, not by replacing an arithmetic norm with an equilibrium norm.

E.3 Original quotient, full-jet support, and adjacent allowance

For $N \geq q - 1$ the literal remainder map $J_N : \mathcal{P}_N(S) \rightarrow E$ is onto and has kernel $\chi \mathcal{P}_{N-q}(S)$; at $N = q - 1$ that kernel is zero. In the coefficient frame $1, S, \dots, S^N$ define its quotient minimum Gram and determinant by

$$G_N = (J_N(H_N^{\text{ar}})^{-1} J_N^*)^{-1}, \quad V_N = \det G_N.$$

To verify the precise ratio, use the frame

$$(\chi, \chi S, \dots, \chi S^{N-q}; 1, S, \dots, S^{q-1}).$$

Reordering it by degree makes a unit-triangular coefficient matrix, because χ is monic; its determinant therefore has modulus one. Positive block elimination gives the first equality below. Gram–Schmidt on the full source and on the relation coefficient space with norm $\|\chi P\|_{\text{ar}}$ gives the next two:

$$\det H_N^{\text{ar}} = \det H_{\text{rel},N} \det G_N, \quad \det H_N^{\text{ar}} = \prod_{d=0}^N \omega_d^{\text{ar}}, \quad \det H_{\text{rel},N} = \prod_{j=0}^{N-q} \nu_j^{\text{ar}}.$$

Empty products are one. Dividing consecutive equalities proves

$$d_{q+r} := \frac{V_{q+r}}{V_{q+r-1}} = T_r^{-1}, \quad r \geq 0. \quad (\text{GRA12})$$

This is a source/relation determinant identity for the original quotient $\mathbb{C}[S]/(\chi)$.

The actual support map remains as follows. Set $E_h = \mathbb{C}[z]/(h)$, $v_h = j_h v_h \in E_h^\times$, and $A_\Sigma = \sum_{i=1}^k M_{z_i}$ on $E_h^{\otimes k}$. Then

$$\eta_{h,k} : E \longrightarrow E_h^{\otimes k}, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad \eta_{h,k} M_S = A_\Sigma \eta_{h,k}. \quad (\text{GRA13})$$

In each local factor the nilpotent has exact order m . In each local tensor block their sum has exact order $1 + k(m-1)$: its power $k(m-1)$ has nonzero coefficient $[k(m-1)!]/[(m-1)!]^k$ on the product of the top nilpotent powers, and its next power vanishes because at least one exponent is m . The real and imaginary signs at the four original centres can be chosen independently, so every pair (a, b) occurs as a sum centre λ_{ab} . Distinct pairs give distinct centres because $\delta, \gamma > 0$. The annihilator of 1 under A_Σ is therefore exactly χ . Multiplication by the full unit preserves this annihilator, proving injectivity of (GRA13); direct multiplication proves its intertwining identity. The original source realization and jets consequently retain their exact commuting formula

$$\mathcal{V}(P) = P\left(\sum_i D_i\right) F_h^{\otimes k}, \quad J^{(k)} \mathcal{V}(P) = \eta_{h,k} J_N P.$$

In particular the relation norm above is measured before its quotient image is zero. No quotient by y^q enters this map.

Here is the complete adjacent calculation in these same norms. Let $p_n(S) = i^n Q_n((S-c)/i)$ be the original monic arithmetic orthogonal polynomial, let $b_n = [p_n]_\chi$, and put

$$R_N = G_N^{-1} = \sum_{n=0}^N b_n b_n^* / \omega_n^{\text{ar}}, \quad \beta_n = \omega_n^{\text{ar}} / \omega_{n-1}^{\text{ar}}.$$

Evenness of w_h , hence of its convolution, gives $Q_n(-y) = (-1)^n Q_n(y)$. Multiplication by S has the original recurrence $S p_n = p_{n+1} + c p_n - \beta_n p_{n-1}$, with the last term absent at zero. Inserting it into R_N cancels all interior adjacent pairs and gives

$$M_S R_N + R_N M_S^* - k R_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N^{\text{ar}}}.$$

Reflection $P(S) \mapsto P(k-S)$ permutes the roots of the even-degree polynomial χ and induces an involution Σ on E . Reflection is an isometry on every original remainder fibre, so the unique minimum implies $\Sigma^* G_N \Sigma = G_N$. Since $\Sigma b_n = (-1)^n b_n$, it follows that $b_N^* G_N b_{N+1} = 0$. For $x = G_N^{1/2} b_N$ and $y = G_N^{1/2} b_{N+1}$ the relative Hermitian control has nonzero eigenvalues $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \|x\|^2 \|y\|^2 / (\omega_N^{\text{ar}})^2.$$

The determinant lemma for adding and removing the last covariance column proves

$$\frac{b_N^* G_N b_N}{\omega_N^{\text{ar}}} = 1 - d_N \quad (N \geq q), \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}^{\text{ar}}} = d_{N+1}^{-1} - 1.$$

At $N = q - 1$ the first q monic remainder columns form a basis and the first fraction is exactly one. Thus no inverse at $q - 2$ is used. Together with (GRA12) these identities prove

$$\epsilon_{q+r-1}^2 = \beta_{q+r}(T_r - 1) \begin{cases} 1, & r = 0, \\ 1 - T_{r-1}^{-1}, & r \geq 1. \end{cases} \quad (\text{GRA14})$$

Only the total pairing vanished. In the original orthogonal kernel–boundary split, $x = x_K \oplus x_B$, $y = y_K \oplus y_B$, one has $x_B^* y_B = -x_K^* y_K$. The mixed exterior term therefore remains

$$\|x_K \otimes y_B - y_K \otimes x_B\|^2 = \|x_K\|^2 \|y_B\|^2 + \|x_B\|^2 \|y_K\|^2 + 2|x_K^* y_K|^2. \quad (\text{GRA15})$$

The classical eigenvalue-sum lineage is Ky Fan [H5], as credited by Overton and Womersley [H4]; the original rank-two exterior case used in GRA16 is calculated explicitly below.

The unchanged exterior benchmark is also explicit. Let E_+ be the sum of the complete primary blocks with $a > k/2$. Since k is odd, $\dim E_+ = q/2$. The nonzero top wedge of a full primary basis has additive M_S eigenvalue $\text{Tr}(M_S|_{E_+})$; all nilpotent parts have trace zero. The relative Hermitian control's additive operator on exterior degree $q/2$ has largest eigenvalue ϵ_N , since it sums distinct members of $\{\epsilon_N, -\epsilon_N, 0, \dots, 0\}$. Its expectation on the stated wedge is $2 \text{Re} \text{Tr}(M_S|_{E_+}) - kq/2$. The wedge stays nonzero under the injective full-jet map (GRA13). Summing the positive odd integers $2a - k$ gives $(k + 1)^2/4$. Therefore

$$0 < \mathcal{L}_{h,k} := 2\delta e_k(k + 1) \frac{(k + 1)^2}{4} = \frac{\delta q(k + 1)}{2} \leq \epsilon_N \quad (N \geq q - 1). \quad (\text{GRA16})$$

Positive covariance gives $0 < d_n \leq 1$. Equation (GRA14) and the strict positive benchmark first force $d_q < 1$ and then $d_n < 1$ at every subsequent degree. Thus all logarithms below are defined for the actual objects.

Finally (GRA9) gives the sharpened finite recurrence enclosure

$$-L_{k,d} \leq \log \frac{\beta_d}{d(d - 1/2)} \leq L_{k,d-1}, \quad d \geq 1. \quad (\text{GRA17})$$

For its lower bound divide the lower bound for ω_d^{ar} by the upper bound for ω_{d-1}^{ar} ; reverse the choices for the upper bound. This is an adjacent norm comparison, not a derivative of a determinant asymptotic.

E.4 Growing ratios and uniformity on every shifted index

The companion moving-parameter reference theorem proves

$$\log(\nu_r^0 / \gamma_{q+r}) = 2q\psi(r/q) + O_h(\log(q + r)), \quad \psi(t) \sim \frac{1}{2t \log t}.$$

The companion full-form theorem proves, throughout $t = r/q \rightarrow \infty$, $\log t = o(q/R_k)$, that its finite guards eventually hold and that $C_\chi = O_h(R_k^2(1 + \log t)^2/q)$ and $\eta = \exp[-q \log 8 + o(q)]$. The constants in the reference estimate are those evaluated at the actual moving support. Insert these proved estimates into (GRA10); for $0 \leq \eta \leq 1/2$, $|\log(1 \pm \eta)| \leq 2\eta$. This proves

$$\log T_r = 2q\psi(t) + O_h \left(k + \log(q + r) + \frac{R_k^2}{q}(1 + \log t)^2 \right). \quad (\text{GRA18})$$

In particular the reference exponent has not been presumed larger than the arithmetic error on the whole of the full-form range.

Now take integers $s = s_k \geq 1$ and suppose

$$t = s/q \longrightarrow \infty, \quad t \log t = o(q/k), \quad H_k = \frac{q}{t \log t}. \quad (\text{GRA19})$$

We prove every needed uniformity directly. The exact formula for q gives $q \geq k^2$, $q \leq C_m k^3$, and $R_k^2/q \leq \delta^2 + \gamma^2$ for $k \geq 1$. Eventually $t \geq e$ and $t \log t \leq q/k$, hence $t \leq q/k \leq C_m k^2$. It follows that

$$\log t = O_m(\log k), \quad \log(q + s + q) = O_m(\log k), \quad (R_k^2/q)(1 + \log t)^2 = O_h((\log k)^2) = o_h(k).$$

Thus the error in (GRA18) is $O_h(k)$, while $H_k/k = q/(kt \log t) \rightarrow \infty$. Also

$$\frac{R_k \log t}{q} = \sqrt{\delta^2 + \gamma^2} \frac{kt \log t}{q} \frac{1}{t} \rightarrow 0,$$

so the narrower range lies inside the proved full-form range.

For every integer $j \in \{-1, 0, \dots, q\}$ put $\tau_j = (s + j)/q$. Uniformly in this entire set,

$$|\tau_j - t| \leq 1, \quad \tau_j/t = 1 + O(1/t), \quad \log \tau_j = \log t + O(1/t), \quad \frac{\tau_j \log \tau_j}{t \log t} = 1 + O(1/t).$$

For example the logarithm estimate follows by the mean value theorem on $[t - 1, t + 1]$ once $t \geq 2$. These bounds prove the two range conditions and the $O_h(k)$ error uniformly for all τ_j . The definition of the scalar limit $2v \log v \psi(v) \rightarrow 1$ implies uniformity on all $v \geq t - 1$: the supremum of its absolute error on that tail tends to zero. Consequently

$$\max_{-1 \leq j \leq q} \left| \frac{\log T_{s+j}}{H_k} - 1 \right| \rightarrow 0. \quad (\text{GRA20})$$

The same argument applied to the finite bounds in (GRA11) shows that their lower and upper endpoints are $H_k(1 + o_h(1))$ uniformly. In particular all the finite lower endpoints are eventually positive; positivity has been proved in this range, not inserted as a new asymptotic premise.

For $r \geq 1$ take logarithms of the exact adjacent identity:

$$2 \log \epsilon_{q+r-1} = \log \beta_{q+r} + \log T_r + \log(1 - T_r^{-1}) + \log(1 - T_{r-1}^{-1}). \quad (\text{GRA21})$$

For $r = s, \dots, s + q$, (GRA17) and the preceding degree bounds give $\log \beta_{q+r} = O_h(k)$. If $M_k = \min_{s-1 \leq r \leq s+q} \log T_r$, then $M_k/H_k \rightarrow 1$ and

$$0 \leq -\log(1 - T_r^{-1}) \leq \frac{e^{-M_k}}{1 - e^{-M_k}} \rightarrow 0$$

uniformly for both adjacent factors. Divide (GRA21) by H_k and use $k/H_k \rightarrow 0$ to obtain

$$\max_{s \leq r \leq s+q} \left| \frac{\log \epsilon_{q+r-1}}{H_k/2} - 1 \right| \rightarrow 0. \quad (\text{GRA22})$$

For a single sequence r this proves $\log \epsilon_{q+r-1} \sim q/(2(r/q) \log(r/q))$ under the same narrower range. Since $\log \mathcal{L}_{h,k} = \log(\delta/2) + \log q + \log(k+1) = O_h(\log k)$, it also proves the original benchmark comparison

$$\log \frac{\epsilon_{q+r-1}}{\delta q(k+1)/2} \sim \frac{q}{2(r/q) \log(r/q)} \rightarrow +\infty. \quad (\text{GRA23})$$

These statements do not extend the allowance asymptotic to degrees for which the arithmetic error in (GRA18) is comparable to its decreasing reference exponent.

E.5 Exact shifted two-window identities and finite enclosures

Put $a = q + s - 1$ and $b = a + q$, and give the rows $N = a + i$, $0 \leq i \leq q$, the weights $w_0 = w_q = 1$, $w_i = 2$ otherwise. Their sum is $2q$. Since $s \geq 1$, $a \geq q$, so every row has both adjacent factors. Define

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}}, & \mathcal{W}_{k,s}^{\text{ar}} &= \log \frac{\omega_b^{\text{ar}} \omega_{b+1}^{\text{ar}}}{\omega_a^{\text{ar}} \omega_{a+1}^{\text{ar}}}, \\ \mathcal{P}_{k,s} &= -\sum_{i=0}^q w_i [\log(1 - d_{a+i}) + \log(1 - d_{a+i+1})], & \mathcal{J}_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}. \end{aligned} \quad (\text{GRA24})$$

For any positive sequence z_n , expansion of the coefficients gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this identity to ω_n^{ar} and V_n in (GRA21), with $r = s + i$ and $\log T_{s+i} = -\log d_{a+i+1}$, proves

$$J_{k,s}^{\text{action}} = \mathcal{B}_{k,s}^{\text{ar}} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \quad \mathcal{B}_{k,s}^{\text{ar}} = \sum_{i=0}^q w_i \log T_{s+i}. \quad (\text{GRA25})$$

The four endpoints are thus exactly $a, a+1, a+q, a+q+1$.

To derive the penalty weights, let $f(x) = -\log(1-x)$ for $0 < x < 1$. The coefficient of $f(d_{a+j})$ is $w_j + w_{j-1}$, taking a missing w as zero. Since $q \geq 2$ for the original packet, the result is

$$\mathcal{P}_{k,s} = \sum_{j=0}^{q+1} c_j f(T_{s-1+j}^{-1}), \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1), \quad \sum_j c_j = 4q. \quad (\text{GRA26})$$

The first coefficient is present because the first row is shifted beyond the exceptional row $q-1$.

For finite bounds suppose all the lower endpoints in (GRA11) on $s-1 \leq r \leq s+q$ are positive, and define $\ell = \min_r \underline{E}_r > 0$ and $u = \max_r \bar{E}_r$. The derivative of f is positive, while e^{-x} decreases, so

$$\sum_{j=0}^{q+1} c_j f(e^{-\bar{E}_{s-1+j}}) \leq \mathcal{P}_{k,s} \leq \sum_{j=0}^{q+1} c_j f(e^{-\underline{E}_{s-1+j}}). \quad (\text{GRA27})$$

The elementary inequalities $x \leq f(x) \leq x/(1-x)$, proved by integrating $1/(1-x)$ between zero and x , give the useful common bounds

$$4qe^{-u} \leq \mathcal{P}_{k,s} \leq \frac{4qe^{-\ell}}{1-e^{-\ell}}. \quad (\text{GRA28})$$

Each individual final term also gives $\mathcal{P}_{k,s} \geq e^{-\bar{E}_{s+q}}$.

Every row has a corresponding explicit finite interval. Put $d = q + r$ and

$$A_r^- = \frac{1}{2} [\log(d(d-1/2)) - L_{k,d} + \underline{E}_r + \log(1 - e^{-\underline{E}_r}) + \log(1 - e^{-\underline{E}_{r-1}})],$$

$$A_r^+ = \frac{1}{2} [\log(d(d-1/2)) + L_{k,d-1} + \bar{E}_r + \log(1 - e^{-\bar{E}_r}) + \log(1 - e^{-\bar{E}_{r-1}})].$$

Each displayed scalar function of E is increasing on $E > 0$. Equations (GRA17) and (GRA21) therefore prove

$$A_r^- \leq \log \epsilon_{q+r-1} \leq A_r^+, \quad \min_{s \leq r \leq s+q} A_r^- \leq \log \min_{a \leq N \leq b} \epsilon_N \leq \min_{s \leq r \leq s+q} A_r^+. \quad (\text{GRA29})$$

The two positive contraction factors are retained in both endpoints.

For the monic part define the literal Gamma window

$$\mathcal{W}_{k,s}^\sigma = \log \frac{\gamma_b \gamma_{b+1}}{\gamma_a \gamma_{a+1}} = \log \frac{(2b)!(2b+2)!}{4^{2q}(2a)!(2a+2)!}. \quad (\text{GRA30})$$

Write $v_n = \log(\omega_n^{\text{ar}}/\gamma_n)$. Equation (GRA9) places v_n in $[\log A_k - L_{k,n}, \log A_k]$. Applying these four intervals with their actual signs, so the four $\log A_k$ terms cancel, proves

$$-L_{k,b} - L_{k,b+1} \leq \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{W}_{k,s}^\sigma \leq L_{k,a} + L_{k,a+1}. \quad (\text{GRA31})$$

Thus the recurrence contribution has four endpoint errors, rather than one independent arithmetic error for every row. In particular, writing P_- and P_+ for the lower and upper quantities in (GRA27), all finite data combine to give

$$\sum_{i=0}^q w_i \underline{E}_{s+i} + \mathcal{W}_{k,s}^\sigma - L_{k,b} - L_{k,b+1} - P_+ \leq J_{k,s}^{\text{action}},$$

$$J_{k,s}^{\text{action}} \leq \sum_{i=0}^q w_i \bar{E}_{s+i} + \mathcal{W}_{k,s}^\sigma + L_{k,a} + L_{k,a+1} - P_-.$$

(GRA32)

E.6 Complete asymptotics and the unchanged mixed receiver

Under (GRA19), (GRA22) directly implies

$$\log \min_{a \leq N \leq a+q} \epsilon_N \sim \frac{H_k}{2} = \frac{q}{2t \log t}. \quad (\text{GRA33})$$

Indeed every row logarithm divided by $H_k/2$ lies between $1 - o(1)$ and $1 + o(1)$, so its minimum does too. This argument does not identify the minimizing row with an endpoint or equate a minimum with a geometric mean.

For the penalty, (GRA20) implies that its minimum and maximum $\log T_r$ are both $H_k(1 + o(1))$. Apply the proof of (GRA28) using these exact minima and maxima to obtain

$$\log(4q) - \max_r \log T_r \leq \log \mathcal{P}_{k,s} \leq \log(4q) - \min_r \log T_r - \log(1 - e^{-\min_r \log T_r}).$$

Here $\log q = O_m(\log k) = o(H_k)$, and the final logarithm tends to zero. Dividing by H_k proves the full logarithmic rate

$$\log \mathcal{P}_{k,s} \sim -H_k = -\frac{q}{t \log t}. \quad (\text{GRA34})$$

All $4q$ coefficients have been retained in the upper estimate.

The separate terms of the action can also be evaluated. From (GRA20) and $\sum_i w_i = 2q$, $\mathcal{B}_{k,s}^{\text{ar}} \sim 2qH_k$. The exact Gamma recurrence gives

$$\mathcal{W}_{k,s}^\sigma = \sum_{i=0}^q w_i \log[(s+q+i)(s+q+i-1/2)].$$

Since $s = qt$, every summand equals $2 \log s + O(1/t)$ uniformly, giving $\mathcal{W}_{k,s}^\sigma = 4q \log(qt) + O(q/t)$. More explicitly, $\sum_i w_i i = q^2$, and $0 \leq \log(1+x) \leq x$ for $x \geq 0$ gives

$$0 \leq \mathcal{W}_{k,s}^\sigma - 4q \log s \leq (6q^2 - q)/s.$$

The four-endpoint estimate (GRA31) now yields

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &\sim 2qH_k, & \mathcal{W}_{k,s}^{\text{ar}} &= 4q \log(qt) + O_h(q/t + k), \\ \mathcal{W}_{k,s}^{\text{ar}} &= o(qH_k), & \mathcal{P}_{k,s} &= \exp[-H_k(1 + o(1))] = o(qH_k), \\ J_{k,s}^{\text{action}} &\sim 2qH_k = \frac{2q^2}{t \log t}. \end{aligned} \quad (\text{GRA35})$$

For the smallness of $\mathcal{W}$, divide its bound by qH_k and use $\log(qt) = O_m(\log k)$, $H_k/k \rightarrow \infty$, and $q \rightarrow \infty$. Equation (GRA25) then proves the last assertion independently of summing the row asymptotics.

It remains to specify exactly how this action reaches the existing mixed calculation. Keep its original surjection $\Lambda : E \twoheadrightarrow B$ and kernel inclusion $I : K \hookrightarrow E$, with the complete divided Taylor unit and original period branches. Explicitly, with the original period map Π and tensor invariant projector P_k , its image inclusion j satisfies $j\Lambda = P_k \Pi^{\otimes k} M_{v_h}^{\otimes k} \iota$, where $\iota[P] = [P(z_1 + \cdots + z_k)]$. For each original source metric G_N^v , put

$$H_{K,N}^v = I^* G_N^v I, \quad Q_{B,N}^v = (\Lambda(G_N^v)^{-1} \Lambda^*)^{-1}, \quad S_N^v = (G_N^v)^{-1} \Lambda^* Q_{B,N}^v.$$

Direct multiplication gives $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$. For a fixed section S_* , the difference $S_N^v - S_*$ has image in K , so the frame change from $[I, S_*]$ to $[I, S_N^v]$ is triangular with identity diagonal. Hence

$$\det H_{K,N}^v \det Q_{B,N}^v = |\det[I, S_*]|^2 \det G_N^v.$$

Taking the return at $a, a+1, b, b+1$ cancels all four frame constants. If $F_K^v, F_B^v, \mathcal{B}^v$ denote these same four-endpoint returns, the result is exactly $F_K^v + F_B^v = \mathcal{B}^v$. Empty determinant factors are one. For the original sources $v = \text{ar}, \sigma, (k)$ set

$$\mathcal{H} = \mathcal{B}^\sigma - \mathcal{B}^{(k)}, \quad \Delta^\sigma = \mathcal{B}^{\text{ar}} - \mathcal{B}^\sigma.$$

The complete shifted allocation and matching budget therefore obey

$$\begin{aligned} \Psi &= F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H} + \Delta^\sigma, \\ C^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s} = \mathcal{B}^{(k)} - F_K^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \\ \Psi + C^{\text{budget}} &= J_{k,s}^{\text{action}}. \end{aligned} \tag{GRA36}$$

This proves the exact map of the new window into the earlier complete action. The common allocation cancels with its matching budget. The new scalar profile consequently assigns no separate leading coefficient to the intrinsic kernel and boundary mixed terms or to their complementary CPQ returns. Those original quantities remain in (GRA15) and (GRA36), with their maps unchanged.

F The original three-equation transfer of every connecting return

Fix the simple quartet, its actual amplitude, period and logarithm branch on the original five-orbit locus. Keep $k = 4l + 1 \geq 9$, $j = k + 4$, $q = (k + 1)^2$, $Q = (j + 1)^2$, $d = Q - q = 8k + 24$, and the full nonreduced conductor

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{s,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v. \quad (\text{PRT1})$$

No conductor factor is removed. Write $B = B_{\text{cof}}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, $\mathbb{J} = \ker \mathbb{B}$, $\mathcal{B} = \text{im } B$ and $W = \mathcal{B} + \mathbb{J}$, as in CPQ2. Define, in this same coefficient space,

$$S_e = \text{im}[B, F_e], \quad K_e = S_e \cap W, \quad T_e = S_e + W \quad (e = 0, 1). \quad (\text{PRT2})$$

These are the CPQ spaces, with S_e abbreviated only to make the two endpoint subscripts legible. The later norm cutoff N is independent of the defining endpoint e . At every $Q - 1 \leq N \leq 2Q$ use exactly

$$D = N - v, \quad \mathcal{D}_N^{(s)} = \text{diag}(D_{j,D}^{(s)}, 4), \quad c'_j = j/2 - 4, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1 \text{ or } j. \quad (\text{PRT3})$$

Thus both actual source pairs $(1, 1)$ and (k, j) are covered in their target order. Their degree- k proper source retains its separate form; none of the identities below substitutes that form for A_{inv} .

F.1 The actual three rows in the original period coefficients

Retain the original polynomials $L_i = (\mathbf{H}(ZW, Z, W, 1)^\top)_i$, $b_{i,a,b} = [Z^a W^b] L_i^4$, the exponential $B_i(z)$ and the translation maps $\mathcal{T}_i, \mathcal{T}_A$ of CPR2. In particular

$$R_i(S) = \frac{\sum_{a,b=0}^4 b_{i,a,b} \chi_j(S + \beta_{4,a,b})}{\chi_k(S)}, \quad b_i = B_i(0), \quad \chi_n(S) = \prod_{a,b=0}^n (S - \beta_{n,a,b}). \quad (\text{PRT4})$$

The quotient is the complete polynomial quotient: root addition makes every numerator vanish at every distinct root of χ_k . Choose the first original index i_* with $b_{i_*} \neq 0$, and put

$$C_i = R_i - (b_i/b_{i_*}) R_{i_*} \quad (i \neq i_*), \quad V = ([S^a] C_i)_{0 \leq a < d, i \neq i_*}. \quad (\text{PRT5})$$

Existence of i_* and independence of these three columns are the complete PDS10–11/CPR17 calculation: the original Bezout identity at zero gives a nonzero b_i ; the four R_i are independent; their leading coefficients are b_i . Subtracting their exact leading multiples therefore gives three independent columns of degree at most $d - 1$. No period coefficient is replaced in this operation.

Let h_b be the original primitive germs used to define the d columns of F_0 in CIT7. They are specified modulo the entire raw analytic source. Define their relation-value matrix by

$$(M_0)_{a,b} = \Lambda_{h_b}(\mathcal{T}_A(\chi_k S^a)), \quad 0 \leq a < d, \quad L = V^\top M_0. \quad (\text{PRT6})$$

Every entry uses the ordinary, complex-linear transpose. The complete annihilator identity CPR9–10 proves that the displayed d independent relation polynomials are the full dual of the raw source quotient. The h_b are a basis of that quotient, so M_0 is invertible. Thus L has rank exactly three. The matrix is computable from the original period arrays by finite polynomial division and pairing; it contains neither a new numerical model nor a selected nonzero period minor.

Choose once a full column basis Z of $\ker L$ and a three-column complement Y so that $[Z, Y]$ is invertible. Then the literal coefficient spaces obey

$$\boxed{S_1 = \mathcal{B} + F_0 \ker L \subset S_0, \quad S_0/S_1 \text{ has the three lifts } F_0 Y.} \quad (\text{PRT7})$$

Here equality is of subspaces of the original $\mathcal{U}$, not only of abstract quotient dimensions. To prove it, use PDS11: truncation of the endpoint-one primitive quotient identifies it with exactly $\ker V^\top$ in endpoint-zero relation-value coordinates. The missing top value is uniquely $-u_*^\top x_0/b_{i_*}$ with the complete u_* from PDS10. For a representative h , set $f = E_A h$. The original target numerator in component i is

$$N_i(f) = \mathbb{V}_j^{-1}(r![z^r](B_i f))_{0 \leq r < Q}, \quad (\mathbb{V}_j)_{r,(a,b)} = \beta_{j,a,b}^r. \quad (\text{PRT8})$$

Those values use only the truncation through degree $Q - 1$. At endpoint one, the additional four equations assert exactly that the same interpolants also have the required next jet; they do not alter a coefficient in PRT8. Division by the complete E_A gives the same target analytic functions, modulo precisely the original lower exponentials. An analytic source change has numerator d and changes the four target numerators by $(L_i^4 d)_i$, hence changes their DNE coordinates by Bx with the full original scalar α_A . This is the coefficient square RRC5–9/CIT8. It proves both inclusions in PRT7, including every analytic-source and lower-gauge change. Finally F_0 induces an injection modulo $\mathcal{B}$ by CIT9; the invertible matrix $[Z, Y]$ therefore proves the asserted dimension and the stated three lifts.

The coefficient change can itself be written explicitly. For the original endpoint-one primitive columns $h_{1,b}$ put $(M_1)_{a,b} = \Lambda_{h_{1,b}}(\mathcal{T}_A(\chi_k S^a))$ for $0 \leq a \leq d$, and let $P_{<d}$ select the first d rows. Then

$$\Gamma = M_0^{-1} P_{<d} M_1, \quad \text{im } \Gamma = \ker L, \quad F_1 = F_0 \Gamma + B H \quad (\text{PRT7a})$$

for the unique matrix H with the displayed equality. Indeed PDS11 proves the first two assertions, including that Γ has rank $d - 3$. The preceding numerator interpolation calculation shows that each difference column lies in $\mathcal{B}$. The original injectivity of B gives its unique coordinate in H . Equivalently $\Gamma = ZT$ for an invertible $(d - 3)$ -square matrix T , so every original fixed frame change in PRT7 is specified by a literal coefficient equation. No conjugation occurs in M_0, M_1, L, Γ . Hermitian adjoints enter only when evaluating the actual metric below.

F.2 An exact sequence of these three directions

PRT7 gives $K_1 \subset K_0$ and $T_1 \subset T_0$. The identity on coefficient representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \xrightarrow{[x] \mapsto [x]} S_0/S_1 \xrightarrow{[x] \mapsto [x]} T_0/T_1 \longrightarrow 0. \quad (\text{PRT9})$$

The first map is injective because $K_0 \cap S_1 = K_1$. If $x \in S_0$ maps to zero on the right, write $x = s + w$ with $s \in S_1$, $w \in W$; then $w = x - s \in K_0$, so its class is the image of K_0/K_1 . Conversely every such class maps to zero. Every element of $T_0 = S_0 + W$ has the class of its S_0 summand modulo $T_1 = S_1 + W$, proving surjectivity. Let

$$a = \dim(T_0/T_1) = \dim(\mathcal{G}_k \cap \mathcal{C}_k) \quad \text{from CPR17–18.} \quad \dim(K_0/K_1) = 3 - a, \quad 0 \leq a \leq 3. \quad (\text{PRT10})$$

The equality with CPR's actual polynomial intersection follows because $\dim(T_e/W) = c_e$ and CPR18 proves $c_0 - c_1 = a$. It can also be calculated directly in the three columns $F_0 Y$: take their images modulo T_1 , retaining the full $\mathbb{B}$, B and L of PRT2/PRT6. Their exact rank is a and their kernel has dimension $3 - a$. This proves a complete original-data rank computation without assuming a generic minor is nonzero. Rank strata, including $a = 0$, are retained.

Give the three terms of PRT9 their original attained metrics

$$A_N = \text{Gram}_{\mathcal{D}_N}(S_0/S_1), \quad B_N = \text{Gram}_{\mathcal{D}_N}(K_0/K_1), \quad C_N = \text{Gram}_{\mathcal{D}_N}(T_0/T_1). \quad (\text{PRT11})$$

Every value frame is fixed in N and in the source-order comparison. For example if X_1 is any fixed independent frame of S_1 , the first one, in the exact three lifts $X_A = F_0 Y$, is

$$A_N = X_A^* \mathcal{D}_N X_A - X_A^* \mathcal{D}_N X_1 (X_1^* \mathcal{D}_N X_1)^{-1} X_1^* \mathcal{D}_N X_A. \quad (\text{PRT12})$$

Select the first independent images of F_0Y modulo T_1 to obtain the a lifts X_C . A full basis of $\ker(F_0Y \bmod T_1)$, followed by subtracting an S_1 representative as in PRT9, gives $3 - a$ lifts X_B in K_0 . Applying PRT12 with (X_1, X_A) replaced by the frames of (K_1, X_B) and (T_1, X_C) gives B_N and C_N . All three are positive definite on their stated quotient values; the determinant of an empty quotient is one.

For completeness every fixed frame factor can be retained before taking a return. For a fixed inclusion $K \subset Y_0$, choose frames P_K, P_{Y_0} and quotient lifts X . Put

$$[P_K, X] = P_{Y_0} R_{Y_0/K}, \quad d_{Y_0/K} = |\det R_{Y_0/K}|^2, \quad g_{Y_0}(N) = \det(P_{Y_0}^* \mathcal{D}_N P_{Y_0}). \quad (\text{PRT13})$$

This is the standard positive-block Schur calculation [H1, §2.1, (2.3)–(2.4)]; the determinant factorization follows from the same block elimination [H2, p. 222, (6a)–(6b)]. Here the block frame is exactly $[P_K, X]$, with its retained factor $d_{Y_0/K}$; neither reference supplies the programme’s connecting sequence or its endpoint comparison.

The full block Schur determinant identity is

$$\det \text{Gram}_{\mathcal{D}_N}(Y_0/K) = d_{Y_0/K} g_{Y_0}(N) / g_K(N). \quad (\text{PRT14})$$

It follows by expressing the restricted form in $[P_K, X]$, taking its determinant, and eliminating its upper-left block. This defines and evaluates every factor from the stated original frames. It does not identify a varying minimizing representative with a fixed lift. CPQ’s frames for each quotient are used in PRT13 when that quotient appears below.

F.3 Evaluation of the difference of the actual CPQ returns

Use the unchanged target return $\mathcal{R}f = -f(Q - 1) - f(Q) + f(2Q - 1) + f(2Q)$ and write

$$U_A = \mathcal{R} \log \det A_N, \quad U_B = \mathcal{R} \log \det B_N, \quad U_C = \mathcal{R} \log \det C_N. \quad (\text{PRT15})$$

These symbols are returns of precisely the three evaluated quotient matrices PRT11–12. They are not freely assignable parameters. The exact CPQ differences are

$$\begin{aligned} V_{\text{conn},0} - V_{\text{conn},1} &= U_A, & V_{h,0} - V_{h,1} &= U_B, \\ V_{s,0} - V_{s,1} &= U_A - U_B, & V_{t,0} - V_{t,1} &= U_C, \\ V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B. \end{aligned} \quad (\text{PRT16})$$

Here V_b, V_I, V_A are independent of the defining primitive endpoint e : their spaces, coefficient maps and frames are the same CPQ3/9 objects. To prove every line, apply PRT14 to $S_e/\mathcal{B}$, $K_e/\mathcal{B}$, S_e/K_e , T_e/W and $\mathcal{U}/T_e$, respectively. For example the logarithmic determinant difference of the latter two endpoint quotients is $\log g_{T_0} - \log g_{T_1}$ plus the explicitly defined fixed frame-factor difference; PRT14 for T_0/T_1 identifies its nonconstant term with $\log \det C_N$. Each frame factor is independent of N , and $\mathcal{R}1 = -1 - 1 + 1 + 1 = 0$, so it cancels only after this full return. The other four differences follow by the same stated determinant identities. Finally substitute them into CPQ10’s $\mathcal{L}_e = V_I + V_{t,e} - V_{s,e} - V_A$.

This also gives the exact propagation, without deleting a complement,

$$\begin{aligned} Z_e &= V_b + V_{\text{rem},e} + V_{s,e} + V_A, & Z_1 - Z_0 &= U_C - U_A + U_B, \\ \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} &= U_C - U_A + U_B, & 4S_{K,j} &= \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1. \end{aligned} \quad (\text{PRT17})$$

In particular a change of primitive endpoint changes the common loss and complementary sum by exactly opposite amounts in their allocation identity. The original total and the complete-action allocation cancellation are preserved.

F.4 Directed finite bounds with the original masses

Use the proved original reconstruction/source constant of FRE1, evaluated at degree j and order s :

$$A_{j,s}^{\text{end}} = \max\{1, Q\mathcal{B}_{2Q-v,s,j,A}^2\}(M_{j,s}^{\text{rec}})^2. \quad (\text{PRT18})$$

The two factors retain DNE13's full mass $(2\pi)^{s/2}$, the full conductor scalar and division coefficients, and LJR16's original fixed-disk reconstruction, phases and Vandermonde. FRE2-4 proves on this exact metric

$$\mathcal{D}_{Q-1}^{(s)} \preceq \mathcal{D}_N^{(s)} \preceq A_{j,s}^{\text{end}} \mathcal{D}_{Q-1}^{(s)}, \quad \log A_{j,s}^{\text{end}} \leq 2Q \log j + O_{\text{actual}}(Q). \quad (\text{PRT19})$$

Substitution in each full affine fibre defining PRT11 and subsequent minimization preserve both inequalities. Original row extension also proves that each of A_N, B_N, C_N increases with N . If G_N is one of these forms of rank r , set $f(N) = \log \det G_N - \log \det G_{Q-1}$. The generalized eigenvalues in PRT19 show $0 \leq f(N) \leq r \log A_{j,s}^{\text{end}}$ and f is increasing. Hence $\mathcal{R} \log \det G = f(2Q-1) + f(2Q) - f(Q)$ is nonnegative and at most $2r \log A_{j,s}^{\text{end}}$. This proves the actual directed intervals

$$\begin{aligned} 0 \leq U_A \leq 6 \log A_{j,s}^{\text{end}}, \quad 0 \leq U_B \leq 2(3-a) \log A_{j,s}^{\text{end}}, \quad 0 \leq U_C \leq 2a \log A_{j,s}^{\text{end}}, \\ \boxed{0 \leq V_{\text{rem},1} - V_{\text{rem},0} \leq 2a \log A_{j,s}^{\text{end}}.} \\ -2(3-a) \log A_{j,s}^{\text{end}} \leq V_{s,0} - V_{s,1} \leq 6 \log A_{j,s}^{\text{end}}, \\ |\mathcal{L}_0 - \mathcal{L}_1| \leq 6 \log A_{j,s}^{\text{end}}, \quad |Z_1 - Z_0| \leq 6 \log A_{j,s}^{\text{end}}. \end{aligned} \quad (\text{PRT20})$$

The last bound uses $\dim B + \dim C = 3$, keeping the exact sequence rather than separately charging two unrelated rank-three errors. These are finite bounds on actual CPQ expressions. For each fixed original period they are $O_{\text{actual}}(Q \log j) = o(kq)$, for both source pairs. At the actual rank stratum $a = 0$, the remaining-target forms are the same quotient in fixed value frames and their returns are exactly equal. At $a = 3$, the kernel spaces coincide and $U_B = 0$ exactly. All intermediate strata remain governed by their actual original three-column matrix. No claim is made that either endpoint's common leading return has now been evaluated; PRT16 evaluates their difference and reduces its calculation to the stated three original directions.

The proof requires the five-orbit coefficient construction, its original distinct quartet roots and $\mu_v \neq 0$. It does not use the additional pole-coordinate gate minors of RPI/RPK. Consequently a period excluded by a pole-coordinate pivot remains covered when it belongs to the original five-orbit locus: PRT6/8/12 construct all maps without that pivot. Rank changes of the three-row image are handled by PRT10. The value v is retained throughout; neither $v \leq 32$ nor a generic value of v has been substituted for an actual coefficient.

G The first original row: exact lower cancellation and its finite bound

Fix the actual simple quartet, original conductor coefficients, period, and branch. Write

$$\begin{aligned}
j &\equiv 1 \pmod{4}, \quad j \geq 13, \quad q = (j+1)^2, \quad q' = (j-7)^2, \quad c' = j/2 - 4, \\
E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \mu_d = E_A^{(d)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \\
\beta_{r,a,b} &= r/2 + (2a-r)\delta + i(2b-r)\gamma, \quad \sigma = (\delta^2 + \gamma^2)^{1/2}, \\
\chi_j(S) &= \prod_{a,b=0}^j (S - \beta_{j,a,b}), \quad \chi'(S') = \chi_{j-8}(S') = \prod_{a,b=0}^{j-8} (S' - \beta_{j-8,a,b}), \\
n &= q - v, \quad m = n - q' = 16j - 48 - v \geq 0, \quad s \in \{1, j\}, \quad \ell = (s-1)/4.
\end{aligned} \tag{FRG1}$$

The prime on χ' names the complete lower root polynomial; it is not differentiation. The original norm is

$$\begin{aligned}
\|P\|_{s,c'}^2 &= \int_{\mathbb{R}} |P(c' + iy)|^2 dm_s(y), \quad dm_s = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{r+1,s} = yp_{r,s} - r(r-1+s/2)p_{r-1,s}, \\
h_{r,s} &= M_s r!(s/2)_r, \quad \phi_r(S') = p_{r,s}((S' - c')/i)/\sqrt{h_{r,s}}.
\end{aligned} \tag{FRG2}$$

These are exactly NSO2,7's native orthonormal polynomials, with the full generating-function and orthogonality proof in SGT2–5; its source mass and recurrence coefficients are unchanged here. The classical family underlying FRG2 is the Meixner–Pollaczek family: $p_{r,s}(y) = r!P_r^{(s/4)}(y/2; \pi/2)$. See Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt, [H21, eqs. (18.19.6)–(18.19.9), (18.22.8), (18.23.7)]; the generating function is also directly available at DLMF (18.23.7). The exact dictionary and qualified source ancestry are retained in HPS1–4: $dy = 2dx$, $M_s = (2\pi)^{s/2}$ and $h_{r,s} = M_s r!(s/2)_r$ are unchanged. This credits the classical polynomial input, not the programme-specific conductor and first-row bounds that follow.

Thus the leading coefficient of ϕ_r is $i^{-r}/\sqrt{h_{r,s}}$. All inner products are conjugate-linear in the first variable, whereas every functional below is complex-linear.

G.1 The actual annihilator and the full first-row leverage

Let $F_b = N_b/E_A$ be the original NSO12 numerator germs, with $N_b(z) = \sum_{a,b'} (Z_{\text{exp},j})_{(a,b'),b} e^{\beta_{j,a,b'} z}$. Keep their original fixed NSO12 frame. Let V_0, T_0 be NSO13's lower and numerator matrices on rows $0, \dots, n-1$, and let v_n, t_n be their next rows. There are q' lower columns and m numerator columns; therefore

$$F_0 = [V_0, T_0] \in \mathbb{C}^{n \times n}, \quad w_n = [v_n, t_n]. \tag{FRG3}$$

The NSO12–13 Vandermonde proof applies at cutoff $n-1 = q-v-1$: if such a combined functional has its first n derivatives zero, multiplication of its germ by E_A makes an upper-grid numerator have its first q derivatives zero. The upper Vandermonde kills this numerator, its class in the fixed conductor quotient vanishes, and the lower Vandermonde kills the remaining lower coefficients. Since the triangular change from derivatives to the ϕ_r values has nonzero diagonal, F_0 is invertible.

Define the original translation and polynomial

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad g = \mathcal{T}_A \chi_j, \quad \lambda = \mu_v \binom{q}{v} \neq 0. \tag{FRG4}$$

Expanding every shift shows that all coefficients above degree $q - v$ vanish and the coefficient at degree $q - v$ is λ : for a monomial S^d , its coefficient at S'^{d-r} is $\binom{d}{r}\mu_r$, and $\mu_0 = \dots = \mu_{v-1} = 0$. Hence $\deg g = n$. For any polynomial P the finite Leibniz identity gives the exact pairing

$$\Lambda_{F_b}(\mathcal{T}_A P) = \Lambda_{E_A F_b}(P) = \Lambda_{N_b}(P). \quad (\text{FRG5})$$

The right side for $P = \chi_j$ is a linear combination of evaluations of χ_j at all its upper roots, so is zero. Also $\beta_{j-8,u,w} + \beta_{8,a,b} = \beta_{j,u+a,w+b}$ lies in the original upper grid. Every summand in $g(\beta_{j-8,u,w})$ is zero. Distinctness of all the lower roots proves $\chi' \mid g$. Consequently all columns in $[V, T]$ annihilate g .

Expand $g = \sum_{r=0}^n c_r \phi_r$. The exact top coefficient and row relation are

$$c_n = i^n \lambda \sqrt{h_{n,s}}, \quad c_{<n}^\top F_0 + c_n w_n = 0. \quad (\text{FRG6})$$

Here the transpose is ordinary because of complex linearity. Since F_0 is square and invertible, $F_0(F_0^* F_0)^{-1} F_0^* = I_n$. Thus

$$\begin{aligned} \zeta_{\text{tot}} &= w_n (F_0^* F_0)^{-1} w_n^* = \frac{\sum_{r < n} |c_r|^2}{|c_n|^2}, \\ 1 + \zeta_{\text{tot}} &= \frac{\|\mathcal{T}_A \chi_j\|_{s,c'}^2}{|\lambda|^2 h_{n,s}}. \end{aligned} \quad (\text{FRG7})$$

No factorial, phase, conductor coefficient, or source mass was removed.

G.2 The exact lower ideal norm

Let

$$\nu_{m,s} = \min_{P \text{ monic, } \deg P = m} \|\chi' P\|_{s,c'}^2. \quad (\text{FRG8})$$

This is the attained positive norm on the original monic affine fibre; it exists because its finite positive Gram has a unique quadratic minimum. A monic degree- n polynomial $u = \sum_{r=0}^n d_r \phi_r$ has $d_n = i^n \sqrt{h_{n,s}}$. Divisibility by χ' is exactly its vanishing at all lower roots, hence exactly

$$V_0^\top d_{<n} = -d_n v_n^\top. \quad (\text{FRG9})$$

The matrix V_0 has full column rank. In these coefficient coordinates the attained minimum is

$$d_{<n,\min} = -d_n \overline{V_0} (V_0^\top \overline{V_0})^{-1} v_n^\top. \quad (\text{FRG10})$$

Multiplication proves the constraint. Its ordinary Hermitian pairing with every vector in $\ker V_0^\top$ vanishes, proving the minimum and its Pythagorean decomposition. Its squared norm is $|d_n|^2 \overline{v_n} (V_0^\top \overline{V_0})^{-1} v_n^\top = |d_n|^2 \zeta_{\text{low}}$, where $\zeta_{\text{low}} = v_n (V_0^* V_0)^{-1} v_n^*$ is real and nonnegative. Thus

$$\boxed{\nu_{m,s} = h_{n,s} (1 + \zeta_{\text{low}}).} \quad (\text{FRG11})$$

This is a typed monic-fibre map, not a change of the original measure. In y coordinates it retains the phase explicitly: write $\chi'(c' + iy) = i^{q'} Q'(y)$, with monic Q' , and $P(c' + iy) = i^m \tilde{P}(y)$, with monic $\tilde{P}$. The forward and inverse maps on monic fibres are $\tilde{P}(y) = i^{-m} P(c' + iy)$ and $P(S') = i^m \tilde{P}((S' - c')/i)$. Their total phase is i^n and has modulus one; their squared norm is precisely FRG8.

Put $A = V_0^* V_0$, $B = V_0^* T_0$, $D_0 = T_0^* T_0 - B^* A^{-1} B$, and $\rho = t_n - v_n A^{-1} B$. Block factorization of $F_0^* F_0$ gives

$$\zeta_{\text{tot}} = \zeta_{\text{low}} + \rho D_0^{-1} \rho^*, \quad r_n = \frac{\rho D_0^{-1} \rho^*}{1 + \zeta_{\text{low}}}. \quad (\text{FRG12})$$

For instance factor the Gram as $\begin{bmatrix} I & 0 \\ B^* A^{-1} & I \end{bmatrix} \text{diag}(A, D_0) \begin{bmatrix} I & A^{-1} B \\ 0 & I \end{bmatrix}$ and multiply its inverse against $[v_n, t_n]$ and its adjoint. Equations FRG7 and FRG11 now prove

$$1 + r_n = \frac{1 + \zeta_{\text{tot}}}{1 + \zeta_{\text{low}}} = \frac{\|\mathcal{T}_A \chi_j\|_{s, c'}^2}{|\mu_v(q)|^2 \nu_{m, s}}. \quad (\text{FRG13})$$

The scalar λ maps g to the unique monic polynomial g/λ in its original one-dimensional coefficient fibre; it is kept in FRG13. In particular, if $P_{m, s}^{\min}$ is the unique monic minimizer in FRG8, then

$$1 + r_n = 1 + \frac{\|g/\lambda - \chi' P_{m, s}^{\min}\|_{s, c'}^2}{\nu_{m, s}} \geq 1. \quad (\text{FRG14})$$

Indeed the difference belongs to $\chi' \mathbb{C}[S']_{< m}$, to which the minimizer is orthogonal. This also characterizes the exact zero-row case: $r_n = 0$ precisely when $g/\lambda = \chi' P_{m, s}^{\min}$.

G.3 An explicit finite bound from the original root products

Define

$$\begin{aligned} \theta &= \pi/2, \quad h = 1/q, \quad \kappa = \theta - h > 0, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ Q_- &= q' + \ell, \quad Q_+ = q + \ell, \quad R' = (j - 8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j + 8)\sigma, 2\ell\}, \quad A_{\text{abs}} = \sum_{a, b=0}^8 |a_{ab}|, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ \epsilon_- &= \left(\theta - \frac{2Q_-}{M} \right) + \frac{Q_-}{M^2} + h(M + 1) + \frac{4q'R'}{M}, \quad C_m = (2m + 1) \binom{2m}{m}^2. \end{aligned} \quad (\text{FRG15})$$

The two inequalities $M \geq 2R'$ and $M \geq M_0$ hold by definition, so the finite bounds below apply at every $j \geq 13$. For each fixed quartet, $M = M_0$ eventually, because M_0 grows quadratically and R' linearly in j . On that eventual range, $F_M = f_\theta(Q_-)$ and $\epsilon_- = \theta^2/(4Q_-) + h(M + 1) + 4q'R'/M$.

For completeness, the Gamma input used in the bound is proved as follows. Gamma recurrence gives

$$dm_s(y) = \beta_s \prod_{r=0}^{\ell-1} (y^2 + (2r + 1/2)^2) d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{FRG16})$$

The factor $4^{-\ell}$ from the recurrence is cancelled exactly by the $2^{2\ell}$ in the original density. Rotate Euler's integral for $\Gamma(a + iy/2)$ through $\text{sgn}(y)(\pi/2 - h)$: absolute integration along that ray proves $|\Gamma(a + iy/2)| \leq \Gamma(a)(\sin h)^{-a} e^{-(\pi/2 - h)|y|/2}$. The small closing arc tends to zero as its radius to the power a , and the large closing arc tends to zero since its cosine is at least $\sin h$. For the lower bound use the exact reflection product $|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$, the same upper bound on the second Gamma factor, and $\cosh(\pi y) \leq e^{\pi|y|}$. Therefore, also at $y = 0$,

$$L_h e^{-(\theta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-\kappa|y|}. \quad (\text{FRG17})$$

For $y \in [M, M + 1]$, every original lower root in the y coordinate has modulus at most R' . Since $R'/y \leq 1/2$, integration of $(1 - t)^{-1} \leq 2$ gives $|\chi'(c' + iy)|^2 \geq y^{2q'} e^{-4q'R'/M}$. Every Gamma

recurrence factor is at least y^2 . Writing $y = M + t$, $0 \leq t \leq 1$, the inequality $\log(1+x) \geq x - x^2/2$ gives

$$2Q_- \log y - \theta y \geq F_M - \left(\theta - \frac{2Q_-}{M} \right) - \frac{Q_-}{M^2}. \quad (\text{FRG18})$$

Thus the complete density $|\chi'(c' + iy)|^2 dm_s/dy$ on this one original-coordinate interval is at least $\beta_s L_h \exp(F_M - \epsilon_-)$.

Here is the exact monic minimum on an interval of length one. Set $L_m(x) = \frac{1}{m!} \frac{d^m}{dx^m} [x^m (x-1)^m]$. Its leading coefficient is $\binom{2m}{m}$. Integrating by parts m times, all boundary terms vanish, proving its orthogonality to degrees less than m . The same integrations give

$$\int_0^1 L_m(x)^2 dx = \binom{2m}{m} \int_0^1 x^m (1-x)^m dx = \frac{1}{2m+1}. \quad (\text{FRG19})$$

The beta integral here is $m!^2/(2m+1)!$, obtained by repeated integration by parts. Subtracting $L_m/\binom{2m}{m}$ from any complex monic polynomial makes an orthogonal lower-degree difference. The least squared norm is exactly C_m^{-1} . Translation of the interval preserves its leading coefficient. The polynomial $P(c' + iy)$ for monic P has leading coefficient i^m ; multiply the displayed minimizing polynomial by that retained phase. Applying this minimum to FRG18 yields

$$\boxed{\nu_{m,s} \geq \frac{\beta_s L_h}{C_m} \exp(F_M - \epsilon_-).} \quad (\text{FRG20})$$

On the entire real line, for every original shift and upper root,

$$|c' + iy + \beta_{8,a,b} - \beta_{j,u,w}| \leq |y| + (j+8)\sigma \leq |y| + B_+. \quad (\text{FRG21})$$

Indeed its constant part is $[(2a-8) - (2u-j)]\delta + i[(2b-8) - (2w-j)]\gamma$. Every original summand in g therefore has modulus at most $(|y| + B_+)^q$, and the complete sum has modulus at most $A_{\text{abs}}(|y| + B_+)^q$. Also $2r + 1/2 \leq 2\ell \leq B_+$ for every recurrence factor when $\ell > 0$, and the empty product for $\ell = 0$ is one. FRG16–17, the substitution $t = y + B_+$ on $y \geq 0$, and enlargement of its integral to $[0, \infty)$ prove

$$\boxed{\|g\|_{s,c'}^2 \leq 2\beta_s U_h A_{\text{abs}}^2 e^{\kappa B_+} \frac{(2Q_+)!}{\kappa^{2Q_++1}}.} \quad (\text{FRG22})$$

The cancellation of coefficients above degree n is still exact in g ; the bound estimated the complete original finite sum before those cancellations and changed no polynomial.

Combining FRG13, FRG20 and FRG22 gives the explicit finite interval

$$\begin{aligned} 0 \leq \log(1+r_n) &\leq E_{j,s}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\mu_v \binom{q}{v}|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) - (2Q_+ + 1) \log \kappa \\ &\quad - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FRG23})$$

The mass factor β_s cancels between these two displayed bounds on the same original norm. In particular FRG23 has not assumed unit mass. Positivity of the exact ratio proves that this computed upper bound is nonnegative throughout its stated domain.

G.4 Evaluation of the first-row scale and the cancelled leading term

All the following limits fix the actual quartet and conductor, including μ_v , and let $j \equiv 1 \pmod{4}$ tend to infinity. They hold for both $s = 1$ and $s = j$. First, $Q_{\pm} \asymp j^2$, $Q_+ - Q_- = 16j - 48$, $B_+ = O(j)$, $R' = O(j)$, and $m = O(j)$. Eventually $M = M_0$, so $F_M = f_{\theta}(Q_-)$. Directly from FRG15, $\epsilon_- = O(j)$; also $\log C_m \leq \log(2m+1) + 4m \log 2 = O(j + \log j)$. Since $2h/\pi \leq \sin h \leq h$, $\log(U_h/L_h) = O(\log q)$. The fixed- v product for $\binom{q}{v}$ makes $|\log |\lambda|| = O(\log q)$.

For every integer $N \geq 1$, integration bounds on the increasing function $\log x$ give

$$N \log N - N + 1 \leq \log(N!) \leq N \log N - N + 1 + \log N. \quad (\text{FRG24})$$

Consequently

$$\begin{aligned} \log((2Q_+)!) - (2Q_+ + 1) \log \kappa &\leq f_\theta(Q_+) + 1 + \log(2Q_+) - \log \theta \\ &\quad + (2Q_+ + 1) \log(\theta/\kappa). \end{aligned} \quad (\text{FRG25})$$

Here $\log(\theta/\kappa) \leq h/(\theta - h)$, so the last term is $O(1)$. Since $f'_\theta(x) = 2 \log(2x/\theta)$, its integral between Q_- and Q_+ is $O(j \log j)$. Every term of the fully finite FRG23 is now controlled, proving

$$\boxed{0 \leq \log(1 + r_n) = O_{\text{actual}}(j \log j).} \quad (\text{FRG26})$$

The leading terms of the two leverages themselves can also be evaluated without taking a difference of uncontrolled remainders. For the upper bound on $\nu_{m,s}$ choose the actual monic trial $P(S') = (S' - c')^m$. Put $B_- = \max\{1, R', 2\ell\}$ and $Q_0 = n + \ell$. The same whole-line integral used in FRG22 now gives

$$\nu_{m,s} \leq 2\beta_s U_h e^{\kappa B_-} \frac{(2Q_0)!}{\kappa^{2Q_0+1}}. \quad (\text{FRG27})$$

The factor $|y|^{2m}$ is bounded by $(|y| + B_-)^{2m}$ and $q' + m = n$, so the exponent is exactly $2Q_0$. Both $Q_- - q$ and $Q_0 - q$ are $O(j)$. Equations FRG20, FRG24–25, FRG27 and the preceding explicit estimates imply

$$\log \nu_{m,s} = \log \beta_s + f_\theta(q) + O_{\text{actual}}(j \log j). \quad (\text{FRG28})$$

To retain the full original factorial normalization, write $b = s/2 = 2\ell + 1/2$ and $u = n + 2\ell$. Then

$$h_{n,s} = \sqrt{2}\beta_s n! \Gamma(u + 1/2), \quad \Gamma(u + 1/2) = \frac{\sqrt{\pi}(2u)!}{4^u u!}. \quad (\text{FRG29})$$

FRG24 in these three factorials shows $\log \Gamma(u + 1/2) = u \log u - u + O(\log u)$. Both $n - q$ and $u - q$ are $O(j)$; integration of $\log x$ between these arguments and q therefore gives

$$\log h_{n,s} = \log \beta_s + 2(q \log q - q) + O_{\text{actual}}(j \log j). \quad (\text{FRG30})$$

Subtracting the two controlled expressions in FRG28 and FRG30, and then using the exact identity FRG13 together with FRG26, proves

$$\boxed{\begin{aligned} \log(1 + \zeta_{\text{low}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j), \\ \log(1 + \zeta_{\text{tot}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j). \end{aligned}} \quad (\text{FRG31})$$

Thus the same positive order- q term belongs to both original leverages and cancels in their exact corrected ratio. This conclusion uses no limiting theorem for a growing-rank determinant and no derivative of an asymptotic error.

Finally the original SFB13–15 four-copy update has budget $\beta_n = 4 \log(1 + r_n)$. Its common signed increment satisfies $|\delta L_n| \leq \beta_n$, either by SFB8 integrated along this positive update, or by the same original projection determinants. For this first original row, FRG26 therefore yields

$$0 \leq \beta_n = O_{\text{actual}}(j \log j), \quad \frac{|\delta L_n|}{jq} \longrightarrow 0. \quad (\text{FRG32})$$

Each original attained quotient in CPQ14 also has its first-row logarithmic determinant increment between 0 and β_n : along the same positive path, its derivative is the trace of the positive relative derivative against the orthogonal projector on its minimum-lift quotient space, which lies between zero and the full trace. Integrating proves the claim with the original quotient kept. Hence each of the five complementary first-row increments is $O_{\text{actual}}(j \log j)$ as well. This evaluates one actual row; FRG32 makes no assertion about the remaining q consecutive rows.

H The first original mixed row in the full Gamma polynomial ideal

Keep the original hypothetical simple quartet, fixed five-orbit period, logarithm branch, complete nonreduced conductor, and both source pairs of NSO and CPQ. This calculation concerns their actual first appended target row, rather than a freely chosen matrix update. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q &= (j + 1)^2, \\ q' &= (j - 7)^2, & c' &= j/2 - 4, & n &= q - v, & m &= n - q' = 16j - 48 - v. \end{aligned} \quad (\text{FMG1})$$

Here $v = \text{ord}_0 E_A$ is the actual conductor order. The retained BTK estimate is $v \leq 32$ on its stated original domain; the earlier $v \leq 80$ already makes $m > 0$ in this calculation. Let $s = 1$ or $s = j$, according to the prescribed source pair. Throughout,

$$\begin{aligned} \beta_{a,u,w} &= a\left(\frac{1}{2} - \delta - i\gamma\right) + 2u\delta + 2wi\gamma, \\ \chi_a(S) &= \prod_{u,w=0}^a (S - \beta_{a,u,w}), \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \\ \mu_r &= E_A^{(r)}(0), \\ (\mathcal{T}_A p)(S') &= \sum_{u,w=0}^8 a_{uw} p(S' + \beta_{8,u,w}), \\ \ell &= \mu_v \binom{q}{v} \neq 0. \end{aligned} \quad (\text{FMG2})$$

The symbol ℓ in FMG2 is a leading coefficient, separate from the integer l in $k = 4l + 1$. No root, period coefficient or conductor factor has been discarded. Define the actual polynomials

$$\chi_- := \chi_{j-8}, \quad g := \mathcal{T}_A \chi_j. \quad (\text{FMG3})$$

For every lower root $\beta_{j-8,a,b}$, each summand defining $g(\beta_{j-8,a,b})$ evaluates χ_j at the upper root $\beta_{j,a+u,b+w}$. Thus χ_- divides g exactly. The first nonzero moment in FMG2 gives $\deg g = n$ and $\text{lc}_{S'} g = \ell$ by the binomial formula. This is a divisibility assertion for the full translated polynomial, including every coefficient.

The original measure and orthonormal polynomials are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, & p_{1,s} &= y, & p_{a+1,s} &= yp_{a,s} - a(a-1+s/2)p_{a-1,s}, \\ h_{a,s} &= M_s a! (s/2)_a, & \phi_a(S') &= p_{a,s}((S' - c')/i) / \sqrt{h_{a,s}}, \\ \langle f_1, f_2 \rangle_s &= \int_{\mathbb{R}} \overline{f_1(c' + iy)} f_2(c' + iy) dm_s(y). \end{aligned} \quad (\text{FMG4})$$

This inner product is conjugate-linear in its first argument. In particular the leading coefficient of ϕ_a in S' is $i^{-a} / \sqrt{h_{a,s}}$.

H.1 The exact original row and its two coupled leverages

Use exactly the NSO12 quotient columns $F_b = N_b/E_A$, $1 \leq b \leq m$. Their numerators lie on the original degree- j grid and vanish to order v at zero. The original lower evaluation and numerator matrices at the preceding cutoff are

$$V_0[a, (u, w)] = \phi_a(\beta_{j-8,u,w}), \quad T_0[a, b] = \Lambda_{F_b}(\phi_a), \quad 0 \leq a < n, \quad (\text{FMG5})$$

where $\Lambda_F(S'^a) = F^{(a)}(0)$ is complex-linear. There are $q' + m = n$ columns in $[V_0, T_0]$. To prove their independence, a zero combination of their first n native values has a generating germ F with $F^{(a)}(0) = 0$ for $0 \leq a < n$, since the polynomial change of basis in FMG4 is invertible. Its numerator $E_A F$ is a degree- j grid exponential polynomial and vanishes through order $n + v - 1 = q - 1$. The original upper Vandermonde is invertible, so that numerator is zero. The NSO12 quotient frame then kills the numerator coefficient, and the lower Vandermonde kills the remaining lower coefficient. This proves that $[V_0, T_0]$ is square and invertible, with its complete original frame.

Write the actual next rows as v_n, t_n . Define

$$\begin{aligned} A_L &= V_0^* V_0, & P_V &= V_0 A_L^{-1} V_0^*, & D_0 &= T_0^* (I - P_V) T_0, \\ \zeta &= v_n A_L^{-1} v_n^*, & \rho &= t_n - v_n A_L^{-1} V_0^* T_0, & r &= \frac{\rho D_0^{-1} \rho^*}{1 + \zeta}. \end{aligned} \quad (\text{FMG6})$$

These are precisely the first SFB12 and NTE1 quantities. Inverses exist by the independence just proved. Set $C_0 = [V_0, T_0]$ and $w_n = [v_n, t_n]$. The total leverage is $\zeta_{\text{all}} = w_n (C_0^* C_0)^{-1} w_n^*$.

Expand $g = \sum_{a=0}^n c_a \phi_a$ with column $c_{<} = (c_0, \dots, c_{n-1})^\top$. The leading coefficient is $c_n = \ell i^n \sqrt{h_{n,s}}$. Every column functional in FMG5 annihilates g . For the numerator columns, the full Leibniz identity gives

$$\Lambda_{F_b}(\mathcal{T}_A \chi_j) = \Lambda_{E_A F_b}(\chi_j) = \sum_{u,w} (N_b)_{uw} \chi_j(\beta_{j,u,w}) = 0.$$

For the lower columns, FMG3 gives the same zero. These are complex-linear equations, so their coefficient row is an ordinary transpose. Therefore

$$w_n = -c_n^{-1} c_{<}^\top C_0, \quad 1 + \zeta_{\text{all}} = \frac{\|g\|_s^2}{|\ell|^2 h_{n,s}}, \quad 1 + r = \frac{1 + \zeta_{\text{all}}}{1 + \zeta}. \quad (\text{FMG7})$$

The middle identity uses the invertibility of C_0 , not a replacement of its Gram by the identity. To prove the last identity with every cross term retained, replace the numerator columns in C_0 by $(I - P_V) T_0$ using the invertible triangular column operation which subtracts $V_0 A_L^{-1} V_0^* T_0$. The preceding Gram then has diagonal blocks A_L, D_0 and the new row is $[v_n, \rho]$. Consequently $\zeta_{\text{all}} = \zeta + \rho D_0^{-1} \rho^*$, which gives FMG7.

H.2 A polynomial minimum for the complete lower denominator

For each integer $a \geq 0$ let

$$\nu_a = \min_{P \text{ monic, deg } P=a} \|\chi_- P\|_s^2 > 0. \quad (\text{FMG8})$$

The positive quadratic minimum used in FMG6–10 is a Schur-complement calculation [H1, §2.1, (2.3)–(2.4)]. Here the eliminated coordinates are the coefficients of the complete lower-root ideal. The following calculation retains the ordinary transpose in its evaluation constraint and the Hermitian adjoint only in its norm; the cited matrix method does not supply the Gamma-polynomial estimates.

The coefficients here are in the unchanged variable S' . This is an attained finite-dimensional minimum in the original measure; strict positivity follows because a nonzero polynomial cannot vanish on the entire integration line. The transformation $P \mapsto \ell \chi_- P$ is an explicitly invertible map from the monic degree- a affine fibre to the fibre with leading coefficient ℓ and factor χ_- . The norm under that map is multiplied by exactly $|\ell|^2$; no scalar is suppressed.

At degree $n = q' + m$, a monic polynomial with factor χ_- has orthonormal coefficient $a_n = i^n \sqrt{h_{n,s}}$. Its old coefficient column $a_{<}$ must satisfy $V_0^\top a_{<} = -a_n v_n^\top$. This is the exact evaluation constraint, including the ordinary transpose. Its unique coefficient minimum is

$$a_{<} = -a_n \overline{V_0} (V_0^\top \overline{V_0})^{-1} v_n^\top. \quad (\text{FMG9})$$

Indeed the displayed column satisfies the constraint, and it is orthogonal to the constraint kernel. Its squared norm is $h_{n,s}v_n A_L^{-1}v_n^* = h_{n,s}\zeta$; the equality follows by conjugating the middle Gram and using that the resulting scalar is real. Adding $|a_n|^2$ proves

$$\boxed{\nu_m = h_{n,s}(1 + \zeta), \quad 1 + r = \frac{\|g\|_s^2}{|\ell|^2\nu_m}.} \quad (\text{FMG10})$$

Thus the lower denominator in the corrected original row is calculated by an actual lower-root ideal minimum. Replacing it by one would remove the explicitly present factor $\nu_m/h_{n,s}$.

H.3 All eight compressed matrices in exact polynomial coordinates

Put $f_a(S') = \chi_-(S')S'^a$ for $0 \leq a < m$, and write $f_a = \sum_{b < n} C_{ba}\phi_b$. Define

$$G = C^*C = (\langle f_a, f_b \rangle_s)_{a,b < m}, \quad H = C^*c_{<} = (\langle f_a, g \rangle_s)_{a < m}, \quad A = C^T T_0 = (\Lambda_{F_b}(f_a))_{a < m, b \leq m}. \quad (\text{FMG11})$$

The column span of C is exactly $\ker V_0^T$: every f_a vanishes on the lower roots; their degrees are distinct; and both spaces have dimension m . Thus

$$I - P_V = \overline{C} \overline{G}^{-1} C^T, \quad D_0 = A^* \overline{G}^{-1} A. \quad (\text{FMG12})$$

The first displayed matrix is self-adjoint, idempotent, and has range $\ker V_0^*$, proving it is exactly the required orthogonal projector. It also proves the second identity. The matrix A is invertible: if its product with x is zero, FMG12 gives $x^* D_0 x = 0$, and the positivity already proved forces $x = 0$. This is the original functional pairing on $\chi_- \mathbb{C}[S']_{<m}$, with its row and column orientations specified in FMG11.

The projection of g onto that polynomial ideal is $CG^{-1}H$. Its perpendicular component has leading coefficient ℓ and is exactly the monic-fibre minimizer multiplied by ℓ . Orthogonality and FMG8 therefore give

$$|\ell|^2\nu_m = \|g\|_s^2 - H^*G^{-1}H, \quad r = \frac{H^*G^{-1}H}{|\ell|^2\nu_m}, \quad \frac{\rho}{\sqrt{1+\zeta}} = \mathbf{b}A, \quad \mathbf{b} = -\frac{H^T \overline{G}^{-1}}{\ell i^n \sqrt{\nu_m}}. \quad (\text{FMG13})$$

For the row identity, substitute FMG7 in the definition of ρ , then apply FMG12 and $c_{<}^T \overline{C} = H^T$. Its denominator is $c_n \sqrt{1+\zeta} = \ell i^n \sqrt{\nu_m}$ by FMG10. This calculation proves both the phase and the single occurrence of the original leading coefficient.

Retain every fixed original CPQ coefficient subspace of $\mathcal{U} = \mathbb{C}^{4m}$:

$$\begin{aligned} I &= \text{im}(B_{\text{cof}}J_0), & B &= \text{im } B_{\text{cof}}, & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T_e &= S + J, & e &= 0, 1. \end{aligned} \quad (\text{FMG14})$$

Their matrices are the original period-dependent maps; none is replaced by a plane having merely the same dimension. Let X denote any fixed independent column frame for one of these eight subspaces. Set

$$\mathcal{A} = \text{diag}(A, 4), \quad \mathcal{G}^\vee = \text{diag}(\overline{G}^{-1}, 4), \quad \lfloor = \text{diag}(\mathbf{b}, 4), \quad Y_X = \mathcal{A}X, \quad (\text{FMG15})$$

$$\boxed{K_{X,n} = \lfloor Y_X (Y_X^* \mathcal{G}^\vee Y_X)^{-1} Y_X^* \rfloor^*.$$

Here $\text{diag}(\mathbf{b}, 4)$ has four one-row blocks, so FMG15 is exactly a 4×4 matrix. If X is empty the result is the zero matrix. To verify FMG15 in the original metric, FMG12 gives $D_0 = \mathcal{A}^* \mathcal{G}^\vee \mathcal{A}$, and FMG13 gives $\mathcal{U}_n = \lfloor \mathcal{A}$. Substituting these identities in NTE1 proves FMG15 as literal equality, with no spectral approximation. The map $x \mapsto \mathcal{A}x$ is an isometry from $(\mathcal{U}, \mathcal{D}_0)$ to $(\mathbb{C}^{4m}, \mathcal{G}^\vee)$, with inverse $\mathcal{A}^{-1}$. It carries sums and intersections bijectively; in particular it preserves both full chains

$$J \subset W_i \subset W \subset T_e, \quad K_i \subset I \subset K_c \subset S. \quad (\text{FMG16})$$

All original invariant, conductor and connecting coefficients are retained in Y_X through their exact pairing $\mathcal{A}$.

Writing $b_X = \log \det(I_4 + K_{X,n})$, the first actual signed row is

$$\delta \mathcal{L}_{e,n} = b_{T_e} + b_{K_c} + b_{W_i} + b_{K_i} - b_W - b_S - b_J - b_I. \quad (\text{FMG17})$$

Equations FMG2–15 evaluate every entry using original polynomial inner products and the actual period coefficient maps. They retain all four eigenvalues of each matrix, including zero eigenvalues. The same equality supplies its threshold contribution by replacing b_X by $\sum_{a=1}^4 \log \max(1, \lambda_{X,a})$ with $\lambda_{X,a}$ the four eigenvalues of FMG15. No simultaneous diagonalization is assumed. Every matrix is positive and bounded above by rI_4 , since FMG15 is the unchanged NTE1 compression and FMG13 evaluates its full relative row norm.

H.4 Finite evaluation from the unchanged Gamma moments

The matrices in FMG11 involve no infinite series. For explicit finite evaluation, expand each polynomial on the original line, $f(c' + iy) = \sum_a d_a(f)y^a$. Define $M_r^{(s)} = \int y^r dm_s(y)$. Then

$$\langle f_1, f_2 \rangle_s = \sum_{a,b} \overline{d_a(f_1)} d_b(f_2) M_{a+b}^{(s)}. \quad (\text{FMG18})$$

These moments can be evaluated directly by the retained recurrence: for a requested degree r , form the $(r+1)$ -square real symmetric Jacobi matrix with off-diagonal entries $J_{a,a+1} = J_{a+1,a} = \sqrt{(a+1)(a+s/2)}$, $0 \leq a < r$, and zero diagonal. Then

$$M_r^{(s)} = M_s(J^r)_{00}. \quad (\text{FMG19})$$

Indeed multiplication by y in the orthonormal frame $p_{a,s}/\sqrt{h_{a,s}}$ has exactly these entries by FMG4. Starting in degree zero, r multiplications never visit a degree larger than r , so the finite truncation changes no path contributing to that matrix entry. Its constant component has the displayed integral, including M_s . Odd moments vanish, while all even moments are finite positive sums of these exact recurrence paths.

The entries of $\mathbf{A}$ have the finite original conductor calculation

$$f_{b,a} = \frac{\nu_{b,a+v} - \sum_{d=v+1}^{a+v} \binom{a+v}{d} \mu_d f_{b,a+v-d}}{\binom{a+v}{v} \mu_v}, \quad \nu_{b,r} = \sum_{u,w} (Z_{\text{exp},j})_{uw,b} \beta_{j,u,w}^r, \quad (\text{FMG20})$$

$$A_{ab} = \sum_r [S'^r] (\chi_-(S') S'^a) f_{b,r}.$$

The triangular equation is exactly the coefficient of z^{a+v} in $E_A F_b = N_b$, retaining every factorial and μ_v . Thus all entries in FMG15 are finite expressions in the original roots, full conductor, actual chosen quotient frame, actual flag frames and Gamma moments. This gives an executable polynomial evaluation of the original first row, with no new choice of period.

H.5 A conductor-moment witness inside the actual residual

The lower root set is symmetric about c' under negation and conjugation. More explicitly

$$i^{-q'} \chi_-(c' + iy) = \prod_{a,b=0}^{j-8} (y - (2b - j + 8)\gamma + i(2a - j + 8)\delta) \quad (\text{FMG21})$$

is real and even as a polynomial in real y . Therefore $|\chi_-(c' + iy)|^2 dm_s(y)$ is an even positive measure. Its monic degree- a orthogonal polynomial in y has parity $(-1)^a$: reflecting it and multiplying by $(-1)^a$ gives another monic minimizer with the same norm, and uniqueness identifies them. Let

P_a be its monic polynomial in S' under the exact map $p(y) \mapsto i^a p((S' - c')/i)$. It has coefficient $[S'^{a-1}]P_a = -ac'$ for $a \geq 1$, while $P_0 = 1$; in every case $\|\chi_- P_a\|_s^2 = \nu_a$.

Define the actual centered conductor moment

$$\eta_A = \frac{\mu_{v+1}}{(v+1)\mu_v} - 4, \quad \beta_m^- = \frac{\nu_m}{\nu_{m-1}} > 0. \quad (\text{FMG22})$$

The coefficient of S^{q-1} in χ_j is $-qj/2$, because its roots have exact mean $j/2$. Using FMG2 gives

$$[S'^{n-1}](g/\ell) = n \left(\frac{\mu_{v+1}}{(v+1)\mu_v} - \frac{j}{2} \right) = -nc' + n\eta_A. \quad (\text{FMG23})$$

The displayed division by ℓ is the explicit affine-fibre map in FMG8; it does not alter the original polynomial g . The coefficient of S'^{n-1} in $\chi_- P_m$ is $-nc'$. Their difference lies in $\chi_- \mathbb{C}[S']_{<m}$ and its coefficient at S'^{n-1} is $n\eta_A$. Its component along the orthogonal column $\chi_- P_{m-1}$ is consequently $n\eta_A$, because every other column has smaller degree. Pythagoras therefore proves

$$r \geq \frac{n^2 |\eta_A|^2 \nu_{m-1}}{\nu_m} = \frac{n^2 |\eta_A|^2}{\beta_m^-}. \quad (\text{FMG24})$$

This is an original-data lower witness, which may vanish for the actual period; no nonvanishing is assumed. Even when it vanishes, FMG13 and FMG15 retain all other orthogonal components and the entire signed mixed row. In particular the exact criterion for $r = 0$ is $H = 0$, equivalently $g = \ell \chi_- P_m$ in the original Gamma inner product. Equality of a single conductor moment alone does not assert this full polynomial identity.

H.6 Position in the full return

The old target cutoff for this row is $N = q_j - 1$, hence native cutoff $n - 1 = q_j - v - 1$. Its new target cutoff is q_j , so its weight in the original four-endpoint return is one. FMG15–17 apply separately to $e = 0$ and $e = 1$ and both prescribed source orders. They have no effect on the coefficients of the other rows. Bounds for this row must therefore be subtracted from the full signed profile by its exact weight; the remaining interior rows still require their original arithmetic calculation. The complete CPQ identity remains

$$4S_{K,j} = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A, \quad (\text{FMG25})$$

and all four complementary returns use the same original rows. The calculation above does not replace any of those returns by a dimension or an assigned constant.

H.7 Receiving the proved finite endpoint estimate in every signed term

The complete companion calculation FRG15–31, using the same actual g, ℓ, ν_m , proves the following finite input. Put

$$\begin{aligned} L_s &= (s-1)/4, \quad \theta = \pi/2, \quad h = 1/q, \quad \kappa = \theta - h, \quad Q_- = q' + L_s, \quad Q_+ = q + L_s, \\ \sigma &= (\delta^2 + \gamma^2)^{1/2}, \quad R' = (j-8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j+8)\sigma, 2L_s\}, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ C_m &= (2m+1) \binom{2m}{m}^2, \quad \epsilon_- = \left(\theta - \frac{2Q_-}{M} \right) + \frac{Q_-}{M^2} + h(M+1) + \frac{4q'R'}{M}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\ell|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) \\ &\quad - (2Q_+ + 1) \log \kappa - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FMG26})$$

With these retained finite constants, that proof gives $0 \leq \log(1+r) \leq E_{j,s}$. In particular $E_{j,s}$ is nonnegative at every original $j \geq 13$, as follows also from the two proved bounds on the positive ratio FMG10. For each fixed original quartet and conductor, the actual choice eventually satisfies $M = M_0$ and $E_{j,s} = O_{\text{actual}}(j \log j)$, for both $s = 1, j$. The companion proof includes the complete Gamma density estimate at $h = 1/q$, the monic minimum on an original unit interval, the whole-line upper integral, and all factorials; the estimate is not an assumed analytic input.

Here is its exact effect on the entire signed row. The two chains in FMG16 imply

$$\begin{aligned} 0 &\leq (b_{T_e} - b_W) + (b_{W_i} - b_J) \leq b_{T_e} - b_J \leq 4 \log(1+r), \\ 0 &\leq (b_S - b_{K_c}) + (b_I - b_{K_i}) \leq b_S - b_{K_i} \leq 4 \log(1+r). \end{aligned} \tag{FMG27}$$

Indeed for nested subspaces the orthogonal projectors in their common original metric are ordered; compressing by the same row preserves this order. Monotonicity of $\log \det(I + K)$ follows by differentiating along their positive connecting segment, or by the ordered eigenvalues. The missing middle increment in each chain is nonnegative, proving the middle bound. The last uses $0 \preceq K_X \preceq rI_4$. Subtraction of these two nonnegative quantities gives the exact signed expression FMG17. It follows that

$$|\delta \mathcal{L}_{e,n}| \leq 4E_{j,s}, \quad |\delta \mathcal{P}_{e,n}| \leq 4E_{j,s}, \quad 0 \leq \delta V_{X,n} \leq 4E_{j,s} \quad (X = b, \text{rem}, s, A). \tag{FMG28}$$

For the threshold quantity $\delta \mathcal{P}$, apply the same argument to $\sum_{a=1}^4 \log \max(1, \lambda_a(K_X))$. The min-max characterization orders every eigenvalue when two Hermitian matrices are ordered: restrict the larger Rayleigh quotient to each test subspace, take its extremum there, and then the same extremum over subspaces. The increasing scalar function $\log \max(1, x)$ preserves those four inequalities. Thus the same two chain estimates hold with upper bound $4 \log \max(1, r) \leq 4 \log(1+r)$. This also retains every zero and every eigenvalue below one.

For the four CPQ returns, their first-row increments are the paired determinant increments on the actual quotients W/W_i , $\mathcal{U}/T_e$, S/K_c , and I/K_i , respectively. For any fixed $Y \supset K$ in this list, the determinant of its attained quotient Gram is $\det G_Y / \det G_K$ times the fixed frame factor, by full block elimination. The factor cancels in a successive increment, leaving $b_Y - b_K$. The same order argument places it between zero and $4 \log(1+r)$, proving the final part of FMG28. For $Y = \mathcal{U}$ its compression is exactly rI_4 . All four quotient fibres and all their cross terms therefore remain in the row-wise CPQ identity.

Since $j = k + 4$ and $q_k = (k + 1)^2$, each quantity in FMG28 divided by kq_k tends to zero. The original endpoint weight is one. Consequently the first original row can be removed from the leading kq_k calculation with the explicit finite error in FMG28, simultaneously in the signed spectral profile and in all four complementary CPQ returns. This statement concerns this one specified row; it leaves all remaining rows in their original correlated expressions.

I The original conductor on every native mixed subquotient

This section receives the proved SGT7–28 estimates on the actual DNE6–9 and NSO12–15 spaces, and then on the fixed CIT/CPQ flags. The exterior quotient argument is already present in CEP37–38; we give its complete proof in the form needed here. The new receiver is the full finite numerator-observation metric, with the conductor multiples retained as its entire quotient denominator, and the rank-dependent SGT error on every original native subquotient.

Fix the original actual hypothetical simple quartet, its fixed original five-orbit period and logarithm branch. Keep $0 < \delta < 1/2$, $\gamma > 2$, every coefficient a_{uw} of the nonzero original conductor, and

$$\begin{aligned} \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma, \quad 0 \leq u, w \leq a, \quad a = 4\ell + 1 \geq 9, \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_n = E_A^{(n)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad \alpha_A = v!/\mu_v, \\ q_a &= (a+1)^2, \quad q'_a = (a-7)^2, \quad m_a = q_a - q'_a - v, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \\ N &\geq q_a - 1, \quad D = N - v, \quad s \in \{1, a\}. \end{aligned} \tag{NCT1}$$

The original domain of the fixed period, including its original rank conditions, is unchanged. The roots in each displayed grid are distinct: equality of two roots gives separately equality of the indices from their real and imaginary parts. Every polynomial space below has complex coefficients. Its Hermitian inner product is conjugate-linear in the first variable. The complex-linear dual consists of linear, rather than conjugate-linear, functionals.

The full original measures, masses, recurrence and frames are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \\ \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c+iy)|^2 dm_s(y). \end{aligned} \tag{NCT2}$$

SGT3–5 proves that these are orthonormal frames on every indicated finite polynomial space. Their leading coefficients are $i^{-n}/\sqrt{h_{n,s}}$; neither this phase nor the mass is removed.

I.1 The exact dual conductor square

Write $\mathcal{P}_d = \mathbb{C}[S]_{\leq d}$, with $\mathcal{P}_{-1} = 0$, and distinguish the target variable S' . The original, unscaled conductor map is

$$\begin{aligned} (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \\ A_{D,s} : (\mathcal{P}_N/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}), \quad [P] \longmapsto \mathcal{T}_A P. \end{aligned} \tag{NCT3}$$

It is the SGT7 isomorphism. For completeness, $\mathcal{T}_A S^n = \sum_{j=0}^n \binom{n}{j} \mu_j S'^{n-j}$. If $n \geq v$, its leading coefficient is $\binom{n}{v} \mu_v \neq 0$, and its degree is $n - v$. Successive elimination in degree gives surjectivity, and the full kernel is $\mathcal{P}_{v-1}$. The quotient is represented isometrically by the original orthogonal complement spanned by $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$.

For an analytic germ F define $\Lambda_F(S'^n) = F^{(n)}(0)$. An upper-variable germ is paired with S^n by the same derivative convention. The contravariant dual map in the direction relevant here is

$$\begin{aligned} A_{D,s}^\vee : (\mathbb{C}[S']_{\leq D})^\vee &\longrightarrow (\mathcal{P}_N/\mathcal{P}_{v-1})^\vee, \quad \Lambda \longmapsto \Lambda \circ A_{D,s}, \\ A_{D,s}^\vee \Lambda_F &= \Lambda_{E_A F}. \end{aligned} \tag{NCT4}$$

To prove the identity, on every $0 \leq n \leq N$ its left side is $\sum_{j=v}^n \binom{n}{j} \mu_j F^{(n-j)}(0)$, precisely $(E_A F)^{(n)}(0)$ by the Leibniz rule. For $n < v$ both sides vanish. The highest derivative of F that occurs is $n - v \leq D$, so the map is well defined on these finite jets; it does not depend on any higher derivative. Conversely the nonzero coefficient $\binom{v+r}{v} \mu_v$ recursively recovers the derivative of order r of F from these products. This also proves the isomorphism on finite jets directly.

Let $w = (\Lambda_F(\phi_{n,s,c_1}))_{n=0}^D$, and let $\tilde{w} = (\Lambda_{E_A F}(\phi_{v+n,s,c_0}))_{n=0}^D$. If $A_{D,s}$ denotes its original orthonormal matrix as in SGT10, then

$$\tilde{w} = A_{D,s}^\top w, \quad \Lambda_{E_A F}(\phi_{n,s,c_0}) = 0 \quad (0 \leq n < v). \quad (\text{NCT5})$$

Indeed the j -th source basis vector has image $\sum_i (A_{D,s})_{ij} \phi_{i,s,c_1}$, and applying the complex-linear functional gives the ordinary transpose in NCT5. There is no complex adjoint in this algebraic square. Squared functional norms are $w^* w$ and $\tilde{w}^* \tilde{w}$. If $A = U \Sigma V^*$ is a singular-value decomposition, then $A^\top = \bar{V} \Sigma U^\top$ is one as well. Thus A and $A^\top$, and their inverses, have exactly the same singular values and every corresponding exterior norm.

No scalar α_A is inserted into NCT3–5. In particular, the literal DNE3 germ is $F_b = \mathcal{N}_b / E_A$, and its image is $\Lambda_{\mathcal{N}_b}$, with coefficient one. The original primary-coordinate maps involving α_A are retained below.

I.2 All numerator observations and all conductor columns

Use the original lexicographic upper-root coefficient order. For $\nu \in \mathbb{C}^{q_a}$ put $\mathcal{N}_\nu(z) = \sum_{u,w} \nu_{uw} e^{\beta_{a,u,w} z}$. Let $\mathcal{N}_{v,a}$ be the coefficient subspace on which $\mathcal{N}_\nu^{(j)}(0) = 0$ for all $0 \leq j < v$. The upper Vandermonde $\mathbb{V}_a = (\beta_{a,u,w}^n)$, $0 \leq n < q_a$, is invertible, so $\dim \mathcal{N}_{v,a} = q_a - v$.

The full original primary conductor C_a , with q'_a rows and q_a columns, is fixed by the literal identity

$$\mathcal{N}_{C_a^\top \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{NCT6})$$

This identity follows by collecting all original shifts, since $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. It retains repeated contributions to the same resulting root. The map $C_a^\top$ is injective: its zero image would make the product in NCT6 identically zero; on a neighborhood where $E_A \neq 0$ the lower exponential sum is zero, hence everywhere by analyticity, and its distinct-root Vandermonde makes $\eta = 0$. Its image lies in $\mathcal{N}_{v,a}$, by the zero order of E_A .

Use exactly the DNE2–3/NSO12 frame:

$$\begin{aligned} W_{v,a} &= \mathbb{V}_a^{-1} P_v, & J_{v,a} &= P_v^\top \mathbb{V}_a C_a^\top, & W_{v,a} J_{v,a} &= C_a^\top, \\ Z_{\text{exp},a} &= W_{v,a} E_{\mathcal{J}_a}, & F_{a,b} &= \mathcal{N}_{Z_{\text{exp},a} e_b} / E_A. \end{aligned} \quad (\text{NCT7})$$

Here P_v inserts zero in the first v positions; $\mathcal{I}_a$ is the first ordered set of q'_a rows with nonzero minor of $J_{v,a}$, and $\mathcal{J}_a$ is its complement. Such a minor exists by the proved injectivity in NCT6. Restricting a relation among columns of $[J_{v,a}, E_{\mathcal{J}_a}]$ first to $\mathcal{I}_a$ and then to its complement proves that they are independent. Consequently $[C_a^\top, Z_{\text{exp},a}]$ is the complete frame of $\mathcal{N}_{v,a}$, and $Z_{\text{exp},a}$ represents its quotient by $\text{im } C_a^\top$. Every actual minor is unchanged.

Define the full upper observation matrix and its covariance by

$$\begin{aligned} \mathbb{E}_{a,N,s}[n, (u, w)] &= \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N, \\ K_{a,N,s} &= \mathbb{E}_{a,N,s}^* \mathbb{E}_{a,N,s}, \\ \hat{D}_{a,D}^{(s)} &= Z_{\text{exp},a}^* [K_{a,N,s} - K_{a,N,s} C_a^\top (\bar{C}_a K_{a,N,s} C_a^\top)^{-1} \bar{C}_a K_{a,N,s}] Z_{\text{exp},a}. \end{aligned} \quad (\text{NCT8})$$

The square covariance $K_{a,N,s}$ is positive definite. Its first q_a observation rows equal an invertible triangular change of the upper Vandermonde, because the polynomials NCT2 have nonzero leading coefficients. Therefore $\mathbb{E}_{a,N,s}$ has full column rank for every stated N , including $N = q_a - 1$. The middle inverse in NCT8 is positive definite by this fact and injectivity of $C_a^\top$.

The same covariance has an exact derivative-coordinate expression. Let $V_{a,N}[n, (u, w)] = \beta_{a,u,w}^n$, and write $S^r = \sum_{n=0}^N B_{s,c_0,N}[n, r] \phi_{n,s,c_0}(S)$. The full monomial Gram is $H_{s,c_0,N} = B^* B$. Applying the complex-linear numerator functional to this identity gives

$$V_{a,N} \nu = B_{s,c_0,N}^T E_{a,N,s} \nu, \quad K_{a,N,s} = V_{a,N}^* \overline{H_{s,c_0,N}^{-1}} V_{a,N}. \quad (\text{NCT8a})$$

Indeed $(B^T)^{-*} (B^T)^{-1} = \overline{(B^* B)^{-1}}$. Thus NCT8 is the full numerator derivative-dual covariance restricted to the vanishing-moment space and minimized over every conductor multiple. This proves the coordinate morphism, including the ordinary transpose and Hermitian conjugation, without changing the original measure or coefficient norm.

The minimum below uses the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)] on the Hermitian Gram of $[C_a^T, Z_{\text{exp},a}]$ in $K_{a,N,s}$. The eliminated block is invertible by the rank argument above, so no pseudoinverse or unstated range condition is needed.

For every original coefficient vector $x \in \mathbb{C}^{m_a}$, the exact squared norm represented by NCT8 is

$$x^* \widehat{D}_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \|E_{a,N,s} (Z_{\text{exp},a} x + C_a^T \eta)\|_2^2. \quad (\text{NCT9})$$

Differentiate the right side with respect to its real and imaginary coordinates, or complete its positive square. The unique minimizer is

$$\eta_{\min} = -(\overline{C_a} K_{a,N,s} C_a^T)^{-1} \overline{C_a} K_{a,N,s} Z_{\text{exp},a} x.$$

Substitution gives NCT8. The quotient is positive definite because $[C_a^T, Z_{\text{exp},a}]$ is independent. The minimum retains every conductor column, including every mixed covariance term.

For $\nu \in \mathcal{N}_{v,a}$ the entries of $E_{a,N,s} \nu$ below row v vanish, since those polynomials use derivatives of order below v . This proves that the ambient polynomial dual norm in NCT9 is exactly the quotient dual norm in NCT4, not an extra truncation of the numerator. All rows $0, \dots, N$ remain in NCT8.

The corresponding original native metric is exactly NSO13:

$$x^* D_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \left\| \Lambda_{\sum_b x_b F_{a,b}} + \sum_{u,w} \eta_{uw} \text{ev}_{\beta_{a-8,u,w}} \right\|_{s,c_1,D,\vee}^2. \quad (\text{NCT10})$$

NCT4 takes every representative inside this minimum to exactly the representative in NCT9. In particular it takes the complete lower evaluation subspace to the complete conductor-multiple subspace, by NCT6. Let Y be the span of all functionals appearing in NCT10, and K its lower evaluation subspace. Then NCT4 induces the isomorphism

$$Y/K \xrightarrow{A_{D,s}^\vee} \Lambda_{\mathcal{N}_{v,a}} / \Lambda_{\text{im } C_a^T}, \quad [F] \mapsto [E_A F]. \quad (\text{NCT11})$$

The two quotient Grams in the same fixed value coordinates x are $D_{a,D}^{(s)}$ and $\widehat{D}_{a,D}^{(s)}$. Finite-jet injectivity follows also directly: vanishing numerator derivatives through $N \geq q_a - 1$ kill all upper coefficients by the Vandermonde. Thus this isomorphism holds on every retained endpoint and on every original coefficient fibre, without a genericity or conditioning assumption.

I.3 Rank-dependent exterior bounds on arbitrary subquotients

We prove the finite-dimensional statement in full. Let $B : H_0 \rightarrow H_1$ be an invertible linear map of complex Hilbert spaces, and let $K \subset Y \subset H_0$ be any subspaces. Both quotients have their attained quotient norms. Put $Y_0 = Y \cap K^\perp$ and $Y_1 = BY \cap (BK)^\perp$. Every $y \in Y$ has the unique decomposition $y = x + k$ with $x \in Y_0$, $k \in K$. The squared norm of a lift of $[y]$ is $\|x\|^2 + \|k'\|^2$, so x is its minimum lift. Under these isometric identifications the induced quotient map is

$$\overline{B} : Y/K \longrightarrow BY/BK, \quad \overline{B} = P_{Y_1} B|_{Y_0}, \quad \overline{B}^{-1} = P_{Y_0} B^{-1}|_{Y_1}. \quad (\text{NCT12})$$

The second formula follows by applying B^{-1} to any representative and subtracting its K component. It is indeed the inverse because the subtracted components lie respectively in BK and K . The projections are contractions on every exterior power: in orthonormal bases their singular values are zeros and ones. Restrictions and inclusions of the stated Hilbert subspaces have exterior norm at most one. Consequently for every $0 \leq r \leq \dim(Y/K)$,

$$\|\bigwedge^r \overline{B}\| \leq \|\bigwedge^r B\|, \quad \|\bigwedge^r \overline{B}^{-1}\| \leq \|\bigwedge^r B^{-1}\|. \quad (\text{NCT13})$$

There is no factor depending on $\dim K$. For any ordered independent quotient frame of length r , its squared wedge norm is its Gram determinant. Applying the first inequality to that wedge and the second to its image gives

$$-2 \log \|\bigwedge^r B^{-1}\| \leq \log \frac{\det G_1}{\det G_0} \leq 2 \log \|\bigwedge^r B\|. \quad (\text{NCT14})$$

These statements include $r = 0$, with determinant and exterior norm one. A change of the common value frame by U multiplies both determinants by $|\det U|^2$, which cancels in their ratio.

For the classical exterior inner product underlying NCT14, see Russell Lyons [H3, §4, (4.1), p. 11 of the cited arXiv version]. Taking both lists there to be the same quotient frame gives its squared wedge norm as its Gram determinant. The quotient map, inverse, and rank-dependent transport bounds are proved here.

Successive restrictions and quotients obey the same rank-only bound. Indeed a subspace of Y/K has a full preimage Y' containing K , and its inherited metric is the quotient metric of Y'/K . A subsequent quotient by a subspace with full preimage K' is Y'/K' , since the union of its original affine fibres is precisely the affine fibre modulo K' . Its minimum over those fibres is the same minimum. Induction reduces any finite sequence to NCT12 with the actual full preimages; the distortion does not accumulate a new factor at each intermediate quotient.

Apply this to $B = A_{D,s}^\top$ using NCT5, and to its orthogonal direct sum of b copies, where $b = 1$ or 4. We now specify its finite constants using the original SGT symbol. To avoid confusing the degree a with a circle radius, denote an admissible SGT radius by $\varrho \in (0, 1)$. Set

$$\begin{aligned} w(t) &= -i \arctan t, & g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), & g(0) &= (-i)^v \mu_v / v!, \\ \rho_n &= \sqrt{n! / (s/2)_n}, & L_{s,D,v} &= \frac{1}{2} \max\{1, s/2 - 1\} \sum_{n=1}^{D+v} \frac{1}{n}. \end{aligned} \quad (\text{NCT15})$$

SGT10 proves the exact original matrix $A_{D,s} = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g) \text{diag}(\rho_v, \dots, \rho_{D+v})$, where $T_D(g)_{i,j} = g_{j-i}$ for $i \leq j$, and zero otherwise. SGT21 proves that the products of the two forward diagonal operator norms and of the two inverse diagonal operator norms are both at most $e^{2L_{s,D,v}}$.

Here is the full list of original finite constants used in SGT19:

$$\begin{aligned}
A_{\text{abs}} &= \sum_{u,w} |a_{uw}|, \quad B_0 = \max\{1, \max_{u,w} |\beta_{8,u,w}|\}, \quad C_v = A_{\text{abs}} B_0^v / |\mu_v|, \\
R_0 &= B_0^{-1} \min\{1, (v+1)/(2eC_v)\}, \quad a_0 = \min\{1/2, \tanh(R_0/2)\}, \\
d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)}, \quad M_0 = \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}, \\
S_D(r) &= \sum_{n=0}^D r^{-n}, \quad G_{0,D} = M_0 S_D(a_0), \\
\mathcal{Z}_\varrho &= \{\zeta : |\zeta| < \varrho, g(\zeta) = 0\}, \quad H_\varrho = \sum_{\zeta \in \mathcal{Z}_\varrho} m_\zeta, \\
A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta-h}}{dt^{m_\zeta-h}} \frac{(t-\zeta)^{m_\zeta}}{g(t)} \Big|_{t=\zeta}, \quad 1 \leq h \leq m_\zeta, \\
W_\varrho &= \operatorname{arctanh} \varrho, \quad m_{E,\varrho} = \min_{|t|=\varrho} |E_A(w(t))|, \quad G_\varrho = A_{\text{abs}} \varrho^{-v} e^{(B_0+4)W_\varrho}, \\
M_\varrho &= \frac{\varrho^v e^{4W_\varrho}}{m_{E,\varrho}} + \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(\varrho - |\zeta|)^h}, \quad P_\varrho = \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(1 - |\zeta|)^h}, \\
F_{\varrho,D} &= G_\varrho S_D(\varrho), \quad V_{\varrho,D} = M_\varrho S_D(\varrho) + P_\varrho, \quad V_{\varrho,D}^{\text{cap}} = \min\{G_{0,D}, V_{\varrho,D}\}.
\end{aligned} \tag{NCT16}$$

Here admissibility means that g has no zero on the indicated circle. The sums retain every pole and its full order. There are finitely many poles in each such closed disk, and there are admissible radii arbitrarily close to one, as proved in SGT14. SGT11–19 proves from these exact original data that

$$\begin{aligned}
\|T_D(g)\| &\leq F_{\varrho,D}, \quad \|T_D(g)^{-1}\| \leq G_{0,D}, \\
T_D(g)^{-1} &= C_{\varrho,D} + R_{\varrho,D}, \quad \operatorname{rank} C_{\varrho,D} \leq H_\varrho, \quad \|R_{\varrho,D}\| \leq V_{\varrho,D}.
\end{aligned} \tag{NCT17}$$

These are proved internal inputs, including SGT11–13's explicit zero-free disk, SGT16–18's exact principal parts and their lower-triangular corrections. In particular NCT17 is a decomposition of the entire inverse; none of its principal parts has been omitted.

For b copies the inverse exceptional rank is at most bH_ϱ ; its global norm and its remainder norm are still bounded by $G_{0,D}$ and $V_{\varrho,D}$, respectively. For a rank- r subquotient put $h_b = \min\{r, bH_\varrho\}$. On the kernel of the exceptional matrix its inverse has norm at most $V_{\varrho,D}$. The singular-value minimum principle therefore bounds every singular value after the first bH_ϱ by that quantity, and every singular value by $G_{0,D}$. Multiplying the largest r singular values, and then the two diagonal factors, gives

$$\begin{aligned}
\left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus b}) \right\| &\leq e^{2rL_{s,D,v}} F_{\varrho,D}^r, \\
\left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus b})^{-1} \right\| &\leq e^{2rL_{s,D,v}} G_{0,D}^{h_b} (V_{\varrho,D}^{\text{cap}})^{r-h_b}.
\end{aligned} \tag{NCT18}$$

For clarity, the minimum principle here follows by diagonalizing the positive matrix of the squared norm. Every subspace of codimension $j-1$ meets the span of its first j eigenvectors, and the complement of its first $j-1$ eigenvectors attains the bound. Apply this with the kernel of the exceptional matrix. This proves the estimate even when its rank bound exceeds the whole matrix dimension, when $h_b = r$.

For a fixed admissible ϱ , suppress that parameter in the notation requested for the receiving forms, and define

$$\begin{aligned}
\mathcal{U}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2r \log^+ F_{\varrho,D}, \\
\mathcal{L}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2h_b \log^+ G_{0,D} + 2(r-h_b) \log^+ V_{\varrho,D}^{\text{cap}}, \\
\mathcal{B}_{a,D,s,r}^{[b]} &= \max\{\mathcal{L}_{a,D,s,r}^{[b]}, \mathcal{U}_{a,D,s,r}^{[b]}\}.
\end{aligned} \tag{NCT19}$$

Here $\log^+ x = \max\{0, \log x\}$; all these constants are finite, nonnegative and exactly zero at $r = 0$. The index a records the original grid and centres; the conductor constants themselves depend only on the fixed original conductor and radius.

Let Q be any original rank- r metric obtained by a finite sequence of restrictions and attained quotients from $(D_{a,D}^{(s)})^{\oplus b}$, and let $\widehat{Q}$ be the identical construction, in the identical coefficient and quotient value frames, from $(\widehat{D}_{a,D}^{(s)})^{\oplus b}$. Equations NCT11–14 and NCT18 prove the finite directed statement

$$\boxed{-\mathcal{L}_{a,D,s,r}^{[b]} \leq \log \det \widehat{Q} - \log \det Q \leq \mathcal{U}_{a,D,s,r}^{[b]}} \quad (\text{NCT20})$$

The full preimage of each restriction and quotient under NCT11 includes the entire conductor-multiple denominator, of dimension bq'_a . NCT13 proves why that dimension contributes no extra volume factor in NCT20. No fixed coefficient map is replaced by an orthogonal surrogate; minimum lifts are used only to prove the bound on its original attained metric.

I.4 The uniform limit and the exact DNE scalar conventions

For fixed $C_D, C_r > 0$, uniformly over $0 \leq D \leq C_D a^2$, $0 \leq r \leq C_r a$, both actual orders, $b = 1, 4$, and every original admissible subquotient,

$$\log \det \widehat{Q} - \log \det Q = o(a^3) = o(aq_a). \quad (\text{NCT21})$$

Here is the limiting proof with the order of choices explicit. Fix an admissible $\varrho < 1$. Since $S_D(\varrho) \leq (D+1)\varrho^{-D}$, NCT16 gives

$$\begin{aligned} \log^+ F_{\varrho,D} &\leq D \log(1/\varrho) + \log(D+1) + \log^+ G_{\varrho}, \\ \log^+ V_{\varrho,D}^{\text{cap}} &\leq D \log(1/\varrho) + \log(D+1) + \log^+(M_{\varrho} + P_{\varrho}), \\ \log^+ G_{0,D} &\leq D \log(1/a_0) + \log(D+1) + \log^+ M_0. \end{aligned}$$

The common term $4rL_{s,D,v}$ is $O(a^2 \log(a+1))$. The exceptional term uses $h_b \leq bH_{\varrho}$, fixed before the limit, and is $O_{\varrho}(a^2 + \log(a+1))$. Consequently the limsup of the uniform absolute error divided by a^3 is at most $2C_r C_D \log(1/\varrho)$. Only after this limit choose admissible radii approaching one. The bound tends to zero. This proves NCT21 without any assertion of a zero-free unit disk, bounded pole count near its boundary, or a coefficient condition on the original subquotients.

The original DNE scalar and transpose square is preserved exactly. Let $A_{\ell,a}$ be DNE4's pairing matrix and Z_a the original low-monomial kernel columns. Keep

$$\begin{aligned} D_{a,D}^{(s)} &= A_{\ell,a}^* \overline{(H_{Y,a,N}^{(s)})^{-1} A_{\ell,a}}, & B_{K,a} &= Z_a^{\top} A_{\ell,a}, \\ Q_{K^{\vee}} &= (B_{K,a} (D_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1} = \overline{(Z_a^* H_{Y,a,N}^{(s)} Z_a)^{-1}}, \\ \widehat{Q}_{K^{\vee}} &= (B_{K,a} (\widehat{D}_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1}, & \widehat{H}_Y &= \overline{A_{\ell,a} (\widehat{D}_{a,D}^{(s)})^{-1} A_{\ell,a}^*}, \\ & & \widehat{Q}_{K^{\vee}} &= \overline{(Z_a^* \widehat{H}_Y Z_a)^{-1}}. \end{aligned} \quad (\text{NCT22})$$

For proof, a functional with coordinate column u on a Gram H has norm $u^* \overline{H^{-1}} u$, since its Riesz vector is $H^{-1} \bar{u}$. Substituting this identity and $B_{K,a} = Z_a^{\top} A_{\ell,a}$ proves every equality in NCT22 by matrix multiplication. The onto map $B_{K,a}$ uses its full fibre, with minimum norm $(B_{K,a} D^{-1} B_{K,a}^*)^{-1}$. This follows by inserting the minimum lift $D^{-1} B_{K,a}^* (B_{K,a} D^{-1} B_{K,a}^*)^{-1} x$ and checking that it is D -orthogonal to $\ker B_{K,a}$. Thus NCT20 applies with $r = \text{rank } B_{K,a}$; inversion changes the sign of the logarithmic determinant difference when it is read on the original restricted kernel Gram.

The primary-coordinate comparison is literally the earlier KRC/RRC map $x \mapsto \alpha_A \mathbf{N}_a Z_{\text{exp},a} x$. For example its four multiplication maps remain

$$M_i = (\alpha_A \mathbf{N}_j Z_{\text{exp},j})^{-1} m_{i,k} (\alpha_A \mathbf{N}_k Z_{\text{exp},k}), \quad j = k + 4. \quad (\text{NCT23})$$

Both scalars and both fixed frame maps are present before their proved cancellation. NCT8–11 acts in the original x coordinates, so the matrices NCT23 are unchanged. If one instead writes the conductor isomorphism as $\alpha_A A_{D,s}$, its ordinary dual is $\alpha_A A_{D,s}^\top$, and its numerator Gram is exactly $|\alpha_A|^2 \widehat{D}$. Every rank- r descendant quotient then has determinant factor $|\alpha_A|^{2r}$, because minimization commutes with this positive scalar. NCT3–22 uses the unscaled map and does not introduce this factor into DNE10. This describes the exact scalar map between the two conventions, rather than silently choosing one.

I.5 All original mixed flags and all four-endpoint signs

For $j = k + 4$, take the original coefficient space $\mathcal{U} = \mathbb{C}^{4m_j}$ and maps of EFP18:

$$\begin{aligned} B &= B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(B_j, 4), \quad J = \ker \mathbb{B}, \quad \mathcal{B} = \text{im } B, \quad I = \text{im}(BJ_0), \\ S &= \text{im}[B, F_e], \quad W = J + \mathcal{B}, \quad K_e = S \cap W, \quad T_e = S + W, \quad K_i = I \cap J, \quad W_i = I + J. \end{aligned} \tag{NCT24}$$

All spaces and the maps NCT23 are the same coefficient objects as before. Replace only their ambient metric by $\text{diag}(\widehat{D}_{j,N-v}^{(s_1)}, 4)$ to define the numerator forms. NCT20 applies to every one of the following original quotients, with its actual rank:

original form	original quotient	rank	
$G_{h,e}$	$K_e/\mathcal{B}$	h_e	
$G_{s,e}$	S/K_e	c_e	
$G_{t,e}$	T_e/W	c_e	
$Q_{\text{rem},e}$	$\mathcal{U}/T_e$	$r_o - c_e$	
A_{inv}	I/K_i	r_I	(NCT25)
H_{inv}	W_i/J	r_I	
Q_b	W/W_i	r_b	
Q_o	$\mathcal{U}/W$	r_o	
$H_{\text{conn},e}$	$S/\mathcal{B}$	r_e	
$H_{B,j,N-v}$	$\mathbb{C}^{m_j}/\ker B_j$	t_j	

The final row uses one copy; every preceding row uses four. For Q_b , any original EFP21 fixed value-frame factor remains on both forms. The table's maps and ranks are the complete EFP18–21 and CPQ data; NCT12 proves their metric transfer by their full preimages under NCT11. It also applies to every additional restriction of these spaces, including all eight signed-profile spaces $T_e, K_e, W_i, K_i, W, S, J, I$, with their exact original frames. Their dimensions are at most $4m_j = O(k)$, so NCT21 applies uniformly without dropping any zero eigenvalue in a later compressed relative operator.

For a fixed-rank original form at the four cutoffs $N_i = (q_j - 1, q_j, 2q_j - 1, 2q_j)$, write $D_i = N_i - v$ and $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$. Abbreviate NCT19's constants at each endpoint by $\mathcal{L}_i$ and $\mathcal{U}_i$, retaining its actual degree, order, rank and number of copies. The exact directed return estimate is

$$\begin{aligned} -\mathcal{U}_0 - \mathcal{U}_1 - \mathcal{L}_2 - \mathcal{L}_3 &\leq \mathcal{R} \log \det \widehat{Q} - \mathcal{R} \log \det Q \\ &\leq \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{U}_2 + \mathcal{U}_3. \end{aligned} \tag{NCT26}$$

This follows by negating the upper bounds at the two negative endpoints and adding the lower bounds at the two positive endpoints for the lower estimate, and doing the reverse for the upper estimate. Every endpoint has $D_i = O(k^2)$, and all ranks in NCT25 are $O(k)$. Thus each displayed return difference is $o(kq_j)$ in both original source pairs $(1, 1)$ and (k, j) . For a source-degree k form evaluated at the same target cutoffs the identical proof uses its original degree k , centre $k/2$, and $s_0 = 1$ or k ; the uniform domain still holds since $q_j = O(k^2)$.

Every finite CPQ determinant-line identity continues to hold for the numerator metrics, because its proof is the same sequence of Schur minima in the same exact coefficient flags. In particular

the original complete complementary returns $V_b, V_{\text{rem}}, V_s, V_A$ all receive NCT26, as do the original equal-rank contraction ratios $\det H_{\text{inv}}/\det A_{\text{inv}}$ and $\det G_t/\det G_s$: subtract the two determinant errors before applying the endpoint signs. For any fixed finite signed sum the absolute error is bounded by the sum of its original summands' $\mathcal{B}$ constants, with multiplicities. No cancellation of an uncomputed summand is asserted. The CPQ affine identities $4S_K = \mathcal{L} + V_b + V_{\text{rem}} + V_s + V_A$ and $P_{\text{rest}} = \mathcal{L} + V_b$ retain their exact forms on these newly specified numerator metrics and differ from their native values by the proved finite errors. The result supplies an exact numerator observation geometry and an $o(kq_j)$ volume transfer; the original mixed numerator matrices and their complete returns remain the quantities to evaluate.

J The original numerator kernel and the complete mixed spectral return

This calculation receives the full conductor-dual transport proved in NCT. It uses the original DNE numerator frame and the original CPQ flags, with their fixed value coordinates. The source of the construction is the accepted cumulative DNE3,6, SGT7–26, CPQ1–20 and SFB12–16. The period, its logarithm branch, the complete nonreduced conductor and its order at zero are fixed as the original tensor degree varies. No arithmetic coefficient or zero packet is supplied by the following calculation.

J.1 The actual finite numerator observations

Fix the original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, and the original permitted period. For $a \equiv 1 \pmod{4}$, $a \geq 9$, retain

$$\begin{aligned} q_a &= (a+1)^2, & q'_a &= (a-7)^2, & m_a &= q_a - q'_a - v, & c_a &= a/2, \\ \beta_{a,u,w} &= a(1/2 - \delta - i\gamma) + 2u\delta + 2wi\gamma, & 0 &\leq u, w \leq a, \\ E_A(z) &= \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, & v &= \text{ord}_0 E_A, & \mu_v &= E_A^{(v)}(0) \neq 0. \end{aligned} \tag{NPK1}$$

The proved conductor bound is $v \leq 32$. Let C_a be the original $q'_a \times q_a$ conductor coefficient matrix, so that its ordinary transpose sends the lower exponential coefficient vector to the numerator of its product with E_A . Write $Z_a = Z_{\text{exp},a}$ for DNE3's fixed $q_a \times m_a$ matrix. Thus $N_{Z_a x}/E_A$ is the original analytic germ for coefficient x , and the columns of $[C_a^\top, Z_a]$ are a basis of the numerator subspace whose first v moments vanish. These matrices include their actual chosen minors and every original coefficient; none is replaced by a generic frame.

For $s \in \{1, a\}$ set $b = s/2$, and use the original recurrence and norms

$$\begin{aligned} p_{0,s}(y) &= 1, & p_{1,s}(y) &= y, & p_{n+1,s}(y) &= yp_{n,s}(y) - n(n-1+b)p_{n-1,s}(y), \\ h_{n,s} &= (2\pi)^{s/2} n!(b)_n, & y_{u,w} &= (\beta_{a,u,w} - c_a)/i = (2w-a)\gamma - i(2u-a)\delta, \\ E_{a,N,s}[n, (u, w)] &= \frac{p_{n,s}(y_{u,w})}{\sqrt{h_{n,s}}}, & 0 &\leq n \leq N, & K_{a,N,s} &= E_{a,N,s}^* E_{a,N,s}. \end{aligned} \tag{NPK2}$$

All square roots in the positive $h_{n,s}$ are positive real. These are functional values of the original Gamma orthonormal polynomials, whose leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$. For a numerator coefficient vector d , the column $E_{a,N,s}d$ is therefore exactly the native functional-value column of N_d at the center $a/2$. It uses the original mass $(2\pi)^{s/2}$.

If $N \geq q_a - 1$, the first q_a rows are an invertible triangular polynomial-frame change times the full original root Vandermonde. The roots are distinct since their real and imaginary index differences vanish separately. Hence $K_{a,N,s}$ is positive definite. Put

$$\widehat{D}_{a,N}^{(s)} = Z_a^* [K_{a,N,s} - K_{a,N,s} C_a^\top (\overline{C}_a K_{a,N,s} C_a^\top)^{-1} \overline{C}_a K_{a,N,s}] Z_a. \tag{NPK3}$$

Its exact value interpretation is

$$x^* \widehat{D}_{a,N}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \|E_{a,N,s}(Z_a x + C_a^\top \eta)\|^2. \tag{NPK4}$$

The block elimination in NPK3–4 is the classical Schur-complement method; see Stephen Boyd, Laurent El Ghaoui, Eric Feron and Venkataramanan Balakrishnan [H1, §2.1, pp. 7–8, (2.3)–(2.4)]. Their real symmetric presentation is used here in its Hermitian form: the two coordinate blocks are $Z_a x$ and $C_a^\top \eta$, and the lower positive block is $\overline{C}_a K_{a,N,s} C_a^\top$. Completing the square with conjugate

adjoints gives exactly the two cross factors displayed above. The conductor and its value map are the programme's separately proved constructions.

The positive normal equation gives NPK3. Independence of $[C_a^T, Z_a]$ and injectivity of E prove strict positivity for $x \neq 0$. This is the complete conductor quotient, including both cross factors and every lower column. By the NCT dual transport, it is the metric on the exact same quotient values as $D_{a,N-v}^{(s)}$ after the map $\Lambda_F \mapsto \Lambda_{E_A F}$. In these literal DNE coordinates the numerator is $Z_a x$. The factor $\alpha_A = v!/\mu_v$ remains in the separate original KRC/RRC primary-coordinate map; it is not inserted into, or removed from, NPK3.

J.2 Evaluation by two original recurrence polynomials

This is the Christoffel–Darboux formula and its confluent form from Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, eqs. (18.2.12)–(18.2.13)], with their $n = N$, their monic polynomials $p_n = p_{n,s}$, and their norm $h_n = h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$. In their prefactor both leading coefficients equal one. The arguments are exactly $z = (\mu - c_a)/i$ and $w = (\lambda - c_a)/i$; hence the row index is conjugated and the column index is not. NPK6 and the degree-zero term below prove the complex polynomial identity with this original mass, and differentiation in z proves the value at every coincident entry.

The entire observation kernel in NPK2 has the following exact formula. For two original nodes λ, μ let $z = (\mu - c_a)/i$ and $w = (\lambda - c_a)/i$. Then

$$K_{a,N,s}[\lambda, \mu] = \begin{cases} \frac{p_{N+1,s}(z)p_{N,s}(w) - p_{N,s}(z)p_{N+1,s}(w)}{h_{N,s}(z-w)}, & z \neq w, \\ \frac{p'_{N+1,s}(w)p_{N,s}(w) - p'_{N,s}(w)p_{N+1,s}(w)}{h_{N,s}}, & z = w. \end{cases} \quad (\text{NPK5})$$

The second case occurs among the actual grid entries: it includes $\mu = \beta_{a,a-u,w_0}$ when $\lambda = \beta_{a,u,w_0}$. It must not be omitted as a nongeneric coincidence.

Here is a direct proof with the original normalization. Write $d_n = n(n-1+b) = h_{n,s}/h_{n-1,s}$ for $n \geq 1$. Subtract the recurrence evaluated at z, w to obtain, for $n \geq 1$,

$$(z-w) \frac{p_{n,s}(z)p_{n,s}(w)}{h_{n,s}} = \frac{p_{n+1,s}(z)p_{n,s}(w) - p_{n,s}(z)p_{n+1,s}(w)}{h_{n,s}} - \frac{p_{n,s}(z)p_{n-1,s}(w) - p_{n-1,s}(z)p_{n,s}(w)}{h_{n-1,s}}. \quad (\text{NPK6})$$

At $n = 0$ the separate equality is $(z-w)/h_{0,s} = (p_{1,s}(z)p_{0,s}(w) - p_{0,s}(z)p_{1,s}(w))/h_{0,s}$; there is no $h_{-1,s}$ term. Summing this equality and NPK6 through N telescopes. The recurrence has real coefficients, so $p_{n,s}(\bar{y}) = \overline{p_{n,s}(y)}$; this identifies the sum with the Hermitian kernel NPK2. The numerator in NPK5 vanishes when $z = w$, and its derivative in z gives the displayed second line. This is a polynomial identity, so the limiting line holds without an analytic exclusion. At $N = 0$ both lines give $1/h_{0,s} = 1/(2\pi)^{s/2}$, confirming the mass and sign.

Consequently NPK3 is specified by the two original recurrence polynomials $p_{N,s}, p_{N+1,s}$ and the original finite matrices C_a, Z_a . It does not require a Taylor division by E_A to evaluate this transported form. NCT supplies the exact map back to the divided native form and its complete finite volume error.

J.3 The original eight flags and their numerator row update

Now fix $j = k + 4$, $k = 4\ell + 1 \geq 9$, and one of the original target orders $s = 1, j$. Put $q = q_j$, and retain the original fixed CPQ coefficient maps on $\mathcal{U} = \mathbb{C}^{4m_j}$:

$$\begin{aligned} I &= \text{im}(B_{\text{cof}} J_0), & B &= \text{im } B_{\text{cof}}, & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \mathbb{B}, \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J. \end{aligned} \quad (\text{NPK7})$$

The primitive index remains $e = 0$ or 1 with its original domain. All spaces in NPK7 are the same coefficient subspaces used in SFB and CPQ; the present construction changes none of their maps. For the proof, choose independent fixed columns X for each nonzero subspace, and use determinant one for the empty space. Set $\widehat{D}_N = \text{diag}(\widehat{D}_{j,N}^{(s)}, 4)$.

For $q \leq N \leq 2q$, let ξ_N be row N of $\mathbf{E}_{j,N,s}$, and evaluate all following old matrices at $N - 1$. Suppressing their j, s indices only in this display, put

$$\begin{aligned} L &= \overline{C}K_{N-1}C^\top, \quad T_0 = \mathbf{E}_{N-1}Z, \quad V_0 = \mathbf{E}_{N-1}C^\top, \\ \widehat{t}_N &= \xi_N Z, \quad \widehat{v}_N = \xi_N C^\top, \quad \widehat{\zeta}_N = \widehat{v}_N L^{-1} \widehat{v}_N^*, \\ \widehat{\rho}_N &= \widehat{t}_N - \widehat{v}_N L^{-1} V_0^* T_0, \quad \widehat{U}_N = \text{diag}(\widehat{\rho}_N / \sqrt{1 + \widehat{\zeta}_N}, 4), \\ \widehat{D}_N &= \widehat{D}_{N-1} + \frac{\widehat{\rho}_N^* \widehat{\rho}_N}{1 + \widehat{\zeta}_N}, \quad \widehat{\mathcal{D}}_N = \widehat{\mathcal{D}}_{N-1} + \widehat{U}_N^* \widehat{U}_N. \end{aligned} \tag{NPK8}$$

The inverse-update identity used in this square completion is recorded by William W. Hager [H2, p. 221, (2)]. In his $A - uv$ notation, set $A = L$, $u = -\widehat{v}_N^*$ and $v = \widehat{v}_N$; the denominator is $1 + \widehat{v}_N L^{-1} \widehat{v}_N^*$. The residual row and quotient metric in NPK8 are computed here.

These formulas retain the complete old conductor Gram L and its scalar denominator. To prove the update, write the old minimizer as $\eta_0 = -L^{-1}V_0^*T_0x$ and replace it by $\eta_0 + h$. The old squared norm is the old minimum plus h^*Lh ; the new row is $\widehat{\rho}_N x + \widehat{v}_N h$. Its remaining minimum is $|\widehat{\rho}_N x|^2 / (1 + \widehat{v}_N L^{-1} \widehat{v}_N^*)$, as follows by solving the positive normal equation, or by substitution of $h = -L^{-1} \widehat{v}_N^* \widehat{\rho}_N x / (1 + \widehat{\zeta}_N)$. This proves every line of NPK8.

The eight actual transported compressed matrices are therefore

$$\widehat{K}_{X,N} = \widehat{U}_N X (X^* \widehat{\mathcal{D}}_{N-1} X)^{-1} X^* \widehat{U}_N^*, \quad \widehat{r}_N = \frac{\widehat{\rho}_N \widehat{D}_{N-1}^{-1} \widehat{\rho}_N^*}{1 + \widehat{\zeta}_N}. \tag{NPK9}$$

For the empty subspace set $\widehat{K} = 0_{4 \times 4}$. The isometry by $\widehat{\mathcal{D}}_{N-1}^{1/2}$ writes $\widehat{K}_X = Y \Pi_X Y^*$, where Π_X is the orthogonal projector onto $\widehat{\mathcal{D}}_{N-1}^{1/2} \text{im } X$ and $Y Y^* = \widehat{r}_N I_4$. Consequently $0 \preceq \widehat{K}_X \preceq \widehat{r}_N I_4$ and inclusion of subspaces gives matrix order. All four eigenvalues, including zeros, are retained. The two original chains are exactly $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$.

Let $\sigma_X = 1$ for $X = T, K_c, W_i, K_i$ and $\sigma_X = -1$ for $X = W, S, J, I$. The determinant lemma applied to NPK8 gives

$$\begin{aligned} \widehat{\ell}_N &= \sum_X \sigma_X \log \det(I_4 + \widehat{K}_{X,N}), \\ \widehat{\mathcal{L}}_e &= \sum_{N=q}^{2q} \vartheta_N \widehat{\ell}_N, \quad \vartheta_q = \vartheta_{2q} = 1, \quad \vartheta_N = 2 \ (q < N < 2q), \\ \widehat{\mathcal{P}}_{k,e} &= \sum_{N=q}^{2q} \vartheta_N \sum_X \sigma_X \sum_{a=1}^4 \log \max(1, \widehat{\lambda}_{X,a,N}). \end{aligned} \tag{NPK10}$$

The native row corresponding to this N is exactly $n = N - v$. The weights sum to $2q$, and NPK10 has the unchanged four endpoints $q-1, q, 2q-1, 2q$. The quantity $\widehat{\mathcal{L}}_e$ is the complete CPQ common return formed with the transported metric: the same quotient determinant identities prove the signed eight-flag expression. No fixed frame determinant is discarded before taking the return.

J.4 Finite and limiting transfer of the complete signed profile

Denote NCT's nonnegative finite volume-error bounds for b_0 copies, quotient rank r , and an admissible original SGT radius ϱ by $\mathcal{U}_{\varrho;a,D,s,r}^{[b_0]}$ and $\mathcal{L}_{\varrho;a,D,s,r}^{[b_0]}$. They retain the actual SGT poles,

residues, zero-free radius, diagonal weights and complete conductor. Define

$$B_{a,D,s,r}^{[b_0]} = \inf_{\substack{0 < \rho < 1 \\ \rho \text{ admissible}}} \max(\mathcal{L}_{\rho;a,D,s,r}^{[b_0]}, \mathcal{U}_{\rho;a,D,s,r}^{[b_0]}),$$

$$\mathcal{N} = (q-1, q, 2q-1, 2q), \quad E_X = \sum_{N \in \mathcal{N}} B_{j,N-v,s,\dim X}^{[4]}, \quad E_{\text{flags}} = \sum_{X \in \{T, K_c, W_i, K_i, W, S, J, I\}} E_X. \quad (\text{NPK11})$$

At every endpoint NCT gives $|\log \det(X^* \widehat{\mathcal{D}}_N X) - \log \det(X^* \mathcal{D}_N X)| \leq B_{j,N-v,s,\dim X}^{[4]}$. Taking the four original signs and then the eight original signs proves

$$|\widehat{\mathcal{L}}_e - \mathcal{L}_e| \leq E_{\text{flags}}. \quad (\text{NPK12})$$

The infimum is over the same original function g and its proved admissible radii. It is nonnegative and finite, since SGT supplies at least one such finite bound. The endpoint inequality holds for every admissible radius, hence also for their infimum; attainment is unnecessary. Each $\dim X \leq 4m_j = O(k)$ and $N = O(k^2)$. For each fixed radius the NCT proof bounds the limsup of the displayed infimum divided by kq_k by a constant times $\log(1/\rho)$, with all remaining terms tending to zero. Only after this limit choose admissible radii tending to one. It follows that $B^{[4]} = o(kq_k)$ and $E_{\text{flags}} = o(kq_k)$. A fixed-radius enclosure itself is not asserted to be little-oh. Any particular finite admissible radius remains a usable, fully specified bound in NPK11–12.

For completeness the threshold error used here has a full direct proof. Put $h(x) = \log(1+x) - \log \max(1, x) = u(x) - v(x)$, where $u(x) = \log(1 + \min(x, 1))$, and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both u, v are increasing, vanish at zero and are at most $\log 2$. On any increasing four-term eigenvalue chain, the alternating sum of disjoint increments of u lies between zero and $u(r)$; adding the missing middle increment proves the upper bound. The same argument holds for v . The alternating h sum therefore lies between $-v(r)$ and $u(r)$. Apply this to each of the four ordered eigenvalues on each of the two chains in NPK9; matrix order implies eigenvalue order by the Rayleigh min–max formula. The difference of the two chain sums has absolute value at most $4(u(r) + v(r)) \leq 8 \log 2$ at each row. After the exact weights, both the original and transported threshold errors are at most $16q \log 2$. Combining this with NPK12 proves the finite receiver

$$|\mathcal{P}_{k,e} - \widehat{\mathcal{P}}_{k,e}| \leq E_{\text{flags}} + 32q_j \log 2 = o(kq_k). \quad (\text{NPK13})$$

This transfers the entire signed profile, with its correlations and zero eigenvalues. It does not assert a rowwise approximation of any individual eigenvalue, or a value for the transported profile.

J.5 All complementary returns and the original arithmetic action

For each of the original CPQ forms Q_Y on the specified quotient values (in particular $G_s, Q_{\text{rem}}, Q_b^E, A_{\text{inv}}$), form $\widehat{Q}_Y$ by the same restriction and full affine minimum using $\widehat{\mathcal{D}}_N$. If its rank is r_Y , NCT and the original four signs give the first line below. Define the transported target quotient and its return explicitly by $\widehat{H}_{B,j,N} = (B_j(\widehat{\mathcal{D}}_{j,N}^{(s)})^{-1} B_j^*)^{-1}$ and $\widehat{S}_{K,j} = \mathcal{R} \log \det \widehat{H}_{B,j,N}$, in the original onto value frame of B_j . Its full four-copy rank is $4t_j$. Then

$$|\widehat{V}_Y - V_Y| \leq E_Y := \sum_{N \in \mathcal{N}} B_{j,N-v,s,r_Y}^{[4]} = o(kq_k),$$

$$4\widehat{S}_{K,j} = \widehat{\mathcal{L}}_e + \widehat{V}_b + \widehat{V}_{\text{rem}} + \widehat{V}_s + \widehat{V}_A, \quad (\text{NPK14})$$

$$4(\widehat{S}_{K,j} - S_{K,j}) = (\widehat{\mathcal{L}}_e - \mathcal{L}_e) + \sum_{Y=b,\text{rem},s,A} (\widehat{V}_Y - V_Y).$$

The second line follows by block Schur factorization in precisely CPQ's fixed frames, so the third is its subtraction from CPQ20. Alternatively NCT applies directly to the rank- $4t_j$ target quotient,

yielding $|4\hat{S}_{K,j} - 4S_{K,j}| \leq E_{\text{target}}$, where $E_{\text{target}} = \sum_{N \in \mathcal{N}} B_{j,N-v,s,4t_j}^{[4]}$. Both enclosures hold for the same value, so their intersection is valid. This retains every complementary term and their exact correlation.

The independent proper-source form Q_{inv} uses its original degree- k frame, cutoff and center. Apply NCT there separately. For the exact relative determinant in NFR the identity is

$$\begin{aligned} \log \frac{\det \hat{A}_{\text{inv}}}{\det \hat{Q}_{\text{inv}}} - \log \frac{\det A_{\text{inv}}}{\det Q_{\text{inv}}} \\ = (\log \det \hat{A}_{\text{inv}} - \log \det A_{\text{inv}}) - (\log \det \hat{Q}_{\text{inv}} - \log \det Q_{\text{inv}}). \end{aligned} \quad (\text{NPK15})$$

The two finite NCT errors add to an enclosure; on the original rank- $O(k)$ and cutoff- $O(k^2)$ domains their four-return is $o(kq_k)$. The two distinct source orders and centers are unchanged in this statement. The EFP same-value maps and HJR inverse-metric identities act on the same quotient coordinates, so NPK14–15 can be substituted into them with their full finite error intervals. In the complete action $J = B + W - P$, transporting only the specified volume term gives exactly $\hat{J} - J = \hat{B} - B$ when the original recurrence window W and penalty P are retained. This supplies an explicit error-bearing map into that action, without assigning any missing intrinsic coefficient or cancelling an uncomputed term.

The result is a concrete finite observation problem on the original root grid: NPK3 and NPK5 calculate all transported forms from the original C_a, Z_a and Gamma recurrence; NPK8–10 calculate their full mixed profile; NPK12–15 prove their quantitative relation to the original divided forms. The surviving large-degree value of these original numerator subquotients is the next arithmetic calculation, and remains present in every displayed identity.

K Original-grid interpolation, relation norms, and an interior mixed row

Keep the exact original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, the fixed original five-orbit period and logarithm branch, and its complete nonreduced conductor. For $a \equiv 1 \pmod{4}$, $a \geq 9$, put

$$\begin{aligned} c &= a/2, \quad q = (a+1)^2, \quad q' = (a-7)^2, \quad \beta_{u,w} = c + (2u-a)\delta + i(2w-a)\gamma, \\ \chi(S) &= \prod_{u,w=0}^a (S - \beta_{u,w}), \quad E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{u,w} z}, \\ \mu_t &= E_A^{(t)}(0) \ (t \geq 0), \quad v = \text{ord}_0 E_A, \quad \mu_v \neq 0, \quad m = q - q' - v, \quad s \in \{1, a\}, \quad N \geq q-1. \end{aligned} \tag{IGI1}$$

Use the lexicographic order of the original grid. A single index λ below denotes a member of this ordered grid. The roots are distinct because real and imaginary parts separately determine both indices. The original bounds on v and the original period domain are unchanged. In particular $v < q$ and $m > 0$.

Retain exactly

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \\ \langle f, g \rangle_s &= \int_{\mathbb{R}} \overline{f(c+iy)} g(c+iy) dm_s(y). \end{aligned} \tag{IGI2}$$

The original SGT/NSO construction proves that these ϕ_n are orthonormal. Their leading coefficient is $i^{-n}/\sqrt{h_{n,s}}$. The inner product is conjugate-linear in its first argument; the functional dual throughout is complex-linear.

K.1 The full original evaluation map and its attained cardinal norm

Define the unchanged evaluation map and its literal cardinal section:

$$\begin{aligned} \text{ev}_N : \mathbb{C}[S]_{\leq N} &\longrightarrow \mathbb{C}^q, \quad P \longmapsto (P(\beta_\lambda))_\lambda, \\ L_\lambda(S) &= \frac{\chi(S)}{(S - \beta_\lambda)\chi'(\beta_\lambda)}, \quad \mathcal{I}x = \sum_{\lambda} x_\lambda L_\lambda(S). \end{aligned} \tag{IGI3}$$

Every displayed derivative is nonzero, since its root is simple. Evaluation at each root proves $\text{ev}_N \mathcal{I} = I_q$. Polynomial division gives, with all coefficients retained,

$$\ker \text{ev}_N = \chi \mathbb{C}[S]_{\leq N-q}, \quad P = \mathcal{I}(\text{ev}_N P) + \chi R, \tag{IGI4}$$

where the relation space is zero at $N = q-1$. Indeed division has a unique remainder of degree below q ; a remainder vanishing at all q distinct roots is zero. Thus IGI4 proves the entire kernel, surjectivity, and the exact relation to the cardinal coordinates.

Let $G_L = (\langle L_\lambda, L_\mu \rangle_s)$. At $N \geq q$ use the complete relation basis $R_j = \chi S^j$, $0 \leq j \leq N-q$, and put $G_R = (\langle R_i, R_j \rangle_s)$ and $B = (\langle R_i, L_\lambda \rangle_s)$. Then the actual value norm is

$$\begin{aligned} x^* Q_N x &= \min_{\text{ev}_N P=x} \|P\|_s^2, \quad Q_N = G_L - B^* G_R^{-1} B, \\ P_{N,x} &= \mathcal{I}x - \sum_j R_j (G_R^{-1} Bx)_j. \end{aligned} \tag{IGI5}$$

The positive quadratic minimum in IGI5 is the Schur-complement method [H1, §2.1, (2.3)–(2.4)] in the original cardinal/relation frame: its three blocks are G_L , B and G_R . The subsequent inverse evaluation metric retains the complex-linear transpose in the observation map.

For $N = q - 1$ put $Q_N = G_L$ and omit the empty second term. Completing the positive square in the coefficient of R proves IGI5: every other lift has squared norm equal to this minimum plus the positive squared norm of its difference in the relation space. The value Gram is positive definite because a zero minimum would be the zero polynomial with zero values. In particular no cross Gram B has been discarded.

The original numerator observation matrix is

$$\mathbf{E}_N[n, \lambda] = \phi_n(\beta_\lambda), \quad 0 \leq n \leq N, \quad K_N = \mathbf{E}_N^* \mathbf{E}_N. \quad (\text{IGI6})$$

Its first q rows are an invertible triangular change of the original Vandermonde, so $K_N > 0$. If $P = \sum z_n \phi_n$, then $\text{ev}_N P = \mathbf{E}_N^T z$, with an ordinary transpose. The unique coefficient minimum for this constraint is

$$z = \overline{\mathbf{E}_N} (\mathbf{E}_N^T \overline{\mathbf{E}_N})^{-1} x, \quad \boxed{Q_N = \overline{K_N^{-1}}}. \quad (\text{IGI7})$$

Multiplication proves the constraint. The displayed column is orthogonal to its kernel; its squared norm is $x^* (\mathbf{E}_N^T \overline{\mathbf{E}_N})^{-1} x$. Since $\mathbf{E}_N^T \overline{\mathbf{E}_N} = \overline{K_N}$, this is exactly IGI7. Equations IGI5 and IGI7 give the exact typed map between the full cardinal Schur minimum and the original functional covariance, including its necessary conjugation.

K.2 All conductor columns, vanishing moments, and quotient duals

Write $C = C_a$ and $Z = Z_{\text{exp},a}$ for the unchanged NCT6–7 matrices. Thus

$$N_{C^T \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}, \quad N_d(z) = \sum_{\lambda} d_{\lambda} e^{\beta_{\lambda} z}. \quad (\text{IGI8})$$

The columns of C^T are independent by injectivity of multiplication by the nonzero germ E_A and the lower Vandermonde. Define

$$G_v[d, \lambda] = \beta_{\lambda}^d \quad (0 \leq d < v), \quad \mathcal{N}_v = \ker G_v, \quad B_0 = [C^T, Z], \quad \mathcal{L}_v = \text{im } G_v^T. \quad (\text{IGI9})$$

At $v = 0$ both G_v and $\mathcal{L}_v$ have zero dimension. The original NCT7 frame proves $\text{im } B_0 = \mathcal{N}_v$ and its columns independent. The rows of G_v are independent by the upper Vandermonde. Moreover $G_v C^T = 0$ by IGI8, so $\mathcal{L}_v \subset \ker C$. Every member of $\mathcal{L}_v$ is the evaluation vector of a polynomial of degree below v , and every such vector occurs, by its displayed monomial columns.

The original numerator coefficient quotient has Gram

$$\begin{aligned} \widehat{D}_N &= Z^* \{K_N - K_N C^T (\overline{C} K_N C^T)^{-1} \overline{C} K_N\} Z, \\ x^* \widehat{D}_N x &= \min_{\eta} (Zx + C^T \eta)^* K_N (Zx + C^T \eta). \end{aligned} \quad (\text{IGI10})$$

Positive square completion gives both the formula and its unique minimum

$$\eta = -(\overline{C} K_N C^T)^{-1} \overline{C} K_N Zx.$$

Strict positivity follows from independence of B_0 .

The exact dual value space is

$$\mathcal{W} = \ker C / \mathcal{L}_v, \quad [z] \mapsto Z^T z \in \mathbb{C}^m. \quad (\text{IGI11})$$

This map is well defined since $Z^T G_v^T = 0$. Its kernel before quotienting is $\ker C \cap \ker Z^T = \ker B_0^T$. Because $\text{im } B_0 = \ker G_v$, the ordinary bilinear annihilator of that image is exactly $\text{im } G_v^T$: inclusion is immediate and both dimensions are v . Thus the kernel is precisely $\mathcal{L}_v$. The rank of the

independent rows of $[C; Z^\top]$ is $q - v$, so any values $(0, t)$ are attained. This proves the isomorphism in IGI11, with its full low-degree denominator retained.

Its attained value norm satisfies exactly

$$\begin{aligned} t^* \widehat{D}_N^{-1} t &= \min_{Cz=0, Z^\top z=t} z^* Q_N z \\ &= \min_{\substack{P \in \mathbb{C}[S]_{\leq N} \\ C \text{ ev}_N P=0, \quad Z^\top \text{ ev}_N P=t}} \|P\|_s^2. \end{aligned} \quad (\text{IGI12})$$

For a direct proof, the minimum under the onto map $A = [C; Z^\top]$ is $(AQ_N^{-1}A^*)^{-1}$ in its value coordinates, as follows from the lift $Q_N^{-1}A^*(AQ_N^{-1}A^*)^{-1}(0, t)^\top$ and orthogonality to $\ker A$. Since $Q_N^{-1} = \overline{K}_N$, the Schur complement of the first block of $AQ_N^{-1}A^*$ is precisely $\widehat{D}_N$. Its bottom block inverse gives IGI12. Taking the complete polynomial minimum IGI5 proves the second line. Thus this is the complex-linear dual of IGI10 with its full metric, not a coordinate dimension comparison.

The conductor condition has the exact polynomial interpretation

$$(C \text{ ev}_N P)_{u,w} = (\mathcal{T}_A P)(\beta_{a-8,u,w}), \quad C \text{ ev}_N P = 0 \iff \chi_{a-8} \mid \mathcal{T}_A P. \quad (\text{IGI13})$$

The first equality is IGI8 coefficient collection and exact root addition. The second uses every distinct lower root and polynomial division. It holds for every P of the stated degree, retaining all cancellations in $\mathcal{T}_A$. The fibre IGI12 minimizes over all resulting low-polynomial values, rather than setting them equal to zero.

For any original fixed coefficient subspace frame X in four copies, its restriction Gram is $X^* \text{diag}(\widehat{D}_N, 4)X$. For a quotient Y/K with fixed independent frame $[X_K, X_Q]$, its Gram is exactly

$$X_Q^* \widehat{D}_N X_Q - X_Q^* \widehat{D}_N X_K (X_K^* \widehat{D}_N X_K)^{-1} X_K^* \widehat{D}_N X_Q, \quad \widehat{D}_N = \text{diag}(\widehat{D}_N, 4). \quad (\text{IGI14})$$

Square completion proves the minimum on its original affine fibre. Applying IGI10–12 in each copy and these exact further minima gives every original CPQ restriction and quotient, with the same frames, full cross terms and values. No choice of orthogonal replacement flag enters IGI14.

K.3 The full relation Christoffel row and the original observation row

Let $\pi_r(S)$ be the unique monic degree- r polynomial minimizing $\|\chi P\|_s^2$, and put

$$\nu_{r,s} = \|\chi \pi_r\|_s^2, \quad \tau_r = \frac{\chi \pi_r}{\sqrt{\nu_{r,s}}}, \quad b_r = (\langle \tau_r, L_\lambda \rangle_s)_\lambda. \quad (\text{IGI15})$$

The monic quadratic minima are orthogonal to all lower-degree directions, by varying each complex coefficient in its real and imaginary parts. Thus the τ_r form an orthonormal basis of the entire relation space in IGI4. Orthogonal projection of every cardinal column onto that complete space gives

$$Q_N = G_L - \sum_{j=0}^{N-q} b_j^* b_j, \quad Q_N = Q_{N-1} - b_r^* b_r \quad (N = q + r). \quad (\text{IGI16})$$

Every row b_j is the actual cardinal/relation cross pairing.

Write $\xi_N = (\phi_N(\beta_\lambda))_\lambda$, the actual next original observation row. The coefficient of ϕ_N in τ_r is

$$t_N = i^N \sqrt{h_{N,s}/\nu_{r,s}}, \quad N = q + r. \quad (\text{IGI17})$$

This follows from their exact leading coefficients. Its old component $z_< = (\langle \phi_n, \tau_r \rangle_s)_{n < N}$ is orthogonal to every old relation, since τ_r is. Its old evaluation values are $-t_N \xi_N^\top$, since the full polynomial τ_r vanishes at every original node. It is therefore the minimum lift IGI7 at cutoff $N - 1$:

$$z_< = -t_N \overline{E_{N-1}} Q_{N-1} \xi_N^\top.$$

Pairing it with each old cardinal column gives $b_r = -\overline{t_N} \overline{\xi_N} Q_{N-1}$. Conjugation and IGI7 yield the exact phase-sensitive row formula

$$\boxed{\xi_N = -i^{-N} \sqrt{\nu_{r,s}/h_{N,s}} \overline{b_r} K_{N-1}.} \quad (\text{IGI18})$$

The missing phase cannot be supplied by a determinant identity alone; it has been computed from the original polynomial frame. Since τ_r has norm one, its old component has squared norm $1 - |t_N|^2$. The same minimum lift gives

$$\begin{aligned} b_r Q_{N-1}^{-1} b_r^* &= 1 - \frac{h_{N,s}}{\nu_{r,s}}, & T_{r,s}^\Gamma &:= \frac{\nu_{r,s}}{h_{N,s}} \geq 1, \\ \xi_N K_{N-1}^{-1} \xi_N^* &= T_{r,s}^\Gamma - 1, & \frac{\det K_N}{\det K_{N-1}} &= T_{r,s}^\Gamma. \end{aligned} \quad (\text{IGI19})$$

For the third formula, substitute IGI18 into the first, or use the norm of $z_<$ and $|t_N|^2 = h_N/\nu_r$. The determinant formula follows by the rank-one update $K_N = K_{N-1} + \xi_N^* \xi_N$. Positivity of Q_N also proves that the first scalar in IGI19 is strictly below one.

K.4 The retained low-moment correction and conductor correction

All matrices in this subsection are evaluated at cutoff $N - 1$. Suppress that cutoff only for $K = K_{N-1}$ and put

$$\begin{aligned} L &= \overline{C} K C^\top, \quad Z_\perp = Z - C^\top L^{-1} \overline{C} K Z, \quad \hat{\zeta}_N = \xi_N C^\top L^{-1} \overline{C} \xi_N^*, \\ \hat{\rho}_N &= \xi_N Z_\perp, \quad \hat{r}_N = \frac{\hat{\rho}_N \hat{D}_{N-1}^{-1} \hat{\rho}_N^*}{1 + \hat{\zeta}_N}, \\ d_N &= \xi_N K^{-1} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1} \xi_N^*. \end{aligned} \quad (\text{IGI20})$$

At $v = 0$ define $d_N = 0$ and omit its inverse. At $v > 0$ that inverse exists because G_v has full row rank. Every factor in d_N has its ordinary Hermitian adjoint as displayed; the defining moment map $G_v d$ itself remains complex-linear.

The exact metric projector identity is

$$B_0 (B_0^* K B_0)^{-1} B_0^* = K^{-1} - K^{-1} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1}. \quad (\text{IGI21})$$

To prove it, apply the isometry $K^{1/2}$ to $\ker G_v$. Its orthogonal complement is the range of $K^{-1/2} G_v^*$. The orthogonal projector onto that complement is $K^{-1/2} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1/2}$; it is self-adjoint, idempotent, and fixes that range. The projector onto the former space is $K^{1/2} B_0 (B_0^* K B_0)^{-1} B_0^* K^{1/2}$. Their sum is the identity, and multiplication on both sides by $K^{-1/2}$ proves IGI21. The empty case is the same formula with zero complement.

Multiply IGI21 by ξ_N and its adjoint. The full restricted numerator leverage is $T_{r,s}^\Gamma - 1 - d_N$. Completing the old conductor/numerator square as in IGI10 decomposes that same quantity into $\hat{\zeta}_N + \hat{\rho}_N \hat{D}_{N-1}^{-1} \hat{\rho}_N^*$. Consequently

$$\boxed{1 + \hat{r}_N = \frac{T_{r,s}^\Gamma - d_N}{1 + \hat{\zeta}_N}, \quad 0 \leq d_N + \hat{\zeta}_N \leq T_{r,s}^\Gamma - 1.} \quad (\text{IGI22})$$

This identity preserves both original corrections. At $v = 0$ the complementary space is zero, so d_N vanishes identically. For $v > 0$ retain the displayed exact expression, including any value zero proved from the original data. Invertibility of the full Vandermonde alone does not eliminate that expression. Substitution of IGI18 into IGI20 evaluates every correction in the same cardinal/relation cross row b_r , with its exact phase.

For the actual target $a = j = k + 4$, let the eight fixed flags be precisely $T_e, K_c, W_i, K_i, W, S, J, I$ of NPK7, with their two nested chains and their original frames X . Then

$$\widehat{U}_N = \text{diag}(\widehat{\rho}_N / \sqrt{1 + \widehat{\zeta}_N}, 4), \quad \widehat{K}_{X,N} = \widehat{U}_N X (X^* \widehat{\mathcal{D}}_{N-1} X)^{-1} X^* \widehat{U}_N^*. \quad (\text{IGI23})$$

These are the same eight NPK9 matrices, now with their exact polynomial expressions IGI5,15,18,20. Set an empty compression equal to the zero 4×4 matrix. The signed row keeps the positive signs T_e, K_c, W_i, K_i and negative signs W, S, J, I . All four eigenvalues, including zero and subthreshold eigenvalues, remain present. No conclusion about these eight orientations is deduced from the scalar $T_{r,s}^\Gamma$ alone: the entire cross row b_r , the original C, Z, G_v , and all eight X are still explicit factors of IGI23.

K.5 A finite calculation at the second original target row

The original $Q_a(y) = i^{-q} \chi(c + iy)$ is real and even. Its root set is invariant under conjugation and negation, by the exact index permutations $u \mapsto a - u$ and $(u, w) \mapsto (a - u, a - w)$, and q is even. Thus $|Q_a(y)|^2 dm_s(y)$ is an even positive measure. The original degree-one monic minimizer is therefore exactly

$$\pi_1(S) = S - c, \quad \nu_{1,s} = \int_{\mathbb{R}} y^2 |Q_a(y)|^2 dm_s(y), \quad T_{1,s}^\Gamma = \frac{\nu_{1,s}}{M_s(q+1)!(s/2)_{q+1}}. \quad (\text{IGI24})$$

Indeed a monic linear polynomial in y is $y + z$; its weighted squared norm is the norm of y plus $|z|^2$ times the positive mass, since the mixed integral of y is zero. Its minimum is at $z = 0$. The map back to S is the exact monic map $p(y) \mapsto ip((S - c)/i)$.

For a finite evaluation with no quadrature, write $Q_a(y) = \sum_{t=0}^q c_t y^t$ using every original root coefficient. For each integer $b \geq 0$ let J_b be the $(b+1)$ -square Jacobi matrix on indices $0, \dots, b$ with diagonal zero and off-diagonal $\sqrt{(n+1)(n+s/2)}$. The original recurrence proves

$$\int y^b dm_s(y) = M_s(J_b^b)_{00}, \quad \nu_{1,s} = M_s \sum_{t,u=0}^q \bar{c}_t c_u (J_{t+u+2}^{t+u+2})_{00}. \quad (\text{IGI25})$$

Starting at degree zero, b applications of the recurrence cannot visit degree greater than b , proving the finite moment identity and its retained mass. Expansion of the original squared polynomial gives the second identity. These are exact finite sums.

At the non-first target row $N = q + 1$, define the fully specified nonnegative number

$$\mathcal{B}_{q+1,s} = \log \frac{T_{1,s}^\Gamma - d_{q+1}}{1 + \widehat{\zeta}_{q+1}}. \quad (\text{IGI26})$$

Here $T_{1,s}^\Gamma$ is IGI24–25, b_1 is the explicit integral of $\chi(S)(S - c)/\sqrt{\nu_{1,s}}$ against each literal cardinal column, and IGI18,20 compute $d_{q+1}, \widehat{\zeta}_{q+1}$ from those original finite coefficients. Its nonnegativity and the finite estimates

$$0 \leq \mathcal{B}_{q+1,s} \leq \log T_{1,s}^\Gamma - \log(1 + \widehat{\zeta}_{q+1}) \leq \log T_{1,s}^\Gamma, \quad |\widehat{\ell}_{q+1}| \leq 4\mathcal{B}_{q+1,s}, \quad |\widehat{p}_{q+1}| \leq 4\mathcal{B}_{q+1,s}, \quad 0 \leq \Delta \widehat{V}_X \leq 4\mathcal{B}_{q+1,s} \quad (X = b, \text{rem}, s, A) \quad (\text{IGI27})$$

follow as follows. IGI22 makes IGI26 exactly $\log(1 + \widehat{r}_{q+1})$, and $d_N \geq 0$ gives the first line. For the second line, ordered projectors along each original flag chain give ordered compressed matrices. The two disjoint increment sums on a chain are nonnegative and at most the full increment $4\log(1 + \widehat{r}_N)$. Their difference is the signed eight-term row. The same argument works with the increasing function $\log \max(1, x)$ on each of the four ordered eigenvalues, giving the threshold bound without omitting zeros. For the four CPQ quotients the exact pairs are $W/W_i, \mathcal{U}/T_e, S/K_c, I/K_i$. Full Schur elimination gives their increment as $b_Y - b_K$, between zero and the same full

four-copy budget. Their original fixed frame factors cancel only in this increment. The interior weight in the original return is two, so its entire weighted contribution is bounded by $8\mathcal{B}_{q+1,s}$. For the native determinant row, write the eight original frames as $\mathcal{X} = \{T_e, K_c, W_i, K_i, W, S, J, I\}$ and define, using exactly the nonnegative NPK11/NCT20 finite error constants,

$$E_{\text{row}}(N) = \sum_{X \in \mathcal{X}} \left(B_{j,N-1-v,s,\dim X}^{[4]} + B_{j,N-v,s,\dim X}^{[4]} \right), \quad N = q + 1. \quad (\text{IGI27a})$$

The two endpoint determinant errors, with their original signs, give $|\delta\mathcal{L}_{e,N-v} - \widehat{\ell}_N| \leq E_{\text{row}}(N)$. The native and transported threshold rows each differ from their own determinant row by at most $8 \log 2$, by the two-chain proof NPK13 before applying its endpoint weights. The full triangle inequality therefore gives

$$\begin{aligned} |\delta\mathcal{P}_{e,N-v} - \widehat{p}_N| &\leq E_{\text{row}}(N) + 16 \log 2, \\ |\delta\mathcal{P}_{e,N-v}| &\leq 4\mathcal{B}_{q+1,s} + E_{\text{row}}(N) + 16 \log 2. \end{aligned} \quad (\text{IGI27b})$$

For each complementary quotient of its original rank r_X , the direct determinant increment error is instead the sum $B_{j,N-1-v,s,r_X}^{[4]} + B_{j,N-v,s,r_X}^{[4]}$; no threshold error is added to a determinant increment. The actual second-row interior weight is two, so every complete bound just displayed, including its error terms, is multiplied by two in the full return. These statements compare the exact scalar increments; they assert no eigenwise approximation of the eight matrices.

K.6 The exact interface to the older arithmetic norm results

The scalar $\nu_{r,s}$ above is measured in dm_s . In GRA1 the original arithmetic measure, at multiplicity one and degree a , is $d\mu_a(y) = w_h^{*a}(y)dy$ with $w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$ and the complete original quartet polynomial h . At order one the present Gamma measure is exactly $dm_1 = d\sigma = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi)$, including its mass $\sqrt{2\pi}$. The identity on the unchanged polynomial space at centre $c = a/2$ is the actual comparison map; its density ratio is literally

$$\frac{d\mu_a}{dm_s}(y) = \frac{w_h^{*a}(y)}{\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2}. \quad (\text{IGI28})$$

The denominator is positive for real y . This is not a claim that the ratio is constant.

For degree cutoff d the proved all-polynomial GRA5 constants are a_a/D_d and A_a , with $D_d = [1 + 4(d + M_h)^2]^{M_h}$ and the unchanged full GRA3 definitions of a_a, A_a, M_h . The proof there uses the actual convolution upper and lower densities and the original Gamma recurrence, and gives $(a_a/D_d)H_1 \preceq H_{\text{ar}} \preceq A_a H_1$. For $L = (s - 1)/4$, Gamma recurrence gives the exact whole-source identity and finite comparison

$$\begin{aligned} dm_s &= \rho_s \prod_{u=0}^{L-1} (y^2 + (2u + 1/2)^2) dm_1, & \rho_s &= \frac{(2\pi)^{2L}}{(1/2)_{2L}}, \\ \mathfrak{a}_s &= \rho_s \prod_{u=0}^{L-1} (2u + 1/2)^2, & \mathfrak{b}_{s,d} &= \rho_s \prod_{u=0}^{L-1} \{4(d + u + 1)^2 + (2u + 1/2)^2\}, \\ \mathfrak{a}_s H_1 &\preceq H_s \preceq \mathfrak{b}_{s,d} H_1, & \frac{a_a}{D_d \mathfrak{b}_{s,d}} H_s &\preceq H_{\text{ar}} \preceq \frac{A_a}{\mathfrak{a}_s} H_s. \end{aligned} \quad (\text{IGI29})$$

For completeness the upper Gamma comparison follows from the orthonormal order-one recurrence: on degree at most d , $\|yp\|_1 \leq 2(d + 1)\|p\|_1$. Multiplication by $iy - t_u$ at step u has squared norm at most $4(d + u + 1)^2 + t_u^2$ times the preceding norm, while pointwise $|iy - t_u|^2 = y^2 + t_u^2 \geq t_u^2$. Multiplying all factors proves the stated comparison, with its exact scalar ρ_s . Combining it with GRA5 in the displayed directions proves the last line. Empty products at $s = 1$ equal one.

Apply these inequalities to every polynomial in the same monic fibre χP , with $d = q + r$, and take minima. This gives

$$\frac{a_a}{D_{q+r} \mathbf{b}_{s,q+r}} \nu_{r,s} \leq \nu_r^{\text{ar}} \leq \frac{A_a}{\mathbf{a}_s} \nu_{r,s}. \quad (\text{IGI30})$$

The identical inequalities apply to all fixed cardinal evaluation fibres IGI5, or the constrained fibres IGI12, by taking the same full minimum on both sides. Inversion and the conjugation in IGI7 give their precise reverse dual inequalities. Thus GRA's arithmetic norm and the numerator Gamma norm have a proved full form map before any scalar conclusion is transported.

On the order-one Gamma relation space, the monic map $U_r P(y) = i^{-r} P(c + iy)$ has inverse $i^r p((S - c)/i)$ and satisfies $U_{q+r}(\chi P) = Q_a U_r P$. Hence the original GRF whole-form comparison applies literally to $\nu_{r,1}$ and to the full relation Gram G_R in IGI5 on precisely its proved finite guards. The GRE reference minima then give its stated scalar bounds, and IGI29 transports any such valid full-form inequality to the other original source order with the displayed constants. The degree-one evaluation IGI24–27 uses its direct proof and does not assert the higher-degree GRF guards at $r = 1$.

The cardinal cross rows b_r remain the pairings IGI15 in these maps. A bound for their single quadratic norm IGI19 determines the complete scalar observation leverage, but does not specify their values on the original conductor and flag columns in IGI20,23. The exact remaining map is IGI18 followed by those literal column maps; IGI22 exhibits its separate nonnegative low-moment and conductor contributions. These retained expressions are the surviving interior mixed calculation, including all complementary CPQ returns, after the scalar relation norms have been supplied by the proved older results.

L The original native interior rows and a growing initial segment

Fix the original simple quartet, complete nonreduced conductor, period, and logarithm branch. Put

$$\begin{aligned}
 j &= k + 4, \quad k = 4l + 1 \geq 9, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \\
 c' &= j/2 - 4, \quad L = q - v, \quad m = L - q', \\
 s &\in \{1, j\}, \quad v = \text{ord}_0 E_A, \quad \mu_t = E_A^{(t)}(0) \ (t \geq 0), \quad \mu_v \neq 0, \\
 E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \tag{NIB1} \\
 \chi_j(S) &= \prod_{u,w=0}^j (S - \beta_{j,u,w}), \quad \chi_-(S') = \chi_{j-8}(S'), \\
 \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma.
 \end{aligned}$$

The polynomial family, weight and norm are the classical Meixner–Pollaczek formulas, with exact parameter and measure transport proved in HPS1–4; see [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)].

All functions and matrices are the original NSO/CPQ objects. The proved original bound $v \leq 80$ suffices here; the stronger BTK bound is retained on its stated domain. Keep the exact source measure, mass, recurrence, and phase

$$\begin{aligned}
 dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\
 h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{h_{n,s}}, \quad \|P\|_s^2 = \int |P(c' + iy)|^2 dm_s(y). \tag{NIB2}
 \end{aligned}$$

The polynomials are the original orthonormal basis. Inner products are conjugate-linear in their first argument, and functional values are complex-linear.

L.1 The complete original annihilator at every subsequent row

Let $F_b = N_b/E_A$, $1 \leq b \leq m$, be the fixed original NSO12 numerator columns. The original functional frame consists of these m functionals together with all q' lower evaluations. Denote its matrix of values on $\phi_0, \dots, \phi_d$ by $\mathcal{F}_d = [V_d, T_d]$, with $L = q' + m$ columns. Explicitly, $V_d[n, (u, w)] = \phi_n(\beta_{j-8,u,w})$ comprises every lower evaluation column, and $T_d[n, b] = \Lambda_{F_b}(\phi_n)$ comprises the original analytic NSO columns, for $0 \leq n \leq d$. At $d = L - 1$ this matrix is invertible: a generating linear combination whose first L derivatives vanish has, after multiplication by E_A , an upper-grid numerator whose first $L + v = q$ derivatives vanish. The upper Vandermonde kills this numerator; the original conductor quotient frame kills the quotient coefficients; the lower Vandermonde kills the remaining coefficients. The triangular polynomial-frame change has nonzero diagonal, proving the same claim for native rows. Appending rows therefore preserves column rank L at every $d \geq L - 1$.

Define the original polynomial translation and its complete relation sequence by

$$\begin{aligned}
 (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \tag{NIB3} \\
 g_t &= \mathcal{T}_A(\chi_j(S) S^t), \quad \ell_t = \mu_v \binom{q+t}{v}, \quad n_t = q + t - v = L + t.
 \end{aligned}$$

Binomial expansion gives $\deg g_t = n_t$ and $\text{lc } g_t = \ell_t \neq 0$. For every original numerator column, the finite Leibniz identity gives

$$\Lambda_{F_b}(g_t) = \Lambda_{N_b}(\chi_j S^t) = 0.$$

Every lower evaluation also annihilates g_t : root addition $\beta_{j-8,a,b} + \beta_{8,u,w} = \beta_{j,a+u,b+w}$ makes each shifted summand zero separately. Thus χ_- divides every g_t , including all of its original coefficients.

At the later native row $n = L + r$, $r \geq 0$, the polynomial map of all L original functional values has rank L , by the preceding old-row argument. Its kernel in degree at most n has dimension $n + 1 - L = r + 1$. The polynomials $g_0, \dots, g_r$ are independent by their distinct degrees and lie in that kernel. Consequently

$$\ker(\Lambda_{\text{original}}|_{\mathbb{C}[S']_{\leq L+r}}) = \text{span}(g_0, \dots, g_r). \quad (\text{NIB4})$$

This identifies the complete annihilator, rather than selecting a proper subspace of relations.

Let $u_t = g_t/\ell_t$ and define its full attained monic norm by

$$\eta_{r,s} = \min_{z_0, \dots, z_{r-1}} \left\| u_r + \sum_{t < r} z_t u_t \right\|_s^2, \quad \nu_{b,s}^- = \min_{P \text{ monic, deg } P=b} \|\chi_- P\|_s^2. \quad (\text{NIB5})$$

The relation minimum NIB5–6 uses classical Schur-complement elimination [H1, §2.1, (2.3)–(2.4)], here on the full Gram of $u_0, \dots, u_r$. This credits the matrix method; the complete annihilator and the later native Gamma bounds are proved in this section.

Both are positive attained minima. For $r = 0$ the first minimum has no variables. For $r > 0$, putting $G_{<} = (\langle u_t, u_h \rangle_s)_{t,h < r}$ and $z = (\langle u_t, u_r \rangle_s)_{t < r}$, its exact value is

$$\eta_{r,s} = \|u_r\|_s^2 - z^* G_{<}^{-1} z, \quad u_r^{\min} = u_r - \sum_{t < r} u_t (G_{<}^{-1} z)_t. \quad (\text{NIB6})$$

Square completion proves this formula, preserving every preceding relation and cross pairing. Scaling by ℓ_t is the displayed invertible map to its monic affine fibre; every ℓ_t remains in NIB3,5,6. Since χ_- divides every g_t , the first fibre is contained in the second monic fibre at $b = m + r$. It follows directly that $\eta_{r,s} \geq \nu_{m+r,s}^-$.

L.2 The exact corrected native row at every interior position

At cutoff $n - 1 = L + r - 1$, retain the original V_0, T_0 , their new rows v_n, t_n , and

$$\begin{aligned} A_L &= V_0^* V_0, \quad D_0 = T_0^* (I - V_0 A_L^{-1} V_0^*) T_0, \\ \zeta_n &= v_n A_L^{-1} v_n^*, \quad \rho_n = t_n - v_n A_L^{-1} V_0^* T_0, \quad r_n = \rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n), \\ \zeta_{\text{all},n} &= [v_n, t_n] ([V_0, T_0]^* [V_0, T_0])^{-1} [v_n, t_n]^*. \end{aligned} \quad (\text{NIB7})$$

All inverses exist by the full column ranks established above. Unlike the first row, the old full matrix need not be square.

The unit relation $\tau_r = u_r^{\min} / \sqrt{\eta_{r,s}}$ has last native coefficient $c_n = i^n \sqrt{h_{n,s} / \eta_{r,s}}$, and annihilates the entire original functional frame. Its old coefficient column is orthogonal to every old annihilator by NIB4,6. If $w_n = [v_n, t_n]$ and $F_0 = [V_0, T_0]$, it is consequently the minimum-norm solution of $F_0^T z = -c_n w_n^T$, namely

$$z = -c_n \overline{F_0} (F_0^T \overline{F_0})^{-1} w_n^T.$$

Its squared norm is $|c_n|^2 \zeta_{\text{all},n}$ by conjugating the positive scalar. Since the full relation has norm one, adding $|c_n|^2$ proves the first formula below. Applying the identical argument to the full lower-evaluation annihilator $\chi_- \mathbb{C}[S']_{\leq m+r}$ proves the second:

$$1 + \zeta_{\text{all},n} = \frac{\eta_{r,s}}{h_{n,s}}, \quad 1 + \zeta_n = \frac{\nu_{m+r,s}^-}{h_{n,s}}. \quad (\text{NIB8})$$

That lower annihilator equality follows from ordinary division by the complete distinct-root polynomial χ_- ; its monic degree n fibre is exactly NIB5 with $b = m + r$.

The invertible old column operation replacing T_0 by $(I - V_0 A_L^{-1} V_0^*) T_0$ makes its Gram block diagonal and its new row $[v_n, \rho_n]$. Thus $\zeta_{\text{all},n} = \zeta_n + \rho_n D_0^{-1} \rho_n^*$, also for the rectangular old matrix. Together with NIB8 this gives

$$1 + r_n = \frac{\eta_{r,s}}{\nu_{m+r,s}}, \quad 0 \leq \log(1 + r_n) \leq \log \frac{\|g_r\|_s^2}{|\ell_r|^2 \nu_{m+r,s}}. \quad (\text{NIB9})$$

The last inequality uses the actual trial u_r in its attained fibre NIB5. It changes neither g_r nor any earlier relation. The measure, lower minimum, and all cross terms in the exact identity remain the original ones.

L.3 All finite constants for the original later row

Define, for every $r \geq 0$,

$$\begin{aligned} L_s &= (s-1)/4, \quad \theta = \pi/2, \quad h = 1/(q+r), \quad \kappa = \theta - h > 0, \quad \sigma = (\delta^2 + \gamma^2)^{1/2}, \\ Q_- &= q' + L_s, \quad Q_+ = q + r + L_s, \quad R' = (j-8)\sigma, \quad M = \max\{2Q_-/\theta, 2R'\}, \\ B_+ &= \max\{1, (j+8)\sigma, 2L_s\}, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|, \quad \beta_s = M_s/(\sqrt{2}\Gamma(s/2)), \\ L_h &= \pi(\sin h)^{3/2}/\Gamma(3/4)^2, \quad U_h = \Gamma(1/4)^2(\sin h)^{-1/2}/(2\pi), \\ F_M &= 2Q_- \log M - \theta M, \quad \epsilon_- = (\theta - 2Q_-/M) + Q_-/M^2 + h(M+1) + 4q'R'/M, \\ C_b &= (2b+1) \binom{2b}{b}^2. \end{aligned} \quad (\text{NIB10})$$

The classical inputs are Euler's integral, the Gamma functional equation, and Euler's reflection formula [H22, eqs. (5.2.1), (5.5.1), (5.5.3)]; HPS5 records the exact reflection specialization.

The Gamma recurrence and the exact Euler/reflection estimates give

$$\begin{aligned} dm_s &= \beta_s \prod_{u=0}^{L_s-1} (y^2 + (2u+1/2)^2) d\sigma(y), \quad d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi), \\ L_h e^{-(\theta+h)|y|} &\leq d\sigma/dy \leq U_h e^{-\kappa|y|}. \end{aligned} \quad (\text{NIB11})$$

Here is the full scalar estimate: rotate Euler's integral for $\Gamma(b + iy/2)$ through $\text{sgn}(y)(\pi/2 - h)$. The integrable bound on that ray is $\Gamma(b)(\sin h)^{-b} e^{-(\pi/2-h)|y|/2}$. The closing arcs vanish because $b > 0$ and their cosines are at least $\sin h > 0$. Squaring this at $b = 1/4$ proves the upper bound. The exact reflection product $|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$, the same upper bound at $3/4$, and $\cosh(\pi y) \leq e^{\pi|y|}$ prove the lower bound. Iterating the Gamma recurrence proves the first identity, with its factor 4^{-L_s} cancelled by the displayed original density's 2^{2L_s} . No mass is suppressed.

On the interval $[M, M+1]$ every original lower root in the y coordinate has modulus at most $R' \leq M/2$. Since $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$, $|\chi_-(c' + iy)|^2 \geq y^{2q'} e^{-4q'R'/M}$ there. The Gamma product contributes at least y^{2L_s} . For $y = M+t$, $0 \leq t \leq 1$, the elementary inequality $\log(1+t/M) \geq t/M - t^2/(2M^2)$ gives

$$2Q_- \log y - \theta y \geq F_M - (\theta - 2Q_-/M) - Q_-/M^2.$$

Together with $e^{-hy} \geq e^{-h(M+1)}$, this gives the lower bound $\beta_s L_h e^{F_M - \epsilon_-}$ for the complete weighted density on this original interval.

The classical Rodrigues and norm identities used for the next minimum are those of the Legendre family; the checked source is Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, eq. (18.5.5), Table 18.5.1, and the Legendre and shifted-Legendre rows of Table 18.3.1]. With their degree b and their argument $u = 2x - 1$, the polynomial below is exactly their $P_b(2x - 1)$.

The change of variables contributes $du = 2dx$, so its norm on $[0, 1]$ is $1/(2b+1)$ and its leading coefficient in x is $\binom{2b}{b}$. The square-completion argument and the explicit S' -coordinate map below give the complex monic minimum in the original fibre, with $b = m + r$.

For a complex monic degree- b polynomial on an interval of length one the exact minimum squared integral is C_b^{-1} . To prove the constant, the Rodrigues polynomial $P_b(x) = \frac{1}{b!}(d/dx)^b[x^b(x-1)^b]$ has leading coefficient $\binom{2b}{b}$. Integrating by parts b times proves its orthogonality to every smaller degree; the same calculation gives $\int_0^1 P_b^2 = \binom{2b}{b} \int_0^1 x^b(1-x)^b dx = 1/(2b+1)$. The last beta integral is $b!^2/(2b+1)!$ by successive integration by parts. Orthogonal square completion proves the complex monic minimum. The map $P(S') \mapsto i^{-b}P(c' + i(M+x))$ is monic in x with inverse $i^b p((S' - c')/i - M)$ and preserves absolute values on the interval. Taking $b = m + r$ therefore proves

$$\nu_{m+r,s}^- \geq \frac{\beta_s L_h}{C_{m+r}} e^{F_M - \epsilon_-}. \quad (\text{NIB12})$$

For the upper estimate keep the original roots fixed. For every real y , upper root and degree-eight shift, $|c' + iy + \beta_{8,u,w} - \beta_{j,a,b}| \leq |y| + (j+8)\sigma$. Also $|c' + iy + \beta_{8,u,w}| \leq |y| + j/2 + 8\sigma \leq |y| + (j+8)\sigma$, because the original $\gamma > 2$ gives $\sigma > 1/2$. Consequently the actual polynomial from NIB3 obeys

$$|g_r(c' + iy)| \leq A_{\text{abs}}(|y| + B_+)^{q+r}.$$

Every Gamma product factor is at most $(|y| + B_+)^2$. Integrating the two half-line bounds from NIB11, substituting $t = |y| + B_+$ and enlarging to $t \geq 0$ gives

$$\|g_r\|_s^2 \leq 2\beta_s U_h A_{\text{abs}}^2 e^{\kappa B_+} \frac{(2Q_+)!}{\kappa^{2Q_++1}}. \quad (\text{NIB13})$$

The last integral is an exact factorial integral by integration by parts. All cancellation in g_r remains exact in NIB3–9; this is an upper inequality for that polynomial's complete norm.

Combining the two estimates in the identical original norm gives the finite bound, valid for every $j \geq 13$ and $r \geq 0$,

$$0 \leq \log(1 + r_{L+r}) \leq E_{j,s,r},$$

$$E_{j,s,r} = 2 \log \frac{A_{\text{abs}}}{|\mu_v^{(q+r)}|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) - (2Q_+ + 1) \log \kappa - F_M + \epsilon_- + \log C_{m+r}. \quad (\text{NIB14})$$

The identical β_s cancels between the two proved bounds; the source measure has not been divided by its mass. The exact ratio in NIB9 is at least one, so the displayed upper constant is necessarily nonnegative.

L.4 Uniform initial interior segment and all complementary terms

Fix any real $C > 0$. Uniformly over integers $0 \leq r \leq Cj$, both original orders $s = 1, j$, and the fixed original quartet and conductor,

$$E_{j,s,r} = O_{\text{actual},C}(j \log j). \quad (\text{NIB15})$$

Here is a direct verification. Eventually $M = 2Q_-/\theta$, since $Q_- \asymp j^2$ while $R' = O(j)$. Then $F_M = f_\theta(Q_-)$, where $f_\theta(x) = 2x \log(2x/\theta) - 2x$, and $\epsilon_- = O(j)$ uniformly. Also $Q_+ = q + r + L_s \asymp j^2$, $Q_+ - Q_- = 16j - 48 + r = O_C(j)$, and $B_+ = O(j)$. The factorial integral estimates

$$b \log b - b + 1 \leq \log(b!) \leq b \log b - b + 1 + \log b \quad (b \geq 1)$$

follow by integrating the increasing logarithm between successive integers. Applied to $b = 2Q_+$, they place $\log((2Q_+)!) - (2Q_+ + 1) \log \kappa$ at $f_\theta(Q_+) + O(\log j)$, because $\log(\theta/\kappa) \leq h/\kappa$ and $Q_+ h = O(1)$. Integration of $f'_\theta(x) = 2 \log(2x/\theta)$ between Q_- and Q_+ now costs $O_C(j \log j)$. Further, $\log C_{m+r} \leq \log(2m + 2r + 1) + 4(m + r) \log 2 = O_C(j)$, and $2h/\pi \leq \sin h \leq h$ gives $\log(U_h/L_h) =$

$O(\log j)$. The fixed v and nonzero μ_v imply $|\log |\mu_v(\frac{q+r}{v})|| = O(\log j)$ uniformly on this range. These estimates prove NIB15 directly from its fully finite expression, with no differentiation of an asymptotic determinant error.

At each of these rows the original four-copy metric update is $\mathcal{D}_1 = \mathcal{D}_0 + U_n^* U_n$, with $U_n = \text{diag}(\rho_n/\sqrt{1+\zeta_n}, 4)$. Its full relative compression is $r_n I_4$. For each original fixed flag X , retain $K_{X,n} = U_n X (X^* \mathcal{D}_0 X)^{-1} X^* U_n^*$ and all four eigenvalues. The two chains $J \subset W_i \subset W \subset T_e$ and $K_i \subset I \subset K_c \subset S$ give, exactly as in FMG27, two nonnegative disjoint increment sums each at most $4\log(1+r_n)$. Their difference is the complete original signed row. Applying the same reasoning to the increasing threshold function on each ordered eigenvalue proves

$$|\delta \mathcal{L}_{e,n}| \leq 4E_{j,s,r}, \quad |\delta \mathcal{P}_{e,n}| \leq 4E_{j,s,r}, \quad 0 \leq \delta V_{X,n} \leq 4E_{j,s,r} \quad (X = b, \text{rem}, s, A). \quad (\text{NIB16})$$

For the last assertion, the original quotients are W/W_i , $\mathcal{U}/T_e$, S/K_c , and I/K_i . Block Schur elimination expresses each successive quotient determinant increment as the difference of its two restriction increments. Nested projections bound that difference between zero and the full update budget. This argument retains all original fixed-frame factors until they cancel in the successive increment, and leaves every quotient cross term in its attained metric.

The original target row for native degree $L+r$ is $N = q+r$. On $0 \leq r \leq q$, its original four-endpoint weights are one at $r=0, q$ and two otherwise. For any integer $0 \leq R \leq \min(q, \lfloor Cj \rfloor)$ let ϑ_r denote those unchanged weights. Summing NIB16 gives the explicit segment bound

$$\begin{aligned} \left| \sum_{r=0}^R \vartheta_r \delta \mathcal{L}_{e,L+r} \right|, \quad \left| \sum_{r=0}^R \vartheta_r \delta \mathcal{P}_{e,L+r} \right| &\leq 4 \sum_{r=0}^R \vartheta_r E_{j,s,r}, \\ 0 \leq \sum_{r=0}^R \vartheta_r \delta V_{X,L+r} &\leq 4 \sum_{r=0}^R \vartheta_r E_{j,s,r} = O_{\text{actual},C}(j^2 \log j) = o(kq_k). \end{aligned} \quad (\text{NIB17})$$

The same last upper order holds for both absolute values in the first line. In particular $r=1$ is an actually non-first original interior row with its complete error bound. The full initial linear segment, together with every complementary CPQ increment, can be removed from the leading kq_k calculation by the stated finite error. All subsequent rows remain in their original correlated expressions, and the complete CPQ identity $4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A$ is preserved. No intrinsic coefficient for the surviving interior profile or conclusion about RH is assigned by this segment estimate.

L.5 Uniform bound on the full original row range and a superlinear prefix

The same finite NIB14 constant has the stronger uniform estimate

$$\boxed{E_{j,s,r} = O_{\text{actual}}((j+r) \log j) \quad (0 \leq r \leq q, \ s \in \{1, j\})}. \quad (\text{NIB18})$$

Here the original quartet, conductor, period and logarithm branch are fixed before the limit. We prove the uniformity from the full finite expression. On this range $q \leq q+r \leq 2q$, so $1/(2q) \leq h \leq 1/q$. Also $Q_- = q' + L_s \asymp j^2$ and $Q_+ = q+r + L_s \asymp j^2$, with constants valid for both allowed orders. The actual lower interval therefore has $M = 2Q_-/\theta$ eventually, independently of r . Thus $F_M = f_\theta(Q_-)$ and $\epsilon_- = \theta^2/(4Q_-) + h(M+1) + 4q'R'/M = O_{\text{actual}}(j)$ uniformly. In particular no new interval or root substitution has been made in the exact finite bound.

The factorial inequalities used in NIB15 give, uniformly here,

$$\log((2Q_+)!) - (2Q_+ + 1) \log \kappa = f_\theta(Q_+) + O(\log j).$$

Indeed the factorial error is $O(\log Q_+)$, the remaining $-\log \kappa$ is bounded, and changing κ to θ costs at most $(2Q_+ + 1)h/\kappa = O(1)$. On the entire interval $[Q_-, Q_+]$, $|f'_\theta(x)| = |2 \log(2x/\theta)| = O(\log j)$. Its length is exactly $16j - 48 + r$. Integration of this derivative therefore gives $f_\theta(Q_+) - f_\theta(Q_-) =$

$O((j+r)\log j)$. The other terms are bounded directly: $\kappa B_+ = O_{\text{actual}}(j)$, $\log C_{m+r} \leq \log(2m + 2r + 1) + 4(m+r)\log 2 = O(j+r)$, and $2h/\pi \leq \sin h \leq h$ gives $\log(U_h/L_h) = O(\log j)$ uniformly for $q \leq q+r \leq 2q$. Finally, because v is fixed and $\mu_v \neq 0$, the finite product for $\binom{q+r}{v}$ has logarithm $O_{\text{actual}}(\log j)$ uniformly on that same range. Every term in NIB14 is now controlled by the claimed bound. Its nonnegativity was already proved in NIB14, completing NIB18.

For every integer $0 \leq R \leq q$ retain exactly the original weights $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ for $0 < r < q$. Define its fully finite prefix budget

$$\mathcal{B}_{j,s}(R) = 4 \sum_{r=0}^R \vartheta_r E_{j,s,r} = O_{\text{actual}}((R+1)(j+R)\log j). \quad (\text{NIB19})$$

The estimate follows by summing NIB18 with $\vartheta_r \leq 2$ and $\sum_{r=0}^R (j+r) = (R+1)j + R(R+1)/2 \leq (R+1)(j+R)$. For every such R , the two absolute prefix sums in NIB17 and each of the four nonnegative CPQ prefix sums are bounded by this same $\mathcal{B}_{j,s}(R)$: this follows by summing the actual inequalities NIB16 with the retained positive weights. Thus NIB19 extends the range of the proved prefix estimate; it changes no original row or endpoint.

In particular take the explicit integer prefix

$$R_j = \left\lfloor \frac{j^{3/2}}{\log j} \right\rfloor, \quad \boxed{\mathcal{B}_{j,s}(R_j) = O_{\text{actual}}\left(j^{5/2} + \frac{j^3}{\log j}\right) = o(kq_k)}. \quad (\text{NIB20})$$

For $j \geq 13$, $\sqrt{j} \log j > 1$, so $0 \leq R_j < j^2 < q$. The point $r = R_j$ is consequently an interior row of the original window, and retains weight two. Also $R_j + 1 \leq j^{3/2}/\log j + 1$ and $j + R_j \leq j + j^{3/2}/\log j$. Multiplying these inequalities and then by $\log j$ gives an upper bound $j^{5/2} + j^3/\log j + j \log j + j^{3/2}$, with the last two terms absorbed in the displayed order. Since $k = j - 4$ and $q_k = (j - 3)^2$, division by kq_k tends to zero. This proves NIB20 for both original source orders and both fixed primitive indices, simultaneously in the signed loss, the signed threshold profile, and all four complementary returns. It is a prefix decomposition of the unchanged rows $0 \leq r \leq q$; the original final endpoint remains $r = q$. Every row beyond R_j remains in its full correlated tail, with no coefficient assigned to that remaining expression.

M The controlled original prefix and its complete remaining return

Keep the original fixed quartet, period, logarithm branch, complete conductor, and the two source pairs of CPQ1. Write $j = k + 4$, $q = q_j = (j + 1)^2$, $L = q - v$, and $s = s_1 \in \{1, j\}$. For every original target cutoff $q - 1 \leq N \leq 2q$, set $D_N = D_{j, N-v}^{(s)}$ and $\mathcal{D}_N = \text{diag}(D_N, 4)$. Every coefficient flag and value frame below is precisely its CPQ frame. NIB proves, for each $0 \leq r \leq q$, the full original update

$$D_{q+r} = D_{q+r-1} + u_r^* u_r, \quad u_r = \frac{\rho_{L+r}}{\sqrt{1 + \zeta_{L+r}}}, \quad a_r = u_r D_{q+r-1}^{-1} u_r^* = \frac{\eta_{r,s}}{\nu_{m+r,s}^-} - 1 \geq 0. \quad (\text{IPR1})$$

Here $\eta_{r,s}$ is the full previous-relation Schur minimum, not the norm of an arbitrarily selected relation, and $\nu_{m+r,s}^-$ is the original lower ideal minimum. The finite NIB14 constant gives $0 \leq \log(1 + a_r) \leq E_{j,s,r}$. Define the retained weights and budget

$$\vartheta_0 = \vartheta_q = 1, \quad \vartheta_r = 2 \ (0 < r < q), \quad \mathfrak{E}_{s,R} = \sum_{r=0}^R \vartheta_r E_{j,s,r} \quad (0 \leq R \leq q). \quad (\text{IPR2})$$

All sums use the existing rows of the original window.

M.1 Exact prefix and tail maps on the original evaluation sites

For a scalar function f of the original cutoff define $\delta_r f = f(q + r) - f(q + r - 1)$ and $\mathcal{R}f = -f(q - 1) - f(q) + f(2q - 1) + f(2q)$. For $0 \leq R < q$ direct telescoping gives

$$\begin{aligned} \mathcal{R}_{\leq R} f &:= \sum_{r=0}^R \vartheta_r \delta_r f = 2f(q + R) - f(q) - f(q - 1), \\ \mathcal{R}_{> R} f &:= \sum_{r=R+1}^q \vartheta_r \delta_r f = f(2q - 1) + f(2q) - 2f(q + R), \\ \mathcal{R}f &= \mathcal{R}_{\leq R} f + \mathcal{R}_{> R} f. \end{aligned} \quad (\text{IPR3})$$

Indeed the first sum is $f(q) - f(q - 1)$ plus twice the successive increments from q to $q + R$. The second has twice the increments from $q + R$ to $2q - 1$ and the final single increment from $2q - 1$ to $2q$. This also proves the formulas at $R = 0$ and $R = q - 1$. At $R = q$ define $\mathcal{R}_{\leq q} = \mathcal{R}$ and $\mathcal{R}_{> q} = 0$; the displayed three-site formulas are not used at that last index. Each operator annihilates constants. Thus the same fixed frame factors that cancel under $\mathcal{R}$ also cancel separately under both operators. No representative depending on N is substituted for a fixed frame.

For a row sequence that is not itself a determinant increment, in particular the original threshold row p_r , the prefix and tail mean exactly $\sum_{r \leq R} \vartheta_r p_r$ and $\sum_{r > R} \vartheta_r p_r$. Formula IPR3 is applied only to an actual scalar potential f , such as a logarithmic determinant.

M.2 Both original allocations and all four complementary quotients

Let $B_j : \mathbb{C}^{m_j} \rightarrow \mathbb{C}^{t_j}$ be the original onto value map and choose fixed independent columns J_j for its kernel and a fixed right inverse X_j . The square matrix $C_j = [J_j, X_j]$ is invertible: applying B_j to $J_j x + X_j y = 0$ gives $y = 0$, and then independence gives $x = 0$. Put

$$H_N = (B_j D_N^{-1} B_j^*)^{-1}, \quad H_N^0 = J_j^* D_N J_j, \quad |\det C_j|^2 \det D_N = \det H_N \det H_N^0. \quad (\text{IPR4})$$

The block minimum in IPR4 is the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)] on the unchanged frame $C_j = [J_j, X_j]$. Its exact determinant factor $|\det C_j|^2$ is retained below.

For fixed value y , every lift is $X_j y + J_j x$. Completing its positive quadratic form gives the full Schur complement of H_N^0 in $C_j^* D_N C_j$. Its inverse is the bottom-right block of $C_j^{-1} D_N^{-1} (C_j^*)^{-1}$, equal to $B_j D_N^{-1} B_j^*$ because $B_j C_j = [0, I]$. This proves both the literal value minimum and IPR4 with its exact frame factor.

Restriction and minimum over the same affine fibre preserve $D_N \succeq D_{N-1}$. Hence the two logarithmic determinant increments in IPR4 are nonnegative; their sum, by the determinant lemma applied to IPR1, is $\log(1 + a_r)$. Consequently, for $Z \leq R$ or $> R$, define the actual partial allocations

$$S_K^Z = \mathcal{R}_Z \log \det H_N, \quad S_R^Z = \mathcal{R}_Z \log \det H_N^0, \quad \mathcal{T}^Z = \mathcal{R}_Z \log \det D_N. \quad (\text{IPR5})$$

They satisfy exactly

$$\begin{aligned} S_K^Z + S_R^Z &= \mathcal{T}^Z, \quad S_K^Z, S_R^Z \geq 0, \\ 0 \leq S_K^{\leq R}, S_R^{\leq R} &\leq \mathcal{T}^{\leq R} = \sum_{r=0}^R \vartheta_r \log(1 + a_r) \leq \mathfrak{E}_{s,R}. \end{aligned} \quad (\text{IPR6})$$

Their full sums are the original HJR32 allocations, with the same value maps and native source order; no allocation fraction is introduced.

For each $e = 0, 1$ let $\mathcal{L}_e(N)$ be $\log(\omega_{\text{conn},e}(N) \omega_{\text{inv}}(N))$. For $X = b, \text{rem}, s, A$ let $Q_X(N)$ be respectively $Q_b^E, Q_{\text{rem},e}, G_{s,e}, A_{\text{inv}}$, and put $V_X^Z = \mathcal{R}_Z \log \det Q_X(N)$ and $\mathcal{L}_e^Z = \mathcal{R}_Z \mathcal{L}_e(N)$. The full fixed-frame determinant identities CPQ7–12 prove the following, separately for the prefix and tail:

$$4S_K^Z = \mathcal{L}_e^Z + V_b^Z + V_{\text{rem}}^Z + V_s^Z + V_A^Z. \quad (\text{IPR7})$$

To see the cancellation explicitly, CPQ8 gives $\log \omega_{\text{conn}} = \log \det G_t - \log \det G_s$. The original invariant quotient gives $\log \omega_{\text{inv}} = \log \det H_{\text{inv}} - \log \det A_{\text{inv}}$ plus its fixed frame constant. The full target factorization gives $4 \log \det H_N = \log \det H_{\text{inv}} + \log \det Q_b^E + \log \det G_t + \log \det Q_{\text{rem}}$ plus its fixed frame constant. Subtraction proves IPR7 after applying either operator in IPR3, since both kill those constants.

Every one of these quotients is taken in its original ambient metric and complete kernel fibre. NIB16's same-row bound and its two nested flag chains give

$$\begin{aligned} |\mathcal{L}_e^{\leq R}| &\leq 4\mathfrak{E}_{s,R}, \quad |\mathcal{P}_e^{\leq R}| \leq 4\mathfrak{E}_{s,R}, \\ 0 \leq V_X^{\leq R} &\leq 4\mathfrak{E}_{s,R} \quad (X = b, \text{rem}, s, A). \end{aligned} \quad (\text{IPR8})$$

Here $\mathcal{P}_e^{\leq R}$ is the weighted original threshold row sum, not the different quantity $\mathcal{P}_e^{\text{rest}}$ in CPQ11. IPR7 retains the correlation between all five terms; IPR6 supplies the stronger direct bound on their combined value $4S_K^{\leq R}$. The terms are not assigned independent extremizers.

The NIB all-row estimate gives $\mathfrak{E}_{s,R} = O_{\text{actual}}((R+1)(j+R) \log j)$ uniformly for $0 \leq R \leq q$. In particular the explicitly evaluated prefix length

$$R_j = \left\lfloor \frac{j^{3/2}}{\log j} \right\rfloor \quad \text{satisfies} \quad \mathfrak{E}_{s,R_j} = O_{\text{actual}}(j^{5/2} + j^3 / \log j) = o(kq_k). \quad (\text{IPR9})$$

For all sufficiently large j it lies below q . The finite budgets IPR2 apply also before that threshold, with the prefix restricted to the original $R \leq q$ domain. Since $kq_k = (j-4)(j-3)^2 \asymp j^3$, each displayed order assertion follows by division. This selects no new programme endpoint: IPR3 is an identity for every original prefix, and IPR9 evaluates one range already controlled by the proved finite inequality.

M.3 Exact transport of the two partial returns

Use the original NCT complex-linear map $\Lambda_F \mapsto \Lambda_{E_A F}$, its ordinary-transpose matrix, and the full lower quotient. For each of the eight original flag frames X and each CPQ value quotient Y , denote its native and transported logarithmic determinants by $f_X, \hat{f}_X$ or $f_Y, \hat{f}_Y$. The original finite NCT20/NPK11 bound is

$$|\hat{f}_Y(N) - f_Y(N)| \leq B_{Y,N}, \quad B_{Y,N} = B_{j,N-v,s,\text{rank } Y}^{[4]} \geq 0. \quad (\text{IPR10})$$

All frames and values on the two sides are identical; the conjugation in the inverse evaluation metric remains IGI7. For $0 \leq R < q$ linearity of IPR3 now proves the finite bounds

$$\begin{aligned} |\hat{V}_Y^{\leq R} - V_Y^{\leq R}| &\leq 2B_{Y,q+R} + B_{Y,q} + B_{Y,q-1} =: E_Y^{\leq R}, \\ |\hat{V}_Y^{> R} - V_Y^{> R}| &\leq B_{Y,2q-1} + B_{Y,2q} + 2B_{Y,q+R} =: E_Y^{> R}. \end{aligned} \quad (\text{IPR11})$$

At $R = 0$, repeated occurrences in the displayed upper bound may also be combined using their actual opposite signs; the uncombined nonnegative bound is valid. This uses only the three evaluation sites and does not sum a determinant error separately for every row. For the signed eight-flag profile sum these bounds over the eight original frames to obtain E_{flags}^Z . For S_K apply them directly to the four-copy target quotient of rank $4t_j$, yielding $|4\hat{S}_K^Z - 4S_K^Z| \leq E_{\text{target}}^Z$. The transported full factorization also gives the exact equality

$$4(\hat{S}_K^Z - S_K^Z) = \hat{\mathcal{L}}_e^Z - \mathcal{L}_e^Z + \sum_{X=b,\text{rem},s,A} (\hat{V}_X^Z - V_X^Z). \quad (\text{IPR12})$$

Both enclosures concern this same value and may be intersected. For the other allocation use the original four-copy kernel frame $\text{diag}(J_j, 4)$. Its restricted determinant is exactly $(\det H_N^0)^4$ in this direct-sum frame. A change to another fixed original kernel frame multiplies it by a cutoff-independent positive factor, annihilated by both operators. Thus this original restriction flag has return $4S_R^Z$, its transported restriction has return $4\hat{S}_R^Z$, and IPR11 at its actual rank $4 \dim \ker B_j$ gives $|4\hat{S}_R^Z - 4S_R^Z| \leq E_{\text{kernel}}^Z$. This supplies the direct same-value transport of the second allocation.

For $0 \leq R < q$, each original or transported threshold row differs from its own signed determinant row by at most $8 \log 2$, as proved by NPK13's two ordered chains. The two row sequences therefore satisfy

$$\begin{aligned} |\hat{\mathcal{P}}_e^{\leq R} - \mathcal{P}_e^{\leq R}| &\leq E_{\text{flags}}^{\leq R} + 16(2R+1) \log 2, \\ |\hat{\mathcal{P}}_e^{> R} - \mathcal{P}_e^{> R}| &\leq E_{\text{flags}}^{> R} + 16\{2(q-R)-1\} \log 2. \end{aligned} \quad (\text{IPR13})$$

Indeed the original prefix weights sum to $1 + 2R$, and the tail weights sum to $2(q-R)-1$. These bounds compare scalar sums; no approximation of individual eigenvalues is asserted. NCT's fixed-radius estimate and subsequent radius limit are uniform over the unchanged cutoff interval $q-1 \leq N \leq 2q$. All ranks here are $O(j)$, so each finite three-site error in IPR11 is $o(kq_k)$ by exactly that order of limits. The extra threshold error is $O(q)$, also $o(kq_k)$. Thus IPR8–9 transport the full controlled prefix, while IPR3 and IPR12 retain the entire uncomputed tail and all its correlated quotients.

M.4 Receiving the estimate in the existing source and action maps

Every equation above holds separately for $s = 1$ and $s = j$ and for $e = 0, 1$. For example

$$\mathcal{L}_{e,s}^{> R} = \mathcal{L}_{e,s} - \mathcal{L}_{e,s}^{\leq R}, \quad V_{X,s}^{> R} = V_{X,s} - V_{X,s}^{\leq R}, \quad S_{K,s}^{> R} = S_{K,s} - S_{K,s}^{\leq R}. \quad (\text{IPR14})$$

Consequently the exact endpoint-difference identities PRT14–20, and the original two-order intervals CPQ19 and HJR33, propagate by subtracting the corresponding prefix on each side. For the positive quotient X , for instance, its order-difference change is in $[-4\mathfrak{E}_{j,R}, 4\mathfrak{E}_{1,R}]$, because

$V_{\bar{X},j}^{\leq R}, V_{\bar{X},1}^{\leq R}$ lie respectively in $[0, 4\mathfrak{E}_{j,R}]$ and $[0, 4\mathfrak{E}_{1,R}]$. For the signed profile its absolute change is at most $4(\mathfrak{E}_{j,R} + \mathfrak{E}_{1,R})$. All these corrections are $o(kq_k)$ at IPR9; none changes a previously proved leading coefficient or supplies the unknown mixed one.

The proper source Q_{inv} still uses degree k , its own centre and its displaced cutoffs. Its exact relation to the target form A_{inv} remains NFR/EFP's logarithmic ratio $\log \det A_{\text{inv}} - \log \det Q_{\text{inv}}$. Applying IPR3 to this difference gives the difference of the two partial returns, with each form evaluated at its original prescribed arguments. No NIB bound for the target is transferred to the proper source without its own proved metric comparison. Writing V_Q for that unchanged proper-source return, the precise change on removing the target prefix is

$$(V_A^{>R} - V_Q) - (V_A - V_Q) = -V_A^{\leq R}.$$

The removed contribution is $V_A^{\leq R}$, bounded by $4\mathfrak{E}_{s,R}$; the proper-source value itself remains.

Finally, in the existing complete action $J = B + W - P$, every use of an IPR3 volume decomposition is made by literal substitution $B = B^{\leq R} + B^{>R}$ for that specified original volume contribution, retaining its original coefficient. Then exactly $J = (B^{>R} + W - P) + B^{\leq R}$. The recurrence window W and penalty P are untouched by this algebraic identity. In particular the controlled prefix cannot be used to cancel an unevaluated tail or penalty. Equations IPR3–14 give the full maps and finite errors needed for that substitution on the original mixed and CPQ terms. The remaining calculation is their correlated interior return at $q + R_j, 2q - 1, 2q$, with their full Schur denominators.

N Human sources for the original Gamma polynomial metric

The Gamma orthogonal polynomials used in NIB2 belong to the classical Meixner–Pollaczek family. The external formulas used here are credited to the human-authored DLMF chapter by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt: the weight and squared norm are equations (18.19.8)–(18.19.9), and the recurrence is equation (18.22.8) [H21]. The DLMF source notes trace these formulas to M. E. H. Ismail’s *Classical and Quantum Orthogonal Polynomials in One Variable*, corrected 2009 reprint, equations (5.9.1), (5.9.8) and (5.9.9). The directly checked formulas in this repair are the DLMF equations; that source trail does not represent a claim that Ismail’s book text was independently inspected here.

Here is the exact dictionary, preserving the original measure and mass. For the original positive source order s , put $\lambda = s/4$ and retain $M_s = (2\pi)^{s/2}$. Write $P_n^{(\lambda)}(x; \varphi)$ for the cited Meixner–Pollaczek polynomial. Then the NIB2 polynomial is

$$p_{n,s}(y) = n!P_n^{(s/4)}(y/2; \pi/2). \quad (\text{HPS1})$$

Indeed the cited recurrence with $x = y/2$ and $\varphi = \pi/2$, multiplied by $n!$, gives

$$p_{n+1,s}(y) = yp_{n,s}(y) - n(n-1+s/2)p_{n-1,s}(y). \quad (\text{HPS2})$$

Its initial values are $p_{0,s} = 1$ and $p_{1,s} = y$, which determine the complete sequence uniquely by induction. The cited leading coefficient $(2\sin\varphi)^n/n!$ becomes exactly one in HPS1.

In the cited convention the squared norm at $\varphi = \pi/2$ is

$$\int_{\mathbb{R}} |\Gamma(\lambda + ix)|^2 |P_n^{(\lambda)}(x; \pi/2)|^2 dx = \frac{2\pi\Gamma(n+2\lambda)}{2^{2\lambda}n!}. \quad (\text{HPS3})$$

The same source gives orthogonality of different degrees. Substitute $y = 2x$, so $dy = 2dx$, and multiply the polynomial by $n!$ as in HPS1. For the unchanged original measure

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy$$

the full norm is therefore

$$\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} 2(n!)^2 \frac{2\pi\Gamma(n+s/2)}{2^{s/2}n!} = M_s n! \frac{\Gamma(n+s/2)}{\Gamma(s/2)} = M_s n!(s/2)_n. \quad (\text{HPS4})$$

This proves the original $h_{n,s}$, including its mass. For complex polynomials the map $f(y) \mapsto f(2x)$ is an invertible polynomial map. The explicit measure factor above describes its norm transport; neither variable change nor polynomial rescaling deletes that factor. In the original complex coordinate the resulting basis remains $\phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{h_{n,s}}$, with unchanged centre c' and phase i^{-n} in its leading coefficient. Thus HPS1–4 apply separately to $s = 1$ and $s = j$ without replacing either order.

The Gamma inputs to NIB11 are also classical human mathematics. Euler’s integral is DLMF (5.2.1); the functional equation and reflection formula are DLMF (5.5.1) and (5.5.3), in the chapter by R. A. Askey and R. Roy [H22]. Its acknowledgements explicitly credit P. J. Davis’s Chapter 6 in Abramowitz and Stegun as a partial source. The substitution $z = 1/4 + iy/2$ in the reflection formula gives

$$|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = \frac{\pi^2}{|\sin(\pi/4 + i\pi y/2)|^2} = \frac{2\pi^2}{\cosh(\pi y)}, \quad (\text{HPS5})$$

because $|\sin(a + ib)|^2 = \sin^2 a + \sinh^2 b$. The points used have real part $1/4$, so none is an excluded integral pole of the cited identity. Iteration of the cited functional equation gives the finite Gamma product in NIB11. Its contour rotation, subsequent finite inequalities, and application to the actual conductor remain the calculations proved in NIB11–20. They do not confer original authorship of the classical Gamma identities or the Meixner–Pollaczek family on this research programme.

R1 Additional complete receiving providers

R2 The original conductor at every source endpoint

Retain the complete simple-quartet conductor of OCF and AKJ, its actual original period matrix, and $k = 4l + 1 \geq 9$ on the proved five-orbit locus. Write

$$\begin{aligned}
q &= (k+1)^2, & k' &= k-8, & q' &= (k-7)^2, & \Delta &= q - q' = 16k - 48, \\
c &= k/2, & c' &= k/2 - 4, & \beta_{ij} &= c + (2i - k)\delta + i(2j - k)\gamma, \\
& & \beta'_{ab} &= c' + (2a - k')\delta + i(2b - k')\gamma, \\
A &= \sum_{r,s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0, & b_{rs} &= 4 + (2r - 8)\delta + i(2s - 8)\gamma, \\
\mu_j &= \sum_{r,s} a_{rs} b_{rs}^j, & v &= \min\{j : \mu_j \neq 0\} \leq 80.
\end{aligned} \tag{CEP1}$$

The factor i before an imaginary term is the imaginary unit; subscript indices are integers in their displayed ranges. The original polynomials are $\chi_k(S) = \prod_{i,j=0}^k (S - \beta_{ij})$ and $\chi_{k'}(S') = \prod_{a,b=0}^{k'} (S' - \beta'_{ab})$. Both have distinct roots. Set $E_k = \mathbb{C}[S]/(\chi_k)$ and $E_{k'} = \mathbb{C}[S']/(\chi_{k'})$. Every source below has the original order $s \in \{1, k\}$ and full mass $M_s = (2\pi)^{s/2}$, with $b = s/2$ and $h_n = M_s n! (b)_n$. In particular the lower grid does not change source order k to $k-8$.

R2.1 The full source, remainder and primary-coordinate square

For every polynomial define the original finite-translation map

$$(\mathcal{T}_A p)(S') = \sum_{r,s=0}^8 a_{rs} p(S' + b_{rs}). \tag{CEP2}$$

For $0 \leq a, b \leq k'$ and $0 \leq r, s \leq 8$, direct substitution gives $\beta'_{ab} + b_{rs} = \beta_{a+r, b+s}$. All indices on the right lie between zero and k . Consequently

$$(\mathcal{T}_A(\chi_k f))(\beta'_{ab}) = 0 \quad \text{for every polynomial } f. \tag{CEP3}$$

Division by the monic $\chi_{k'}$ and evaluation at its distinct roots then show $\chi_{k'} \mid \mathcal{T}_A(\chi_k f)$. Thus CEP2 induces the specified complex-linear map $\bar{\mathcal{T}}_A : E_k \rightarrow E_{k'}$.

In the original ordered primary coordinates its matrix is

$$C_{(a,b),(i,j)} = \begin{cases} a_{i-a, j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad C \in \text{Mat}_{q' \times q}(\mathbb{C}). \tag{CEP4}$$

Evaluation of CEP2 proves this entry formula, including its absence of conjugation or binomial coefficients. Its ordinary transpose $C^\top$ is the exact biform coefficient multiplication by the nonzero polynomial A . That multiplication is injective because the biform ring embeds in a polynomial integral domain. Therefore C has rank q' and is surjective. With the original ordered Vandermonde matrices $V_\beta, V_{\beta'}$, the polynomial coordinate matrix is exactly $V_{\beta'}^{-1} C V_\beta$. The kernel is the AKJ15 space

$$V_A = \{[p] \in E_k : \chi_{k'} \mid \mathcal{T}_A p\}, \quad \dim V_A = \Delta. \tag{CEP5}$$

CEP3 proves independence of the representative, and CEP4 proves the dimension. The original common kernel $K = \ker \Lambda$ is contained in V_A by the full conductor inclusion OCF17.

For every endpoint $N \in \{q-1, q, 2q-1, 2q\}$, put $d = N - v$ and define

$$\begin{aligned}
\mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, & \mathcal{P}'_d &= \mathbb{C}[S']_{\leq d}, \\
U_N : \mathcal{P}_N &\rightarrow E_k, & p &\mapsto [p], & V_d : \mathcal{P}'_d &\rightarrow E_{k'}, & g &\mapsto [g].
\end{aligned}$$

The top horizontal map is precisely $\mathcal{T}_{A,N} : \mathcal{P}_N \longrightarrow \mathcal{P}'_{N-v}$. Taylor's finite identity and the first nonzero moment show that CEP2 lowers every degree at least v by exactly v and kills exactly $\mathbb{C}[S]_{<v}$. The exact commutative square is

$$\begin{array}{ccc} \mathcal{P}_N & \xrightarrow{\mathcal{T}_{A,N}} & \mathcal{P}'_{N-v} \\ U_N \downarrow & & \downarrow V_{N-v} \\ E_k & \xrightarrow{\bar{\mathcal{T}}_A} & E_{k'}. \end{array} \quad (\text{CEP6})$$

The lower remainder is surjective because $N - v \geq q - 1 - v \geq q' - 1$; the latter inequality is $\Delta \geq v$, and $\Delta \geq 96 > 80 \geq v$.

The map is complex-linear with its following full action defect. If $M_S, M_{S'}$ denote the original multiplication operators, then

$$CM_S - M_{S'}C = C_b, \quad (C_b)_{(a,b),(i,j)} = \begin{cases} a_{i-a,j-b}b_{i-a,j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{CEP7})$$

Indeed each entry is multiplied by $\beta_{ij} - \beta'_{ab} = b_{i-a,j-b}$. In the original centred coordinates $Y = (M_S - cI)/i$, $Y' = (M_{S'} - c'I)/i$, this becomes $CY - Y'C = (C_b - 4C)/i$. Thus the exact relation to multiplication retains its coefficient defect.

R2.2 The literal finite inverse at the larger cutoffs

On $\mathcal{P}_N$, define the literal finite symbol

$$B_{N+1}(z) = 1 + \sum_{j=v+1}^N \frac{v! \mu_j}{j! \mu_v} z^{j-v}, \quad \mathcal{T}_{A,N} = \frac{\mu_v}{v!} \partial_S^v B_{N+1}(\partial_S). \quad (\text{CEP8})$$

The expression on the right is evaluated in the output variable S' . Taylor expansion of each term in CEP2 through degree N proves this identity coefficient by coefficient. Its constant term one makes $B_{N+1}(\partial_S)$ invertible: its difference from the identity lowers degree, so the inverse is its finite nilpotent Neumann sum through power N . A right inverse of ∂_S^v on the polynomial spaces exists by integration of each monomial; composition proves surjectivity of the top map in CEP6.

The relation to the earlier literal B_q of AKJ14 is

$$\begin{aligned} D_{N,q} &= B_{N+1}(\partial_S) - B_q(\partial_S) = \sum_{j=q}^N \frac{v! \mu_j}{j! \mu_v} \partial_S^{j-v}, \\ B_{N+1}(\partial_S)^{-1} - B_q(\partial_S)^{-1} &= -B_{N+1}(\partial_S)^{-1} D_{N,q} B_q(\partial_S)^{-1}, \\ \mathcal{T}_{A,N} - \frac{\mu_v}{v!} \partial_S^v B_q(\partial_S) &= \sum_{j=q}^N \frac{\mu_j}{j!} \partial_S^j. \end{aligned} \quad (\text{CEP9})$$

Empty sums are zero. The first two differences have image contained in $\mathbb{C}[S]_{\leq N-q+v}$; the last has image contained in $\mathbb{C}[S']_{\leq N-q}$. The inverse formula follows by multiplication, and its degree assertion follows since both inverse factors preserve every degree prefix. At $N = q - 1$ all differences vanish. At the other three endpoints these are the full retained defects; their ranks are not assigned the cutoff-independent bound v .

For comparison, the entire symbol $B_\infty(z) = v! \sum_{r,s} a_{rs} e^{b_{rs}z} / (\mu_v z^v)$, with removable value one at zero, obeys at each cutoff

$$B_\infty(\partial_S) - B_{N+1}(\partial_S) = \sum_{n=N+1-v}^N \frac{v! \mu_{n+v}}{(n+v)! \mu_v} \partial_S^n. \quad (\text{CEP10})$$

Its image and the image of the corresponding inverse difference lie in $\mathbb{C}[S]_{<v}$. This follows from the first derivative degree $N + 1 - v$, and from the inverse identity with the same order as AIE15. Thus the rank-at-most- v comparison is with the correctly growing finite symbol in CEP8.

For completeness the original-source inverse estimate extends directly to these literal symbols. Retain the exact constants $A_*^{\text{abs}}, B_*^{\text{sh}}, C_v, R_0, \rho_0$ of AIE4. For every $|z| \leq R_0$ the finite coefficient sum in CEP8 is at most the full absolute moment sum:

$$|B_{N+1}(z) - 1| \leq \frac{v!}{|\mu_v|} \sum_{j=v+1}^N \frac{|\mu_j|}{j!} |z|^{j-v} \leq C_v \frac{B_*^{\text{sh}} |z|}{v+1} e^{B_*^{\text{sh}} |z|} \leq \frac{1}{2}. \quad (\text{CEP11})$$

The middle inequality follows by writing $j = v+1+h$ and using $v!/(v+1+h)! \leq 1/((v+1)h!)$. Hence $F_{N+1}(z) = 1/B_{N+1}(-i \arctan z)$ has coefficient absolute values at most $2\rho_0^{-a}$ on degree $a \geq 1$, and constant coefficient one. This is Cauchy's formula on $|z| = \rho_0$, since $|\arctan z| \leq \operatorname{arctanh} \rho_0 = R_0$.

In the original orthonormal Gamma basis, the lowering matrix $L_s \varphi_n = \sqrt{w_n/w_{n-1}} \varphi_{n-1}$, $w_n = n!/(b)_n$, satisfies $\partial_y = \arctan L_s$ and $L_s^{N+1} = 0$. Evaluation of the power-series inverse in this nilpotent matrix therefore gives the exact finite operator $F_{N+1}(L_s)$, with entries $[z^{n-m}]F_{N+1}(z) \sqrt{w_n/w_m}$. The chain rule $S = c + iy$ supplies the retained $-i$. Bounding the absolute row and column sums gives

$$\|B_{N+1}(\partial_S)^{-1}\|_{s,c} \leq \alpha_{N,s} \left(1 + 2 \sum_{a=1}^N \rho_0^{-a}\right), \quad \alpha_{N,s} = \max_{0 \leq m \leq n \leq N} \sqrt{w_n/w_m}. \quad (\text{CEP12})$$

This proves the larger-cutoff assertion for the actual finite symbol rather than substituting N into an identity for B_q . Likewise the explicit AIE18 raising right inverse on these degree spaces, followed by i^v , $B_{N+1}(\partial_S)^{-1}$ and $v!/\mu_v$, is a right inverse of CEP8. Its comparison with the monomial integral remains exactly the degree-below- v correction in AIE25, now with this literal inverse.

R2.3 The exact source operator matrix and its full determinant

Use the unchanged original orthonormal polynomials $u_n(S) = p_n((S - c)/i)/\sqrt{h_n}$ in $\mathcal{P}_N$ and $u'_m(S') = p_m((S' - c')/i)/\sqrt{h_m}$ in $\mathcal{P}'_{N-v}$. Let A_N denote the matrix of $\mathcal{T}_{A,N}$ in these two frames. Define the analytic coefficient series at zero

$$t(z) = \sum_{r,s} a_{rs} \exp\left(\frac{b_{rs} - 4}{i} \arctan z\right) = \sum_{\ell \geq 0} t_\ell z^\ell. \quad (\text{CEP13})$$

Then its exact entries are

$$(A_N)_{m,n} = \begin{cases} t_{n-m} \sqrt{w_n/w_m}, & n \geq m, \\ 0, & n < m, \end{cases} \quad 0 \leq m \leq N - v, \quad 0 \leq n \leq N. \quad (\text{CEP14})$$

To prove this, the original generating identity $\sum p_n(y) z^n/n! = (1 + z^2)^{-b/2} e^{y \arctan z}$ shows that translation by η multiplies the series by $e^{\eta \arctan z}$. Extracting its finite coefficients and dividing by the complete $\sqrt{h_n}$ gives $\sqrt{w_n/w_m}$ after the two copies of M_s cancel at this specified step. Evaluation of CEP2 at $c' + iy$ uses $p(c + i(y + (b_{rs} - 4)/i))$, proving the exact centre shift in CEP13.

For $\ell < v$, the shifted moment $\sum a_{rs} (b_{rs} - 4)^\ell$ is zero by the binomial theorem and the definition of v ; at $\ell = v$ it equals μ_v . Since $\arctan z = z + O(z^3)$, this gives

$$t_0 = \cdots = t_{v-1} = 0, \quad t_v = \frac{(-i)^v \mu_v}{v!} \neq 0. \quad (\text{CEP15})$$

The first v columns of A_N vanish. Its remaining square block $\hat{A}_N$, with columns $v, \dots, N$, is upper triangular with diagonal $t_v \sqrt{w_{m+v}/w_m}$. Thus its rank is $N - v + 1$, its kernel in the source is

exactly the first v orthonormal coordinates, and

$$\begin{aligned}\det \hat{A}_N &= \left(\frac{(-i)^v \mu_v}{v!} \right)^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ \det(A_N A_N^*) &= \left| \frac{\mu_v}{v!} \right|^{2(N-v+1)} \prod_{m=0}^{N-v} \frac{w_{m+v}}{w_m}.\end{aligned}\tag{CEP16}$$

This retains the complex determinant phase and the original moment. The operator norm also has the exact original-source upper bound

$$\|A_N\| \leq M_{A,N} := \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_N+2)}, \quad H_N = \sum_{j=1}^N 1/j.\tag{CEP17}$$

Indeed AKJ22 bounds each exact translation in CEP13 by its displayed factor, and the triangle inequality gives the sum.

There is a closed four-endpoint expression for CEP16. For the signs $\sigma = (1, 1, -1, -1)$ at the ordered endpoints, write $\mathcal{D}_A = \sum_N \sigma_N \log \det(A_N A_N^*)$. Cancellation of the full products gives

$$\begin{aligned}\mathcal{D}_A &= -4q \log \left| \frac{\mu_v}{v!} \right| + \mathcal{W}_{v,s}, \\ \mathcal{W}_{v,s} &= \sum_{a=0}^{v-1} (\log w_{q-1-a} + \log w_{q-a} - \log w_{2q-1-a} - \log w_{2q-a}).\end{aligned}\tag{CEP18}$$

For $v = 0$ the second sum is zero. The product identity used here is $\prod_{m=0}^{N-v} w_{m+v}/w_m = \prod_{a=0}^{v-1} w_{N-a}/\prod_{a=0}^{v-1} w_a$, obtained by cancellation of consecutive products; it remains valid when the two index intervals overlap.

Put $c_b^{\log} = \max\{1, b-1\}$. For every $j \geq 1$ and $b \geq 1/2$,

$$\left| \log \frac{j}{b+j-1} \right| \leq \frac{c_b^{\log}}{j}.\tag{CEP19}$$

For $b \geq 1$ use $\log(1 + (b-1)/j) \leq (b-1)/j$. For $1/2 \leq b < 1$, put $a = 1 - b \leq 1/2$ and use $-\log(1 - a/j) \leq a/(j-a) \leq 1/j$; the last inequality is $aj \leq j-a$, which follows from $a \leq 1/2$ and $j \geq 1$. Summing CEP19 over the two intervals of length q in each summand of CEP18 proves the completely explicit estimate

$$|\mathcal{W}_{v,s}| \leq \frac{2v c_b^{\log} q}{q-v}, \quad |\mathcal{D}_A| \leq 4q \left| \log \left| \frac{\mu_v}{v!} \right| \right| + \frac{2v c_b^{\log} q}{q-v}.\tag{CEP20}$$

All denominators are positive since $q \geq 100 > v$. For either original source and the fixed actual quartet and period, the displayed bound is $O_{\text{actual}}(q+k) = o(kq)$.

R2.4 The induced original quotient metric and its correction

Let the matrices of the two remainder maps in the orthonormal source frames and original primary target frames be denoted by the same U_N, V_d as their maps. Explicitly

$$(U_N)_{(i,j),n} = \frac{p_n((\beta_{ij} - c)/i)}{\sqrt{h_n}}, \quad (V_d)_{(a,b),m} = \frac{p_m((\beta'_{ab} - c')/i)}{\sqrt{h_m}}.\tag{CEP21}$$

They have sizes q by $N+1$ and q' by $N-v+1$, respectively. Their first q , respectively q' , columns have nonzero evaluation Vandermonde determinant, so both have full row rank. The square CEP6 is the precise matrix identity

$$CU_N = V_d A_N, \quad R_N = U_N U_N^*, \quad R'_d = V_d V_d^*, \quad G_N = R_N^{-1}, \quad G'_d = (R'_d)^{-1}.\tag{CEP22}$$

Here G_N is the original upper primary quotient metric of OKM8, and G'_d is the lower-grid metric in the same source order s . The induced metric on E_k/V_A , in its specified target coordinates $E_{k'}$, is

$$Q_N = (CG_N^{-1}C^*)^{-1} = (V_d A_N A_N^* V_d^*)^{-1}. \quad (\text{CEP23})$$

To prove the first assertion, the minimum of $x^* G_N x$ under $Cx = y$ is attained at $x = G_N^{-1} C^* (CG_N^{-1} C^*)^{-1} y$. Every other lift differs by an element of $\ker C$ orthogonal to this lift in the G_N metric, proving the minimum and its Gram matrix. The second equality is CEP22.

The exact correction to the lower original Gamma quotient metric is

$$\begin{aligned} W_d &= V_d^* (R'_d)^{-1/2}, \quad W_d^* W_d = I_{q'}, \\ K_{A,N} &= W_d^* A_N A_N^* W_d > 0, \quad Q_N = (R'_d)^{-1/2} K_{A,N}^{-1} (R'_d)^{-1/2}, \\ \log \det Q_N &= \log \det G'_d - \log \det K_{A,N}. \end{aligned} \quad (\text{CEP24})$$

All square roots are the unique positive Hermitian square roots. These formulas follow by multiplying out the definition of $K_{A,N}$ and inverting. In particular the passage between the induced metric and the lower original source metric retains this actual compressed covariance.

If Z_d is any orthonormal basis of $\ker V_d$, completing W_d to a unitary matrix, the further exact determinant factorization is

$$\det K_{A,N} = \det(A_N A_N^*) \det(Z_d^* (A_N A_N^*)^{-1} Z_d). \quad (\text{CEP25})$$

For a proof put $D = A_N A_N^* > 0$ and write its block matrix in the unitary frame $[W_d, Z_d]$. The upper left block is $K_{A,N}$. Block Gaussian elimination gives $\det D = \det K_{A,N} \det(D_{22} - D_{21} K_{A,N}^{-1} D_{12})$. The lower right block of D^{-1} is the inverse of that Schur complement, proving CEP25. The empty-complement case has determinant one. The dimension of this retained complement is $N - v + 1 - q'$.

The positive-block elimination in CEP25–27 is the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)]. The proof retains the full complement and each original frame determinant before applying the endpoint signs.

Choose any fixed full frame $I_V : \mathbb{C}^\Delta \hookrightarrow E_k$ of $\ker C$ and a fixed right section L_C of C . Then $F_V = [I_V, L_C]$ is invertible and

$$\det(I_V^* G_N I_V) = |\det F_V|^2 \frac{\det G_N}{\det Q_N} = |\det F_V|^2 \frac{\det G_N}{\det G'_d} \det K_{A,N}. \quad (\text{CEP26})$$

Indeed $CF_V = [0, I]$, so the lower right block of $(F_V^* G_N F_V)^{-1}$ is $CG_N^{-1} C^* = Q_N^{-1}$. The Schur determinant argument just used gives CEP26. The frame factor is independent of source cutoff and cancels only after applying the four signs.

For the actual inclusion $K \subset V_A$, choose its original coefficient frame, and let $\Xi : \mathbb{C}^\Delta \rightarrow \mathbb{C}^{\Delta-r}$ have its coordinate kernel, $r = 8k - 16$, as in AKJ18. With $H_{V,N} = I_V^* G_N I_V$, the quotient metric is $Q_{\Xi,N} = (\Xi H_{V,N}^{-1} \Xi^*)^{-1}$. The same proof with a frame of $\ker \Xi$ and a right section gives, with all constant frame factors cancelled by the endpoint signs,

$$F_K^{(s)} = \sum_N \sigma_N \log \det G_N - \sum_N \sigma_N \log \det G'_{N-v} + \sum_N \sigma_N \log \det K_{A,N} - \sum_N \sigma_N \log \det Q_{\Xi,N}. \quad (\text{CEP27})$$

Thus the conductor quotient and the remaining original observation quotient have both been retained in the determinant relation. The lower source cutoffs in this equation are exactly $q - 1 - v, q - v, 2q - 1 - v, 2q - v$. Relative to the lower grid's own four cutoffs they have the respective shifts $\Delta - v, \Delta - v, 2\Delta - v, 2\Delta - v$.

R2.5 An exact graph quotient with controlled exterior metric error

In original orthonormal coordinates define

$$S_N : \mathcal{P}_N \longrightarrow \mathbb{C}^v \oplus \mathcal{P}'_{N-v}, \quad S_N p = (P_{<v} p, \mathcal{T}_{A,N} p). \quad (\text{CEP28})$$

Here $P_{<v}$ records the first v original upper orthonormal coefficients, and the target carries the orthogonal direct-sum norm with those coordinates and the unchanged lower Gamma source. The first v source coordinates span the polynomials of degree below v . Thus CEP14–16 give its exact block matrix $S_N = \text{diag}(I_v, \hat{A}_N)$ in the ordered source split at degree v . In particular it is invertible, and

$$\begin{aligned} \|S_N\| &\leq M_N := \max\{1, M_{A,N}\}, \\ |\det S_N| &= \left| \frac{\mu_v}{v!} \right|^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ |\log |\det S_N|| &\leq L_N := (N-v+1) \left| \log \left| \frac{\mu_v}{v!} \right| \right| + v c_b^{\log} H_N. \end{aligned} \quad (\text{CEP29})$$

The last bound follows by the consecutive-product identity preceding CEP19 and $|\log w_j| \leq c_b^{\log} H_j$: the two lists of v logarithms are each bounded by $v c_b^{\log} H_N$, followed by the factor $1/2$.

The upper and lower source ideals at this cutoff are

$$I_N = \chi_k \mathbb{C}[S]_{\leq N-q}, \quad I'_N = \chi_{k'} \mathbb{C}[S']_{\leq N-v-q'}. \quad (\text{CEP30})$$

A negative degree specifies the zero subspace. Let $\mathcal{W}_N = U_N^{-1}(V_A)$. The exact image and ideal graph under CEP28 are

$$\begin{aligned} S_N \mathcal{W}_N &= \mathbb{C}^v \oplus I'_N, \\ S_N I_N &= \{(P_{<v} h, \mathcal{T}_{A,N} h) : h \in I_N\} \subset \mathbb{C}^v \oplus I'_N, \\ V_A &\xrightarrow{\simeq} (\mathbb{C}^v \oplus I'_N) / (S_N I_N), \quad U_N p \longmapsto [S_N p]. \end{aligned} \quad (\text{CEP31})$$

The first equality follows because $\mathcal{W}_N$ is the preimage of I'_N under the top map of CEP6, and S_N is invertible. CEP3 proves the ideal inclusion. If representatives differ by I_N , their images differ by $S_N I_N$, so the last map is well defined. Conversely every class in the target has a preimage under S_N in $\mathcal{W}_N$, and the kernel consists exactly of I_N . This proves the isomorphism. Its norm on the left is the restriction of G_N : all lifts of a member of V_A already belong to $\mathcal{W}_N$, so their minimum source norm is the quotient norm of $\mathcal{W}_N / I_N$. The low-degree graph component $P_{<v} h$ is explicitly retained.

For reference the ideal dimensions at the four original endpoints are

N	$\dim I_N$	$\dim I'_N$	$\dim((\mathbb{C}^v \oplus I'_N) / S_N I_N)$
$q-1$	0	$\Delta - v$	Δ
q	1	$\Delta + 1 - v$	Δ
$2q-1$	q	$q + \Delta - v$	Δ
$2q$	$q+1$	$q + \Delta + 1 - v$	Δ .

(CEP32)

These follow by counting monomials in CEP30; S_N preserves dimensions. Moreover $\mathcal{T}_A|_{I_N}$ is injective because a nonzero member has degree at least $q > v$ and its image has degree exactly v less. Hence the second coordinate of the retained ideal graph already has dimension $\dim I_N$.

Here is the complete exterior norm estimate for this exact map. For an invertible complex matrix S of dimension n and $0 \leq t \leq n$, its singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$ give

$$\left\| \bigwedge^t S^{-1} \right\| = \frac{\left\| \bigwedge^{n-t} S \right\|}{|\det S|}. \quad (\text{CEP33})$$

Indeed the numerator is $\sigma_1 \cdots \sigma_{n-t}$ and the denominator is the product of all singular values; their ratio is the product of the t largest singular values of S^{-1} . This includes $t = 0, n$. For $n = N + 1$, CEP29 therefore gives the finite bounds

$$\begin{aligned} \log^+ \left\| \bigwedge^t S_N \right\| &\leq t \log M_N, \\ \log^+ \left\| \bigwedge^t S_N^{-1} \right\| &\leq (N + 1 - t) \log M_N + L_N. \end{aligned} \tag{CEP34}$$

Here $\log^+ x = \max\{0, \log x\}$. The right sides are nonnegative. Thus for all these ranks both bounds are at most

$$E_N = (N + 1) \log M_N + L_N. \tag{CEP35}$$

With $R_{\text{sh}} = \max_{r,s} |b_{rs} - 4|$ and $A_*^{\text{abs}} = \sum |a_{rs}|$, equation CEP17 gives $\log M_N \leq \log^+ A_*^{\text{abs}} + R_{\text{sh}}(H_N + 2)$. For the fixed original quartet and period, v, R_{sh}, μ_v and A_*^{abs} are fixed while $c_b^{\log} = O(k)$. Since $N \leq 2q$ and $H_N \leq 1 + \log N$, all four explicit bounds in CEP35 satisfy

$$E_N = O_{\text{actual}}(q \log(q + 1) + k \log(q + 1)) = o(kq). \tag{CEP36}$$

This uses only the forward norm and the exact determinant of the original conductor map.

We now prove the exact relation to the metric of the original kernel, including the quotient projection. In a finite-dimensional Hilbert space let $I \subset W$ and let S be invertible. The induced map $W/I \rightarrow SW/SI$ sends a minimal lift $x \in W \cap I^\perp$ to the orthogonal projection of Sx off SI . Orthogonal projection is a contraction on every exterior power: in an orthonormal basis its singular values are zeros and ones. Therefore, on any t -dimensional subspace of W/I , the exterior norm of the induced map is at most $\|\bigwedge^t S\|$. Applying the identical argument to S^{-1} gives the inverse bound. The minimal-lift identification is isometric since every other lift differs by a perpendicular element of I , whose squared norm adds. This proves the quotient assertion with no choice of coefficient norm.

Apply this result with $W = \mathcal{W}_N$, $I = I_N$, and the actual rank- r subspace $K \subset V_A$. The target subspace is the explicit image of K under CEP31; denote its Gram matrix in the transported original frame by $\tilde{H}_{K,N}$. Let $I_K : \mathbb{C}^r \hookrightarrow E_k$ be that original primary frame, and write its original minimum source lift as $Y_N = U_N^* G_N I_K$. Let Z_N be the matrix in the original orthonormal source coordinates of the independent columns $\chi_k(S) S^a$, $0 \leq a \leq N - q$. If there are no columns, all following products involving them are zero. The exact target orthogonal projection and Gram matrix are

$$\begin{aligned} P_N^{\text{graph}} &= I - S_N Z_N (Z_N^* S_N^* S_N Z_N)^{-1} Z_N^* S_N^*, \\ \tilde{H}_{K,N} &= Y_N^* S_N^* P_N^{\text{graph}} S_N Y_N, \quad H_{K,N} = Y_N^* Y_N = I_K^* G_N I_K. \end{aligned} \tag{CEP37}$$

The inverted Gram matrix is positive definite because its columns are independent and S_N is invertible. Multiplication shows that P_N^{graph} is Hermitian, idempotent, kills $S_N I_N$ and fixes its orthogonal complement. Thus CEP37 is precisely the minimum quotient Gram of CEP31, and its kernel frame has full rank because CEP31 is injective. Replacing the lift Y_N by any lift of the same primary frame adds columns in I_N , which the projection kills; hence every choice of lift gives this same matrix.

Applying the proved exterior quotient estimate and squaring volumes gives

$$\begin{aligned} -2 \log \left\| \bigwedge^r S_N^{-1} \right\| &\leq \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \\ &\leq 2 \log \left\| \bigwedge^r S_N \right\|, \\ \left| \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \right| &\leq 2E_N. \end{aligned} \tag{CEP38}$$

In detail, divide the chosen frame by its positive Gram square root so it is orthonormal in the source quotient. The squared norm of its exterior image is exactly the determinant ratio. The

forward and inverse exterior estimates prove the two inequalities, and CEP34–35 prove the last assertion.

Consequently the following reduction of the actual four-endpoint kernel determinant is proved:

$$\begin{aligned}\tilde{F}_K^{(s)} &= \sum_N \sigma_N \log \det \tilde{H}_{K,N}, \\ |F_K^{(s)} - \tilde{F}_K^{(s)}| &\leq 2 \sum_{N \in \{q-1, q, 2q-1, 2q\}} E_N = o(kq).\end{aligned}\tag{CEP39}$$

The frame is the original fixed frame of K at every endpoint; CEP37 gives all four transported matrices explicitly. Their target spaces retain the original ideal graphs, the different source cutoffs, all conductor coefficients, and the full original Gamma masses. CEP39 proves that the leading kq coefficient is unchanged by this particular graph-quotient transport. It supplies no value for that coefficient: its remaining determinant is the displayed, completely specified four-endpoint graph quotient.

R2.6 All ideal graph entries as original polynomial products

Each individual shift in CEP2 induces the unital algebra map $\sigma_{rs} : E_k \rightarrow E_{k'}, [p] \mapsto [p(S' + b_{rs})]$. The divisibility in CEP3, applied to the individual shifted polynomial, proves independence of the representative; substitution preserves multiplication and one. Thus $\bar{\mathcal{T}}_A = \sum a_{rs} \sigma_{rs}$ is its exact linear combination. For any $x, y \in E_k$, the complete multiplicative defect of that linear combination is

$$\bar{\mathcal{T}}_A(xy) - \bar{\mathcal{T}}_A(x)\bar{\mathcal{T}}_A(y) = \sum_{r,s} a_{rs} \sigma_{rs}(x) \sigma_{rs}(y) - \sum_{r,s,t,u} a_{rs} a_{tu} \sigma_{rs}(x) \sigma_{tu}(y).\tag{CEP40}$$

This follows by applying the proved individual multiplicativity and expanding the product of the two complete sums.

There is a complete finite product for the map on the source ideal. For each original shift put

$$d_{rs}(S') = \prod_{\substack{0 \leq i,j \leq k \\ (i,j) \notin \{r, \dots, r+k'\} \times \{s, \dots, s+k'\}}} (S' + b_{rs} - \beta_{ij}).\tag{CEP41}$$

This is monic of degree Δ , and $\chi_k(S' + b_{rs}) = \chi_{k'}(S') d_{rs}(S')$. Indeed CEP3's root addition identifies exactly the q' factors in the indicated rectangle with the factors of $\chi_{k'}$. All remaining factors are precisely the product in CEP41. Consequently for every $h \in \mathbb{C}[S]_{\leq L}$, $L = N - q$, the exact ideal map is

$$\begin{aligned}\mathcal{T}_{A,N}(\chi_k h) &= \chi_{k'} \mathcal{U}_{A,L} h, \\ (\mathcal{U}_{A,L} h)(S') &= \sum_{r,s} a_{rs} d_{rs}(S') h(S' + b_{rs}) \in \mathbb{C}[S']_{\leq L + \Delta - v}.\end{aligned}\tag{CEP42}$$

The equality follows by the product identity just proved. The degree statement follows from CEP8: if h has degree t and leading coefficient $h_t \neq 0$, the leading coefficient of its displayed image is $h_t \binom{q+t}{v} \mu_v$ at degree $t + \Delta - v$. This follows because $\chi_k, \chi_{k'}$ are monic and $\mathcal{T}_A$ lowers degree exactly by v with its stated leading derivative. Thus $\mathcal{U}_{A,L}$ is injective. At $L = -1$ the domain and image are zero.

This gives a further exact sequence, with all maps specified:

$$0 \longrightarrow \mathbb{C}[S]_{<v} \longrightarrow V_A \longrightarrow \mathbb{C}[S']_{\leq L + \Delta - v} / \mathcal{U}_{A,L} \mathbb{C}[S]_{\leq L} \longrightarrow 0.\tag{CEP43}$$

The first map is the original quotient class. For the second, lift a member of V_A to $p \in \mathcal{W}_N$, divide $\mathcal{T}_{AP}$ by $\chi_{k'}$, and take its class in the last space. A change of lift by $\chi_k h$ adds exactly $\mathcal{U}_{A,L} h$

by CEP42, proving independence. Every target polynomial times $\chi_{k'}$ has a preimage under the onto map $\mathcal{T}_{A,N}$, proving surjectivity. If the image is zero, subtract the corresponding $\chi_k h$; the remaining polynomial is killed by $\mathcal{T}_A$ and hence has degree below v . Its class injects because $v < q$. This proves exactness. The last space has dimension $\Delta - v$ at every original endpoint, as is also obtained by subtracting the two polynomial dimensions.

For a literal matrix version of the ideal graph, equip $\mathbb{C}[S']_{\leq L+\Delta-v}$ with the ordered lower orthonormal Gamma coefficient basis, and write $M_{\chi',N}$ for multiplication by the full original $\chi_{k'}$ into $\mathcal{P}'_{N-v}$. It is injective. The map

$$D_{\chi',N} = \text{diag}(I_v, M_{\chi',N}) : \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq L+\Delta-v} \longrightarrow \mathbb{C}^v \oplus I'_N \quad (\text{CEP44})$$

is an isomorphism onto the indicated subspace. Its pullback metric is exactly $G_{\chi',N} = \text{diag}(I_v, M_{\chi',N}^* M_{\chi',N})$; this follows by taking the squared norm of each component. For the domain polynomial coefficient frame of h , let $Z_{\chi,N} h$ be the original upper orthonormal coefficients of $\chi_k h$. In these coordinates the complete graph relation matrix is

$$\mathcal{Z}_{A,N} = \begin{pmatrix} P_{<v} Z_{\chi,N} \\ [\mathcal{U}_{A,L}] \end{pmatrix}, \quad D_{\chi',N} \mathcal{Z}_{A,N} = S_N Z_{\chi,N}. \quad (\text{CEP45})$$

The first component is the original low-degree projection, and the second consists of the exact finite products and translations in CEP41–42. The equality is CEP28 and CEP42. It proves that the graph quotient in CEP31 is precisely the quotient by $\text{im } \mathcal{Z}_{A,N}$ in the displayed metric $G_{\chi',N}$. All matrix inverses and minimum projections in CEP37 can therefore be computed using these full original polynomial products. No coefficient of the ideal, source metric or low-degree graph component is discarded by this additional coordinate isomorphism.

R2.7 The compressed conductor correction is subleading at every endpoint

The exact exterior estimate also controls the large compressed covariance in CEP24 without multiplying a scalar inverse bound by q' . Write $n' = N - v + 1$ and retain $A_N = [0, \hat{A}_N]$ with its exact square block in CEP16. Since $W_d^* W_d = I_{q'}$, the coefficient matrix of its image under $\hat{A}_N^*$ gives

$$K_{A,N} = (\hat{A}_N^* W_d)^* (\hat{A}_N^* W_d). \quad (\text{CEP46})$$

The squared norm of the exterior product of the q' columns on the right is its determinant. The norm of the exterior product of the columns of W_d is one, because those columns are orthonormal. The forward estimate therefore gives $\det K_{A,N} \leq \|\hat{A}_N\|^{2q'} \leq M_N^{2q'}$. For the reverse inequality apply $\bigwedge^{q'} \hat{A}_N^*$ to the same image columns; their image is the orthonormal column wedge of W_d . Thus

$$\det K_{A,N} \geq \left\| \bigwedge^{q'} \hat{A}_N^{-1} \right\|^{-2}. \quad (\text{CEP47})$$

The singular values of an adjoint are the same, so the adjoint inverse in this argument has exactly the displayed exterior norm. The singular complement identity CEP33 in dimension n' gives $\|\bigwedge^{q'} \hat{A}_N^{-1}\| \leq M_N^{n'-q'} / |\det \hat{A}_N|$. CEP29 bounds the absolute logarithm of the last determinant by L_N . Consequently the complete two-sided estimate is

$$\begin{aligned} -2((N - v + 1 - q') \log M_N + L_N) &\leq \log \det K_{A,N} \leq 2q' \log M_N, \\ |\log \det K_{A,N}| &\leq 2E_N = o(kq) \quad \text{at every original endpoint and source order.} \end{aligned} \quad (\text{CEP48})$$

All ranks in this argument are their actual ranks, $q' \leq N - v + 1$ as proved after CEP6. In particular the estimate applies to the complete original lower quotient, whose rank grows quadratically.

Define the fully specified total-source difference and the remaining original observation quotient return by

$$\begin{aligned}\mathcal{T}_k^{(s)} &= \sum_N \sigma_N (\log \det G_N - \log \det G'_{N-v}), \\ F_{\Xi,k}^{(s)} &= \sum_N \sigma_N \log \det Q_{\Xi,N}, \\ \mathcal{E}_{A,k}^{(s)} &= \sum_N \sigma_N \log \det K_{A,N}.\end{aligned}$$

Equations CEP26–27 and CEP48 prove the exact identities and bounds

$$\begin{aligned}F_{V_A}^{(s)} &= \mathcal{T}_k^{(s)} + \mathcal{E}_{A,k}^{(s)}, & F_K^{(s)} &= \mathcal{T}_k^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)}, \\ |\mathcal{E}_{A,k}^{(s)}| &\leq 2 \sum_N E_N = o(kq), \\ \frac{F_K^{(s)} - \mathcal{T}_k^{(s)} + F_{\Xi,k}^{(s)}}{kq} &\longrightarrow 0.\end{aligned}\tag{CEP49}$$

Here $F_{V_A}^{(s)}$ uses its fixed original coefficient frame, and $Q_{\Xi,N}$ is the exact quotient metric of CEP27. The frame factors cancel with coefficients $1 + 1 - 1 - 1$ only in these returns, as proved in CEP26. Thus the leading metric effect of the full conductor source transport has now been controlled. The remaining rank- $8k - 32$ observation quotient and the shifted lower-grid source determinants are still their explicit original matrices; CEP49 carries them forward into the next calculation.

R3 The fixed-period problem and the result proved here

The current handoff already proves the absolute $O(kq)$ kernel estimate, the conductor frame, its residual observation, and the finite inverse. The new calculation here removes the $O(kq)$ *exterior-inverse allowance* left by AIE29. It replaces that allowance by an explicit $O_{\text{actual}}(q \log q) = o(kq)$ error and transfers the actual four-endpoint determinant to the smaller conductor coefficient space. It does not identify this transfer error with the original kernel volume, and it does not evaluate the latter's leading coefficient.

Retain an actual hypothetical simple quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha), \quad v_h = 2\xi/h.$$

The original full unit and period observation are

$$U = M_{j_h v_h}, \quad j\Lambda_k = P_k \Pi^{\otimes k} U^{\otimes k} \iota_k, \quad P_k = \frac{1}{5} \sum_{b=0}^4 (C^{-b})^{\otimes k}. \quad (\text{FEX1})$$

Fix one original period and logarithm branch on the proved five-orbit locus, for example any fixed point with $|u| \geq R_*$. Here R_* is the unchanged RPC/AMT radius. Its formulas and the proof of the rank are in the supplied complete source; no new rank premise is imposed. Throughout the principal application,

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad q = (k+1)^2, \quad c = k/2, \quad k' = k - 8, \quad c' = c - 4, \\ q' &= (k-7)^2, \quad a = q - q' = 16k - 48, \quad r = 8k - 16, \quad j_k = 8k - 32, \\ \beta_{ab}^{(k)} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi = \prod_{a,b=0}^k (S - \beta_{ab}^{(k)}), \quad \psi = \chi_{k'}. \end{aligned} \quad (\text{FEX2})$$

The letter a without indices denotes the degree difference, not an amplitude. Let $K = \ker \Lambda_k \subset E = \mathbb{C}[S]/(\chi)$, using its fixed original coefficient inclusion. The actual ranks satisfy $a = r + j_k$. Every class below has its unique representative of degree less than q whenever a polynomial representative is required.

Retain the complete original conductor biform and its coefficients

$$A = \sum_{r_0, s_0=0}^8 a_{r_0 s_0} X_1^{r_0} X_0^{8-r_0} Y_1^{s_0} Y_0^{8-s_0}.$$

Write $\nu = (r_0, s_0)$ for an index in this sum. Define

$$\begin{aligned} b_\nu &= 4 + (2r_0 - 8)\delta + i(2s_0 - 8)\gamma, \quad \mu_j = \sum_\nu a_\nu b_\nu^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80, \\ (\mathcal{T}_{AP})(S') &= \sum_\nu a_\nu p(S' + b_\nu), \quad \mathcal{C}_A = \frac{v!}{\mu_v} \mathcal{T}_A, \\ B_q(X) &= \sum_{n=0}^{q-1-v} \frac{v! \mu_{v+n}}{(v+n)! \mu_v} X^n. \end{aligned} \quad (\text{FEX3})$$

The coefficients are those of the actual $\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}$, fixed as k changes. The original distinct-node Vandermonde proves the stated finite v and $\mu_v \neq 0$. In particular, $B_q(0) = 1$. The scalar in $\mathcal{C}_A$ is displayed explicitly; it is not a renormalization of an arithmetic source or observation.

Only the two original Gamma sources are used:

$$\begin{aligned} s &\in \{1, k\}, \quad b = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^b}{4\pi \Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy, \\ h_n &= M_s n! (b)_n, \quad w_n = \frac{n!}{(b)_n}, \quad \|p\|_{s, c_0}^2 = \int |p(c_0 + iy)|^2 dm_{0,s}(y). \end{aligned} \quad (\text{FEX4})$$

In particular $s = k'$ is never inserted as a source order. The polynomial ψ in FEX2 will enter as a literal multiplication factor in these unchanged measures.

The main conclusion will be the finite interval FEX35 and the consequence

$$\boxed{F_K^{(s)} = \widehat{F}_k^{(s)} + O_{H,\delta,\gamma}(q \log q), \quad s = 1, k.} \quad (\text{FEX5})$$

Here $\widehat{F}$ is defined in FEX29–30 using the actual residual observation and all original transformed relation columns. The error is proved, not postulated. At the end, the unchanged arithmetic comparison transfers the same leading-limit question to F_K^{ar} .

R4 A full-flag determinant bound

Let $T : (V, g) \rightarrow (V', g')$ be an isomorphism of n -dimensional complex Hilbert spaces. Write $\|T\| \leq M$ and let $J > 0$ be its squared whole-space Jacobian. For $R \subset W \subset V$, set $t = \dim W - \dim R$. Give W/R and TW/TR their actual Hilbert quotient norms. For any coefficient basis of W/R , let D_Q be the ratio of its target and source Gram determinants under the induced map. Then

$$\boxed{JM^{-2(n-t)} \leq D_Q \leq M^{2t}.} \quad (\text{FEX6})$$

All empty factors have determinant one. The upper bound follows by applying T to each lift and taking the infimum over the same fibre. For the lower bound, use the flag $0 \subset R \subset W \subset V$. Choose any basis adapted to it, on both sides using its transported basis. Successive block elimination of the complete positive Gram gives

$$J = D_R D_Q D_{V/W}. \quad (\text{FEX7})$$

Each block elimination uses its full Schur complement. The other two induced maps also have norm at most M , by restriction or minimization, and their dimensions total $n - t$. Thus their squared determinant product is at most $M^{2(n-t)}$, which proves FEX6. This proves the assertion without asserting orthogonality of a fixed coefficient section.

An equivalent inverse form is useful. If T is an endomorphism of a single Hilbert space and $|\det T| = 1$, then, for every full-rank n by t coefficient matrix X ,

$$\boxed{M^{-2t} \leq \frac{\det(X^* T^{-*} T^{-1} X)}{\det(X^* X)} \leq M^{2(n-t)}.} \quad (\text{FEX8})$$

For example, this follows by taking $W = T^{-1} \text{im } X$, $R = 0$ in FEX6 and inverting the determinant ratio. It also follows by taking the product of the largest t inverse singular values and cancelling against the full product. The statement is a bound: $\det T = 1$ does *not* say that a restricted Gram determinant is unchanged. For $T = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$, restriction to the second coordinate has squared volume 17.

R5 Polynomial forward control replaces the exponential exterior loss

For completeness, the derivative bound used here follows directly from the original polynomial norms. The recurrence and generating identity are

$$\begin{aligned} p_0 &= 1, & p_1 &= y, & p_{n+1} &= yp_n - n(b + n - 1)p_{n-1}, \\ \sum_{n \geq 0} \frac{p_n(y)}{n!} z^n &= (1 + z^2)^{-b/2} e^{y \arctan z}, & \|p_n\|^2 &= h_n. \end{aligned} \quad (\text{FEX9})$$

They are the supplied original Gamma identities. Differentiation of the finite coefficient at zero, followed by the full normalization, gives

$$\partial_y \varphi_n = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \frac{(-1)^{(j-1)/2}}{j} \sqrt{\frac{w_n}{w_{n-j}}} \varphi_{n-j}, \quad \varphi_n = p_n / \sqrt{h_n}. \quad (\text{FEX10})$$

The mass M_s cancels from this ratio, not from the underlying norms. Let $H_D = \sum_{j=1}^D 1/j$ and $H_0 = 0$. For $n > m \geq 1$, $b \geq 1/2$ gives $w_n/w_m \leq \sqrt{n/m}$ and $w_n \leq 2\sqrt{n}$. Indeed square each factor $j/(j-1/2)$ and use $j(j-1) \leq (j-1/2)^2$; the product telescopes. Apply the weighted Schur estimate to the absolute matrix in FEX10 with weights $w_n^{-1/2}$. Its row sum is at most H_D ; its column sum at n is at most $H_n + 4$. The excess for $1 \leq m < n$ is bounded by

$$\frac{\sqrt{n/m} - 1}{n - m} = \frac{1}{\sqrt{m}(\sqrt{n} + \sqrt{m})} \leq \frac{1}{\sqrt{nm}},$$

whose sum is at most two; the $m = 0$ term is at most $2/\sqrt{n} \leq 2$. Enlarging the odd-index sums is legitimate because all terms are nonnegative. Cauchy–Schwarz in each weighted row and then summing the column estimate proves

$$\|\partial_y\|_{\mathcal{P}_D} \leq \sqrt{H_D(H_D + 4)} \leq H_D + 2, \quad \|f(y + z)\| \leq e^{|z|(H_D+2)} \|f\|. \quad (\text{FEX11})$$

At $D = 0$ the derivative is zero. Translation is the exact finite Taylor exponential. The coordinate substitution $S = c_0 + iy$ gives $\partial_S = -i\partial_y$ with the same norm.

Retain every actual coefficient in

$$A_*^{\text{abs}} = \sum_{\nu} |a_{\nu}|, \quad B_*^{\text{sh}} = \max_{\nu} |b_{\nu}|, \quad C_v = \frac{A_*^{\text{abs}}(B_*^{\text{sh}})^v}{|\mu_v|} \geq 1, \quad P_D = C_v e^{B_*^{\text{sh}}(H_D+2)}. \quad (\text{FEX12})$$

Since $|\mu_{v+n}| \leq A_*^{\text{abs}}(B_*^{\text{sh}})^{v+n}$ and $v!/(v+n)! \leq 1/n!$, the triangle inequality in FEX10–11 gives

$$\|B_q(\partial_S)\|_{s,c_0;\mathcal{P}_D} \leq P_D. \quad (\text{FEX13})$$

This bound holds for the literal fixed polynomial B_q on any explicitly specified finite degree space. On $D \leq q-1$ it is the original AIE operator, whose matrix is triangular with diagonal one. FEX8 therefore proves its new exterior estimate

$$P_D^{-2t} \leq \frac{\det(X^*(B_q^{-1})^* B_q^{-1} X)}{\det(X^* X)} \leq \min\{N_{D,s}^{2t}, P_D^{2(D+1-t)}\}. \quad (\text{FEX14})$$

Here X is any original t -plane in $\mathcal{P}_D$ and $N_{D,s}$ is the unchanged AIE10 inverse bound. The new bound requires neither a new zero-free disk nor a lower bound on an unevaluated restricted minor. For $D = q-1$, $t = r$, its upper logarithm is at most

$$2(q-r)\{\log C_v + B_*^{\text{sh}}(H_{q-1} + 2)\} = O_{H,\delta,\gamma}(q \log q) = o(kq). \quad (\text{FEX15})$$

Thus the factor rD in the old exterior allowance is unnecessary. The forward norm, together with the full Jacobian, controls the product; raising the worst inverse singular value to the r th power loses that information.

There is a direct high-source extension of this statement, distinguished from changing AIE's cutoff. For a fixed k use the same polynomial $B_q(\partial_S)$ on each $\mathcal{P}_N$. Map the exact flag

$$\chi \mathcal{P}_{N-q} \subset I_K \mathbb{C}^r + \chi \mathcal{P}_{N-q}$$

to its literal image under B_q . Its relation space is $B_q(\partial_S)(\chi \mathcal{P}_{N-q})$, not the original ideal by assumption. Its whole Jacobian is one, and FEX13 is valid on this newly specified domain. If $\tilde{F}_K$

is the signed kernel determinant using the native source norm on these transported flags, FEX6 gives

$$-2(2q+1)\log P_{2q} \leq \tilde{F}_K - F_K \leq 2(4q+1)\log P_{2q}. \quad (\text{FEX16})$$

Indeed at endpoint N its logarithmic Gram change belongs to $[-2(N+1-r)\log P_N, 2r\log P_N]$. Insert the four original signs and use $P_N \leq P_{2q}$. This is an exact extension and a signed estimate; no commutation with the original remainder map has been assumed.

R6 The literal conductor at every source endpoint

The previous high-source extension is useful, but the actual conductor provides a more informative reduction. On $\mathcal{P}_N$, $N \geq q-1$, its exact Taylor identity is

$$\begin{aligned} \mathcal{C}_A &= \partial_S^v B^{[N]}(\partial_S), & B^{[N]}(X) &= \sum_{n=0}^{N-v} \frac{v!\mu_{v+n}}{(v+n)!\mu_v} X^n, \\ B^{[q-1]} &= B_q, & L &:= \ker(\mathcal{C}_A|_{\mathcal{P}_N}) = \mathcal{P}_{v-1}, & \mathcal{C}_A : \mathcal{P}_N/L &\xrightarrow{\sim} \mathcal{P}_{N-v}. \end{aligned} \quad (\text{FEX17})$$

For $v=0$, $L=0$. These assertions follow either from the leading nonzero derivative or from the commuting invertible triangular factor $B^{[N]}$. Thus the operator on the high endpoints is defined by the original translations themselves. It is not obtained by silently changing the cutoff in the definition of B_q .

The domain has its original quotient Gamma norm at centre c ; the codomain has its native Gamma norm at centre c' . Define

$$M_N = \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{\nu} |a_{\nu}| e^{[b_{\nu}-4](H_N+2)} \right\}, \quad m_N = \log M_N. \quad (\text{FEX18})$$

The difference of centres is essential: $S' + b_{\nu} = c + iy + (b_{\nu} - 4)$. FEX11 proves $\|\mathcal{C}_A p\|_{s,c'} \leq M_N \|p\|_{s,c}$. Taking the infimum over the same L -coset gives the same bound for the induced isomorphism in FEX17.

Its squared whole-space Jacobian is explicitly

$$\mathcal{J}_{N,s} = \prod_{d=v}^N \frac{w_d}{w_{d-v}} = \frac{\prod_{d=N-v+1}^N w_d}{\prod_{d=0}^{v-1} w_d}. \quad (\text{FEX19})$$

Products are one when $v=0$. To prove the formula, take the orthonormal quotient basis represented by $\varphi_v((S-c)/i), \dots, \varphi_N((S-c)/i)$, with the normalization in FEX10, and the corresponding native basis at c' . The triangular diagonal of the map is $i^{-v}(d)_v \sqrt{h_{d-v}/h_d}$, where $(d)_v = d!/(d-v)!$ is a falling factorial in this formula. Its squared modulus is w_d/w_{d-v} . The source mass is present in both h factors before cancellation; translation of the centre affects only strictly lower coefficients. Telescoping proves the second equality, including overlapping product ranges. For the unnormalized $\mathcal{T}_A$ the squared Jacobian is $|\mu_v/v!|^{2(N+1-v)} \mathcal{J}_{N,s}$.

Put $\zeta_{N,s} = \log \mathcal{J}_{N,s}$. The following finite bounds will suffice:

$$\begin{aligned} 0 &\leq \zeta_{N,1} \leq v \log(2\sqrt{N}), & s &= 1, \\ 0 &\leq -\zeta_{N,s} \leq v(b-1)H_N, & s &= k \geq 9. \end{aligned} \quad (\text{FEX20})$$

For the first use monotonicity of w_n and $1 \leq w_n \leq 2\sqrt{n}$. For the second, w_n decreases and $-\log w_n = \sum_{j=1}^n \log(1 + (b-1)/j) \leq (b-1)H_n$; then use the second product formula in FEX19. These proofs also show directly that the Jacobian error at the tensor source is $O(vk \log q)$, rather than an uncontrolled determinant of dimension q .

The old explicit AIE inverse is still compatible with this estimate. For example, let R_A be that right inverse for the unnormalized $\mathcal{T}_A : \mathcal{P}_D \rightarrow \mathcal{P}_{D-v}$, and let $R_{\min}$ be its actual minimum-norm

right inverse. Their difference takes values in L and is orthogonal to $R_{\min}$. On a t -plane X in the target,

$$\det((R_A X)^* R_A X) \leq \det((R_{\min} X)^* R_{\min} X)(1 + M_{\text{raw}}^2 C_{A,D,s}^2)^{\min(t,v)}. \quad (\text{FEX21})$$

Here $\|\mathcal{T}_A\| \leq M_{\text{raw}}$ and $\|R_A\| \leq C_{A,D,s}$ are the original finite forward and AIE22 bounds. Indeed the excess Gram has rank at most v , and $R_{\min}^* R_{\min} \succeq M_{\text{raw}}^{-2} I$. The operator $R_A - R_{\min}$ is an orthogonal projection of R_A , so its norm is at most $C_{A,D,s}$. Apply the matrix determinant lemma, or congruence with the minimum Gram, to obtain FEX21. FEX6 bounds the minimum inverse determinant by $J_{\text{raw}}^{-1} M_{\text{raw}}^{2(D+1-v-t)} \det(X^* X)$. Only the fixed rank v , not t , multiplies the exponential allowance for the raw low-degree correction. The main reduction below avoids that raw correction by retaining its exact two quotient metrics instead.

R7 The residual observation and the smaller coefficient space

AKJ12–25 identifies the original conductor preimage as

$$V_A = \{p \in \mathcal{P}_{q-1} : \psi \mid \mathcal{T}_A p\}. \quad (\text{FEX22})$$

Its complete degree-adapted basis is $1, S, \dots, S^{v-1}$ followed by

$$P_h = \frac{(q' + h + v)!}{(q' + h)!} B_q (\partial_S)^{-1} \mathcal{J}_v(\psi(S) S^h), \quad 0 \leq h < a - v, \quad \mathcal{J}_v S^d = \frac{d!}{(d + v)!} S^{d+v}. \quad (\text{FEX23})$$

The variable in this inverse coefficient expression is identified with S' after applying the translation map. Each P_h is monic of degree $q' + v + h$. Its normalized conductor image is exactly

$$\mathcal{C}_A P_h = d_h \psi(S') S'^h, \quad d_h = \frac{(q' + h + v)!}{(q' + h)!}. \quad (\text{FEX24})$$

The full original residual matrix in this basis is $\Xi = (\Xi_{\text{lo}}, \Xi_{\text{hi}})$, of size j_k by a and rank j_k . Its kernel maps isomorphically onto the original K . These are the supplied actual coefficient maps, not independent matrix parameters. Put

$$r_{\text{lo}} = \text{rank } \Xi_{\text{lo}}, \quad t_0 = v - r_{\text{lo}} = \dim(K \cap L), \quad t = r - t_0. \quad (\text{FEX25})$$

Choose a quotient coordinate map Q_{lo} with kernel exactly $\text{im } \Xi_{\text{lo}}$, and a full-column-rank basis W_k of

$$\ker(Q_{\text{lo}} \Xi_{\text{hi}}) \subset \mathbb{C}^{a-v}. \quad (\text{FEX26})$$

The residual map has rank $j_k - r_{\text{lo}}$ because the full matrix has rank j_k . Hence the space in FEX26 has dimension t . These choices are made using actual nonzero minors and then kept fixed as the source endpoint changes. No generic minor or generic rank is substituted. Define the polynomial column matrix

$$Z_k = \text{diag}(d_h) W_k \quad \text{in the frame } (1, S', \dots, S'^{a-v-1}). \quad (\text{FEX27})$$

Then $\mathcal{C}_A K = \psi \text{ im } Z_k$. In fact, for a high coordinate z in FEX26 the equation $\Xi_{\text{lo}} \ell + \Xi_{\text{hi}} z = 0$ has a solution. Its different solutions differ by $K \cap L$; FEX24 kills that ambiguity and gives precisely ψZ_k . Thus this is the actual image, including the original unit and period dependence in Ξ and the conductor coefficients. The inverse finite-symbol factor no longer appears in the columns Z_k .

For every $j \geq 0$ define the exact transformed relation polynomial

$$R_j^A(S') = \frac{\mathcal{C}_A(\chi S^j)(S')}{\psi(S')} = \frac{v!}{\mu_v} \sum_{\nu} a_{\nu} \frac{\chi(S' + b_{\nu})}{\psi(S')} (S' + b_{\nu})^j. \quad (\text{FEX28})$$

Every displayed quotient is a polynomial. Indeed, if S' is the (a', b') inner-grid root, then $S' + b_{r_0 s_0}$ is the original root with indices $(a' + r_0, b' + s_0)$. All these indices are between 0 and k . The degree of R_j^A is $a - v + j$ and its leading coefficient is $(q + j)_v$; this follows from the first nonzero derivative in FEX17. It is nonzero, and these distinct degrees prove linear independence of the relation columns. In particular the high-source terms have not been replaced by an unjustified continuation of the low-degree symbol.

At endpoint N put $m_N^{\text{pol}} = N - v - q'$ and $d_N^{\text{rel}} = N + 1 - q$. Form the complete polynomial Gram $H_{N,s}^\psi$ of $(1, S', \dots, S'^{m_N^{\text{pol}}})$ in the measure $|\psi(c' + iy)|^2 dm_{0,s}(y)$ at $S' = c' + iy$. Zero-extend Z_k to this frame and let R_N^A have columns $R_0^A, \dots, R_{d_N^{\text{rel}}-1}^A$. Define

$$\begin{aligned}\widehat{H}_{K,N}^{(s)} &= Z_k^* H_{N,s}^\psi Z_k \\ &\quad - Z_k^* H_{N,s}^\psi R_N^A ((R_N^A)^* H_{N,s}^\psi R_N^A)^{-1} (R_N^A)^* H_{N,s}^\psi Z_k, \\ \widehat{F}_k^{(s)} &= \sum_{i=0}^3 \epsilon_i \log \det \widehat{H}_{K,N_i}^{(s)}, \\ (N_0, N_1, N_2, N_3) &= (q-1, q, 2q-1, 2q), \\ (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) &= (1, 1, -1, -1).\end{aligned}\tag{FEX29}$$

The empty relation inverse is absent at $N = q - 1$. The four relation ranks are still exactly $0, 1, q, q + 1$. They have not been truncated to the smaller kernel dimension. Positivity follows because a nonzero linear combination of Z_k has degree at most $a - v - 1$, whereas every nonzero combination of the R_j^A has degree at least $a - v$. The complete Gram is positive in the original positive Gamma density. Minimization over an increasing relation span proves

$$\widehat{H}_{K,q-1}^{(s)} \succeq \widehat{H}_{K,q}^{(s)} \succeq \dots, \quad \widehat{F}_k^{(s)} \geq 0.\tag{FEX30}$$

A change of the fixed basis W_k gives the same four-endpoint value; its four identical Gram factors cancel with the original signs.

R8 A finite error for the actual compressed determinant

Let $K^+ = K + L \subset E$. This is only an auxiliary space for an exact metric calculation, not an enlargement of the observation kernel. All quotient and restriction metrics below are induced from the same original $G_N^{(s)}$ and their indicated fixed coefficient frames. Let $F_{K^+/L}$ and $F_{K^+/K}$ denote their four-endpoint volumes. The two block Gram factorizations of K^+ give

$$\boxed{F_K = F_{K^+/L} + F_L - F_{K^+/K}}.\tag{FEX31}$$

Before summing there are fixed basis Jacobians; they cancel only after all four signs. Both factorizations use the actual Schur complements. The two subquotient dimensions are $\dim L = v$ and $\dim(K^+/K) = v - t_0$. The existing full AKS covariance inequality passes by restriction and minimization to every fixed subquotient. Thus, with $\mathcal{L}_s = \mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}$ from AKS,

$$0 \leq F_L \leq v \mathcal{L}_s, \quad 0 \leq F_{K^+/K} \leq (v - t_0) \mathcal{L}_s.\tag{FEX32}$$

No angle condition between K and L is used.

For a fully explicit existing allowance one may use the AKS Newton bound in place of a sharper available $\mathcal{L}_s$. Write

$$\begin{aligned}R &= k \sqrt{\delta^2 + \gamma^2}, \quad t_q = \pi/2 - 1/q, \quad a_q = 2 \log 3 / t_q, \quad s_q = (a_q^q - 1) / (a_q - 1), \\ b_{s,q} &= [(9/25) \sin(1/q)]^{-s/2} \exp(2R(\log 3 + t_q/2)), \\ \mathcal{L}_{s,j} &= \log \left(1 + b_{s,q} s_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(s/2)_d} (25/16)^d \right).\end{aligned}\tag{FEX33}$$

These are the original complete coefficient bounds with both masses retained before norm-ratio cancellation. They satisfy $\mathcal{L}_s = O_{\delta,\gamma}(q)$ for $s = 1, k$ along this family. Indeed $\log s_q = O(q)$, the last geometric-weighted sum has logarithm $O(q + \log q)$ because $d!/(s/2)_d \leq 2\sqrt{d}$, and $\log b_{s,q} = O(k \log q + R) = o(q) + O(q)$.

Apply FEX6 to the source $\mathcal{P}_N/L$ and its conductor image. The relation subspace is $(\chi\mathcal{P}_{N-q} + L)/L$. The larger flag term is generated also by K , giving quotient dimension t . By FEX28, its image is the span of ψR_N^A and ψZ_k . Multiplication by ψ is an exact isometry from the indicated weighted coefficient space onto this polynomial image. The source quotient metric is exactly the metric of K^+/L , since successive minimization first by $\chi\mathcal{P}_{N-q}$ and then by L is minimization over their sum. Thus, in the fixed transported frames, put

$$\begin{aligned} e_N &= \log \det \widehat{H}_{K,N}^{(s)} - \log \det H_{K^+/L,N}^{(s)}, & n_N &= N + 1 - v, \\ e_{N,s}^- &= \zeta_{N,s} - 2(n_N - t)m_N, & e_{N,s}^+ &= 2tm_N, & e_{N,s}^- &\leq e_N \leq e_{N,s}^+, \\ D_s^- &= e_{q-1,s}^- + e_{q,s}^- - e_{2q-1,s}^+ - e_{2q,s}^+, & D_s^+ &= e_{q-1,s}^+ + e_{q,s}^+ - e_{2q-1,s}^- - e_{2q,s}^-. \end{aligned} \quad (\text{FEX34})$$

The signs on $e_{N,s}^\pm$ denote new finite scalar bounds; they do not redefine an original source scalar or covariance. All endpoint quantities are computed from FEX18–20 and the actual finite ranks in FEX25. Original frame factors have already been accounted for by transported quotient frames. Combining the four inequalities and FEX31–32 proves the central finite result:

$$\boxed{\widehat{F}_k^{(s)} - D_s^+ - (v - t_0)\mathcal{L}_s \leq F_K^{(s)} \leq \widehat{F}_k^{(s)} - D_s^- + v\mathcal{L}_s.} \quad (\text{FEX35})$$

This is a bound for the actual kernel determinant, not for a different ambient determinant. Every residual coefficient and relation cross term remains in FEX29.

For a simpler explicit error let

$$J_{\max,s} = \max_{i=0}^3 |\zeta_{N_i,s}|, \quad m_{\max} = m_{2q}, \quad \mathcal{E}_{k,s} = v\mathcal{L}_s + 2J_{\max,s} + 2(4q + 1)m_{\max}. \quad (\text{FEX36})$$

Then

$$\boxed{|F_K^{(s)} - \widehat{F}_k^{(s)}| \leq \mathcal{E}_{k,s} = O_{H,\delta,\gamma}(q \log q) = o(kq).} \quad (\text{FEX37})$$

To check the finite factors, the lower signed bound for $\sum \epsilon_i e_{N_i}$ is at least $-2J_{\max,s} - 2(2q + 1 - 2v)m_{\max}$, and the upper bound is at most $2J_{\max,s} + 2(4q + 1 - 2v)m_{\max}$. Enlarge to FEX36 and use FEX32. Also

$$m_{\max} \leq \log \max\{1, v!A_*^{\text{abs}}/|\mu_v|\} + 8\sqrt{\delta^2 + \gamma^2}(H_{2q} + 2). \quad (\text{FEX38})$$

Together with FEX20 and FEX33 this proves the order claim. All constants are fixed at the actual fixed period. Dividing by kq uses $\log q/k \rightarrow 0$. A uniform claim over periods approaching a locus where μ_v vanishes is not made: the displayed inverse coefficient must remain in the finite bound. Uniformity holds on a fixed coefficient stratum with this coefficient ratio bounded and the same v .

It follows in particular that $F_K^{(s)}/(kq)$ and $\widehat{F}_k^{(s)}/(kq)$ have exactly the same limit points. This does *not* assert that either sequence has a unique limit. The old uniform ceiling for F_K remains true, while FEX35–37 constitute a new, fixed-period transfer at strictly smaller error scale.

R9 The original arithmetic and signed receiving identities

Let $[L_s^{\text{old}}, U_s^{\text{old}}]$ denote the current MRI4e interval, including its first compressed increment and the minimum of its existing finite upper bounds. The following is an add-only improvement at the same four endpoints:

$$\begin{aligned} L_s^{\text{new}} &= \max\{L_s^{\text{old}}, 0, \widehat{F}_k^{(s)} - D_s^+ - (v - t_0)\mathcal{L}_s\}, \\ U_s^{\text{new}} &= \min\{U_s^{\text{old}}, \widehat{F}_k^{(s)} - D_s^- + v\mathcal{L}_s\}, \\ L_s^{\text{new}} &\leq F_K^{(s)} \leq U_s^{\text{new}}, \quad s = 1, k. \end{aligned} \quad (\text{FEX39})$$

The intersection need not improve every finite instance. The comparison of scales, however, is unconditional at each fixed actual period in the stated domain.

The full original Gamma multiplier is

$$f_l(S) = \prod_{j=0}^{l-1} (S - c - 2j - \tfrac{1}{2}), \quad dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}. \quad (\text{FEX40})$$

Its transported full-jet kernel, conditional covariance and mixed blocks are AMT23–30, unchanged. The existing difference interval is $-r\Omega_k^{\text{low}} \leq F_K^{(1)} - F_K^{(k)} \leq r\Omega_k^{\text{high}}$, where

$$\begin{aligned} \Pi_l &= \prod_{j=0}^{l-1} (2j + \tfrac{1}{2})^2, \quad U_{l,N}^{\text{poly}} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + (2j + \tfrac{1}{2})^2], \\ \Omega_k^{\text{low}} &= \log \frac{U_{l,q-1}^{\text{poly}} U_{l,q}^{\text{poly}}}{\Pi_l^2}, \quad \Omega_k^{\text{high}} = \log \frac{U_{l,2q-1}^{\text{poly}} U_{l,2q}^{\text{poly}}}{\Pi_l^2}. \end{aligned} \quad (\text{FEX41})$$

Consequently one may further intersect the tensor-source interval with $[L_1^{\text{new}} - r\Omega_k^{\text{high}}, U_1^{\text{new}} + r\Omega_k^{\text{low}}]$. The fixed-source interval has the corresponding reverse intersection. No independent error assignment is substituted for their common total.

For the arithmetic measure retain exactly

$$w_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 / (2\pi), \quad dm_{h,k}(y) = w_h^{*k}(y) dy, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma. \quad (\text{FEX42})$$

Let $H_N^{\text{ar}}, H_N^\sigma$ be the full coefficient-source Grams. AMT6 retains $(a_k/D_N)H_N^\sigma \preceq H_N^{\text{ar}} \preceq A_k H_N^\sigma$. For $m = 1$ its original constants are $p = 7/2$, $B_h = 50$, $M = 25$, $\alpha = \pi/2$, $D_N = [1 + 4(N + M)^2]^M$, and

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{FEX43})$$

Here $\vartheta_h = \int_{-1}^1 w_h$ and $M_\alpha = \int e^{\alpha y} w_h(y) dy$ are the original integrals, and the named positive envelope constants are the unchanged TW/BT constants. In particular none is inferred from a diagnostic packet.

Keep the AMT exact deficits

$$\begin{aligned} L_N^{\text{ar}} &= \log(A_k D_N / a_k), \quad d_{E,N} = q \log A_k - \log \frac{\det G_N^{\text{ar}}}{\det G_N^\sigma}, \\ c_{X,N} &= \min\{r_X L_N^{\text{ar}}, d_{E,N}\}, \quad C_X^- = c_{X,q-1} + c_{X,q}, \quad C_X^+ = c_{X,2q-1} + c_{X,2q}, \quad X = K, B. \end{aligned} \quad (\text{FEX44})$$

The source-evaluated caps replacing $d_{E,N}$ by $\Xi_{k,N}$ remain available. The complete nested positive deficits and their exact common sum are those of AMT13–18. Using either version of the caps, FEX39 therefore gives precisely the new MRI4f interval

$$\begin{aligned} L_{\text{ar}}^{\text{new}} &= \max\{0, L_1^{\text{new}} - C_K^-, L_1^{\text{new}} + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^{\text{new}} &= \min\{\mathcal{B}_k^{\text{ar}}, U_1^{\text{new}} + C_K^+, U_1^{\text{new}} + \delta_k^\sigma + C_B^-\}, \\ L_{\text{ar}}^{\text{new}} &\leq F_K^{\text{ar}} \leq U_{\text{ar}}^{\text{new}}. \end{aligned} \quad (\text{FEX45})$$

Subtracting the appropriate kernel endpoints from the unchanged total $\mathcal{B}_k^{(s)}$ or $\mathcal{B}_k^{\text{ar}}$ gives the exact boundary interval. This keeps correlation with the actual signed arithmetic value. Its QRI/QGQ finite intervals, including the simple-multiplicity nonzero radius, are not altered by this calculation.

The BRI6d identity remains, with its original orientation,

$$\begin{aligned} F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k, \\ L_k^{\text{new}} + \mathcal{H}_k + \delta_k^\sigma &\leq F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} \leq U_k^{\text{new}} + \mathcal{H}_k + \delta_k^\sigma. \end{aligned} \quad (\text{FEX46})$$

The already supplied source differences are $o(kq)$, and the complete Gamma return has $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ with the unchanged positive equilibrium constant. Consequently the new reduction has the exact leading-scale receiving form

$$\boxed{\frac{F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}}{kq} = \frac{\widehat{F}_k^{(1)}}{kq} - \frac{\mathcal{C}_\Gamma}{4} + o(1).} \quad (\text{FEX47})$$

In particular $F_K^{\text{ar}}/(kq)$, $F_K^{(1)}/(kq)$, $F_K^{(k)}/(kq)$, and both new $\widehat{F}$ ratios have the same limit points. The leading value of FEX29 is still needed to decide the sign in FEX47. Neither the old ceiling nor the new $o(kq)$ transfer assigns that value.

R10 Actions, representatives, and the retained mixed-support maps

The conductor is a coefficient calculation, not a replacement for the arithmetic action or the original observation. Its precise action defect on polynomials is

$$\mathcal{C}_A(Sp) - S'\mathcal{C}_{AP} = \frac{v!}{\mu_v} \sum_{\nu} a_{\nu} b_{\nu} p(S' + b_{\nu}). \quad (\text{FEX48})$$

For the representative of degree less than q , write $\tau(p) = [S^{q-1}]p$. The original action is $A_{\chi}[p] = [Sp - \chi\tau(p)]$. Its transported polynomial defect is the right side of FEX48 minus $\mathcal{C}_A(\chi\tau(p))$. This formula is in the full polynomial target; divisibility by ψ is not asserted for an action image outside the actual conductor preimage. In particular K is not assumed invariant, and the original OKA9 observation-action defect is retained.

Nor is the auxiliary L an invariant subspace of that action when $v > 0$. On its top monomial S^{v-1} , multiplication followed by $\mathcal{C}_A$ gives the nonzero constant $v!$, whereas all its lower monomials give zero. This is the explicit rank-one defect. No quotient Euler action on K^+/L was needed in the metric proof.

One can retain all coefficients while using the calculation. Let $R_{C,N}$ be any of the explicitly specified finite right inverses of $\mathcal{C}_A$ on the full space $\mathcal{P}_{N-v}$, for example $(B^{[N]}(\partial_S))^{-1}\mathcal{J}_v$. Then

$$p \mapsto (p - R_{C,N}\mathcal{C}_{AP}, \mathcal{C}_{AP}) \in L \oplus \mathcal{P}_{N-v}, \quad (\ell, g) \mapsto \ell + R_{C,N}g \quad (\text{FEX49})$$

are inverse coefficient maps. This explicitly retains the low component rather than treating it as zero. The original remainder map, full unit, and literal period observation may be composed with the inverse in FEX49; they are then the same original maps on the same class, not newly assigned maps on a smaller fibre. The volume identity FEX31 restores both low-component quotient metrics exactly.

If an original minimum section is changed by χQ , its original cochain primitive is still the AKP primitive. Namely divide $\chi(s_1 + \dots + s_k)$ successively by the original $h(s_j)$ to obtain $\sum_j h(s_j)Q_j(\mathbf{s})$. Then use the original tensor differential and the primitive

$$\sum_{j=1}^k (-1)^{j-1} Q_j(\mathbf{D}) Q(D_1 + \dots + D_k) (F_h^{\otimes(j-1)} \otimes \phi_* \otimes F_h^{\otimes(k-j)}). \quad (\text{FEX50})$$

Its differential is the literal difference of the original theta representatives, because $h(D)F_h = \Theta\phi_*$. Minus-only legs use the unchanged lift $(0, -\mathcal{F}_{\text{Fourier}}\phi)$ and transported $1-D$ action. This statement applies to an actual relation difference; no element of K or L is newly declared a theta boundary. All support-face insertions and signed coefficient permutations commute with the reconstructed linear maps componentwise. A supported zero stays at its original label and is not identified with the external τ . The additional proper-source Taylor kernels from the mixed-primary lane remain, not being replaced by a conductor kernel.

The linear-algebra and Gamma-polynomial lemmas apply at any finite multiplicity with their actual dimensions. The explicit OCF conductor, its four additional orbit factors, and the rank $8k - 16$ used for FEX22–47 belong only to the stated simple-quartet case. No such rank or frame is assigned to the full-multiplicity OMG case. Its nilpotents, row-residue annihilator, and observed-iterate kernel remain the supplied ones. This continuation makes no Frobenius identification or arithmetic purity assertion.

R11 Scope of verification and the next exact quantity

The analytic assertions proved here are FEX6–21 and the exact transfer FEX22–47 from the explicitly named supplied inputs. Finite exact fixtures test the flag identity, its inverse determinant bounds, the source Jacobian, and the full four-endpoint reduction with nonzero cross terms. They do not evaluate actual xi zeros or actual period coefficients. The accompanying execution record distinguishes those computations from the written proofs and from inherited formal certificates.

The next unresolved quantity is the fixed-period determinant FEX29, with the actual Z_k , R_j^A , and weighted original Gamma Gram. Its leading value is *not* established by this paper. It is now separated from the finite-symbol inverse with an explicit error smaller than kq . In particular, repeating an operator inverse bound of exponential size and raising it to r is no longer the remaining task. The complete original high relation ranks, their mixed Gram blocks, and the period position of $\text{im } Z_k$ are the data that still need estimation. No RH contradiction or disproof follows from the result proved here.

Source dependencies

The fixed objects and current ranks are the supplied OCS/OCF/OKM/OKA, AKJ12–25, and RPC/AMT statements in `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex`. The retained inverse is AIE1–29; the original covariance bound is AKS20–32. The signed source transfer is AMT13–49 and the unchanged GEL/QGT/QRI/QGQ provider in `02_FULL_GAMMA_PROOFS(1).tex`. All complete current source files and their exact hashes accompany this continuation. PR32’s residue and finite observation certificates are inherited inputs only; no new Lean execution is claimed here.

R12 The original weighted relation metric as a minimum dual extension

Retain the original simple quartet, fixed five-orbit period and logarithm branch, all coefficients and shifts of ROG and RDS, and the same two source orders $s \in \{1, k\}$. Put

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & \Delta &= q - q', & \mathbf{m} &= \Delta - v, & c' &= k/2 - 4, \\ N &\in \{q-1, q, 2q-1, 2q\}, & L &= N - q, & d &= N - v, & M &= N - v - q' = L + \mathbf{m}. \end{aligned} \quad (\text{DNE1})$$

Here $0 \leq v \leq 80$, $\mu_v \neq 0$, and χ' is the complete lower-grid polynomial χ_{k-8} , never a derivative. All spaces of degree at most -1 are zero. The source norm is the original norm at $S' = c' + iy$, with mass $M_s = (2\pi)^{s/2}$.

R12.1 A fixed original exponential quotient frame

We specify a finite frame using only the actual original matrices. Let $\mathbb{V}$ be the upper-root Vandermonde with rows $a = 0, \dots, q-1$ and entries $\mathbb{V}_{a,(i,j)} = \beta_{ij}^a$. Let $P_v : \mathbb{C}^{q-v} \rightarrow \mathbb{C}^q$ insert zero in the first v positions, and let C be the exact primary conductor of CEP4, with q' rows and q columns. Define

$$W_v = \mathbb{V}^{-1}P_v, \quad J_v = P_v^\top \mathbb{V} C^\top, \quad W_v J_v = C^\top. \quad (\text{DNE2})$$

The Vandermonde is invertible by the original distinct roots. The first v rows of $\mathbb{V} C^\top$ vanish because the corresponding exponential columns are $E_A e^{\beta'_{ab} z}$, whose zero order is at least v . This proves the last equality. Consequently W_v is an isomorphism onto the coefficient vectors of $\mathcal{N}_v$, and J_v has rank q' , since $C^\top$ is injective.

Choose the first q' row positions $\mathcal{I}$ in a fixed ordering of all subsets for which $\det(J_v)_{\mathcal{I}} \neq 0$, and let $\mathcal{J}$ be their complement. Such a subset exists by the proved rank. Let $E_{\mathcal{J}}$ insert these $\mathbf{m}$ complementary coordinates and put

$$Z_{\text{exp}} = W_v E_{\mathcal{J}}, \quad \mathcal{N}_a(z) = \sum_{i,j} (Z_{\text{exp}})_{(i,j),a} e^{\beta_{ij} z}, \quad F_a(z) = \frac{\mathcal{N}_a(z)}{E_A(z)} \quad (1 \leq a \leq \mathbf{m}). \quad (\text{DNE3})$$

The columns of $[J_v, E_{\mathcal{J}}]$ are independent: restriction of a relation to $\mathcal{I}$ first kills its J_v coefficients, and then its other coefficients vanish. Thus the $\mathcal{N}_a$ give a basis of the exact quotient $\mathcal{N}_v / E_A \mathcal{E}_{k-8}$. Every root value and actual minor defining this frame is retained. The frame is fixed as the endpoint N and source order s change.

For an analytic germ F at zero define the complex-linear functional Λ_F by $\Lambda_F(S'^n) = F^{(n)}(0)$. Let ℓ_a be the functional from RDS8. Then

$$\ell_a(g) = \Lambda_{F_a}(\chi' g), \quad f_{\ell_a} = \chi'(D) F_a, \quad A_{\ell,ba} = \ell_a(S'^b) \quad (0 \leq b < \mathbf{m}). \quad (\text{DNE4})$$

Indeed multiplication by χ' in the functional is precisely the constant-coefficient operator $\chi'(D)$ on its generating series. RDS8 proves that these functionals form the entire dual of the relation quotient. RDS2 identifies that quotient with the increasing monomials below degree $\mathbf{m}$, so A_{ℓ} is invertible. This supplies its existence without assuming a nonzero unevaluated minor.

R12.2 The exact original source extension problem

On $\mathbb{C}[S']_{\leq M}$ use the original weighted norm $\|g\|_{\chi',s,c'} = \|\chi' g\|_{s,c'}$. Let G_M be its full monomial Gram, and let $\mathbf{U}_L$ be the complete ROG4 relation matrix. The quotient

$$Y_N = \mathbb{C}[S']_{\leq M} / \text{im } \mathcal{U}_L \simeq \mathbb{C}[S']_{< \mathbf{m}}$$

has Gram, in its fixed low-monomial representative frame,

$$H_{Y,N}^{(s)} = E_{< \mathbf{m}}^* G_M E_{< \mathbf{m}} - E_{< \mathbf{m}}^* G_M \mathbf{U}_L (\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M E_{< \mathbf{m}}. \quad (\text{DNE5})$$

The second term is absent when $L = -1$. Distinct leading degrees of the columns of $\mathcal{U}_L$ show that the low representatives and all relation columns form a basis. Positivity of G_M and minimization over the full relation columns therefore prove DNE5 and its strict positivity.

For $x \in \mathbb{C}^{\mathbf{m}}$ write $\mathcal{N}_x = \sum_a x_a \mathcal{N}_a$ and $F_x = \mathcal{N}_x / E_A$. Its functional ℓ_x on Y_N has the exact norm

$$\|\ell_x\|_{Y_N^\vee}^2 = \min_{\eta \in \mathbb{C}^{q'}} \left\| \Lambda_{F_x} + \sum_{a,b} \eta_{ab} \text{ev}_{\beta'_{ab}} \right\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee}^2. \quad (\text{DNE6})$$

Here the dual consists of complex-linear functionals. To prove the equality, multiply by χ' to identify the weighted polynomial space isometrically with $I_N = \chi' \mathbb{C}[S']_{\leq M} \subset \mathbb{C}[S']_{\leq d}$. The restriction of Λ_{F_x} to I_N is exactly $g \mapsto \ell_x(g)$ by DNE4, and it annihilates $\chi' \text{im } \mathcal{U}_L$ by RDS5.

A functional annihilating a subspace has the same norm on the Hilbert quotient: the minimum lift of each quotient class has the same functional value, and every other lift has larger norm. A functional on a Hilbert subspace has a unique extension of minimum norm to the ambient space, namely its composition with the orthogonal projection onto that subspace. Every other extension adds a functional on the perpendicular complement; their squared dual norms add. These two facts reduce its quotient dual norm to the minimum over all ambient extensions of its restriction.

The difference of two extensions annihilates I_N . That annihilator has dimension $(d+1) - (M+1) = q'$. Evaluation at each original lower root annihilates I_N , and these q' evaluation functionals are independent on $\mathbb{C}[S']_{\leq d}$, by the first q' monomials and the distinct-root Vandermonde. They therefore constitute its whole annihilator. This proves DNE6 and retains every permitted extension.

Let

$$\phi_{n,s,c'}(S') = \frac{p_n((S' - c')/i)}{\sqrt{h_n}}, \quad h_n = M_s n! (s/2)_n, \quad 0 \leq n \leq d$$

be the unchanged original orthonormal source frame. Form the complete finite matrices

$$\begin{aligned} T_{N,s;na} &= \Lambda_{F_a}(\phi_{n,s,c'}), & V_{N,s;n,(i,j)} &= \phi_{n,s,c'}(\beta'_{ij}), \\ \mathcal{D}_N^{(s)} &= T_{N,s}^* [I - V_{N,s}(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^*] T_{N,s}. \end{aligned} \quad (\text{DNE7})$$

The matrix $V_{N,s}$ has rank q' by the same Vandermonde argument, since the displayed orthonormal polynomials span all degrees at most d . In an orthonormal basis the squared norm of a complex-linear functional is the sum of the squared moduli of its values. Thus the minimum in DNE6 is $\min_\eta \|T_{N,s}x + V_{N,s}\eta\|_2^2$. The minimizing vector is $-(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^* T_{N,s}x$. Substitution proves DNE7, including both mixed factors, and gives $\|\ell_x\|^2 = x^* \mathcal{D}_N^{(s)} x$. Positivity follows because the ℓ_a are an independent dual frame of Y_N .

R12.3 Complex-linear duality and the actual fixed kernel columns

For a positive Gram H and a complex-linear functional $y \mapsto a^\top y$, its squared dual norm is $a^\top H^{-1} \bar{a} = a^* \overline{H^{-1} a}$. Indeed its Riesz vector is $H^{-1} \bar{a}$, and evaluation of its squared norm gives the first expression; the second is its equal real conjugate. Therefore the precise metric identity and determinant consequence of DNE4–7 are

$$\mathcal{D}_N^{(s)} = A_\ell^* (\overline{H_{Y,N}^{(s)}})^{-1} A_\ell, \quad \det \mathcal{D}_N^{(s)} = \frac{|\det A_\ell|^2}{\det H_{Y,N}^{(s)}}. \quad (\text{DNE8})$$

The conjugation records the ordinary complex-linear dual coordinates; it is not discarded by changing the source frame. In the four-endpoint sum, the fixed factor $|\det A_\ell|^2$ cancels with signs $(1, 1, -1, -1)$.

Retain the actual fixed matrix Z_k of FEX27 and FGC9, with $t = r - t_0$ independent columns in the low-monomial frame of dimension $\mathbf{m}$. Multiplying every relation column by the same nonzero

scalar $\alpha_A = v!/\mu_v$ does not change its span or its Schur complement. Consequently its actual native FEX29 matrix and the exact dual quotient map are

$$\widehat{H}_{K,N}^{(s)} = Z_k^* H_{Y,N}^{(s)} Z_k, \quad B_K = Z_k^\top A_\ell, \quad Q_{K^\vee,N}^{(s)} = \left[B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* \right]^{-1} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}. \quad (\text{DNE9})$$

The map B_K evaluates each relation-annihilating functional on the actual columns Z_k . It is onto, because A_ℓ is invertible and Z_k has rank t . The first inverse-Gram formula is the minimum dual metric on that same evaluation target. Substitution of DNE8 yields

$$B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* = Z_k^\top \overline{H_{Y,N}^{(s)}} \bar{Z}_k = \widehat{H}_{K,N}^{(s)},$$

which proves the final equality with every transpose in its original direction. Empty rank has determinant one. In particular the full fixed-column return is exactly

$$\widehat{F}_k^{(s)} = - \sum_N \sigma_N \log \det Q_{K^\vee,N}^{(s)}. \quad (\text{DNE10})$$

Its relation to the original $F_K^{(s)}$ retains both low quotient volumes and the full source-map error of FGC12–13 and FEX34–37; DNE10 removes none of those terms.

R12.4 A finite source bound before the high-order derivative

Use the actual R, R', B, C_v and R_0 of ROG6 and RDS11, and put

$$\rho = \tanh(R_0/4), \quad C_{F,k,A} = \frac{2R^v}{|\mu_v|} e^{RR_0}. \quad (\text{DNE11})$$

For any $\mathcal{N} = \sum \nu_{ij} e^{\beta_{ij}z} \in \mathcal{N}_v$, RDS12 proves $|\mathcal{N}/E_A| \leq C_{F,k,A} \|\nu\|_1$ on the complete disk $|z| \leq R_0$. Apply the original Gamma generating identity directly to the functional $\Lambda_{\mathcal{N}/E_A}$:

$$\sum_{n \geq 0} \Lambda_{\mathcal{N}/E_A} (p_n((S' - c')/i)) \frac{z^n}{n!} = (1 + z^2)^{-s/4} e^{ic' \arctan z} \frac{\mathcal{N}(-i \arctan z)}{E_A(-i \arctan z)}. \quad (\text{DNE12})$$

The quotient at zero is removable. On $|z| = \rho$, $|\arctan z| \leq R_0/4$. Cauchy's coefficient estimate therefore gives the exact bound

$$\begin{aligned} \|\Lambda_{\mathcal{N}/E_A}\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee} &\leq \mathcal{B}_{d,s,k,A} \|\nu\|_1, \\ \mathcal{B}_{d,s,k,A}^2 &= \frac{C_{F,k,A}^2}{M_s} (1 - \rho^2)^{-s/2} e^{|c'|R_0/2} \sum_{n=0}^d \frac{n!}{(s/2)_n} \rho^{-2n}. \end{aligned} \quad (\text{DNE13})$$

Indeed the bound for the n th orthonormal coefficient is the square root of its displayed summand times the common factor and $\|\nu\|_1$; summing squared coefficients gives the dual norm. The full Gamma mass remains explicitly in this formula. No application of $\chi'(D)$ and no $q!$ factor is needed in this exact extension norm.

For a fully specified comparison reference, use Euclidean norm in the original upper exponential coefficient order ν_{ij} , with its actual quotient by $\text{im } C^\top$. Its positive Gram in the fixed DNE3 frame is

$$G_{\text{exp}} = Z_{\text{exp}}^* [I - C^\top (\bar{C} C^\top)^{-1} \bar{C}] Z_{\text{exp}}, \quad 0 \prec \mathcal{D}_N^{(s)} \preceq q \mathcal{B}_{d,s,k,A}^2 G_{\text{exp}}. \quad (\text{DNE14})$$

The reference is explicitly specified here for comparison; the actual weighted Gamma quotient remains DNE7. The inverse in G_{exp} exists by the full row rank of C . Its expression is the orthogonal-projection formula, and is positive on this quotient frame by DNE3. Every representative $\nu + C^\top \eta$ remains in $\mathcal{N}_v$ and changes $\mathcal{N}/E_A$ by precisely a lower exponential, by RDS7. Combining DNE6 and DNE13, and then using $\|w\|_1 \leq \sqrt{q} \|w\|_2$, gives

$$x^* \mathcal{D}_N^{(s)} x \leq \mathcal{B}_{d,s,k,A}^2 \inf_\eta \|Z_{\text{exp}} x + C^\top \eta\|_1^2 \leq q \mathcal{B}_{d,s,k,A}^2 x^* G_{\text{exp}} x.$$

This proves DNE14 without a bound on an omitted frame inverse.

At fixed actual period and quartet the finite constant satisfies

$$\log(1 + q\mathcal{B}_{d,s,k,A}^2) = O_{\text{actual}}(q), \quad s = 1, k, \quad d = N - v. \quad (\text{DNE15})$$

For detail, R_0, ρ, v, μ_v are fixed, $R = O(k)$, $|c'| = O(k)$ and $s \leq k$. Thus $\log C_{F,k,A} = O_{\text{actual}}(k + \log(k + 1))$. The original Gamma ratios obey $n!/(s/2)_n \leq 2\sqrt{n}$ for $n \geq 1$ and are one at $n = 0$, as proved by the product estimate in FEX10–11. Hence the sum in DNE13 is at most $(d + 1) \max\{1, 2\sqrt{d}\} \rho^{-2d}$. Since $d \leq 2q$, its logarithm is $O_{\text{actual}}(q)$. The remaining factors have logarithms $O_{\text{actual}}(k)$, with the original positive mass retained before estimation. These bounds prove DNE15.

Restriction of DNE14 to any specified plane, and minimization under any specified onto quotient map, preserve its scalar upper bound. In particular it applies to the actual map B_K with the full reference quotient $(B_K G_{\text{exp}}^{-1} B_K^*)^{-1}$. This is a finite norm comparison of the original metric with a completely retained reference form. The reference contains the full original Vandermonde and conductor data; neither its determinant nor the original return in DNE10 has been assigned a leading value by this upper bound.

R13 Compatible high jets and the original positive return

Retain all fixed original quartet, amplitude, period and branch data of LJR, DNE, PEC and OVR. In particular the arithmetic domain remains $|u| \geq R_*$, the actual period is fixed while $k = 4l + 1$ grows, and the source order is exactly $s = 1$ or $s = k$. All asymptotic uses of OFV retain its proved eventual threshold. Write

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & n &= q-v, & \mathbf{m} &= q-q'-v, & d_0 &= q-v-1, \\ q-1 &\leq N \leq 2q, & D &= N-v, & e_N &= N-q+1, & D+1 &= n+e_N. \end{aligned} \quad (\text{HJR1})$$

The polynomial degree D is the actual native source degree after the conductor. The original centers remain $c = k/2$ and $c' = c - 4$. No change of s is made between different endpoints.

R13.1 The exact compatible high-jet graph

Let $B_{s,D}$ be the original matrix of S'^a in the complete orthonormal Gamma frame $\phi_0, \dots, \phi_D$, as in LJR2-3, with $h_a = M_s a! (s/2)_a$ and $M_s = (2\pi)^{s/2}$. Define

$$\begin{aligned} (\mathbb{V}_N)_{j,(a,b)} &= \beta_{ab}^j \quad (0 \leq j \leq N), \\ (J_{E,N})_{ja} &= \binom{j}{a} \mu_{j-a} \quad (0 \leq j \leq N, \ 0 \leq a \leq D), \\ \tilde{J}_{E,N} &= (J_{E,N})_{\{v,\dots,N\},\{0,\dots,D\}}, \quad P_{v,N} : \mathbb{C}^{D+1} \longrightarrow \mathbb{C}^{N+1}. \end{aligned} \quad (\text{HJR2})$$

Here $P_{v,N}$ inserts v initial zero coordinates, and an out-of-range term in the convolution is zero. The square $\tilde{J}_{E,N}$ is lower triangular with diagonal $\mu_v \binom{v+a}{a}$ and is therefore invertible.

On the original coefficient subspace $\mathcal{N}_v$ put

$$\mathbf{F}_{N,s} = B_{s,D}^{-\top} \tilde{J}_{E,N}^{-1} P_{v,N}^\top \mathbb{V}_N, \quad \mathbf{E}_{N,s} = \mathbf{F}_{N,s} \mathbf{R}_s = \begin{pmatrix} I_n \\ H_{N,s} \end{pmatrix}. \quad (\text{HJR3})$$

The right factor $\mathbf{R}_s$ is the exact low reconstruction LJR4. Thus $\mathbf{F}_{N,s}$ takes original upper exponential coefficients to their original native functional values through degree D ; $\mathbf{E}_{N,s}$ extends the first n such values. For a literal formula for the inverse convolution, if $u_j = (\mathbb{V}_N \nu)_j$, then

$$f_a = \frac{v!}{\mu_v} a! \sum_{r=0}^a \kappa_{a-r} \frac{u_{v+r}}{(v+r)!}, \quad \mathbf{F}_{N,s} \nu = B_{s,D}^{-\top} (f_0, \dots, f_D)^\top. \quad (\text{HJR4})$$

The coefficients κ_a are all the actual inverse-symbol coefficients of LJR6, with their proved finite recurrence. The source coefficient transformation in HJR4 contains the original phase i^{-a} and every h_a .

To prove these statements, use the analytic germs

$$\mathcal{N}(z) = \sum \nu_{ab} e^{\beta_{ab} z}, \quad E_A(z) = (\mu_v/v!) z^v B_\infty(z).$$

The order condition $\nu \in \mathcal{N}_v$ makes the quotient analytic at zero. Formal Taylor multiplication gives HJR4 and HJR2, and the triangular $B_{s,D}^{-\top}$ changes ordinary moments to orthonormal functional values. Its first n moments and orthonormal values depend only on the first q jets u_j . LJR6 therefore proves that the top block of HJR3 is I_n . The bottom block $H_{N,s}$ is the specified remaining e_N rows.

For a fully intrinsic description, write $\chi(S) = S^q + \sum_{a=0}^{q-1} \chi_a S^a$. Define $\mathcal{U}_{N,s} \subset \mathbb{C}^{D+1}$ by requiring that $u = J_{E,N} B_{s,D}^\top w$ satisfy

$$u_{q+j} + \sum_{a=0}^{q-1} \chi_a u_{a+j} = 0, \quad 0 \leq j < e_N. \quad (\text{HJR5})$$

Then $\mathcal{U}_{N,s} = \text{im } \mathbf{E}_{N,s}$, and its dimension is n . Indeed each original root is annihilated by χ , so HJR3 satisfies every recurrence. Conversely the first q entries of u , with its first v zeros, determine a unique ν by the invertible upper Vandermonde. The monic recurrences determine every later u_j uniquely and give $\mathbf{V}_N \nu$. The invertible lower convolution block then determines w uniquely. This proves the graph equality, its dimension, and the independence of its e_N constraints. The free ambient later functional values have not been identified with upper exponential coefficients; the exact identification uses HJR5.

Let $V_{N,s}$ denote the $(D+1)$ -by- q' lower evaluation matrix $(V_{N,s})_{a,\beta'} = \phi_a(\beta')$, and let $V = V_{q-1,s}$. The complete lower-exponential identity gives

$$\mathbf{E}_{N,s} V = V_{N,s}, \quad \mathcal{U}_{N,s} / \text{im } V_{N,s} \simeq \mathcal{N}_v / \text{im } C^\top. \quad (\text{HJR6})$$

For its proof use $\nu = C^\top \eta$ in HJR4: $\mathcal{N}/E_A$ is exactly the sum of the lower exponentials with coefficients η , at all orders. Since $\mathbf{E}$ is injective, its induced quotient kernel is exactly $\text{im } V$. This proves the typed quotient identification with the original denominators.

R13.2 One lower-evaluation minimum over the complete window

Keep precisely DNE's fixed coefficient quotient frame Z_{exp} . Put $T_N = \mathbf{F}_{N,s} Z_{\text{exp}}$, $T_0 = T_{q-1}$, and

$$\begin{aligned} A &= V^* V, & K_0 &= A^{-1} V^* T_0, & Z_0 &= T_0 - V K_0, & Q_0 &= V A^{-1/2}, \\ D_0 &= Z_0^* Z_0 = \mathcal{D}_{q-1}^{(s)}, & V^* Z_0 &= 0, & Q_0^* Q_0 &= I_{q'}. \end{aligned} \quad (\text{HJR7})$$

The matrices A and its positive square root are the actual lower-evaluation Gram, retaining the complete root and Gamma conditioning. No replacement evaluation metric is made. With $H = H_{N,s}$, define

$$P_N = H Z_0, \quad Q_N = H Q_0, \quad \mathcal{C}_N = (I_{e_N} + Q_N Q_N^*)^{-1/2} P_N. \quad (\text{HJR8})$$

At $e_N = 0$ all these matrices have zero rows. The exact native metric at every endpoint is

$$\boxed{\mathcal{D}_N^{(s)} = D_0 + P_N^* (I_{e_N} + Q_N Q_N^*)^{-1} P_N = D_0 + \mathcal{C}_N^* \mathcal{C}_N.} \quad (\text{HJR9})$$

In particular its increment is positive and has rank $\text{rank } P_N \leq \min\{\mathbf{m}, e_N\}$. The first endpoint update $N = q$ has rank at most one. The last two endpoints have $e_{2q-1} = q$ and $e_{2q} = q+1$, respectively.

Here is a proof with the complete minimizer. Subtract the fixed lower extension $\mathbf{E} V K_0$ from $T_N = \mathbf{E} T_0$. Every remaining allowed extension is $\mathbf{E}(Z_0 x + Q_0 t)$, with one vector $t \in \mathbb{C}^{q'}$ used for both its low and high rows. Its squared norm is

$$x^* D_0 x + \|t\|^2 + \|P_N x + Q_N t\|^2. \quad (\text{HJR10})$$

The minimum occurs at $t = -(I + Q_N^* Q_N)^{-1} Q_N^* P_N x$. Completing the square leaves $x^* D_0 x + x^* P_N^* [I - Q_N (I + Q_N^* Q_N)^{-1} Q_N^*] P_N x$. Multiplication by $I + Q_N Q_N^*$ shows that the bracket is $(I + Q_N Q_N^*)^{-1}$, proving HJR9. DNE7 proves that these and only these lower evaluations are the permitted native extension differences. Thus HJR10 is exactly its full minimum, without separate minima on different row blocks.

R13.3 The original kernel and annihilator updates

Write $B = B_K$ and $J = J_0$ for the actual maps in PEC19,24, so B is onto, J is injective, and $\text{im } J = \ker B$. Define the full initial forms and minimum lift by

$$\begin{aligned} D_{B,0} &= (B D_0^{-1} B^*)^{-1}, & D_{J,0} &= J^* D_0 J, \\ L_0 &= D_0^{-1} B^* D_{B,0}, & W_0 &= J (D_{J,0})^{-1/2}. \end{aligned} \quad (\text{HJR11})$$

Then $BL_0 = I$, $BW_0 = 0$, $W_0^*D_0W_0 = I$, and $W_0^*D_0L_0 = 0$. These are literal original kernel and annihilator maps; the displayed W_0 is a coefficient matrix for $\ker B$ with its full D_0 metric. The induced exact updates are

$$\begin{aligned} D_{J,N} &= D_{J,0} + (\mathcal{C}_N J)^*(\mathcal{C}_N J), \\ D_{B,N} &= D_{B,0} + (\mathcal{C}_N L_0)^*[I + \mathcal{C}_N W_0 W_0^* \mathcal{C}_N^*]^{-1}(\mathcal{C}_N L_0). \end{aligned} \quad (\text{HJR12})$$

The first is restriction of HJR9. For the second, every B -lift of y is $L_0 y + W_0 t$. Its low squared norm is $y^* D_{B,0} y + \|t\|^2$, and its added squared norm is $\|\mathcal{C}_N L_0 y + \mathcal{C}_N W_0 t\|^2$. The same completion as HJR10 proves the second formula. It preserves both the original lower-evaluation denominator in $\mathcal{C}_N$ and the original kernel denominator in HJR12. Empty source, kernel or target ranks use their unique empty matrices and determinant one.

The exact identities DNE9 and PEC25 are

$$D_{B,N} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}, \quad D_{J,N} = \overline{Q_{0,N}^{-1}}. \quad (\text{HJR13})$$

Thus HJR12 is a full original native kernel and complementary residual calculation, rather than a freely chosen coefficient compression.

R13.4 Exact one-row updates and their allocation

The original orthonormal polynomial sequence is independent of its finite cutoff. At any M with $q-1 \leq M < 2q$, let $d = M - v$, and write T, V for the complete matrices through degree d . Their next rows, at degree $d+1$, are t, v_+ . Every entry of t is $\Lambda_{N_a/E_A}(\phi_{d+1})$, computed by HJR4 and the original monomial matrix $B_{s,d+1}$ through degree $d+1$; every entry of v_+ is $\phi_{d+1}(\beta')$. Set

$$\begin{aligned} A_M &= V^* V, \quad K_M = A_M^{-1} V^* T, \quad r_M = t - v_+ K_M, \\ \lambda_M &= 1 + v_+ A_M^{-1} v_+^* > 0, \quad h_M = \lambda_M^{-1/2} r_M. \end{aligned} \quad (\text{HJR14})$$

Then the exact update is

$$D_{M+1} = D_M + h_M^* h_M, \quad D_M = \mathcal{D}_M^{(s)}. \quad (\text{HJR15})$$

To prove this, subtract the attained old minimum from the new extension problem. The old residual is orthogonal to all old lower evaluations, so the remaining minimization for coefficient correction η is $\eta^* A_M \eta + \|r_M x - v_+ \eta\|^2$. Its minimizer is $(A_M + v_+^* v_+)^{-1} v_+^* r_M x$. The same identity used in HJR10 gives the remaining form $r_M^* r_M / (1 + v_+ A_M^{-1} v_+^*)$, proving HJR15. Every evaluation row has participated in the old A_M ; the update does not reset the lower-evaluation coefficient.

At this endpoint put $D_{B,M} = (B D_M^{-1} B^*)^{-1}$, $L_M = D_M^{-1} B^* D_{B,M}$ and $D_{J,M} = J^* D_M J$. The exact inverse-metric decomposition is

$$D_M^{-1} = J(D_{J,M})^{-1} J^* + L_M(D_{B,M})^{-1} L_M^*. \quad (\text{HJR16})$$

Indeed $[J, L_M]$ is a full coefficient frame whose Gram in D_M is $\text{diag}(D_{J,M}, D_{B,M})$. The off-diagonal blocks vanish because $BJ = 0$. Inverting that frame Gram proves HJR16. Define the two actual nonnegative scalar row energies

$$x_M = h_M J(D_{J,M})^{-1} J^* h_M^*, \quad y_M = h_M L_M(D_{B,M})^{-1} L_M^* h_M^*. \quad (\text{HJR17})$$

Both retain λ_M^{-1} from HJR14 and every original period-dependent kernel coordinate. The determinant lemma, proved by extending a nonzero vector to a basis or by expanding a rank-one determinant, now gives

$$\begin{aligned} g_M &= \log \frac{\det D_{M+1}}{\det D_M} = \log(1 + x_M + y_M), \\ g_{0,M} &= \log \frac{\det D_{J,M+1}}{\det D_{J,M}} = \log(1 + x_M), \\ g_{K,M} &= \log \frac{\det D_{B,M+1}}{\det D_{B,M}} = \log \left(1 + \frac{y_M}{1 + x_M} \right), \quad g_M = g_{0,M} + g_{K,M}. \end{aligned} \quad (\text{HJR18})$$

The restriction update gives the second equality directly. For the quotient use the one-row instance of HJR12: its added form is $(h_M L_M)^*(h_M L_M)/(1+x_M)$. Its determinant relative to $D_{B,M}$ proves the third. This proves the allocation of each full original row increment between the actual kernel quotient and its annihilator, with its complete positive denominator.

R13.5 The four original signs and an exterior bound at scale kq

For X equal to the full space, K , or 0, define

$$I_{X,N} = \log \frac{\det D_{X,N}}{\det D_{X,q-1}}, \quad I_N = I_{\text{full},N}. \quad (\text{HJR19})$$

All these numbers are nonnegative and increase with N . At each N , completing J by any fixed right inverse of B factors the full Gram into its restriction and quotient. The fixed frame determinant cancels between endpoints, so

$$I_N = I_{K,N} + I_{0,N}, \quad I_{X,N} = \sum_{M=q-1}^{N-1} g_{X,M}. \quad (\text{HJR20})$$

In the sum for the full space $g_{\text{full},M} = g_M$. The source dual identities HJR13 therefore give exactly

$$\begin{aligned} \widehat{F}_k^{(s)} &= I_{K,2q-1} + I_{K,2q} - I_{K,q}, \\ F_{0,k,s} &= I_{0,2q-1} + I_{0,2q} - I_{0,q}, \\ \mathcal{T}_{k,s} &:= \widehat{F}_k^{(s)} + F_{0,k,s} = I_{2q-1} + I_{2q} - I_q. \end{aligned} \quad (\text{HJR21})$$

Equivalently every line is the weighted sum of its original row increments, with weights

$$w_M = \begin{cases} 1, & M = q-1 \text{ or } M = 2q-1, \\ 2, & q \leq M \leq 2q-2. \end{cases} \quad (\text{HJR22})$$

For example I_{2q-1} contains all rows through $2q-2$, I_{2q} adds the last row, and I_q removes one copy of the first. This proves all endpoint signs. Writing $a = I_q$, $b = I_{2q-1} - I_q$ and $c = I_{2q} - I_{2q-1}$ gives $\mathcal{T} = a + 2b + c$ with $a, b, c \geq 0$. In particular

$$0 \leq I_N \leq I_{2q} \leq \mathcal{T}_{k,s} \quad (q-1 \leq N \leq 2q). \quad (\text{HJR23})$$

To give its finite original bound, retain $\delta_{K,k,s} = \widehat{F}_k^{(s)} - F_K^{(s)}$ with $|\delta_{K,k,s}| \leq \mathcal{E}_{k,s}^{\text{FEX}}$, and precisely $\delta_{0,k,s}$ and $F_{\Xi(L)}^{(s)}$ from PEC29. The latter is the original restriction return called $F_{W_0}^{(s)}$ there. OAR3 and PEC29 give the exact equality

$$\mathcal{T}_{k,s} = \mathcal{F}_{\partial,k}^{(s)} + \mathfrak{e}_{k,s} + \delta_{K,k,s} - F_{\Xi(L)}^{(s)} - \delta_{0,k,s}. \quad (\text{HJR24})$$

The original finite signs are consequently

$$\begin{aligned} \max\{0, \mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - \mathcal{E}_{k,s}^{\text{FEX}} - E_{k,s}^{\text{tr}} - (v - t_0)\mathcal{L}_s\} &\leq \mathcal{T}_{k,s} \\ &\leq U_{k,s}^{\text{jet}} := \mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ + \mathcal{E}_{k,s}^{\text{FEX}} + E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{HJR25})$$

All errors on the right and left are the explicit accepted FEX, OAR and PEC expressions. The inequality uses $0 \leq F_{\Xi(L)}^{(s)} \leq (v - t_0)\mathcal{L}_s$ and $|\delta_0| \leq E^{\text{tr}}$ only after retaining their actual values in HJR24. In particular

$$\mathcal{T}_{k,s} = \Delta q \mathfrak{E}_{\partial} + o(kq), \quad U_{k,s}^{\text{jet}} = \Delta q \mathfrak{E}_{\partial} + o(kq). \quad (\text{HJR26})$$

This follows from the fully proved OFV/OVR outer first variation and the stated orders of the finite errors. The actual period remains fixed; no uniformity over a moving period is asserted.

Now take any paired restriction, quotient, or subquotient of the common coefficient space at the two metrics (D_0, D_N) . Let its rank be t and its determinant increment be $I_{t,N}$. These are attained restrictions and quotients along the same algebraic flag, retaining all its complete metrics. Since $D_N \geq D_0$, every such form increases. Full flag factorization, with nonnegative log increments on both complementary flags, proves

$$0 \leq I_{t,N} \leq I_N \leq U_{k,s}^{\text{jet}} = O_{\text{actual}}(kq). \quad (\text{HJR27})$$

For the same reason all generalized eigenvalues for (D_N, D_0) are at least one; the sum of all their logarithms is I_N . Hence every exterior degree of the identity between these two metrics has norm at most $\exp(U_{k,s}^{\text{jet}}/2)$. The same statement holds on every specified paired flag. This is a full exterior bound with no additional rank multiplier and with the complete lower-evaluation and kernel conditioning retained.

R13.6 Exact cancellation of the low leading scale

Retain LJR's actual finite remainder $\mathcal{R}_{t,k,s}$, including its full-space formula LJR28 and the two-sided flag bounds LJR31. Keep the fixed G_{exp} quotient or restriction associated with this same flag. Then HJR19 gives the exact identity at every endpoint

$$\log \frac{\det G_t}{\det D_{t,N}} = 2tq \log k + \mathcal{R}_{t,k,s} - I_{t,N}. \quad (\text{HJR28})$$

HJR27 therefore propagates the same low leading scale $2tq \log k$ to all four endpoints, with remainder $O_{\text{actual}}(kq)$. It does so by the recurrence-compatible graph and the full Schur minima, not by assigning the later free jets the low dimension.

Let $p_{t,N}, n_{t,N}$ be PEC's two nonnegative logarithmic spectral parts for $(D_{t,N}, G_t)$. The same identity is

$$n_{t,N} = 2tq \log k + \mathcal{R}_{t,k,s} - I_{t,N} + p_{t,N}, \quad 0 \leq p_{t,N} \leq \varepsilon_{N,s}^{\vee} = o(kq). \quad (\text{HJR29})$$

At the actual kernel quotient and annihilator restriction, write $X = K, 0$ respectively. Substituting HJR29 with the four original signs cancels the four copies of both $2tq \log k$ and $\mathcal{R}_{t,k,s}$ exactly and gives

$$\begin{aligned} A_{X,k,s} &= I_{X,2q-1} + I_{X,2q} - I_{X,q} - P_{X,k,s}, \\ P_{X,k,s} &= p_{X,2q-1} + p_{X,2q} - p_{X,q-1} - p_{X,q}, \quad 0 \leq P_{X,k,s} \leq E_{k,s}^{\vee}. \end{aligned} \quad (\text{HJR30})$$

The signs of the p terms follow by substituting their displayed positions in HJR29. Their monotonicity and finite bound are PEC21,26 on these same maps. Together with HJR18 and HJR22 this supplies the explicit original calculation to carry forward:

$$\begin{aligned} A_{K,k,s} &= \sum_{M=q-1}^{2q-1} w_M \log \left(1 + \frac{y_M}{1 + x_M} \right) - P_{K,k,s}, \\ A_{0,k,s} &= \sum_{M=q-1}^{2q-1} w_M \log(1 + x_M) - P_{0,k,s}. \end{aligned} \quad (\text{HJR31})$$

The same original x_M, y_M sum in the full row energy $\log(1 + x_M + y_M)$. Every numerator, denominator and moment used to compute these scalars is fixed in HJR2–4, HJR14, HJR16–17. Their weighted total has the proved leading value HJR26; their separate allocation is retained by the exact formulas HJR31. Thus the next-order kq problem is the specified allocation of these original Gamma row energies, with all low factorial and Vandermonde terms canceled only through the exact equality HJR28.

The two actual row allocations in the original Gamma orders. Keep the original base degree $a = 4l + 1 \geq 9$, $q_a = (a + 1)^2$, the standard endpoints $N_i = q_a - 1, q_a, 2q_a - 1, 2q_a$ and $d_i = N_i - v$. This paragraph uses the degree- a standard window, not a degree- k source on a degree- j target window. Let $\Omega_a^{[a],\text{lo/hi}}$ mean the full NSR2 constants at these d_i . The exact ranks are $t_a = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$. Then

$$\begin{aligned} S_{K,a}^{(s)} &= \sum_{M=q_a-1}^{2q_a-1} w_M \log \left(1 + \frac{y_M^{(s)}}{1 + x_M^{(s)}} \right) = \widehat{F}_a^{(s)}, \\ S_{R,a}^{(s)} &= \sum_{M=q_a-1}^{2q_a-1} w_M \log(1 + x_M^{(s)}) = F_{0,a,s}, \quad S_{K,a}^{(s)} + S_{R,a}^{(s)} = \mathcal{T}_{a,s}. \end{aligned} \tag{HJR32}$$

The row h_M in HJR14 still contains $(1 + v_+(V^*V)^{-1}v_+^*)^{-1/2}$ and the full old lower projection. The inverse-metric decomposition HJR16 in the full frame $[J, L_M]$ proves the two scalar energies; HJR18 gives their determinant increments. Telescoping with the original weights $w_M = 1$ at the extremes and 2 inside proves HJR32, including all zero-energy rows. Applying the full NSO same-fibre comparison to these exact restriction and quotient values now gives

$$\begin{aligned} -t_a \Omega_a^{[a],\text{hi}} &\leq S_{K,a}^{(a)} - S_{K,a}^{(1)} \leq t_a \Omega_a^{[a],\text{lo}}, \\ -j_{0,a} \Omega_a^{[a],\text{hi}} &\leq S_{R,a}^{(a)} - S_{R,a}^{(1)} \leq j_{0,a} \Omega_a^{[a],\text{lo}}. \end{aligned} \tag{HJR33}$$

The ranks belong to the final coefficient quotient and restriction; the full raw polynomial plane has not been assigned these ranks. The common mass is retained before the signed return, so these order differences are $O_{\text{actual}}(a^2 \log q_a) = o(aq_a)$.

At the actual target degree $a = j = k + 4$, the original fourfold quotient flag of CPQ2–8 further gives

$$\begin{aligned} 4S_{K,j}^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ 4S_{R,j}^{(s_1)} &= 4\mathcal{T}_{j,s_1} - V_I - V_b - V_t - V_{\text{rem}}, \\ 4S_{K,j}^{(s_1)} &= \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \end{aligned} \tag{HJR34}$$

The first line is the full target Gram factorization with all its fixed frame determinants before the return; the second subtracts it from four copies of HJR32. For the third use $\mathcal{L}_e = V_I + V_t - V_s - V_A$. Every positive quotient has its literal four-row formula CPQ14–18. This transports the same original row allocation into the connecting and invariant calculation without assigning any individual large-row concentration or leading coefficient.

The common-angle interval returned to the two original allocations. At the same target degree $j = k + 4$, $k = 4l + 1 \geq 9$, retain the original two source orders and every quotient of HJR34. Put $Z_e = V_b + V_{\text{rem}} + V_s + V_A$. Each summand is its complete attained four-endpoint return: the respective spaces are $W/W_i, \mathcal{U}/T, S/K_c$ and I/K_i . The exact value maps and frames are CPQ4–8; in particular the source term V_A still uses the target order and its own fibre. Let L_-, L_+ be the explicit SFC9–10 expressions of the initial common determinant and three original positive row budgets. Define

$$\begin{aligned} K_-^{(s_1)} &= \max\{0, (L_- + Z_e)/4\}, \\ K_+^{(s_1)} &= \min\{\mathcal{T}_{j,s_1}, U_{j,s_1}^{\text{native}}, (L_+ + Z_e)/4\}, \\ K_-^{(s_1)} &\leq S_{K,j}^{(s_1)} \leq K_+^{(s_1)}, \quad \mathcal{T}_{j,s_1} - K_+^{(s_1)} \leq S_{R,j}^{(s_1)} \leq \mathcal{T}_{j,s_1} - K_-^{(s_1)}. \end{aligned} \tag{HJR35}$$

The U^{native} entry is used only on the original RQB35 domain and is omitted elsewhere. HJR34 gives the exact affine map $S_K = (\mathcal{L}_e + Z_e)/4$. Applying it to CIT35 proves the bounds before intersection. Positivity of both allocations, their exact total HJR32 and the applicable native

cap prove the stated intersection. Subtracting from that same total reverses the bounds and proves the residual interval. No summand of Z_e is removed. HJR33 then intersects the two order intervals with the asymmetric translations displayed and proved in SFC19. This gives the complete finite receiving calculation on the original rows, including their lower-Gram denominators, without choosing a new period or source order.

The refined comparison returned to both complete allocations. Keep the full $Z_e^p = V_b^p + V_{\text{rem}}^p + V_s^p + V_A^p$ in SFC18 and HJR35. Replace their common-angle endpoints by the proved EFR4 endpoints and define

$$\widehat{K}_-^p = \max\{0, (\widetilde{L}_-^p + Z_e^p)/4\}, \quad \widehat{K}_+^p = \min\{\mathcal{T}_j^p, U_{j,s_1}^{\text{native}}, (\widetilde{L}_+^p + Z_e^p)/4\}.$$

The original cap appears only on its original proved domain. The exact affine identity $4S_{K,j}^p = \mathcal{L}_e^p + Z_e^p$ proves this enclosure. The quotient $H_{B,j,D} = (B_j D_{j,D}^{-1} B_j^*)^{-1}$ has rank t_j , while the restriction $J_j^* D_{j,D} J_j$ has rank $j_{0,j}$. Their original inverse-metric return identities are HJR32. Applying EFP16 to both and using $\Delta S_K + \Delta S_R = \Delta \mathcal{T}_j$ gives

$$\begin{aligned} A_K^- &= \max\{t_j \Phi_j^-, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^+\}, \\ A_K^+ &= \min\{t_j \Phi_j^+, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^-\}. \end{aligned} \tag{EFR5}$$

Thus the complete replacement for the NSR-width translation in SFC19 is

$$\begin{aligned} \max\{\widehat{K}_-^1, \widehat{K}_-^0 + A_K^-\} &\leq S_{K,j}^1 \leq \min\{\widehat{K}_+^1, \widehat{K}_+^0 + A_K^+\}, \\ \max\{\widehat{K}_-^0, \widehat{K}_-^1 - A_K^+\} &\leq S_{K,j}^0 \leq \min\{\widehat{K}_+^0, \widehat{K}_+^1 - A_K^-\}. \end{aligned} \tag{EFR6}$$

Adding the actual difference interval to pair zero and subtracting it from pair one prove the two lines. Subtraction in reversed endpoint order from each same-pair $\mathcal{T}_j^p$ gives the residual intervals. Every low correction and all four positive returns remain; no individual allocation is prescribed.

R14 Native transport of the bounded original invariant multipliers

Keep the complete original fixed quartet, period, branch, conductor and Gamma measures. Use the actual BIM1 family $\mathcal{H}$ on its proved domain $k \equiv 1 \pmod{4}$, $k \geq 37$. For $h \in \mathcal{H}$ put $d_h = \deg h = 4r_h$ and $\ell_h = k - d_h \geq 9$. The earlier finite degrees keep their original matrices. The two source-order pairs used below are exactly

$$(s_{\ell_h}, s_k) = (1, 1) \quad \text{or} \quad (\ell_h, k), \quad c'_j = j/2 - 4, \quad D \geq q_k - v - 1, \quad q_j = (j+1)^2, \quad q'_j = (j-7)^2. \quad (\text{MNT1})$$

The polynomial cutoff D is the same integer in the source and target of each map. Its relation to their different original endpoint windows is calculated below.

R14.1 The literal finite translation sum and its two centres

Use every actual coefficient h_{ab} from BIM5 and set

$$B_h(z) = \mathcal{N}_h(z) = \sum_{a,b=0}^{d_h} h_{ab} e^{\beta_{d_h,a,b} z}, \quad (\text{MNT2})$$

$$\beta_{d,a,b} = d/2 + (2a - d)\delta + i(2b - d)\gamma, \quad \eta_{h,a,b} = \beta_{d_h,a,b} - d_h/2.$$

For $h = 1$ the sole coefficient is one, $d_h = 0$, and $B_h = 1$. All other arrays are the literal original multinomial coefficients of $h(\text{Ht})$. Multiplication of numerators has the exact dual map

$$\mathcal{A}_h : \mathbb{C}[S']_{\leq D} \longrightarrow \mathbb{C}[S']_{\leq D}, \quad (\mathcal{A}_h p)(S') = \sum_{a,b} h_{ab} p(S' + \beta_{d_h,a,b}), \quad \Lambda_{B_h F}(p) = \Lambda_F(\mathcal{A}_h p). \quad (\text{MNT3})$$

To prove the identity apply both sides to S'^n . Leibniz' rule on $(B_h F)^{(n)}(0)$ gives precisely $\sum_{a,b} h_{ab} \sum_{j=0}^n \binom{n}{j} \beta_{d_h,a,b}^{n-j} F^{(j)}(0)$, the right side. All germs used here are analytic at zero: the individual source domain is the complete original low-jet subspace $\mathcal{N}_{v,\ell_h}$, so $F = \mathcal{N}/E_A$ is analytic. This does not assign a functional to an individual summand outside that subspace in BIM's larger coupled domain.

For $p_k(y) = p(c'_k + iy)$ the two original centres satisfy $c'_k - c'_{\ell_h} = d_h/2$. Thus the exact pullback is

$$(\mathcal{A}_h p)(c'_{\ell_h} + iy) = \sum_{a,b} h_{ab} p_k(y + \eta_{h,a,b}/i). \quad (\text{MNT4})$$

This retains the phase $1/i = -i$ and the full center difference. Let $H_D = \sum_{j=1}^D 1/j$, with $H_0 = 0$, and put

$$K_h(D) = \sum_{a,b} |h_{ab}| \exp(|\eta_{h,a,b}|(H_D + 2)). \quad (\text{MNT5})$$

AKJ21–23 proves $\|\partial_y\|_{\leq D,s} \leq H_D + 2$ for every original source order $s \geq 1$: its ratio estimate uses only $s/2 \geq 1/2$. The exact finite Taylor identity $p(y+z) = \exp(z\partial_y)p(y)$ therefore gives $\|p(y+z)\|_s \leq e^{|z|(H_D+2)}\|p\|_s$. Apply this to each term of MNT4 and then use the triangle inequality. For either unchanged common source order this proves the norm bound $K_h(D)$ between the displayed centres. In particular $|\eta_{h,a,b}| \leq d_h \sqrt{\delta^2 + \gamma^2}$ and $H_D \leq 1 + \log D$ for $D \geq 1$. Thus $K_h(D)$ is a fully specified polynomial bound in $D+1$ at the fixed actual data and degree $d_h \leq 28$.

R14.2 The complete source-order change

For the degree-order pair put $s = \ell_h$ and $r = r_h$. The full original measure is

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}. \quad (\text{MNT6})$$

The Gamma recurrence, applied exactly r times, proves

$$\frac{dm_{s+4r}}{dm_s}(y) = \frac{(2\pi)^{2r}}{(s/2)_{2r}} \prod_{j=0}^{r-1} (y^2 + (s/2 + 2j)^2) \geq a_{s,r} := (2\pi)^{2r} \prod_{j=0}^{r-1} \frac{s + 4j}{s + 4j + 2} > 0. \quad (\text{MNT7})$$

For detail the squared Gamma ratio is $4^{-r} \prod_j (y^2 + (s/2 + 2j)^2)$. The factor $2^{(s+4r)/2}/2^{s/2} = 4^r$ cancels it, the mass ratio is $(2\pi)^{2r}$, and the remaining denominator ratio is $(s/2)_{2r}$. Pairing its consecutive factors proves the displayed minimum. For $r = 0$ every product and the constant equal one. This computation retains the actual full source mass.

Define

$$\kappa_h(D) = \begin{cases} K_h(D), & (s_{\ell_h}, s_k) = (1, 1), \\ K_h(D)a_{\ell_h, r_h}^{-1/2}, & (s_{\ell_h}, s_k) = (\ell_h, k). \end{cases} \quad (\text{MNT8})$$

First apply MNT4–5 in order s_{ℓ_h} . Then use $\|p_k\|_{s_{\ell_h}} \leq a_{\ell_h, r_h}^{-1/2} \|p_k\|_{s_k}$, which follows by integrating MNT7. This proves the complete primal bound

$$\|\mathcal{A}_h p\|_{s_{\ell_h}, c'_{\ell_h}} \leq \kappa_h(D) \|p\|_{s_k, c'_k}. \quad (\text{MNT9})$$

Since $\ell_h \geq 9$ and $r_h \leq 7$, the positive factors a_{ℓ_h, r_h} are bounded above and away from zero by explicit constants obtained by replacing ℓ_h in each ratio of MNT7 by infinity or by nine. Together with MNT5 this yields $\log \max\{1, \kappa_h(D)\} = O_{\text{actual}}(\log(D + 1))$, uniformly over the finite family and both source-order choices.

R14.3 Every original lower evaluation in the quotient map

Let $\mathcal{E}_j$ denote the complete original degree- j exponential space. BIM7 gives

$$B_h \mathcal{N}_{v, \ell_h} \subset \mathcal{N}_{v, k}, \quad B_h(E_A \mathcal{E}_{\ell_h - 8}) \subset E_A \mathcal{E}_{k - 8}. \quad (\text{MNT10})$$

The first inclusion follows by multiplying an analytic numerator with its first v zero jets by the entire B_h . The second is the exact root-addition identity in BIM6–7. It follows that multiplication induces the typed map

$$\mathbf{M}_h : \mathcal{N}_{v, \ell_h} / E_A \mathcal{E}_{\ell_h - 8} \longrightarrow \mathcal{N}_{v, k} / E_A \mathcal{E}_{k - 8}, \quad [\mathcal{N}] \longmapsto [B_h \mathcal{N}]. \quad (\text{MNT11})$$

Its kernel is exactly the source classes whose product belongs to $E_A \mathcal{E}_{k-8}$. No assertion that this kernel vanishes is needed or used. The polynomial and low-jet multiplication kernels in BIM and the full nonreduced component kernels in KRC remain the same kernels after this explicit coordinate map.

At the common polynomial degree D , use the full original minimum extension metric $D_{j,D}^{(s_j)}$ on the quotient in degree j , defined by

$$\|[\mathcal{N}]\|_{j,D}^2 = \min_{\eta \in \mathbb{C}^{q'_j}} \left\| \Lambda_{\mathcal{N}/E_A} + \sum_{\beta' \in \text{grid}(j-8)} \eta_{\beta'} \text{ev}_{\beta'} \right\|_{(\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s_j, c'_j})^\vee}^2. \quad (\text{MNT12})$$

The first complete jet window has $D \geq q_j - v - 1$ here, so DNE's Vandermonde and relation-annihilator argument proves strict positivity and the exact quotient kernel. Its proof applies at every larger finite D ; it makes no use of an upper cutoff.

Choose the attained source minimum in MNT12. Precompose its functional by $\mathcal{A}_h$. Its first term becomes $\Lambda_{B_h \mathcal{N}/E_A}$ by MNT3. Every old lower evaluation becomes the full sum of target lower evaluations with roots $\beta'_{\ell_h-8} + \beta_{d_h, a, b}$ by MNT3 and BIM6. Thus this is an allowed target extension, with all its coefficients retained. The target minimum is no larger. The dual form of MNT9 then proves

$$\|\mathbf{M}_h[\mathcal{N}]\|_{k,D} \leq \kappa_h(D) \|[\mathcal{N}]\|_{\ell_h, D}. \quad (\text{MNT13})$$

This proves the metric transport through the complete original lower-evaluation denominators. It neither resets those coefficients nor minimizes different row blocks independently.

For exact native coordinates let $T_j = \alpha_A N_j Z_{\text{exp},j}$ be KRC5, where $\alpha_A = v!/\mu_v$ is the same constant for all degrees because E_A is fixed. The DNE coefficient matrix of MNT11 is $T_k^{-1} M_h^{\text{strip}} T_{\ell_h}$. Equivalently multiply the original upper coefficient array by h_{ab} and reduce by the full target $C_k^\top$ to the $Z_{\text{exp},k}$ coordinates. The two factors α_A cancel only in this conjugation of the exact multiplication map. If one instead uses the scaled FGC dual metric, both corresponding Grams acquire $|\alpha_A|^{-2}$, so MNT13 remains the same inequality by that proved common scalar map.

R14.4 The finite family reaches the native kernel image

Restrict each source of MNT11 to its actual invariant low-jet part $\mathcal{J}_{v,\ell_h}$, using precisely the DNE matrix J_{ℓ_h} and native form $J_{\ell_h}^* D_{\ell_h,D} J_{\ell_h}$. Their direct sum has its full block diagonal native Gram $\mathcal{G}_D$. Let Φ_k be the complete coefficient matrix of the sum of these restricted maps, and let $\Psi_k = B_k \Phi_k$ be its composition with the actual onto kernel-dual map. Put

$$\mathcal{K}(D) = \left(\sum_{h \in \mathcal{H}} \kappa_h(D)^2 \right)^{1/2}. \quad (\text{MNT14})$$

Applying MNT13, the triangle inequality in the one target Hilbert space and then Cauchy–Schwarz gives $\|\Phi_k x\|_{k,D} \leq \mathcal{K}(D) \sqrt{x^* \mathcal{G}_D x}$. Taking the further B_k quotient can only decrease the target norm, so the same bound holds for Ψ_k . Its exact image has codimension at most the full BIM11 defect $e_k \leq (|\mathcal{H}| - 1)v$: the quotient of the target by that image is an image of the full coefficient quotient by $\text{im } \Phi_k$. This is the precise native use of the individually constrained image. No analytic germ was assigned to the larger coupled tuple’s nonanalytic summands.

To specify both metrics on this actual image, choose an injective frame U_k of $\text{im } \Psi_k$ and its exact left inverse L_{U_k} , and set

$$\begin{aligned} \tilde{\Psi}_k &= L_{U_k} \Psi_k, & Q_D^{\text{source}} &= (\tilde{\Psi}_k \mathcal{G}_D^{-1} \tilde{\Psi}_k^*)^{-1}, \\ H_D^{\text{image}} &= U_k^* (B_k D_{k,D}^{-1} B_k^*)^{-1} U_k, & H_D^{\text{image}} &\preceq \mathcal{K}(D)^2 Q_D^{\text{source}}. \end{aligned} \quad (\text{MNT15})$$

The matrix $\tilde{\Psi}_k$ is onto. Its minimum source lift has exactly the first displayed Gram, by the complete Hilbert quotient formula proved in KRC19. Apply the preceding norm inequality to that attained lift to obtain MNT15. This specifies the exact source quotient and target image metrics. Every exterior degree has norm at most $\max\{1, \mathcal{K}(D)\}^{\text{rank } U_k}$ for this identity, since each singular value is at most $\mathcal{K}(D)$. On $D = O(q_k)$ its logarithm is $O_{\text{actual}}(k \log q_k) = o(kq_k)$. This is an upper expansion bound for the specified operator; no inverse bound is inferred from it.

At the four original target cutoffs, the missing kernel quotient in MNT15 has rank at most e_k . Its full native determinant return is bounded by FRE10 with that rank, so the complete kernel return differs from the return of H_D^{image} by a nonnegative $O_{\text{actual}}(q_k \log k) = o(kq_k)$ term. This uses the actual image and its remaining quotient and does not assume evenly distributed row energies.

R14.5 The exact displaced source endpoints

The source polynomial cutoff in every preceding map is the literal target cutoff $D = N - v$. For each h define

$$b_{h,k} = q_k - q_{\ell_h} = d_h(2k - d_h + 2). \quad (\text{MNT16})$$

Thus the four source endpoint indices used in MNT12 are exactly

$$q_{\ell_h} - 1 + b_{h,k}, \quad q_{\ell_h} + b_{h,k}, \quad 2q_{\ell_h} - 1 + 2b_{h,k}, \quad 2q_{\ell_h} + 2b_{h,k}. \quad (\text{MNT17})$$

No source cutoff has been replaced by its degree’s usual endpoint. For any fixed source flag, write $F_h(N) = \log \det D_{X,\ell_h,N-v}$ and define the native positive return using its original signs on the dual

metric. Its shifted return minus its usual return is exactly

$$\begin{aligned} & [F_h(2q_{\ell_h} - 1 + 2b_{h,k}) - F_h(2q_{\ell_h} - 1)] + [F_h(2q_{\ell_h} + 2b_{h,k}) - F_h(2q_{\ell_h})] \\ & - [F_h(q_{\ell_h} - 1 + b_{h,k}) - F_h(q_{\ell_h} - 1)] - [F_h(q_{\ell_h} + b_{h,k}) - F_h(q_{\ell_h})]. \end{aligned} \quad (\text{MNT18})$$

This is direct subtraction of the two four-endpoint expressions. Each bracket is an actual non-negative sum of source row increments, by the same minimum extension monotonicity used in MNT12. The exact signed combination retains their potential concentration. No subleading order for MNT18 is asserted.

There is likewise an exact finite relative operator on the matched image,

$$\begin{aligned} \mathcal{R}_D &= (Q_D^{\text{source}})^{-1/2} H_D^{\text{image}} (Q_D^{\text{source}})^{-1/2}, \\ 0 \prec \mathcal{R}_D \preceq \mathcal{K}(D)^2 I, \quad \log \det H_D^{\text{image}} &= \log \det Q_D^{\text{source}} + \log \det \mathcal{R}_D. \end{aligned} \quad (\text{MNT19})$$

Conjugating MNT15 proves the inequality and taking determinants proves the equality. The native image's four-return therefore contains the same four signed relative determinants, with high endpoints positive for the dual return and low endpoints negative. Their negative spectral parts and the displaced source windows MNT18 remain exact finite quantities. The proof supplies the actual bounded-degree native operator, its polynomial upper expansion, and the subleading unmatched quotient. It does not assign a value to the remaining matched relative determinant by omitting either of these quantities.

R14.6 The original displaced windows with their achieved finite bound

In MNT18 retain the same fixed source degree $j = k - d_h$, source order 1 or j , and common cutoff $N - v$. The complete NW/OW/WNR proofs below now bound its actual displaced difference, with the original two high brackets positive and two low brackets negative. Use the literal WNR10 quantities to define $\Omega_{h,k,s}^{\text{lo}} = U_{h,s}^{\text{lo}}(0) + U_{h,s}^{\text{lo}}(1)$ and $\Omega_{h,k,s}^{\text{hi}} = U_{h,s}^{\text{hi}}(q_j) + U_{h,s}^{\text{hi}}(q_j + 1)$ for $d_h > 0$, with both zero for $d_h = 0$. Then

$$-\Omega_{h,k,s}^{\text{lo}} \leq \mathcal{W}_{h,k,s} \leq \Omega_{h,k,s}^{\text{hi}}, \quad \mathcal{W}_{h,k,s} = o(kq_k) \quad \text{on WNR2.} \quad (\text{MNT20})$$

Here the two finite Ω are the exact WNR sums of NW6's original outer-volume differences, two CTV tail bounds and $4(E_a + E_b)$ at each interval. On $j \geq 4096$ and $j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(j+1)^2$, OW10 and OW18 give their full explicit unequal endpoint errors. NW7 proves the same upper increment bound on every specified native flag by Schur factorization; no rank factor is added. Applying it separately to the four literal intervals proves the finite lower and upper signs in MNT20. OW's $O_{\text{actual}}(q_j \log(q_j + 1))$ bounds and $j/k \rightarrow 1$ prove the displayed little-oh. Before that exact domain use NW6's actual outer determinants. If $d_h = 0$ all four increments vanish identically. NBG/NKC/NBR additionally transport the complete degree-four source and source-kernel gauge determinants into its actual invariant image; MNT19's full matched denominator remains.

This calculation uses the full NBG1–32 and NKC1–17 native maps, the DFX1–35 degree-four multiplication and branch kernels, and the KRC1–27 original annihilator coordinates. It addresses the further target-invariant quotient in NBG30–32. Neither its kernel nor its metric loss is deleted. The original period, logarithm branch, conductor, phases and both source-order families remain fixed. The final section strengthens the error in ITL20–25 from $O_{\text{actual}}(q_k \log q_k)$ to $O_{\text{actual}}(k \log q_k)$ by retaining the identical rank-at-most- r update in the two original compatibility comparisons.

R15 Original spaces, maps and two positive forms

Fix the actual simple quartet and its proved five-orbit period data. Let $k = 4l + 1 \geq 9$, $0 \leq v = \text{ord}_0 E_A \leq 80$, $q_j = (j + 1)^2$, $q'_j = (j - 7)^2$, and $D \geq q_{k+4} - v - 1$. Retain precisely

$$(s_0, s_1) = (1, 1) \text{ or } (k, k + 4), \quad c'_j = j/2 - 4, \quad dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{ITL1})$$

The full original native dual Grams are $D_{k,D}^{(s_0)}$ and $D_{k+4,D}^{(s_1)}$. They are the attained minima over all original lower evaluations in MNT12; the lower coefficients are not minimized separately across jet rows. Use the exact KRC/DFX matrices

$$\begin{aligned} J_0 &= \mathbf{J}_k, \quad J_1 = \mathbf{J}_{k+4}, \quad B_1 = \mathbf{B}_{k+4}, \quad \ker B_1 = \text{im } J_1, \\ M_i &= (\alpha_A N_{k+4} Z_{\text{exp}, k+4})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp}, k}), \quad \alpha_A = v!/\mu_v, \quad m_{i,k}[f] = [x_i^4 f], \\ G &= J_0^* D_{k,D}^{(s_0)} J_0, \quad \mathcal{D} = \text{diag}(D_{k+4,D}^{(s_1)}, \dots, D_{k+4,D}^{(s_1)}), \\ F &= \begin{bmatrix} M_1 J_0 \\ M_2 J_0 \\ M_3 J_0 \\ M_4 J_0 \end{bmatrix}, \quad \mathcal{B} = \text{diag}(B_1, B_1, B_1, B_1), \quad W = \text{diag}(J_1, J_1, J_1, J_1), \quad X = \mathcal{B}F. \end{aligned} \quad (\text{ITL2})$$

All stars in metric calculations are complex adjoints. The ordinary transposes inside the original $\mathbf{B}_j, \mathbf{J}_j$ definitions are still those of complex-linear duality and are not changed. The scalar α_A cancels only in the displayed conjugation for M_i . The two source masses in ITL1 are retained in $G, \mathcal{D}$.

Let $V = \mathbb{C}^{j_0, k}$ be the invariant source coordinates. The map F is injective: NBG21–27 proves that the four original target conductor minima give a strictly positive pullback on the full conductor quotient, hence on this invariant restriction. Equivalently DFX21 proves the full conductor tuple's injectivity before its target-invariant quotient. Set

$$A = F^* \mathcal{D} F \succ 0, \quad E = \ker X = \{a \in V : Fa \in \text{im } W\}. \quad (\text{ITL3})$$

The equality follows from $\ker \mathcal{B} = \text{im } W$ in ITL2. It is the exact NBG30 kernel, not an assumption of injectivity after the invariant quotient.

Choose fixed full-column matrices Z_E, Z_R such that Z_E is a basis of E and $Z = [Z_E, Z_R]$ is an invertible basis of V . Write $e = \dim E$, $r = \dim V - e$. All choices are independent of D . Put

$$F_E = F Z_E, \quad F_R = F Z_R, \quad U = X Z_R. \quad (\text{ITL4})$$

Then U is an injective frame of $\text{im } X$. For any positive form P on V define its quotient form in these same lifts by

$$P/E = Z_R^* P Z_R - Z_R^* P Z_E (Z_E^* P Z_E)^{-1} Z_E^* P Z_R. \quad (\text{ITL5})$$

When $e = 0$ the subtracted block is zero. This is the original minimum over $Z_R b + E$: completing its square proves ITL5, with its unique least-norm representative. Let $Q = G/E$ and $A_E = A/E$. The matched target form is

$$H = U^* (\mathcal{B} \mathcal{D}^{-1} \mathcal{B}^*)^{-1} U. \quad (\text{ITL6})$$

The inverse in ITL6 is exactly the four copies of $H_{B,D}$ from NBG30. Thus Q, H are the NBG31 two image metrics, in the particular fixed image frame $U = XZ_R$. If its earlier frame is $U_{\mathcal{X}}$, write $U = U_{\mathcal{X}}C$ with the unique invertible coordinate matrix C . Both earlier Grams are conjugated by C ; their determinant ratio and generalized eigenvalues are unchanged by this proved common coordinate map. No new metric has been selected.

R16 An explicit original coefficient determinant for the target loss

For a full-column matrix Y in the target product, let

$$P_Y = Y(Y^*\mathcal{D}Y)^{-1}Y^*\mathcal{D} \quad (\text{ITL7})$$

be its $\mathcal{D}$ -orthogonal projection. If Y is empty, $P_Y = 0$ and its Gram determinant is one. Choose any fixed coefficient complement W_R such that $\widehat{W} = [F_E, W_R]$ is a basis of $\text{im } W$. It exists since F is injective and $F_E \subset \text{im } W$. This complement keeps every other original invariant direction. Set

$$\begin{aligned} Y &= (I - P_{F_E})F_R, & Z_T &= (I - P_{F_E})W_R, \\ A_E &= Y^*\mathcal{D}Y, & C_T &= Z_T^*\mathcal{D}Z_T, & L &= Z_T^*\mathcal{D}Y, \\ H &= A_E - L^*C_T^{-1}L. \end{aligned} \quad (\text{ITL8})$$

The first Gram equality follows from ITL5 with $P = A$. For the last equality, all target lifts of Ub are precisely $F_Rb + \text{im } W$. Indeed $\mathcal{B}(F_Rb + w) = Ub$ if and only if $w \in \ker \mathcal{B} = \text{im } W$. Minimize first over F_E and then over W_R ; completing the two squares gives ITL8. Since $F_Rb \in \text{im } W$ implies $Z_Rb \in E$, it forces $b = 0$. Consequently $H \succ 0$. Also $C_T \succ 0$ if its rank is nonzero: a linear combination of W_R projecting into F_E would contradict the complement. These statements justify every inverse and include their empty-rank conventions.

Define the additional target-invariant loss matrix

$$\Omega_{\text{inv},D} = A_E^{-1/2} H A_E^{-1/2} = I - A_E^{-1/2} L^* C_T^{-1} L A_E^{-1/2}, \quad 0 \prec \Omega_{\text{inv},D} \preceq I. \quad (\text{ITL9})$$

The square roots only express the identity map between its two specified Hilbert coordinate spaces. ITL8 is its unaltered coefficient formula. In particular

$$\begin{aligned} \omega_{\text{inv},D} &:= \det \Omega_{\text{inv},D} = \frac{\det \begin{pmatrix} C_T & L \\ L^* & A_E \end{pmatrix}}{\det C_T \det A_E} \\ &= \frac{\det([\widehat{W}, F_R]^*\mathcal{D}[\widehat{W}, F_R]) \det(F_E^*\mathcal{D}F_E)}{\det(\widehat{W}^*\mathcal{D}\widehat{W}) \det([F_E, F_R]^*\mathcal{D}[F_E, F_R])} > 0. \end{aligned} \quad (\text{ITL10})$$

To prove the second line, eliminate the F_E block from the three indicated complete Grams. Their Schur blocks are respectively the block in the first line, C_T , and A_E . The determinant of the F_E block appears once on each required side and gives precisely ITL10. No normalization of $\widehat{W}$ is used. Changing its complement or the fixed source lifts changes the adapted block determinants by the same squared determinant factors, which cancel in the displayed ratio, or leaves the minimum forms unchanged.

For a direct finite spectral expression put $T_D = C_T^{-1/2} L A_E^{-1/2}$. The singular values $\rho_i(T_D)$ lie in $[0, 1)$ by ITL9, and

$$-\log \omega_{\text{inv},D} = \sum_i -\log(1 - \rho_i(T_D)^2). \quad (\text{ITL11})$$

Zero singular values contribute zero. This is the precise remaining invariant-target loss, evaluated after the complete old lower gauges; it is not the earlier NBG independent-gauge loss and it is not identified with a rank fraction.

R17 Return of both lower-gauge corrections, including the actual kernel

Let C_D be NBG27's correlated pullback on the invariant source V . It is obtained by restricting the full correlated native norm to the original J_0 ; no independent target invariant quotient is taken in it. NBG23 and 27 prove

$$K_-^{-2}G \preceq C_D \preceq K_+^2G, \quad 0 \prec A \preceq C_D. \quad (\text{ITL12})$$

Retain exactly the original constants NBG12. For the two fixed restrictions V and E define

$$\omega_{V,D} = \frac{\det A}{\det C_D}, \quad \omega_{E,D} = \frac{\det(Z_E^* A Z_E)}{\det(Z_E^* C_D Z_E)}. \quad (\text{ITL13})$$

These are the NBG29 restriction losses $\det \Omega_{V,D}$ and $\det \Omega_{E,D}$, respectively. The zero-dimensional restriction has determinant one. For every positive P , the block determinant in the fixed frame Z gives

$$\det(P/E) = \frac{|\det Z|^2 \det P}{\det(Z_E^* P Z_E)}. \quad (\text{ITL14})$$

It follows in particular that

$$\frac{\det(A/E)}{\det(C_D/E)} = \frac{\omega_{V,D}}{\omega_{E,D}} \leq 1. \quad (\text{ITL15})$$

The inequality also follows by minimizing ITL12 over the same affine fibres; it is not inferred from the separate inequalities $\omega_V, \omega_E \leq 1$ alone.

Let $\mathcal{R}_{X,D} = Q^{-1/2} H Q^{-1/2}$, the exact matched relative operator NBG32, and put $e_{\text{cor},D} = \log \det(C_D/E) - \log \det(G/E)$. Combining ITL8–15 proves the exact identity

$$\log \det \mathcal{R}_{X,D} = e_{\text{cor},D} + \log \omega_{V,D} - \log \omega_{E,D} + \log \omega_{\text{inv},D}. \quad (\text{ITL16})$$

Quotienting the first pair of inequalities in ITL12 proves the explicit finite estimate

$$-2r \log \max(1, K_-) \leq e_{\text{cor},D} \leq 2r \log \max(1, K_+). \quad (\text{ITL17})$$

This follows by bounding each positive generalized eigenvalue and multiplying all r of them. At $D = O(q_{k+4})$ its absolute value is $O_{\text{actual}}(k \log q_{k+4})$, because $r \leq j_{0,k} = 8k - 32 - v + t_{0,k} = O(k)$ and NBG13 gives $\log \max(1, K_+, K_-) = O_{\text{actual}}(\log(D+1))$. No lower bound for either other loss has been assumed.

R18 Exact cancellation of the common compatibility denominator

Apply NKC6–16 to the two actual source restrictions V and E , using their full DNE functional images in the ambient finite dual quotient. Their correlated isometric images satisfy $\mathcal{A}_E \subset \mathcal{A}_V$, since $E \subset V$ and the same map $I - P_C$ is used. Here the calligraphic subscripts name subspaces, not a change in the Gram A . Let P_V, P_E be the corresponding native orthogonal projectors. Retain NKC's single lower-gauge coefficient complement Z_c , complete evaluation map $\mathcal{V}$, compatibility matrix $\mathcal{K}$, next evaluation map $\mathcal{V}_2$, and Gram G_c . In those exact coordinates let

$$\begin{aligned} \mathcal{H}_{V,D} &= Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c + Z_c^* \mathcal{V}^* (P_T - P_C - P_V) \mathcal{V} Z_c, \\ \mathcal{H}_{E,D} &= \mathcal{H}_{V,D} + Z_c^* \mathcal{V}^* (P_V - P_E) \mathcal{V} Z_c. \end{aligned} \quad (\text{ITL18})$$

Nested orthogonal images imply $P_V - P_E$ is an orthogonal projector of rank r . Thus the second difference is positive semidefinite of rank at most r , and both complete Grams are positive by NKC14. This retains the entire source complement even when the invariant source is a proper ambient subspace.

Set $b_{k,D} = r_{\text{comp},k} \max\{-\log c_-, \log c_+\}$, with the actual NKC13 constants and $r_{\text{comp},k} = 3k^2 - 10k - 13$. NKC15–16 gives, separately for $Y = V, E$,

$$\log \omega_{Y,D} = \log \det \mathcal{H}_{Y,D} - \log \det G_c + \eta_{Y,D}, \quad |\eta_{Y,D}| \leq b_{k,D}. \quad (\text{ITL19})$$

Subtract the two exact equalities before estimating their errors: the common $\log \det G_c$ cancels. Therefore

$$\log \det \mathcal{R}_{X,D} = \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} + \log \omega_{\text{inv},D} + \epsilon_D, \quad (\text{ITL20})$$

where the finite bound, with the original rank r , is

$$|\epsilon_D| \leq 2r \max\{\log \max(1, K_-), \log \max(1, K_+)\} + 2b_{k,D}. \quad (\text{ITL21})$$

On the original endpoint range this is $O_{\text{actual}}(q_k \log q_k) = o(kq_k)$. Both displayed determinant ratios in ITL20 are at most one. Their four-endpoint signed sums are not thereby nonpositive. If $P_V - P_E = YY^*$ is an isometric factor, a smaller exact formula for the first ratio is

$$\frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} = \det \left(I_r + Y^* \mathcal{V} Z_c \mathcal{H}_{V,D}^{-1} Z_c^* \mathcal{V}^* Y \right)^{-1}. \quad (\text{ITL22})$$

The determinant lemma proves this directly from ITL18. All coefficients in its r by r matrix come from the specified original source-complement projector and full Gram; no bound on the inverse is asserted without calculating it.

R19 Branch defect and all four original endpoint signs

The source kernel E is not replaced by zero in any preceding formula. DFX24–28 supplies the exact inclusion

$$E \subset \mathcal{E}_{v,k} = \mathcal{J}_{v,k} \cap \ker \mathbf{j}_k, \quad \mathcal{J}_{v,k}/E \rightarrow \mathcal{J}_{v,k}/\mathcal{E}_{v,k} \simeq \text{im } \mathcal{B}_{\{1,2,3,4\},k}, \quad (\text{ITL23})$$

with kernel $\mathcal{E}_{v,k}/E$. All source coordinates are transported by the original $\mathbf{T}_k J_0$ in this display. For the actual full biforms $B_a = \psi_2(Q_a)$, let $\mathcal{L} = \text{lcm}(B_1, B_2, B_3, B_4)$ have bidegree (b_z, b_w) . DFX26–27 gives

$$0 \leq e \leq \dim \mathcal{E}_{v,k} \leq (k - b_z + 1)(k - b_w + 1) - (k - 7)^2. \quad (\text{ITL24})$$

These bounds keep the maximum factor multiplicities in the lcm and all shared components. They do not identify the actual kernel with the full shared-component space; its exact finite coefficient kernel remains DFX31–35.

At $N = q_{k+4} - 1, q_{k+4}, 2q_{k+4} - 1, 2q_{k+4}$ use $D = N - v$ and signs $\varepsilon_N = -1, -1, +1, +1$. Summing ITL20 gives the original further-quotient return

$$\begin{aligned} \sum_N \varepsilon_N \log \det \mathcal{R}_{X,N-v} &= \sum_N \varepsilon_N \log \frac{\det \mathcal{H}_{V,N-v}}{\det \mathcal{H}_{E,N-v}} \\ &\quad + \sum_N \varepsilon_N \log \omega_{\text{inv},N-v} + \sum_N \varepsilon_N \epsilon_{N-v}. \end{aligned} \quad (\text{ITL25})$$

The last term has the explicit sum of four ITL21 bounds, and is $o(kq_k)$. The other two full native determinants are ITL10 and ITL18/22. This is a finite quantitative reduction of the additional invariant quotient, not a value assigned to its remaining leading coefficient. The accepted NW1–13 displacement estimate applies to fixed original source flags when subsequently returning their cutoffs to their own degree; it does not itself estimate either long signed sum in ITL25.

When $r = 0$, both matched Grams are empty and every determinant in the matched ratio is one. Then $E = V$, the two compatibility Grams coincide, and ITL16 and ITL20 give exactly zero with zero actual error; a nonzero upper error bound is not used as a replacement for this identity. No RH consequence, native invariant isometry, or new Lean execution is claimed.

R20 The identical native update sharpens the error to the image rank

The preceding separate use of NKC15 is valid, but the two native comparisons share more data than that estimate uses. Retain the same endpoint, the same lower-gauge isometry W_c of NKC6, and the same nested source projectors $P_E \preceq P_V$ of ITL18. Choose an isometric frame Y for their difference, so that $YY^* = P_V - P_E$ and Y has exactly r columns. This is a coordinate expression of the original difference subspace; it does not replace either native metric. Define

$$\mathcal{J}_D = Y^* W_c : \mathbb{C}^{r_{\text{comp}}, k} \longrightarrow \mathbb{C}^r.$$

For $Z = V, E$ retain the two exact NKC10 and NKC13 matrices

$$\Xi_{Z,D} = I - W_c^* P_Z W_c, \quad \Gamma_{Z,D} = W_c^* (\mathcal{R}^* \mathcal{R} + P_T - P_C - P_Z) W_c.$$

Their differences are literally the same Gram:

$$\begin{aligned} \Xi_{E,D} &= \Xi_{V,D} + \mathcal{J}_D^* \mathcal{J}_D, & \Gamma_{E,D} &= \Gamma_{V,D} + \mathcal{J}_D^* \mathcal{J}_D, \\ 0 &\prec c_- \Xi_{V,D} \preceq \Gamma_{V,D} \preceq c_+ \Xi_{V,D}, & 0 &< c_- \leq 1 \leq c_+. \end{aligned} \tag{ITL26}$$

The first two identities follow by subtracting the displayed definitions, using $P_V - P_E = YY^*$. The last two facts are NKC13–14 with their original constants $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$. All inverses below are therefore positive-matrix inverses already justified by the exact native intersection theorem.

In the actual r -dimensional image-difference coordinates put

$$T_\Xi = \mathcal{J}_D \Xi_{V,D}^{-1} \mathcal{J}_D^*, \quad T_\Gamma = \mathcal{J}_D \Gamma_{V,D}^{-1} \mathcal{J}_D^*.$$

The determinant lemma applied twice to ITL26 gives

$$\begin{aligned} \log \omega_{V,D} - \log \omega_{E,D} &= -\log \det(I_r + T_\Xi), \\ \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} &= -\log \det(I_r + T_\Gamma). \end{aligned} \tag{ITL27}$$

For the first line use $\det \Omega_{Z,D} = \det \Xi_{Z,D}$ from NKC10. For the second use the exact common conjugation $\Gamma_{Z,D} = G_c^{-1/2} \mathcal{H}_{Z,D} G_c^{-1/2}$ from NKC16, so its denominator cancels before the determinant lemma is applied. Thus neither line estimates or omits a kernel determinant. In particular the second line is precisely ITL22, expressed through the complete original gauge isometry.

Inversion reverses the positive form comparison ITL26. Congruence by $\mathcal{J}_D^*$ consequently gives

$$\frac{1}{c_+} T_\Xi \preceq T_\Gamma \preceq \frac{1}{c_-} T_\Xi, \quad \frac{1}{c_+} (I_r + T_\Xi) \preceq I_r + T_\Gamma \preceq \frac{1}{c_-} (I_r + T_\Xi). \tag{ITL28}$$

For the second pair, the first pair supplies the T_Ξ term, while $c_+^{-1} I_r \preceq I_r \preceq c_-^{-1} I_r$ supplies the identity term. This argument does not require T_Ξ, T_Γ to commute. Congruence by $(I_r + T_\Xi)^{-1/2}$ puts each of the r positive generalized eigenvalues between c_+^{-1} and c_-^{-1} . Multiplication and logarithms prove the finite asymmetric bound

$$\begin{aligned} \vartheta_D &:= (\log \omega_{V,D} - \log \omega_{E,D}) - \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} = \log \det(I_r + T_\Gamma) - \log \det(I_r + T_\Xi), \\ &-r \log c_+ \leq \vartheta_D \leq -r \log c_-. \end{aligned} \tag{ITL29}$$

The source-complement matrices and their inverses remain in ITL27; their possibly small eigenvalues are not replaced by an absolute conditioning bound. ITL29 measures only the difference between the two specified evaluations of the same rank-at-most- r update, not the size of either negative logarithm.

Combine ITL29 with the exact ITL16 and the finite ITL17. The very same ϵ_D in ITL20 now has the sharper bounds

$$\begin{aligned} \epsilon_D &= e_{\text{cor},D} + \vartheta_D, & -\alpha_D &\leq \epsilon_D \leq \beta_D, \\ \alpha_D &= 2r \log \max(1, K_-) + r \log c_+, & \beta_D &= 2r \log \max(1, K_+) - r \log c_-. \end{aligned} \quad (\text{ITL30})$$

Both constants are nonnegative and retain exactly the two original source-order families. Since $r = O(k)$ and all the displayed logarithmic constants are $O_{\text{actual}}(\log(D+1))$ by NBG13/NKC3–4, this gives $\epsilon_D = O_{\text{actual}}(k \log q_k)$ at the four original endpoints. The previous larger bound ITL21 remains true; ITL30 is the bound to use for the same error in ITL20 and ITL25.

Write $D_i = N_i - v$ for the four ordered target endpoints of ITL25. With all their signs retained,

$$-\beta_{D_1} - \beta_{D_2} - \alpha_{D_3} - \alpha_{D_4} \leq \sum_{i=1}^4 \varepsilon_i \epsilon_{D_i} \leq \alpha_{D_1} + \alpha_{D_2} + \beta_{D_3} + \beta_{D_4}. \quad (\text{ITL31})$$

Thus the remainder of that exact four-return is now $O_{\text{actual}}(k \log q_k) = o(kq_k)$. No estimate for either leading long determinant sum in ITL25 is asserted by this sharpening. The original source kernel E and its nonreduced enclosing defect in ITL23–24 are unaltered. If $r = 0$, $\mathcal{J}_D$ has zero rows, both determinants in ITL27 are one, and ITL29–31 are exactly zero.

Current arithmetic rows and simultaneous spectrum. The original invariant-target term in ITL25 now has the full arithmetic representation

$$\sum_i \varepsilon_i \log \omega_{D_i} = \sum_{n=d+1}^{d+q_j+1} w_n^{\text{row}} (\gamma_{H,n} - \gamma_{A,n}), \quad \left| \sum_i \varepsilon_i \log \omega_{D_i} \right| \leq \sum_n w_n^{\text{row}} b_n^{\text{row}}. \quad (\text{ITL32})$$

Here $d = q_j - v - 1$ and w_n^{row} is exactly the weight ϑ_n of IRR25/NFR3, namely $1, 2, \dots, 2, 1$. The quantity b_n^{row} is the full four-row budget β_n of FAB/NFR4. The original comparison errors ϑ_D and β_D in ITL29–31 retain their definitions. Every row and both positive determinants are IRR7–25. The full-spectrum paired-plane proof IF and FAB gives $b_n^{\text{row}} = 4 \log(1 + \rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n))$, with no image-rank multiplier. NFR1–5 proves the literal map to ITL20/25 and retains the asymmetric ITL30–31 error. The budget and its signed row difference are distinct specified quantities; no asymptotic value is assigned.

The complete residual boundary and the retained target budget. Set $j = k + 4$, keep $(s_0, s_1) = (1, 1)$ or (k, j) , and use $D = N - v$ at each of the four original endpoints. In the original coefficient spaces put $B_{\text{cof}} = \text{stack}_i M_{i,k}$, $\mathbb{B} = \text{diag}(\mathbf{B}_j, 4)$, $\mathbb{J} = \ker \mathbb{B}$, $V_b = \mathbb{C}^{m_k}$, $m_k = 16k - 48 - v$, $J_b = \text{im } J_0$, $\Phi_b = \mathbb{B} B_{\text{cof}}$, $\mathcal{L}_b = \ker \Phi_b$, and $Y_b = J_b + \mathcal{L}_b$. The map $[x] \mapsto [B_{\text{cof}} x]$ gives exactly

$$V_b/Y_b \simeq \frac{\mathbb{J} + \text{im } B_{\text{cof}}}{\mathbb{J} + \text{im}(B_{\text{cof}} J_0)}, \quad r_b = t_{K,k} - \dim \mathcal{L}_b + \dim(J_b \cap \mathcal{L}_b). \quad (\text{ITL33})$$

Indeed an image is zero precisely when $B_{\text{cof}}(x - J_0 a) \in \mathbb{J}$ for some a , which is $x \in Y_b$. The full nonreduced branch calculation RQB5–12 computes this kernel and its shared-component constraint; it retains $t_{0,k}$ in $t_{K,k}$.

In RQB13's exact quotient lifts let Q_b^E be the attained target Gram after all lower and invariant coefficients have been minimized. It is related to the earlier CIT25 lift by $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$, with its fixed invertible quotient-coordinate matrix C_b retained in RQB15. Write $\mathcal{V}_b = \mathcal{R} \log \det Q_b^E$ and let Q_o be the complete remaining quotient of the fourfold kernel-dual space by $\text{im } \Phi_b$. Sequential minima in the full frame $C_T = [U_R, U_b, U_o]$ prove

$$\begin{aligned} |\det C_T|^2 (\det H_{B,j,D})^4 &= \det H_{\text{inv}}(D) \det Q_b^E(D) \det Q_o(D), \\ \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_b + \mathcal{V}_o &= 4\widehat{F}_j^{(s_1)}, \quad 0 \leq \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_b \leq 4\widehat{F}_j^{(s_1)} \leq 4U_{j,s_1}^{\text{native}}. \end{aligned} \quad (\text{ITL34})$$

Here H_{inv} is exactly ITL6, and $\mathcal{R}f = -f(q_j - 1 - v) - f(q_j - v) + f(2q_j - 1 - v) + f(2q_j - v)$ when f is indexed by D . The three returns are nonnegative: the original native form increases by its complete arithmetic rows, and each minimum uses the same fixed affine fibre. The fixed frame determinant cancels under precisely those four signs. RQB16–28 proves the full row and quotient formulas. The finite $U_{j,s_1}^{\text{native}}$ is RQB35’s minimum retaining the full ROQ observation return and both exact FEX source corrections, not a replacement by the old uncorrected kernel metric. The source quotient Q_{inv} , the matched relative operator, and the two comparison errors of ITL29–32 retain their original definitions.

R21 The native target loss in the earlier matched quotient

Keep the actual simple quartet, period, branch and nonreduced conductor, $k = 4l+1 \geq 9$, $j = k+4$, and $D \geq q_j - v - 1$. The exact original source-order pairs are $(s_0, s_1) = (1, 1)$ or (k, j) . All original DNE, KRC, DFX and NBG maps are those in ITL1–6:

$$\begin{aligned} \mathsf{T}_a &= \alpha_A N_a Z_{\text{exp},a}, \quad \alpha_A = v!/\mu_v, \quad M_i = \mathsf{T}_j^{-1} m_{i,k} \mathsf{T}_k, \quad F = \text{stack}_i(M_i J_0), \\ \mathcal{B} &= \text{diag}(B_1, 4), \quad \ker \mathcal{B} = \text{im diag}(J_1, 4), \quad E = \ker(\mathcal{B}F), \\ G &= J_0^* D_{k,D}^{(s_0)} J_0, \quad \mathcal{D}_D = \text{diag}(D_{j,D}^{(s_1)}, 4), \quad A = F^* \mathcal{D}_D F, \quad Q = G/E. \end{aligned} \tag{NFR1}$$

Here $\text{diag}(P, 4)$ means four copies, and every quotient P/E is its attained minimum in the fixed lifts $[Z_E, Z_R]$ of ITL4. The actual image frame is $U = \mathcal{B}FZ_R$. Its native target form is $H = U^*(\mathcal{B}\mathcal{D}_D^{-1}\mathcal{B}^*)^{-1}U$ and its source pullback form before the last target quotient is $A_E = A/E$. A lift of Ub is exactly $FZ_R b + \ker \mathcal{B}$. Its complete minimum over that kernel gives ITL8's H . The source minimum permits just FZ_E . This proves the typed identification of ITL9's $\omega_D = \det H / \det A_E$ with the last target loss in NBG31–32, retaining the full source kernel.

Let C_D be the original correlated pullback on the invariant source. Schur factorization in $[Z_E, Z_R]$ gives

$$\begin{aligned} \log \det(Q^{-1/2} H Q^{-1/2}) &= e_{\text{cor},D} + \log \omega_{V,D} - \log \omega_{E,D} + \log \omega_D \\ &= \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} + \log \omega_D + \epsilon_D, \quad -a_D \leq \epsilon_D \leq b_D, \\ a_D &= 2r \log \max(1, K_-) + r \log c_+, \quad b_D = 2r \log \max(1, K_+) - r \log c_-, \quad r = \dim V - \dim E. \end{aligned} \tag{NFR2}$$

In the first line $e_{\text{cor}} = \log \det(C_D/E) - \log \det(G/E)$ and the two restriction losses are exactly ITL13. Their determinant ratio equals $\det(A/E)/\det(C_D/E)$ because the same full frame factor cancels. The complete compatibility forms in the second line are ITL18. Their common denominator G_c cancels before estimating. Their identical positive update has rank at most r ; compression of NKC's two-sided comparison bounds its logdet difference between $-r \log c_+$ and $-r \log c_-$. Adding the correlated quotient interval ITL17 proves exactly NFR2. This is ITL26–30 applied to the original matched quotient, not a replacement of its target metric.

R22 Every original row and both signs in the long return

Put $F_E = FZ_E$, $F_R = FZ_R$, and let $\widehat{W} = [F_E, W_R]$ span $\ker \mathcal{B}$. The four fixed flags are the images of $Y_K = F_E$, $Y_S = [F_E, F_R]$, $Y_W = \widehat{W}$, $Y_T = [\widehat{W}, F_R]$. They obey $S \cap W = K$, because $Fa \in \ker \mathcal{B}$ is exactly $a \in E$. Their raw frames are $Z_{Y,D} = [\mathcal{V}_D, \mathcal{T}_D Y]$, using all original lower rows and all numerator rows IRR7–11. Analytic numerator division is the literal recursion

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v},$$

and the polynomial coefficients and complete row formulas remain IRR8–9, with i^{-n} , $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$ and center $j/2 - 4$. Thus there is no unspecified substitute evaluator.

Write $G_{Y,D} = Z_{Y,D}^* Z_{Y,D}$ and $L_{Y,n} = z_{Y,n} G_{Y,n-1}^{-1} z_{Y,n}^*$ for the original new four-row block. Eliminating the entire lower block and then the actual source or target kernel gives $\omega_D = \det G_{T,D} \det G_{K,D} / (\det G_{W,D} \det G_{S,D})$. The determinant lemma for each raw Gram

therefore proves

$$\begin{aligned}
& \log \omega_n - \log \omega_{n-1} = \gamma_{H,n} - \gamma_{A,n}, \\
& \gamma_{H,n} = \log \det(I_4 + L_{T,n}) - \log \det(I_4 + L_{W,n}) \geq 0, \\
& \gamma_{A,n} = \log \det(I_4 + L_{S,n}) - \log \det(I_4 + L_{K,n}) \geq 0, \\
& \mathcal{L}_{\text{inv}} = \sum_{n=d+1}^{d+q_j+1} \vartheta_n(\gamma_{H,n} - \gamma_{A,n}), \quad d = q_j - v - 1, \\
& \vartheta_{d+1} = \vartheta_{d+q_j+1} = 1, \quad \vartheta_n = 2 \quad (d+2 \leq n \leq d+q_j).
\end{aligned} \tag{NFR3}$$

For positivity, transform each pair (K, S) or (W, T) to its old minimum-section columns. The larger raw covariance is $L_Y = L_X + R_X Q_X^{-1} R_X^*$; factoring its positive determinant gives IRR21's nonnegative increment. The common raw lower minimum is never taken separately for different rows. Finally telescope over $[d, d+q_j]$ and $[d+1, d+q_j+1]$. Their overlap gives the exact displayed weights.

R23 The simultaneous native budget and the exact signed receiver

For the same original ordered generalized eigenvalues $\lambda_i(D)$ of $(H_D, A_{E,D})$ put $u_i(D) = \text{arcosh}(\lambda_i(D)^{-1/2})$. On the positive path joining two actual adjacent metrics, the original minimum lifts give orthonormal paired vectors p_i, q_i with $p_i^* q_m = \sqrt{\lambda_m} \delta_{im}$. Consequently $J_i = q_i q_i^* - p_i p_i^*$ has eigenvalues $\pm \sqrt{1 - \lambda_i}$ on mutually orthogonal planes. The full repeated-eigenvalue and zero-angle arguments IF5–6 and FAB6–9 retain the ordered spectrum through crossings and give, almost everywhere,

$$(\log \lambda_i)' = \text{Tr}(B J_i), \quad \sum_i |(\log \lambda_i)'| \leq \text{Tr } B, \quad \sum_i |u_i'| \leq \frac{1}{2} \text{Tr } B.$$

Indeed sign maximization of the trace takes place on those disjoint planes, so its operator norm is at most one regardless of their count. Here $B = \mathcal{D}^{-1/2} \Delta \mathcal{D}^{-1/2} \succeq 0$. Its integral is the exact ambient logdet increment. The complete lower row minimization gives, with all rows through $n-1$, $C_L = V^* V$, $D_0 = T^*(I - V C_L^{-1} V^*) T$, $\zeta_n = v_n C_L^{-1} v_n^*$ and $\rho_n = t_n - v_n C_L^{-1} V^* T$. Completing its square proves the one-target update $\rho_n^* \rho_n / (1 + \zeta_n)$ and hence

$$\begin{aligned}
& \beta_n = 4 \log \left(1 + \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n} \right), \\
& \sum_i |\log \lambda_i(n) - \log \lambda_i(n-1)| \leq \beta_n, \quad \sum_i |u_i(n) - u_i(n-1)| \leq \beta_n/2, \quad |\mathcal{L}_{\text{inv}}| \leq \sum_n \vartheta_n \beta_n.
\end{aligned} \tag{NFR4}$$

No image-rank factor multiplies this budget.

At the four ordered endpoints $D_i = d, d+1, d+q_j, d+q_j+1$, $\varepsilon = (-1, -1, +1, +1)$, substituting NFR3 into NFR2 gives

$$\begin{aligned}
& \sum_i \varepsilon_i \log \det \mathcal{R}_{X, D_i} = \sum_i \varepsilon_i \log \frac{\det \mathcal{H}_{V, D_i}}{\det \mathcal{H}_{E, D_i}} + \sum_n \vartheta_n (\gamma_{H,n} - \gamma_{A,n}) + \mathcal{E}_{\text{fin}}, \\
& -b_{D_1} - b_{D_2} - a_{D_3} - a_{D_4} \leq \mathcal{E}_{\text{fin}} \leq a_{D_1} + a_{D_2} + b_{D_3} + b_{D_4}.
\end{aligned} \tag{NFR5}$$

The signs follow by negating the two low endpoint errors. Only this finite comparison error is $O_{\text{actual}}(k \log q_k)$. The two positive row sums and the full compatibility ratio are retained in their exact source metrics. The $q_j + 1$ row degrees are the original long interval, and neither its budget nor its signed difference is assigned an arithmetic leading coefficient. When $r = 0$, both quotient forms are empty and all actual loss terms vanish exactly. The common target-source mass cancels from a ratio only through the equal quotient ranks; it remains in every row and absolute native Gram in NFR1–4.

The exact positive target budget in the matched relative return. Keep NFR1–5’s source quotient Q_{inv} , target H_{inv} , and matched relative operator $\mathcal{R}_X$ in their original fixed quotient-value coordinates. Their literal determinant equality is $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$. Use RQB3’s actual residual boundary quotient $\mathbb{C}^{m_k}/(\text{im } J_0 + \ker \Phi_b)$, where $\Phi_b = \mathbb{B}B_{\text{cof}}$, and its full target minimum Q_b^E from RQB14. Completing the unchanged image $[U_R, U_b]$ to the full target kernel-dual frame gives the remaining quotient Q_o in RQB27. Therefore

$$\begin{aligned} \mathcal{R} \log \det \mathcal{R}_X &= 4\widehat{F}_j^{(s_1)} - \mathcal{V}_b - \mathcal{V}_o - \mathcal{R} \log \det Q_{\text{inv}}, \\ 0 &\leq \mathcal{R} \log \det Q_{\text{inv}} + \mathcal{R} \log \det \mathcal{R}_X = \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4U_{j,s_1}^{\text{native}} - \mathcal{V}_b. \end{aligned} \quad (\text{NFR6})$$

The first equality substitutes the exact NFR determinant ratio into RQB28; the finite second line uses RQB35. Every occurrence of $\mathcal{R}$ has NFR5’s ordered four endpoints and signs, with the shift $D = N - v$ retained. The quotient-value frame factors cancel only in that return. The full original source Gram Q_{inv} , the positive residual return $\mathcal{V}_b$, and the positive missing target return $\mathcal{V}_o$ are all present. The row difference, compatibility errors and asymmetric interval in NFR5 remain exact; the displayed bound does not assign an individual leading coefficient to the common long return.

Both source orders in the complete matched target allocation. Use $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (k, j)$, $j = k + 4$. Keep the exact original Q_{inv} and the fixed quotient values Z_R of NFR1–6. Write $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = \mathcal{R} \log \det \mathcal{R}_X^p$. The original remaining quotient now splits under the actual map $E_e = [0 \ 0 \ I]C_T^{-1}\mathbb{B}F_e$ as in CPQ4–8: $V_o^p = V_t^p + V_{\text{rem}}^p$, where V_t is its connecting restriction and V_{rem} its full further quotient. Hence NFR6 receives the exact refinement

$$\begin{aligned} \mathcal{M}^p &= 4\widehat{F}_j^{(s_1)} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p, \quad \mathcal{M}^p + V_Q^p = V_I^p \geq 0, \\ \mathcal{M}^p &= \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p, \\ V_A^p &= V_Q^p + \mathcal{R} \log \frac{\det A_{\text{inv}}^p}{\det Q_{\text{inv}}^p}. \end{aligned} \quad (\text{NFR7})$$

The first identity follows from the full RQB27 determinant and CPQ8’s second attained quotient. The second substitutes $\mathcal{L}_e = V_I + V_t - V_s - V_A$. The final identity uses the same Z_R values for the two actual source forms, so the ratio is the determinant of their positive relative operator. The target pullback A_{inv} and independent degree- k source Q_{inv} thus both remain.

Let $r_I = j_{0,k} - \epsilon$, and evaluate all NSR2 widths at $D_i = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$. Then

$$\begin{aligned} \Lambda^- &= -r_I(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) \leq \Delta\mathcal{M} \leq r_I(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}) = \Lambda^+, \\ U_{\mathcal{M}}^p &= 4U_{j,s_1}^{\text{native}} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p, \quad -V_Q^p \leq \mathcal{M}^p \leq U_{\mathcal{M}}^p. \end{aligned} \quad (\text{NFR8})$$

The source degree is k , with its original centre, at these same target-based D_i ; its displacement is $b = 8k + 24$ at the two lows and $2b$ at the two highs. The comparison of V_I has rank r_I in target order j , that of V_Q has the same rank in source order k . Subtracting their two directed intervals gives $\Lambda^\pm$. The upper endpoint retains the full RQB35 cap on its proved finite domain, together with both missing factors. Consequently both order intervals can be used simultaneously:

$$\begin{aligned} \max\{-V_Q^1, -V_Q^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{U_{\mathcal{M}}^1, U_{\mathcal{M}}^0 + \Lambda^+\}, \\ \max\{-V_Q^0, -V_Q^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{U_{\mathcal{M}}^0, U_{\mathcal{M}}^1 - \Lambda^-\}. \end{aligned} \quad (\text{NFR9})$$

Translate the complete pair-zero interval by $[\Lambda^-, \Lambda^+]$ and intersect with pair one. Reversing that difference gives the other line. This is the exact same matched return, not a replacement of its source quotient by an invariant source image.

The earlier full compatibility error remains in its own original coordinates. Put $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$ and retain NFR2’s endpoint constants

$$a_i^p = 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \quad b_i^p = 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i).$$

Then, with all original source-complement projections retained,

$$\begin{aligned} E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, & E_p^+ &= a_0^p + a_1^p + b_2^p + b_3^p, \\ \Lambda^- - r_I W_j - E_1^+ + E_0^- &\leq \Delta\mathcal{C} \leq \Lambda^+ + r_I W_j - E_1^- + E_0^+. \end{aligned} \quad (\text{NFR10})$$

Indeed NFR5 gives $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Subtract this identity in both orders and use the rank- r_I same-target contraction width $r_I W_j$. This proves every low and high error sign in NFR10.

The same common-angle enclosure in the matched source quotient. For each original pair $p = 0, 1$ keep the full NFR7 identity and put $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$. Let L_-^p, L_+^p be SFC9–10 evaluated in that pair, with all four actual source and target Grams in SFC2. The matched return therefore satisfies

$$L_-^p + D_e^p \leq \mathcal{M}^p \leq L_+^p + D_e^p. \quad (\text{NFR11})$$

This is the image of CIT35 under the exact translation $\mathcal{M}^p = \mathcal{L}_e^p + D_e^p$, proved in NFR7. In particular the determinant of the relative positive operator A_{inv}^p versus Q_{inv}^p remains in D_e^p . The coefficient values agree through Z_R , but their two source metrics and their centres remain the original ones. Intersect NFR11 with NFR8 on the cap's original domain. Writing the resulting endpoints as M_-^p, M_+^p , the complete two-order refinement is

$$\begin{aligned} \max\{M_-^1, M_-^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{M_+^1, M_+^0 + \Lambda^+\}, \\ \max\{M_-^0, M_-^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{M_+^0, M_+^1 - \Lambda^-\}. \end{aligned} \quad (\text{NFR12})$$

The original interval $\Lambda^- \leq \mathcal{M}^1 - \mathcal{M}^0 \leq \Lambda^+$ is NFR8, with degree- k source widths at the displaced target cutoffs. Adding it to pair zero, or subtracting it from pair one, gives the two lines. Every intersection contains its actual matched return and retains NFR10's complete compatibility errors.

The equilibrium interval with the original displaced source. Keep NFR7–12's $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = V_I^p - V_Q^p$ in the same Z_R values. Use $\Phi_j^\pm$ at target degree j and $\Phi_k^\pm$ at source degree k , both at the actual target-window integers D_i . The source polynomial degree is $D_i - q'_k$, with maximum $k^2 + 34k + 1 - v$. For $k \geq 37$,

$$2q_k - (k^2 + 34k + 1 - v + (k-1)/4) = k^2 - \frac{121}{4}k + \frac{5}{4} + v \geq 251 + v.$$

Thus EFP28 verifies the required degree range; the original GPA radius and the individual EOR tests remain explicit further tests. The source shifts are $8k + 24$ at the lows and $16k + 48$ at the highs. Both rank- r_I determinants then give

$$\Lambda_{\text{new}}^- = r_I(\Phi_j^- - \Phi_k^+) \leq \Delta\mathcal{M} \leq r_I(\Phi_j^+ - \Phi_k^-) = \Lambda_{\text{new}}^+. \quad (\text{EFR7})$$

This subtracts the actual independent source interval from the target interval. It keeps the distinct centres c'_k, c'_j and, before the four return, the scalar $-r_I \log(\kappa_j/\kappa_k)$.

Use the translated common interval $[\tilde{L}_-^p + D_e^p, \tilde{L}_+^p + D_e^p]$, $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$, and intersect with NFR8 on its cap domain. Calling the resulting endpoints $\widehat{M}_-^p, \widehat{M}_+^p$, NFR12 becomes exactly EFP30 with $\Lambda_{\text{new}}^\pm$. The full relative determinant $\mathcal{R} \log(\det A_{\text{inv}}/\det Q_{\text{inv}})$ remains in D_e^p by NFR7. Finally NFR10's actual endpoint errors $E_p^\pm$ give

$$\Lambda_{\text{new}}^- - r_I W_j - E_1^+ + E_0^- \leq \Delta\mathcal{C} \leq \Lambda_{\text{new}}^+ + r_I W_j - E_1^- + E_0^+. \quad (\text{EFR8})$$

This follows by subtracting the same-target invariant contraction and the full error difference from the exact NFR5 identity. No error constant is replaced by a Gamma-order scalar.

R24 The surviving boundary quotient and its original arithmetic kernel

Retain the actual simple quartet, its proved five-orbit period and logarithm branch, every coefficient of the original nonreduced conductor, and the complete CIT, CNC, DFX, KRC, DNE, NBG and PCJ maps. Put

$$\begin{aligned} k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad m_a = 16a - 48 - v, \\ 0 \leq v = \text{ord}_0 E_A \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \\ (s_0, s_1) = (1, 1) \text{ or } (k, j), \quad c'_a = a/2 - 4, \quad D \geq q_j - v - 1. \end{aligned} \quad (\text{RQB1})$$

All coefficient spaces and dual functionals are complex-linear. No star replaces an ordinary transpose in an original algebraic map. Metric stars are complex adjoints. The degree progression and the common cutoff D are literal, including the two distinct source measures. Write $s_k = s_0$ and $s_j = s_1$ when source orders have degree subscripts.

R24.1 The exact coefficient kernel, rank and original kernel quotient

Let $V = \mathbb{C}^{m_k}$, $U = \mathbb{C}^{4m_j}$ and retain

$$\begin{aligned} B_0 = \mathbf{B}_k, \quad B_1 = \mathbf{B}_j, \quad J_0 = \mathbf{J}_k, \quad J_1 = \mathbf{J}_j, \quad \ker B_a = \text{im } J_a \quad (a = 0, 1), \\ M_i = (\alpha_A N_j Z_{\text{exp}, j})^{-1} m_{i, k} (\alpha_A N_k Z_{\text{exp}, k}), \quad m_{i, k}[f] = [x_i^4 f], \\ B_{\text{cof}} = \text{stack}_{i=1}^4 M_i, \quad \mathbb{B} = \text{diag}(B_1, 4), \quad X_{\mathbb{J}} = \text{diag}(J_1, 4), \quad \mathbb{J} = \text{im } X_{\mathbb{J}}, \\ \Phi = \mathbb{B} B_{\text{cof}}, \quad J = \text{im } J_0, \quad \mathcal{L} = \ker \Phi, \quad Y = J + \mathcal{L}, \quad E = \ker(\Phi J_0). \end{aligned} \quad (\text{RQB2})$$

Here $\text{diag}(P, 4)$ means four copies. The original KRC bijections specify the inverses onto their full images; both occurrences of α_A are retained before their proved cancellation in M_i . CNC6 or DFX21 proves B_{cof} injective on the entire V . All $\mathbf{B}_a$ are onto their actual kernel-dual coordinate spaces. Write $t_{K, a}$ for their ranks and $j_{0, a}$ for the column ranks of $\mathbf{J}_a$; the original exact values are $t_{K, a} = 8a - 16 - t_{0, a}$ and $j_{0, a} = 8a - 32 - v + t_{0, a}$.

Set exactly as in CIT24

$$W = \mathbb{J} + \text{im } B_{\text{cof}}, \quad W_{\text{inv}} = \mathbb{J} + \text{im}(B_{\text{cof}} J_0).$$

The literal map and its full kernel are

$$\begin{aligned} \pi_b : V \longrightarrow W/W_{\text{inv}}, \quad x \longmapsto [B_{\text{cof}} x], \quad \ker \pi_b = Y, \\ V/Y \xrightarrow{\sim} W/W_{\text{inv}} \xrightarrow{\sim} \text{im } \Phi / \text{im}(\Phi J_0), \quad [x] \longmapsto [B_{\text{cof}} x] \longmapsto [\Phi x]. \end{aligned} \quad (\text{RQB3})$$

Indeed $B_{\text{cof}} x \in W_{\text{inv}}$ precisely when $B_{\text{cof}}(x - J_0 a) \in \mathbb{J}$ for some a , equivalently $x - J_0 a \in \ker \Phi$. This proves the first kernel and both surjections. The second arrow has kernel exactly $\mathbb{J}$ before the displayed quotient, because $\ker \mathbb{B} = \mathbb{J}$. Thus it is injective on the stated quotient.

Put $\ell = \dim \mathcal{L}$, $\epsilon = \dim E$ and $r_b = \dim(W/W_{\text{inv}})$. Since J_0 is injective, it identifies E with $J \cap \mathcal{L}$. Consequently

$$\begin{aligned} r_b = m_k - j_{0, k} - \ell + \epsilon = t_{K, k} - \ell + \epsilon = \text{rank } \Phi - \text{rank}(\Phi J_0), \\ 0 \longrightarrow B_0 \mathcal{L} \longrightarrow \mathbb{C}^{t_{K, k}} \xrightarrow{[B_0 x] \mapsto [B_{\text{cof}} x]} W/W_{\text{inv}} \longrightarrow 0, \quad \dim B_0 \mathcal{L} = \ell - \epsilon. \end{aligned} \quad (\text{RQB4})$$

The middle arrow is well defined because changing x by $\ker B_0 = J$ changes its image by W_{inv} . Its kernel is $B_0 Y = B_0 \mathcal{L}$. Under KRC4–5 this is the exact quotient of $(K_k / (K_k \cap L_k))^\vee$, where $L_k = \mathbb{C}[S]_{<v}$ is the original polynomial low space. Thus the original low overlap $t_{0, k}$ remains in both the domain rank and its actual quotient.

Here is a finite calculation of $\mathcal{L}$ using the input polynomials. Use the exact Segre coefficient map $\Psi_a : S_{x,a} \rightarrow T_{a,a}$ of DFX13, the selector U_a^{inv} of degree- a monomials with $\sum i\nu_i \equiv 0 \pmod{5}$, and the complete original primary conductor convolution $C_a^\top$. Let $\text{quo}(G)$ be the following exact row-pivot quotient: take the first independent columns F of G , the first row set $\mathcal{R}$ with $F_{\mathcal{R}}$ invertible, and send x to $(x - FF_{\mathcal{R}}^{-1}x_{\mathcal{R}})_{\mathcal{R}^c}$. It is onto and has kernel $\text{im } G$, since the subtracted vector agrees with x on $\mathcal{R}$ and a zero remainder means x is that same column combination. Empty and full ranks give the identity and zero map respectively. Define

$$\begin{aligned} P_j^{\text{inv}} &= \text{quo}[C_j^\top, \Psi_j U_j^{\text{inv}}], \\ (H_{i,k}\nu)_{ab} &= \sum_{r,t=0}^4 b_{i,r,t} \nu_{a-r,b-t}, \quad b_{i,r,t} = [Z^r W^t] L_i(Z, W)^4, \\ \mathcal{A}_k &= \text{stack}_{i=1}^4 (P_j^{\text{inv}} H_{i,k} Z_{\text{exp},k}), \quad \mathcal{L} = \ker \mathcal{A}_k, \quad E = \ker(\mathcal{A}_k J_0). \end{aligned} \tag{RQB5}$$

Every off-rectangle coefficient in the convolution is zero. Membership of $H_{i,k} Z_{\text{exp},k} x$ in the displayed column space means its full conductor class has an invariant representative. Its first v jets already vanish by Leibniz multiplication; the conductor columns have zero such jets, so the invariant representative has zero jets as well. It is therefore exactly membership in $\mathcal{J}_{v,j}$, which KRC5 identifies with $\text{im } J_1$. This proves RQB5 and shows that the two computed kernels are those in RQB2, including all exceptional coefficient minors. In particular the first line of RQB4 is an exact finite rank formula in the original coefficients, without a chosen generic rank.

R24.2 A branch constraint for arbitrary source numerators

Work first in $S_x = \mathbb{C}[x_1, x_2, x_3, x_4]$, with the original H , $Q = Q_0(H^{-1}x)$, $Q_a = g^a Q$ and $\mathcal{A} = \prod_{a=1}^4 Q_a$, where

$$gf(x) = f(\text{diag}(\zeta^{-1}, \zeta^{-2}, \zeta^{-3}, \zeta^{-4})x).$$

The primary order in the Segre map is unchanged. In the polynomial UFD $T = \mathbb{C}[z_0, z_1, w_0, w_1]$ put

$$\psi f = f(H(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top), \quad \ell_i = \psi x_i, \quad B_a = \psi Q_a, \quad A^{\text{hom}} = \prod_{a=1}^4 B_a. \tag{RQB6}$$

The restriction map is onto $T_{d,d}$ with kernel $Q S_{x,d-2}$: each biform monomial has the lift in KRC13, and two lifts with the same exponents differ by the relation $t_1 t_4 - t_2 t_3$. Grouping terms with the same image proves the full kernel, as in DFX13. Hence the complete branch quotient is $T_{d,d}/B_a T_{d-2,d-2}$; no radical is taken.

The four ℓ_i are linearly independent $(1,1)$ biforms, because their coefficient matrix is the actual invertible H . Any irreducible common factor of three has bidegree $(1,0)$, $(0,1)$ or $(1,1)$: factors of a bihomogeneous polynomial are bihomogeneous, as follows by comparing least and greatest degree in each grading in a domain. Multiples of a fixed factor of these types inside $T_{1,1}$ have dimensions two, two or one. Three independent rows cannot be contained there. It follows that any irreducible factor divides at most two ℓ_i , and therefore

$$\gcd\{\ell_i^4 \ell_t^4 : 1 \leq i < t \leq 4\} = 1 \quad \text{in } T. \tag{RQB7}$$

For each irreducible factor, choose two indices outside the at most two rows divisible by it. Their pair product is not divisible by it. This proves RQB7 in the biform polynomial UFD, without transferring factorization to the homogeneous quadric quotient ring.

Let $[f] \in \mathcal{C}_{v,k}$ be the image of a vector of $\mathcal{L}$. For each i there is an invariant p_i such that $x_i^4 f - p_i \in (Q, \mathcal{A})$. Fix $a \in \{1, 2, 3, 4\}$ and reduce modulo (Q, Q_a) . The ideal $(Q, \mathcal{A})$ maps to zero there; applying g^a to it also maps to zero, because $g^a Q = Q_a$ and $g^a \mathcal{A}$ contains Q as the factor with original index $5 - a$. Since $g^a p_i = p_i$, the two reductions give

$$x_i^4 f = \lambda_{i,a} x_i^4 g^a f, \quad \lambda_{i,a} = \zeta^{-4ia} \pmod{(Q, Q_a)}. \tag{RQB8}$$

For fixed nonzero a the four $\lambda_{i,a}$ are nonzero and pairwise distinct. Multiply the equation with index i by x_t^4 and that with index t by x_i^4 , then subtract. It follows first that $x_i^4 x_t^4 g^a f = 0$, and then that $x_i^4 x_t^4 f = 0$ in this same full branch quotient. On applying ψ , every irreducible factor P of B_a , with its actual exponent e_P , therefore satisfies

$$e_P \leq \text{ord}_P(\ell_i^4 \ell_t^4 \psi f) \quad \text{for every } i < t.$$

RQB7 supplies a pair whose multiplier has exponent zero. Thus $\text{ord}_P \psi f \geq e_P$ for every P , so B_a divides ψf with all its multiplicities. Repeating for all four a proves the new arbitrary-source constraint

$$\mathsf{T}_k \mathcal{L} \subset \mathcal{H}_{v,k} := \mathcal{C}_{v,k} \cap \ker \mathsf{j}_k, \quad \mathsf{T}_k = \alpha_A N_k Z_{\text{exp},k}. \quad (\text{RQB9})$$

DFX24 proves its invariant-source specialization. The proof above retains the arbitrary f and $g^a f$ as two different branch values until the pair equations eliminate them.

The constraint is computable on the complete nonreduced factor scheme. Let

$$\mathcal{L}_{\text{br}} = \text{lcm}(B_1, B_2, B_3, B_4), \quad (b_z, b_w) = \text{bideg } \mathcal{L}_{\text{br}}, \quad G_{\text{br}} = A^{\text{hom}} / \mathcal{L}_{\text{br}}.$$

Retain the actual scalar in A^{hom} and compensating factor units in this equation. Divisibility gives the exact kernel

$$\ker \mathsf{j}_k \simeq T_{k-b_z, k-b_w} / G_{\text{br}} T_{k-8, k-8}, \quad [u] \mapsto [\mathcal{L}_{\text{br}} u], \quad h_k = (k - b_z + 1)(k - b_w + 1) - (k - 7)^2. \quad (\text{RQB10})$$

Indeed all four branch classes vanish exactly when their biform representative is divisible by this lcm. Such representatives differ by A^{hom} exactly by the stated denominator. Nonzero biform multiplication is injective, proving the dimension.

Choose a quotient monomial basis in RQB10 by the specified row-pivot rule, multiply by $\mathcal{L}_{\text{br}}$, take its affine coefficient array and apply the original jet rows $\beta_{k,a,b}^n$, $0 \leq n < v$. Call this full matrix $\mathcal{J}_{\text{br},v}$. The denominator is killed since it is a multiple of E_A after the original exponential substitution. Thus, with $h_{v,k} = \dim \mathcal{H}_{v,k}$,

$$h_{v,k} = h_k - \text{rank } \mathcal{J}_{\text{br},v}, \quad d_{v,k} = \dim(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k}), \quad 0 \leq \ell - \epsilon \leq h_{v,k} - d_{v,k}. \quad (\text{RQB11})$$

For example $d_{v,k}$ is exactly the nullity of the four branch quotient matrices on $\mathsf{T}_k J_0$, computed with DFX30/KRC13. The last inequality follows from the injective map $\mathcal{L}/(\mathcal{L} \cap J) \rightarrow \mathcal{H}_{v,k}/(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k})$ induced by the original T_k . Its kernel is precisely $\mathcal{L} \cap J$, so no dimension of an overlapping low space is dropped. The resulting finite rank control is

$$\max\{0, t_{K,k} - h_{v,k} + d_{v,k}\} \leq r_b \leq t_{K,k}, \quad r_b = t_{K,k} - \text{rank}(B_0 Z_{\mathcal{L}}), \quad (\text{RQB12})$$

where $Z_{\mathcal{L}}$ is the exact kernel frame from RQB5. This preserves every shared-component defect. It gives full rank exactly on the coefficient strata where $B_0 Z_{\mathcal{L}} = 0$; that criterion is a computed matrix equality, not a replacement of a nonreduced conductor by its reduced support.

R24.3 Full native and raw Gamma realizations

Use the actual fixed ITL4 complement Z_R of E , so $I_R = B_{\text{cof}} J_0 Z_R$. Choose a fixed complement E_b of Y in V by exact coefficient elimination, and set

$$R_b = B_{\text{cof}} E_b, \quad Y_{\text{inv}} = [X_{\mathbb{J}}, I_R], \quad Y_W = [X_{\mathbb{J}}, I_R, R_b]. \quad (\text{RQB13})$$

These are full frames of W_{inv} and W : applying $\mathbb{B}$ kills $X_{\mathbb{J}}$, makes $\Phi J_0 Z_R$ independent, and RQB3 makes the last quotient columns independent. In particular R_b represents exactly r_b quotient values. For the full target DNE form set

$$\begin{aligned} \mathcal{D}_D &= \text{diag}(D_{j,D}^{(s_1)}, 4), \quad Q_b^E(D) = R_b^* \mathcal{D}_D R_b - R_b^* \mathcal{D}_D Y_{\text{inv}} (Y_{\text{inv}}^* \mathcal{D}_D Y_{\text{inv}})^{-1} Y_{\text{inv}}^* \mathcal{D}_D R_b, \\ \mathcal{H}_D &= \text{diag}(H_{B,j,D}, 4), \quad H_{B,a,D} = (\mathbf{B}_a(D_{a,D}^{(s_a)})^{-1} \mathbf{B}_a^*)^{-1} = \widehat{H}_{K,a,D}^{-1}, \\ U_R &= \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad Q_b^E(D) = U_b^* \mathcal{H}_D U_b - U_b^* \mathcal{H}_D U_R (U_R^* \mathcal{H}_D U_R)^{-1} U_R^* \mathcal{H}_D U_b. \end{aligned} \quad (\text{RQB14})$$

Both formulas are attained minima: the first minimizes over the complete W_{inv} ; the second first minimizes over $\ker \mathbb{B} = \mathbb{J}$, then over its remaining image U_R . The fibres are identical by RQB3, so the formulas agree. Every inverse is justified by the full frames and DNE positivity. Empty image ranks omit their subtracted blocks.

If R_b^{CIT} is the particular earlier CIT25 lift, there is a unique invertible C_b on quotient values and some A_b with $R_b^{\text{CIT}} = R_b C_b + Y_{\text{inv}} A_b$. The exact minimum gives

$$Q_b^{\text{CIT}} = C_b^* Q_b^E C_b, \quad \det Q_b^{\text{CIT}} = |\det C_b|^2 \det Q_b^E. \quad (\text{RQB15})$$

The coefficients are obtained by solving the full frame Y_W . They are fixed as D or the source-order pair changes. Thus this new calculation transports to the original CIT25 factor with its actual squared frame determinant; it does not silently choose a different quotient volume.

For complete arithmetic rows at target degree j keep every original moment $\mu_d = \sum_{u,w=0}^8 a_{u,w} \beta_{8,u,w}^d$, and put

$$\begin{aligned} \beta_{j,a,b} &= j(1/2 - \delta - i\gamma) + 2a\delta + 2bi\gamma, & \nu_{a,n} &= \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{-1,t} &= 0, \quad b_{0,t} = \mathbf{1}_{t=0}, \quad b_{n+1,t} = i^{-1}(b_{n,t-1} - c'_j b_{n,t}) - n(n-1 + s_1/2) b_{n-1,t}, \\ t_n &= \left(\frac{\sum_t b_{n,t} f_{a,t}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, & v_n &= \left(\frac{\sum_t b_{n,t} \beta_{j-8,u,w}^t}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,w}. \end{aligned} \quad (\text{RQB16})$$

The first recursion follows by differentiating $E_A F_a = N_a$ $n + v$ times and isolating the term with μ_v . The second is the original monic Gamma recurrence after $y = (S' - c'_j)/i$, with $1/i = -i$ retained. Thus t_n, v_n are the complex-linear evaluations in the original orthonormal frame. The measure is exactly

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad s = s_1 \in \{1, j\}.$$

No α_A is inserted in the division rows: it belongs to the KRC coefficient transport already specified in RQB2.

Stack these rows through D as T_D, V_D , and put $\mathcal{T}_D = \text{diag}(T_D, 4)$, $\mathcal{V}_D = \text{diag}(V_D, 4)$. Define the literal full raw frames

$$Z_{\text{inv},D} = [\mathcal{V}_D, \mathcal{T}_D Y_{\text{inv}}], \quad Z_{W,D} = [Z_{\text{inv},D}, \mathcal{T}_D R_b], \quad G_{X,D} = Z_{X,D}^* Z_{X,D}. \quad (\text{RQB17})$$

Every lower evaluation occurs. Both frames are independent: a zero raw value modulo the lower evaluations gives a zero coefficient class by the original upper Vandermonde at $D + v \geq q_j - 1$, and then the coefficient frames in RQB13 are independent. Full elimination, with the complete old lower and invariant coefficients, therefore proves

$$Q_b^E(D) = (\mathcal{T}_D R_b)^* (I - P_{Z_{\text{inv},D}}) \mathcal{T}_D R_b, \quad \det Q_b^E(D) = \frac{\det G_{W,D}}{\det G_{\text{inv},D}}, \quad P_Z = Z(Z^* Z)^{-1} Z^*. \quad (\text{RQB18})$$

This is the full raw Gamma realization of the original residual factor. It has no independent row-by-row gauge minimization.

R24.4 The exact original source denominator and compatibility transport

Let Z_Y be a fixed full basis of Y and retain the complement E_b from RQB13. On the source put $G_0 = D_{k,D}^{(s_0)}$. Its quotient form in these fixed residual lifts is

$$Q_0(D) = E_b^* G_0 E_b - E_b^* G_0 Z_Y (Z_Y^* G_0 Z_Y)^{-1} Z_Y^* G_0 E_b. \quad (\text{RQB19})$$

Choose the first independent columns Z_Λ of $B_0 Z_{\mathcal{L}}$ and write $C_0 = [Z_\Lambda, B_0 E_b]$, an invertible full frame of $\mathbb{C}^{t_{K,k}}$. The exact same fibre in RQB4 gives

$$\det Q_0(D) = \frac{|\det C_0|^2 \det H_{B,k,D}}{\det(Z_\Lambda^* H_{B,k,D} Z_\Lambda)}. \quad (\text{RQB20})$$

Indeed the source quotient first minimizes over J to obtain the original $H_{B,k,D}$ and then over $B_0 \mathcal{L}$. The adapted full Gram has determinant $|\det C_0|^2 \det H_{B,k,D}$; eliminating its first block proves RQB20. In full matrix form Q_0 is the Schur complement in the lifts $B_0 E_b$. This preserves both the source low overlap $t_{0,k}$ and its further hidden kernel of rank $\ell - \epsilon$.

Let $A_D = B_{\text{cof}}^* \mathcal{D}_D B_{\text{cof}}$ and let C_D be the full correlated pullback from NBG27 on V . The notation A_D/Y , C_D/Y means the attained quotient in the fixed lifts E_b . By RQB3 the further target minimum satisfies

$$0 \prec Q_b^E \preceq A_D/Y \preceq C_D/Y, \quad K_-^{-2} Q_0 \preceq C_D/Y \preceq K_+^2 Q_0. \quad (\text{RQB21})$$

To prove the first comparison, all representatives in $B_{\text{cof}} E_b b + B_{\text{cof}} Y$ are allowed in $B_{\text{cof}} E_b b + W_{\text{inv}}$, while $W_{\text{inv}} \cap \text{im } B_{\text{cof}} = B_{\text{cof}} Y$. The native form on the former fibre is A_D/Y . The remaining comparisons are NBG27 minimized on the same source affine fibre; this establishes their exact typed maps.

Define

$$\omega_b(D) = \frac{\det Q_b^E(D)}{\det(A_D/Y)}, \quad e_b^{\text{cor}}(D) = \log \frac{\det(C_D/Y)}{\det Q_0(D)}, \quad \omega_Z(D) = \frac{\det A_D|_Z}{\det C_D|_Z} \quad (Z = V, Y).$$

The restriction determinants in the last ratio use identical fixed frames. The adapted determinant identity cancels those frames and gives

$$\log \frac{\det Q_b^E}{\det Q_0} = e_b^{\text{cor}} + \log \omega_V - \log \omega_Y + \log \omega_b, \quad 0 < \omega_b \leq 1. \quad (\text{RQB22})$$

Apply the complete PCJ13–16 shared-update proof to the actual nested original source images $Y \subset V$; these are their DNE functional classes modulo the original source lower evaluations. Its source difference rank is exactly r_b by RQB4. The full compatibility-plus-source-complement forms are the unchanged PCJ1 matrices; call their nonnegative logarithms χ_Y, χ_V . Consequently

$$\begin{aligned} \log \frac{\det Q_b^E(D)}{\det Q_0(D)} &= -(\chi_Y(D) - \chi_V(D)) + \log \omega_b(D) + \varepsilon_b(D), \\ -2r_b \log \max(1, K_-) - r_b \log c_+ &\leq \varepsilon_b(D) \leq 2r_b \log \max(1, K_+) - r_b \log c_-. \end{aligned} \quad (\text{RQB23})$$

Here $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$ are the actual NKC3–4/13 constants at this same k, D and source pair. For the signs, PCJ16 says $-\log(\omega_V/\omega_Y) = \chi_Y - \chi_V + \nu$ with $r_b \log c_- \leq \nu \leq r_b \log c_+$. The correlated quotient error is in $[-2r_b \log \max(1, K_-), 2r_b \log \max(1, K_+)]$ by NBG28. Subtracting ν proves exactly RQB23. Since $r_b \leq t_{K,k} = O(k)$, the explicit error is $O_{\text{actual}}(k \log q_j) = o(k q_k)$ at the original four target endpoints. The two full logarithmic losses remain. In particular the proved one-sided native bound is $Q_b^E \preceq K_+^2 Q_0$, retaining RQB20's original source denominator; an inverse estimate has not been assumed.

R24.5 Four original rows and the complete retained target budget

For a new row degree $n \geq q_j - v$, use degree-major row order and

$$\begin{aligned} z_{\text{inv},n} &= [\text{diag}(v_n, 4), \text{diag}(t_n, 4) Y_{\text{inv}}], \quad z_{W,n} = [z_{\text{inv},n}, \text{diag}(t_n, 4) R_b], \\ L_{X,n} &= z_{X,n} G_{X,n-1}^{-1} z_{X,n}^*, \\ R_n &= \text{diag}(t_n, 4) R_b - z_{\text{inv},n} G_{\text{inv},n-1}^{-1} z_{\text{inv},n-1}^* \mathcal{T}_{n-1} R_b. \end{aligned} \quad (\text{RQB24})$$

The row permutation from copy-major order is fixed and unitary. Eliminate the old Z_{inv} block in the full raw frame: its Gram becomes $\text{diag}(G_{\text{inv}}, Q_b^E)$ and its new row becomes $[z_{\text{inv}}, R_n]$. Thus

$$\begin{aligned} L_{W,n} &= L_{\text{inv},n} + R_n Q_b^E (n-1)^{-1} R_n^*, \\ Q_b^E(n) &= Q_b^E(n-1) + R_n^* (I_4 + L_{\text{inv},n})^{-1} R_n, \\ \gamma_{b,n} &:= \log \frac{\det Q_b^E(n)}{\det Q_b^E(n-1)} = \log \det(I_4 + L_{W,n}) - \log \det(I_4 + L_{\text{inv},n}) \geq 0. \end{aligned} \quad (\text{RQB25})$$

For the middle equality, minimize the added squared row norm over the old eliminated variables. Completing that square gives the displayed inverse $I_4 + L_{\text{inv}}$, as can also be checked by multiplication. The first identity and the rectangular determinant lemma give the last one. All zero-energy rows and rank-zero quotients are included; in the latter case $\gamma_{b,n} = 0$.

Put $d = q_j - v - 1$ and retain the original endpoint signs. The exact return is

$$\begin{aligned} \mathcal{V}_b &= -\log \det Q_b^E(d) - \log \det Q_b^E(d+1) + \log \det Q_b^E(d+q_j) + \log \det Q_b^E(d+q_j+1) \\ &= \sum_{n=d+1}^{d+q_j+1} w_n \gamma_{b,n} \geq 0, \quad w_{d+1} = w_{d+q_j+1} = 1, \quad w_n = 2 \text{ otherwise.} \end{aligned} \quad (\text{RQB26})$$

Telescoping both low-to-high intervals proves every weight and sign. The same return for Q_b^{CIT} is equal by RQB15: its constant $2 \log |\det C_b|$ cancels only under these four signs.

Complete $[U_R, U_b]$ by any fixed columns U_o to a full frame $C_T = [U_R, U_b, U_o]$ of the fourfold original kernel-dual space $\mathbb{C}^{4t_{K,j}}$. Define $Q_o(D)$ as its full quotient metric modulo $\text{im } \Phi$ in the lifts U_o . The first restriction in this frame is the exact ITL6 Gram $H_{\text{inv}}(D) = U_R^* \mathcal{H}_D U_R$. Sequential attained minima give

$$|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}}(D) \det Q_b^E(D) \det Q_o(D). \quad (\text{RQB27})$$

For proof, first eliminate the U_R block in $C_T^* \mathcal{H}_D C_T$; its next diagonal Schur block is RQB14. Eliminate that block; the remaining minimum allows precisely $\text{im}[U_R, U_b] = \text{im } \Phi$ and is Q_o . Thus every fixed-frame determinant is displayed. The missing rank is $4t_{K,j} - \text{rank } \Phi$, with no image-completeness assumption.

Each restriction and quotient in RQB27 increases as D increases: the original DNE metric increases by retaining additional orthonormal rows, and taking minima on the same affine fibres preserves that inequality. Therefore all three original weighted returns are nonnegative. The same is true row by row, and HJR13,18,21 give

$$\begin{aligned} \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_o &= 4\widehat{F}_j^{(s_1)}, \\ \mathcal{V}_{\text{inv}}^{\text{target}} &= \mathcal{R} \log \det H_{\text{inv}}, \quad \mathcal{V}_o = \mathcal{R} \log \det Q_o, \quad 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\widehat{F}_j^{(s_1)}. \end{aligned} \quad (\text{RQB28})$$

Here $\mathcal{R}$ has exactly the signs and cutoffs of RQB26. Its $\widehat{F}_j$ is the original fixed-column kernel return, since $H_{B,j,D} = \widehat{H}_{K,j,D}^{-1}$. The inverse changes the primal determinant sign exactly; no equality with the earlier F_K has been imposed. The frame constant in RQB27 disappears only because the four signs sum to zero. The inequality also holds on every single row, with the right side $4g_{K,N}$ for that target row. Thus the full aggregate budget retains zero-energy rows and does not require a hypothesis about concentration.

For an explicit finite cap already proved for these same measures, retain HJR24–25 at base degree j :

$$\begin{aligned} \widehat{F}_j^{(s_1)} + F_{0,j,s_1} &= \mathcal{F}_{\partial,j}^{(s_1)} + \mathbf{e}_{j,s_1} + \delta_{K,j,s_1} - F_{\Xi(L),j}^{(s_1)} - \delta_{0,j,s_1}, \\ |\delta_{K,j,s_1}| &\leq \mathcal{E}_{j,s_1}^{\text{FEX}}, \quad |\delta_{0,j,s_1}| \leq E_{j,s_1}^{\text{tr}}, \quad -e_{j,s_1}^- \leq \mathbf{e}_{j,s_1} \leq e_{j,s_1}^+, \quad F_{\Xi(L),j}^{(s_1)} \geq 0, \\ \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4U_{j,s_1}^{\text{jet}}, \quad U_{j,s_1}^{\text{jet}} = \mathcal{F}_{\partial,j}^{(s_1)} + e_{j,s_1}^+ + \mathcal{E}_{j,s_1}^{\text{FEX}} + E_{j,s_1}^{\text{tr}}. \end{aligned} \quad (\text{RQB29})$$

Every symbol here is the unchanged finite original OAR/FEX/PEC quantity in the complete HJR proof. The exact equality is retained before applying bounds. Both F_{0,j,s_1} and the omitted positive restriction are nonnegative by the original native row calculation. Together with RQB28 this proves RQB29. HJR26's already proved outer first variation gives

$$\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\Delta_j q_j \mathfrak{C}_\partial + o(kq_k) = O_{\text{actual}}(kq_k), \quad \Delta_j = 16j - 48. \quad (\text{RQB30})$$

The actual period is fixed, and $jq_j/(kq_k) \rightarrow 1$. This is an aggregate upper bound on two specified positive native volumes, with their full missing quotient retained in RQB28. It does not assign an allocation or a leading value to either one.

Finally transport the calculation directly into the accepted CIT25–26. In its original lifts use RQB15 and choose the fixed full frame X_W there, so $[X_{\mathbb{J}}, I_R, R_b^{\text{CIT}}] = X_W C_W$. Then the earlier complete equality remains literally

$$|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E. \quad (\text{RQB31})$$

Substitution into CIT26 replaces its one retained factor $\log \det Q_b^{\text{CIT}}$ by $\log \det Q_b^E + 2 \log |\det C_b|$. The raw calculation is RQB16–18, its exact original source quotient is RQB19–20, and its two independent native losses are RQB23. In the signed four-return the frame constants cancel and the residual contribution obeys the directed interval

$$-4\widehat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} \leq -\mathcal{V}_b \leq 0. \quad (\text{RQB32})$$

Indeed RQB28 says $\mathcal{V}_b \leq 4\widehat{F}_j - \mathcal{V}_{\text{inv}}^{\text{target}}$. This is precisely the sign of the residual factor in CIT26. Its kernel and source rank are computed by RQB4–12, and all lower-gauge, source-complement, invariant-overlap and fixed-frame terms remain in the exact transport. The common long connecting loss is not replaced by this bound.

R24.6 The directed original kernel cap and its source correction

There is a sharper finite version of RQB29 which uses the full original positive observation return before bounding it. All quantities in this paragraph have base degree j , source $s = s_1 \in \{1, j\}$ and $N_0, N_1, N_2, N_3 = q_j - 1, q_j, 2q_j - 1, 2q_j$. Retain the primal sign vector $\sigma = (1, 1, -1, -1)$ in FEX. With $t = t_{K,j}$ and $L_j = \mathbb{C}[S]_{<v}$, the exact FEX31/34 source correction is

$$\delta_{K,j,s} := \widehat{F}_j^{(s)} - F_K^{(s)} = \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_j}^{(s)} + F_{(K_j+L_j)/K_j}^{(s)}. \quad (\text{RQB33})$$

The endpoint scalar $e_{N,s}$ is the actual logarithmic Jacobian between the full original $(K_j + L_j)/L_j$ source and its conductor image in FEX34. It is not a deleted low direction. For full numerical coefficient expressions retain

$$\begin{aligned} M_N^A &= \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{a,b=0}^8 |a_{ab}| e^{|\beta_{8,a,b}-4|(H_N+2)} \right\}, \quad m_N^A = \log M_N^A, \\ w_{d,s} &= d!/(s/2)_d, \quad \zeta_{N,s} = \sum_{d=v}^N \log(w_{d,s}/w_{d-v,s}), \\ e_{N,s}^- &= \zeta_{N,s} - 2(N+1-v-t)m_N^A, \quad e_{N,s}^+ = 2tm_N^A, \\ D_s^+ &= e_{q_j-1,s}^+ + e_{q_j,s}^+ - e_{2q_j-1,s}^- - e_{2q_j,s}^-, \\ \mathcal{E}_{j,s}^{\text{FEX}} &= v\mathcal{L}_s + 2 \max_i |\zeta_{N_i,s}| + 2(4q_j+1)m_{2q_j}^A. \end{aligned} \quad (\text{RQB34})$$

Here H_N is the harmonic sum and $\mathcal{L}_s$ is the original AKS two-window constant at degree j . The explicit alternative FEX33 is available with its unchanged target degree and measure. FEX34

proves $e_{N,s}^- \leq e_{N,s} \leq e_{N,s}^+$ using the whole Gamma Jacobian and the actual conductor norm. The original two low quotient metrics give $0 \leq F_{L_j} \leq v\mathcal{L}_s$ and $0 \leq F_{(K_j+L_j)/K_j} \leq (v-t_{0,j})\mathcal{L}_s$. Thus RQB33 proves $\delta_{K,j,s} \leq D_s^+ + (v-t_{0,j})\mathcal{L}_s$. FEX36–37 independently proves $|\delta_{K,j,s}| \leq \mathcal{E}_{j,s}^{\text{FEX}}$. These two bounds retain the original source mass in their proved Gamma norm ratios and keep $t_{0,j}$ before estimating.

Let $\Phi_{j,s}$ be exactly OAR4's full positive sum $\log(1 + \eta_{q_j-1}) + 2 \sum_{N=q_j}^{2q_j-2} \log(1 + \eta_N) + \log(1 + \eta_{2q_j-1})$. All η_N are the original ROQ constrained rows with their complete conductor-observation covariance. If $U_{j,s}^{\text{old}}$ denotes the retained AKS/KAF kernel cap, OAR3/7 and RQB33–34 give the finite number

$$\begin{aligned} U_{j,s}^{\text{native}} &:= \min\{U_{j,s}^{\text{jet}}, \min\{U_{j,s}^{\text{old}}, \mathcal{F}_{\partial,j}^{(s)} + e_{j,s}^+ - \Phi_{j,s}\} \\ &\quad + \min\{D_s^+ + (v-t_{0,j})\mathcal{L}_s, \mathcal{E}_{j,s}^{\text{FEX}}\}\}, \\ 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4\widehat{F}_j^{(s)} \leq 4U_{j,s}^{\text{native}}. \end{aligned} \quad (\text{RQB35})$$

Indeed the inner first minimum bounds the original F_K from above, using its full residual subtraction. The second minimum bounds the exact difference $\widehat{F}_j - F_K$. Their sum bounds $\widehat{F}_j$; HJR25 supplies the other cap. This proves the outer minimum without identifying the two different kernel metrics or assuming a new conditioning estimate. Replacing the full Φ by only its first term recovers the earlier OAR6 partial cap, whereas RQB35 retains all computed positive terms.

Finally FEX37 transfers the proved OVR9 limit superior to this exact fixed-column kernel return. Therefore, with the original $\kappa_* = 32 \log(25 \log 3/(4\pi))$ and the same $\mathfrak{C}_{\partial}$ as in RQB30,

$$0 \leq \limsup_{k \rightarrow \infty} \frac{\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}}}{kq_k} \leq 4 \min\{\kappa_*, 16\mathfrak{C}_{\partial}\}. \quad (\text{RQB36})$$

For proof divide RQB28 by kq_k , insert $jq_j/(kq_k) \rightarrow 1$ and $(\widehat{F}_j - F_K)/(jq_j) \rightarrow 0$, then apply OVR9 separately in each unchanged source order. This proves only the displayed upper bound on the sum of two nonnegative exact volumes. Both remain present in the identity RQB28, with its full missing quotient; no individual limit or allocation is presumed.

The complete missing target quotient and its connecting restriction. Keep every block of the original $C_T = [U_R, U_b, U_o]$ in RQB27. At both original source-order pairs use $\pi_o = [0 \ 0 \ I_{r_o}]C_T^{-1}\mathbb{B}$, $E_e = \pi_o F_e$, $c_e = \text{rank } E_e$. The kernel of π_o is exactly W by the equation $\mathbb{B}z = \Phi x$, and its restriction to $\mathcal{S}_e$ has kernel $K_e = \mathcal{S}_e \cap W$. Choose the original fixed complement Z_c of $\ker E_e$ and $C_o = [E_e Z_c, R_e^o]$. The complete two target minima are

$$\begin{aligned} G_{t,e} &= (E_e Z_c)^* Q_o (E_e Z_c), \quad Q_{\text{rem},e} = \text{Gram}(\mathcal{U}/T_e) \text{ in the lifts } R_e^o, \quad T_e = \mathcal{S}_e + W, \\ |\det C_T|^2 |\det C_o|^2 (\det H_{B,j,D})^4 &= \det H_{\text{inv}} \det Q_b^E \det G_{t,e} \det Q_{\text{rem},e}, \\ \text{rank } G_{t,e} &= c_e, \quad \text{rank } Q_{\text{rem},e} = r_o - c_e. \end{aligned} \quad (\text{RQB37})$$

The quotient form on $\mathcal{U}/W$ first minimizes over every lower evaluation and $\mathbb{J}$, then over $\text{im } \Phi$, and is exactly Q_o . On the connecting values its restriction is therefore the first line. The further minimum over the whole connecting image gives the stated quotient on $\mathcal{U}/T_e$. Eliminating the first block of $C_o^* Q_o C_o$ proves $|\det C_o|^2 \det Q_o = \det G_t \det Q_{\text{rem}}$. Multiplication by the original RQB27 identity proves every absolute factor in RQB37.

Set $V_t = \mathcal{R} \log \det G_{t,e}$ and $V_{\text{rem}} = \mathcal{R} \log \det Q_{\text{rem},e}$. The entire original budget now reads

$$\begin{aligned} V_o &= V_t + V_{\text{rem}}, \quad V_t, V_{\text{rem}} \geq 0, \\ V_I + V_b + V_t + V_{\text{rem}} &= 4\widehat{F}_j^{(s_1)}, \\ -4\widehat{F}_j^{(s_1)} + V_I + V_t &\leq -V_b \leq 0, \\ -(r_o - c_e)\Omega_j^{\text{hi}} &\leq \Delta V_{\text{rem}} \leq (r_o - c_e)\Omega_j^{\text{lo}}, \quad -c_e\Omega_j^{\text{hi}} \leq \Delta V_t \leq c_e\Omega_j^{\text{lo}}. \end{aligned} \quad (\text{RQB38})$$

The original increasing row metric preserves order under both complete minima, proving positivity. Taking the four signs of RQB37 cancels only its fixed frame determinants. The third line uses $V_{\text{rem}} \geq 0$ after retaining its exact value in the second. The last line uses NSO's same-fibre comparison at the two actual ranks, with $D = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$ and $s_1 = 1, j$. It retains the original low overlaps in r_o .

For a literal calculation of the new quotient choose full coefficient columns X_T spanning T_e and fixed lifts X_Q under $\mathbb{B}$ of $U_o R_e^o$. Through the old degree D put

$$\begin{aligned} Z &= [\mathcal{V}_D, \mathcal{T}_D X_T], \quad G = Z^* Z, \quad Q = (\mathcal{T}_D X_Q)^* (I - ZG^{-1}Z^*) \mathcal{T}_D X_Q, \\ a &= [\text{diag}(v_{D+1}, 4), \text{diag}(t_{D+1}, 4) X_T], \quad b = \text{diag}(t_{D+1}, 4) X_Q, \\ R &= b - aG^{-1}Z^* \mathcal{T}_D X_Q, \quad L = aG^{-1}a^*, \\ Q^+ &= Q + R^*(I_4 + L)^{-1}R, \quad \log \frac{\det Q^+}{\det Q} = \log \frac{\det(I_4 + L + RQ^{-1}R^*)}{\det(I_4 + L)} \geq 0. \end{aligned} \tag{RQB39}$$

The original RQB16 rows retain all division moments, the phase $1/i$, the centres and $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$. DNE positivity and the full coefficient frames prove that Z and $[Z, \mathcal{T}_D X_Q]$ are independent even at the first cutoff. Their minimum is exactly Q_{rem} . Subtracting the old attained section leaves $\eta^* G \eta + \|Rx - a\eta\|^2$; its minimum proves the last line, including the original old lower Gram. The return uses weights one at the first and last rows and two inside. Thus no new row model or independent rowwise gauge minimum was substituted.

R25 Source-order intervals in the original connecting and invariant returns

This section receives the accepted NSO1–31 proof into the exact CIT28, NCR20, LCC36, CIT30/RQB and NFR6 formulas. It preserves their full source and target quotients. The scalar polynomial estimate remains the already proved AMT23,29–30 estimate, with the exact native dual square NSO7–15; it is not a new analytic bound. Keep the actual simple quartet, the proved five-orbit domain $|u| \geq R_*$, fixed logarithm branch, nonreduced conductor, all original coefficients and

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad c'_a = a/2 - 4, \\ 0 \leq v \leq 80, \quad \mu_v \neq 0, \quad m_a &= 16a - 48 - v, \quad t_a = 8a - 16 - t_{0,a}, \quad j_{0,a} = 8a - 32 - v + t_{0,a}, \\ p &= 0 : (s_0, s_1) = (1, 1), \quad p = 1 : (s_0, s_1) = (k, j), \\ D_0 &= q_j - v - 1, \quad D_1 = q_j - v, \quad D_2 = 2q_j - v - 1, \quad D_3 = 2q_j - v, \\ \mathcal{R}f &= -f(D_0) - f(D_1) + f(D_2) + f(D_3). \end{aligned} \tag{NSR1}$$

Here $t_{0,a}$ is the actual dimension of the original kernel's intersection with the low polynomial space. A superscript p always means this complete original pair, and $\Delta Z = Z^1 - Z^0$. The D_i are polynomial cutoffs. They remain the same for source degree k and target degree j .

R25.1 The finite native intervals and their exact value spaces

For $s = 4\ell_s + 1$ retain the full NSO constants

$$\begin{aligned} \rho_s &= \frac{(2\pi)^{2\ell_s}}{(1/2)^{2\ell_s}}, \quad \Pi_s = \prod_{h < \ell_s} (2h + \tfrac{1}{2})^2, \quad U_{s,D} = \prod_{h < \ell_s} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\}, \\ \mathbf{a}_s &= \rho_s \Pi_s, \quad \mathbf{b}_{s,D} = \rho_s U_{s,D}, \quad \omega_{s,D} = \log(\mathbf{b}_{s,D}/\mathbf{a}_s), \\ \Omega_s^{\text{lo}} &= \omega_{s,D_0} + \omega_{s,D_1}, \quad \Omega_s^{\text{hi}} = \omega_{s,D_2} + \omega_{s,D_3}, \quad W_s = \Omega_s^{\text{lo}} + \Omega_s^{\text{hi}}. \end{aligned} \tag{NSR2}$$

All $s = 1$ products equal one and all its widths are zero. The underlying measures remain $dm_s = (2\pi)^{s/2} 2^{s/2} |\Gamma(s/4 + iy/2)|^2 dy / (4\pi \Gamma(s/2))$.

For a specified rank- r native dual restriction or quotient $G_X^{(s)}(D)$ at one fixed degree and center, put

$$L_X(D) = \log \frac{\det G_X^{(s)}(D)}{\det G_X^{(1)}(D)}. \quad -r \log \mathbf{b}_{s,D} \leq L_X(D) \leq -r \log \mathbf{a}_s, \quad -r \Omega_s^{\text{hi}} \leq \mathcal{R}L_X \leq r \Omega_s^{\text{lo}}. \tag{NSR3}$$

These are exactly NSO16 and NSO19, at the explicitly specified D_i . To recall their precise map, let $S'^n = \sum_h (\mathbf{B}_{s,c,D})_{hn} \phi_{h,s,c}$ with $\phi_{h,s,c} = p_{h,s}((S' - c)/i) / \sqrt{(2\pi)^{s/2} h! (s/2)_h}$. The derivative values of a complex-linear functional are $\mathbf{B}_s^T w_s$, and its Taylor values are $U_s w_s$, $U_s = \text{diag}(1/n!) \mathbf{B}_s^T$. Consequently

$$w_s = (\mathbf{B}_s^T)^{-1} \mathbf{B}_1^T w_1, \quad G_s^\vee = \overline{H_s^{-1}}, \quad \mathbf{b}_{s,D}^{-1} G_1^\vee \preceq G_s^\vee \preceq \mathbf{a}_s^{-1} G_1^\vee. \tag{NSR4}$$

The same map carries every numerator and every lower evaluation. On a fixed fibre $Xx + Zz$ the full minimum is $X^*GX - X^*GZ(Z^*GZ)^{-1}Z^*GX$ in the value x . Applying NSR4 to all members of that fibre preserves its two constants. Every restriction, subsequent quotient and fixed direct sum therefore has the same inequality at its final rank. Relative eigenvalues prove the first interval in NSR3. Writing $L_X = -r \log \mathbf{a}_s - u_D$ with $0 \leq u_D \leq r \omega_{s,D}$ gives $\mathcal{R}L_X = u_{D_0} + u_{D_1} - u_{D_2} - u_{D_3}$ and proves the second interval with its directed signs. This recalls the full established transport, without deleting any raw gauge.

All scalar and frame factors in NSR3 remain until this same value space is identified. In particular a fixed change R of quotient lifts changes the Gram by R^*GR , giving $|\det R|^2$ in its

determinant; the factor cancels only in the proved same-frame order ratio or in $\mathcal{R}$. NSO6 gives, for $s = k, j$ and every D_i here, $\log \mathfrak{b}_{s,D_i} = O(k \log(q_j + 1))$. Thus all ensuing rank- $O(k)$ intervals are $o(kq_k)$. This statement does not give that conclusion on an $O(q_k)$ -rank raw polynomial plane.

R25.2 The two low source and image determinants at their actual ranks

For $e = 0, 1$ put $D_e = q_j - v - 1 + e$ and $r_e = 8k + 24 - 3e$. Retain the actual CIT primitive Taylor complement P_e^{Taylor} . In the two source orders $P_e^{(s_0)} = U_{s_0, c'_k, D_e}^{-1} P_e^{\text{Taylor}}$. The source norm minimizes over all columns $[V_0, T_0]$. The target coefficient F_e is uniquely determined by

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Taylor}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}. \quad (\text{NSR5})$$

The matrix $\text{Mult}(B_i)[n, h]$ is $B_i^{(n-h)}(0)/(n-h)!$ for $n \geq h$, and zero otherwise. The two Taylor columns on the right are the same actual analytic germs and lower exponentials in both orders. If two representations differ, multiplication by E_A makes the numerator vanish through $D_e + v \geq q_j - 1$; the full original upper Vandermonde kills its coefficient class, and the lower Vandermonde then kills the lower difference. This proves uniqueness of F_e . Existence is the original primitive condition: all jets are allowed at $e = 0$, while at $e = 1$ the four rows $\Lambda(\mathcal{T}_{B_i} \mathcal{T}_A \chi_j)$ impose exactly the missing target coordinate in each copy. Their rank four gives $(D_e + 1 - 4e) - (q_k - v) = r_e$. The positive connecting image has this same rank by CIT9's full kernel $\text{im } \mathbb{T} \cap \mathcal{L}_1 = \mathbb{T} \mathcal{L}_0$.

Write $L_{\text{src},e} = \Delta \log \det H_{\text{src},e}$ and $L_{\text{conn},e} = \Delta \log \det H_{\text{conn},e}$ at this defining cutoff. Define the literal four endpoints

$$\begin{aligned} s_e^- &= -r_e \log \mathfrak{b}_{k,D_e}, & s_e^+ &= -r_e \log \mathfrak{a}_k, & t_e^- &= -r_e \log \mathfrak{b}_{j,D_e}, & t_e^+ &= -r_e \log \mathfrak{a}_j, \\ s_e^- &\leq L_{\text{src},e} \leq s_e^+, & t_e^- &\leq L_{\text{conn},e} \leq t_e^+. \end{aligned} \quad (\text{NSR6})$$

The source fibre is $P_e x + T_0 \xi + V_0 \eta$; the image fibre first minimizes every target lower evaluation, then $F_e x + B_{\text{cof}} \xi$. Equation NSR4 proves both intervals on these complete fibres. In particular the number multiplying the logarithm is r_e , not the number of raw columns used to implement the minimum.

Retain the complete original CIT17 errors, with no new constant substituted:

$$\begin{aligned} \mathcal{X}_e^p &= \chi_{X_k, D_e}^p - \chi_{Y_e, D_e}^p, & A_e^p &= \log \det H_{\text{conn},e}^p - \log \det H_{\text{src},e}^p = -\mathcal{X}_e^p + \epsilon_e^p, \\ \epsilon_e^{p,-} &= -2r_e \log \max(1, K_-^p) - r_e \log c_+^p, & \epsilon_e^{p,+} &= 2r_e \log \max(1, K_+^p) - r_e \log c_-^p, \\ c_-^p &= \min(1, (L_-^p)^{-2}), & c_+^p &= \max(1, (L_+^p)^2), & b_e^- &= \epsilon_e^{1,-} - \epsilon_e^{0,+}, & b_e^+ &= \epsilon_e^{1,+} - \epsilon_e^{0,-}. \end{aligned} \quad (\text{NSR7})$$

The $K_{\pm}, L_{\pm}$ are the original NGB/NKC constants, including their degree- $k + 8$ comparison, all coefficients, centers and full measures. They give precisely the CIT17/PCJ16 interval for ϵ_e^p .

CIT28, NCR20 and the second line of LCC36 now have the following full finite receiving:

$$\begin{aligned} L_{\text{conn},e} &= -\Delta \mathcal{X}_e + \zeta_e, & \zeta_e &= L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0, \\ s_e^- + b_e^- &\leq \zeta_e \leq s_e^+ + b_e^+, \\ s_e^- + b_e^- - t_e^+ &\leq \Delta \mathcal{X}_e \leq s_e^+ + b_e^+ - t_e^-. \end{aligned} \quad (\text{NSR8})$$

Subtracting the two exact equalities in NSR7 proves the first line. Add the source and error intervals to get the second. Solve the first line for $\Delta \mathcal{X}_e$ and use both actual target endpoints in NSR6 to obtain the third. Thus both low determinant ratios and this complete compatibility difference are $o(kq_k)$ with the displayed finite bounds. At $e = 0$ only, $\chi_{Y_0} = 0$ because its source complement is zero. The four constraints and χ_{Y_1} remain at $e = 1$.

The earlier PSR26 finite source interval also remains valid on its own domain

$$k \geq \max(9, v - 1, \lceil (\pi/2) \sqrt{\delta^2 + \gamma^2} \rceil + 1)$$

and its original admissible SGT14 radius $0 < a < 1$. On that domain replace $[s_e^-, s_e^+]$ in NSR8 by its intersection with PSR26's full centered interval, retaining D_e^{ord} and both PSR errors. Both

intervals contain the same source ratio; their intersection does too. No PSR domain is imposed on NSR6 itself. The factors in the source determinant are unchanged since

$$(C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(n!) P_e^{\text{Tay}}. \quad (\text{NSR9})$$

This follows by canceling the exact $\mathbb{B}_s^\top$ with its inverse from U_s^{-1} . It proves equality of the full Θ_e and the actual complement/frame factors in PSR23 before any order comparison.

R25.3 Exact connecting and full-cohomology contraction return

Set $\mathcal{V} = \mathbb{C}^{m_k}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, and keep

$$\begin{aligned} B &= B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(B_j, 4), \quad \mathbb{J} = \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \\ \mathcal{B} &= \text{im } B, \quad \mathcal{S}_e = \text{im}[B, F_e], \quad W = \mathbb{J} + \mathcal{B}, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W, \\ c_e &= \text{rank}[\Phi, \mathbb{B}F_e] - \text{rank } \Phi, \quad h_e = r_e - c_e. \end{aligned} \quad (\text{NSR10})$$

Here B is injective on the whole original conductor coefficient source, as proved by CNC/DFX, so $\dim \mathcal{B} = m_k$ and $\dim \mathcal{S}_e = m_k + r_e$. The map

$$\mathcal{S}_e \longrightarrow \text{im}[\Phi, \mathbb{B}F_e] / \text{im } \Phi, \quad z \longmapsto [\mathbb{B}z] \quad (\text{NSR11})$$

is onto and its kernel is exactly K_e : its value is zero precisely when $z - Bx \in \ker \mathbb{B} = \mathbb{J}$ for some x . Thus $\dim(\mathcal{S}_e/K_e) = c_e$ and $\dim(K_e/\mathcal{B}) = h_e$. The identical representative map identifies $\mathcal{S}_e/K_e$ with T_e/W . Both are the exact quotient-value spaces in CIT21. In particular $0 \leq c_e \leq r_e$.

At every later cutoff the common ambient form is $\mathcal{D}_D^{(s_1)} = \text{diag}(D_{j,D}^{(s_1)}, 4)$. Let $G_{e,S}^p, G_{e,T}^p$ be its two full quotient metrics on $\mathcal{S}_e/K_e$ and T_e/W , in the same ordered values under NSR11. Then

$$\omega_{\text{conn},e}^p(D) = \frac{\det G_{e,T}^p(D)}{\det G_{e,S}^p(D)}, \quad |\Delta \log \omega_{\text{conn},e}(D)| \leq c_e \omega_{j,D}, \quad |\Delta \mathcal{R} \log \omega_{\text{conn},e}| \leq c_e W_j. \quad (\text{NSR12})$$

To prove the middle inequality, each quotient in NSR11 has rank c_e and the constants in NSR3. Subtract its two endpoint log intervals. The common term $-c_e \log \mathfrak{a}_j$ cancels, leaving $[-c_e \omega_{j,D}, c_e \omega_{j,D}]$. For the last inequality apply the original signs $(-, -, +, +)$ to these four symmetric intervals. The entire lower gauges and the two different larger fibres K_e, W are retained in this argument. There is no supposition that their minimum representatives or angles coincide. This is an explicit finite bound for the common-return order difference in CIT21/NCR14.

The full low cohomology has the fixed cycle space $\mathcal{Z}_e \supset \mathcal{S}_e$, with $\dim(\mathcal{Z}_e/\mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0)$. Its boundary remains $\mathcal{B}$. On the same native form, the source-order log ratio of its full cohomology Gram has rank $r_e + \kappa_e$, and that of its remaining quotient by the connecting image has rank κ_e :

$$\begin{aligned} -(r_e + \kappa_e) \log \mathfrak{b}_{j,D} &\leq \Delta \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{B}) \leq -(r_e + \kappa_e) \log \mathfrak{a}_j, \\ -\kappa_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{S}_e) \leq \kappa_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{NSR13})$$

The exact connecting inclusion identifies its cohomology quotient with $\ker U_e$ by LCC13; these are the attained metrics of that same flag. Completing a fixed frame of $\mathcal{S}_e/\mathcal{B}$ gives the full determinant as its restriction determinant times this last quotient determinant and the fixed adapted-frame factor. Thus NSR13 retains the whole quotient in the last paragraph of LCC36, rather than erasing it.

There is also a rank-exact comparison of the contractions. Put $K_e^{\text{full}} = \mathcal{Z}_e \cap W$ and $T_e^{\text{full}} = \mathcal{Z}_e + W$. The representative map gives

$$0 \longrightarrow K_e^{\text{full}}/K_e \longrightarrow \mathcal{Z}_e/\mathcal{S}_e \longrightarrow T_e^{\text{full}}/T_e \longrightarrow 0, \quad d_K + d_T = \kappa_e. \quad (\text{NSR14})$$

The kernel proof is $z \in \mathcal{Z}_e$ maps into T_e exactly when $z - s \in W$ for some $s \in \mathcal{S}_e$; hence its kernel modulo $\mathcal{S}_e$ is $(\mathcal{S}_e + K_e^{\text{full}})/\mathcal{S}_e \simeq K_e^{\text{full}}/K_e$. The adapted determinant identities for the two cross

ratios then express $\log \omega_e - \log \omega_{\text{conn},e}$ as the sum of the T_e^{full}/T_e and K_e^{full}/K_e log determinants, minus that of $\mathcal{Z}_e/\mathcal{S}_e$, with their fixed frame constant. Since $d_T + d_K = \kappa_e$, NSR3 gives

$$\begin{aligned} |\Delta(\log \omega_e - \log \omega_{\text{conn},e})(D)| &\leq \kappa_e \omega_{j,D}, \\ |\Delta \mathcal{R}(\log \omega_e - \log \omega_{\text{conn},e})| &\leq \kappa_e W_j, \quad |\Delta \mathcal{R} \log \omega_e| \leq (c_e + d_T)W_j \leq (c_e + \kappa_e)W_j. \end{aligned} \quad (\text{NSR15})$$

For the first bound subtract the three intervals in NSR3: their common scalar has coefficient $d_T + d_K - \kappa_e = 0$, and both remaining one-sided widths are $\kappa_e \omega_{j,D}$. Taking the four signs proves the second. The full quotient map $\mathcal{Z}_e/(\mathcal{Z}_e \cap W) \rightarrow (\mathcal{Z}_e + W)/W$ has rank $c_e + d_T$ by the exact sum inclusion in NSR14. Applying the same two-minimum proof as NSR12 at that actual rank gives the third inequality; $d_T \leq \kappa_e$ gives its last bound. These new order intervals coexist with NCR19's $|\mathcal{R} \log \omega_e^p - \mathcal{R} \log \omega_{\text{conn},e}^p| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}}$ for each original pair. In particular the middle upper bound in NSR15 can be replaced by the minimum of its displayed value and $2\kappa_e(\log A_{j,j}^{\text{end}} + \log A_{j,1}^{\text{end}})$. Both concern that same actual scalar difference.

The sharper tail ranks in the same source-order returns. Retain all NSR1–15 objects on their original five-orbit domain, including both complete native pairs, their fixed coefficient frames and the displaced polynomial cutoffs D_0, D_1, D_2, D_3 . The widths are exactly those of NSR2:

$$\begin{aligned} \omega_{j,D} &= \log(\mathfrak{b}_{j,D}/\mathfrak{a}_j), \quad \Omega_j^{\text{lo}} = \omega_{j,D_0} + \omega_{j,D_1}, \\ \Omega_j^{\text{hi}} &= \omega_{j,D_2} + \omega_{j,D_3}, \quad W_j = \sum_{a=0}^3 \omega_{j,D_a} = \Omega_j^{\text{lo}} + \Omega_j^{\text{hi}}. \end{aligned}$$

The complete BTK25–26 calculation gives $v \leq 32$ and $\kappa_e \leq \bar{\kappa}_e := 3 \max(v - e - 2, 0)$. In NSR14, its first quotient also has the exact presentation

$$\frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{\sim} K_e^{\text{full}}/K_e, \quad [j] \mapsto [j].$$

An element of K_e^{full} is $b + j$ with $b \in \mathcal{B} \subset \mathcal{S}_e$ and $j \in \mathcal{Z}_e \cap \mathbb{J}$; this proves surjectivity. Its kernel is exactly $\mathcal{S}_e \cap \mathbb{J}$, proving the isomorphism. The following arrow of NSR14 is the identity $[z] \mapsto [z]$ onto $(\mathcal{Z}_e + W)/(\mathcal{S}_e + W)$. It is onto since the terms of W disappear in the target; its kernel is precisely the first quotient, because $z = s + b + j$ implies $[z] = [j]$. Consequently $d_K + d_T = \kappa_e$. The full quotient-value map is

$$\Psi_e : \mathcal{Z}_e/(\mathcal{Z}_e \cap W) \longrightarrow \mathbb{B}\mathcal{Z}_e/\text{im } \Phi, \quad [z] \longmapsto [\mathbb{B}z].$$

It is onto by the displayed target, and its kernel is zero: $\mathbb{B}z = \Phi x$ means $z - Bx \in \mathbb{J}$, hence $z \in \mathcal{Z}_e \cap W$. Thus its rank is $\dim((\mathcal{Z}_e + W)/W) = c_e + d_T$, by the sum-quotient sequence with subspace $(\mathcal{S}_e + W)/W$ of dimension c_e . With the unchanged $\Delta Z = Z^1 - Z^0$ and $\mathcal{R}f = -f(D_0) - f(D_1) + f(D_2) + f(D_3)$, the sharpened finite intervals are

$$\begin{aligned} 0 &\leq d_T \leq \kappa_e \leq \bar{\kappa}_e, \\ |\Delta \mathcal{R} \log \omega_e| &\leq (c_e + d_T)W_j \leq (c_e + \bar{\kappa}_e)W_j, \\ |\Delta \mathcal{R}(\log \omega_e - \log \omega_{\text{conn},e})| &\leq \kappa_e W_j \leq \bar{\kappa}_e W_j, \\ -\bar{\kappa}_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{S}_e) \leq \bar{\kappa}_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{RKR3})$$

To verify the constants and signs, for any one of the fixed quotients X of rank r put $L_X(D) = \Delta \log \det \text{Gram}(X)$. The full native form inequality NSR4 and its same affine minima give $L_X(D) = -r \log \mathfrak{a}_j - u_X(D)$ with $0 \leq u_X(D) \leq r \omega_{j,D}$. The complete contraction of rank $c_e + d_T$ has two such quotient forms in the same value coordinates; subtracting their intervals cancels the common scalar and gives the pointwise width $(c_e + d_T)\omega_{j,D}$. The adapted determinant identity for the difference between the full and connecting contractions uses the quotients of ranks d_K, d_T, κ_e with

signs $(+, +, -)$. Its same-frame order difference is $-u_K - u_T + u_{\mathcal{X}/\mathcal{S}}$, since $d_K + d_T - \kappa_e = 0$. It lies in $[-\kappa_e \omega_{j,D}, \kappa_e \omega_{j,D}]$. Applying the four original signs proves the two symmetric intervals in RKR3. For the single quotient of rank κ_e , those signs give $u(D_0) + u(D_1) - u(D_2) - u(D_3)$, hence the directed interval $[-\kappa_e \Omega_j^{\text{hi}}, \kappa_e \Omega_j^{\text{lo}}]$. Every width is nonnegative. Substituting BTK25 therefore proves the last inequalities without changing their directions.

The middle interval may also be intersected with the original two-family bound from CIT23. Its absolute value is at most

$$\min \left\{ \bar{\kappa}_e W_j, 2\bar{\kappa}_e (\log A_{j,j}^{\text{end}} + \log A_{j,1}^{\text{end}}) \right\}.$$

Both entries bound the same scalar in the same frames, by the triangle inequality applied to the two original family returns. For $v - e \leq 2$, the actual spaces satisfy $\mathcal{Z}_e = \mathcal{S}_e$, so $d_K = d_T = 0$ and $\omega_e^p(D) = \omega_{\text{conn},e}^p(D)$ at every retained cutoff, $p = 0, 1$. The tail terms in RKR3 are then zero. The first NSR13 interval retains its exact rank $r_e + \kappa_e$; the present calculation inserts no value for the remaining connecting rank c_e or its absolute loss.

R25.4 The complete boundary ranks, metrics and positive returns

Write $J = \text{im } J_0 = \ker B_k$ and retain the full original coefficient kernels

$$\begin{aligned} \mathcal{L} &= \ker \Phi, \quad \ell = \dim \mathcal{L}, \quad E = \ker(\Phi J_0), \quad \epsilon = \dim E, \quad Y = J + \mathcal{L}, \\ r_I &= j_{0,k} - \epsilon, \quad r_b = t_k - \ell + \epsilon, \quad r_o = 4t_j - m_k + \ell, \quad r_I + r_b + r_o = 4t_j. \end{aligned} \tag{NSR16}$$

No symbol ℓ here replaces the source-order index ℓ_s in NSR2. Since J_0 is injective, $J \cap \mathcal{L} = J_0 E$ and $\dim Y = j_{0,k} + \ell - \epsilon$. Hence $\text{rank } \Phi = m_k - \ell$, $\text{rank } \Phi J_0 = r_I$, and its image quotient has dimension r_b . Completing its full image in $\mathbb{C}^{4t_j}$ gives r_o and the last equality. All three ranks are nonnegative actual ranks, with both original low intersections and every hidden coefficient kernel present.

Keep the original full lift frames Z_R of the invariant source modulo E , E_b of $\mathcal{V}/Y$, and U_o completing the image frame

$$U_R = \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad C_T = [U_R, U_b, U_o] \text{ invertible.} \tag{NSR17}$$

The target form is four copies of the original $H_{B,j,D}^{(s_1)} = [B_j(D_{j,D}^{(s_1)})^{-1} B_j^*]^{-1}$. Its restriction on U_R is H_{inv} , its next quotient in lifts U_b is Q_b^E , and its quotient modulo $\text{im } \Phi$ in lifts U_o is Q_o . Their ranks are r_I, r_b, r_o . Before any return,

$$|\det C_T|^2 (\det H_{B,j,D}^{(s_1)})^4 = \det H_{\text{inv}}^p(D) \det Q_b^{E,p}(D) \det Q_o^p(D). \tag{NSR18}$$

Successively minimize the first and then the second coordinate blocks in this fixed frame to prove the identity. In original conductor coordinates the same fibres are $\mathbb{J} + \text{im}(B J_0)$ and $\mathbb{J} + \text{im } B$, because $\ker \mathbb{B} = \mathbb{J}$. Thus all lower and invariant gauges are included. The map $\mathcal{V}/Y \rightarrow W/W_{\text{inv}}$ is $[x] \mapsto [Bx]$; its full zero fibre is $x - J_0 z \in \mathcal{L}$. This also proves the original RQB quotient identification used in CIT29.

Let $\mathcal{V}_I^p = \mathcal{R} \log \det H_{\text{inv}}^p$, $\mathcal{V}_b^p = \mathcal{R} \log \det Q_b^{E,p}$ and $\mathcal{V}_o^p = \mathcal{R} \log \det Q_o^p$. Then the complete original budget and its order receiving are

$$\begin{aligned} \mathcal{V}_I^p + \mathcal{V}_b^p + \mathcal{V}_o^p &= 4\widehat{F}_j^{(s_1)}, \quad \mathcal{V}_I^p, \mathcal{V}_b^p, \mathcal{V}_o^p \geq 0, \\ -r_X \Omega_j^{\text{hi}} &\leq \Delta \mathcal{V}_X \leq r_X \Omega_j^{\text{lo}} \quad (X = I, b, o), \\ -4t_j \Omega_j^{\text{hi}} &\leq 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}) \leq 4t_j \Omega_j^{\text{lo}}, \\ \Delta \mathcal{V}_I + \Delta \mathcal{V}_b + \Delta \mathcal{V}_o &= 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}). \end{aligned} \tag{NSR19}$$

Here $(r_X) = (r_I, r_b, r_o)$. The budget is RQB27–28: the conjugate inverse $H_{B,j,D} = \overline{\widehat{H}_{K,j,D}^{-1}}$ changes the primal sign exactly, and the fixed frame factor in NSR18 cancels under $\mathcal{R}$. Increasing original

Gamma row cutoffs increases each native form and preserves that order under its complete fibre minimum; this proves positivity of all returns. NSR3 now proves each order interval at its actual rank. The last equality retains their exact correlation.

For the earlier CIT lifts, $R_b^{\text{CIT}} = R_b^E C_b + Y_{\text{inv}} A_b$ with C_b invertible and fixed. Hence $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$ by the same minimum; all finite determinant identities retain $|\det C_b|^2$. If $[X_{\mathbb{J}}, I_R, R_b^{\text{CIT}}] = X_W C_W$, then

$$|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E. \quad (\text{NSR20})$$

This is the original CIT25/RQB31 equality. Its two fixed coordinate determinants cancel in the order ratios and in $\mathcal{R}$, because NSR5 and NSR17 retain the same ordered value coordinates.

The residual source denominator remains the full quotient $Q_0 = D_{k,D}^{(s_0)}/Y$ in lifts E_b , of rank r_b . Equivalently it is the original kernel-dual quotient by $B_k \mathcal{L}$, whose rank is $\ell - \epsilon$. With the fixed original frame $C_0 = [Z_{\Lambda}, B_k E_b]$ it satisfies

$$\begin{aligned} \det Q_0(D) &= \frac{|\det C_0|^2 \det H_{B,k,D}}{\det(Z_{\Lambda}^* H_{B,k,D} Z_{\Lambda})}, \\ -r_b \Omega_k^{\text{hi}} &\leq \Delta \mathcal{R} \log \det Q_0 \leq r_b \Omega_k^{\text{lo}}, \\ -r_b(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) &\leq \Delta \mathcal{R} \log \frac{\det Q_b^E}{\det Q_0} \leq r_b(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}). \end{aligned} \quad (\text{NSR21})$$

To prove the first equality, minimize first over $J = \ker B_k$ and then over $B_k \mathcal{L}$; the complete fibre is exactly Y . The independent frame C_0 gives its Schur determinant. NSR3 proves the second, and subtraction of the two directed rank- r_b return intervals proves the third. No inverse bound for the residual multiplier has entered this calculation.

R25.5 The common contraction in CIT30 with all boundary factors

The original ITL loss is $\omega_{\text{inv}}^p(D) = \det H_{\text{inv}}^p(D) / \det A_{\text{inv}}^p(D)$, where $A_{\text{inv}} = F^* \mathcal{D}_D F / E$, $F = B J_0$ and the quotient values are Z_R . Its two attained target-fibre forms have the same rank r_I . Therefore, by the proof of NSR12,

$$|\Delta \log \omega_{\text{inv}}(D)| \leq r_I \omega_{j,D}, \quad |\Delta \mathcal{R} \log \omega_{\text{inv}}| \leq r_I W_j. \quad (\text{NSR22})$$

The source fibre here is $F Z_R x + F E$ and the target fibre is $F Z_R x + \mathbb{J}$; $F E = \text{im } F \cap \mathbb{J}$ by its complete coefficient kernel. Thus this compares the same original two minima, not the independent source-order metric Q_{inv} used below.

Put exactly as in CIT30

$$\begin{aligned} \mathcal{P}_e^p &= \mathcal{R}(\log g_{T_e} + \log g_{K_e} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}})^p, \\ \mathcal{L}_e^p &= \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}})^p, \quad C_e^{\text{ord}} = (c_e + r_I) W_j. \end{aligned} \quad (\text{NSR23})$$

Substitution of NSR20 in the full CIT26 cross ratio proves

$$\begin{aligned} \mathcal{L}_e^p &= \mathcal{P}_e^p - \mathcal{V}_b^p, \quad \mathcal{P}_e^p - 4\widehat{F}_j^{(s_1)} + \mathcal{V}_I^p \leq \mathcal{L}_e^p \leq \mathcal{P}_e^p, \\ -C_e^{\text{ord}} &\leq \Delta \mathcal{L}_e \leq C_e^{\text{ord}}, \\ -C_e^{\text{ord}} - r_b \Omega_j^{\text{hi}} &\leq \Delta \mathcal{P}_e \leq C_e^{\text{ord}} + r_b \Omega_j^{\text{lo}}, \\ \max\{\mathcal{P}_e^1 - 4\widehat{F}_j^{(j)} + \mathcal{V}_I^1, \mathcal{P}_e^0 - \mathcal{V}_b^0 - C_e^{\text{ord}}\} &\leq \mathcal{L}_e^1 \\ &\leq \min\{\mathcal{P}_e^1, \mathcal{P}_e^0 - \mathcal{V}_b^0 + C_e^{\text{ord}}\}. \end{aligned} \quad (\text{NSR24})$$

The first directed bound is the unchanged RQB32 inequality for the negative residual factor: use $0 \leq \mathcal{V}_b^p \leq 4\widehat{F}_j^{(s_1)} - \mathcal{V}_I^p$. The second line adds NSR12 and NSR22, with exact ranks c_e, r_I . Since $\Delta \mathcal{P}_e = \Delta \mathcal{L}_e + \Delta \mathcal{V}_b$, NSR19 proves the third line with its asymmetric widths. The last two lines intersect the original pair-one bound with $\mathcal{L}_e^0 + [-C_e^{\text{ord}}, C_e^{\text{ord}}]$ and substitute the exact pair-zero residual identity. Both intervals contain the same $\mathcal{L}_e^1$, so their intersection does also. The equality $\mathcal{V}_b + \mathcal{V}_I + \mathcal{V}_o = 4\widehat{F}_j$ remains exact throughout; no missing quotient is assigned zero. Every displayed order interval is $o(kq_k)$, while neither common value in the first line is evaluated.

R25.6 Both source-order intervals in the matched invariant return

Keep NFR1's full original source quotient

$$\begin{aligned} Q_{\text{inv}}^p(D) &= (J_0^* D_{k,D}^{(s_0)} J_0)/E \quad \text{in the same lifts } Z_R, \quad \text{rank } Q_{\text{inv}} = r_I, \\ \mathcal{R}_X^p(D) &= (Q_{\text{inv}}^p(D))^{-1/2} H_{\text{inv}}^p(D) (Q_{\text{inv}}^p(D))^{-1/2}, \\ V_Q^p &= \mathcal{R} \log \det Q_{\text{inv}}^p, \quad \mathcal{M}^p = \mathcal{R} \log \det \mathcal{R}_X^p. \end{aligned} \tag{NSR25}$$

The kernel $E = \ker(\Phi J_0)$ and complement Z_R are independent of the measures. Hence these are the actual NFR/ITL matched quotient-value spaces. At every endpoint $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, even though its inverse square roots depend on the source metric. This proves the exact matrix-to-determinant map.

The complete NFR6 return and the new finite order interval are

$$\begin{aligned} \mathcal{M}^p &= 4\widehat{F}_j^{(s_1)} - \mathcal{V}_b^p - \mathcal{V}_o^p - V_Q^p, \quad \mathcal{M}^p + V_Q^p = \mathcal{V}_I^p \geq 0, \\ -r_I \Omega_k^{\text{hi}} &\leq \Delta V_Q \leq r_I \Omega_k^{\text{lo}}, \\ \Delta \mathcal{M} &= 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}) - \Delta \mathcal{V}_b - \Delta \mathcal{V}_o - \Delta V_Q, \\ \Lambda^- &:= -r_I(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) \leq \Delta \mathcal{M} \leq r_I(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}) =: \Lambda^+. \end{aligned} \tag{NSR26}$$

The first equality substitutes NSR18–19 in the exact matched determinant. The second follows from NSR3 on the rank- r_I source quotient. For the last interval use $\Delta \mathcal{M} = \Delta \mathcal{V}_I - \Delta V_Q$, then subtract the directed intervals for these same two rank- r_I values. The preceding line preserves its full target-budget realization with all three target factors. Using the actual invariant rank before estimating avoids assigning the much larger compatibility-plane rank to this matched quotient.

In NSR26, the source degree is still k , with center c'_k , at the displayed D_i based on q_j . Its displacement from its degree- k standard endpoints is $b = q_j - q_k = 8k + 24$ at both lows and $2b$ at both highs. The constants ω_{k,D_i} in NSR2 use exactly those displaced cutoffs. No window displacement is silently absorbed into $\Lambda^\pm$.

Let $U_{j,1}^{\text{native}}, U_{j,j}^{\text{native}}$ be the complete RQB35 caps, with their original positive observation sum, full FEX source correction and all finite-domain restrictions of those providers. Define

$$U_{\mathcal{M}}^p = 4U_{j,s_1}^{\text{native}} - \mathcal{V}_b^p - \mathcal{V}_o^p - V_Q^p. \tag{NSR27}$$

Both original source-order intervals are retained explicitly:

$$\begin{aligned} -V_Q^0 &\leq \mathcal{M}^0 \leq U_{\mathcal{M}}^0, \quad -V_Q^1 \leq \mathcal{M}^1 \leq U_{\mathcal{M}}^1, \\ \max\{-V_Q^1, -V_Q^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{U_{\mathcal{M}}^1, U_{\mathcal{M}}^0 + \Lambda^+\}, \\ \max\{-V_Q^0, -V_Q^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{U_{\mathcal{M}}^0, U_{\mathcal{M}}^1 - \Lambda^-\}. \end{aligned} \tag{NSR28}$$

For each pair, nonnegativity of $\mathcal{V}_I$ gives its lower bound, and $\widehat{F}_j^{(s_1)} \leq U_{j,s_1}^{\text{native}}$ in the first line of NSR26 gives the upper bound with the full missing return retained. Adding Λ^-, Λ^+ to the pair-zero interval and intersecting with the pair-one interval proves the second line. Reversing the difference and intersecting in the other direction proves the third. In particular this refines the displayed NFR6 upper bound by keeping its exact nonnegative $\mathcal{V}_o$, while preserving Q_{inv} and $\mathcal{V}_b$ on both sides.

The preexisting NFR5 error also has an exact receiving. Let $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$, with the full original compatibility and source-complement forms, and let $\mathcal{A}^p = \mathcal{R} \log \omega_{\text{inv}}^p$. At each cutoff NFR2's constants, denoted here by a_i^p, b_i^p , retain their actual rank r_I :

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p, \\ \mathcal{M}^p &= \mathcal{C}^p + \mathcal{A}^p + \mathcal{E}^p, \quad E_p^- \leq \mathcal{E}^p \leq E_p^+, \\ \Lambda^- - r_I W_j - E_1^+ + E_0^- &\leq \Delta \mathcal{C} \leq \Lambda^+ + r_I W_j - E_1^- + E_0^+. \end{aligned} \tag{NSR29}$$

The first four lines are NFR2/5 with the two low errors negated and the two high errors unnegated. The exact difference is $\Delta\mathcal{C} = \Delta\mathcal{M} - \Delta\mathcal{A} - \Delta\mathcal{E}$. Insert NSR22, NSR26 and these two error intervals to prove the last line. Thus the full shared compatibility ratio also receives a finite $o(kq_k)$ order-difference bound, without replacing either individual large compatibility determinant or either original source-complement projection.

R25.7 The existing arithmetic allocation comparison and its exact map

For the original degree- a standard window only, denote its cutoffs by $d_i^{[a]} = q_a - v - 1, q_a - v, 2q_a - v - 1, 2q_a - v$; write $\Omega_a^{[a],\text{lo/hi}}$ for NSR2 evaluated there. Keep HJR16–22's original inverse-metric decomposition with $D = D_{a,d}^{(s)}$, $J = J_a$ and $B = B_a$:

$$D^{-1} = J(J^*DJ)^{-1}J^* + L(BD^{-1}B^*)L^*, \quad L = D^{-1}B^*(BD^{-1}B^*)^{-1}. \quad (\text{NSR30})$$

Indeed $[J, L]$ is a full coefficient frame with $BJ = 0$, $BL = I$, and block Gram

$$\text{diag}(J^*DJ, (BD^{-1}B^*)^{-1}).$$

Inverting that exact frame Gram proves NSR30. The complete original new row h gives

$$\begin{aligned} x &= hJ(J^*DJ)^{-1}J^*h^*, \\ y &= hL(BD^{-1}B^*)L^*h^*. \end{aligned}$$

The original old lower Schur denominator remains inside h . The row returns are $\log(1+x)$ and $\log(1+y/(1+x))$, with the original endpoint weights one at the two extremes and two inside. Their exact sums are

$$\begin{aligned} S_K^{(s)} &= \mathcal{R}_a \log \det H_{B,a,d}^{(s)} = \widehat{F}_a^{(s)}, \quad S_R^{(s)} = \mathcal{R}_a \log \det (J_a^* D_{a,d}^{(s)} J_a) = F_{0,a,s}, \\ S_K^{(s)} + S_R^{(s)} &= \mathcal{T}_{a,s}, \\ -t_a \Omega_a^{[a],\text{hi}} &\leq S_K^{(a)} - S_K^{(1)} \leq t_a \Omega_a^{[a],\text{lo}}, \\ -j_{0,a} \Omega_a^{[a],\text{hi}} &\leq S_R^{(a)} - S_R^{(1)} \leq j_{0,a} \Omega_a^{[a],\text{lo}}. \end{aligned} \quad (\text{NSR31})$$

This is the same native restriction/quotient transport NSO15–19 applied to the exact preexisting HJR maps. All zero rows remain in their aggregate. It gives no rowwise or averaged allocation value.

To connect these actual sums to the earlier AMT arithmetic objects, retain the original FEX/RQB33 identity and its complete low correction:

$$\begin{aligned} S_K^{(s)} &= F_K^{(s)} + \delta_{K,a,s}, \quad \delta_{K,a,s} = \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_a}^{(s)} + F_{(K_a+L_a)/K_a}^{(s)}, \quad \sigma = (1, 1, -1, -1), \\ S_R^{(s)} &= \mathcal{T}_{a,s} - F_K^{(s)} - \delta_{K,a,s}, \\ S_K^{(a)} - S_K^{(1)} &= -\Delta_K^{\Gamma,[a]} + \delta_{K,a,a} - \delta_{K,a,1}, \\ S_R^{(a)} - S_R^{(1)} &= \mathcal{T}_{a,a} - \mathcal{T}_{a,1} + \Delta_K^{\Gamma,[a]} - \delta_{K,a,a} + \delta_{K,a,1}. \end{aligned} \quad (\text{NSR32})$$

Here $N_i = d_i^{[a]} + v$ and the original low space is $L_a = \mathbb{C}[S]_{<v}$. The scalar $e_{N_i,s}$ is the full original conductor source Jacobian; its exact finite bounds remain RQB34/FEX34. The existing AMT44 convention is $\Delta_K^{\Gamma,[a]} = F_K^{(1)} - F_K^{(a)}$. Its proved interval is

$$-(8a-16)\Omega_{\Gamma,a}^{\text{low}} \leq \Delta_K^{\Gamma,[a]} \leq (8a-16)\Omega_{\Gamma,a}^{\text{high}}, \quad \Delta_K^{\Gamma} + \Delta_B^{\Gamma} = \mathcal{H}_a, \quad (\text{NSR33})$$

where $\Omega_{\Gamma,a}^{\text{low}} = \omega_{a,q_a-1} + \omega_{a,q_a}$ and $\Omega_{\Gamma,a}^{\text{high}} = \omega_{a,2q_a-1} + \omega_{a,2q_a}$ use the original primal, unshifted cutoffs of AMT43. For the exact underlying map, AMT7–9 minimizes the unchanged original polynomial

remainder map, restricts to K_a and takes the original observation quotient; AMT42 compares their two full original source Grams in those same values. Their adapted determinant gives the retained sum with $\mathcal{H}_a$. Subtraction of these already proved equalities is precisely NSR32. Thus the original primal rank $8a - 16$, the native rank $t_a = 8a - 16 - t_{0,a}$, the residual rank $j_{0,a}$, and every low correction stay in the typed comparison. No new analytic bound or new equality of those distinct source presentations is claimed.

All receiving formulas above apply their finite intervals before limits. The surviving order differences are $o(kq_k)$ on the fixed actual programme, while CIT30, RQB28 and NFR6 continue to retain the common long connecting/ITL loss, the positive residual and missing returns, and the original S_K, S_R allocation.

The finite analytic comparison in the same native coefficient form. At an original actual degree a , keep $D = N - v$, the original centre c'_a , the complete lower-evaluation minimum and both source orders $1, a$. Use the explicitly guarded endpoints $\lambda_{a,D}^\pm$ of EFP4–11. These intersect the NSO constants with QOT and, only where both EOR parity tests hold, the EOR constants on the same original form:

$$e^{\lambda_{a,D}^-} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq e^{\lambda_{a,D}^+} D_{a,D}^{(1)}. \quad (\text{EFR1})$$

The polynomial degree in the QOT/EOR entries is exactly $N - v - q'_a$. Taking the maximum of two scalar lower bounds and minimum of their upper bounds proves the intersection. Every candidate vector in a fixed affine fibre satisfies it; minimization over that same fibre proves its transfer to the whole lower, relation, kernel and quotient minima, as in the explicit EFP13 Schur formula. The ordinary-transpose Taylor change and the conjugated inverse in NSO/NSR remain unchanged. On a fixed rank- r restriction or attained quotient, the determinant difference lies in $[r\lambda^-, r\lambda^+]$. At the four original endpoints it lies in $[r\Phi_a^-, r\Phi_a^+]$, with $\Phi_a^\pm$ given in EFP15 using signs $(-, -, +, +)$. The four identical $-r \log \kappa_a$ contributions cancel only in that return; no source mass is dropped at an endpoint. Where QOT's original guard fails, EFP11 retains the full NSO comparison. The defining primitive source uses EFP's separately proved relation restriction and Θ_e map, with degree $24k - 25 - v + e$.

R26 Native Gamma-order transport and its equilibrium-support refinement

Source and presentation. This chapter retains the complete refined QOT and EOR source, with every proof, original constant, coordinate map, finite domain and bibliography. Explicit equation tags preserve the source equation numbers. Local symbols, theorem environments, labels and bibliography keys are scoped or namespaced for the cumulative edition. A precision sentence at the EOR flags records that the coefficient space is fixed while its Gamma metric depends on source order. The delivered source and every reversible presentation change remain preserved in the source closure.

Scope of the complete delivered calculation. For the original fixed simple-quartet family, a full quadratic-form comparison controls the change from Gamma order one to the original degree-dependent order. It acts on every original coefficient direction before either relation or observation minimization. The difference of the two native kernel allocations, and separately of the two residual allocations, is $O(q_k)$, with limiting absolute allowance $8\log(2 + \sqrt{3})$ after division by q_k . The common connecting/invariant long loss has source-order difference $O(q_k)$ with allowance $32\log(2 + \sqrt{3})$. At the two defining primitive cutoffs, the paired source-to-connecting-image order correction is $o(q_k)$. Its scalar shift is evaluated exactly, and the original compatibility difference is retained and compared in the same coordinates. The individual kq_k -scale allocation coefficients and the absolute common long loss are not evaluated here. No offcritical packet is excluded by these comparisons alone.

Refinement retained in this edition. The subsequent EOR calculation, included in full below, replaces the high-degree symmetric enclosure by the actual equilibrium-support enclosure. For $\Delta S_X = S_X(k; k) - S_X(k; 1)$ it gives the limiting interval $[-8\log(\pi v_*/4), -8\log(\pi u_*/4)]$ at scale q_k . The common source-order allowance becomes $16\log(v_*/u_*)$. The original QOT estimates remain valid; these are subsequent strict refinements, not a reattribution of the older results. The original source bytes are preserved. Two truncated norm control sequences in its source have been restored here, with an exact reversible record.

R26.1 Original data and the result's scope

Fix the stipulated hypothetical actual simple quartet, its full unit, one original five-orbit period and its logarithm branch. Put

$$\begin{aligned}
 k = 4l + 1 \geq 9, \quad q_k &= (k + 1)^2, \quad q'_k = (k - 7)^2, \quad \Delta_k = 16k - 48, \quad c'_k = k/2 - 4, \\
 \beta_{n,a,b} &= n/2 + (2a - n)\delta + i(2b - n)\gamma, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\
 \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), \quad \psi_k = \chi_{k-8}, \\
 E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v.
 \end{aligned} \tag{QOT1}$$

The a_{ab} are the actual conductor coefficients. Its nonreduced factors and exact v remain. The fixed bound $v \leq 80$ is sufficient; no exact matrix replaces v by that bound. The polynomial ψ_k is a lower-grid polynomial, not a derivative.

At degree k the two original source orders are $s = 1, k$. The complete measures, masses, and

monic norms are

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad p_{0,s} = 1, \quad p_{1,s} = y, \\ p_{n+1,s} &= y p_{n,s} - n(n-1 + s/2) p_{n-1,s}. \end{aligned} \tag{QOT2}$$

All comparisons below identify the same original polynomial coefficients or quotient values, not their different orthonormal coordinate vectors. Source orders are never altered inside an endpoint telescoping sum.

The quantitative root/Gamma input used below is the supplied full-form GPA proof. Its original finite domain is

$$k \geq 29, \quad R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33} q_k. \tag{QOT3}$$

For paired degree $j = k + 4$ statements impose the corresponding domain at j as well. Every fixed actual quartet eventually satisfies both conditions. Arithmetic statements retain the original period domain $|u| \geq R_*$, or the specified original exact period tests. There is no uniform assertion over a varying period.

The available native primal forms are the attained minima

$$\begin{aligned} z^* H_{Y,k,N}^{(s)} z &= \min_h \int_{\mathbb{R}} |(Ez + \mathbf{U}_N h)(c'_k + iy)|^2 |\psi_k(c'_k + iy)|^2 dm_s(y), \\ H_{K,k,N}^{(s)} &= Z_k^* H_{Y,k,N}^{(s)} Z_k, \\ H_{R,k,N}^{(s)} &= [P_Y (H_{Y,k,N}^{(s)})^{-1} P_Y^*]^{-1}. \end{aligned} \tag{QOT4}$$

Here E , $\mathbf{U}_N$, Z_k , and P_Y are the complete original GPA/FGC/SAR maps. Each relation column is $\alpha_A \mathcal{T}_A(\chi_k S^d)/\psi_k$, where $\mathcal{T}_A p(S') = \sum a_{ab} p(S' + \beta_{8,a,b})$. Their domains, all coefficients and both minima in (QOT4) remain unchanged. The ranks and endpoint signs are

$$\begin{aligned} \mathfrak{m}_k &= \Delta_k - v, \quad t_0 = \dim(K \cap \mathbb{C}[S]_{<v}), \quad t_K = 8k - 16 - t_0, \quad t_R = 8k - 32 - v + t_0, \\ N &= (q_k - 1, q_k, 2q_k - 1, 2q_k), \quad \sigma = (+1, +1, -1, -1), \\ S_{X,k,s} &= \sum_N \sigma_N \log \det H_{X,k,N}^{(s)}, \quad X = K, R. \end{aligned} \tag{QOT5}$$

The actual dual coefficient metric is

$$D_{k,N-v}^{(s)} = A_{\ell,k}^* \overline{(H_{Y,k,N}^{(s)})^{-1}} A_{\ell,k}. \tag{QOT6}$$

The same invertible, source-order-independent $A_{\ell,k}$ appears in both metrics. The conjugation is required by complex-linear duality.

R26.2 A complete power-weight comparison

For $\alpha > 0$, $\vartheta > 0$, let

$$\|p\|_\alpha^2 = \int_{\mathbb{R}} |p(y)|^2 |y|^\alpha e^{-\vartheta|y|} dy.$$

Lemma QOT2.1 (Multiplication with its next-degree row retained). *For $n \geq 0$ define*

$$\begin{aligned} c_n &= \frac{(\sqrt{n + \alpha + 1/2} - \sqrt{n + 1/2})^2}{\vartheta}, \\ C_n &= \frac{(\sqrt{n + \alpha + 1/2} + \sqrt{n + 1/2})^2}{\vartheta}. \end{aligned} \tag{QOT7}$$

Every complex polynomial of degree at most n satisfies

$$c_n \|p\|_\alpha \leq \|yp\|_\alpha \leq C_{n+1} \|p\|_\alpha. \tag{QOT8}$$

Proof. For the classical Laguerre norm and recurrence, with the standard phase retained, see Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, Table 18.3.1; eq. (18.5.5) and Table 18.5.1; eq. (18.5.12); eq. (18.9.1) and Table 18.9.1]. Here their parameter is α , their argument is $x = \vartheta y$, and their degree is i ; hence the squared norm is $\vartheta^{-\alpha-1} \Gamma(i + \alpha + 1)/i!$. The positive square root of this full norm yields precisely the negative off-diagonal entry displayed next. The next-degree row required for the multiplication norm remains in the local proof.

On $(0, \infty)$, the normalized Laguerre polynomials for $y^\alpha e^{-\vartheta y} dy$ give multiplication matrix

$$(J_n)_{ii} = (2i + \alpha + 1)/\vartheta, \quad (J_n)_{i,i+1} = -\sqrt{(i+1)(i+\alpha+1)}/\vartheta \quad (0 \leq i < n).$$

Its entries follow from Rodrigues' coefficient formula and integration by parts, as in the supplied GPA/LVM proof. For completeness, $L_n^{(\alpha)}(x) = \sum_{r=0}^n (-1)^r \binom{n+\alpha}{n-r} x^r / r!$ has squared norm $\Gamma(n + \alpha + 1)/n!$ and obeys

$$xL_n^{(\alpha)} = (2n + \alpha + 1)L_n^{(\alpha)} - (n+1)L_{n+1}^{(\alpha)} - (n+\alpha)L_{n-1}^{(\alpha)}.$$

Coefficient comparison proves the recurrence; the stated norms give J_n , including its negative off-diagonal entries.

The function $f(x) = \sqrt{x(x+\alpha)}$ is concave for $x > 0$, since $f''(x) = -\alpha^2/[4(x(x+\alpha))^{3/2}]$. Continuity includes $x = 0$. Hence $f(i) + f(i+1) \leq 2f(i+1/2)$. Bounding the cross terms of the tridiagonal quadratic form by the adjacent squared moduli gives

$$c_n I \preceq J_n \preceq C_n I.$$

Indeed the lower row expression $2i + \alpha + 1 - 2f(i+1/2)$ decreases with i , and the upper expression increases. Omitting a missing boundary off-diagonal only improves these bounds. Cauchy–Schwarz now gives $\|yp\| \|p\| \geq \langle yp, p \rangle \geq c_n \|p\|^2$ on this half-line.

For the upper norm bound the multiplication map is rectangular: its last row is $-\sqrt{(n+1)(n+\alpha+1)} e_n^\top / \vartheta$. It is a restriction of J_{n+1} , not J_n . Thus its norm is at most C_{n+1} . Apply both inequalities also to $p(-y)$ on $y > 0$ and add their squares. The sign of multiplication on the negative half-line has no effect on its norm. This proves (QOT8) for the entire original coefficient direction, without a parity restriction. $\square$

For integers $a, \ell \geq 0$ put

$$\begin{aligned} d_{\alpha,a,\ell}^- &= 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha}, \\ d_{\alpha,a,\ell}^+ &= 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha}. \end{aligned} \tag{QOT9}$$

Empty sums are zero.

Theorem QOT2.2 (Centred weight shift). *On every polynomial of degree at most a ,*

$$(\alpha/\vartheta)^{2\ell} e^{-d_{\alpha,a,\ell}^-} \|p\|_\alpha^2 \leq \|p\|_{\alpha+2\ell}^2 \leq (\alpha/\vartheta)^{2\ell} e^{d_{\alpha,a,\ell}^+} \|p\|_\alpha^2. \tag{QOT10}$$

Proof. Iterate (QOT8) in the fixed α norm on $p, yp, \dots, y^{\ell-1}p$, with degree bounds $a, a+1, \dots, a+\ell-1$. The resulting lower product is $\prod_{r=0}^{\ell-1} c_{a+r}^2$ and the upper product is $\prod_{r=1}^{\ell} C_{a+r}^2$. Also $\|y^\ell p\|_\alpha^2 = \|p\|_{\alpha+2\ell}^2$ exactly. Finally

$$c_n = (\alpha/\vartheta) e^{-2 \operatorname{arsinh} \sqrt{(n+1/2)/\alpha}}, \quad C_n = (\alpha/\vartheta) e^{2 \operatorname{arsinh} \sqrt{(n+1/2)/\alpha}}.$$

Substitution proves both sides. The displayed central scalar is not an exact equality of forms; for $p = 1, \ell = 1$, the actual moment ratio is $(\alpha+1)(\alpha+2)/\vartheta^2$, not α^2/ϑ^2 . $\square$

R26.3 Finite constants in the original Gamma comparison

We record the full inherited GPA constants so the comparison does not use an unspecified conditioning constant. Abbreviate $q = q_k$, $q' = q'_k$, $R = R_k$, $\epsilon = 2^{-32}$, $\vartheta = \pi/2$, and $h = 1/q$. Put

$$\begin{aligned} \ell_s &= (s-1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \\ C_L &= \frac{\pi}{\Gamma(3/4)^2}, \quad K_q = \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\ \mathcal{E}_k &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q), \\ r_{\ell_s} &= \frac{\sum_{r=0}^{\ell_s-1} (2r+1/2)^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\ L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ K_{a,Q} &= 2Q + 1 + 2\{a + \sqrt{a(a+2Q)}\}. \end{aligned} \tag{QOT11}$$

The logarithmic endpoints are

$$\begin{aligned} b_{k,a,s}^- &= -\mathcal{E}_k + \log L_h - K_{a,Q_s} \log(1 + h/\vartheta), \\ b_{k,a,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a,Q_s} \log \frac{1}{1 - h/\vartheta}. \end{aligned} \tag{QOT12}$$

The supplied full proof gives, for every $\deg p \leq a \leq 2q$,

$$\begin{aligned} \beta_s e^{b_{k,a,s}^-} \|p(c'_k + i \cdot)\|_{2Q_s}^2 &\leq \int_{\mathbb{R}} |\psi_k(c'_k + iy) p(c'_k + iy)|^2 dm_s(y) \\ &\leq \beta_s e^{b_{k,a,s}^+} \|p(c'_k + i \cdot)\|_{2Q_s}^2. \end{aligned} \tag{QOT13}$$

This uses precisely the degree bound in the proof of GPA12–15, not only its originally listed endpoints. In all applications here $a + \ell_s \leq 2q$ also holds on the stated eventual domains. The proof pairs the original opposite roots outside $[-\epsilon q, \epsilon q]$, bounds the entire inner polynomial integral by K_q times the reference outside integral, and retains the full source recurrence

$$dm_s = \beta_s \prod_{r=0}^{\ell_s-1} (y^2 + (2r+1/2)^2) d\sigma, \quad d\sigma = dm_1.$$

It then uses the whole-line Gamma sandwich $L_h e^{-(\vartheta+h)|y|} \leq d\sigma/dy \leq U_h e^{-(\vartheta-h)|y|}$ and the exact polynomial dilation norm bound. Thus none of the original roots, the inner region, or the source mass is removed by (QOT13). At fixed actual data, $b_{k,a,s}^\pm = O_{\text{actual}}(\log q)$ uniformly for these degrees and both original orders.

Define

$$\begin{aligned} \alpha_k &= 2q'_k, \quad \ell = (k-1)/4, \quad \kappa_k = \beta_k (4q'_k/\pi)^{(k-1)/2}, \\ l_{k,a} &= b_{k,a,k}^- - b_{k,a,1}^+ - d_{\alpha_k,a,\ell}^-, \quad u_{k,a} = b_{k,a,k}^+ - b_{k,a,1}^- + d_{\alpha_k,a,\ell}^+, \quad w_{k,a} = u_{k,a} - l_{k,a} \geq 0. \end{aligned} \tag{QOT14}$$

Theorem QOT3.1 (Full original source-order transport). *On the degree- a original coefficient space, and on every fixed restriction or attained quotient of it,*

$$\kappa_k e^{l_{k,a}} H^{(1)} \preceq H^{(k)} \preceq \kappa_k e^{u_{k,a}} H^{(1)}. \tag{QOT15}$$

In particular it holds for all three forms in (QOT4) at $a = N - v - q'_k$. On their original complex-linear dual coefficient metric it becomes

$$\kappa_k^{-1} e^{-u_{k,a}} D^{(1)} \preceq D^{(k)} \preceq \kappa_k^{-1} e^{-l_{k,a}} D^{(1)}. \tag{QOT16}$$

The same dual inequality passes to every specified fixed flag subquotient, including the fourfold target space.

Proof. Apply (QOT10) with $\alpha = 2q'$ and $\ell = (k-1)/4$ to the complete $p(c' + iy)$ direction. Combine its lower bound with the lower order- k inequality and upper order-one inequality in (QOT13); combine the three opposite inequalities for the upper side. This gives exactly (QOT14)–(QOT15). On a restriction substitute the same coefficient columns. On a quotient every candidate in the same affine fibre obeys both inequalities, so its minimum does as well. This preserves the entire relation and kernel Schur corrections. Inversion reverses positive-form inequalities, complex conjugation preserves them, and the fixed congruence (QOT6) proves (QOT16). Apply the same restriction/minimum argument to its subsequent fixed flags. $\square$

R26.4 The two-half-line transport is literal

To specify the coefficient comparison completely, for $\alpha = 2Q_s$ and $0 \leq j \leq r \leq a$ let

$$A_{j,r} = (-1)^j \binom{r}{j} \Gamma(r + \alpha + 1) \sqrt{\frac{j!}{\Gamma(j + \alpha + 1)}} \vartheta^{-r-(\alpha+1)/2}, \quad P_{r,b} = \binom{b}{r} (c'_k)^{b-r} i^r. \quad (\text{QOT17})$$

Other triangular entries are zero. These are respectively the monomial-to-orthonormal-Laguerre and original affine coefficient maps. The exact reflection is

$$R_{j,r} = (-1)^j 2^{r-j} \binom{\alpha + r}{r-j} \sqrt{\frac{r! \Gamma(j + \alpha + 1)}{j! \Gamma(r + \alpha + 1)}}, \quad R^2 = I, \quad W = \begin{bmatrix} \text{AP} \\ \text{RAP} \end{bmatrix}. \quad (\text{QOT18})$$

The Gram W^*W is the complete reference integral over both half-lines. Reflection is complex-linear: it does not conjugate a period coefficient. For any actual relation or invariant column x , the two row blocks are $(\text{AP})x$ and $(\text{RAP})x$. The source change in Theorem QOT3.1 is on these full integrals before minimization; it is not a separate fit on each half-line. In particular the same U_N , Z_k , and P_Y enter their two quadratic forms. All original factorials and the source factor β_s have been displayed, rather than assigned to an unknown inverse.

R26.5 An evaluated allowance at scale q_k

For $X = K, R$ let $a_N = N - v - q'_k$. Theorem QOT3.1 and the four original signs give

$$\begin{aligned} t_X(l_{k,a_{q-1}} + l_{k,a_q} - u_{k,a_{2q-1}} - u_{k,a_{2q}}) &\leq S_{X,k,k} - S_{X,k,1} \\ &\leq t_X(u_{k,a_{q-1}} + u_{k,a_q} - l_{k,a_{2q-1}} - l_{k,a_{2q}}). \end{aligned} \quad (\text{QOT19})$$

The four copies of $t_X \log \kappa_k$ cancel only in this signed sum.

In fact the scalar is an explicit positive rational number at every original degree:

$$\boxed{\kappa_{4l+1} = 2^{10l} \frac{(2l)!}{(4l)!} (k-7)^{4l}}. \quad (\text{QOT20})$$

This follows from $\Gamma(2l + 1/2) = \sqrt{\pi}(4l)!/[4^{2l}(2l)!]$. The powers of π in the original mass and the central power weight cancel only in this exact scalar identity.

At a low endpoint, $a = \Delta_k - v - 1$ or $\Delta_k - v$ is $O(k)$. Since $\text{arsinh } x \leq x$ for $x \geq 0$, (QOT9) gives $d^\pm = O(\sqrt{k})$, hence $l, u = O_{\text{actual}}(\sqrt{k} + \log q_k)$. At either high endpoint, $a = q_k + \Delta_k - v + O(1)$ and $\alpha_k = 2q'_k$. Uniformly over the $\ell = O(k)$ summands,

$$\frac{a + r + 1/2}{\alpha_k} \longrightarrow \frac{1}{2}.$$

Therefore

$$\frac{l_{k,a}}{k} \longrightarrow -a_*, \quad \frac{u_{k,a}}{k} \longrightarrow a_*, \quad a_* = \operatorname{arsinh}(1/\sqrt{2}) = \frac{1}{2} \log(2 + \sqrt{3}). \quad (\text{QOT21})$$

Since $t_X/k \rightarrow 8$ and $q_k/k^2 \rightarrow 1$, (QOT19) proves

$$\limsup_{k \rightarrow \infty} \frac{|S_{X,k,k} - S_{X,k,1}|}{q_k} \leq 8 \log(2 + \sqrt{3}), \quad X = K, R. \quad (\text{QOT22})$$

Sharper receiving interval (EOR24–26). In the same original source and quotient coordinates, use the finite endpoints (EOR24) wherever their guards hold, intersected as in (EOR21). The resulting limiting interval is

$$-C_+ \leq \liminf \frac{S_{X,k,k} - S_{X,k,1}}{q_k} \leq \limsup \frac{S_{X,k,k} - S_{X,k,1}}{q_k} \leq C_-,$$

$$C_+ = 8 \log(\pi v_*/4), \quad C_- = 8 \log(4/(\pi u_*)).$$

Here $C_+ < 8.881817865597$ and $C_- < 5.773751910974$; neither is an individual allocation coefficient. The full four endpoint scalars still cancel with their original signs.

This is an allowance for the difference of the two source-order allocations, not the asymptotic coefficient of either allocation. In particular their kq_k -normalized limit points coincide.

There is a genuine evaluated low-endpoint order-change term. Stirling's elementary factorial bounds give

$$\log \kappa_k = \frac{k}{2} \log k + \frac{k}{2}(1 + 4 \log 2) + O(\log k). \quad (\text{QOT23})$$

It can also be obtained from the integer/half-integer Gamma formula and integral bounds for $\sum \log n$. At either low endpoint, Theorem QOT3.1 gives an error $O_{\text{actual}}(k^{3/2} + k \log q_k) = o(q_k)$ after taking a rank- t_X determinant. Thus

$$\log \frac{\det H_{X,k,q_k-1+e}^{(k)}}{\det H_{X,k,q_k-1+e}^{(1)}} = 4q_k \log k + (4 + 16 \log 2)q_k + o(q_k),$$

$$e = 0, 1, \quad X = K, R. \quad (\text{QOT24})$$

No next-order term for the complete four-return is assigned: its high determinants still require their actual allocation calculation.

R26.6 The defining primitive source and connecting image

Put $j = k + 4$, $Q = q_j$, $D_e = Q - v - 1 + e$, and $d_e = 8k + 24 + e$. The original PDS relations are

$$r_a = \mathcal{T}_A(\chi_k S^a), \quad 0 \leq a < d_e, \quad \deg r_a = q_k + a - v.$$

Lemma QOT6.1 (The original lower factor divides every primitive column). *Every r_a is divisible by the full ψ_k . Its quotient degree is $\Delta_k - v + a$.*

Proof. For a root $\beta_{k-8,u,w}$ of ψ_k , every summand in $r_a(\beta_{k-8,u,w})$ evaluates χ_k at $\beta_{k-8,u,w} + \beta_{8,b,c} = \beta_{k,u+b,w+c}$, one of its original upper roots. Each summand is zero, without a cancellation or support assumption. The lower roots are distinct, so their complete product divides r_a . The degree follows from the first nonzero conductor moment μ_v . $\square$

In particular the whole relation Gram H_e is a restriction of the original weighted form in Theorem QOT3.1 with degree bound

$$a_s = \Delta_k - v + d_e - 1 = 24k - 25 - v + e. \quad (\text{QOT25})$$

This bound is within $2q_k$ on the stated domain. No leading-monomial approximation or unproved inverse-conductor estimate enters.

We next identify the same primitive coordinates in both source orders. Let R_e^{pol} be the full coefficient matrix of the r_a , let B_s be the monomial-to-original-Gamma coefficient matrix, and let P_e^{Tay} be CIT's fixed Taylor complement. Then

$$\begin{aligned} U_s &= \text{diag}(1/r!)B_s^\top, \quad P_e^{(s)} = U_s^{-1}P_e^{\text{Tay}}, \quad C_e^{(s)} = B_s R_e^{\text{pol}}, \\ \Theta_e &= (C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(r!)P_e^{\text{Tay}}. \end{aligned} \quad (\text{QOT26})$$

The actual Θ_e is independent of s ; every mass, affine phase, and factorial cancels here against its exact inverse, not against an assumed norm. At $e = 0$ it is invertible; at $e = 1$ it has image the same rank- r_e primitive constraint space, where $r_e = d_e - 4e = 8k + 24 - 3e$. PDS/CIT's equality of affine fibres gives

$$H_{\text{src},e}^{(s)} = \Theta_e^* \overline{(H_e^{(s)})^{-1}} \Theta_e.$$

Put $l_s = l_{k,a_s}$, $u_s = u_{k,a_s}$. Theorem QOT3.1 therefore proves

$$-r_e u_s \leq \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} + r_e \log \kappa_k \leq -r_e l_s. \quad (\text{QOT27})$$

This restriction retains all four original primitive constraints at $e = 1$.

The corresponding CIT image columns are also the same coefficient columns in both source pairs. In ordinary Taylor coordinates their equation is

$$\text{stack}_i \text{Mult}(B_i)P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4)F_e + \text{diag}(\tilde{V}_j, 4)E_e^{\text{low}}.$$

All Taylor matrices are those of the same specified germs. The original analytic quotient is injective at this target cutoff modulo the full lower evaluations, so its coefficient F_e is unique. This proves equality of the two F_e and of B_{cof} in their common value coordinates. It is the exact PSR23–24/CIT coefficient argument, not an inference from equal leading determinant orders.

The connecting Gram is the complete fixed quotient

$$\begin{aligned} H_{\text{conn},e}(N) &= F_e^* \mathcal{D}_N F_e - F_e^* \mathcal{D}_N B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_N B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_N F_e, \\ \mathcal{D}_N &= \text{diag}(D_{j,N-v}^{(s_1)}, 4). \end{aligned} \quad (\text{QOT28})$$

At its defining low cutoff $N = Q - 1 + e$, let

$$a_t = \Delta_j - v - 1 + e = 16k + 15 - v + e, \quad l_t = l_{j,a_t}, \quad u_t = u_{j,a_t}. \quad (\text{QOT29})$$

The dual comparison and the identical complete boundary minimum give

$$-r_e u_t \leq \log \frac{\det H_{\text{conn},e}^{(k,j)}}{\det H_{\text{conn},e}^{(1,1)}} + r_e \log \kappa_j \leq -r_e l_t. \quad (\text{QOT30})$$

All four constants l_s, u_s, l_t, u_t are $O_{\text{actual}}(\sqrt{k} + \log q_k)$.

R26.6.1 The exact scalar cancellation and its compatibility return

The Gamma recurrence gives the elementary identity

$$\boxed{\begin{aligned} \frac{\kappa_{k+4}}{\kappa_k} &= \frac{256(k-3)^{k+3}}{k(k+2)(k-7)^{k-1}}, \\ \log \frac{\kappa_{k+4}}{\kappa_k} &= 2 \log k + 8 \log 2 + 4 + O(k^{-1}). \end{aligned}} \quad (\text{QOT31})$$

Indeed $\beta_{k+4}/\beta_k = 16\pi^2/[k(k+2)]$ and the two remaining powers supply $16/\pi^2$ and the displayed integer powers. Every original source mass is included in this cancellation.

Define the difference of the two original source-to-connecting-image logarithmic Jacobians by

$$\mathcal{J}_e^{\text{ord}} = \log \frac{\det H_{\text{conn},e}^{(k,j)}}{\det H_{\text{src},e}^{(k)}} - \log \frac{\det H_{\text{conn},e}^{(1,1)}}{\det H_{\text{src},e}^{(1)}}.$$

Subtracting (QOT27) from (QOT30), in the same original primitive values, proves

$$\boxed{r_e(l_s - u_t) \leq \mathcal{J}_e^{\text{ord}} + r_e \log(\kappa_j/\kappa_k) \leq r_e(u_s - l_t), \quad \mathcal{J}_e^{\text{ord}} = o(q_k).} \quad (\text{QOT32})$$

The error before division by q_k is $O_{\text{actual}}(k^{3/2} + k \log q_k)$. This bound can exceed the displayed $O(k \log k)$ scalar centre; the centre is not asserted to be the next asymptotic term. Separately, the two low order changes in (QOT27) and (QOT30) each equal

$$-4q_k \log k - (4 + 16 \log 2)q_k + o(q_k). \quad (\text{QOT33})$$

Thus the common large term cancels only in their paired Jacobian. Neither absolute low Gram is declared close to its order-one value.

Write $\mathcal{X}_e^{(s_0, s_1)} = \chi_{X_k, D_e} - \chi_{Y_e, D_e}$ for CIT's full proper-source compatibility difference. Its exact identity is

$$\log \det H_{\text{conn},e} - \log \det H_{\text{src},e} = -\mathcal{X}_e + \varepsilon_e,$$

where the original finite endpoints are

$$\varepsilon_e^- = -2r_e \log \max(1, K_-) - r_e \log c_+, \quad \varepsilon_e^+ = 2r_e \log \max(1, K_+) - r_e \log c_-.$$

The constants $K_{\pm}, c_{\pm}$ are exactly the original multiplier and compatibility constants at that cutoff and source pair. They are not redefined by the present comparison; their supplied bounds are $O_{\text{actual}}(k \log q_k)$ after the displayed rank factors. Let $\Delta \mathcal{X}_e = \mathcal{X}_e^{(k,j)} - \mathcal{X}_e^{(1,1)}$. Then

$$\begin{aligned} r_e(l_t - u_s) + \varepsilon_{e,(k,j)}^- - \varepsilon_{e,(1,1)}^+ &\leq \Delta \mathcal{X}_e - r_e \log(\kappa_j/\kappa_k) \\ &\leq r_e(u_t - l_s) + \varepsilon_{e,(k,j)}^+ - \varepsilon_{e,(1,1)}^-. \end{aligned} \quad (\text{QOT34})$$

In particular $\Delta \mathcal{X}_e = o(q_k)$. Only the difference between source-order pairs is small. The absolute compatibility loss is not bounded here. At $e = 0$ the original χ_Y is zero by its proved full-source statement; at $e = 1$ all four constraints and the complete χ_Y remain.

R26.7 Source-order control for the original common long loss

At every target endpoint use the fixed coefficient flags

$$I = \text{im}(B_{\text{cof}} J_0), \quad B = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B},$$

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T = S + J.$$

The original losses in the common value frames are

$$\omega_i = \frac{\det \text{Gram}(W_i/J)}{\det \text{Gram}(I/K_i)}, \quad \omega_c = \frac{\det \text{Gram}(T/W)}{\det \text{Gram}(S/K_c)}. \quad (\text{QOT35})$$

These are ITL's invariant and CIT's connecting losses. Put $r_i = \dim(I/K_i)$, $r_c = \dim(S/K_c)$, and $R_e = r_i + r_c$. The complete kernel of the original multiplication/observation map is retained in K_i, K_c . In particular

$$R_e \leq (8k - 32 - v + t_0) + (8k + 24 - 3e) \leq 16k - 8.$$

All the flags are literally independent of source order, by their polynomial maps and the uniqueness argument preceding (QOT28).

A general immediate consequence of (QOT16) is useful. If two matched fixed subquotients have the same rank r , their determinant ratio obeys

$$\left| \log \omega^{(j)}(N) - \log \omega^{(1)}(N) \right| \leq r w_{j,N-v-q'_j}. \quad (\text{QOT36})$$

Indeed the logarithm of each determinant's order change belongs to $[-r \log \kappa_j - ru, -r \log \kappa_j - rl]$. Subtracting these intervals cancels the scalar before bounding. This proof does not posit a lower singular-value gap for an observation map.

Let $\mathcal{R}$ be the original native-dual return at $N = q_j - 1, q_j, 2q_j - 1, 2q_j$ with signs $(-, -, +, +)$, and define

$$\mathcal{L}_e^{(s_0, s_1)} = \mathcal{R} \log(\omega_c \omega_i).$$

The endpoint-first estimate is

$$\boxed{\begin{aligned} |\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}| &\leq R_e \sum_N w_{j,N-v-q'_j} = O_{\text{actual}}(q_k), \\ \limsup_{k \rightarrow \infty} \frac{|\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}|}{q_k} &\leq 32 \log(2 + \sqrt{3}). \end{aligned}} \quad (\text{QOT37})$$

Sharper common-loss transport (EOR28–29). Replace the displayed width by the intersection width in (EOR28). On the same paired original families this gives

$$\limsup \frac{|\mathcal{L}_e^{(k,k+4)} - \mathcal{L}_e^{(1,1)}|}{q_k} \leq 16 \log(v_*/u_*) < 29.311139553141.$$

This does not evaluate the absolute order-one common loss. Both target gauges, Q_b , Q_o and the original source denominator remain in the receiving identity.

For the last constant, the two low widths are $o(j)$ and each high width divided by j tends to $2a_* = \log(2 + \sqrt{3})$. Now $R_e/k \leq 16 + o(1)$ and $j/k \rightarrow 1$. This proof compares only four endpoint forms; adding a per-row source-order error over q_j rows would lose the result.

The original IRR/CIT rows are unchanged. Explicitly

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n,$$

$$b_{n+1,m} = i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m},$$

$$t_n = \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{h_{n,s_1}}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} (\beta')^m}{\sqrt{h_{n,s_1}}} \right)_{\beta'}.$$

Each source family uses its own rows with the same original coefficients. For every fixed flag Y , the raw frame is $[\text{diag}(V_D, 4), \text{diag}(T_D, 4)Y]$. Its positive update retains the complete old Gram and both lower-evaluation and further-quotient Schur terms. Hence the original long weights $1, 2, \dots, 2, 1$ and the signed difference $\gamma_{H,n} - \gamma_{A,n}$ telescope to exactly the four endpoint forms used above. Neither positive term is discarded, and (QOT37) does not assign a sign or value to their absolute long sum.

R26.8 Residual boundary and original arithmetic receivers

The RQB/CIT identity is retained separately in each source pair:

$$\mathcal{V}_b + \mathcal{V}_{\text{inv}} + \mathcal{V}_o = 4\widehat{F}_j^{(s_1)}, \quad \mathcal{V}_b, \mathcal{V}_{\text{inv}}, \mathcal{V}_o \geq 0, \quad \mathcal{L}_e = P_e^{\text{rest}} - \mathcal{V}_b. \quad (\text{QOT38})$$

Each term is the same original fixed flag return. For a rank- r target flag form Q_X , (QOT16) gives the finite endpoint bound

$$-r \log \kappa_j - r u_{j,a_N} \leq \log \frac{\det Q_X^{(j)}(N)}{\det Q_X^{(1)}(N)} \leq -r \log \kappa_j - r l_{j,a_N}.$$

For its dual-sign long return the resulting lower bound is $r(\sum_{\text{low}} l - \sum_{\text{high}} u)$ and the upper bound is $r(\sum_{\text{low}} u - \sum_{\text{high}} l)$. The source scalar cancels, and the actual ranks of Q_b, H_{inv}, Q_o are retained.

In particular the full source denominator of Q_b is not replaced by a Euclidean norm. It is the original minimum on $V_b/(\text{im } J_0 + \ker \Phi_b)$, and the target is the minimum on W/W_i . At a displaced source cutoff $N_s \leq 2q_{k+4}$, its original polynomial quotient degree is at most $2q_{k+4} - v - q'_k = k^2 + 34k + 1 - v$. For $k \geq 37$ this is at most $2q_k$, so the same full-form theorem also bounds that source minimum. The exact fixed congruence between the CIT and RQB value frames cancels only in matched ratios or the four-return. Thus (QOT38) and CIT's P_e^{rest} retain the source denominator, A_{inv}, Q_o , and both target gauge losses. The present theorem controls their source-order change, not their separate leading values.

For a proved interval $[L_{X,1}, U_{X,1}]$ for an order-one native allocation, add the explicit two endpoints of (QOT19) to obtain an order- k interval; intersect it with all existing intervals. Couple the pair by the exact equality $S_R = \mathcal{T} - S_K$. No independence of the two error terms is assumed. The original arithmetic receiving identities remain

$$\begin{aligned} F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \\ \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \\ \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}}. \end{aligned} \quad (\text{QOT39})$$

The unequal flag endpoints, the full CEP/ α_A transfer, both low metrics, the arithmetic source comparison, and the correlated mixed pairing remain. In particular the matching residual in C_k^{budget} still cancels the same term in Ψ_k . A sharper source-order comparison of a kernel volume alone does not lower the exactly combined untilted action allowance.

R26.9 Exact constants, verification, and remaining calculation

The constants in the two new limiting allowances satisfy

$$\begin{aligned} 10.535663175398 &< 8 \log(2 + \sqrt{3}) < 10.535663175399, \\ 42.142652701594 &< 32 \log(2 + \sqrt{3}) < 42.142652701595, \\ 15.090354888959 &< 4 + 16 \log 2 < 15.090354888960. \end{aligned} \quad (\text{QOT40})$$

They have an elementary rational certificate. Enclose $\sqrt{3}$ between

$$\frac{1732050807568877293527446341505}{10^{30}}$$

and that rational plus 10^{-30} , checking their squares. Use

$$\log(2 + \sqrt{3}) = \frac{2}{\sqrt{3}} \sum_{n \geq 0} \frac{3^{-n}}{2n+1},$$

whose tail after m terms is at most $3^{-m}/[(2m+1)(1-1/3)]$. Fifty terms suffice. For $\log 2$ use $2\sum_{n\geq 0}(1/3)^{2n+1}/(2n+1)$ and its geometric tail. All endpoints in (QOT40) follow by rational arithmetic. These are constants, not certified actual period inputs.

The new results are the full native order transport, the explicit $O(q_k)$ allocation and common-long-loss order allowances, the evaluated low-endpoint q_k coefficient, and the $o(q_k)$ paired primitive-to-connecting-image and compatibility differences. They use the complete original fibre maps. The older source-band and total first-variation theorems are not claimed as new results.

The original order-one common long loss, the signed combination P_e^{rest} with its remaining quotients, and the individual kq_k -scale coefficients of S_K, S_R remain uncomputed. The present result removes an order-transport obstruction: a subsequent order-one leading interval transfers to the original paired orders with a now specified lower-order error. It does not itself furnish an unconditional narrower numerical leading interval for an individual allocation, an RH exclusion, or a certified offcritical zero. No independent mathematical review or Lean certification is asserted. Execution and PDF inspection records, when delivered, have their separate declared scope.

R26.10 Original quantities, inherited inputs and exact comparison domains

Fix the actual hypothetical simple quartet, the full arithmetic unit, one original five-orbit period and its logarithm branch. Throughout

$$\begin{aligned}
k = 4l + 1 &\geq 29, \quad q = (k + 1)^2, \quad q' = (k - 7)^2, \quad \Delta = 16k - 48, \quad c' = k/2 - 4, \\
\beta_{a,b}^{(n)} &= n/2 + (2a - n)\delta + i(2b - n)\gamma, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\
\chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{a,b}^{(n)}), \quad \psi = \chi_{k-8}, \quad E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{a,b}^{(8)} z}, \\
v = \text{ord}_0 E_A, \quad \mu_v &= E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \quad \mathfrak{m} = \Delta - v.
\end{aligned} \tag{EOR1}$$

The a_{ab} are the actual fixed-period coefficients, not free numerical inputs. Their nonreduced conductor is unchanged; the exact v occurs in all cutoffs. The fixed bound $v \leq 80$ suffices. No higher-multiplicity metric statement is claimed.

The two source orders are exactly $s = 1, k$ with

$$\begin{aligned}
dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
h_{n,s} &= M_s n!(s/2)_n, \quad \vartheta = \pi/2.
\end{aligned} \tag{EOR2}$$

Retain the complete original relation matrix $\mathbf{U}_N$, whose columns are $\alpha_A \mathcal{T}_A(\chi_k S^d)/\psi$ for $0 \leq d \leq N - q$, where $\mathcal{T}_A p(S') = \sum a_{ab} p(S' + \beta_{a,b}^{(8)})$. The increasing low-monomial inclusion is E ; Z_k and P_Y are the original kernel frame and residual quotient map. Define the same attained forms

$$\begin{aligned}
z^* H_{Y,N}^{(s)} z &= \min_h \int_{\mathbb{R}} |(Ez + \mathbf{U}_N h)(c' + iy)|^2 |\psi(c' + iy)|^2 dm_s(y), \\
H_{K,N}^{(s)} &= Z_k^* H_{Y,N}^{(s)} Z_k, \quad H_{R,N}^{(s)} = [P_Y (H_{Y,N}^{(s)})^{-1} P_Y^*]^{-1}.
\end{aligned} \tag{EOR3}$$

Both minima are over the full original affine fibres. At the four endpoints

$$\begin{aligned}
(N_0, N_1, N_2, N_3) &= (q - 1, q, 2q - 1, 2q), \quad (\sigma_0, \sigma_1, \sigma_2, \sigma_3) = (+, +, -, -), \\
a_N &= N - v - q', \quad t_0 = \dim(K \cap \mathbb{C}[S]_{<v}), \\
t_K &= 8k - 16 - t_0, \quad t_R = 8k - 32 - v + t_0, \\
S_X(k; s) &= \sum_{i=0}^3 \sigma_i \log \det H_{X,N_i}^{(s)}, \quad \Delta S_X = S_X(k; k) - S_X(k; 1), \quad X = K, R.
\end{aligned} \tag{EOR4}$$

Zero-dimensional determinants are one. The notation ΔS_X is a source-order difference, not the root-strip dimension Δ .

We use GPA's full-form theorem, not a determinant consequence. Its finite domain is

$$R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q, \quad k \geq 29. \tag{EOR5}$$

The new high-degree localization has additional explicit guards in Section R26.12. All hold eventually for each fixed stipulated packet. The old finite QOT bounds remain when a new guard is not satisfied. Arithmetic maps retain their original period domain $|u| \geq R_*$ or the supplied exact period tests. No moving-period uniformity is inferred.

R26.10.1 The complete inherited finite logarithmic constants

For later use reproduce the constants of GPA/QOT. Let $\epsilon = 2^{-32}$, $h_q = 1/q$, $\ell_s = (s-1)/4$, $Q_s = q' + \ell_s$, and

$$\begin{aligned}
\beta_s &= \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \quad C_L = \pi/\Gamma(3/4)^2, \\
K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \quad \mathcal{E}_k = \frac{qR_k^2}{\epsilon^2 q^2 - R_k^2} + \log(1 + K_q), \\
r_{\ell_s} &= \frac{\sum_{a=0}^{\ell_s-1} (2a+1/2)^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\
L_h &= C_L (\sin h_q)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h_q)^{-1/2}, \\
K_{a,Q} &= 2Q + 1 + 2a + 2\sqrt{a(a+2Q)}, \\
b_{k,a,s}^- &= -\mathcal{E}_k + \log L_h - K_{a,Q_s} \log(1 + h_q/\vartheta), \\
b_{k,a,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a,Q_s} \log(1/(1 - h_q/\vartheta)).
\end{aligned} \tag{EOR6}$$

Write $\|p\|_Q^2 = \int_{\mathbb{R}} |p(y)|^2 |y|^{2Q} e^{-\vartheta|y|} dy$. For every original polynomial of degree at most a in the proved degree domain, GPA gives

$$\beta_s e^{b_{k,a,s}^-} \|p(c' + i \cdot)\|_{Q_s}^2 \leq \|\psi p\|_{s,c'}^2 \leq \beta_s e^{b_{k,a,s}^+} \|p(c' + i \cdot)\|_{Q_s}^2. \tag{EOR7}$$

These are bounds on the entire polynomial norm. The original inside-region error, root pairs and source mass are in (EOR6); none is reinterpreted as a small inverse. At the degrees used here $b^\pm = O_{\text{actual}}(\log q)$. This remains the inherited analytic input, not a new proof attribution.

R26.11 The original equilibrium and an explicit exterior gap

Use the proved EIQ equilibrium for $V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. For clarity denote its squared support endpoints by A, B and its density by ϱ . Thus

$$\begin{aligned}
A &= u^2, \quad B = w^2, \quad uK(\kappa) = \alpha\pi/\beta, \quad wE(\kappa) = (\alpha+2)\pi/\beta, \quad \kappa^2 = 1 - u^2/w^2, \\
\mathcal{P}(z) &= \int_A^B \log|z - t| \varrho(t) dt, \quad D(x) = V(x) - 2\mathcal{P}(x) - \lambda \geq 0, \quad D|_{[A,B]} = 0, \\
\mathfrak{h}(x) &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(A+s)(B+s)}} > 0, \\
D'(x) &= \sqrt{(x-A)(x-B)} \mathfrak{h}(x), \quad x \notin [A, B].
\end{aligned} \tag{EOR8}$$

The last square root is negative on $0 < x < A$ and positive on $x > B$. These are EIQ10–19's actual density and resolvent, including the strict left exterior inequality. EIQ3–6 and 29 prove smooth positive endpoint dependence on the positive parameter quadrant.

For each such parameter pair define explicit positive constants

$$\begin{aligned}
g &= B - A, \quad h_* = \frac{\beta}{2\pi(B+1)(B+2)(B+3)}, \quad c_* = \frac{2}{3} h_* \sqrt{g}, \\
\kappa_* &= \min \left\{ Ah_* \sqrt{g}, \frac{\beta\sqrt{B}}{6\pi(3B+2)} \right\}, \quad \eta_* = \min\{A/2, 1\}, \\
L &= \max_{x \in [A,B]} |\beta/(2\sqrt{x}) - \alpha/x|, \quad C_*^2 = 32e^{3L/2}.
\end{aligned} \tag{EOR9}$$

The stars here label local gap constants, not any canonical allowance from the programme. The maximum L is an elementary finite maximum: evaluate the absolute expression at A, B and at $(4\alpha/\beta)^2$ if that point belongs to $[A, B]$.

Lemma QOT11.1 (Both exterior tails, with a global upper derivative bound). *For $0 < \eta \leq \eta_*$,*

$$\begin{aligned} D(x) &\geq c_* \eta^{3/2} \quad (0 < x \leq A - \eta \text{ or } x \geq B + \eta), \\ xD'(x) &\geq \kappa_* \sqrt{\eta} \quad (x \geq B + \eta). \end{aligned} \tag{EOR10}$$

Proof. In the integral in (EOR8), restrict s to $[1, 2]$. For $x \in [A/2, B+1]$, $x \leq B+1$, $x+s \leq B+3$ and $\sqrt{(A+s)(B+s)} \leq B+2$, whereas $\sqrt{s} \geq 1$. Thus $\mathfrak{h}(x) \geq h_*$. On either endpoint segment of length η , $|D'(x)| \geq h_* \sqrt{g} \sqrt{\text{dist}(x, [A, B])}$. Integrating proves the first assertion at the segment endpoints. Monotonicity from (EOR8) proves it on both entire exterior tails.

For $B + \eta \leq x \leq B + 1$ the same bound gives $xD'(x) \geq Ah_* \sqrt{g} \sqrt{\eta}$. For $x \geq B + 1$ restrict the integral instead to $s \in [x, 2x]$ to obtain

$$\mathfrak{h}(x) \geq \frac{\beta}{6\pi\sqrt{x}(B+2x)}, \quad xD'(x) \geq \frac{\beta\sqrt{x}(x-B)}{6\pi(B+2x)} \geq \frac{\beta\sqrt{B}}{6\pi(3B+2)}.$$

The last ratio is minimized at $x = B + 1$ after the factor $\sqrt{x}$ is bounded below by $\sqrt{B}$. Since $\sqrt{\eta} \leq 1$, this proves the second assertion with the displayed common constant. All inequalities concern the same EIQ potential and its complete positive half-line. $\square$

R26.12 Uniform polynomial localization and a growing moment shift

Let m be a positive integer and P any complex polynomial of degree at most m . Set $\|P\|_{m,V}^2 = \int_0^\infty |P(x)|^2 e^{-mV(x)} dx$. The following complete evaluation bound is the argument of OOQ7, restated to specify its constants for the parameter pair used here:

$$|P(x)|^2 e^{-mV(x)} \leq C_*^2 m^4 e^{-mD(x)} \|P\|_{m,V}^2, \quad x \notin [A, B], \quad m \geq \max\{1, 2/g\}. \tag{EOR11}$$

For completeness, let $M = \max_{[A,B]} |P| e^{-mV/2}$, attained at x_0 . One side of x_0 has length $1/m$ inside the support. There $|P(x)| \leq e^{L/2} |P(x_0)|$. A Chebyshev expansion on an interval I proves $\|P'\|_{\infty,I} \leq 4m^3 |I|^{-1} \|P\|_{\infty,I}$: the nonconstant coefficients have modulus at most $2\|P\|_\infty$ and $|T'_j| \leq j^2$, so the sum and affine derivative factor give the bound. It follows that $|P| \geq |P(x_0)|/2$ on a segment of length $(8e^{L/2}m^4)^{-1}$. The weight there is at least $e^{-L}e^{-mV(x_0)}$. Hence $\|P\|_{m,V}^2 \geq M^2/(32e^{3L/2}m^4)$. Finally $\log |P(z)| - m\mathcal{P}(z)$ is subharmonic off the support, where $\mathcal{P}$ is the logarithmic potential in (EOR8). The equilibrium equality bounds its boundary values by $\log M + m\lambda/2$. It extends subharmonically at infinity, since $\deg P \leq m$. The maximum principle proves (EOR11). Zeros give only negative singularities. This uses the logarithmic potential, not a partition-function error estimate.

Define

$$\begin{aligned} \eta_m &= \left(\frac{\log(C_*^2(B+2)) + 10\log(m+1)}{c_* m} \right)^{2/3}, \quad \varepsilon_m = (m+1)^{-6}, \\ m &\geq \max\{1, 2/g\}, \quad \eta_m \leq \eta_*, \quad m\kappa_* \sqrt{\eta_m} \geq \ell + 2. \end{aligned} \tag{EOR12}$$

The last line gives all additional finite guards; $\ell \geq 0$ may be real.

Theorem QOT12.1 (Uniform norm comparison from the actual support). *Under (EOR12), every degree-at-most- m complex polynomial satisfies*

$$(A - \eta_m)^\ell (1 - \varepsilon_m) \|P\|_{m,V}^2 \leq \int_0^\infty x^\ell |P(x)|^2 e^{-mV(x)} dx \leq (B + \eta_m)^\ell (1 + \varepsilon_m) \|P\|_{m,V}^2.$$

(EOR13)

Proof. For any $0 < \eta \leq \eta_*$, (EOR11) and the left gap give

$$\frac{\int_0^{A-\eta} |P|^2 e^{-mV} dx}{\|P\|_{m,V}^2} \leq C_*^2 m^4 A e^{-mc_* \eta^{3/2}}.$$

Consequently the numerator in (EOR13) is at least $(A - \eta)^\ell$ times one minus this mass fraction. For the upper tail integrate the derivative inequality in (EOR10):

$$D(x) \geq c_* \eta^{3/2} + \kappa_* \sqrt{\eta} \log \frac{x}{B + \eta}, \quad x \geq B + \eta.$$

If $m\kappa_* \sqrt{\eta} > \ell + 1$, direct integration gives

$$\int_{B+\eta}^{\infty} x^\ell e^{-mD(x)} dx \leq \frac{(B + \eta)^{\ell+1} e^{-mc_* \eta^{3/2}}}{m\kappa_* \sqrt{\eta} - \ell - 1}.$$

The portion up to $B + \eta$ is bounded by $(B + \eta)^\ell \|P\|_{m,V}^2$; use (EOR11) on the rest. At the chosen η_m ,

$$e^{-mc_* \eta_m^{3/2}} = [C_*^2 (B + 2)(m + 1)^{10}]^{-1}.$$

The lower omitted mass and the upper added fraction, after division by $(B + \eta_m)^\ell$, are both at most $(m + 1)^{-6}$: $A < B + 2$, $B + \eta_m < B + 2$, $m^4 \leq (m + 1)^4$, and the upper denominator is at least one. This proves both full integral inequalities, without assuming that P avoids either tail. $\square$

On any compact positive parameter rectangle the constants in (EOR9) have uniform positive lower bounds and finite upper bounds. This follows from the proved smooth positive endpoints; it is not a condition on an unspecified polynomial. Thus, uniformly for $\ell = O(\sqrt{m})$,

$$\eta_m = O((\log(m + 1)/m)^{2/3}), \quad \ell \eta_m = o(1), \quad \varepsilon_m = O(m^{-6}). \quad (\text{EOR14})$$

All guards hold eventually. In particular the logarithmic endpoints in (EOR13) are $\ell \log A + o(1)$ and $\ell \log B + o(1)$. These are uniform enclosures; equality with either endpoint has not been asserted.

R26.13 Both square-map sheets and the finite improved native interval

Apply the localization to the high polynomial degrees in (EOR4). For a general polynomial $p_c(y) = p(c' + iy)$ of degree at most a , write exactly

$$p_c(y) = P_0(y^2/q^2) + (y/q)P_1(y^2/q^2), \quad m_0 = \lfloor a/2 \rfloor, \quad m_1 = \lfloor (a - 1)/2 \rfloor. \quad (\text{EOR15})$$

This defines both P_ϵ from the same original coefficients, with no reality assumption. The odd cross terms integrate to zero in the even reference weight, so

$$\|p_c\|_{q'}^2 = q^{2q'+1} \sum_{\epsilon=0}^1 \int_0^\infty |P_\epsilon(x)|^2 x^{q'+\epsilon-1/2} e^{-\vartheta q \sqrt{x}} dx. \quad (\text{EOR16})$$

The factor $q^{2q'+1}$ includes both half-lines and their Jacobians. After shifting q' to $q' + \ell$, the right side has the extra factor $q^{2\ell} x^\ell$. Both parity blocks remain.

For each $\epsilon = 0, 1$ set

$$\alpha_\epsilon = \frac{q' + \epsilon - 1/2}{m_\epsilon}, \quad \beta_\epsilon = \frac{\vartheta q}{m_\epsilon}, \quad [A_\epsilon, B_\epsilon] = \text{supp } \mu_{\alpha_\epsilon, \beta_\epsilon}. \quad (\text{EOR17})$$

Use the localization only when $m_\epsilon \geq 1$, both parameter pairs belong to $[3/2, 5/2] \times [\pi/2, 3\pi/2]$, and (EOR12) holds in both blocks. These tests are explicit in the actual integers and the unique

EIQ endpoints. For the two original high cutoffs they hold eventually. Below that domain keep QOT's finite Laguerre bounds unchanged.

Let $\eta_\epsilon, \varepsilon_\epsilon$ be (EOR12)'s values. Define

$$\begin{aligned}
F_{k,a}^- &= \min_{\epsilon=0,1} \{ \ell \log(A_\epsilon - \eta_\epsilon) + \log(1 - \varepsilon_\epsilon) \}, \\
F_{k,a}^+ &= \max_{\epsilon=0,1} \{ \ell \log(B_\epsilon + \eta_\epsilon) + \log(1 + \varepsilon_\epsilon) \}, \\
L_{k,a}^E &= b_{k,a,k}^- - b_{k,a,1}^+ + 2\ell \log \frac{q\vartheta}{2q'} + F_{k,a}^-, \\
U_{k,a}^E &= b_{k,a,k}^+ - b_{k,a,1}^- + 2\ell \log \frac{q\vartheta}{2q'} + F_{k,a}^+, \\
\kappa_k &= \beta_k (4q'/\pi)^{2\ell} = 2^{10\ell} \frac{(2\ell)!}{(4\ell)!} (k-7)^{4\ell}.
\end{aligned} \tag{EOR18}$$

The comparison centre is exactly the earlier rational κ_k ; changing the bounding endpoints does not change its meaning or absorb a mass.

Theorem QOT13.1 (Full high-endpoint source-order comparison). *For every degree-at-most- a original polynomial in the stated joint domain,*

$$\kappa_k e^{L_{k,a}^E} \|\psi p\|_{1,c'}^2 \leq \|\psi p\|_{k,c'}^2 \leq \kappa_k e^{U_{k,a}^E} \|\psi p\|_{1,c'}^2. \tag{EOR19}$$

It holds after every original fixed restriction or attained quotient, in particular all three forms in (EOR3) at the actual cutoff $a = a_N$. On their original complex-linear dual metric it is reversed:

$$\kappa_k^{-1} e^{-U_{k,a}^E} D^{(1)} \preceq D^{(k)} \preceq \kappa_k^{-1} e^{-L_{k,a}^E} D^{(1)}. \tag{EOR20}$$

Proof. Apply (EOR13) separately to the exact two summands in (EOR16). Taking the minimum lower constant and maximum upper constant yields $e^{2\ell \log q + F^-} \|p_c\|_{q'}^2 \leq \|p_c\|_{q'+\ell}^2 \leq e^{2\ell \log q + F^+} \|p_c\|_{q'}^2$. Combine the lower inequality with the lower order- k and upper order-one sides of (EOR7), and combine the opposite three inequalities for the upper side. This is exactly (EOR18)–(EOR19). For each affine fibre every candidate lift satisfies the two inequalities; taking both infima preserves them. For a restriction substitute the full fixed columns. These operations keep all relation, kernel and residual mixed blocks. Positive-form inversion reverses the order, and the source-independent congruence $D = A_{\ell,k}^* \overline{H_Y^{-1}} A_{\ell,k}$ gives (EOR20). It too passes to the same fixed target flag subquotients by the identical minimum argument. No norm is imposed on an observed coordinate space. $\square$

For a guaranteed intersection with the earlier finite bounds reproduce QOT's endpoints. For $\alpha_k = 2q'$ put

$$d_{k,a}^- = 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha_k}, \quad d_{k,a}^+ = 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha_k},$$

$$L_{k,a}^Q = b_{k,a,k}^- - b_{k,a,1}^+ - d_{k,a}^-, \quad U_{k,a}^Q = b_{k,a,k}^+ - b_{k,a,1}^- + d_{k,a}^+.$$

At an endpoint where the new guards hold, use

$$\bar{b}_{k,a} = \max\{L_{k,a}^Q, L_{k,a}^E\}, \quad \bar{b}_{k,a} = \min\{U_{k,a}^Q, U_{k,a}^E\}. \tag{EOR21}$$

Else use the QOT endpoints alone. The new finite enclosure never discards a stronger earlier one. QOT's low bounds remain part of this definition.

R26.14 Evaluated asymmetric constants at scale q_k

Let u_*, v_* denote the original unsquared EIQ endpoints at $(2, \pi)$:

$$u_* K(\kappa) = 2, \quad v_* E(\kappa) = 4, \quad \kappa^2 = 1 - u_*^2/v_*^2. \quad (\text{EOR22})$$

They are not period variables, and no independent Gamma order is introduced. At either high cutoff $a_N = q + \Delta - v + O(1)$, so

$$m_\epsilon = q/2 + O(k), \quad (\alpha_\epsilon, \beta_\epsilon) = (2, \pi) + O(k^{-1}).$$

Smooth endpoint dependence, (EOR14), and $b^\pm = O_{\text{actual}}(\log q)$ now give

$$\begin{aligned} L_{k,a_N}^E &= ka_- + O_{\text{actual}}(\log q), \quad a_- = \frac{1}{2} \log(\pi u_*/4), \\ U_{k,a_N}^E &= ka_+ + O_{\text{actual}}(\log q), \quad a_+ = \frac{1}{2} \log(\pi v_*/4). \end{aligned} \quad (\text{EOR23})$$

For example $F^- = (k/4) \log(u_*^2) + O(1)$ because $\ell \eta_{m_\epsilon} = o(1)$ and the endpoint displacement costs $O(\ell/k) = O(1)$. The remaining centre is $2\ell \log(\pi q/(4q')) = (k/2) \log(\pi/4) + O(1)$. The upper proof is identical. These are asymptotics of proved bounding constants, not a claim about the unknown eigenvalues of the actual high compressed forms.

Taking rank- t_X determinants and the four original signs in (EOR19) yields the finite interval

$$\begin{aligned} t_X(\underline{b}_{k,a_{N_0}} + \underline{b}_{k,a_{N_1}} - \bar{b}_{k,a_{N_2}} - \bar{b}_{k,a_{N_3}}) &\leq \Delta S_X \\ &\leq t_X(\bar{b}_{k,a_{N_0}} + \bar{b}_{k,a_{N_1}} - \underline{b}_{k,a_{N_2}} - \underline{b}_{k,a_{N_3}}). \end{aligned} \quad (\text{EOR24})$$

The four identical rank factors multiplying $\log \kappa_k$ cancel exactly here, not before the endpoint forms are defined. At a low endpoint $a_N = O(k)$ the retained QOT endpoints are $O_{\text{actual}}(\sqrt{k} + \log q)$; this follows directly from $\text{arsinh } x \leq x$ in its sums. Since $t_X = 8k + O(1)$,

Theorem QOT14.1 (Narrower source-order allocation interval). *For either original allocation $X = K, R$ at a fixed stipulated packet and period,*

$$\boxed{-C_+ \leq \liminf_{k \equiv 1 \pmod{4}} \frac{\Delta S_X}{q_k} \leq \limsup_{k \equiv 1 \pmod{4}} \frac{\Delta S_X}{q_k} \leq C_-,} \quad (\text{EOR25})$$

where

$$C_+ = 8 \log(\pi v_*/4), \quad C_- = 8 \log(4/(\pi u_*)). \quad (\text{EOR26})$$

More explicitly the finite interval (EOR24) implies $-C_+ q_k - O_{\text{actual}}(k^{3/2} + k \log q_k) \leq \Delta S_X \leq C_- q_k + O_{\text{actual}}(k^{3/2} + k \log q_k)$. No unique normalized limit is asserted.

Proof. Use (EOR23) at the two high endpoints and the preceding low bound. The lower centre is $-2t_X ka_+$, the upper centre is $-2t_X ka_-$. Replacing $t_X k$ by $8q_k$ costs only $O(k)$ because t_0, v are bounded. Substitution of $a_\pm$ gives the stated constants and finite order of the remaining allowance. The explicit finite endpoints, rather than an unspecified big- O constant, are available in (EOR24). $\square$

R26.15 The original common long loss and its fixed-frame receiver

Use $j = k + 4$ and the original paired source orders $(1, 1)$ and (k, j) . In target coordinates the source-independent metric is $\mathcal{D}_{j,N} = \text{diag}(D_{j,N-v}, 4)$. Here “source-independent” refers to the fixed coefficient space and flags; the metric is $\mathcal{D}_{j,N}^{(s_1)} = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$ and depends on the original source order s_1 . The complete original flags are

$$\begin{aligned} I &= \text{im}(B_{\text{cof}} J_0), \quad B_0 = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B}, \\ K_i &= I \cap J, \quad W_i = I + J, \quad W = B_0 + J, \quad K_c = S \cap W, \quad T = S + J, \\ \omega_i &= \frac{\det \text{Gram}(W_i/J)}{\det \text{Gram}(I/K_i)}, \quad \omega_c = \frac{\det \text{Gram}(T/W)}{\det \text{Gram}(S/K_c)}, \quad R_e = \dim(I/K_i) + \dim(S/K_c) \leq 16k - 8. \end{aligned} \quad (\text{EOR27})$$

All Grams are their attained native quotient forms in matched value frames. The fixedness of F_e under the source presentation follows from the same Taylor cochain equation and uniqueness modulo all lower evaluation columns, as in QOT's full primitive map. No corresponding denominator or multiplier kernel is changed by the present scalar comparison.

Let $\mathcal{R}$ have signs $(-, -, +, +)$ at $N = q_j - 1, q_j, 2q_j - 1, 2q_j$ and write $\mathcal{L}_e = \mathcal{R} \log(\omega_c \omega_i)$. With $w_{j,a} = \bar{b}_{j,a} - \underline{b}_{j,a}$, subtraction of the matched quotient determinant intervals in (EOR20) proves

$$\boxed{|\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}| \leq R_e \sum_N w_{j,N-v-q'_j}.} \quad (\text{EOR28})$$

The common factor κ_j^{-r} cancels in each rank- r quotient contraction at each endpoint. Thus a small observation inverse is not needed. This estimate is applied at four full endpoint forms, not summed as an independent error over q_j rows. By (EOR23), each high width equals $\frac{j}{2} \log(v_*/u_*) + O_{\text{actual}}(\log q_j)$, and both low widths are $O_{\text{actual}}(\sqrt{j} + \log q_j)$. Hence

$$\boxed{\limsup \frac{|\mathcal{L}_e^{(k,k+4)} - \mathcal{L}_e^{(1,1)}|}{q_k} \leq C_{\text{com}} := 16 \log(v_*/u_*) = 2(C_+ + C_-).} \quad (\text{EOR29})$$

The error about this bound is $O_{\text{actual}}(k^{3/2} + k \log q_k)$, with the complete finite version (EOR28). This does not replace the earlier one-budget estimate for the absolute order-one common long loss by an evaluated value.

R26.15.1 The actual rows, half-lines and all remaining terms

The unchanged source-coordinate pullback is $(P_{c',a})_{r,n} = \binom{n}{r} (c')^{n-r} i^r$. The full power-exponential reference rows remain

$$\begin{aligned} A_{b,r} &= (-1)^b \binom{r}{b} \Gamma(r + 2Q_s + 1) \sqrt{\frac{b!}{\Gamma(b + 2Q_s + 1)}} \vartheta^{-r-(2Q_s+1)/2}, \\ R_{b,r} &= (-1)^b 2^{r-b} \binom{2Q_s + r}{r-b} \sqrt{\frac{r! \Gamma(b + 2Q_s + 1)}{b! \Gamma(r + 2Q_s + 1)}}, \\ W &= \begin{bmatrix} \text{AP}_{c',a} \\ \text{RAP}_{c',a} \end{bmatrix}. \end{aligned} \quad (\text{EOR30})$$

Zero triangular entries are understood. Formula (EOR16) is merely a second exact coefficient presentation of this same complete two-half-line norm. For every original column x , both row blocks of Wx remain; only their integrated even/odd cross terms cancel by symmetry of the reference density. Original period coefficients are not conjugated by reflection. The new theorem acts on these full quadratic forms before either Schur minimum.

The native rows are still calculated from the original symbol and Gamma recurrence:

$$\begin{aligned} f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} (\beta_{u,w}^{(j)})^n, \\ b_{n+1,r} &= i^{-1} (b_{n,r-1} - c'_j b_{n,r}) - n(n-1 + s_1/2) b_{n-1,r}, \\ t_n &= \left(\frac{\sum_r b_{n,r} f_{a,r}}{\sqrt{M_{s_1} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_r b_{n,r} (\beta')^r}{\sqrt{M_{s_1} n! (s_1/2)_n}} \right)_{\beta'}. \end{aligned} \quad (\text{EOR31})$$

In a new step the preceding complete lower Gram is V^*V and the corrected row is $\rho_n = t_n - v_n(V^*V)^{-1}V^*T$. Its update denominator is $1 + v_n(V^*V)^{-1}v_n^*$. The subsequent boundary or invariant minimum also remains. None of these rows or denominators is approximated by its rank in (EOR28).

For an order-one interval $[L_{X,1}, U_{X,1}]$, add the two finite endpoints in (EOR24), intersect with all earlier bounds, and couple by the exact $S_R = \mathcal{T} - S_K$. In the common loss add and subtract the explicit allowance in (EOR28). Retain separately

$$\begin{aligned} \mathcal{L}_e &= P_e^{\text{rest}} - \mathcal{V}_b, & \mathcal{V}_b + \mathcal{V}_{\text{inv}} + \mathcal{V}_o &= 4\hat{F}_j, \\ F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, & \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \\ \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, & J_k^{\text{action}} &= \Psi_k + C_k^{\text{budget}}. \end{aligned} \quad (\text{EOR32})$$

The original source denominator, A_{inv} , Q_b , Q_o , both target gauges, both low metrics, unequal flag errors and the complete source-transfer correction are unchanged. The matching residual cancellation in the full action budget is not altered by narrowing a source-order interval. The previously proved low primitive $o(q_k)$ paired source/image comparison is retained without being reattributed as a new theorem.

R26.16 Certified constants and the precise remaining task

Let $r = u_*/v_* \in (0, 1)$ and define the normalized elliptic integrals

$$\bar{K} = \frac{2}{\pi} K(\sqrt{1-r^2}), \quad \bar{E} = \frac{2}{\pi} E(\sqrt{1-r^2}).$$

The endpoint equation is $r\bar{K}/\bar{E} = 1/2$, and

$$C_+ = 8 \log(2/\bar{E}), \quad C_- = 8 \log \bar{K}, \quad C_{\text{com}} = -16 \log r. \quad (\text{EOR33})$$

Thus no numerical enclosure of π is required for these three constants. The accompanying integer/rational calculation certifies

$$\begin{aligned} 0.16010167095771835588456745 &< r < 0.16010167095771835588456746, \\ 8.881817865596 &< C_+ < 8.881817865597, \\ 5.773751910973 &< C_- < 5.773751910974, \\ 29.311139553140 &< C_{\text{com}} < 29.311139553141. \end{aligned} \quad (\text{EOR34})$$

Here is the rigorous basis of that certificate. For $m = 1 - r^2$ and $c_n = \binom{2n}{n}^2 / 16^n$,

$$\bar{K} = \sum_{n \geq 0} c_n m^n, \quad \bar{E} = 1 - \sum_{n \geq 1} \frac{c_n m^n}{2n-1}.$$

These follow by binomial expansion in the defining elliptic integrals and the integral of $\sin^{2n} \theta$. If $t_n = c_n m^n$, then $t_{n+1}/t_n = m((2n+1)/(2n+2))^2 \leq m$. After degree N , the omitted $\bar{K}$ tail is at most $t_N m / (1-m)$ and the omitted $1 - \bar{E}$ tail at most $t_N m / ((1-m)(2N+1))$. At both rational endpoints in (EOR34), use $N = 4096$, enclosing each recurrence operation by integer floor and ceiling at a fixed denominator 10^{80} . The resulting intervals certify opposite signs of $2r\bar{K} - \bar{E}$. EIQ's strict monotonicity then encloses the unique root. Monotonicity in m encloses both normalized elliptic integrals between these endpoint evaluations. Finally, for $x > 1$ and $z = (x-1)/(x+1)$, use

$$\log x = 2 \sum_{n=0}^{M-1} \frac{z^{2n+1}}{2n+1} + R_M, \quad 0 < R_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

Integer interval arithmetic with $M = 400$ proves the displayed decimal brackets. The full script records its actual rational endpoints and rounding directions. These are universal-constant certificates, not actual-period evaluations.

The new allocation interval $[-C_+, C_-]$ is strictly contained in the earlier $[-8 \log(2 + \sqrt{3}), 8 \log(2 + \sqrt{3})]$ interval. The new common source-order allowance is strictly below

$32 \log(2 + \sqrt{3})$. These are genuine narrower q_k -scale enclosures for the same differences. They do not assert optimality of the enclosures or existence of either normalized source-order limit.

What remains is the absolute order-one common long loss and its allocation between the actual kernel and residual. Neither an individual kq_k -scale coefficient nor a narrower unconditional interval for that coefficient is assigned here. The current result makes a subsequent order-one estimate more precisely transferable; it does not substitute for that estimate. The source comparison is uniform in all polynomial directions on its stated finite domain; the later arithmetic and geometric maps retain their fixed-period and simple-quartet hypotheses. No RH conclusion, actual offcritical-zero certificate, independent referee acceptance, or Lean execution is claimed.

References

- [1] Split-Zero research programme. (2026). *Current kernel and multiplicity proofs* (supplied 498-page edition). Complete GPA1–23, LVM1–19, DNE, SAR, CIT/ITL receiving providers. Original source filename: `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS(6).tex`.
- [2] Split-Zero research programme. (2026). *Full primitive connecting proofs*. Complete PDS1–24, CIT1–26 and PDI1–16 providers in the supplied 13-page source; the same-coordinate argument is also recorded in PSR23–24 of the later 40-page supplement.
- [3] Split-Zero research programme. (2026). *Primitive-band continuation* (549-page edition and 40-page supplement). Repository path `workbenches/splitzero-tandem/continuations/20260915-primitive-band-continuation`; consulted at commit `1efd53337561d8f67cf0ac1d119acdaa222644c3`. Its evaluated totals and later RQB/PSR identities retain their own proof provenance.

R27 The complete original action relative to its arithmetic total

This is a local reconstruction of the complete delivered response following the 16 September 2026, 15:02:50 prompt in `ChatGPT-Residual Lower Bound Continuation-20260916-2150.md`, lines 7043–7730. The response reported that its build tools timed out: no original standalone archive, compiled PDF, successful fresh Lean run or actual-period numerical certificate is inferred from it. This section supplies the additive proof in full. Its input source is the unchanged `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex`; the original AMT, MCE and OMR formulas below are rederived at the points where used. The fixed arithmetic density-envelope constants are precisely AMT4–6’s proved constants. Their defining bounds and domains are displayed, rather than replaced by newly selected measures or a new positivity premise.

R27.1 Domain, unchanged arithmetic objects and exact minimum metrics

Use the original stipulated full zero-quartet domain: fixed $0 < \delta < 1/2$, $\gamma > 2$, and complete common order $m \geq 1$ at the four distinct centres. The statement below concerns these original objects; it does not assert that such an off-line zero quartet exists. Write

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, \quad \mathcal{R} = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad h(s) = \prod_{\zeta \in \mathcal{R}} (s - \zeta)^m, \quad g = 2\xi, \quad v_h = g/h, \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_{h,b} = w_h^{*b}, \quad \mu_h = \int_{\mathbb{R}} w_h(y) dy, \\ k &= 4\ell + 1 \geq 9, \quad c = k/2, \quad e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \beta_{ab})^{e_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \end{aligned} \tag{CAI1}$$

The arithmetic source is $L^2(\mathbb{R}, m_{h,k}(y) dy)$ with its mass μ_h^k , evaluated in the literal coordinate $S = c + iy$. Its inner product is conjugate-linear in the first argument. For each $N \geq 0$ let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and let $p_n(S) = i^n Q_n((S - c)/i)$ be the monic orthogonal polynomials; Q_n is monic in the real variable y . Their full squared norms are $\omega_{n,k}^{\text{ar}} > 0$. The substitution has inverse $P(S) = f((S - c)/i)$ and is never a change of the original source mass. In formulas below ω_n abbreviates this same arithmetic norm.

The exact remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto for $N \geq q - 1$, with kernel $\chi \mathcal{P}_{N-q}$, where $\mathcal{P}_{-1} = 0$. Put $b_n = [p_n]_{\chi}$ in the frame $1, S, \dots, S^{q-1}$. Then

$$R_N = \sum_{n=0}^N b_n b_n^* / \omega_n, \quad G_N = R_N^{-1}, \quad V_N = \det G_N. \tag{CAI2}$$

The first q columns form a unit-triangular monic remainder matrix, so $R_N > 0$. In any source coefficient frame with Gram H_N , the actual minimum section is $H_N^{-1} J_N^* G_N$. Its composition with J_N is the identity. A vector in $\ker J_N$ pairs to zero with this section; therefore the squared norm of another lift is its minimum squared norm plus the squared norm of that relation. This proves the interpretation of G_N without changing the admitted relation subspace.

The original period observation is the AMT3 map $\Lambda : E \twoheadrightarrow B$, with actual kernel inclusion $I : K \hookrightarrow E$. Explicitly, if $E_h = \mathbb{C}[s]/(h)$, $U = M_{j_h v_h}$, $\iota[P] = [P(s_1 + \dots + s_k)]$, Π is the original period matrix and P_k its original tensor invariant projector, then $j\Lambda = P_k \Pi^{\otimes k} U^{\otimes k} \iota$, where $j : B \hookrightarrow (\mathbb{C}^{4m})^{\otimes k}$ is the image inclusion. No period branch or kernel frame is changed. The action calculation below is independent of that choice because it uses the complete quotient E ; the exact relation to the original observation is the following splitting. In the fixed kernel and image frames set

$$H_{K,N} = I^* G_N I, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad S_N = G_N^{-1} \Lambda^* Q_{B,N}. \tag{CAI3}$$

Direct multiplication proves $\Lambda S_N = 1$ and $I^* G_N S_N = 0$. For any fixed section S_* of Λ , the frame $[I, S_*]$ is invertible and $S_N - S_*$ has image in K . Thus changing $[I, S_*]$ to $[I, S_N]$ has determinant one. Orthogonality proves the exact identity

$$\det H_{K,N} \det Q_{B,N} = |\det[I, S_*]|^2 V_N. \quad (\text{CAI4})$$

Empty factors have determinant one, so this also covers zero-dimensional kernel or image. In particular no kernel invariance under A is needed.

For any one of the original sources $a = \text{ar}, 1, k$, define its full two-window return and the kernel and boundary returns by

$$\begin{aligned} \mathcal{B}_k^a &= \log \frac{V_{q-1}^a V_q^a}{V_{2q-1}^a V_{2q}^a}, \\ F_K^a &= \log \frac{\det H_{K,q-1}^a \det H_{K,q}^a}{\det H_{K,2q-1}^a \det H_{K,2q}^a}, \quad F_B^a = \log \frac{\det Q_{B,q-1}^a \det Q_{B,q}^a}{\det Q_{B,2q-1}^a \det Q_{B,2q}^a}, \\ F_K^a + F_B^a &= \mathcal{B}_k^a, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma, \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^{(k)}. \end{aligned} \quad (\text{CAI5})$$

Here $a = 1$ denotes the original fixed Gamma source σ . The equality follows by multiplying CAI4 at the two lower degrees and dividing by its two upper-degree instances. Its four identical frame constants cancel exactly, rather than being set equal to one.

R27.2 The actual rank-two control, reflection and exterior benchmark

The quotient v_h is entire at the selected centres because their complete zero orders have been divided out. The exact equalities $\xi(1-s) = \xi(s)$ and $h(1-s) = h(s)$ show that w_h and every $m_{h,b}$ are even. Monic orthogonality consequently gives $Q_n(-y) = (-1)^n Q_n(y)$. Self-adjoint multiplication by y , orthogonality to degrees below $n-1$, and pairing with Q_{n-1} prove the real monic recurrence. In the original S coordinate it is

$$Sp_n = p_{n+1} + cp_n - (\omega_n/\omega_{n-1})p_{n-1}, \quad (\text{CAI6})$$

with the lower term absent at $n = 0$. Inserting its quotient columns into CAI2 cancels every interior adjacent pair and gives

$$\begin{aligned} AR_N + R_N A^* - kR_N &= (b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N, \\ D_N^{\text{ctrl}} &= A^* G_N + G_N A - kG_N, \quad G_N^{-1/2} D_N^{\text{ctrl}} G_N^{-1/2} = (yx^* + xy^*)/\omega_N, \\ x &= G_N^{1/2} b_N, \quad y = G_N^{1/2} b_{N+1}. \end{aligned} \quad (\text{CAI7})$$

The symbol D_N^{ctrl} is distinct from the scalar source constant D_N below. Reflection $P(S) \mapsto P(k-S)$ preserves χ because it permutes (a, b) with $(k-a, k-b)$ and q is even. It therefore induces an involution Σ on E , with $\Sigma A \Sigma = kI - A$. Changing variables $y \mapsto -y$ is an isometry of each entire remainder fibre. Uniqueness of the minimum proves $\Sigma^* G_N \Sigma = G_N$. Parity gives $\Sigma b_n = (-1)^n b_n$, and hence $b_N^* G_N b_{N+1} = -b_N^* G_N b_{N+1} = 0$. Thus the only possibly nonzero eigenvalues of the relative Hermitian control in CAI7 are $\epsilon_N, -\epsilon_N$, where

$$\epsilon_N^2 = \frac{\|x\|^2 \|y\|^2}{\omega_N^2}, \quad \phi_N = 0. \quad (\text{CAI8})$$

For comparison with the original tilted formulas, before specialization the exact expression is $(\|x\|^2 \|y\|^2 - |x^* y|^2)/\omega_N^2$ and $x^* y = i\omega_N \phi_N$. At the present untilted source only the *total* pairing vanishes. In the orthogonal kernel-boundary split write $x = x_K \oplus x_B$, $y = y_K \oplus y_B$, and $z_K = x_K^* y_K$, $z_B = x_B^* y_B$. Then $z_B = -z_K$, not generally zero. In particular the mixed exterior summand remains

$$\|x_K \otimes y_B - y_K \otimes x_B\|^2 = \|x_K\|^2 \|y_B\|^2 + \|x_B\|^2 \|y_K\|^2 + 2|z_K|^2. \quad (\text{CAI9})$$

Here is the original source-to-class map and its lower allowance. The complete Taylor unit $v_h = j_h v_h$ is invertible in E_h , and

$$\eta : E \longrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad A_\Sigma = \sum_i M_{s_i}, \quad \eta A = A_\Sigma \eta. \quad (\text{CAI10})$$

In each local tensor factor the nilpotent has order m . The sum of the k nilpotents has exact order e_k : the power $e_k - 1 = k(m - 1)$ has nonzero top coefficient $[k(m - 1)]! / [(m - 1)!]^k$, and every term of its next power has one exponent at least m . Every centre β_{ab} occurs, and distinct pairs have distinct centres. The annihilator of A_Σ on its cyclic unit is therefore exactly χ ; this proves injectivity of CAI10, including all jets. Reflection fixes v_h and intertwines this inclusion with the original tensor reflection $s_i \mapsto 1 - s_i$.

Let E_+ be the direct sum of all complete primary blocks with $a > k/2$. Its dimension is $t_+ = q/2$. A primary idempotent is explicitly $\chi_\beta \sum_{j=0}^{e_k-1} (1/\chi_\beta)^{(j)}(\beta)(S - \beta)^j / j! \pmod{\chi}$, where $\chi_\beta = \chi / (S - \beta)^{e_k}$. Its product with the powers $(S - \beta)^d$, $0 \leq d < e_k$, gives a complete primary basis. The top wedge ξ_+ of these bases is nonzero, and remains nonzero under $\bigwedge^{t_+} \eta$. Its additive action eigenvalue is $\text{Tr}(A|_{E_+})$; nilpotents contribute zero to that trace. The relative Hermitian additive action on any exterior degree $1 \leq t \leq q - 1$ has largest eigenvalue ϵ_N , since its eigenvalues are sums of distinct members of $\{\epsilon_N, -\epsilon_N, 0, \dots, 0\}$. Its expectation on ξ_+ is $2 \text{Re Tr}(A|_{E_+}) - kt_+$. Summing the positive odd numbers $2a - k$ gives $(k + 1)^2/4$, so

$$0 < L_{h,k} := 2\delta e_k(k + 1) \frac{(k + 1)^2}{4} = \frac{\delta q(k + 1)}{2} \leq \epsilon_N \quad (N \geq q - 1). \quad (\text{CAI11})$$

This is a calculation on the same stipulated arithmetic class, not a claim of its existence and not an assumed desired upper estimate.

R27.3 Exact cancellation before any estimates

Set $d_n = V_n/V_{n-1}$ for $n \geq q$, and $b_n^{\text{rec}} = \omega_n/\omega_{n-1}$; neither is the centre displacement δ or the quotient column b_n . From the rank-one covariance update and the determinant lemma,

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - d_N \quad (N \geq q), \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = d_{N+1}^{-1} - 1.$$

The first equality follows by removing the last covariance column; the second follows by adding the next one. At $N = q - 1$ the first q columns are the full quotient basis and $b_{q-1}^* G_{q-1} b_{q-1} = \omega_{q-1}$, without using any inverse at $q - 2$. Thus CAI8 gives the original MCE23/OMR18 formulas

$$\epsilon_{q-1}^2 = b_q^{\text{rec}}(d_q^{-1} - 1), \quad \epsilon_N^2 = b_{N+1}^{\text{rec}}(1 - d_N)(d_{N+1}^{-1} - 1) \quad (N \geq q). \quad (\text{CAI12})$$

Positive covariances give $0 < d_n \leq 1$, and CAI11–12 force $d_n < 1$ at every degree appearing here. All following logarithms therefore exist. Put $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ for $q \leq N \leq 2q - 2$; their sum is $2q$. Retain exactly

$$\begin{aligned} \mathcal{W}_k^{\text{ar}} &= \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q}, \\ \mathcal{P}_- &= -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \\ \mathcal{P}_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n), \\ J_k^{\text{action}} &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N, \quad \Psi_k = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}, \quad C_k^{\text{budget}} = F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \end{aligned} \quad (\text{CAI13})$$

Multiply CAI12 over $N = q - 1, \dots, 2q - 2$ and over $N = q, \dots, 2q - 1$ separately. The two exact products are

$$\begin{aligned} \prod_{N=q-1}^{2q-2} \epsilon_N^2 &= \frac{\omega_{2q-1} V_{q-1}}{\omega_{q-1} V_{2q-1}} (1 - d_{2q-1}) \prod_{n=q}^{2q-2} (1 - d_n)^2, \\ \prod_{N=q}^{2q-1} \epsilon_N^2 &= \frac{\omega_{2q} V_q}{\omega_q V_{2q}} (1 - d_q)(1 - d_{2q}) \prod_{n=q+1}^{2q-1} (1 - d_n)^2. \end{aligned} \quad (\text{CAI14})$$

Indeed $d_{N+1}^{-1} - 1 = (1 - d_{N+1})/d_{N+1}$; norm ratios telescope, reciprocal d factors give the displayed volume ratios, and the endpoint and interior contraction exponents are precisely those written. Taking logarithms and using CAI5 proves

$$\boxed{J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+}. \quad (\text{CAI15})$$

Equivalently $\Psi_k = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$ and $C_k^{\text{budget}} = \mathcal{B}_k^{(k)} - F_K^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+$. The cancellation uses the exact joint identities, not independent estimates for the kernel, residual, low metrics or transfer errors.

R27.4 All-degree multiplication bounds on the actual arithmetic blocks

The AMT1 domain is every convolution order $b \geq 3$, not just the later tensor subsequence. Its constants, without alteration, are

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy > 0, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy \in (0, \infty), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_b &= \frac{c_h \vartheta_h^{b-3} e^{-\alpha(b-3)} (b-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_b = \frac{C_h^{\text{tilt}}}{c_\Gamma} b^{p+1} M_\alpha^{b-1}, \quad D_N = [1 + 4(N+M)^2]^M. \end{aligned} \quad (\text{CAI16})$$

Here c_h is the original TW7 lower-envelope constant and C_h^{tilt} the original BT12 compact/tail maximum, as recorded in AMT4. Their already proved defining inequalities, for every $b \geq 3$, are

$$\begin{aligned} m_{h,b}(y) &\geq c_h \vartheta_h^{b-3} e^{-\alpha(|y|+b-3)} (b-2+|y|)^{-B_h}, \\ m_{h,b}(y) &\leq b C_h^{\text{tilt}} M_\alpha^{b-1} e^{-\alpha|y|} (1+|y|/b)^{-p}, \\ c_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2} &\leq \frac{d\sigma}{dy} \leq C_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2}, \quad d\sigma = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi). \end{aligned} \quad (\text{CAI17})$$

The factors $b-2+|y| \leq (b-2)(1+|y|)$, $(1+|y|)^{B_h} \leq 2^{B_h/2} (1+y^2)^M$, and $1+|y|/b \geq (1+|y|)/b$, with $p > 1/2$, give $a_b(1+y^2)^{-M} d\sigma/dy \leq m_{h,b} \leq A_b d\sigma/dy$. The orthonormal Gamma recurrence is $ye_j = \sqrt{(j+1)(j+1/2)}e_{j+1} + \sqrt{j(j-1/2)}e_{j-1}$. It gives $\|yP\|_\sigma \leq 2(N+1)\|P\|_\sigma$ on degree N . Iterate at the successive degrees and expand $(1+y^2)^M$ to get $\int |P|^2 (1+y^2)^M d\sigma \leq D_N \|P\|_\sigma^2$. Cauchy–Schwarz applied with the two weights $(1+y^2)^{\pm M/2}$ therefore proves, for every complex polynomial of degree at most N ,

$$\frac{a_b}{D_N} \|P\|_\sigma^2 \leq \|P\|_{m_{h,b}}^2 \leq A_b \|P\|_\sigma^2. \quad (\text{CAI18})$$

These are precisely the full-polynomial AMT6 inequalities; in particular they apply at $b = 3, 4, 5$ with no new source-order restriction.

The fixed Gamma monic polynomials have full norms $\gamma_n = \sqrt{2\pi} n!(1/2)_n$. Their recurrence gives the generating series $F(y, z) = \sum_{n \geq 0} p_n^\sigma(y) z^n / n!$:

$$(1+z^2)\partial_z F = (y-z/2)F, \quad F(y, 0) = 1, \quad F = (1+z^2)^{-1/4} e^{y \arctan z}. \quad (\text{CAI19})$$

One may verify the differential equation coefficient by coefficient, and integration at zero uniquely gives the displayed analytic function on $|z| < 1$. On $|z| = 1/4$, its prefactor has modulus at most $(16/15)^{1/4}$ and $|\arctan z| \leq \sum_{j \geq 0} |z|^{2j+1}/(2j+1) = \operatorname{arctanh}(1/4)$. Consequently Cauchy's coefficient bound and $n!/(1/2)_n = 4^n/\binom{2n}{n} \leq 2n+1$ imply

$$|P(y)|^2 \leq C_\sigma(N+1)(2N+1)16^N e^{\nu_0|y|} \|P\|_\sigma^2, \quad C_\sigma = \frac{4}{\sqrt{15}\sqrt{2\pi}}, \quad \nu_0 = \log(5/3). \quad (\text{CAI20})$$

For the binomial inequality the central coefficient is the maximum of the $2n+1$ positive coefficients summing to 4^n . For the polynomial inequality expand in the complete degree- N orthonormal basis and apply Cauchy–Schwarz to the coefficient vector; then bound each of the $N+1$ summands by $(2N+1)16^N$. This covers complex coefficients.

Put $d_0 = \alpha - \nu_0 > 0$. Positivity and convolution give $\int e^{\alpha y} m_{h,b}(y) dy = M_\alpha^b$ by Tonelli, and evenness gives $\int e^{\alpha|y|} m_{h,b}(y) dy \leq 2M_\alpha^b$. For $x > R \geq 0$, $x^2 e^{-d_0 x} \leq (16/d_0^2) e^{-d_0 R/2}$; the maximum of $x^2 e^{-d_0 x/2}$ is $16/(e^2 d_0^2) \leq 16/d_0^2$. Moreover $N+M \leq M(N+1)$ since $M \geq 1$, so $D_N \leq (1+4M^2)^M (N+1)^{2M}$. Combining these facts with CAI18–20, and $2N+1 \leq 2(N+1)$, proves for $b = 3, 4, 5$

$$\begin{aligned} \int_{|y|>R} y^2 |P(y)|^2 m_{h,b}(y) dy &\leq H_h(N+1)^{2M+2} 16^N e^{-d_0 R/2} \|P\|_{m_{h,b}}^2, \\ H_h &= \frac{64C_\sigma(1+4M^2)^M}{d_0^2} \max_{b=3,4,5} \frac{M_\alpha^b}{a_b}. \end{aligned} \quad (\text{CAI21})$$

All constants are fixed- h data, including the unscaled source masses. Choose

$$T_h = \frac{2}{d_0} \{\log 16 + 2M + 3 + \log^+ H_h\}, \quad C_{\text{mult}}(h) = \sqrt{T_h^2 + 1}. \quad (\text{CAI22})$$

Here $\log^+ x = \max(0, \log x)$. At $R = T_h(N+1)$ the logarithm of the factor in CAI21 is at most $\log H_h + (2M+2)N + N \log 16 - (\log 16 + 2M + 3 + \log^+ H_h)(N+1) \leq 0$, using $\log(N+1) \leq N$. The inner interval has $y^2 \leq T_h^2(N+1)^2$. Adding its integral and the tail, and using $(N+1)^2 \geq 1$, proves

$$\|yP\|_{m_{h,b}} \leq C_{\text{mult}}(h)(N+1)\|P\|_{m_{h,b}}, \quad b = 3, 4, 5. \quad (\text{CAI23})$$

R27.5 Total-degree tensor compression, including the output face

For every integer $k \geq 3$, write $r_k = \lfloor k/3 \rfloor$ and partition k into r_k blocks of sizes 3, 4, 5. If $k = 3r_k$ take all threes; if $k = 3r_k + 1$ or $3r_k + 2$, replace one three by four or five. This also covers $k = 3, 4, 5$. Convolution proves the exact isometry

$$L^2(m_{h,k}) \longrightarrow \bigotimes_{i=1}^{r_k} L^2(m_{h,b_i}), \quad f(y) \longmapsto f(y_1 + \cdots + y_{r_k}), \quad \sum_i (b_i/2 + i y_i) = k/2 + i \sum_i y_i. \quad (\text{CAI24})$$

The norm identity follows first by Tonelli for $|f|^2$ and then by the definition of convolution. Its image is the sum-dependent subspace, not the whole product space; total masses multiply to μ_h^k . In each block let $\varphi_n^{(b)}$ be the real orthonormal polynomial with positive leading coefficient. Its exact recurrence is

$$y\varphi_n^{(b)} = a_{n+1}^{(b)}\varphi_{n+1}^{(b)} + a_n^{(b)}\varphi_{n-1}^{(b)}, \quad a_0^{(b)} = 0, \quad 0 < a_n^{(b)} \leq C_{\text{mult}}(h)n \quad (n \geq 1).$$

There is no diagonal term by evenness. The bound follows by taking the coefficient of φ_n in $y\varphi_{n-1}$ and applying CAI23.

Use all product basis vectors of total degree at most $N+1$, and let M_{N+1} be the compression of multiplication by $\sum_i y_i$ to their span. It is a real symmetric matrix. At a vertex $\nu = (\nu_1, \dots, \nu_{r_k})$ of total degree at most N , its full absolute row sum is at most $C_{\text{mult}} \sum_i (\nu_i + (\nu_i + 1)) \leq C_{\text{mult}}(2N + r_k)$.

At total degree $N + 1$ only downward entries remain in the compression, so that row sum is at most $C_{\text{mult}}(N + 1) \leq C_{\text{mult}}(2N + r_k)$. Symmetry gives the same bound for column sums and hence the operator norm: directly, for any vector z , pair the equal off-diagonal entries and apply $2|z_\nu z_\mu| \leq |z_\nu|^2 + |z_\mu|^2$ to its quadratic form. Its absolute Rayleigh quotient is bounded by the maximal row sum.

The image of a univariate polynomial of degree at most N under CAI24 has total degree at most N . Multiplication by $\sum_i y_i$ has total degree at most $N + 1$, so no output is lost by the compression. Therefore

$$\|yP\|_{m_{h,k}} \leq C_{\text{mult}}(h)(2N + r_k)\|P\|_{m_{h,k}}, \quad \boxed{\frac{\omega_{N+1,k}^{\text{ar}}}{\omega_{N,k}^{\text{ar}}} \leq C_{\text{mult}}(h)^2(2N + r_k)^2} \quad (k \geq 3, N \geq 0). \quad (\text{CAI25})$$

For the last step yQ_N is an actual degree- $N + 1$ monic competitor; its orthogonal monic minimum has no greater norm. The S -monic phases have modulus one, so precisely the original norms occur. This proof uses the full total-degree space, not independent degree- N cutoffs in each factor, and yields $N + k$ rather than kN .

R27.6 Both original contraction penalties on their complete window

Return now to the stipulated packet in CAI1. By CAI11–12 and $1 - d_N \leq 1$,

$$d_n \leq \frac{b_n^{\text{rec}}}{b_n^{\text{rec}} + L_{h,k}^2} \quad (q \leq n \leq 2q).$$

For $n = q$ this is the endpoint formula; for $n \geq q + 1$ it follows from the formula at $N = n - 1$. CAI25 bounds all these recurrence ratios by

$$B_{h,k} = C_{\text{mult}}(h)^2(4q + r_k - 2)^2, \quad z_{h,k} = \frac{B_{h,k}}{L_{h,k}^2} = \frac{4C_{\text{mult}}(h)^2(4 + (r_k - 2)/q)^2}{\delta^2(k + 1)^2}. \quad (\text{CAI26})$$

The map $b \mapsto b/(b + L^2)$ is increasing for $b > 0$, whence

$$0 < d_n \leq \frac{z_{h,k}}{1 + z_{h,k}}, \quad \frac{1}{1 + z_{h,k}} \leq 1 - d_n < 1, \quad q \leq n \leq 2q. \quad (\text{CAI27})$$

It is the complementary factors $1 - d_n$, not d_n themselves, that approach one. In CAI13 the coefficient sum in $\mathcal{P}_-$ is $1 + 2(q - 1) = 2q - 1$, and that in $\mathcal{P}_+$ is $2 + 2(q - 1) = 2q$. Thus the exact finite bound is

$$\boxed{0 \leq \mathcal{P}_- + \mathcal{P}_+ \leq \Pi_{h,k} := (4q - 1) \log(1 + z_{h,k}).} \quad (\text{CAI28})$$

Since $q = e_k(k + 1)^2$ and $r_k \leq k/3$, $z_{h,k} = O_h(k^{-2})$ and $\log(1 + z) \leq z$ show that $\Pi_{h,k} = O_h(q/k^2) = O_h(e_k)$. This is $O_h(1)$ for $m = 1$, and $O_h(k) = o(q)$ for any fixed $m > 1$. No early- k smallness assumption was used in CAI27–28. At an interior degree $N \geq q$, writing $\tau_N = -\log d_{N+1}$ gives the exact pointwise comparison $\epsilon_N^2/(b_{N+1}^{\text{rec}} e^{\tau_N}) = (1 - d_N)(1 - d_{N+1})$, whose logarithm lies in $[-2 \log(1 + z_{h,k}), 0]$ on the window. At $N = q - 1$ the one remaining factor is $1 - d_q$.

R27.7 The four-norm factorial window and its finite remainder

Minimize CAI18 over all monic polynomials, retaining the literal degree- n coefficient one. It gives $(a_k/D_n)\gamma_n \leq \omega_{n,k}^{\text{ar}} \leq A_k\gamma_n$. Define

$$\begin{aligned} \Lambda_{k,n} &= \log(A_k D_n / a_k), \quad E_{\text{lo}} = \Lambda_{k,q-1} + \Lambda_{k,q}, \quad E_{\text{hi}} = \Lambda_{k,2q-1} + \Lambda_{k,2q}, \\ \mathcal{W}_q^\sigma &= \log \frac{\gamma_{2q-1} \gamma_{2q}}{\gamma_{q-1} \gamma_q}, \quad -E_{\text{hi}} \leq \mathcal{W}_k^{\text{ar}} - \mathcal{W}_q^\sigma \leq E_{\text{lo}}. \end{aligned} \quad (\text{CAI29})$$

For the lower bound put both numerator norms at their lower bounds and both denominator norms at their upper bounds; for the upper bound reverse those choices. The common a_k, A_k terms are thereby counted exactly twice. This comparison occurs *after* the window has telescoped, not once at each of its q steps. More explicitly, the original AMT22 constants give

$$\mathfrak{s}_h = \alpha + \log(M_\alpha/\vartheta_h), \quad \mathfrak{c}_h = -\log M_\alpha + 3 \log \vartheta_h - 3\alpha + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \quad (\text{CAI30})$$

$$\log(A_k/a_k) = k\mathfrak{s}_h + (p+1) \log k + B_h \log(k-2) + \mathfrak{c}_h.$$

Since h is fixed and $\log D_n = O_h(1 + \log(n+1))$, the two errors in CAI29 are $O_h(k + \log q)$.

The identity $(1/2)_n = (2n)!/(4^n n!)$ gives $\gamma_n = \sqrt{2\pi}(2n)!/4^n$. Cancelling these four literal norms, including their masses, gives

$$\mathcal{W}_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}. \quad (\text{CAI31})$$

Here the endpoint factor results from $(4q-2)! = (4q)!/[(4q)(4q-1)]$ and $(2q-2)! = (2q)!/[(2q)(2q-1)]$; it is not discarded.

For a fully finite enclosure put $C(q) = 4q \log q + (8 \log 2 - 4)q$ and $\eta_q = 1/(16q) + 1/(4q-2)$. For $f(x) = \log x$, integration by parts twice on each unit interval $[j, j+1]$ gives

$$\frac{f(j) + f(j+1)}{2} - \int_j^{j+1} f(x) dx = \frac{1}{2} \int_j^{j+1} (x-j)(j+1-x) f''(x) dx.$$

Since $f''(x) = -x^{-2}$ and $(x-j)(j+1-x)/2 \leq 1/8$, summing from $j = 2q$ to $4q-1$ bounds the composite trapezoidal error by $-\frac{1}{8} \int_{2q}^{4q} x^{-2} dx = -1/(32q) \leq \xi_q \leq 0$. The factorial sum is the trapezoidal sum plus $[f(4q) - f(2q)]/2$. Substitution in CAI31 and exact integration give

$$\mathcal{W}_q^\sigma = C(q) - \log 2 + 2\xi_q + \log(1 - 1/(4q-1)).$$

Finally $-x/(1-x) \leq \log(1-x) \leq 0$ for $0 < x < 1$, obtained by integrating $-1/(1-x)$, proves for every integer $q \geq 1$

$$C(q) - \log 2 - \eta_q \leq \mathcal{W}_q^\sigma \leq C(q) - \log 2. \quad (\text{CAI32})$$

R27.8 Complete finite interval, action mean and exact receiving scope

Combining CAI15, CAI28 and CAI29 proves the entirely specified interval

$$\boxed{\mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma - E_{\text{hi}} - \Pi_{h,k} \leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma + E_{\text{lo}}.} \quad (\text{CAI33})$$

In particular CAI32 gives

$$\begin{aligned} \mathcal{B}_k^{\text{ar}} + C(q) - \log 2 - \eta_q - E_{\text{hi}} - \Pi_{h,k} &\leq J_k^{\text{action}}, \\ J_k^{\text{action}} &\leq \mathcal{B}_k^{\text{ar}} + C(q) - \log 2 + E_{\text{lo}}. \end{aligned} \quad (\text{CAI34})$$

For fixed h and its complete multiplicity, $e_k = O_h(k)$ and $q = e_k(k+1)^2$. All noncentral terms in CAI34 are $O_h(k + \log q) = o(q)$. Therefore

$$\boxed{J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + 4q \log q + (8 \log 2 - 4)q + O_h(k + \log q).} \quad (\text{CAI35})$$

The finer finite centre in CAI34 retains $-\log 2$ and every error term. The coefficient $8 \log 2 - 4$ is a coefficient of $J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}}$, not a coefficient of J_k^{action} after replacing $\mathcal{B}_k^{\text{ar}}$ by only its leading asymptotic. The exact signed correction $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma$ remains unevaluated at scale q . Nothing in CAI35 assigns either individual kernel/residual allocation, or the absolute order-one long loss.

The original weighted arithmetic geometric mean is exactly

$$\mathcal{G}_k = \prod_{N=q-1}^{2q-1} \epsilon_N^{w_N/(2q)} = \exp(J_k^{\text{action}}/(4q)).$$

Its finite bounds are the exponentials of CAI34 divided by $4q$. Consequently, since $(k + \log q)/q = O_h(1/k)$ for fixed m ,

$$\boxed{\frac{\mathcal{G}_k}{q \exp(\mathcal{B}_k^{\text{ar}}/(4q))} = \frac{4}{\exp(1)}(1 + O_h(k^{-1})) \longrightarrow \frac{4}{\exp(1)}}. \quad (\text{CAI36})$$

This uses $e^x = 1 + O(x)$ as $x \rightarrow 0$, which follows directly by integrating the exponential derivative on a bounded interval. It also proves $(J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}})/(kq) \rightarrow 0$ since $\log q = O_h(\log k)$.

The exact exterior receiver is unchanged:

$$\begin{aligned} 4q \log L_{h,k} &\leq J_k^{\text{action}}, \\ L_{h,k} &\leq \min_{q-1 \leq N \leq 2q-1} \epsilon_N \leq \mathcal{G}_k, \\ \frac{\mathcal{G}_k}{L_{h,k}} &= \frac{8}{\exp(1)\delta(k+1)} \exp(\mathcal{B}_k^{\text{ar}}/(4q))(1 + O_h(k^{-1})). \end{aligned} \quad (\text{CAI37})$$

The first two inequalities follow term by term from CAI11 and the weights' sum $2q$; a minimum is bounded by, not identified with, the weighted mean. In the original simple-quartet baseline domain alone, AMT40 and the complete BSL/HBL source proof give $\mathcal{B}_k^{\text{ar}}/q^2 \rightarrow C_B > 769/17010 > 0$. Substitution into CAI35 proves $J_k^{\text{action}}/q^2 \rightarrow C_B$, and CAI37 then tends to infinity exponentially in q . This last consequence uses that already proved baseline only on its original $m = 1$ domain; the new interval, penalty bound and relative expansion retain every complete multiplicity.

For the existing original simple-quartet outer receiver, retain its literal finite identity $\Psi_k = \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \epsilon_{k,k}$, with the original $-e_{k,k}^- \leq \epsilon_{k,k} \leq e_{k,k}^+$. Substituting CAI15 gives

$$\Phi_{k,k} \leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log L_{h,k}. \quad (\text{CAI38})$$

This is exactly OMR26's arithmetic backward receiver. Every signed low metric and error remains where its original identity puts it. CAI33–34 can be substituted for the *complete* action; they do not justify independently improving one summand while holding its correlated budget fixed. No failure of the broader programme or exclusion at other admitted degrees follows from the size of this particular mean.

Continuation to the derived growing-window endpoints (RGI8–10). The complete action in this section has the following exact continuation to the original allowance rows $q-1, \dots, q+r-1$, for each integer $r \geq 1$. Use weight one at the two ends and two in the interior; equivalently write these weights as w_j for the row $q+j-1$. Their sum is $2r$. The original adjacent relations are $d_{q+j} = V_{q+j}/V_{q+j-1} = 1/T_j^{\text{ar}}$ and $\beta_{q+j} = \omega_{q+j}/\omega_{q+j-1}$. Taking logarithms of the exact reflection identity RGC24 gives

$$\log \epsilon_{q+j-1}^2 = \log \beta_{q+j} - \log d_{q+j} - a_{q+j} - \mathbf{1}_{j \geq 1} a_{q+j-1}, \quad a_d = -\log(1 - d_d). \quad (\text{RGI8})$$

On the positive-allowance domain supplied by RGC25–26 all these logarithms are defined. For any positive sequence z_n , direct cancellation of each interior exponent gives

$$\sum_{j=0}^r w_j (\log z_{q+j} - \log z_{q+j-1}) = \log z_{q+r-1} + \log z_{q+r} - \log z_{q-1} - \log z_q.$$

For $r = 1$ the occurrences of z_q cancel on the right as well. Apply this identity separately to the original ω and V in RGI8. Thus

$$\mathcal{J}_{q,r} = \mathcal{B}_{q,r} + \mathcal{W}_{q,r} - \mathcal{P}_{q,r}, \quad \mathcal{B}_{q,r} = \log \frac{V_{q-1}V_q}{V_{q+r-1}V_{q+r}}, \quad \mathcal{W}_{q,r} = \log \frac{\omega_{q+r-1}\omega_{q+r}}{\omega_{q-1}\omega_q}. \quad (\text{RGI9})$$

Here $\mathcal{J}_{q,r} = \sum_j w_j \log \epsilon_{q+j-1}^2$ and the exact penalty is RGC29. Its coefficients are 2, 1 for $r = 1$ and 3, 4, ..., 4, 3, 1 for $r \geq 2$, with total $4r - 1$. Inserting the proved intersection bound RGI7 and the four original AMT endpoint errors RGC32 yields

$$\begin{aligned} \mathcal{B}_{q,r} + \mathcal{W}_{q,r}^\sigma - \Lambda_{k,q+r-1} - \Lambda_{k,q+r} - P_{q,r}^{\text{cap}} &\leq \mathcal{J}_{q,r}, \\ \mathcal{J}_{q,r} &\leq \mathcal{B}_{q,r} + \mathcal{W}_{q,r}^\sigma + \Lambda_{k,q-1} + \Lambda_{k,q}. \end{aligned} \quad (\text{RGI10})$$

This follows by adding the lower, respectively upper, endpoint inequalities to RGI9 and using $0 \leq \mathcal{P}_{q,r} \leq P_{q,r}^{\text{cap}}$. Only the four endpoint errors are charged. At $r = q$ the endpoints and weights are exactly those of the earlier complete action. Its correlated allocation cancellation remains the same identity. The intrinsic signed mixed profile and the CPQ returns retain their original receiving maps; no value for them is assigned by these canonical allowance bounds.

The moving-degree result in the complete action (GRB9–10). The new growing-ratio proof also gives an exact receiver at shifted rows. Keep q, χ, E , all three source forms, and every kernel/boundary map fixed as in this section. Choose an integer $s \geq 1$, put $a = q + s - 1$, $b = a + q$, and give rows $N = a + i$ the weights $w_0 = w_q = 1$ and $w_i = 2$ otherwise. Their relation index is exactly $r = s + i$ in RGI8. Every row now has the preceding contraction, because $a \geq q$. Expanding the weighted differences for any positive sequence z gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this separately to the original V and ω in the exact adjacent identity proves

$$\begin{aligned} J_{k,s}^{\text{action}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}} + \log \frac{\omega_b \omega_{b+1}}{\omega_a \omega_{a+1}} - \sum_{j=0}^{q+1} c_j [-\log(1 - T_{s-1+j}^{-1})], \\ J_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}, \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1). \end{aligned} \quad (\text{GRB9})$$

Indeed $c_j = w_j + w_{j-1}$ with absent weights zero, so their total is $2 \sum_i w_i = 4q$. These four endpoints and every coefficient are derived from the original maps. The full finite scalar enclosures, four AMT endpoint errors and their proof are GRA27–32; GRB1–4 intersects them with all eligible earlier bounds. For $t = s/q \rightarrow \infty$ and $t \log t = o(q/k)$ the complete proof GRA19–35 consequently gives

$$\log \min_{a \leq N \leq b} \epsilon_N \sim \frac{q}{2t \log t}, \quad \log \mathcal{P}_{k,s} \sim -\frac{q}{t \log t}, \quad J_{k,s}^{\text{action}} \sim \frac{2q^2}{t \log t}. \quad (\text{GRB10})$$

The proof controls the full adjacent index interval $s - 1, \dots, s + q$; its error is uniform on that interval, as shown there before summing.

The exact transfer to the mixed spaces retains $I : K \hookrightarrow E$ and $\Lambda : E \twoheadrightarrow B$. For each unchanged positive metric G_N^v its minimum section is $S_N^v = (G_N^v)^{-1} \Lambda^* (\Lambda (G_N^v)^{-1} \Lambda^*)^{-1}$. It satisfies $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$ by direct multiplication. Relative to any fixed section S_* , the frame change from $[I, S_*]$ to $[I, S_N^v]$ has identity diagonal because $\Lambda(S_N^v - S_*) = 0$. Its determinant is one. Block orthogonality therefore gives

$$\det(I^* G_N^v I) \det(\Lambda (G_N^v)^{-1} \Lambda^*)^{-1} = |\det[I, S_*]|^2 \det G_N^v.$$

The four-endpoint return cancels that same frame constant and proves $F_K^v + F_B^v = \mathcal{B}^v$. Thus the complete allocation and its matching budget in GRA36 still sum to GRB9: the occurrence of $F_K^{(k)}$ in one has its opposite occurrence in the other. Every intrinsic mixed profile and complementary CPQ return retains its original map and coefficient; the new scalar estimates do not assign those separate coefficients.

R28 Original objects and the finite comparison data

This calculation composes the full-form comparisons proved in NSO1–31, QOT and EOR with the actual coefficient maps of NSR, CPQ and SFC. The analytic localization proof remains the complete QOT/EOR provider. Here every subsequent map, determinant operation, endpoint sign and interval intersection is proved explicitly. The comparison constants below are their finite constants, not their asymptotic limits.

Fix the original actual hypothetical simple quartet, its conductor, one original five-orbit period and its branch. Keep $0 < \delta < 1/2$, $\gamma > 2$, the complete conductor coefficients a_{uv} , $v = \text{ord}_0 E_A \leq 80$, $\mu_v = E_A^{(v)}(0) \neq 0$, and $\alpha_A = v!/\mu_v$. Keep the original rank domain, including $|u| \geq R_*$. For an actual degree $a = 4l + 1 \geq 9$, set

$$\begin{aligned} q_a &= (a+1)^2, & q'_a &= (a-7)^2, & c'_a &= a/2 - 4, & m_a &= 16a - 48 - v, \\ t_a &= 8a - 16 - t_{0,a}, & j_{0,a} &= 8a - 32 - v + t_{0,a}, & m_a &= t_a + j_{0,a}, \\ D &= N - v, & A &= D - q'_a, & R_a &= a\sqrt{\delta^2 + \gamma^2}. \end{aligned} \quad (\text{EFP1})$$

The symbol Δ on a single-degree form means its order- a value minus its order-one value. On a paired object it means the value in $p = 1 : (k, k+4)$ minus its value in $p = 0 : (1, 1)$, at the same cutoffs and in the same coefficient or quotient-value frames. The capital A in this section is the polynomial degree bound, not a coefficient matrix or an assigned allocation. Each cutoff and each source order retains its original centre. The full measures are

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1, a. \quad (\text{EFP2})$$

The native metric is $D_{a,D}^{(s)} = A_{\ell,a}^* \overline{(H_{Y,a,N}^{(s)})^{-1}} A_{\ell,a}$. The invertible $A_{\ell,a}$ is the same actual coefficient map in both orders. H_Y is the complete quotient of the original polynomial norm by its relation columns. In particular its polynomial degree is $A = N - v - q'_a$. Complex conjugation in this duality formula is retained.

Write $\ell = (a-1)/4$ and $\tau_r = 2r + 1/2$. NSO's finite full-monomial comparison, transferred by its exact ordinary-transpose Taylor equation and its whole lower-evaluation minimum, gives

$$\begin{aligned} \rho_a &= \frac{(2\pi)^{2\ell}}{(1/2)_{2\ell}}, & \mathfrak{a}_a &= \rho_a \prod_{r=0}^{\ell-1} \tau_r^2, \\ \mathfrak{b}_{a,D} &= \rho_a \prod_{r=0}^{\ell-1} \{4(D+r+1)^2 + \tau_r^2\}, \\ \mathfrak{b}_{a,D}^{-1} D_{a,D}^{(1)} &\preceq D_{a,D}^{(a)} \preceq \mathfrak{a}_a^{-1} D_{a,D}^{(1)}. \end{aligned} \quad (\text{EFP3})$$

This applies throughout the original NSO domain, irrespective of the additional QOT/EOR tests below.

Here are the finite QOT inputs with the actual degree a and polynomial bound A written separately. They are used when

$$a \geq 29, \quad R_a \leq 2^{-33} q_a, \quad 0 \leq A \leq 2q_a. \quad (\text{EFP4})$$

The applications below also verify the provider's stronger $A + \ell \leq 2q_a$ range. In the standard degree- a window, $A_{\min} = 16a - 49 - v \geq 335$ for $a \geq 29$ and $v \leq 80$, while $A_{\max} = 2q_a - v - q'_a$ and $2q_a - (A_{\max} + \ell) = q'_a + v - \ell \geq 477 + v$. The last expression increases for $a \geq 29$. Put $q = q_a$,

$q' = q'_a$, $\epsilon_0 = 2^{-32}$, $\vartheta = \pi/2$, $h = 1/q$, and

$$\begin{aligned}
\ell_s &= (s-1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \\
C_L &= \pi/\Gamma(3/4)^2, \quad K_q = \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\
\mathcal{E}_a &= \frac{qR_a^2}{\epsilon_0^2 q^2 - R_a^2} + \log(1 + K_q), \\
r_{\ell_s} &= \frac{\sum_{r=0}^{\ell_s-1} (2r+1/2)^2}{\epsilon_0^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\
L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\
K_{A,Q} &= 2Q + 1 + 2A + 2\sqrt{A(A+2Q)}, \\
b_{a,A,s}^- &= -\mathcal{E}_a + \log L_h - K_{A,Q_s} \log(1 + h/\vartheta), \\
b_{a,A,s}^+ &= \mathcal{E}_a + r_{\ell_s} + \log U_h + K_{A,Q_s} \log \frac{1}{1 - h/\vartheta}, \\
d_{a,A}^- &= 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(A+r+1/2)/(2q')}, \quad d_{a,A}^+ = 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(A+r+1/2)/(2q')}, \\
L_{a,A}^Q &= b_{a,A,a}^- - b_{a,A,1}^+ - d_{a,A}^-, \quad U_{a,A}^Q = b_{a,A,a}^+ - b_{a,A,1}^- + d_{a,A}^+, \\
\kappa_a &= \beta_a (4q'/\pi)^{2\ell} = 2^{10\ell} \frac{(2\ell)!}{(4\ell)!} (a-7)^{4\ell}.
\end{aligned} \tag{EFP5}$$

In particular κ_a is a displayed source mass and scale, retained without a change of measure or a change of coefficient vectors.

We specify EOR's finite tests and endpoints as well. Put $m_0 = \lfloor A/2 \rfloor$, $m_1 = \lfloor (A-1)/2 \rfloor$. In parity block $\varepsilon = 0, 1$ require $m_\varepsilon \geq 1$ and

$$\alpha_\varepsilon = (q' + \varepsilon - 1/2)/m_\varepsilon, \quad \beta_\varepsilon = \vartheta q/m_\varepsilon, \quad (\alpha_\varepsilon, \beta_\varepsilon) \in [3/2, 5/2] \times [\pi/2, 3\pi/2]. \tag{EFP6}$$

For this pair let $[A_\varepsilon, B_\varepsilon]$ be the EIQ equilibrium support for $\beta_\varepsilon \sqrt{x} - \alpha_\varepsilon \log x$. It is specified by the unique positive endpoints $A_\varepsilon = u_\varepsilon^2$, $B_\varepsilon = w_\varepsilon^2$ satisfying

$$u_\varepsilon K(\zeta_\varepsilon) = \alpha_\varepsilon \pi / \beta_\varepsilon, \quad w_\varepsilon E(\zeta_\varepsilon) = (\alpha_\varepsilon + 2) \pi / \beta_\varepsilon, \quad \zeta_\varepsilon^2 = 1 - u_\varepsilon^2 / w_\varepsilon^2.$$

$K(z) = \int_0^{\pi/2} (1 - z^2 \sin^2 t)^{-1/2} dt$ and $E(z) = \int_0^{\pi/2} (1 - z^2 \sin^2 t)^{1/2} dt$. For each block separately, suppress its index and set

$$\begin{aligned}
g &= B - A, \quad h_* = \frac{\beta}{2\pi(B+1)(B+2)(B+3)}, \quad c_* = \frac{2}{3} h_* \sqrt{g}, \\
\kappa_* &= \min\{A h_* \sqrt{g}, \beta \sqrt{B}/[6\pi(3B+2)]\}, \quad \eta_* = \min\{A/2, 1\}, \\
L_* &= \max_{A \leq x \leq B} |\beta/(2\sqrt{x}) - \alpha/x|, \quad C_*^2 = 32e^{3L_*/2}, \\
\eta_m &= \left\{ \frac{\log(C_*^2(B+2)) + 10 \log(m+1)}{c_* m} \right\}^{2/3}, \quad \varepsilon_m = (m+1)^{-6}, \\
m &\geq \max\{1, 2/g\}, \quad \eta_m \leq \eta_*, \quad m \kappa_* \sqrt{\eta_m} \geq \ell + 2.
\end{aligned} \tag{EFP7}$$

The maximum defining L_* is evaluated at A, B and, when inside this interval, $(4\alpha/\beta)^2$; differentiating the expression proves this finite candidate list. Retain both parity blocks, their actual $\eta_\varepsilon, \varepsilon_\varepsilon$,

and define

$$\begin{aligned}
F_{a,A}^- &= \min_{\varepsilon=0,1} [\ell \log(A_\varepsilon - \eta_\varepsilon) + \log(1 - \varepsilon_\varepsilon)], \\
F_{a,A}^+ &= \max_{\varepsilon=0,1} [\ell \log(B_\varepsilon + \eta_\varepsilon) + \log(1 + \varepsilon_\varepsilon)], \\
L_{a,A}^E &= b_{a,A,a}^- - b_{a,A,1}^+ + 2\ell \log(q\vartheta/(2q')) + F_{a,A}^-, \\
U_{a,A}^E &= b_{a,A,a}^+ - b_{a,A,1}^- + 2\ell \log(q\vartheta/(2q')) + F_{a,A}^+.
\end{aligned} \tag{EFP8}$$

At a cutoff satisfying (EFP4), use $\underline{b}_{a,A} = L_{a,A}^Q$, $\bar{b}_{a,A} = U_{a,A}^Q$, intersecting with the EOR endpoints only when (EFP6)–(EFP7) hold in both blocks:

$$\underline{b}_{a,A} = \max\{L_{a,A}^Q, L_{a,A}^E\}, \quad \bar{b}_{a,A} = \min\{U_{a,A}^Q, U_{a,A}^E\}. \tag{EFP9}$$

The proved QOT/EOR full-form inequalities give, on the same original dual coefficient metric,

$$\kappa_a^{-1} e^{-\bar{b}_{a,A}} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq \kappa_a^{-1} e^{-\underline{b}_{a,A}} D_{a,D}^{(1)}. \tag{EFP10}$$

These are the two accepted analytic inputs used below. Their full proofs are QOT's full original source-order theorem and EOR's full high-endpoint theorem, including its exact two-sheet square map. No analytically unproved bound is used in the propagation.

R29 Exact intersection before any restriction or quotient

Define $\lambda_{a,D}^- = -\log \mathfrak{b}_{a,D}$ and $\lambda_{a,D}^+ = -\log \mathfrak{a}_a$ on the whole native domain. On every native endpoint define $w_{a,D} = \lambda_{a,D}^+ - \lambda_{a,D}^-$, using the final endpoints prescribed here. At an endpoint where (EFP4) holds replace those endpoints by

$$\begin{aligned}
\lambda_{a,D}^- &= \max\{-\log \mathfrak{b}_{a,D}, -\log \kappa_a - \bar{b}_{a,D-q'_a}\}, \\
\lambda_{a,D}^+ &= \min\{-\log \mathfrak{a}_a, -\log \kappa_a - \underline{b}_{a,D-q'_a}\}, \\
w_{a,D} &= \lambda_{a,D}^+ - \lambda_{a,D}^-.
\end{aligned} \tag{EFP11}$$

Then $e^{\lambda_{a,D}^-} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq e^{\lambda_{a,D}^+} D_{a,D}^{(1)}$. For any vector, both original lower inequalities hold on its same quadratic form, so their larger scalar is still a lower bound; the upper proof uses the smaller scalar. Applying to a nonzero vector shows $\lambda^- \leq \lambda^+$, hence $w \geq 0$. In the guarded domain one can record the unchanged source scalar by writing

$$\begin{aligned}
\underline{b}_{a,D}^* &= \max\{\log \mathfrak{a}_a - \log \kappa_a, \underline{b}_{a,D-q'_a}\}, \quad \bar{b}_{a,D}^* = \min\{\log \mathfrak{b}_{a,D} - \log \kappa_a, \bar{b}_{a,D-q'_a}\}, \\
\lambda^- &= -\log \kappa_a - \bar{b}^*, \quad \lambda^+ = -\log \kappa_a - \underline{b}^*.
\end{aligned} \tag{EFP12}$$

These equalities change the written scalar endpoints only. They do not multiply any original matrix by κ_a .

Here is the full typed restriction and quotient argument. Let Y be a fixed coefficient subspace, $K \subset Y$ a fixed subspace, X_K an independent frame of K and X_Q fixed lifts of a basis of Y/K . For each of the two metrics $G^{(s)}$ define

$$\begin{aligned}
Q^{(s)} &= X_Q^* G^{(s)} X_Q - X_Q^* G^{(s)} X_K (X_K^* G^{(s)} X_K)^{-1} X_K^* G^{(s)} X_Q, \\
x^* Q^{(s)} x &= \min_z (X_Q x + X_K z)^* G^{(s)} (X_Q x + X_K z).
\end{aligned} \tag{EFP13}$$

The positive Hessian makes the minimum attained and unique in z . Its normal equation gives the displayed formula. For every z the two scalar metric inequalities hold on the same vector $X_Q x + X_K z$. Taking their minima therefore gives $e^{\lambda^-} Q^{(1)} \preceq Q^{(a)} \preceq e^{\lambda^+} Q^{(1)}$. Restriction is the case $K = 0$. An iterated quotient has the same property by successive applications to the original affine fibres. A block diagonal sum of four copies has the same constants by addition of the four

quadratic inequalities. In particular the coefficient dimensions do not introduce an extra factor of four in these constants.

For a rank- r such form, conjugating by the inverse positive square root $(Q^{(1)})^{-1/2}$ for the proof of the determinant inequality bounds each of its r positive relative eigenvalues between e^{λ^-} and e^{λ^+} . It follows that

$$r\lambda_{a,D}^- \leq \log \det Q^{(a)} - \log \det Q^{(1)} \leq r\lambda_{a,D}^+. \quad (\text{EFP14})$$

All determinant ratios here use the same value frame in both orders. Replacing that common frame by C multiplies both determinants by $|\det C|^2$ and hence leaves the displayed difference exactly unchanged. This proof does not change the original forms.

R30 Four endpoints and equal-rank contractions

For any of the actual four cutoffs N_i let $D_i = N_i - v$. Use exactly $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$ and put

$$\begin{aligned} \Phi_a^- &= -\lambda_{a,D_0}^+ - \lambda_{a,D_1}^+ + \lambda_{a,D_2}^- + \lambda_{a,D_3}^-, \\ \Phi_a^+ &= -\lambda_{a,D_0}^- - \lambda_{a,D_1}^- + \lambda_{a,D_2}^+ + \lambda_{a,D_3}^+, \\ W_a &= \sum_{i=0}^3 w_{a,D_i}. \end{aligned} \quad (\text{EFP15})$$

For any fixed rank- r subquotient over this window,

$$r\Phi_a^- \leq \Delta \mathcal{R} \log \det Q \leq r\Phi_a^+, \quad \Phi_a^+ - \Phi_a^- = W_a. \quad (\text{EFP16})$$

Indeed negate the upper endpoint at $i = 0, 1$ for the lower bound, and add the lower endpoints at $i = 2, 3$; the upper proof reverses these choices. This verifies every sign. Where (EFP12) applies at all four endpoints, the four copies of $-r \log \kappa_a$ cancel because $-1 - 1 + 1 + 1 = 0$. The lower and upper expressions become, respectively, $r(\underline{b}_0^* + \underline{b}_1^* - \bar{b}_2^* - \bar{b}_3^*)$ and $r(\bar{b}_0^* + \bar{b}_1^* - \underline{b}_2^* - \underline{b}_3^*)$. The same cancellation is valid if the NSO-only endpoints are written algebraically with this same scalar. It requires no new analytic comparison at an unguarded endpoint.

For two same-rank source and target quotients in matched value coordinates let $\omega = \det Q_t / \det Q_s$. Both receive (EFP14). Subtracting the two directed intervals yields

$$-rw_{a,D} \leq \Delta \log \omega(N) \leq rw_{a,D}, \quad |\Delta \mathcal{R} \log \omega| \leq rW_a. \quad (\text{EFP17})$$

The common scalar cancels already in this same-endpoint ratio. For two contractions of ranks r_1, r_2 in the same ambient metric the product consequently has return difference bounded by $(r_1 + r_2)W_a$. No lower singular-value estimate for a separate observation matrix has been used.

R31 The five original CPQ quotients and their full frames

Take $j = k + 4$, fix $e \in \{0, 1\}$, and use the actual pairs $p = 0 : (1, 1)$ and $p = 1 : (k, j)$. All target cutoffs here are $N = (q_j - 1, q_j, 2q_j - 1, 2q_j)$. The fixed coefficient maps are

$$\begin{aligned} \mathcal{U} &= \mathbb{C}^{4m_j}, \quad B = B_{\text{cof}} : \mathbb{C}^{m_k} \hookrightarrow \mathcal{U}, \quad \mathbb{B} = \text{diag}(B_j, 4) : \mathcal{U} \twoheadrightarrow \mathbb{C}^{4t_j}, \\ J &= \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \quad \mathcal{B} = \text{im } B, \quad S = \text{im}[B, F_e], \\ W &= J + \mathcal{B}, \quad K_e = S \cap W, \quad T_e = S + W. \end{aligned} \quad (\text{EFP18})$$

The original Taylor cochain equation of NSR5 makes F_e the same fixed coefficient map in both source presentations, and identifies $\mathbb{C}^{r_e}$ with $S/\mathcal{B}$, $r_e = 8k + 24 - 3e$. Keep $C_T = [U_R, U_b, U_o]$ and $\pi_o = [0 \ 0 \ I]C_T^{-1}\mathbb{B}$. For $z \in \mathcal{U}$, $\pi_o z = 0$ is equivalent to $\mathbb{B}z \in \text{im}(\mathbb{B}B)$, hence to $z \in W$. Surjectivity follows from that of $\mathbb{B}$. Thus π_o induces the exact isomorphism $\mathcal{U}/W \rightarrow \mathbb{C}^{r_o}$. Writing $E_e = \pi_o F_e$, its restriction gives $S/K_e \rightarrow \text{im } E_e$ and $T_e/W \rightarrow \text{im } E_e$. The identity on representatives is the

exact isomorphism $S/K_e \rightarrow T_e/W$. Its target fibre is larger, so its attained target norm is at most its attained source norm.

Choose the original fixed Z_h spanning $\ker E_e$, complement Z_c , $C_P = [Z_h, Z_c]$, $A_e = E_e Z_c$ and $C_o = [A_e, R_e^o]$. Keep $F = BJ_0$, $E = \ker(\Phi J_0)$ and the original lifts Z_R . The five quotient maps, with their full original coefficient fibres, are

form	Y/K	value lifts X_Q	rank
$G_{h,e}$	$K_e/\mathcal{B}$	$F_e Z_h$	h_e
$G_{s,e}$	S/K_e	$F_e Z_c$	c_e
$G_{t,e}$	T_e/W	$F_e Z_c$	c_e
$Q_{\text{rem},e}$	$\mathcal{U}/T_e$	fixed lifts of $U_o R_e^o$ under $\mathbb{B}$	$r_o - c_e$
A_{inv}	$\text{im } F/FE$	$F Z_R$	r_I

(EFP19)

Here $c_e = \text{rank } E_e$, $h_e = r_e - c_e$, $r_I = j_{0,k} - \dim \ker(\Phi J_0)$, $r_o = 4t_j - m_k + \dim \ker \Phi$, and $r_b = t_k - \dim \ker \Phi + \dim \ker(\Phi J_0)$. For the fourth row π_o followed by the quotient by $\text{im } A_e$ is the value map. For the fifth row injectivity of B identifies $\text{im } F/FE$ with the original coefficient quotient in Z_R . Every row is therefore precisely (EFP13), evaluated in $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$. In particular the fourth rank is not replaced by the whole r_o .

With $V_X^p = \mathcal{R} \log \det X^p$, (EFP16) gives

$$\begin{aligned} h_e \Phi_j^- &\leq \Delta V_h \leq h_e \Phi_j^+, & c_e \Phi_j^- &\leq \Delta V_s \leq c_e \Phi_j^+, \\ c_e \Phi_j^- &\leq \Delta V_t \leq c_e \Phi_j^+, & (r_o - c_e) \Phi_j^- &\leq \Delta V_{\text{rem}} \leq (r_o - c_e) \Phi_j^+, \\ r_I \Phi_j^- &\leq \Delta V_A \leq r_I \Phi_j^+. \end{aligned} \quad (\text{EFP20})$$

The identical proof also applies to $V_I, V_b, V_o, V_{\text{conn}}$ with ranks r_I, r_b, r_o, r_e . These inequalities retain the exact correlations $V_{\text{conn}} = V_h + V_s$ and $V_o = V_t + V_{\text{rem}}$. They refine CPQ19 without replacing any summand.

For clarity the absolute frame identities that precede every return are still

$$\begin{aligned} |\det C_P|^2 \det H_{\text{conn}} &= \det G_h \det G_s, & |\det C_o|^2 \det Q_o &= \det G_t \det Q_{\text{rem}}, \\ |\det C_T|^2 (\det H_{B,j,D})^4 &= \det H_{\text{inv}} \det Q_b^E \det Q_o, \\ P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem}} - \log \det G_s - \log \det A_{\text{inv}} + \log \mathfrak{c}_e, \\ \mathfrak{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}. \end{aligned} \quad (\text{EFP21})$$

These follow by the Schur eliminations in (EFP13) in the displayed full frames and the original $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$. Thus each constant has its original place before $\mathcal{R}1 = 0$ cancels it. The propagation uses the same minimum over the original lower-evaluation columns. For example, for the raw old kernel Gram G_K^{raw} , appended kernel rows a_n , quotient rows b_n and preceding projection coefficients C_n , the unchanged increment is

$$Q^+ = Q + R_n^* (I + a_n (G_K^{\text{raw}})^{-1} a_n^*)^{-1} R_n, \quad R_n = b_n - a_n C_n. \quad (\text{EFP22})$$

It follows by minimizing $\eta^* G_K^{\text{raw}} \eta + \|R_n x - a_n \eta\|^2$. Thus neither the old lower Gram nor the mixed kernel denominator is removed by the finite scalar comparisons.

R32 The common return and the complete allocation intervals

Put $I = \text{im}(BJ_0)$, $K_i = I \cap J$, $W_i = I + J$. The identity maps $I/K_i \rightarrow W_i/J$ and $S/K_e \rightarrow T_e/W$ are bijections: each kernel is the stated intersection and each image is the stated sum modulo its denominator. Their Gram ratios in the actual matched frames are $\omega_{\text{inv}} = \det H_{\text{inv}} / \det A_{\text{inv}}$ and $\omega_{\text{conn},e} = \det G_t / \det G_s$. Consequently for the actual common return $\mathcal{L}_e = \mathcal{R} \log(\omega_{\text{inv}} \omega_{\text{conn},e})$,

$$|\Delta \mathcal{L}_e| \leq B_e, \quad B_e = (r_I + c_e) W_j, \quad \mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b. \quad (\text{EFP23})$$

This is (EFP17) applied to both exact quotient bijections. In particular the common projection cross block in SFC5 is retained; a product determinant does not require that cross block to vanish.

Let L_-^p, L_+^p be the actual SFC9–10 finite expressions in pair p . They are determined from $\omega(N_0) = \det G_t \det H_{\text{inv}} / (\det G_s \det A_{\text{inv}})$, $x = \text{arcosh}(\omega(N_0)^{-1/2})$, and $\eta = \frac{1}{2} \log(\det \mathcal{D}_{N_1} / \det \mathcal{D}_{N_0})$, $b = \frac{1}{2} \log(\det \mathcal{D}_{N_2} / \det \mathcal{D}_{N_1})$, $c = \frac{1}{2} \log(\det \mathcal{D}_{N_3} / \det \mathcal{D}_{N_2})$, all in that same pair. To make their finite meaning explicit, put $f(t) = -2 \log \cosh t$, $A_c = e^{2c}$, $B_b = e^{2b}$. Let $z_* \geq 1$ be the unique root

$$A_c^2 B_b z^3 - \{A_c B_b (A_c + 1) - 3A_c\} z^2 - (3A_c B_b - A_c - 1) z - 1 = 0.$$

Then $t_* = b + c + \frac{1}{2} \log z_*$ and $\hat{t} = \min\{x + \eta, \max(\max(0, x - \eta), t_*)\}$ give

$$L_- = f(x + \eta + b) + f(x + \eta + b + c) - f(x) - f(x + \eta), \quad L_+ = f((\hat{t} - b)_+) + f((\hat{t} - b - c)_+) - f(\hat{t}) - f(x).$$

Their complete proof is SFC9–16; they are actual scalar evaluations, not independently assigned angle bounds.

Use the following exact two-pair intersections:

$$\begin{aligned} \tilde{L}_-^1 &= \max\{L_-^1, L_-^0 - B_e\}, & \tilde{L}_+^1 &= \min\{L_+^1, L_+^0 + B_e\}, \\ \tilde{L}_-^0 &= \max\{L_-^0, L_-^1 - B_e\}, & \tilde{L}_+^0 &= \min\{L_+^0, L_+^1 + B_e\}. \end{aligned} \tag{EFP24}$$

Each contains its actual $\mathcal{L}_e^p$: combine CIT35 in pair p with (EFP23) and CIT35 in the other pair. Consequently the earlier signed remainder lies in $[V_b^p + \tilde{L}_-^p, V_b^p + \tilde{L}_+^p]$. This interval may also be intersected with the already proved CIT33 bounds. In source-order differences the exact identity gives $\Delta \mathcal{P}_e^{\text{rest}} \in [r_b \Phi_j^- - B_e, r_b \Phi_j^+ + B_e]$.

Write $\mathcal{T}_j^p = \mathcal{T}_{j,s_1}$ in its actual pair. Keep the complete $Z_e^p = V_b^p + V_{\text{rem}}^p + V_s^p + V_A^p$ and the actual native cap $U_{j,s_1}^{\text{native}}$ only on its proved domain. Define

$$\begin{aligned} \hat{K}_-^p &= \max\{0, (\tilde{L}_-^p + Z_e^p)/4\}, \\ \hat{K}_+^p &= \min\{\mathcal{T}_j^p, U_{j,s_1}^{\text{native}}, (\tilde{L}_+^p + Z_e^p)/4\}. \end{aligned} \tag{EFP25}$$

Omit the cap entry outside its original domain. The exact CPQ20 affine identity $4S_{K,j}^p = \mathcal{L}_e^p + Z_e^p$ and $S_{R,j}^p = \mathcal{T}_j^p - S_{K,j}^p$ prove $\hat{K}_-^p \leq S_{K,j}^p \leq \hat{K}_+^p$ and the reversed residual interval. All four summands in Z_e^p remain.

The native allocation maps are $S_{K,j} = \mathcal{R} \log \det H_{B,j,D}$ and $S_{R,j} = \mathcal{R} \log \det (J_j^* D_{j,D} J_j)$, of ranks t_j and $j_{0,j}$, with J_j the full fixed frame of $\ker B_j$. The first is the attained quotient by that kernel; the second is its restriction. Equation (EFP16) therefore controls both, with the original inverse-metric signs. Writing the actual $\Delta \mathcal{T}_j = \mathcal{T}_j^1 - \mathcal{T}_j^0$, put

$$\begin{aligned} A_K^- &= \max\{t_j \Phi_j^-, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^+\}, \\ A_K^+ &= \min\{t_j \Phi_j^+, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^-\}. \end{aligned} \tag{EFP26}$$

The first pair of entries is the direct kernel interval. The second follows by subtracting the residual interval from the exact equality $\Delta S_K + \Delta S_R = \Delta \mathcal{T}_j$. They contain the actual ΔS_K and so are ordered. The fully propagated SFC19/HJR35 receiver is

$$\begin{aligned} \max\{\hat{K}_-^1, \hat{K}_-^0 + A_K^-\} &\leq S_{K,j}^1 \leq \min\{\hat{K}_+^1, \hat{K}_+^0 + A_K^+\}, \\ \max\{\hat{K}_-^0, \hat{K}_-^1 - A_K^+\} &\leq S_{K,j}^0 \leq \min\{\hat{K}_+^0, \hat{K}_+^1 - A_K^-\}. \end{aligned} \tag{EFP27}$$

To prove the first line add the actual difference interval to the pair-zero interval and intersect with pair one. For the second line subtract that difference from pair one and intersect with pair zero. Subtract these endpoints in reversed order from each same-pair $\mathcal{T}_j^p$ to obtain the complete residual allocation bounds. No allocation value is selected by this interval composition.

R33 The degree- k source at the displaced NFR cutoffs

The proper source quotient is $Q_{\text{inv}}^p = (J_0^* D_{k,D}^{(s_0)} J_0)/E$ in the original Z_R values, while the invariant target form is H_{inv}^p . They have the same rank r_I and retain different centres c'_k, c'_j . Let $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = V_I^p - V_Q^p$. For the source constants evaluate (EFP11) at degree k and at the actual $D_i = N_i - v$ used by the target. Relative to the standard degree- k window these cutoffs have displacement $b = q_j - q_k = 8k + 24$ at the two lows and $2b$ at the two highs.

For these source cutoffs the largest polynomial degree is

$$A_{k,\text{max}} = 2q_j - v - q'_k = k^2 + 34k + 1 - v. \quad (\text{EFP28})$$

For $k \geq 37$ one has $2q_k - A_{k,\text{max}} = k^2 - 30k + 1 + v \geq 260 + v$, and $2q_k - (A_{k,\text{max}} + (k-1)/4) = k^2 - (121/4)k + 5/4 + v \geq 251 + v$. Both right sides increase for $k \geq 37$ by differentiation, or by taking their exact positive finite differences at successive $4l + 1$. The smallest degree is $q_j - v - 1 - q'_k = 24k - 25 - v \geq 0$. Thus the stated $k \geq 37$ guard, together with the actual QOT root test at k , validates the polynomial degree range at all four displaced source endpoints. It is a sufficient finite guard; no minimality is claimed. EOR is added at an endpoint only when its own two-block tests hold with this degree- k value of A .

Let $\Phi_k^\pm$ and W_k now mean precisely these displaced degree- k values, while $\Phi_j^\pm, W_j$ use the target degree. Then

$$\Lambda_{\text{new}}^- = r_I(\Phi_j^- - \Phi_k^+) \leq \Delta\mathcal{M} \leq r_I(\Phi_j^+ - \Phi_k^-) = \Lambda_{\text{new}}^+. \quad (\text{EFP29})$$

This follows by subtracting the full rank- r_I source-return interval from its target-return interval. At a single cutoff their log-determinant difference retains the centre $-r_I \log(\kappa_j/\kappa_k)$ wherever both comparisons are guarded. Only after the four endpoint sum do the four copies cancel. Replacing the degree- k endpoints by degree- j endpoints would change this exact source map and is not used.

For each pair define the exact translation $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$. By NFR7, $\mathcal{M}^p = \mathcal{L}_e^p + D_e^p$. Thus (EFP24) gives the interval $[\tilde{L}_-^p + D_e^p, \tilde{L}_+^p + D_e^p]$. Intersect it with $-V_Q^p \leq \mathcal{M}^p$ and, on the cap domain, with

$$\mathcal{M}^p \leq 4U_{j,s_1}^{\text{native}} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p.$$

Call these resulting endpoints $\widehat{M}_-^p, \widehat{M}_+^p$. The earlier NFR12 interval becomes

$$\begin{aligned} \max\{\widehat{M}_-^1, \widehat{M}_-^0 + \Lambda_{\text{new}}^-\} &\leq \mathcal{M}^1 \leq \min\{\widehat{M}_+^1, \widehat{M}_+^0 + \Lambda_{\text{new}}^+\}, \\ \max\{\widehat{M}_-^0, \widehat{M}_-^1 - \Lambda_{\text{new}}^+\} &\leq \mathcal{M}^0 \leq \min\{\widehat{M}_+^0, \widehat{M}_+^1 - \Lambda_{\text{new}}^-\}. \end{aligned} \quad (\text{EFP30})$$

Its proof is the same addition and subtraction of exact directed intervals used in (EFP27), now with both actual sources.

Finally retain the full proper-source compatibility error. Use the original endpoint constants

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p. \end{aligned}$$

For $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$, NFR5 is the exact identity $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Equations (EFP17) and (EFP29) therefore give

$$\Lambda_{\text{new}}^- - r_I W_j - E_1^+ + E_0^- \leq \Delta\mathcal{C} \leq \Lambda_{\text{new}}^+ + r_I W_j - E_1^- + E_0^+. \quad (\text{EFP31})$$

Subtract the target invariant-loss difference and the error difference from $\Delta\mathcal{M}$ to verify each of these signs. The original source-complement projections, constants, and both old low cutoffs remain in this formula.

R34 The defining primitive source: exact finite intersection

Keep the original fixed hypothetical simple quartet, the complete nonreduced conductor, the full arithmetic unit, the fixed five-orbit period and logarithm branch, and the original period domain $|u| \geq R_*$ (or its specified exact period tests). Write

$$k = 4\ell + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad c'_a = a/2 - 4,$$

$$D_e = q_j - v - 1 + e, \quad d_e = 8k + 24 + e, \quad r_e = d_e - 4e = 8k + 24 - 3e, \quad e \in \{0, 1\}.$$

Here $v = \text{ord}_0 E_A$ is the actual conductor order, $0 \leq v \leq 80$, and $\mu_v = E_A^{(v)}(0) \neq 0$. Every statement below fixes e ; subtraction is between the source-order pairs $p = 0 : (1, 1)$ and $p = 1 : (k, j)$, never between $e = 0$ and $e = 1$. The source centre is c'_k and the target centre is c'_j .

The inherited finite constants are exactly NSO3-4/NSR2 and QOT11-14:

$$\ell_a = (a - 1)/4, \quad \rho_a = \frac{(2\pi)^{2\ell_a}}{(1/2)^{2\ell_a}}, \quad \mathbf{a}_a = \rho_a \prod_{h=0}^{\ell_a-1} (2h + \tfrac{1}{2})^2,$$

$$\mathbf{b}_{a,D} = \rho_a \prod_{h=0}^{\ell_a-1} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\},$$

$$\kappa_a = 2^{10\ell_a} \frac{(2\ell_a)!}{(4\ell_a)!} (a - 7)^{4\ell_a}, \quad l_{a,b} = b_{a,b,a}^- - b_{a,b,1}^+ - 4 \sum_{h=0}^{\ell_a-1} \text{arsinh} \sqrt{\frac{b + h + 1/2}{2q'_a}},$$

$$u_{a,b} = b_{a,b,a}^+ - b_{a,b,1}^- + 4 \sum_{h=1}^{\ell_a} \text{arsinh} \sqrt{\frac{b + h + 1/2}{2q'_a}}.$$

The $b^\pm$ in the last two formulas are the complete GPA constants QOT11-12, equivalently EOR6; the original inside-region error, source mass, root radius, and trigonometric factors are retained. No new analytic estimate replaces those constants. In particular κ_a is the original full-mass scalar of QOT20.

The exact source map. Let

$$E_A(z) = \sum_{b,c=0}^8 a_{bc} e^{\beta_{8,b,c} z}, \quad \mathcal{T}_{AP}(S') = \sum_{b,c=0}^8 a_{bc} p(S' + \beta_{8,b,c}),$$

and keep the unscaled original relation columns

$$r_h = \mathcal{T}_A(\chi_k S^h), \quad 0 \leq h < d_e, \quad \psi_k = \chi_{k-8}.$$

At every lower root $\beta_{k-8,u,w}$ and every summand,

$$\beta_{k-8,u,w} + \beta_{8,b,c} = \beta_{k,u+b,w+c}.$$

Thus χ_k vanishes term by term. The lower roots are distinct, since $0 < \delta < 1/2$ and $\gamma > 2$, and their complete product ψ_k divides every r_h . For a polynomial of degree n ,

$$\mathcal{T}_{AP} = \sum_{t=0}^n \frac{\mu_t}{t!} p^{(t)}.$$

The first nonzero summand has $t = v$, so the leading coefficient of r_h is $\binom{q_k+h}{v} \mu_v \neq 0$ and $\deg r_h = q_k + h - v$. Consequently the full relation columns are independent, their maximum degree is D_e , and the polynomial degree to which the weighted QOT theorem is applied is exactly

$$a_{\text{src},e} = \max_{h < d_e} \deg(r_h/\psi_k) = q_k - v + d_e - 1 - q'_k = 24k - 25 - v + e. \quad (\text{EFP32})$$

No leading-term replacement was used to obtain the restriction map: the entire polynomial r_h/ψ_k is retained in that map.

Write R_e^{pol} for the columns of the unscaled original r_h (not the divided polynomials r_h/ψ_k) in $1, S', \dots, S'^{D_e}$, and use the exact native monomial matrix B_{s, c'_k, D_e} of NSO7. Put

$$C_e^{(s)} = B_{s, c'_k, D_e} R_e^{\text{pol}}, \quad H_e^{(s)} = (C_e^{(s)})^* C_e^{(s)}, \quad U_s = \text{diag}(1/n!) B_{s, c'_k, D_e}^\top.$$

The Gram $H_e^{(s)}$ is precisely the integral Gram of all $r_h(c'_k + iy)$ for the original measure

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy.$$

The underlying derivative values are complex-linear; hence the transpose in U_s is ordinary transpose.

For completeness, the relation annihilator is exactly the analytic plane used in the original primitive quotient. If a native analytic column has generating germ G , then $E_A G$ is an upper-grid exponential numerator. For the lower columns $G = e^{\beta_{k-8,u,w} z}$ the same assertion follows from the displayed root addition. Therefore

$$\Lambda_G(r_h) = \left[(\chi_k(\partial) \partial^h)(E_A G) \right]_{z=0} = 0.$$

The full native columns $[V_0, T_0]$ have rank $q_k - v$: if their Taylor combination vanishes through D_e , its product with E_A is an upper-grid numerator vanishing through $D_e + v \geq q_k - 1$; the upper Vandermonde makes its coefficient vector zero, the fixed numerator quotient of NSO12 makes its analytic class zero, and the lower Vandermonde makes its lower coefficients zero. These are exactly the same complete columns as NSO12–13 and NSO21. Since $D_e + 1 - d_e = q_k - v$ and $C_e^{(s)}$ has rank d_e , this proves

$$\ker(C_e^{(s)})^\top = \text{im}[V_0, T_0].$$

The original Taylor primitive plane is the whole jet plane for $e = 0$; for $e = 1$ it is the kernel of all four independent original rows $\Lambda(\mathcal{T}_{B_i} \mathcal{T}_A \chi_j)$. Its analytic subspace is the preceding complete plane. Thus the original Taylor complement P_e^{Tay} and its native lifts $P_e^{(s)} = U_s^{-1} P_e^{\text{Tay}}$ give

$$\Theta_e = (C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(n!) P_e^{\text{Tay}}. \quad (\text{EFP33})$$

This matrix is literally independent of s , including every factorial, affine phase, mass and original coefficient. It has rank r_e ; it is square invertible for $e = 0$, and has the original rank- r_e constraint image for $e = 1$.

For a value x in this fixed rank- r_e frame, the complete affine fibre is

$$P_e^{(s)} x + \text{im}[V_0, T_0] = \{w : (C_e^{(s)})^\top w = \Theta_e x\}.$$

Every member remains in the primitive plane, because the analytic plane is contained in it. The Euclidean least norm on an onto fibre $Aw = b$ is $b^*(AA^*)^{-1}b$: the minimizing vector is $A^*(AA^*)^{-1}b$, and its inner product with every vector of $\ker A$ is zero. Here

$$AA^* = (C_e^{(s)})^\top \overline{C_e^{(s)}} = \overline{H_e^{(s)}}.$$

We have consequently proved the exact original source identity

$$H_{\text{src},e}^{(s)} = \Theta_e^* \overline{(H_e^{(s)})^{-1}} \Theta_e. \quad (\text{EFP34})$$

This is the QOT26–27 map, with the identical NSO21,26/NSR9 value frame. In particular the primitive source has not been identified with H_Y . Although $D_e - q'_k = a_{\text{src},e}$ numerically at this defining cutoff, that equality alone proves no map of forms and no extension to later primitive source cutoffs.

All finite guards. NSO25 and NSR6 hold on their original domain $k \geq 9$ without a GPA radius guard. Add the QOT source interval only where

$$k \geq 29, \quad R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k;$$

add its defining target interval only where the corresponding conditions hold for j . Retain those two conditions explicitly in paired use. The actual polynomial degrees are

$$a_{\text{src},e} = 24k - 25 - v + e, \quad a_{\text{tgt},e} = D_e - q'_j = 16k + 15 - v + e = 16j - 49 - v + e.$$

The requirements $0 \leq a_{\text{src},e}$, $a_{\text{src},e} + \ell_k \leq 2q_k$ and their target analogues hold for $k, j \geq 29$: positivity follows using $v \leq 80$, while the upper expressions are at most $25k$ and $17j$, respectively, and these are below $2(k+1)^2$ and $2(j+1)^2$ on that domain. The exact degrees, not these bounds, are used in all constants.

The EOR21 endpoint rule additionally requires, for both parity blocks,

$$m_\epsilon = \lfloor (a - \epsilon)/2 \rfloor \geq 1, \quad \alpha_\epsilon = (q'_b + \epsilon - \tfrac{1}{2})/m_\epsilon \in [3/2, 5/2], \quad \beta_\epsilon = \frac{\pi q_b}{2m_\epsilon} \in [\pi/2, 3\pi/2],$$

as well as all EOR12 localization guards with the EOR9 constants. Here (b, a) would be $(k, a_{\text{src},e})$ or $(j, a_{\text{tgt},e})$. Neither application is permitted on the present GPA domain. Indeed $\gamma > 2$ gives

$$2k < R_k \leq 2^{-33}q_k, \quad q_k > 2^{34}k.$$

At the source the upper rectangle guard would imply

$$q_k \leq 3m_\epsilon \leq \tfrac{3}{2}a_{\text{src},e} \leq 36k - 36 < 2^{34}k,$$

a contradiction. At the target it would imply

$$q_j \leq 3m_\epsilon \leq \tfrac{3}{2}a_{\text{tgt},e} \leq 24k + 24 < 2^{34}k < q_j,$$

again a contradiction. The middle strict inequality holds for $k \geq 29$. Therefore EOR21's actual fallback here is exactly QOT's low interval. No high-endpoint equilibrium constants are inserted into this primitive comparison. Its high target-flag applications remain separate.

The componentwise intersection and its proof. In the joint GPA domain define

$$\begin{aligned} \lambda_{\text{src},e} &= \max\{l_{k,a_{\text{src},e}}, \log \mathfrak{a}_k - \log \kappa_k\}, & v_{\text{src},e} &= \min\{u_{k,a_{\text{src},e}}, \log \mathfrak{b}_{k,D_e} - \log \kappa_k\}, \\ \lambda_{\text{tgt},e} &= \max\{l_{j,a_{\text{tgt},e}}, \log \mathfrak{a}_j - \log \kappa_j\}, & v_{\text{tgt},e} &= \min\{u_{j,a_{\text{tgt},e}}, \log \mathfrak{b}_{j,D_e} - \log \kappa_j\}. \end{aligned} \quad (\text{EFP35})$$

Let $L_{\text{src},e}$ and $L_{\text{conn},e}$ denote the logarithmic source-order determinant ratios of NSR6. The exact finite result is

$$\begin{aligned} S_e^- &= -r_e(\log \kappa_k + v_{\text{src},e}) \leq L_{\text{src},e} \leq -r_e(\log \kappa_k + \lambda_{\text{src},e}) = S_e^+, \\ T_e^- &= -r_e(\log \kappa_j + v_{\text{tgt},e}) \leq L_{\text{conn},e} \leq -r_e(\log \kappa_j + \lambda_{\text{tgt},e}) = T_e^+. \end{aligned} \quad (\text{EFP36})$$

Equivalently, without moving the scalar centres,

$$\begin{aligned} S_e^- &= \max\{-r_e \log \mathfrak{b}_{k,D_e}, -r_e(\log \kappa_k + u_{k,a_{\text{src},e}})\}, \\ S_e^+ &= \min\{-r_e \log \mathfrak{a}_k, -r_e(\log \kappa_k + l_{k,a_{\text{src},e}})\}, \\ T_e^- &= \max\{-r_e \log \mathfrak{b}_{j,D_e}, -r_e(\log \kappa_j + u_{j,a_{\text{tgt},e}})\}, \\ T_e^+ &= \min\{-r_e \log \mathfrak{a}_j, -r_e(\log \kappa_j + l_{j,a_{\text{tgt},e}})\}. \end{aligned}$$

Where one GPA guard fails, omit that component's QOT candidate and use its original NSR6 interval. This is an unconditional finite rule on the original NSO domain, with explicitly guarded additional proved bounds.

To prove the source statement, apply QOT15 to the fixed complete columns r_h/ψ_k of degree at most $a_{\text{src},e}$:

$$\kappa_k e^{l_{k,\text{src},e}} H_e^{(1)} \preceq H_e^{(k)} \preceq \kappa_k e^{u_{k,\text{src},e}} H_e^{(1)}.$$

Inversion reverses this inequality, coefficient conjugation preserves it, and the same full-rank Θ_e in (EFP34) gives the rank- r_e dual determinant interval QOT27. The whole-source minimum NSO21 independently gives NSO25/NSR6 on the identical affine fibre. Each lower endpoint bounds the same scalar from below, and each upper endpoint bounds it from above. Their maximum and minimum give $S_e^\pm$. Their interval is nonempty because it contains that actual scalar.

For the target, the ordinary Taylor multiplier equation NSO23/NSR5 is

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}.$$

All its columns represent the same actual germs. If two representations differ, multiplying the zero Taylor difference by E_A produces an upper-grid numerator zero through $D_e + v \geq q_j - 1$; the upper Vandermonde and the fixed numerator quotient kill the analytic class, and the lower Vandermonde kills the remaining lower class. Thus F_e is literally the same coefficient matrix in the two pairs. The full multiplier kernel of NSO24 gives its quotient by the complete boundary image rank r_e . Its metric is the attained minimum

$$x^* H_{\text{conn},e}^{(s_1)} x = \min_{\xi} (F_e x + B_{\text{cof}} \xi)^* \text{diag}(D_{j,D_e}^{(s_1)}, 4) (F_e x + B_{\text{cof}} \xi).$$

This includes all four lower minima inside D_{j,D_e} and the full boundary minimum over ξ . QOT16 on that target degree $a_{\text{tgt},e}$ therefore gives QOT30; NSO24–25 independently gives the target NSR6 interval. Taking their intersection proves $T_e^\pm$. No source-centre identification with c'_j occurs in this argument.

The strongest propagation from these component intervals. Retain the exact NSR7 identity, with all original constants,

$$\begin{aligned} A_e^p &= \log \det H_{\text{conn},e}^p - \log \det H_{\text{src},e}^p = -\mathcal{X}_e^p + \epsilon_e^p, & \mathcal{X}_e^p &= \chi_{X_k,D_e}^p - \chi_{Y_e,D_e}^p, \\ c_e^{p,-} &= -2r_e \log \max(1, K_-^p) - r_e \log c_+^p, & c_e^{p,+} &= 2r_e \log \max(1, K_+^p) - r_e \log c_-^p, \\ c_-^p &= \min(1, (L_-^p)^{-2}), & c_+^p &= \max(1, (L_+^p)^2), & b_e^- &= \epsilon_e^{1,-} - \epsilon_e^{0,+}, & b_e^+ &= \epsilon_e^{1,+} - \epsilon_e^{0,-}. \end{aligned}$$

These $K_\pm^p, L_\pm^p$ retain the original NBG/NKC domains, the degree $k+8$ comparison, both centres, the shift $(\beta_{4,u,w} - 2)/i$, every coefficient and both full Gamma masses. They are not replaced by the scalar order-comparison constants.

Set $\mathcal{J}_e^{\text{ord}} = A_e^1 - A_e^0$ and $\Delta \mathcal{X}_e = \mathcal{X}_e^1 - \mathcal{X}_e^0$. Then

$$\begin{aligned} T_e^- - S_e^+ &\leq \mathcal{J}_e^{\text{ord}} \leq T_e^+ - S_e^-, \\ \zeta_e &= L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0, & S_e^- + b_e^- &\leq \zeta_e \leq S_e^+ + b_e^+, \\ L_{\text{conn},e} &= -\Delta \mathcal{X}_e + \zeta_e, & S_e^- - T_e^+ + b_e^- &\leq \Delta \mathcal{X}_e \leq S_e^+ - T_e^- + b_e^+. \end{aligned} \tag{EFP37}$$

In the joint GPA domain the scalar-bearing forms of the two principal intervals are exactly

$$\begin{aligned} r_e(\lambda_{\text{src},e} - v_{\text{tgt},e}) &\leq \mathcal{J}_e^{\text{ord}} + r_e \log(\kappa_j/\kappa_k) \leq r_e(v_{\text{src},e} - \lambda_{\text{tgt},e}), \\ r_e(\lambda_{\text{tgt},e} - v_{\text{src},e}) + b_e^- &\leq \Delta \mathcal{X}_e - r_e \log(\kappa_j/\kappa_k) \leq r_e(v_{\text{tgt},e} - \lambda_{\text{src},e}) + b_e^+. \end{aligned} \tag{EFP38}$$

Here the exact inherited centre is

$$\frac{\kappa_j}{\kappa_k} = \frac{256(k-3)^{k+3}}{k(k+2)(k-7)^{k-1}}.$$

This is QOT31, including the full source masses; it is not absorbed into an error interval.

For proof, subtract the two component determinant ratios to get $\mathcal{J}_e^{\text{ord}} = L_{\text{conn},e} - L_{\text{src},e}$. The smallest value on their interval rectangle is $T_e^- - S_e^+$, and the largest is $T_e^+ - S_e^-$. Subtracting the two exact NSR7 identities gives $\Delta\mathcal{X}_e = L_{\text{src},e} - L_{\text{conn},e} + \epsilon_e^1 - \epsilon_e^0$, which proves the remaining bounds by the same endpoint calculation. Substitution of the scalar-bearing expressions (EFP36) proves (EFP38) with both displayed signs.

These are the smallest enclosures obtained from the component intervals and these exact linear identities: a linear function attains each bound at the indicated rectangle vertex. This is a statement about the information in those intervals, not about realizability or optimality for the actual correlated matrices. In particular this componentwise procedure retains mixed NSO/QOT endpoint choices that can be lost by merely intersecting the two separately propagated paired intervals. The common variables in $\zeta_e = L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0$ and $\Delta\mathcal{X}_e = \zeta_e - L_{\text{conn},e}$ remain shared. Their marginal intervals must not subsequently be treated as independent when recovering $L_{\text{conn},e} = \zeta_e - \Delta\mathcal{X}_e$.

The earlier PSR26 full centered interval is also retained exactly as in NSR8–9: denote its unchanged endpoints by P_e^-, P_e^+ , and, only on its original domain

$$k \geq \max\{9, v - 1, \lceil (\pi/2)\sqrt{\delta^2 + \gamma^2} \rceil + 1\}$$

with the same original admissible SGT14 radius $0 < a < 1$, replace S_e^- by $\max\{S_e^-, P_e^-\}$ and S_e^+ by $\min\{S_e^+, P_e^+\}$ in (EFP37). Here $P_e^\pm$ are the literal preexisting PSR26 endpoints, including its D_e^{ord} and both errors, not newly defined analytic constants. No PSR guard is imposed on NSO or QOT. The proof is again intersection of bounds on the same source determinant ratio; the original Θ_e and complement/frame factors remain as in (EFP33).

At $e = 0$ the established equality $\chi_{Y_0, D_0} = 0$ may be used. At $e = 1$ the complete four-row primitive constraint and χ_{Y_1, D_1} stay in every formula. Since these intersections are subsets of QOT27,30,32,34, they preserve the already proved $O_{\text{actual}}(k^{3/2} + k \log q_k)$ paired error and the conclusions $\mathcal{J}_e^{\text{ord}} = o(q_k)$ and $\Delta\mathcal{X}_e = o(q_k)$ at fixed actual data. Indeed the original low l, u are $O_{\text{actual}}(\sqrt{k} + \log q_k)$, the rank is $O(k)$, $\log(\kappa_j/\kappa_k) = O(\log k)$, and NSO27 gives $\epsilon_e^{p,\pm} = O_{\text{actual}}(k \log q_k)$. These inherited order-difference conclusions assign no absolute compatibility value and no later primitive source extension.

R35 Consequences and exact scope

Every new interval is an intersection of finite inequalities on the same original coefficient directions or matched quotient values. Consequently it is contained in the corresponding NSO interval and, where available, in the QOT/EOR interval. Empty coefficient blocks have rank zero, determinant one and zero return, so all formulas retain those cases.

The EOR limiting allowances already proved in the complete provider therefore remain valid for the direct allocation differences and the common return after these further finite intersections: the two allocation endpoints are $-8 \log(\pi v_*/4)$ and $8 \log(4/(\pi u_*))$ after division by q_a , and the common difference allowance after division by q_k is $16 \log(v_*/u_*)$, on their same fixed-actual-data eventual domains. This follows by containment of intervals, not by a new limit exchange. Here $u_* K(\zeta) = 2$, $v_* E(\zeta) = 4$ and $\zeta^2 = 1 - u_*^2/v_*^2$ as in EOR.

The finite expressions retain the actual initial common angle, all four returns in Z_e , the original independent source quotient, and every proper-source error. No numerical individual allocation, no uniform estimate in a moving period, and no exclusion of an off-line quartet follows solely from these identities. The mathematical output is the proved finite propagation through the displayed exact maps, now available at every stated receiving site.

Source status. This is the complete accepted integration derivative supplied by the publication owner; the captured original and the exact derivative change are preserved separately. The independent written review is recorded in the source manifest. Its statements about review, execution and publication describe the captured source state. This typesetting conversion supplies no additional mathematical acceptance. Captured pull request: <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/37>; head 834bc0ea2b2510ac39862d9bd176200e78254c41. The source is preserved byte for byte in ECT_PUBLIC_RESEARCH_NOTE.md. Every displayed equation is assigned a unique ECT tag; the exact source-to-tag correspondence accompanies this body.

R36 Uniform exponential suppression of the original adjacent ratios

16 September 2026. Written proof contribution requiring independent review. No new Lean certificate, actual off-critical zero example, or RH conclusion is asserted.

R36.1 Source and review

Base main: 1efd53337561d8f67cf0ac1d119acdaa222644c3. AMT1–6 were directly reread in workbenches/splitzero-tandem/continuations/20260915-primitive-band-continuation/13_ARITHMETIC_MIXED_TRANSFER.tex, blob b6cfec2ac9517acc378aa02686ada6e7923ef381.

The user supplied the complete 540-line Pastedmarkdown(2)(8).md: grouped-convolution recurrence, contraction penalties, and the exact full-action cancellation. Its recurrence proof checks out on the stated AMT inputs. Fixed arithmetic blocks 3,4,5 have a degree-linear multiplication bound; their tensor product is compressed at TOTAL degree $N+1$, retaining the full output face. This proves the stated bound $\omega_{N+1}/\omega_N \leq C_{\text{mult}}(h)^{2(2N+\text{floor}(k/3))}$, and the exterior benchmark gives its polynomial penalty bound $O_{-h}(1+k(m-1))$. The original 4q-1 penalty coefficient count is correct. Only the prose needs clarification: d_n tends to zero; $1-d_n$ is close to one.

The other new transcript announces an exponential full-band candidate but contains no complete quantile proof. The argument below bypasses that unfinished proof. It requires only a lower bound for the actual relation minimum, not a sharp minimum asymptotic. The EOR source-order theorem is not an input to this continuation.

R36.2 1. Actual source and original relation norm

Fix the complete stipulated quartet h , its multiplicity m , and amplitude $v_h=2\text{xi}/h$. Retain

$$e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \quad c = k/2, \quad R_k = k\sqrt{\delta^2 + \gamma^2}, \quad \alpha = \pi/2. \quad (\text{ECT1})$$

Every centered root of the original monic degree- q polynomial χ has modulus at most R_k . Its full multiplicities remain. Write $\mu_k = w_h^{\{k\}}$. AMT gives

$$\mu_k(y) \geq \ell_{h,k} e^{-\alpha|y|} (k-2+|y|)^{-B_h}, \quad \mu_k \leq A_k \sigma, \quad (\text{ECT2})$$

$$B_h = 42 + 8m, \quad \ell_{h,k} = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}, \quad A_k = (C_h^{\text{tilt}}/c_\Gamma) k^{p+1} M_\alpha^{k-1}, \quad p = 8m - 9/2. \quad (\text{ECT3})$$

All constants are the original fixed- h constants in AMT4; no source mass is normalized away. The Gamma monic norm is

$$g_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} (2n)! / 4^n. \quad (\text{ECT4})$$

Let ω_n be the original arithmetic monic norm and define

$$h_r^\chi = \inf_{P \text{ monic, deg } P=r} \int |\chi(c+iy)P(c+iy)|^2 \mu_k(y) dy. \quad (\text{ECT5})$$

The original source/relation determinant factorization gives

$$\boxed{d_n := V_n/V_{n-1} = \omega_n/h_{n-q}^\chi \quad (n \geq q).} \quad (\text{ECT6})$$

Proof: V_N is the source Hankel determinant at $N+1$ divided by the relation Hankel determinant at $N-q+1$. Taking the adjacent ratio yields (ECT6). The denominator is the norm of an original relation BEFORE its quotient value becomes represented zero. No $G_{-(q-2)}$ occurs. The substitution $Q(y)=i^{\wedge\{-r\}}P(c+iy)$ is monic in y and preserves absolute values, so the following polynomial estimate uses the original coordinate through an explicit isometry, not a new arithmetic quotient.

R36.3 2. A two-band lower bound for every monic direction

Take $q \geq 1$, $0 \leq r \leq q$, $n=q+r$, and set

$$a = 20n/63, \quad b = 40n/21, \quad E_n = [-b, -a] \cup [a, b], \quad d\nu_n(y) = \frac{|y| dy}{\pi \sqrt{(y^2 - a^2)(b^2 - y^2)}}. \quad (\text{ECT7})$$

This is a probability measure; y squared pushes it to the arcsine law on $[a^2, b^2]$, with equal mass on the two half-lines. For a monic complex polynomial Q of degree r ,

$$\int \log |Q| d\nu_n \geq r \log \frac{\sqrt{b^2 - a^2}}{2}. \quad (\text{ECT8})$$

For each root z , symmetry turns its logarithmic integral into one half the arcsine integral of $\log|z^{\wedge 2} - x|$. On an interval with midpoint C and half-width D , factoring the quadratic in $\exp(i\theta)$ and applying the circle Jensen formula gives $\log(D/2)$ plus nonnegative log-max root terms. Hence this integral is at least $\log((b^2 - a^2)/4)$; summing proves (ECT8). Roots on the bands are handled by logarithmic integrability or a limit.

The same arcsine substitution gives exactly

$$\int \log |y| d\nu_n = \log((a+b)/2), \quad \int \log(d\nu_n/dy) d\nu_n = \log \frac{2}{\pi(b-a)}. \quad (\text{ECT9})$$

For the entropy formula use that each of the expectations of $\log(x - a^{\wedge 2})$, $\log(b^{\wedge 2} - x)$ equals $\log((b^2 - a^2)/4)$.

Let

$$E_0 = \int_0^{\pi/2} \sqrt{\sin^2 t + \frac{1}{36} \cos^2 t} dt. \quad (\text{ECT10})$$

Then $\int |y| d\nu_n = 2b E_0/\pi$. Here $E_0 < 21/20$ has an elementary proof. Put $\rho=1/6$ and $t_0=\arcsin \rho$. On $[0, t_0]$, the integrand minus $\sin t$ is at most $\rho \cos t$, whose integral is ρ squared. On $[t_0, \pi/2]$, rationalization bounds the difference by ρ squared $\cos$ squared $t/(2 \sin t)$. Therefore

$$E_0 - 1 \leq \frac{1}{36} + \frac{1}{72} [\log(6 + \sqrt{35}) - \sqrt{35}/6] < \frac{1}{20}. \quad (\text{ECT11})$$

The last inequality uses $\log 12 < 5/2$ and $\sqrt{35}/6 > 9/10$. The first seven positive Taylor terms of $\exp(5/2)$ already exceed 12, so no numerical quadrature is needed. Consequently $\alpha \int |y| d\nu_n = b E_0 < 2n$.

Apply probability Jensen to $f/(d\nu_n/dy)$, where $f=|Q|^2 \exp(-\alpha|y|)$. Its logarithm is integrable. Equations (ECT8)–(ECT9) prove

$$\begin{aligned}
\int_{E_n} |Q(y)|^2 |y|^{2q} e^{-\alpha|y|} dy &\geq \frac{\pi(b-a)}{2} \left(\frac{b^2-a^2}{4} \right)^r \left(\frac{a+b}{2} \right)^{2q} e^{-2n} \\
&= \frac{50\pi n}{63} n^{2n} e^{-2n} (500/567)^r (100/81)^q \\
&\geq \frac{50\pi n}{63} n^{2n} e^{-2n} e^{c_* q}, \quad c_* := \log(50000/45927) > 0.
\end{aligned} \tag{ECT12}$$

The last step uses $r \leq q$ and $500/567 < 1$. An exact rational logarithm-series certificate gives

$$0.084969826010092 < c_* < 0.084969826010093. \tag{ECT13}$$

This holds for ALL monic coefficient directions, including $r=0$. No r/q bounded away from zero, limiting quantile construction, or compressed spectral gap is assumed.

R36.4 3. Restore the original roots, arithmetic density and mass

Assume $q \geq 7R_k$. On E_n , $q \leq n \leq 2q$, every original root gives $|iy-z| \geq |y| - R_k$. Thus

$$|\chi(c + iy)| \geq |y|^q (1 - 63R_k/(20n))^q. \tag{ECT14}$$

The factor is positive and at least $1/2$. Since $|y| \leq 2n$, the ORIGINAL density lower bound and (ECT12) imply

$$h_{n-q}^\chi \geq \ell_{h,k} (k-2+2n)^{-B_h} (1 - 63R_k/(20n))^{2q} \frac{50\pi n}{63} n^{2n} e^{-2n} e^{c_* q}. \tag{ECT15}$$

The finite positive-real remainder used here is the one in Askey and Roy [H22, eq. (5.11.1), §5.11(ii)]. For $x > 0$ it gives $\log \Gamma(x) = (x - \frac{1}{2}) \log x - x + \frac{1}{2} \log(2\pi) + r(x)$, where $0 \leq r(x) \leq 1/(12x)$. Taking $x = 2n$, with $n \geq 1$, and using $(2n)! = 2n\Gamma(2n)$ in the retained $g_n = \sqrt{2\pi}(2n)!/4^n$ gives exactly $g_n \leq 2\pi\sqrt{2n} n^{2n} e^{-2n} e^{1/(24n)}$. The arithmetic comparison factor, when present, remains the original A_k .

By the original monic upper comparison and the positive-real Stirling remainder,

$$\omega_n \leq A_k g_n \leq 2\pi\sqrt{2n} A_k n^{2n} e^{-2n} e^{1/(24n)}. \tag{ECT16}$$

Divide (ECT16) by (ECT15). For a uniform bound, $-\log(1-x) \leq 2x$ shows that the root-factor logarithmic cost is at most $63R_k/5$. Define the explicit finite certificate

$$U_{h,k}^{\text{ctr}} = \frac{63\sqrt{2}A_k}{25c_h\sqrt{q}} \vartheta_h^{-(k-3)} (k-2+4q)^{B_h} \exp \left(\alpha(k-3) + \frac{63}{5}R_k + \frac{1}{24q} - c_* q \right). \tag{ECT17}$$

Then

$$\boxed{0 < d_n \leq U_{h,k}^{\text{ctr}} \quad (q \leq n \leq 2q).} \tag{ECT18}$$

It is still a true, possibly uninformative, bound when $U_{\text{ctr}} \geq 1$. At fixed actual h and full multiplicity,

$$\log U_{h,k}^{\text{ctr}} = -c_* q + O_h(k + \log q). \tag{ECT19}$$

All original masses, the full amplitude, convolution order and primary multiplicities are retained. The reference bands are only integration subsets. No quotient was replaced by the monomial quotient. The period observation is not needed for this bound.

R36.5 4. The penalties are exponentially small

Keep exactly the original endpoint multiplicities:

$$P_- = -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \quad (\text{ECT20})$$

$$P_+ = -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n). \quad (\text{ECT21})$$

Their combined coefficient count is $4q-1$. If $U_{\text{ctr}} < 1$,

$$0 \leq P_- + P_+ \leq -(4q-1) \log(1 - U_{h,k}^{\text{ctr}}) \leq \frac{(4q-1)U_{h,k}^{\text{ctr}}}{1 - U_{h,k}^{\text{ctr}}}. \quad (\text{ECT22})$$

Hence $P_- + P_+ = O_h(q \exp(-c_* q/2))$, improving the submitted $O_h(e_k)$. For finite use intersect (ECT22) with the submitted polynomial allowance; retain the older bound where either new guard fails.

The complete exact receiver remains

$$J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - P_- - P_+. \quad (\text{ECT23})$$

Thus $J_{\text{action}} = B_{\text{ar}} + W_{\text{ar}}$ plus a nonpositive exponentially small error. The separate monic-window comparison $W_{\text{sigma-E}_{\text{hi}}} \leq W_{\text{ar}} \leq W_{\text{sigma-E}_{\text{lo}}}$ still has error $O_h(k + \log q)$; that error has NOT become exponentially small.

The submission's exact Gamma window and finite enclosure check out:

$$W_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}, \quad (\text{ECT24})$$

$$C(q) - \log 2 - \left(\frac{1}{16q} + \frac{1}{4q-2} \right) \leq W_q^\sigma \leq C(q) - \log 2, \quad C(q) = 4q \log q + (8 \log 2 - 4)q. \quad (\text{ECT25})$$

The new action enclosure replaces the old penalty allowance by the minimum of the old bound and (ECT22). It does not evaluate a further coefficient of B_{ar} or of a separate allocation.

R36.6 5. This also controls the minimum, not only the action mean

At the original untilted source, total parity gives

$$\epsilon_{q-1}^2 = b_q(1 - d_q)/d_q, \quad \epsilon_N^2 = b_{N+1}(1 - d_N)(1 - d_{N+1})/d_{N+1} \quad (N \geq q), \quad (\text{ECT26})$$

where $b_n = \omega_n / \omega_{n-1}$. This is total parity, not an assertion that the mixed kernel and boundary pairings vanish separately.

AMT's lower monic comparison gives

$$b_n \geq \frac{a_k}{A_k D_n} n(n-1/2), \quad D_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}. \quad (\text{ECT27})$$

Let $b_{\text{low}} = (a_k / (A_k D_{2q})) q(q-1/2)$. Equation (ECT18) proves, for $U_{\text{ctr}} < 1$,

$$\min_{q-1 \leq N \leq 2q-1} \epsilon_N \geq \sqrt{b_{\text{low}}} \frac{1 - U_{h,k}^{\text{ctr}}}{\sqrt{U_{h,k}^{\text{ctr}}}}. \quad (\text{ECT28})$$

The first endpoint has only one factor $1-d_q$ and gives at least this bound. Since $\log b_{\text{low}} = O_h(k + \log q)$,

$$\liminf \frac{1}{q} \log \min_{q-1 \leq N \leq 2q-1} \epsilon_N \geq c_*/2. \quad (\text{ECT29})$$

For the original exterior benchmark $L_{\text{hk}} = \delta q(k+1)/2$ on $k=4l+1$,

$$\boxed{\liminf \frac{1}{q} \log \frac{\min_{q-1 \leq N \leq 2q-1} \epsilon_N}{L_{h,k}} \geq c_*/2 > 0.} \quad (\text{ECT30})$$

Conditional on the original arithmetic packet/source inputs, every canonical allowance in THIS window is exponentially larger than the polynomial benchmark. This is stronger than the earlier geometric-mean obstruction. It does not identify the arithmetic spectrum with the allowance, negate RH, or prove that the whole tau programme fails.

R36.7 6. Extension to any fixed proportional window

For any fixed $C > 1$ choose small fixed $\rho > 0$ and let

$$H(\rho) = 1 + \rho^2 + \frac{\rho^2}{2} \left[\log \frac{1 + \sqrt{1 - \rho^2}}{\rho} - \sqrt{1 - \rho^2} \right]. \quad (\text{ECT31})$$

The preceding integral proof gives $E(\rho) \leq H(\rho)$. Use $b = 2n/H(\rho)$, $a = \rho b$. The exponential cost is $\leq 2n$ and the two normalized factors are $(1 + \rho)/H(\rho)$, $\sqrt{1 - \rho^2}/H(\rho)$. For $q \leq n \leq Cq$ the gap is at least $c_C(\rho)q$, where

$$c_C(\rho) = 2 \log(1 + \rho) + (C - 1) \log(1 - \rho^2) - 2C \log H(\rho). \quad (\text{ECT32})$$

As $H(\rho) - 1 = O(\rho^2 \log(1/\rho))$, $c_C(\rho) = 2\rho + O_C(\rho^2 \log(1/\rho)) > 0$ for sufficiently small ρ . The band has inner radius a fixed positive constant times n , so the actual root and density costs remain $O_C(h, C)(k + \log q)$. This proves the analogous exponential bound for every fixed C , and the minimum-allowance conclusion follows as above. Constants depend on C . No uniform assertion is made for C growing with k .

A mere replacement of $2q$ by another fixed multiple therefore does not repair this canonical small-allowance strategy. A further RH-directed estimate needs a genuinely different admitted degree scaling or another same-class comparison. The exact mixed objects remain useful, but a cancelled allocation cannot be improved while its correlated action budget is held fixed.

R36.8 Verification and scope

The local executable checker uses exact rational and Gaussian-rational arithmetic for the rate certificate, Gamma norms/orthogonality and generating ODE, all $3/4/5$ partitions $k=3..101$, the complete top face, original quotient ratios, three actual canonical reference constructions, parity, and the weighted action product. Its reference mass is seven, not silently one. Separate 180-digit tests of monic exponential-weight minima and Gamma-window bounds are explicitly NON-INTERVAL diagnostics.

Normal and Python -O executions completed with byte-identical success records. All four false-formula controls were rejected in both modes. An initial combined shell batch timed out during the controls; the controls were subsequently rerun in separate modes and all eight expected errors recorded. No timeout is called a proof failure or a passing execution.

No new Lean run, independent second-reader approval, actual zeta packet, actual-period integral, or full cumulative-source audit is claimed. The infinite theta quotient, represented-zero/absence distinction, mixed supports, full arithmetic units and multiplicities are unchanged. The proof requires the explicit original AMT density envelopes and standard positive-real Stirling bounds. It does not assume the desired exterior upper estimate.

Primary literature: NIST Digital Library of Mathematical Functions, sections 18.23.7 and 18.19 (the Meixner–Pollaczek generating function and normalization), 5.11.1/5.11(ii) (positive-real Stirling remainder), and 19.2 (elliptic-integral conventions). The two Jensen calculations are included above; no novelty claim is made for those classical ingredients.

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R37 The established low-polynomial bound in the complete packet action

This section propagates the existing AKS26–32 and FEX31–33 estimates through the exact inverse-dual map TAC6b and TAC9a. The estimate itself belongs to those earlier calculations. The contribution here is its explicit insertion into TAC10, TAC15–18 and FTR31, with all finite constants, frames and four endpoint signs retained. The programme’s recorded origin and the creators of its deposited notes remain credited [H19, H20].

R37.1 The original metric and the finite bound

Keep the actual hypothetical simple quartet, its fixed period and branch, and the complete conductor of TAC1b. Write

$$\begin{aligned} a = 4l + 1 \geq 9, \quad q = (a + 1)^2, \quad s \in \{1, a\}, \quad c = a/2, \quad R_a = a\sqrt{\delta^2 + \gamma^2}, \\ \beta_{a,u,w} = c + (2u - a)\delta + i(2w - a)\gamma, \quad \chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad (\text{LRP1}) \\ E_a = \mathbb{C}[S]/(\chi_a), \quad L_v = \mathbb{C}[S]_{<v} \subset E_a, \quad 0 \leq v \leq 80. \end{aligned}$$

The original inclusion of L_v in the monomial coefficient frame of E_a is $P_{<v}$. Let $G_N^{(s)}$ be the quotient Gram of $\mathbb{C}[S]_{\leq N} \rightarrow E_a$, for the original Gamma source

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \quad S = c + iy. \quad (\text{LRP2})$$

Thus $G_{L,N}^{(s)} = P_{<v}^* G_N^{(s)} P_{<v}$. The map and metric are the ones in FEX31 and TAC6b; the target’s centre $c - 4$ is not substituted for the centre c of this whole numerator packet.

For completeness the finite covariance estimate needed here is derived on the actual source. Set $b_s = s/2$. Its original orthogonal polynomials and their norms are

$$\begin{aligned} p_0 = 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(d-1+b_s)p_{d-1}, \quad h_{d,s} = M_s d!(b_s)_d, \\ u_d(S) = p_d((S-c)/i)/\sqrt{h_{d,s}}, \quad \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d = (1+z^2)^{-b_s/2} e^{y \arctan z}. \quad (\text{LRP3}) \end{aligned}$$

The precise classical dictionary is $p_d(y) = d! P_d^{(s/4)}(y/2; \pi/2)$. The authored NIST DLMF formulas give its weight, norms and generating function, including this change of variable [H21, eqs. (18.19.8)–(18.19.9), (18.23.7)]. The mass and phase calculation is given in HPS and TAC1a; AKS15–16 also derives the generating function from this recurrence and the norm from the original transform $\int e^{ty} dm_s = M_s (\cos t)^{-b_s}$, $|t| < \pi/2$.

Let C_s have the original monomial coefficient columns $[u_0], \dots, [u_{q-1}]$. It is invertible, with determinant $i^{-q(q-1)/2} \prod_{d < q} h_{d,s}^{-1/2}$. Put

$$v_j = C_s^{-1} [u_{q+j}], \quad K_j = I_q + \sum_{r=0}^{j-1} v_r v_r^*, \quad T_s = C_s^{-1} P_{<v}, \quad 0 \leq j \leq q+1. \quad (\text{LRP4})$$

The full source map in its orthonormal basis is $C_s [I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$: it has the required image and is orthogonal to the full relation kernel. Pythagoras therefore proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := G_{L,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s. \quad (\text{LRP5})$$

This is the original AKS26 compression, with the literal inclusion $P_{<v}$ in place of a general fixed kernel frame.

Here are the complete finite constants from AKS3 and AKS30, with κ_0 denoting AKS30's scalar c_0 , to avoid confusion with the source centre:

$$\begin{aligned}
t_q &= \pi/2 - 1/q, \quad \alpha_q = 2 \log 3/t_q, \quad \Sigma_q = (\alpha_q^q - 1)/(\alpha_q - 1), \\
B_{s,q} &= [(9/25) \sin(1/q)]^{-s/2} \exp\{2R_a(\log 3 + t_q/2)\}, \\
\mathcal{L}_{s,j} &= \log \left(1 + B_{s,q} \Sigma_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(s/2)_d} (25/16)^d \right), \\
\alpha_\infty &= 4 \log 3/\pi, \quad \kappa_0 = \log \frac{25 \log 3}{4\pi} > 0, \\
E_{s,q}(R_a) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R_a \left(\log 3 + \frac{\pi}{4} \right) \\
&\quad + \log((q+1)(4q+1)) - 2 \log(\alpha_\infty - 1).
\end{aligned} \tag{LRP6}$$

An empty sum has value zero. To verify the covariance bound including every root coefficient, put $P(y) = i^{-q} \chi_a(c + iy)$ and list its roots $\omega_\nu = (\beta_\nu - c)/i$ in their original order. All satisfy $|\omega_\nu| \leq R_a$. Newton interpolation gives, for complex z ,

$$\text{rem}_P e^{zy} = \sum_{r=0}^{q-1} [\omega_1, \dots, \omega_{r+1}] e^{z \cdot} \prod_{\nu=1}^r (y - \omega_\nu).$$

The divided difference equals the integral of $z^r e^{z \sum \theta_\nu \omega_\nu}$ over the standard r -simplex of volume $1/r!$. This identity follows by expanding the exponential: the simplex moments are $\prod n_\nu!/(r + \sum n_\nu)!$, and the Newton divided difference of y^{r+n} is the complete homogeneous polynomial of degree n in the nodes. Hence its absolute value is at most $|z|^r e^{R_a|z|}/r!$.

For $0 < t < \pi/2$ the positive cosh series in the original transform gives $\|y^r\| \leq \sqrt{M_s}(\cos t)^{-s/4}(2/t)^r r!$. Expanding the entire Newton product and bounding each elementary symmetric coefficient gives

$$\left\| \prod_{\nu=1}^r (y - \omega_\nu) \right\| \leq \sqrt{M_s}(\cos t)^{-s/4} (2/t)^r r! e^{R_a t/2}.$$

For a complex circle $|w| = z < 1$, use $|\arctan w| \leq \text{arctanh } z$ and $|1 + w^2| \geq 1 - z^2$ in the finite remainder of LRP3. Cauchy coefficient extraction, followed by the preceding full-vector norm bound and division by $h_{d,s}$, yields

$$\begin{aligned}
\|v_{d-q}\|^2 &\leq \frac{d!}{(s/2)_d} z^{-2d} [(1 - z^2) \cos t]^{-s/2} e^{2R_a(\text{arctanh } z + t/2)} \\
&\quad \times \left[\sum_{r=0}^{q-1} (2 \text{arctanh } z/t)^r \right]^2.
\end{aligned} \tag{LRP7}$$

This extraction is performed on a finite remainder vector holomorphic in $|w| < 1$; it requires no integral of the generating function over the original source on that large circle. Parseval in its low-degree orthonormal basis identifies the left side with the full remainder norm divided by $h_{d,s}$. The two original masses thus cancel by this exact equality. Setting $z = 4/5$, $t = t_q$ and summing the complete positive rank-one matrices proves

$$I_q \preceq K_j \preceq e^{\mathcal{L}_{s,j}} I_q. \tag{LRP8}$$

The finite estimate $\mathcal{L}_{s,j} \leq 2\kappa_0 q + E_{s,q}(R_a)$ for $j = q, q+1$ follows as in AKS31. Explicitly, $d!/(s/2)_d \leq 4^d / \binom{2d}{d} \leq 2d + 1$, so the sum in LRP6 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also

$$\Sigma_q \leq \frac{\alpha_q^q}{\alpha_\infty - 1}, \quad 2q \log \alpha_q \leq 2q \log \alpha_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first holds since $\alpha_q \geq \alpha_\infty > 1$; the second uses $-\log(1-x) \leq x/(1-x)$, and the third is the sine chord bound. The exponential in LRP6 uses $t_q \leq \pi/2$. The order- q terms add to $2\kappa_0 q$, and $\log(1+e^x) \leq \log 2 + \max(0, x)$ gives the stated finite bound. Here $q \geq 100$ satisfies AKS31's guard $q \geq 36$.

For $v > 0$ set $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Inversion and compression in LRP8 show $0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}$ and $\bar{d}_j \leq v \mathcal{L}_{s,j}$. Define

$$\begin{aligned} V_s &= \|v_0\|^2, & \theta_s &= v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0, \\ \ell_{a,s} &= \log \frac{1 + V_s}{1 + V_s - \theta_s} = d_1, & \mathcal{B}_{a,s}^L &= v[4\kappa_0 q + 2E_{s,q}(R_a)], & u_{a,s} &= \mathcal{B}_{a,s}^L - \ell_{a,s}. \end{aligned} \quad (\text{LRP9})$$

The matrix defining θ_s is the orthogonal projection onto $\text{im } T_s$, so $0 \leq \theta_s \leq V_s$. The inverse identity $(I + v_0 v_0^*)^{-1} = I - v_0 v_0^* / (1 + V_s)$ and its compressed rank-one determinant prove the formula for d_1 . For $v = 0$ set $\ell_{a,s} = u_{a,s} = \mathcal{B}_{a,s}^L = 0$; all low determinants then equal one.

The exact primal four-endpoint return is

$$F_{L_a}^{(s)} = \log \frac{\det G_{L,q-1}^{(s)} \det G_{L,q}^{(s)}}{\det G_{L,2q-1}^{(s)} \det G_{L,2q}^{(s)}} = d_q + d_{q+1} - d_1, \quad (\text{LRP10})$$

$$\boxed{0 \leq \ell_{a,s} \leq F_{L_a}^{(s)} \leq u_{a,s} \leq \mathcal{B}_{a,s}^L.}$$

The lower bound uses $d_q, d_{q+1} \geq d_1$ and the upper bound uses LRP8 and LRP6 at both high increments. This retains the sharpened AKS28 interval in addition to FEX32's bound.

For fixed quartet and conductor, $R_a = O_{\delta,\gamma}(a)$ and $v \leq 80$. The displayed constants give $E_{1,q}(R_a) = O_{\delta,\gamma}(a + \log q)$ and $E_{a,q}(R_a) = O_{\delta,\gamma}(a \log q)$. Consequently

$$0 \leq F_{L_a}^{(s)} \leq \mathcal{B}_{a,s}^L = O_{\delta,\gamma}(q) = o(aq), \quad s = 1, a. \quad (\text{LRP11})$$

This holds along the original discrete family; its finite value is never set to zero. The difference between source orders has the directed bound

$$\ell_{a,a} - u_{a,1} \leq F_{L_a}^{(a)} - F_{L_a}^{(1)} \leq u_{a,a} - \ell_{a,1}. \quad (\text{LRP12})$$

No sign for that difference is required or inferred.

R37.2 The exact dual map and propagation through the receiving equations

TAC6b proves on the complete original dual packet that $Q_{v,N}^{(s)} = \overline{(G_{L,N}^{(s)})}^{-1}$ in the paired coefficient frames. Its proof minimizes the full dual norm over the annihilator of L_v ; this is the inverse restriction metric. The Schur minimum and block factorization used there agree with the classical positive-matrix formulas [H23, Appendices A.5.5, p. 650, and C.4.1, p. 673]. Since each determinant is positive real, its logarithm changes sign under this conjugate inverse. With the original dual sign operator $\mathcal{R}f = -f(q-1) - f(q) + f(2q-1) + f(2q)$ this proves

$$\mathcal{V}_{a,s} = \mathcal{R} \log \det Q_{v,N}^{(s)} = -\mathcal{R} \log \det G_{L,N}^{(s)} = F_{L_a}^{(s)}. \quad (\text{LRP13})$$

Thus LRP10 bounds the actual final factor in TAC10, without changing the analytic middle quotient or the conductor restriction.

Retain the exact complete action TAC15 and put

$$A_{a,s,R}^0 = \mathcal{C}_{a,s} + \delta_{a,s}^{\text{ar}} + \mathcal{T}_{a,s,R} + \mathcal{W}_q^\sigma. \quad (\text{LRP14})$$

All four terms keep TAC's original metric and cutoffs. In particular $\delta_{a,1}^{\text{ar}} = \delta_a^\sigma$ and $\delta_{a,a}^{\text{ar}} = \mathcal{H}_a + \delta_a^\sigma$. Substituting LRP10 in the two sides of TAC16 gives

$$\begin{aligned} A_{a,s,R}^0 + \ell_{a,s} + \Xi_{a,s}^- - E_{\text{hi}} - \Pi_{h,a} &\leq J_a, \\ J_a &\leq A_{a,s,R}^0 + u_{a,s} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}. \end{aligned} \quad (\text{LRP15})$$

This is a finite interval for the same action, proved by scalar inequalities on the same actual matrices. At $a = j = k + 4$ and the original R_j , FTR28's three joint controls give the further upper bound

$$\begin{aligned} J_j \leq & \mathcal{C}_{j,s} + \delta_{j,s}^{\text{ar}} + \mathcal{W}_{q_j}^\sigma + \mathfrak{U}_R + u_{j,s} \\ & + \Xi_{j,s}^+ + \mathfrak{E}_{s,R} + E_{10}. \end{aligned} \tag{LRP16}$$

Indeed substitute $\mathcal{T}_{j,s,R} \leq \mathfrak{U}_R$ in LRP15. Every coefficient remains one, exactly as in FTR31. Since $jq_j/(kq_k) \rightarrow 1$, LRP11 also controls this term on the original comparison scale kq_k .

For the allocation identity TAC17 set $\mathcal{E}_{a,s}^{\text{src}} = \sum_{i=0}^3 \sigma_i e_{N_i,s}$, with $\sigma = (1, 1, -1, -1)$, and keep the actual observation kernel K_a and $t_0 = \dim(K_a \cap L_v)$. FEX32 passes the same full covariance inequality by restriction and minimization to $(K_a + L_v)/K_a$, of dimension $v - t_0$. Therefore

$$\begin{aligned} -u_{a,s} & \leq \delta_{K,a,s} - \mathcal{E}_{a,s}^{\text{src}} = -F_{L_a}^{(s)} + F_{(K_a+L_v)/K_a}^{(s)} \\ & \leq (v - t_0)(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \ell_{a,s}. \end{aligned} \tag{LRP17}$$

Both bounds are $O_{\delta,\gamma}(q)$. All source Jacobians remain in $\mathcal{E}_{a,s}^{\text{src}}$; this statement makes no estimate for them. At $s = a$, TAC18 still cancels the exact two occurrences of $\delta_{K,a,a}$ and the exact five allocation terms with coefficients $+1/4$ and $-1/4$. Applying separate upper bounds before that identity is unnecessary; the cancellation remains exact.

The two original Gamma source orders also give a useful exact check on the same arithmetic action. Write $\Delta X = X_{a,a} - X_{a,1}$ for its source-order difference and retain the whole native return $\mathcal{T}_{a,s} = \mathcal{T}_{a,s}^{\leq R} + \mathcal{T}_{a,s,R}$. Subtracting the two TAC15 identities cancels J_a , W_a and both arithmetic penalties. TAC13 gives $\delta_{a,a}^{\text{ar}} - \delta_{a,1}^{\text{ar}} = \mathcal{H}_a$. Hence

$$\begin{aligned} \Delta \mathcal{C} + \Delta \mathcal{T} + \Delta \Xi + \mathcal{H}_a & = -(F_{L_a}^{(a)} - F_{L_a}^{(1)}), \\ \ell_{a,1} - u_{a,a} & \leq \Delta \mathcal{C} + \Delta \mathcal{T} + \Delta \Xi + \mathcal{H}_a \leq u_{a,1} - \ell_{a,a}. \end{aligned} \tag{LRP18}$$

The combined expression is $O_{\delta,\gamma}(q) = o(aq)$ by LRP11. Its conductor, native and source-correction terms retain their exact values and correlation. The calculation compares source orders at the same packet degree a ; it does not substitute for the separate degree- k proper-source return of PRD and NFR.

The cumulative receiving calculation therefore has an evaluated subleading bound for its low-polynomial factor. Its exact value, the independent proper degree- k source, the complete conductor restriction $\mathcal{C}_{a,s}$, the correlated native tail, the arithmetic correction and all nine action blocks remain in their proved formulas. LRP11 evaluates only this identified low factor; the finite controls for the remaining factors retain their actual values.

R38 The absolute common kernel: the new finite result

This complete incoming calculation sharpens the current absolute common kernel estimate. Its original comparison was against the preceding logarithmic bound; the intervening KAF chapter already established the $O(kq)$ scale. The new finite envelope reduces its uniform limsup constant from $16(4 + 10 \log 2)$ to $32 \log(25 \log 3/(4\pi))$. AKJ below takes the pointwise minimum of both finite bounds. No limiting kernel coefficient is inferred from these upper bounds.

Retain the actual hypothetical complete quartet and its entire quotient

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha)^m, \quad v_h = 2\xi/h.$$

The full primary unit is $U = M_{j_h v_h}$. The original tensor order, centre, polynomial, and coefficient quotient are

$$\begin{aligned} k &= 4l + 1, \quad l \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \lambda_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \quad (\text{AKS1})$$

For the principal conclusion $m = 1$. The source-operator estimates below also hold with the stated repeated-root list for $m > 1$, but no simple-quartet rank is assigned to those multiplicities.

Let $\Lambda_k(u) : E \rightarrow B_k(u)$ be the original observation, $K(u) = \ker \Lambda_k(u)$, and $I : K(u) \hookrightarrow E$ its original coefficient inclusion. For a source $s \in \{1, k\}$, let $G_N^{(s)}$ be the original minimum-remainder Gram, in the frame $(1, S, \dots, S^{q-1})$. Define $H_{K,N}^{(s)} = I^* G_N^{(s)} I$ and retain exactly

$$F_K^{(s)} = \log \det H_{K,q-1}^{(s)} + \log \det H_{K,q}^{(s)} - \log \det H_{K,2q-1}^{(s)} - \log \det H_{K,2q}^{(s)}. \quad (\text{AKS2})$$

Every empty determinant is one. All conjugate transposes refer to the specified coordinate matrices, not to a bilinear residue pairing.

Here are entirely explicit positive numbers used in the bound:

$$\begin{aligned} R &= k\sqrt{\delta^2 + \gamma^2}, \quad b_s = s/2, \quad t_q = \frac{\pi}{2} - \frac{1}{q}, \quad \mathbf{a}_q = \frac{2 \log 3}{t_q}, \quad \mathbf{s}_q = \frac{\mathbf{a}_q^q - 1}{\mathbf{a}_q - 1}, \\ \mathbf{b}_{s,q} &= \left[\frac{9}{25} \sin \frac{1}{q} \right]^{-b_s} \exp \left(2R \left(\log 3 + \frac{t_q}{2} \right) \right), \\ \mathcal{Z}_{s,j} &= 1 + \mathbf{b}_{s,q} \mathbf{s}_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(b_s)_d} \left(\frac{25}{16} \right)^d, \quad \mathcal{L}_{s,j} = \log \mathcal{Z}_{s,j} \quad (0 \leq j \leq q + 1). \end{aligned} \quad (\text{AKS3})$$

The empty sum gives $\mathcal{Z}_{s,0} = 1$. In the admitted family $q \geq 36$, so $t_q \in (0, \pi/2)$ and $\mathbf{a}_q > 1$. These parameters are radii in the existing Laplace and generating functions; neither is a new source order.

Main theorem (absolute finite bound). For the original simple quartet, every original logarithm branch, and every $|u| \geq R_*$, where R_* is precisely RPC21 or RPC23 (recorded in Section R44), put $r = 8k - 16$. Then

$$\boxed{0 \leq F_K^{(s)}(k, u) \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}), \quad s \in \{1, k\}.} \quad (\text{AKS4})$$

There is a stronger, fully period-dependent interval in AKS28 below. In particular, with

$$\kappa_* = 32 \log \left(\frac{25 \log 3}{4\pi} \right), \quad 25.02078097649 < \kappa_* < 25.02078097650, \quad (\text{AKS5})$$

one has the uniform statement

$$\boxed{0 \leq \limsup_{\substack{k \rightarrow \infty \\ k \equiv 1 \pmod{4}}} \sup_{\substack{|u| \geq R_* \\ \text{original branches}}} \frac{F_K^{(s)}(k, u)}{kq} \leq \kappa_*}. \quad (\text{AKS6})$$

Here the source may be $s = 1$ for every k , or $s = k$. The same bound holds for the actual arithmetic kernel. Its exact finite allowances are in AKS34–36. Thus the absolute kernel is $O_h(kq)$ on this domain, not only $O_h(kq \log q)$. Neither AKS5 nor AKS6 calls κ_* the actual leading coefficient.

R39 An exact coefficient realization of the actual lost directions

R39.1 The period matrix and invariant restriction, with their full coefficients

For $m = 1$, order the one-factor roots as $(++)$, $(+-)$, $(-+)$, $(--)$, and let $\mathcal{E}_{\text{prim}}$ have their original Lagrange idempotents as columns. Set $\zeta = \exp(2\pi i/5)$ and $F_{jt} = \zeta^{jt} - 1$, $1 \leq j, t \leq 4$. The exact inverse is $(F^{-1})_{tj} = \zeta^{-tj}/5$. Retain

$$\begin{aligned} \Pi[P]_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, & \Phi'_h &= h, & \Phi_h(0) &= 0, \\ \Gamma_j &= -\ell_0 + \ell_j, & \arg \ell_j &= (\arg u + \pi + 2\pi j)/5, \\ \mathbf{H} &= F^{-1} \Pi U \mathcal{E}_{\text{prim}}, & D &= \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \\ Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_1 t_4 - t_2 t_3, & Q_a(x) &= Q(D^{-a}x). \end{aligned} \quad (\text{AKS7})$$

The integral always uses the degree-less-than-four representative. Invertibility of Π , all contour phases, and its full determinant are the PDE/PQO results in the supplied proofs. The four unit values are $(a, \bar{a}, \bar{a}, a)$, with the actual $a = v_h(\rho) \neq 0$; no value of a is selected. Hence $\mathbf{H}$ is the original invertible matrix.

The following formulas make the coefficient realization directly computable. Write

$$\mathcal{I}_{k,0} = \{\nu \in \mathbb{Z}_{\geq 0}^4 : |\nu| = k, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}.$$

For $\nu \in \mathcal{I}_{k,0}$ define $B_{\nu;(a,b)}$ by summing over 4×4 arrays $n_{t,\alpha} \geq 0$ with row sums ν_t , total plus-real count a , and total plus-imaginary count b :

$$B_{\nu;(a,b)} = \sum_{\substack{\sum_{\alpha} n_{t,\alpha} = \nu_t \\ \sum_{t,\alpha: \epsilon_{\alpha}=+} n_{t,\alpha} = a \\ \sum_{t,\alpha: \eta_{\alpha}=+} n_{t,\alpha} = b}} \left(\prod_{t=1}^4 \frac{\nu_t!}{\prod_{\alpha} n_{t,\alpha}!} \right) \prod_{t,\alpha} \mathbf{H}_{t,\alpha}^{n_{t,\alpha}}. \quad (\text{AKS8})$$

Let $\mathbf{V}_{(a,b),d} = \lambda_{a,b}^d$ for $0 \leq d < q$ and set $\mathbf{A} = \mathbf{B}\mathbf{V}$. All entries of $\mathbf{A}$ are now finite expressions in the actual periods and unit. There is no generic period matrix in this definition.

For completeness, put $t_{\alpha}(w) = \exp((\eta_{\alpha}\gamma - i\epsilon_{\alpha}\delta)w)$ and $x(w) = \mathbf{H}t(w)$. Expansion gives

$$x(w)^{\nu} = \sum_{a,b} B_{\nu;(a,b)} e^{\omega_{a,b}w}, \quad \omega_{a,b} = (2b - k)\gamma - i(2a - k)\delta.$$

In the original primary algebra, $e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b}w} \varepsilon_{a,b}$, where $Y = (M_S - cI)/i$. The exponentials are independent by the Vandermonde of the distinct $\omega_{a,b}$. OCS9's unscaled symmetric word basis therefore identifies the original observation with

$$j\Lambda_k = \mathcal{F}_0 \mathbf{A}, \quad \ker \mathbf{A} = K(u) \quad \text{in the original coefficient frame.} \quad (\text{AKS9})$$

Every multinomial in AKS8 belongs to its actual ordered words; the word multiplicity is not inserted a second time. The exact inherited row Gram of $\mathcal{F}_0$ is the restriction to $\mathcal{I}_{k,0}$ of

$$\mathcal{G}_{\nu,\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}. \quad (\text{AKS10})$$

This follows by taking the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T F^* F y)^k$, since $F^* F = 5(I + \mathbf{1}\mathbf{1}^*)$. It is positive definite and includes every row cross term.

Here is the precise role of the orbit ideal. The map sending a homogeneous polynomial f to the covector specified by $\Theta_k(f)(e^{wY}[1]) = f(x(w))$ has kernel $Q\mathbb{C}[x]_{k-2}$ and is onto $E^\vee$. Indeed equal exponential indices correspond exactly to monomial differences divisible by $t_1 t_4 - t_2 t_3$; the displayed Vandermonde proves that there are no other relations. Its invariant image is $\text{im } \Lambda_k^\vee$. On the five-member domain, the five Q_a are distinct prime quadrics. An invariant polynomial divisible by Q is divisible by each Q_a , and therefore by their full product; its quotient is invariant by cancellation in the polynomial ring. Consequently

$$\mathbb{C}[x]^{\langle D \rangle} \cap (Q) = \left(\prod_{a=0}^4 Q_a \right) \mathbb{C}[x]^{\langle D \rangle}, \quad (\text{AKS11})$$

$$K(u)^\vee \simeq \mathbb{C}[x]_k / (\mathbb{C}[x]_k^{\langle D \rangle} + Q\mathbb{C}[x]_{k-2}),$$

$$b := \text{rank } \mathbf{A} = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad r = q - b = 8k - 16.$$

Negative degrees are zero. The count is $d_{5,d,0} = ((\binom{d+3}{3} + 4\epsilon_d)/5$, $\epsilon_d = \mathbf{1}_{d \equiv 0(5)} - \mathbf{1}_{d \equiv 1(5)}$; subtracting the cubics gives the stated expression for $k \geq 10$, and $k = 5, 9$ give 12, 44 directly. This is the inherited OQX calculation, repeated to specify which kernel AKS9 realizes. The source metric is still to be applied to this kernel, not to the invariant ring itself.

R39.2 A determinant formula requiring no assumed nonzero pivot

Let $\mathcal{G}_0$ be the full positive row Gram in AKS10 and set

$$W = \mathbf{A}^* \mathcal{G}_0 \mathbf{A}, \quad \sigma_j = [z^j] \det(I_q + zW), \quad P_K = \frac{1}{\sigma_b} \sum_{j=0}^b (-1)^j \sigma_{b-j} W^j. \quad (\text{AKS12})$$

Since W has exactly b positive eigenvalues, $\sigma_b > 0$. The scalar polynomial in AKS12 is one at zero and zero at every nonzero eigenvalue: it is the normalized product of the factors $\lambda_i - z$ for the b positive eigenvalues. Hence $P_K = P_K^* = P_K^2$ and $\text{im } P_K = K(u)$. This is the projector derived from the literal observation, not a freely chosen projection. Replacing $\mathcal{G}_0$ by any other positive coordinate Gram leaves this same coefficient-orthogonal kernel projector, since its image and orthogonal complement are unchanged. No such replacement is needed here.

At every original endpoint define

$$D_{K,N}^{(s)} = [z^r] \det(I_q + z P_K G_N^{(s)} P_K). \quad (\text{AKS13})$$

These numbers are strictly positive for $r > 0$ and are one for $r = 0$. To prove their exact metric meaning, choose an orthonormal coefficient basis U_K of $\text{im } P_K$ just for the proof. In a unitary extension, $P_K G_N P_K$ has its positive block $U_K^* G_N U_K$ and a zero block. Thus $D_{K,N} = \det(U_K^* G_N U_K)$. Write the original inclusion as $I = U_K T$; then $|\det T|^2 = \det(I^* I)$ and

$$\det H_{K,N}^{(s)} = \det(I^* I) D_{K,N}^{(s)}, \quad \boxed{F_K^{(s)} = \log \frac{D_{K,q-1}^{(s)} D_{K,q}^{(s)}}{D_{K,2q-1}^{(s)} D_{K,2q}^{(s)}}}. \quad (\text{AKS14})$$

This supplies an equivalent exact coefficient determinant formula for all the $8k - 16$ lost directions, including the entire period dependence. The original frame factor is present at each endpoint and cancels only with its four actual signs. It is unnecessary to suppose any particular minor nonzero. Alternatively, selecting the first independent r columns of P_K gives an explicit coefficient frame; the determinant formula avoids even that chart choice.

For $m > 1$ replace AKS8–9 by OCS12–18’s full confluent matrix $\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}$ and its primary isomorphism. AKS12–14 are then identical with the actual rank, not with the rank in AKS11. All primary nilpotents and all unit derivatives remain in that matrix.

R40 A bounded exact remainder operator on the original source

R40.1 The complete source norms and coefficient map

For either existing source set $M_s = (2\pi)^{s/2}$ and retain

$$dm_{0,s}(y) = \frac{M_s 2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad \int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}, \quad |t| < \pi/2. \quad (\text{AKS15})$$

The supplied IGO1–3 proof derives this transform by the beta integral, Fourier inversion and a contour shift within $|\operatorname{Im} x| < \pi/2$. In particular it includes the exact masses $M_1 = \sqrt{2\pi}$ and $M_k = (2\pi)^{k/2}$ and exponential domination on every smaller strip. The original real monic polynomials satisfy

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d &= (1 + z^2)^{-b_s/2} e^{y \arctan z}, \quad h_d^{(s)} = \|p_d\|^2 = M_s d! (b_s)_d. \end{aligned} \quad (\text{AKS16})$$

The generating formula follows by differentiating in z . To check the normalization directly, integrate the product of two generating functions at small real z, w . AKS15 and the cosine addition identity give $M_s(1 - zw)^{-b_s}$. Taking any finite pair of derivatives at zero, justified by the stated exponential domination, gives orthogonality and the norms in AKS16. Thus the leading phase in $u_d(S) = p_d((S - c)/i)/\sqrt{h_d^{(s)}}$ is exactly i^{-d} . The same generating function agrees with NIST DLMF 18.23.7 after the original parameter rescaling; it is derived here from the supplied recurrence, not substituted for it.

List the original roots $\omega_\nu = (\lambda_{a,b} - c)/i$ in lexicographic order, repeating each exactly e times, and put

$$P(y) = i^{-q} \chi(c + iy) = \prod_{\nu=1}^q (y - \omega_\nu), \quad \Phi : [f(y)] \mapsto [f((S - c)/i)]. \quad (\text{AKS17})$$

The inverse is substitution $S = c + iy$, and $P((S - c)/i) = i^{-q} \chi(S)$. This is an exact algebra isomorphism, including the complete confluent orders; there is no monomial comparison divisor. Let C_s have columns $[u_0], \dots, [u_{q-1}]$. Its nonzero determinant is $i^{-q(q-1)/2} \prod_{d < q} (h_d^{(s)})^{-1/2}$. Define

$$v_j = C_s^{-1} [u_{q+j}], \quad \mathbf{K}_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q + 1. \quad (\text{AKS18})$$

Parseval in the low-degree original orthonormal polynomial basis proves

$$\|v_j\|_2^2 = \frac{\|\operatorname{rem}_P p_{q+j}\|_{m_{0,s}}^2}{M_s (q+j)! (b_s)_{q+j}}. \quad (\text{AKS19})$$

This is the full norm of the actual remainder, not a diagonal entry of an approximated quotient Gram.

R40.2 Newton interpolation with all roots and their orders

For $0 \leq a < q$ define $N_a(y) = \prod_{\nu=1}^a (y - \omega_\nu)$, with $N_0 = 1$. For every complex v , the unique entire-function remainder is

$$\text{rem}_P(e^{vy}) = \sum_{a=0}^{q-1} [\omega_1, \dots, \omega_{a+1}] e^{v \cdot} N_a(y), \quad |[\omega_1, \dots, \omega_{a+1}] e^{v \cdot}| \leq \frac{|v|^a}{a!} e^{R|v|}. \quad (\text{AKS20})$$

Here divided differences at repeated nodes mean Hermite divided differences, not evaluation with a zero denominator.

A proof covering the confluent case is useful. On the standard a -simplex with barycentric coordinates $\theta_1, \dots, \theta_{a+1}$,

$$[\omega_1, \dots, \omega_{a+1}] e^{v \cdot} = v^a \int_{\Delta_a} e^{v \sum \theta_\nu \omega_\nu} d\theta.$$

The integral of $\prod \theta_\nu^{n_\nu}$ is $\prod n_\nu! / (a + \sum n_\nu)!$, as follows by iterated integration of integer beta integrals. Expanding the exponential therefore gives $\sum_{n \geq 0} v^{a+n} h_n(\omega_1, \dots, \omega_{a+1}) / (a+n)!$. For monomials, the divided difference of y^{a+n} is exactly h_n : this follows by multiplying the geometric series $\prod (1 - \omega_\nu z)^{-1}$ and the Newton division recurrence. It is a polynomial identity in the nodes and so also holds at repeated nodes. This proves the integral formula. Iterated Newton division for polynomials, followed by the convergent exponential series, proves the first equality in AKS20, including every required derivative. Finally the simplex has volume $1/a!$ and $|\sum \theta_\nu \omega_\nu| \leq R$, proving its stated bound.

For any $0 < t < \pi/2$, put $\tau = 2/t$. Positivity of every term of the cosh series in AKS15 gives

$$\|y^a\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a!.$$

Indeed its square is at most $M_s (\cos t)^{-b_s} (2a)! t^{-2a}$ and $(2a)! \leq 4^a (a!)^2$. Expanding the *whole* Newton polynomial, then using $|e_j(\omega_1, \dots, \omega_a)| \leq \binom{a}{j} R^j$, proves

$$\begin{aligned} \|N_a\|_{m_{0,s}} &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \sum_{j=0}^a \binom{a}{j} R^{a-j} \tau^j j! \\ &= \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! \sum_{n=0}^a \frac{(R/\tau)^n}{n!} \\ &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! e^{R/\tau}. \end{aligned} \quad (\text{AKS21})$$

There are no omitted polynomial coefficients. Combining AKS20–21 gives for every complex v

$$\|\text{rem}_P(e^{vy})\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} e^{R(|v|+t/2)} \sum_{a=0}^{q-1} (2|v|/t)^a. \quad (\text{AKS22})$$

R40.3 Cauchy extraction in the actual Gamma generating function

Fix $0 < z < 1$ and write $a_z = \text{arctanh } z$. For $|w| = z$ the defining power series gives $|\arctan w| \leq a_z$ and $|1 + w^2| \geq 1 - z^2$. Apply the literal finite-dimensional remainder map to AKS16 and use Cauchy's formula for its d th coefficient. The norm triangle inequality on that circle and AKS22 give

$$\begin{aligned} \frac{\|\text{rem}_P p_d\|_{m_{0,s}}}{d!} &\leq z^{-d} (1 - z^2)^{-b_s/2} \sqrt{M_s} (\cos t)^{-b_s/2} \\ &\quad \times e^{R(a_z+t/2)} \sum_{a=0}^{q-1} (2a_z/t)^a. \end{aligned} \quad (\text{AKS23})$$

Cauchy's formula is legitimate for the finite remainder vector: its entries are holomorphic for $|w| < 1$ by the actual matrix generating identity OPG7. The source norm is taken only after this finite-vector identity; there is no unjustified integral of a nonintegrable generating function on the large circle.

Substitution of AKS23 into AKS19 retains the factorial denominator and cancels the two occurrences of the original mass, giving

$$\|v_j\|_2^2 \leq \frac{d!}{(b_s)_d} z^{-2d} [\cos t(1 - z^2)]^{-b_s} e^{2R(az+t/2)} \left[\sum_{a=0}^{q-1} (2a_z/t)^a \right]^2, \quad d = q + j. \quad (\text{AKS24})$$

Adding the *complete* rank-one positive matrices in AKS18 now proves $K_j \preceq \mathcal{Z}_{s,j} I_q$ after setting $z = 4/5$ and $t = t_q$. Indeed $a_{4/5} = \log 3$, $1 - z^2 = 9/25$, and $\cos t_q = \sin(1/q)$, so the exact scalar obtained is AKS3. The high index $j = q + 1$ includes the original polynomial of degree $2q$.

There is also a root-by-root finite refinement. Put $\mu_{s,d} = \int y^{2d} dm_{0,s}$, $R_a = \max_{\nu \leq a+1} |\omega_\nu|$, and

$$B_{s,a}^{\text{root}} = \sum_{d=0}^a |e_{a-d}(\omega_1, \dots, \omega_a)| \sqrt{\mu_{s,d}}, \quad (\text{AKS25})$$

$$J_s^{\text{root}}(z) = \frac{(1 - z^2)^{-b_s}}{M_s} \left[\sum_{a=0}^{q-1} \frac{a_z^a}{a!} e^{a_z R_a} B_{s,a}^{\text{root}} \right]^2.$$

Replace the scalar multiplying $d!z^{-2d}/(b_s)_d$ in AKS24 by $J_s^{\text{root}}(z)$ to get a valid sharper bound, by the same proof before AKS21 is relaxed. Every moment is exactly $M_s(2d)! [w^{2d}] (\cos w)^{-b_s}$. Thus this refinement keeps every actual root coefficient and a finite original moment formula. AKS21 bounds it by the explicit scalar in AKS24. The proof never replaces the source determinant by a product of diagonal moments: AKS24 is a full-vector operator bound, and the next step restricts the inverse of the full covariance.

R41 The four original kernel Grams and the evaluated scale

The original orthonormal source map at degree $q + j - 1$ is $C_s[I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$. It has that image and is orthogonal to the full relation kernel, so Pythagoras proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := H_{K,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s, \quad T_s = C_s^{-1} I. \quad (\text{AKS26})$$

This verifies the source-minimum step in the coefficient determinant AKS13–14. Neither a cross block nor any original relation is removed.

Since $I_q \preceq K_j \preceq \mathcal{Z}_{s,j} I_q$, inversion and compression give $\mathcal{Z}_{s,j}^{-1} H_0 \preceq H_j \preceq H_0$, where $H_0 = T_s^* T_s$. Define $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Congruence and all r eigenvalues give

$$0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}, \quad 0 \leq d_j \leq r \mathcal{L}_{s,j}, \quad F_K^{(s)} = d_q + d_{q+1} - d_1. \quad (\text{AKS27})$$

The identity uses precisely the four endpoints in AKS2. In particular it proves AKS4, as well as the sharper interval

$$d_1 \leq F_K^{(s)} \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1. \quad (\text{AKS28})$$

For a directly calculable expression for its lower endpoint set $V = \|v_0\|^2$ and $\theta = v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0$. Then

$$0 \leq \theta \leq V, \quad d_1 = \log \frac{1 + V}{1 + V - \theta}. \quad (\text{AKS29})$$

Indeed the inverse of $I + v_0 v_0^*$ is $I - v_0 v_0^*/(1 + V)$; take its full compressed determinant. For $r = 0$ set $\theta = d_1 = 0$. In GMB/OPR notation $V = t_0$ and $V - \theta = \xi_0$, so this is exactly the first original section-corrected kernel increment, not an independently chosen kernel observation.

Here is an evaluated finite remainder for AKS6. Put

$$\begin{aligned} a_\infty &= \frac{4 \log 3}{\pi}, \quad c_0 = \log \frac{25 \log 3}{4\pi} > 0, \\ E_{s,q}(R) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R \left(\log 3 + \frac{\pi}{4} \right) \\ &\quad + \log((q+1)(4q+1)) - 2 \log(a_\infty - 1). \end{aligned} \tag{AKS30}$$

Then for both source orders, every $q \geq 36$, and the same actual kernel,

$$\boxed{\mathcal{L}_{s,q}, \mathcal{L}_{s,q+1} \leq 2c_0 q + E_{s,q}(R), \quad 0 \leq F_K^{(s)} \leq 4rc_0 q + 2rE_{s,q}(R).} \tag{AKS31}$$

To verify every factor, $b_s \geq 1/2$ gives

$$\frac{d!}{(b_s)_d} \leq \frac{4^d}{\binom{2d}{d}} \leq 2d + 1.$$

The last inequality uses that the central binomial is the largest of the $2d + 1$ positive coefficients whose sum is 4^d . Thus the sum in AKS3 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also $\mathfrak{s}_q \leq \mathfrak{a}_q^q/(a_\infty - 1)$ and

$$2q \log \mathfrak{a}_q \leq 2q \log a_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first follows from $-\log(1-x) \leq x/(1-x)$ at $x = 2/(\pi q)$; the second is the chord bound for sine on $[0, \pi/2]$. Use $t_q \leq \pi/2$ in the remaining exponential. The order- q terms combine to $4q \log(5/4) + 2q \log a_\infty = 2c_0 q$. The upper bound for the logarithm of the added positive scalar is positive, so $\log(1 + e^x) \leq \log 2 + \max(0, x)$ proves the first bound in AKS31. AKS28 proves the second.

On the original simple family $q = (k+1)^2$, $R = O_h(k)$ and $r = 8k - 16$. Consequently $E_{1,q}(R) = O_h(k + \log q)$ and $E_{k,q}(R) = O_h(k \log q)$. Dividing AKS31 by kq proves AKS6, with $32c_0 = \kappa_*$. In fact the fixed-source upper bound is

$$F_K^{(1)} \leq (32k - 64)c_0 q + (16k - 32)E_{1,q}(R), \tag{AKS32}$$

whose explicit second term is $O_h(k^2)$. The tensor-source version has an $O_h(k^2 \log q)$ upper remainder. These are remainders of upper bounds, not assertions about the actual signed error or leading value. The proof is uniform in all the period parameters in the stated domain: no norm or condition number of a period matrix occurs in AKS30.

The precise earlier loss. OKV15 had already retained $\|v_j\|^2 \leq q^2 2^{6q} (2D_* q)^{2d} / (d!(b_s)_d)$, $d = q + j$. Replacing its denominator by $1/2$ caused the old $q \log q$ exponent. Even without AKS20–24 one obtains the valid repair

$$\|K_{q+1}\| \leq 1 + (q+1)q^2 2^{6q} \frac{(4D_* q)^{4q}}{(4q)!}, \quad D_* \geq 3. \tag{AKS33}$$

Here $d!(b_s)_d \geq (2d)!/4^d$ and the remaining factorial ratio increases up to $d = 2q$ for $D_* \geq 3$. The elementary inequality $\log n! \geq n \log n - n + 1$ already makes the logarithm of AKS33 $O_h(q)$. The Newton calculation strengthens this repair to the packet-independent limsup constant AKS5, while retaining R in its finite remainder.

Exact enclosure of the displayed numerical ceiling. For $x > 1$ and $z = (x - 1)/(x + 1)$ the integrated geometric series gives

$$0 \leq \log x - 2 \sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{81}}{81(1-z^2)}.$$

For $0 < x < 1$, the alternating sum through $j = 39$ for $\arctan x$ has error between zero and $x^{81}/81$. The tangent addition formula, with the angle in $(0, \pi/2)$, proves $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is exactly one. Use the resulting rational lower and upper bounds for π and for $\log 3$, then apply the first displayed bound to the two endpoints of $25 \log 3/(4\pi)$. Multiplication by 32 gives, with outward decimal rounding, the interval

$$\kappa_* \in [25.0207809764994871, 25.0207809764994872].$$

The accompanying rational-arithmetic implementation checks these inequalities by integer cross multiplication, rather than a floating-point logarithm. This proves the less finely rounded strict interval AKS5.

R42 The complete arithmetic and mixed receivers

R42.1 The unchanged finite arithmetic allowances

Use exactly AMT4's constants. To prevent a loss hidden in order notation, write them here:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - 9/2, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \\ D_N &= [1 + 4(N + M)^2]^M, \quad L_N^{\text{ar}} = \log(A_k D_N / a_k), \\ E_k^+ &= L_{q-1}^{\text{ar}} + L_q^{\text{ar}}, \quad E_k^- = L_{2q-1}^{\text{ar}} + L_{2q}^{\text{ar}}. \end{aligned} \tag{AKS34}$$

The constants c_h, C_h^{tilt} are the original TW/BT envelope constants, not new choices. Their actual inequalities and complete source construction are preserved in AMT4–6 and the current source closure. Minimize AMT6 on the same remainder fibre and then restrict to $I(K)$. AMT9–20 gives, with the original arithmetic source w_h^{*k} ,

$$-rE_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq rE_k^-.$$

Let $\underline{U}_s = d_1^{(s)}$ and $\overline{U}_s = r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}$. The new finite conclusion is

$$\boxed{\max\{0, \underline{U}_1 - rE_k^+\} \leq F_K^{\text{ar}} \leq \overline{U}_1 + rE_k^-} \tag{AKS35}$$

This improves AMT37's absolute upper bound. For the sharper correlated version retain $d_{E,N} = q \log A_k - \log(\det G_N^{\text{ar}} / \det G_N^{(1)})$ and $c_{X,N} = \min\{r_X L_N^{\text{ar}}, d_{E,N}\}$, where $r_K = r$, $r_B = q - r$. Put $C_X^- = c_{X,q-1} + c_{X,q}$, $C_X^+ = c_{X,2q-1} + c_{X,2q}$. AMT18 then gives

$$\begin{aligned} \max\{0, \underline{U}_1 - C_K^-, \underline{U}_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq \overline{U}_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \end{aligned} \tag{AKS36}$$

One may replace $d_{E,N}$ in the caps by the original full-source $\Xi_{k,N}$ or its proved finite majorant. The scalar $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$ retains exactly AMT21:

$$\begin{aligned} \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\} &\leq \delta_k^\sigma \\ &\leq \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}. \end{aligned} \quad (\text{AKS37})$$

Since $E_k^\pm = O_h(k + \log q)$ by the exact AMT22 expression, AKS35 and AKS32 prove the arithmetic version of AKS6, with an explicit $O_h(k^2)$ upper remainder. Nonnegativity follows from the actual nested source minima, as in AMT15, not from comparison with a possibly negative lower endpoint.

R42.2 The original Gamma multiplier, not a replacement higher source

Retain $t_j = 2j + 1/2$, $f_l(S) = \prod_{j=0}^{l-1}(S - c - t_j)$ and

$$dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad T_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + t_j^2]. \quad (\text{AKS38})$$

The exact source map is $\mathcal{P}_N \rightarrow f_l \mathcal{P}_N \subset \mathcal{P}_{N+l}$, $P \mapsto f_l P$, with inverse division on this specified subspace. Let E_χ be the complete divided-jet map and U_f its Leibniz multiplication matrix, $(U_f)_{r,a} = f_l^{(r-a)}(\lambda)/(r-a)!$ for $r \geq a$. AMT25–28 gives the precise transported observation and kernel

$$\begin{aligned} \tilde{\Lambda} &= \Lambda E_\chi^{-1} U_f^{-1}, \quad \tilde{I} = U_f E_\chi I, \quad \ker \tilde{\Lambda} = U_f E_\chi K, \\ C_N &= \mathcal{K}_{\chi, N+l} - \mathcal{K}_{\chi, f; N+l} \mathcal{K}_{f, N+l}^{-1} \mathcal{K}_{f, \chi; N+l}, \\ H_{K,N}^{(k)} &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}. \end{aligned} \quad (\text{AKS39})$$

Each covariance is the original finite sum of full polynomial jet outer products divided by its original monic norm. These formulas retain the extra f constraints and their full cross covariance. In particular $M_f K = K$ is not assumed. AKS26 calculates the same tensor metric directly in its own orthonormal source; the two representations agree by their identical source norm and identical constrained minimum.

Define

$$w_N = \log(U_{l,N}/T_l), \quad \Omega^{\text{low}} = w_{q-1} + w_q, \quad \Omega^{\text{high}} = w_{2q-1} + w_{2q}.$$

The already proved same-coefficient comparison AMT42–44 retains all mass factors before cancellation and gives

$$-r\Omega^{\text{low}} \leq \Delta_K^\Gamma := F_K^{(1)} - F_K^{(k)} \leq r\Omega^{\text{high}}. \quad (\text{AKS40})$$

It therefore sharpens the two independent absolute bounds by

$$\begin{aligned} \max\{\underline{U}_k, \underline{U}_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \\ &\leq \min\{\overline{U}_k, \overline{U}_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKS41})$$

The original finite estimate $0 \leq w_N \leq l \log[1 + 16(N + l)^2]$ proves that AKS40 is $o(kq)$ on the actual rank- r domain. Along with AKS35 this preserves the completed fact that all three source kernels differ by $o(kq)$. AKS4, not AKS40, is the new absolute-scale calculation.

R42.3 Boundary, QGQ and arithmetic receiving identities

The minimum and determinant factorization in AKS42 use the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)]. The fixed original kernel frame and right section below specify the complete block change of coordinates.

At each endpoint retain the original minimum section $S_N^{(s)} = (G_N^{(s)})^{-1} \Lambda^* (\Lambda (G_N^{(s)})^{-1} \Lambda^*)^{-1}$. The full Schur identity is

$$\det(I^* G_N^{(s)} I) \det Q_N^{(s)} = |\det[I, S_*]|^2 \det G_N^{(s)}, \quad Q_N^{(s)} = (\Lambda (G_N^{(s)})^{-1} \Lambda^*)^{-1}. \quad (\text{AKS42})$$

Thus the new finite absolute boundary interval is

$$\max\{0, \mathcal{B}_k^{(s)} - \overline{U}_s\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)} - \underline{U}_s. \quad (\text{AKS43})$$

The baselines in this formula are the *complete* original BSL/HBL objects, with their full finite relation errors. No q^2 coefficient is assigned to K ; its proved coefficient there remains zero.

Write $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k$. The exact full identities still read

$$\begin{aligned} F_B^{(1)} - F_B^{(k)} &= \mathcal{H}_k - \Delta_K^\Gamma, \\ \boxed{\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma.} \end{aligned} \quad (\text{AKS44})$$

The Gamma interval is not re-estimated in this argument. For either complete QRI1 or QRI2 interval $[\omega^-, \omega^+]$ for its already calculated centre W_k , retain its exact source errors $e_0 \in [-a_0, b_0]$ and $e_q + e_{q+1} \in [-b_k^+, b_k^-]$. Then

$$\underline{\mathcal{H}}_k = -\omega^+ - b_k^- - a_0, \quad \overline{\mathcal{H}}_k = -\omega^- + b_k^+ + b_0, \quad \mathcal{H}_k \in [\underline{\mathcal{H}}_k, \overline{\mathcal{H}}_k]. \quad (\text{AKS45})$$

These are exactly QRI3, not new unspecified error choices. The complete QRI1–9, QGT and QGQ definitions remain verbatim in each updated receiver; the new interval is intersected with all their retained tighter finite intervals. In particular the simple-quartet nonzero error radius stays

$$\limsup \left| \frac{\mathcal{H}_k + lq\mathcal{C}_\Gamma}{q} + \frac{\partial_\alpha \mathcal{L}(2, \pi)}{8} - \frac{\log 2}{4} \right| \leq \frac{A_{\text{QGT}}}{\sqrt{2}}. \quad (\text{AKS46})$$

No zero-radius or $o(q)$ arithmetic assertion is introduced.

For example the complete finite absolute defect relative to the tensor boundary, with the actual Gamma and arithmetic scalars retained, is

$$\underline{U}_k + \mathcal{H}_k + \delta_k^\sigma \leq \mathcal{B}_k^{\text{ar}} - F_B^{(k)} \leq \overline{U}_k + \mathcal{H}_k + \delta_k^\sigma. \quad (\text{AKS47})$$

Inserting AKS41 makes this sharper. Replacing $\mathcal{H}_k$ by the appropriate endpoints of AKS45 and δ_k^σ by AKS37 gives its fully inherited finite scalar enclosure with the same signs. Also $F_B^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}$; subtract the interval AKS35 or AKS36 to get its new absolute boundary interval. This retains the exact correlations rather than taking independent hypothetical source errors.

The existing limit is $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ and $\delta_k^\sigma/(kq) \rightarrow 0$. Therefore the new interval implies

$$\begin{aligned} \liminf_k \inf_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\geq -\mathcal{C}_\Gamma/4, \\ \limsup_k \sup_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\leq \kappa_* - \mathcal{C}_\Gamma/4. \end{aligned} \quad (\text{AKS48})$$

All original branches are included in the infima and suprema. This does not decide its sign. The already proved boundary *difference* limits of AMT49 are unchanged. Every subsequential limit of the absolute normalized kernel lies in $[0, \kappa_*]$ and is shared by its three source versions when they are evaluated on the same sequence of period parameters. Existence of a single limit is not proved by boundedness.

Finally the HC residual remains exactly $\mathcal{R}_k = \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k)$, with its original D_h and original nonnegative contraction, phase and trace terms. AKS44 changes none of them. The supplied positive q^2 baseline and its complete Gamma correction are not re-presented as new results here.

R43 Actions, support faces, and the actual arithmetic class

The coefficient formula AKS12 supplies all four blocks of the same sum action: $P_K Y P_K$, $P_K Y (I - P_K)$, $(I - P_K) Y P_K$ and $(I - P_K) Y (I - P_K)$. There is no claim that either off-diagonal block vanishes. In the actual minimum-section coordinates the corresponding formula remains

$$[I, S_N]^{-1} Y [I, S_N] = \begin{pmatrix} H_{K,N}^{-1} I^* G_N Y I & H_{K,N}^{-1} I^* G_N Y S_N \\ \Lambda Y I & \Lambda Y S_N \end{pmatrix}. \quad (\text{AKS49})$$

Its proof is multiplication by the inverse coordinate maps $H_{K,N}^{-1} I^* G_N$ and Λ . In the invariant-ring model, the same action is the derivation $\mathcal{D}f = (\mathbf{H} \Omega \mathbf{H}^{-1} x) \cdot \nabla f$, $\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta)$, with $\mathcal{D}Q = 0$. For the actual conductor $\mathcal{A}_Q = \prod_{a=1}^4 Q_a$, its full defect is $\mathcal{D}\mathcal{A}_Q = \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{b \neq a, 1 \leq b \leq 4} Q_b$, exactly OQC22. Thus this coefficient realization preserves the original action leaving the observed dual.

In particular the section-corrected vector is still OPR2's

$$C_s r_j^{\ker} = e_{q+j} - S_{q+j-1}^{(s)} \Lambda e_{q+j} = I(\zeta_{q+j} - \kappa S_{q+j-1}^{(s)} \Lambda e_{q+j}),$$

with its full denominator $1 + \xi_j$ in the kernel increment. The operator proof estimates the complete covariance and its actual restriction, so none of these coupled terms is replaced by a raw fixed-section vector. The all-iterate kernel stays $\bigcap_{a=0}^{q-1} \ker(\Lambda Y^a)$; nothing in AKS4 declares it zero.

On the existing arithmetic coefficient realization retain $\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V}_{h,k} = \eta \text{rem}_\chi$, where $\eta = U^{\otimes k} \iota$ is the original injection. AKS17 and the Newton coefficients compute precisely this same remainder. Any two polynomial lifts differ by χQ in the original relation space, so their original arithmetic cohomology classes agree. The specified primitive of the difference is EP.48, with the full differential and cochain signs proved in AKP3–5 below; it depends on the lift difference. The source comparison to w_h^{*k} is the identity on original polynomial coefficients, followed by the actual minima as in AMT8; it is not a replacement arithmetic source.

The scalar object stays $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $0_R^\bullet$, injective amplitude/support pair, and infinite integer target. Every coefficient map above is applied on each original present support label and outer leg. It commutes with the specified coefficient inclusions because both compositions apply the same linear map to the same coefficient; a present zero is not converted into absence. No new base object, purity condition, or identification of the sum action with a Frobenius action is used. The actual distinguished class and its higher-jet directions remain in the full fibre, even when the observation kills them.

The geometric source of the comparison problem. Pierre Deligne's work is a substantive source of the programme's weight and common-image comparison problem. Its precise geometric results include *La conjecture de Weil. I*, Theorem (1.6), p. 276, for smooth projective varieties over a finite field, with ℓ different from its characteristic, and *Weil. II*, Theorem 3.3.1, p. 204, and Corollaries 3.3.4–3.3.5, p. 206, on compact-support upper weights and the associated dual bounds [H17, H18]. The trace-formula framework recalled in *Weil. I*, §1.5, is Alexander Grothendieck's; Deligne also explicitly credits André Weil's conjectures, Bernard Dwork's rationality theorem and Nicholas M. Katz's notes in his account of the antecedents.

The present passage proves coefficient and arithmetic-class maps on the original analytic source. It leaves the further geometric comparison as a mathematical task: construct an actual target cohomology and common-image map, identify the action on the same class, and verify the geometric and weight hypotheses before applying the cited theorems. The finite coefficient identities above are retained as inputs to that task. A citation to Deligne is neither an assumption that the missing map exists nor a reason to discard that comparison problem.

R44 Exact period domain and the interior alternative

For reference the inherited radius used above is completely specified as follows; these are AMT33–34/RPC17–23, with no new constants. Put

$$\begin{aligned}
\beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\rho) \neq 0, \\
c_3 &= (a^{-2} - \bar{a}^{-2})/(4i\delta\gamma), \quad c_1 = (a^{-2} + \bar{a}^{-2})/2 - (\delta^2 - \gamma^2)c_3, \\
g_j &= 5^{j/5-1}e^{\pi ij/5}\Gamma(j/5), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{j < 4} |g_{j+1}/g_j|, \\
E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr, \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_* = \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta.
\end{aligned} \tag{AKS50}$$

Then

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_K B_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \tag{AKS51}$$

The actual unit implies $(c_1, c_3) \neq (0, 0)$ and all integrals converge. RPC’s full period comparison proves a nonzero $(4, 1)$ or $(2, 1)$ commutator entry, respectively, for $|u| \geq R_*$ on every branch. It therefore proves the five-member orbit needed in AKS11.

Inside this radius the same coefficient determinant AKS12–14 and the operator bound AKS24 remain valid. The orbit is decided by the original noninvariant quadratic coefficients, not by the radius alone. Since Q is nondegenerate, its projective orbit has either one or five members. A one-member orbit must have character one: its character has both fourth and fifth powers equal to one, by the quadratic determinant and $D^5 = I$. In that case $Q = a'x_1x_4 + b'x_2x_3$ with $a'b' \neq 0$, and

$$b = d_{5,k,0} - d_{5,k-2,0} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}, \quad r = q - b. \tag{AKS52}$$

If a noninvariant coefficient is nonzero, AKS11 applies instead. These are the complete OQC alternatives. The bound AKS31 always uses that actual r ; it does not extend the $8k - 16$ specialization to an unevaluated exceptional point. This calculation does not assert that the interior exceptional set is empty.

R45 Verification scope and the remaining absolute coefficient

The new mathematical content is AKS20–32 and its application to the actual coefficient determinant AKS12–14, followed by the finite receiving improvements AKS35–48. The complete source and orbit identifications used in that application are the supplied OCS/OQC/OQX, RPC, OPG/OPR and AMT results; their exact predecessors are preserved. AKS33 pinpoints a repair obtainable immediately from the earlier proof, whereas the Newton bound supplies the sharper explicit constant.

The bound *at* scale kq is now finite and uniform. What is not proved is a matching lower coefficient, a unique limit for $F_K^{(1)}/(kq)$, or a sign for the interval AKS48. Those questions involve the orientation of the original period kernel within the inverse remainder covariance; replacing that compression by a scalar bound loses precisely that information. This is the finite-bounds outcome allowed by the current request, not a claim to have evaluated an asymptotic coefficient. The source differences are already completed inputs and are not renamed as the remaining problem.

The accompanying exact checks distinguish finite algebraic diagnostics from the analytic proof for the full family. They do not assert that a diagnostic root is a zero of ξ , that a diagnostic matrix equals an actual period matrix, or that a new Lean theorem has been checked. No RH conclusion is drawn from this kernel bound.